

# Applied Anatomy, Physiology and Physics MCQ – Chapter 1

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 1: Fundamental Concepts (Chapter 1)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. The larynx extends from the root of the tongue to the trachea and lies opposite which vertebral levels?**

- A. C1 to C3
- B. C3 to C6
- C. C5 to T1
- D. C6 to T2

**Ans: B**

**Q2. What is the distance between the teeth and the vocal cords?**

- A. 12–15 cm
- B. 10–15 cm
- C. 16–20 cm
- D. 5–8 cm

**Ans: A**

**Q3. Which of the following is an UNPAIRED cartilage of the larynx?**

- A. Arytenoid
- B. Corniculate
- C. Cuneiform
- D. Cricoid

**Ans: D**

**Q4. Which is the narrowest part of the larynx in adults?**

- A. The glottis
- B. The subglottis
- C. The cricoid ring
- D. The epiglottis

**Ans: A**

**Q5. All muscles of the larynx are supplied by the recurrent laryngeal nerve EXCEPT which muscle?**

- A. Posterior cricoarytenoid
- B. Lateral cricoarytenoid
- C. Cricothyroid
- D. Thyroarytenoid

**Ans: C**

**Q6. Sensory supply above the vocal cords is provided by which nerve?**

- A. Recurrent laryngeal nerve
- B. Internal branch of superior laryngeal nerve
- C. External branch of superior laryngeal nerve
- D. Glossopharyngeal nerve

**Ans: B**

**Q7. In Semon's law, in bilateral partial paralysis of the recurrent laryngeal nerve, which muscles are affected first?**

- A. The adductors
- B. The cricothyroid
- C. The abductors (posterior cricoarytenoids)
- D. The tensors

**Ans: C**

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| <p><b>Q8. During general anesthesia with muscle relaxants, the vocal cords are held in which position?</b></p> <p>A. Fully adducted position<br/>B. Fully abducted position<br/>C. Cadaveric (mid) position<br/>D. Paramedian position</p> | <p><b>Ans: C</b></p> |
| <p><b>Q9. What is the length of the trachea?</b></p> <p>A. 5–8 cm<br/>B. 10–12 cm<br/>C. 15–18 cm<br/>D. 20–22 cm</p>                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q10. The trachea consists of how many incomplete rings?</b></p> <p>A. 10–12<br/>B. 12–14<br/>C. 16–20<br/>D. 22–24</p>                                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q11. What is the distance of the carina from the upper incisors?</b></p> <p>A. 28–30 cm<br/>B. 20–22 cm<br/>C. 15–18 cm<br/>D. 32–35 cm</p>                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q12. Up to what age is the angle of both right and left bronchus the same (55°) in children?</b></p> <p>A. 1 year<br/>B. 2 years<br/>C. 3 years<br/>D. 5 years</p>                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q13. What is the total number of alveoli and their surface area?</b></p> <p>A. 100 million, 40 m<sup>2</sup><br/>B. 300 million, 70 m<sup>2</sup><br/>C. 500 million, 100 m<sup>2</sup><br/>D. 200 million, 50 m<sup>2</sup></p>     | <p><b>Ans: B</b></p> |
| <p><b>Q14. Which cells of the alveolar epithelium secrete surfactant?</b></p> <p>A. Type I cells<br/>B. Type II cells<br/>C. Clara cells<br/>D. Goblet cells</p>                                                                           | <p><b>Ans: B</b></p> |

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| <p><b>Q15. According to Table 1.1, what is the length and angle of the right main bronchus?</b></p> <p>A. Length 5 cm, angle 45°<br/> B. Length 2.5 cm, angle 25°<br/> C. Length 2.5 cm, angle 45°<br/> D. Length 5 cm, angle 25°</p>                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q16. In the supine position, the most commonly involved segment in foreign body aspiration is:</b></p> <p>A. Posterior basal segment of right lower lobe<br/> B. Posterior segment of upper lobe<br/> C. Apical segment of right lower lobe<br/> D. Lateral basal segment of left lower lobe</p> | <p><b>Ans: C</b></p> |
| <p><b>Q17. Ciliary activity of the respiratory mucosa is inhibited by all inhalational agents EXCEPT:</b></p> <p>A. Halothane<br/> B. Ether<br/> C. Isoflurane<br/> D. Sevoflurane</p>                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q18. Sympathetic supply to the bronchial tree (T2 to T5) causes:</b></p> <p>A. Bronchoconstriction<br/> B. Bronchodilatation<br/> C. Increased secretions<br/> D. No effect on airway caliber</p>                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q19. The pneumotaxic center, which has an inhibitory effect on the apneustic center, is located in the:</b></p> <p>A. Upper pons<br/> B. Lower pons<br/> C. Medulla<br/> D. Midbrain</p>                                                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q20. Central respiratory centers are most sensitive to changes in:</b></p> <p>A. Arterial pO<sub>2</sub><br/> B. CSF pH (influenced by pCO<sub>2</sub>)<br/> C. Blood temperature<br/> D. Plasma bicarbonate</p>                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q21. Peripheral chemoreceptors present in the carotid body and aortic arch are very sensitive to:</b></p> <p>A. Hypercarbia<br/> B. Acidosis<br/> C. Hypoxia<br/> D. Hyperthermia</p>                                                                                                            | <p><b>Ans: C</b></p> |

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| <p><b>Q22. Which is the most important muscle of inspiration, moving 1.5 cm in quiet respiration and 6–10 cm in deep breathing?</b></p> <p>A. External intercostals<br/>B. Scalene muscles<br/>C. Diaphragm<br/>D. Sternomastoids</p> | Ans: C |
| <p><b>Q23. During anesthesia with inhalational agents, expiration becomes active and is mediated by which muscles?</b></p> <p>A. Abdominal muscles<br/>B. External intercostals<br/>C. Diaphragm<br/>D. Scalene muscles</p>           | Ans: A |
| <p><b>Q24. Maximum airway resistance to airflow occurs in which part of the airway?</b></p> <p>A. Terminal bronchi<br/>B. Trachea and then main bronchus<br/>C. Alveolar ducts<br/>D. Segmental bronchi</p>                           | Ans: B |
| <p><b>Q25. What is the normal average ventilation/perfusion (V/Q) ratio?</b></p> <p>A. 0.3<br/>B. 2.1<br/>C. 0.8<br/>D. 1.5</p>                                                                                                       | Ans: C |
| <p><b>Q26. The normal alveolar to arterial (A-a) oxygen difference is:</b></p> <p>A. 3–5 mm Hg<br/>B. 10–15 mm Hg<br/>C. 0.3–0.5 mm Hg<br/>D. 20–25 mm Hg</p>                                                                         | Ans: A |
| <p><b>Q27. What is the normal anatomical dead space (as a fraction of tidal volume)?</b></p> <p>A. 10% of tidal volume<br/>B. 30% of tidal volume<br/>C. 50% of tidal volume<br/>D. 60% of tidal volume</p>                           | Ans: B |
| <p><b>Q28. Anatomical dead space is DECREASED by which of the following?</b></p> <p>A. Neck extension<br/>B. Atropine<br/>C. Intubation and tracheostomy<br/>D. Anesthesia mask and circuits</p>                                      | Ans: C |

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| <p><b>Q29. What is the volume of alveolar dead space in the standing position?</b></p> <p>A. Zero<br/>B. 60–80 mL<br/>C. 150 mL<br/>D. 2 mL/kg</p>                                                                                                                                                                   | Ans: B |
| <p><b>Q30. Endotracheal tubes and tracheostomy affect anatomical dead space by:</b></p> <p>A. Increasing it by adding circuit volume<br/>B. Decreasing it by bypassing the upper airways<br/>C. Having no effect on it<br/>D. Doubling it</p>                                                                        | Ans: B |
| <p><b>Q31. What is the normal oxygen uptake?</b></p> <p>A. 1000 mL/min<br/>B. 250 mL/min<br/>C. 500 mL/min<br/>D. 100 mL/min</p>                                                                                                                                                                                     | Ans: B |
| <p><b>Q32. What volume of oxygen does 1 gram of hemoglobin carry?</b></p> <p>A. 0.003 mL<br/>B. 1.34 mL<br/>C. 20 mL<br/>D. 0.34 mL</p>                                                                                                                                                                              | Ans: B |
| <p><b>Q33. At a pO<sub>2</sub> of 60 mm Hg, the oxygen saturation of hemoglobin is approximately:</b></p> <p>A. 97%<br/>B. 90%<br/>C. 75%<br/>D. 50%</p>                                                                                                                                                             | Ans: B |
| <p><b>Q34. What is the P<sub>50</sub> (partial pressure at which oxygen saturation is 50%) and is it affected by anesthetics?</b></p> <p>A. 26 mm Hg; not affected by anesthetics<br/>B. 40 mm Hg; affected by anesthetics<br/>C. 60 mm Hg; not affected by anesthetics<br/>D. 50 mm Hg; affected by anesthetics</p> | Ans: A |
| <p><b>Q35. According to the Bohr effect, alkalosis shifts the oxygen dissociation curve:</b></p> <p>A. To the right<br/>B. To the left<br/>C. Upward<br/>D. Downward</p>                                                                                                                                             | Ans: B |

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| <p><b>Q36. Which of the following shifts the oxygen dissociation curve to the RIGHT?</b></p> <ul style="list-style-type: none"> <li>A. Carbon monoxide poisoning</li> <li>B. Hypothermia</li> <li>C. Inhalational anesthetics</li> <li>D. Decreased 2,3 DPG</li> </ul>                                                                                      | <p><b>Ans: C</b></p> |
| <p><b>Q37. What is the oxygen flux (amount of oxygen leaving the left ventricle per minute)?</b></p> <ul style="list-style-type: none"> <li>A. 250 mL/min</li> <li>B. 1000 mL/min</li> <li>C. 500 mL/min</li> <li>D. 5000 mL/min</li> </ul>                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q38. In what form is the major portion of carbon dioxide transported in blood?</b></p> <ul style="list-style-type: none"> <li>A. As bicarbonate (90%)</li> <li>B. Dissolved (5%)</li> <li>C. As carbamino compounds</li> <li>D. Attached to hemoglobin as carboxyhemoglobin</li> </ul>                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q39. Tracheal tug, a downward movement of the trachea during deep inspiration, is seen in:</b></p> <ul style="list-style-type: none"> <li>A. Flail chest</li> <li>B. Deep anesthesia with inhalational agents and partially curarized patients</li> <li>C. Hiccups</li> <li>D. Bronchospasm</li> </ul>                                                | <p><b>Ans: B</b></p> |
| <p><b>Q40. Hypoxic pulmonary vasoconstriction (HPV) is maximally blunted by which inhalational agent?</b></p> <ul style="list-style-type: none"> <li>A. Ether</li> <li>B. Nitrous oxide</li> <li>C. Halothane</li> <li>D. Sevoflurane</li> </ul>                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q41. What is the normal tidal volume?</b></p> <ul style="list-style-type: none"> <li>A. 400–500 mL (10 mL/kg)</li> <li>B. 1200–1500 mL</li> <li>C. 2400–2600 mL</li> <li>D. 100–150 mL</li> </ul>                                                                                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q42. Functional residual capacity (FRC) is the sum of which two volumes?</b></p> <ul style="list-style-type: none"> <li>A. Inspiratory reserve volume + tidal volume</li> <li>B. Tidal volume + expiratory reserve volume</li> <li>C. Expiratory reserve volume + residual volume</li> <li>D. Inspiratory reserve volume + residual volume</li> </ul> | <p><b>Ans: C</b></p> |

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| <p><b>Q43. During general anesthesia, the functional residual capacity (FRC) decreases by approximately:</b></p> <ul style="list-style-type: none"> <li>A. 5%</li> <li>B. 15–20%</li> <li>C. 30–40%</li> <li>D. 50%</li> </ul>                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q44. Which lung volumes/capacities CANNOT be measured by spirometry?</b></p> <ul style="list-style-type: none"> <li>A. Tidal volume and vital capacity</li> <li>B. Inspiratory and expiratory reserve volume</li> <li>C. Residual volume, functional residual capacity and total lung capacity</li> <li>D. Inspiratory capacity</li> </ul>   | <p><b>Ans: C</b></p> |
| <p><b>Q45. A normal breath-holding time on the simple bedside test is:</b></p> <ul style="list-style-type: none"> <li>A. &gt; 25 seconds</li> <li>B. 15–25 seconds (borderline)</li> <li>C. &lt; 15 seconds</li> <li>D. 5–10 seconds</li> </ul>                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q46. Closing capacity (the volume at which the airway closes) is normally:</b></p> <ul style="list-style-type: none"> <li>A. 1 litre more than functional residual capacity</li> <li>B. 1 litre less than functional residual capacity</li> <li>C. Equal to functional residual capacity</li> <li>D. Equal to total lung capacity</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q47. What is the normal total lung compliance (lung and chest wall acting in opposite directions)?</b></p> <ul style="list-style-type: none"> <li>A. 0.2 L/cm H<sub>2</sub>O</li> <li>B. 0.1 L/cm H<sub>2</sub>O</li> <li>C. 0.4 L/cm H<sub>2</sub>O</li> <li>D. 2.5 L/cm H<sub>2</sub>O</li> </ul>                                          | <p><b>Ans: B</b></p> |
| <p><b>Q48. According to Boyle's law, at a constant temperature the volume of a gas is:</b></p> <ul style="list-style-type: none"> <li>A. Directly proportional to pressure</li> <li>B. Inversely proportional to pressure</li> <li>C. Directly proportional to temperature</li> <li>D. Inversely proportional to temperature</li> </ul>            | <p><b>Ans: B</b></p> |
| <p><b>Q49. Laminar flow is more dependent on which property, according to the Hagen-Poiseuille law?</b></p> <ul style="list-style-type: none"> <li>A. Density</li> <li>B. Temperature</li> <li>C. Viscosity</li> <li>D. Molecular weight</li> </ul>                                                                                                | <p><b>Ans: C</b></p> |



**Q50. The Poynting effect explains which phenomenon used in the Entonox cylinder?**

- A. Mixing of liquid nitrous oxide at high pressure with oxygen leads to formation of gaseous nitrous oxide
- B. Oxygen liquefies when mixed with nitrous oxide
- C. Nitrous oxide separates from oxygen at high pressure
- D. Both gases remain in liquid state in the cylinder

**Ans: A**

# Answer Key & Explanations

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## Q1. Answer: B — C3 to C6

Page 3: The larynx is the organ of voice extending from the root of the tongue to the trachea and lies opposite C3 to C6.

Topic: Airway Anatomy

## Q2. Answer: A — 12–15 cm

Page 3: The distance between teeth and vocal cords is 12–15 cm, and the distance between vocal cords and carina is 10–15 cm.

Topic: Airway Anatomy

## Q3. Answer: D — Cricoid

Page 3: The larynx consists of 3 paired cartilages (arytenoid, corniculate, cuneiform) and 3 unpaired cartilages (thyroid, cricoid and epiglottis).

Topic: Airway Anatomy

## Q4. Answer: A — The glottis

Page 3: The glottis is the narrowest part in adults, while the subglottis (at the level of cricoid) is the narrowest part in children up to the age of 6 years.

Topic: Airway Anatomy

## Q5. Answer: C — Cricothyroid

Page 3: All muscles of the larynx are supplied by the recurrent laryngeal nerve except cricothyroid, which is supplied by the external branch of the superior laryngeal nerve.

Topic: Larynx Nerve Supply

## Q6. Answer: B — Internal branch of superior laryngeal nerve

Page 3: Sensory supply up to the vocal cords is by the internal branch of the superior laryngeal nerve, and below the vocal cords by the recurrent laryngeal nerve.

Topic: Larynx Nerve Supply

## Q7. Answer: C — The abductors (posterior cricoarytenoids)

Page 4: In bilateral partial paralysis of recurrent laryngeal nerve, the abductors (posterior cricoarytenoids) are affected first (Semon's Law), causing the cords to be held in adduction producing respiratory stridor.

Topic: Laryngeal Nerve Paralysis

## Q8. Answer: C — Cadaveric (mid) position

Page 4: In complete paralysis of both recurrent and superior laryngeal nerves, the cords are held in mid-position (cadaveric position). The cords are also in cadaveric position during anesthesia with muscle relaxants.

Topic: Laryngeal Nerve Paralysis

## Q9. Answer: B — 10–12 cm

Page 4: The length of the trachea is 10–12 cm. It starts from the cricoid ring (C6) and ends at the carina (T5).

Topic: Trachea

**Q10. Answer: C — 16–20**

Page 4: The trachea consists of 16–20 incomplete rings. The diameter of the trachea is 1.2 cm.

Topic: Trachea

**Q11. Answer: A — 28–30 cm**

Page 4: At the carina the trachea divides into right and left main bronchus. The distance of the carina from the upper incisors is 28–30 cm.

Topic: Bronchial Tree

**Q12. Answer: C — 3 years**

Page 4: In children the angle of both right and left bronchus is the same, i.e. 55° up to the age of 3 years. After this the right bronchus becomes shorter, wider and less acute.

Topic: Bronchial Tree

**Q13. Answer: B — 300 million, 70 m<sup>2</sup>**

Page 4: The total number of alveoli is 300 million with a surface area of 70 m<sup>2</sup>. The thickness of the alveolar capillary membrane is 0.3 mm.

Topic: Bronchial Tree

**Q14. Answer: B — Type II cells**

Page 4: The alveolar epithelium has type I and type II cells. Type II cells secrete surfactant.

Topic: Bronchial Tree

**Q15. Answer: B — Length 2.5 cm, angle 25°**

Page 5, Table 1.1: The right bronchus is shorter (length 2.5 cm), wider, and the angle with vertical is 25°. The left bronchus is longer (5 cm), narrower, with an angle of 45°.

Topic: Bronchial Tree

**Q16. Answer: C — Apical segment of right lower lobe**

Page 4: Due to the shorter, wider and less acute angle of the right bronchus, foreign body aspiration is more common on the right side. In supine position the most commonly involved segment is the apical segment of right lower lobe.

Topic: Foreign Body Aspiration

**Q17. Answer: B — Ether**

Page 5: The mucosa has ciliated epithelium up to terminal bronchioles. Ciliary activity is inhibited by all inhalational agents except ether.

Topic: Respiratory Mucosa

**Q18. Answer: B — Bronchodilatation**

Page 6: Nerve supply is parasympathetic by vagus (causes bronchoconstriction) and sympathetic by T2 to T5 (causes bronchodilatation).

Topic: Bronchial Nerve Supply

**Q19. Answer: A — Upper pons**

Page 6: The pneumotaxic center is in the upper pons and the apneustic center is in the lower pons. The pneumotaxic center normally has an inhibitory effect on the apneustic center.

Topic: Regulation of Respiration

**Q20. Answer: B — CSF pH (influenced by pCO<sub>2</sub>)**

Page 6: Central respiratory centers are most sensitive to changes in CSF pH, which is in turn influenced by pCO<sub>2</sub>. Increase in pCO<sub>2</sub> stimulates respiration while decrease in pCO<sub>2</sub> inhibits respiration.

*Topic: Regulation of Respiration*

**Q21. Answer: C — Hypoxia**

Page 6: Peripheral chemoreceptors are present in the carotid body and aortic arch. Carotid body receptors are very sensitive to hypoxia.

*Topic: Regulation of Respiration*

**Q22. Answer: C — Diaphragm**

Page 6: The diaphragm is the most important muscle of inspiration (moves 1.5 cm in quiet respiration and 6–10 cm in deep breathing).

*Topic: Muscles of Respiration*

**Q23. Answer: A — Abdominal muscles**

Page 6: Expiration is normally passive. During anesthesia with inhalational agents, expiration is active, mediated by abdominal muscles.

*Topic: Muscles of Respiration*

**Q24. Answer: B — Trachea and then main bronchus**

Page 6: Maximum airway resistance to airflow occurs in trachea and then main bronchus, and minimum in terminal bronchi. Turbulent flow (seen in trachea and main bronchi) resistance is proportional to the square of flow rates.

*Topic: Airway Resistance*

**Q25. Answer: C — 0.8**

Page 6: Both ventilation and perfusion are more at bases compared to apex, but perfusion at base is comparatively higher, decreasing V/Q ratio towards the base (from 2.1 at apex to 0.3 at base, average 0.8).

*Topic: Ventilation/Perfusion*

**Q26. Answer: A — 3–5 mm Hg**

Page 6: The V/Q mismatch creates alveolar to arterial [(A-a) pO<sub>2</sub> difference] which is normally 3–5 mm Hg. This A-a difference is increased in lung pathologies affecting alveoli.

*Topic: Ventilation/Perfusion*

**Q27. Answer: B — 30% of tidal volume**

Page 7: Anatomical dead space is constituted by air present in nose, trachea and bronchial tree (up to terminal bronchioles). Normally it is 30% of tidal volume or 2 mL/kg or 150 mL.

*Topic: Dead Space*

**Q28. Answer: C — Intubation and tracheostomy**

Page 7: Anatomical dead space is decreased by intubation (nasal cavity bypassed if tube diameter is less than airway diameter), tracheostomy (upper airways and nasal cavity bypassed), hyperventilation, neck flexion and bronchoconstrictors. It is increased by old age, neck extension, jaw protrusion, bronchodilators, atropine, and anesthesia mask/circuits.

*Topic: Dead Space*

**Q29. Answer: B — 60–80 mL**

Page 7: Alveolar dead space is constituted by alveoli which are only ventilated but not perfused. It is 60–80 mL in standing position and zero in lying position (perfusion is equal in all parts of lung when lying).

Topic: Dead Space

**Q30. Answer: B — Decreasing it by bypassing the upper airways**

Page 7: Endotracheal tubes and tracheostomy decrease the anatomical dead space by bypassing the upper airways. However, all anesthesia circuits, masks and humidifiers increase the anatomical dead space.

Topic: Anesthesia and Dead Space

**Q31. Answer: B — 250 mL/min**

Page 7: Normal oxygen uptake is 250 mL/min. Oxygen is mainly carried in blood attached to hemoglobin (1 g of Hb carries 1.34 mL of oxygen).

Topic: Oxygen in Blood

**Q32. Answer: B — 1.34 mL**

Page 7: 1 g of Hb carries 1.34 mL of oxygen. A very small amount, 0.003 mL/dL/mm Hg, is carried as dissolved fraction. Oxygen content of arterial blood is 20 mL/dL and that of venous blood is 15 mL/dL.

Topic: Oxygen in Blood

**Q33. Answer: B — 90%**

Page 7: Normally Hb is 97% saturated at normal pO<sub>2</sub> of 95–98 mm Hg. At 60 mm Hg saturation is still 90%. After this point there is a sudden drop in oxygen saturation (cyanosis appears when pO<sub>2</sub> falls below 50–60 mm Hg).

Topic: Oxygen Dissociation Curve

**Q34. Answer: A — 26 mm Hg; not affected by anesthetics**

Page 8: P<sub>50</sub> is the partial pressure at which oxygen saturation is 50%. The partial pressure of oxygen for 50% saturation is 26 mm Hg. P<sub>50</sub> is not affected by anesthetics.

Topic: Oxygen Dissociation Curve

**Q35. Answer: B — To the left**

Page 8: Bohr effect — alkalosis shifts the O<sub>2</sub> dissociation curve to the left and acidosis to the right.

Topic: Oxygen Dissociation Curve

**Q36. Answer: C — Inhalational anesthetics**

Page 8: Shift to the right is seen with acidosis, high pCO<sub>2</sub>, increased 2,3 DPG, hyperthermia and inhalational anesthetics. Shift to the left is seen with alkalosis, low pCO<sub>2</sub>, decreased 2,3 DPG, carbon monoxide poisoning, abnormal hemoglobins, hypophosphatemia and hypothermia.

Topic: Oxygen Dissociation Curve

**Q37. Answer: B — 1000 mL/min**

Page 8: Oxygen flux is the amount of oxygen leaving the left ventricle per minute. It is 1,000 mL/min.

Topic: Oxygen in Blood

**Q38. Answer: A — As bicarbonate (90%)**

Page 8: Carbon dioxide is transported in blood as bicarbonate (90%), dissolved (5%), and as carbamino compounds. Deoxygenated blood has more CO<sub>2</sub> content at a given pCO<sub>2</sub> — this is called the Haldane effect.

Topic: Carbon Dioxide

**Q39. Answer: B — Deep anesthesia with inhalational agents and partially curarized patients**

Page 8: Tracheal tug is a downward movement of trachea during deep inspiration. It is seen in deep anesthesia (by inhalational agents) and in partially curarized patients.

Topic: Abnormalities of Chest Movements

**Q40. Answer: C — Halothane**

Page 9: HPV is a protective mechanism — whenever there is hypoxia, vasoconstriction occurs shunting blood to well-perfused areas. All inhalational agents blunt this hypoxic pulmonary vasoconstriction (maximum with halothane).

Topic: Hypoxic Pulmonary Vasoconstriction

**Q41. Answer: A — 400–500 mL (10 mL/kg)**

Page 9: Tidal volume is the volume of gas inspired or expired in each breath during normal quiet respiration. It is 400–500 mL (10 mL/kg).

Topic: Lung Volumes

**Q42. Answer: C — Expiratory reserve volume + residual volume**

Page 9: Functional residual capacity (FRC) is the volume of gas in lungs after end expiration. It is ERV + RV and equals 2400–2600 mL. During general anesthesia FRC decreases by 15–20%.

Topic: Lung Volumes

**Q43. Answer: B — 15–20%**

Page 9: During general anesthesia FRC decreases by 15–20%. FRC is the volume of gas in lungs after end expiration (ERV + RV).

Topic: Lung Volumes

**Q44. Answer: C — Residual volume, functional residual capacity and total lung capacity**

Page 10: Spirometry cannot measure residual volume; therefore any lung volume which requires measurement of residual volume — i.e. functional residual capacity and total lung capacity — cannot be measured by spirometry. Body plethysmography, helium dilution and nitrogen washout are used for these.

Topic: Pulmonary Function Tests

**Q45. Answer: A — > 25 seconds**

Page 9–10: Breath-holding time is a very simple and useful bedside test. Normal is > 25 seconds. Patients with breath-holding time of 15–25 seconds are considered borderline cases and breath holding time < 15 seconds indicates severe pulmonary dysfunction.

Topic: Simple Bedside Test

**Q46. Answer: B — 1 litre less than functional residual capacity**

Page 10: Closing capacity is the volume at which airway closes, which is normally 1 litre less than functional residual capacity. If FRC falls below closing capacity there will be significant hypoventilation and V/Q abnormalities.

Topic: Pulmonary Function Tests

**Q47. Answer: B — 0.1 L/cm H<sub>2</sub>O**

Page 10: Lung compliance is 0.2 L/cm H<sub>2</sub>O, chest wall compliance is 0.2 L/cm H<sub>2</sub>O, and total compliance is 0.1 L/cm H<sub>2</sub>O (lung and chest wall compliance act in opposite direction).

Topic: Lung Compliance

**Q48. Answer: B — Inversely proportional to pressure**

Page 11: Boyle's law states that at a constant temperature, the volume of a gas is inversely proportional to the pressure. Charles's law states that at constant pressure, volume of gas is directly proportional to temperature.

*Topic: Physics Related to Anesthesia*

**Q49. Answer: C — Viscosity**

Page 11: Laminar flow is produced when gas passes through a straight tube and is smooth. Laminar flow is more dependent on viscosity. At laminar flow, Hagen-Poiseuille law applies (flow rate directly proportional to pressure gradient and fourth power of radius, inversely proportional to viscosity and length). Turbulent flow is more dependent on density.

*Topic: Physics Related to Anesthesia*

**Q50. Answer: A — Mixing of liquid nitrous oxide at high pressure with oxygen leads to formation of gaseous nitrous oxide**

Page 11: The Poynting effect — mixing of liquid nitrous oxide at low pressure with oxygen at high pressure (in Entonox) leads to formation of gas of nitrous oxide. Therefore oxygen and nitrous oxide are both present in gaseous state in the Entonox cylinder.

*Topic: Physics Related to Anesthesia*

# Chapters 1–5 Combined MCQ

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Sections 1–3: Applied Anatomy, Equipment & Quintessential (Chapters 1–5)

**100 Questions • 60 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

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**Q1. The larynx extends from the root of the tongue to the trachea and lies opposite which vertebral levels?**

- A. C5 to T1
- B. C3 to C6
- C. C6 to T2
- D. C1 to C3

**Ans: B**

**Q2. All muscles of the larynx are supplied by the recurrent laryngeal nerve EXCEPT which muscle?**

- A. Thyroarytenoid
- B. Cricothyroid
- C. Posterior cricoarytenoid
- D. Lateral cricoarytenoid

**Ans: B**

**Q3. In Semon's law, in bilateral partial paralysis of the recurrent laryngeal nerve, which muscles are affected first?**

- A. The abductors (posterior cricoarytenoids)
- B. The tensors
- C. The cricothyroid
- D. The adductors

**Ans: A**

**Q4. What is the total number of alveoli and their surface area?**

- A. 200 million, 50 m<sup>2</sup>
- B. 300 million, 70 m<sup>2</sup>
- C. 100 million, 40 m<sup>2</sup>
- D. 500 million, 100 m<sup>2</sup>

**Ans: B**

**Q5. The pneumotaxic center, which has an inhibitory effect on the apneustic center, is located in the:**

- A. Lower pons
- B. Medulla
- C. Midbrain
- D. Upper pons

**Ans: D**

**Q6. What is the normal average ventilation/perfusion (V/Q) ratio?**

- A. 2.1
- B. 1.5
- C. 0.8
- D. 0.3

**Ans: C**

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| <p><b>Q7. What is the normal anatomical dead space as a fraction of tidal volume?</b></p> <p>A. 50% of tidal volume<br/>B. 30% of tidal volume<br/>C. 10% of tidal volume<br/>D. 60% of tidal volume</p>                                                                                         | Ans: B |
| <p><b>Q8. What is the P50 (partial pressure at which oxygen saturation is 50%) and is it affected by anesthetics?</b></p> <p>A. 40 mm Hg; affected by anesthetics<br/>B. 50 mm Hg; affected<br/>C. 26 mm Hg; not affected by anesthetics<br/>D. 60 mm Hg; not affected</p>                       | Ans: C |
| <p><b>Q9. Functional residual capacity (FRC) is the sum of which two volumes?</b></p> <p>A. Inspiratory reserve volume + tidal volume<br/>B. Expiratory reserve volume + residual volume<br/>C. Tidal volume + expiratory reserve volume<br/>D. Inspiratory reserve volume + residual volume</p> | Ans: B |
| <p><b>Q10. According to Boyle's law, at a constant temperature the volume of a gas is:</b></p> <p>A. Directly proportional to temperature<br/>B. Inversely proportional to temperature<br/>C. Directly proportional to pressure<br/>D. Inversely proportional to pressure</p>                    | Ans: D |
| <p><b>Q11. In children, which is the narrowest part of the larynx?</b></p> <p>A. The subglottis (at the level of cricoid)<br/>B. The epiglottis<br/>C. The vocal cords<br/>D. The glottis</p>                                                                                                    | Ans: A |
| <p><b>Q12. In complete paralysis of both recurrent and superior laryngeal nerves, the vocal cords are held in which position?</b></p> <p>A. Fully abducted position<br/>B. Cadaveric (mid) position<br/>C. Paramedian position<br/>D. Fully adducted position</p>                                | Ans: B |
| <p><b>Q13. The trachea starts from the cricoid ring (C6) and ends at the carina, which corresponds to which vertebral level?</b></p> <p>A. T7<br/>B. T3<br/>C. T2<br/>D. T5</p>                                                                                                                  | Ans: D |

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| <p><b>Q14. The right main bronchus compared to the left is:</b></p> <ul style="list-style-type: none"> <li>A. Same length, same width, angle 55°</li> <li>B. Longer (5 cm), wider, angle 25°</li> <li>C. Shorter (2.5 cm), wider, angle 25°</li> <li>D. Longer, narrower, angle 45°</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q15. Ciliary activity of the respiratory mucosa is inhibited by all inhalational agents EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Ether</li> <li>D. Sevoflurane</li> </ul>                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q16. Peripheral chemoreceptors present in the carotid body are very sensitive to:</b></p> <ul style="list-style-type: none"> <li>A. Hyperthermia</li> <li>B. Hypercarbia</li> <li>C. Hypoxia</li> <li>D. Acidosis</li> </ul>                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q17. Maximum airway resistance to airflow occurs in which part of the airway?</b></p> <ul style="list-style-type: none"> <li>A. Segmental bronchi</li> <li>B. Trachea and then main bronchus</li> <li>C. Alveolar ducts</li> <li>D. Terminal bronchi</li> </ul>                          | <p><b>Ans: B</b></p> |
| <p><b>Q18. The normal alveolar to arterial (A-a) oxygen difference is:</b></p> <ul style="list-style-type: none"> <li>A. 10–15 mm Hg</li> <li>B. 0.3–0.5 mm Hg</li> <li>C. 3–5 mm Hg</li> <li>D. 20–25 mm Hg</li> </ul>                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q19. What is the volume of alveolar dead space in the standing position?</b></p> <ul style="list-style-type: none"> <li>A. 2 mL/kg</li> <li>B. 60–80 mL</li> <li>C. 150 mL</li> <li>D. Zero</li> </ul>                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q20. How much oxygen does 1 gram of hemoglobin carry?</b></p> <ul style="list-style-type: none"> <li>A. 0.003 mL</li> <li>B. 1.34 mL</li> <li>C. 20 mL</li> <li>D. 0.34 mL</li> </ul>                                                                                                    | <p><b>Ans: B</b></p> |

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| <p><b>Q21. At a pO<sub>2</sub> of 60 mm Hg, the oxygen saturation of hemoglobin is approximately:</b></p> <p>A. 97%<br/>B. 75%<br/>C. 50%<br/>D. 90%</p>                                                                                                 | Ans: D |
| <p><b>Q22. Carbon dioxide in blood is mainly transported as:</b></p> <p>A. Dissolved form (5%)<br/>B. Carboxyhemoglobin<br/>C. Bicarbonate (90%)<br/>D. Carbamino compounds</p>                                                                          | Ans: C |
| <p><b>Q23. During general anesthesia, the functional residual capacity (FRC) decreases by approximately:</b></p> <p>A. 15–20%<br/>B. 50%<br/>C. 30–40%<br/>D. 5%</p>                                                                                     | Ans: A |
| <p><b>Q24. Closing capacity is normally how much less than functional residual capacity?</b></p> <p>A. 1 litre<br/>B. 3 litres<br/>C. 0.5 litre<br/>D. 2 litres</p>                                                                                      | Ans: A |
| <p><b>Q25. Hypoxic pulmonary vasoconstriction (HPV) is maximally blunted by which inhalational agent?</b></p> <p>A. Sevoflurane<br/>B. Ether<br/>C. Halothane<br/>D. Nitrous oxide</p>                                                                   | Ans: C |
| <p><b>Q26. Which valve prevents delivery of a hypoxic gas mixture by cutting off or proportionately reducing nitrous oxide flow when oxygen pressure falls?</b></p> <p>A. Check valve<br/>B. Interlock valve<br/>C. APL valve<br/>D. Fail-safe valve</p> | Ans: D |
| <p><b>Q27. What minimum FiO<sub>2</sub> does the oxygen-nitrous proportioning system guarantee?</b></p> <p>A. 33%<br/>B. 25%<br/>C. 21%<br/>D. 15%</p>                                                                                                   | Ans: B |

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| <p><b>Q28. In the Mapleson classification, which circuit is the circuit of choice for spontaneous ventilation?</b></p> <p>A. Type F (Jackson-Rees)<br/>B. Type E (Ayre's T piece)<br/>C. Type D (Bain circuit)<br/>D. Type A (Magill circuit)</p>            | <p><b>Ans: D</b></p> |
| <p><b>Q29. The Bain circuit is a coaxial modification of which Mapleson type?</b></p> <p>A. Type D<br/>B. Type B<br/>C. Type A<br/>D. Type C</p>                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q30. Compound A, a nephrotoxic product, is produced by the reaction of which agent with sodalime and barylime?</b></p> <p>A. Halothane<br/>B. Desflurane<br/>C. Isoflurane<br/>D. Sevoflurane</p>                                                      | <p><b>Ans: D</b></p> |
| <p><b>Q31. Desiccated sodalime produces carbon monoxide most significantly with which agent?</b></p> <p>A. Desflurane<br/>B. Sevoflurane<br/>C. Isoflurane<br/>D. Halothane</p>                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q32. The standard traditional composition of sodalime includes Ca(OH) 94%, NaOH 5% and:</b></p> <p>A. Ba(OH) 20%<br/>B. MgOH 2%<br/>C. LiOH 5%<br/>D. KOH 1%</p>                                                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q33. The Jackson-Rees circuit (Mapleson F) is most commonly used for which patient population?</b></p> <p>A. Adults &gt; 70 kg<br/>B. Children &lt; 6 years or &lt; 20 kg<br/>C. Only for controlled ventilation in adults<br/>D. Elderly patients</p> | <p><b>Ans: B</b></p> |
| <p><b>Q34. The first anesthesia machine was invented by Edmund Gaskin Boyle in which year?</b></p> <p>A. 1917<br/>B. 1930<br/>C. 1907<br/>D. 1923</p>                                                                                                        | <p><b>Ans: A</b></p> |

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| <p><b>Q35. The color of the oxygen cylinder is:</b></p> <ul style="list-style-type: none"> <li>A. Black body with white shoulders</li> <li>B. Grey body with black shoulders</li> <li>C. Blue body with white shoulders</li> <li>D. White body with black shoulders</li> </ul>                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q36. Contents of a nitrous oxide cylinder are measured by:</b></p> <ul style="list-style-type: none"> <li>A. Volume displacement</li> <li>B. Weight (not by pressure gauge)</li> <li>C. Flow rate measurement</li> <li>D. Pressure gauge reading</li> </ul>                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q37. Entonox is a mixture of 50% oxygen and 50% nitrous oxide stored at what pressure?</b></p> <ul style="list-style-type: none"> <li>A. 1000 psi</li> <li>B. 760 psi</li> <li>C. 2000 psi</li> <li>D. 75 psi</li> </ul>                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q38. The pin index positions for oxygen on the yoke of the anesthesia machine are:</b></p> <ul style="list-style-type: none"> <li>A. 3, 5</li> <li>B. 2, 5</li> <li>C. 1, 5</li> <li>D. 2, 6</li> </ul>                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q39. The Diameter Index Safety System (DISS) is used to prevent:</b></p> <ul style="list-style-type: none"> <li>A. Wrong fitting of central supply pipelines to the machine</li> <li>B. Wrong cylinder placement on the yoke</li> <li>C. Backflow in breathing circuits</li> <li>D. Overfilling of gas cylinders</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q40. The oxygen flush (emergency oxygen flush) delivers oxygen at what rate and pressure?</b></p> <ul style="list-style-type: none"> <li>A. 50 L/min at 100 psi</li> <li>B. 35 L/min at 45–60 psi</li> <li>C. 20 L/min at 30 psi</li> <li>D. 10 L/min at 15 psi</li> </ul>                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q41. The 1st stage (primary) pressure regulator reduces cylinder pressure to:</b></p> <ul style="list-style-type: none"> <li>A. 100–150 psi</li> <li>B. 45–60 psi</li> <li>C. 15–35 psi</li> <li>D. 5–8 psi</li> </ul>                                                                                                      | <p><b>Ans: B</b></p> |

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| <p><b>Q42. The flowmeter tubes in a rotameter are also called:</b></p> <ul style="list-style-type: none"> <li>A. Magill tubes</li> <li>B. Heidbrink tubes</li> <li>C. Connell tubes</li> <li>D. Thorpe's tubes</li> </ul>                                                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q43. As a safety measure, the oxygen flow control knob on a rotameter is differentiated by being:</b></p> <ul style="list-style-type: none"> <li>A. Square shaped</li> <li>B. Same as other knobs but red colored</li> <li>C. More fluted, more prominent and larger in diameter</li> <li>D. Smaller and smooth</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q44. In a rotameter, the oxygen flowmeter tube should be positioned:</b></p> <ul style="list-style-type: none"> <li>A. Downstream (last in sequence)</li> <li>B. Upstream (first in sequence)</li> <li>C. Position does not matter</li> <li>D. In the middle</li> </ul>                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q45. Vaporizers are made of copper because copper possesses:</b></p> <ul style="list-style-type: none"> <li>A. High specific heat and high thermal conductivity</li> <li>B. Low cost and low density</li> <li>C. Resistance to corrosion only</li> <li>D. High elasticity</li> </ul>                                       | <p><b>Ans: A</b></p> |
| <p><b>Q46. Which vaporizer is specifically designed for desflurane due to its very low boiling point (23.5°C)?</b></p> <ul style="list-style-type: none"> <li>A. Tec 6</li> <li>B. Tec 5</li> <li>C. Goldman vaporizer</li> <li>D. Tec 7</li> </ul>                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q47. The color code for the dial knob on a halothane vaporizer is:</b></p> <ul style="list-style-type: none"> <li>A. Blue</li> <li>B. Purple</li> <li>C. Yellow</li> <li>D. Red</li> </ul>                                                                                                                                 | <p><b>Ans: D</b></p> |
| <p><b>Q48. When two vaporizers are mounted on the same machine, the agent with the LOWER boiling point should be placed:</b></p> <ul style="list-style-type: none"> <li>A. First (nearest to rotameter)</li> <li>B. It does not matter</li> <li>C. Second (downstream)</li> <li>D. After the check valve</li> </ul>              | <p><b>Ans: A</b></p> |

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| <p><b>Q49. In the open system of anesthesia delivery, which mask was used for pouring inhalational agent directly over the patient?</b></p> <p>A. Mary Carterall mask<br/>B. Venturi mask<br/>C. Hudson mask<br/>D. Schimmelbusch mask</p>                                                          | <p><b>Ans: D</b></p> |
| <p><b>Q50. The fresh gas flow for Bain circuit during spontaneous ventilation is:</b></p> <p>A. <math>1.6 \times</math> minute volume<br/>B. <math>2.5 \times</math> minute volume<br/>C. <math>3 \times</math> minute volume<br/>D. Equal to minute volume</p>                                     | <p><b>Ans: B</b></p> |
| <p><b>Q51. If Magill circuit is used for controlled ventilation, the required fresh gas flow is:</b></p> <p>A. <math>&gt; 3 \times</math> minute volume (unpredictable)<br/>B. <math>1.6 \times</math> minute volume<br/>C. <math>2.5 \times</math> minute volume<br/>D. Equal to minute volume</p> | <p><b>Ans: A</b></p> |
| <p><b>Q52. In human beings, the closed circuit system was first introduced by:</b></p> <p>A. Bain in 1972<br/>B. Jackson-Rees in 1950<br/>C. Waters in 1923<br/>D. Magill in 1920</p>                                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q53. The negative pressure (suction bulb) leak test of the anesthesia machine can detect leaks as small as:</b></p> <p>A. 100 mL<br/>B. 200 mL<br/>C. 50 mL<br/>D. 30 mL</p>                                                                                                                  | <p><b>Ans: D</b></p> |
| <p><b>Q54. The Laryngeal Mask Airway (LMA) is classified as which type of airway device?</b></p> <p>A. Infraglottic airway device<br/>B. Subglottic airway device<br/>C. Nasopharyngeal airway device<br/>D. Supraglottic airway device</p>                                                         | <p><b>Ans: D</b></p> |
| <p><b>Q55. The ProSeal LMA differs from the classic LMA in that it:</b></p> <p>A. Is made of PVC<br/>B. Is disposable only<br/>C. Can only be used in children<br/>D. Has a drain tube allowing passage of a gastric tube</p>                                                                       | <p><b>Ans: D</b></p> |



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| <p><b>Q56. The appropriate length of a Guedel airway is the distance between:</b></p> <ul style="list-style-type: none"> <li>A. Angle of mouth to earlobe</li> <li>B. Upper incisors to angle of mandible</li> <li>C. Tip of nose to tragus plus 1 cm</li> <li>D. Chin to thyroid notch</li> </ul>                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q57. Facemasks used in anesthesia are available in sizes from:</b></p> <ul style="list-style-type: none"> <li>A. 1 to 4</li> <li>B. 0 (smallest) to 6 (largest)</li> <li>C. 2 to 8</li> <li>D. 1 to 5</li> </ul>                                                                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q58. AMBU stands for:</b></p> <ul style="list-style-type: none"> <li>A. Automatic Mechanical Breathing Utility</li> <li>B. Artificial Manual Breathing Unit</li> <li>C. Assisted Mechanical Backup Unit</li> <li>D. Advanced Manual Breathing Unit</li> </ul>                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q59. Which is the most commonly used laryngoscope?</b></p> <ul style="list-style-type: none"> <li>A. Macintosh</li> <li>B. Bullard</li> <li>C. McCoy</li> <li>D. Miller</li> </ul>                                                                                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q60. For neonatal intubation, which laryngoscope blade is preferred and why?</b></p> <ul style="list-style-type: none"> <li>A. Bullard blade – for fiberoptic view</li> <li>B. Macintosh curved blade – for better tongue control</li> <li>C. Magill straight blade – neonatal epiglottis is large, leafy and more anterior</li> <li>D. McCoy blade – for movable tip</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q61. Video laryngoscopy as a rescue modality after failed direct laryngoscopy has reported intubation success rates of:</b></p> <ul style="list-style-type: none"> <li>A. 70–80%</li> <li>B. 94–99%</li> <li>C. 80–85%</li> <li>D. 50–60%</li> </ul>                                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q62. The most common complication of laryngoscopy involving dental injury primarily affects:</b></p> <ul style="list-style-type: none"> <li>A. Premolars</li> <li>B. Canines</li> <li>C. Lower incisors</li> <li>D. Upper incisors</li> </ul>                                                                                                                                    | <p><b>Ans: D</b></p> |

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| <p><b>Q63. The classical LMA was discovered by:</b></p> <ul style="list-style-type: none"> <li>A. Waters</li> <li>B. Archie Brain</li> <li>C. Magill</li> <li>D. Macintosh</li> </ul>                                                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q64. Which LMA is contraindicated in patients with a full stomach?</b></p> <ul style="list-style-type: none"> <li>A. Only first generation LMA</li> <li>B. All types of LMA</li> <li>C. Only ProSeal LMA</li> <li>D. Only I-Gel</li> </ul>                                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q65. The Intubating LMA (LMA Fastrach) can guide up to what size endotracheal tube?</b></p> <ul style="list-style-type: none"> <li>A. 9 no.</li> <li>B. 6 no.</li> <li>C. 8 no.</li> <li>D. 7 no.</li> </ul>                                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q66. The I-Gel supraglottic device differs from other LMAs because its cuff is:</b></p> <ul style="list-style-type: none"> <li>A. Made of silicone only</li> <li>B. Prefilled with gel, avoiding air cuff complications</li> <li>C. Air-filled with high volume</li> <li>D. Not present; it is cuffless</li> </ul>                                  | <p><b>Ans: B</b></p> |
| <p><b>Q67. The chief advantage of PVC endotracheal tubes over red rubber tubes is that PVC tubes have:</b></p> <ul style="list-style-type: none"> <li>A. Low pressure, high volume cuff causing less tracheal injury</li> <li>B. No cuff available</li> <li>C. Higher pressure, lower volume cuff</li> <li>D. Better resistance to autoclaving</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q68. For a 5-year-old child, the endotracheal tube size is calculated as:</b></p> <ul style="list-style-type: none"> <li>A. Age/2 + 3</li> <li>B. Age/4 + 4.5</li> <li>C. Age/3 + 3.5</li> <li>D. Age/4 + 4</li> </ul>                                                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q69. The cuff pressure of an endotracheal tube should be kept less than what value to prevent tracheal ischemia?</b></p> <ul style="list-style-type: none"> <li>A. 15 cm H<sub>2</sub>O</li> <li>B. 50 cm H<sub>2</sub>O</li> <li>C. 20 cm H<sub>2</sub>O</li> <li>D. 30 cm H<sub>2</sub>O</li> </ul>                                               | <p><b>Ans: D</b></p> |

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| <p><b>Q70. Capnography (measuring end tidal CO<sub>2</sub>) is considered the surest sign for confirming:</b></p> <ul style="list-style-type: none"> <li>A. Adequacy of muscle relaxation</li> <li>B. Depth of anesthesia</li> <li>C. Correct position of endotracheal tube</li> <li>D. Blood pressure stability</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q71. The maximum permissible time for which an endotracheal tube can be kept in situ is:</b></p> <ul style="list-style-type: none"> <li>A. 14 days</li> <li>B. 10 days</li> <li>C. 21 days</li> <li>D. 7 days</li> </ul>                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q72. The most common postoperative complication of intubation is:</b></p> <ul style="list-style-type: none"> <li>A. Sore throat (pharyngitis, laryngitis)</li> <li>B. Subglottic edema</li> <li>C. Vocal cord granuloma</li> <li>D. Tracheal stenosis</li> </ul>                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q73. Contraindications for nasal intubation include all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Oral surgery</li> <li>B. Nasal polyp</li> <li>C. Basal skull fractures</li> <li>D. Bleeding disorders</li> </ul>                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q74. Non-metallic anesthesia items such as endotracheal tubes and facemasks are best sterilized by:</b></p> <ul style="list-style-type: none"> <li>A. Ethylene oxide (ETO) gas sterilization</li> <li>B. Autoclaving</li> <li>C. UV radiation</li> <li>D. Boiling</li> </ul>                                          | <p><b>Ans: A</b></p> |
| <p><b>Q75. Heat and moisture exchanger (HME) filter is also called:</b></p> <ul style="list-style-type: none"> <li>A. Heated humidifier</li> <li>B. Active humidifier</li> <li>C. Artificial nose or condenser humidifier</li> <li>D. Nebulizer</li> </ul>                                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q76. ASA physical status classification I denotes:</b></p> <ul style="list-style-type: none"> <li>A. Severe systemic disease</li> <li>B. Incapacitating disease</li> <li>C. Mild systemic disease</li> <li>D. Normal healthy patient</li> </ul>                                                                       | <p><b>Ans: D</b></p> |

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| <p><b>Q77. In Mallampati Class I, which structures are visible?</b></p> <ul style="list-style-type: none"> <li>A. Soft palate only</li> <li>B. Faucial pillars, soft palate and uvula</li> <li>C. Soft palate and base of uvula</li> <li>D. Hard palate only</li> </ul>                                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q78. The recommended fasting period for clear fluids before elective surgery in adults is:</b></p> <ul style="list-style-type: none"> <li>A. 2 hours</li> <li>B. 4 hours</li> <li>C. 6 hours</li> <li>D. 1 hour</li> </ul>                                                                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q79. Which benzodiazepine is the drug of choice for premedication in current day practice?</b></p> <ul style="list-style-type: none"> <li>A. Diazepam</li> <li>B. Midazolam</li> <li>C. Lorazepam</li> <li>D. Chlordiazepoxide</li> </ul>                                                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q80. Glycopyrrolate is preferred over atropine as an anticholinergic premedicant because it:</b></p> <ul style="list-style-type: none"> <li>A. Causes more tachycardia</li> <li>B. Crosses the blood-brain barrier easily</li> <li>C. Does not cross the blood-brain barrier, so no central side effects</li> <li>D. Has longer duration of action on the heart</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q81. ASA classification VI refers to:</b></p> <ul style="list-style-type: none"> <li>A. Patient with severe systemic disease</li> <li>B. Moribund patient</li> <li>C. Emergency surgery patient</li> <li>D. Brain dead patients for organ donation</li> </ul>                                                                                                              | <p><b>Ans: D</b></p> |
| <p><b>Q82. The major limitation of the original ASA classification was that it did NOT include:</b></p> <ul style="list-style-type: none"> <li>A. Medical history</li> <li>B. Age, obesity, and risk associated with surgical procedure</li> <li>C. Physical examination findings</li> <li>D. Drug allergy status</li> </ul>                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q83. Standard dose estrogen-containing oral contraceptive pills should be stopped how long before surgery?</b></p> <ul style="list-style-type: none"> <li>A. 1 week</li> <li>B. 48 hours</li> <li>C. 2 weeks</li> <li>D. 4 weeks</li> </ul>                                                                                                                                | <p><b>Ans: D</b></p> |

|                                                                                                                                                                                                                                                                             |                      |
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| <p><b>Q84. Warfarin should be stopped how many days prior to elective surgery, and what INR value must be achieved?</b></p> <p>A. 7 days; INR &lt; 1.0<br/> B. 1 day; INR &lt; 2.5<br/> C. 2 days; INR &lt; 2.0<br/> D. 4 days; INR &lt; 1.5</p>                            | <p><b>Ans: D</b></p> |
| <p><b>Q85. Regarding aspirin in the perioperative period, the current recommendation is to:</b></p> <p>A. Replace with clopidogrel<br/> B. Continue aspirin as stopping carries risk of ischemia<br/> C. Stop 7 days before surgery<br/> D. Stop 14 days before surgery</p> | <p><b>Ans: B</b></p> |
| <p><b>Q86. Smoking should ideally be stopped how many weeks before surgery for restoration of ciliary activity?</b></p> <p>A. 3–4 weeks<br/> B. 12 weeks<br/> C. 1–2 weeks<br/> D. 6–8 weeks</p>                                                                            | <p><b>Ans: D</b></p> |
| <p><b>Q87. Antibiotic prophylaxis must be given within what time frame before skin incision?</b></p> <p>A. 60 minutes<br/> B. 30 minutes<br/> C. 2 hours<br/> D. 4 hours</p>                                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q88. The recommended fasting period for breast milk in neonates and infants is:</b></p> <p>A. 4 hours<br/> B. 8 hours<br/> C. 6 hours<br/> D. 2 hours</p>                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q89. Which of the following is a congenital cause of difficult intubation associated with hypoplastic mandible?</b></p> <p>A. Ludwig's angina<br/> B. Acromegaly<br/> C. Rheumatoid arthritis<br/> D. Pierre Robin syndrome</p>                                       | <p><b>Ans: D</b></p> |
| <p><b>Q90. The normal thyromental distance (with fully extended neck) is:</b></p> <p>A. 2 cm<br/> B. &lt; 4 cm<br/> C. 10 cm<br/> D. 6.5 cm or more (&gt; 3 finger breadths)</p>                                                                                            | <p><b>Ans: D</b></p> |

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| <p><b>Q91. Cormack-Lehane Grade I laryngoscopic view indicates:</b></p> <ul style="list-style-type: none"> <li>A. No recognizable structures</li> <li>B. Only epiglottis visible</li> <li>C. Only posterior glottis visible</li> <li>D. Glottis fully visible</li> </ul>                                                                                                      | <p><b>Ans: D</b></p> |
| <p><b>Q92. Which technique is considered the gold standard for managing an anticipated difficult airway?</b></p> <ul style="list-style-type: none"> <li>A. Awake fiberoptic intubation</li> <li>B. Surgical tracheostomy</li> <li>C. Retrograde intubation</li> <li>D. Blind nasal intubation</li> </ul>                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q93. In a 'cannot intubate, cannot ventilate' emergency, the immediate surgical airway procedure is:</b></p> <ul style="list-style-type: none"> <li>A. Fiberoptic intubation</li> <li>B. Jet ventilation through nasal cannula</li> <li>C. Tracheostomy</li> <li>D. Cricothyroidotomy</li> </ul>                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q94. In Mallampati Class II, the structures visible on opening the mouth and protruding the tongue are:</b></p> <ul style="list-style-type: none"> <li>A. Faucial pillars, soft palate and uvula</li> <li>B. Only soft palate</li> <li>C. Only hard palate</li> <li>D. Base of uvula and soft palate</li> </ul>                                                         | <p><b>Ans: D</b></p> |
| <p><b>Q95. The normal mentohyoid distance is:</b></p> <ul style="list-style-type: none"> <li>A. &gt; 5 cm (2 finger breadths)</li> <li>B. &gt; 10 cm</li> <li>C. &gt; 6.5 cm (3 finger breadths)</li> <li>D. &gt; 3 cm (1 finger breadth)</li> </ul>                                                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q96. Cormack-Lehane Grade III indicates visualization of:</b></p> <ul style="list-style-type: none"> <li>A. Only the epiglottis</li> <li>B. Full glottis</li> <li>C. Posterior part of glottis</li> <li>D. No structures at all</li> </ul>                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q97. In the difficult airway algorithm, when Plan A (reattempt intubation) fails in a 'cannot intubate but can ventilate' situation, Plan B involves:</b></p> <ul style="list-style-type: none"> <li>A. Revert to facemask only</li> <li>B. Wake the patient up</li> <li>C. Surgery under LMA or intubation via ILMA</li> <li>D. Immediate cricothyroidotomy</li> </ul> | <p><b>Ans: C</b></p> |

**Q98. For cervical spine injury, intubation should be done with neck in which position?**

- A. Full flexion
- B. Lateral rotation
- C. Neutral position with neck stabilization
- D. Full extension

**Ans: C**

**Q99. Klippel-Feil syndrome causes difficult intubation due to:**

- A. Laryngeal edema
- B. Macroglossia
- C. Cervical vertebrae fusion restricting neck movements
- D. Temporomandibular joint ankylosis

**Ans: C**

**Q100. In every patient, which three parameters of airway assessment must be evaluated?**

- A. CT scan, MRI and blood tests
- B. ECG, blood pressure and pulse oximetry
- C. Mallampati grading, thyromental distance and neck movements
- D. Chest X-ray, spirometry, ABG

**Ans: C**

# Answer Key & Explanations

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## Q1. Answer: B — C3 to C6

Page 26: The larynx is the organ of voice extending from the root of the tongue to the trachea and lies opposite C3 to C6.

Topic: Airway Anatomy

## Q2. Answer: B — Cricothyroid

Page 26: All muscles of the larynx are supplied by recurrent laryngeal nerve except cricothyroid, which is supplied by the external branch of the superior laryngeal nerve.

Topic: Larynx Nerve Supply

## Q3. Answer: A — The abductors (posterior cricoarytenoids)

Page 27: In bilateral partial paralysis of recurrent laryngeal nerve, the abductors (posterior cricoarytenoids) are affected first (Semon's Law), causing the cords to be held in adduction producing respiratory stridor.

Topic: Laryngeal Nerve Paralysis

## Q4. Answer: B — 300 million, 70 m<sup>2</sup>

Page 27: Total number of alveoli is 300 million with surface area of 70 m<sup>2</sup>. The thickness of alveolar capillary membrane is 0.3 mm.

Topic: Respiratory Anatomy

## Q5. Answer: D — Upper pons

Page 29: The pneumotaxic center is in the upper pons and the apneustic center is in the lower pons. The pneumotaxic center normally has an inhibitory effect on the apneustic center.

Topic: Regulation of Respiration

## Q6. Answer: C — 0.8

Page 29: Both ventilation and perfusion are more at bases compared to apex, but perfusion at base is comparatively higher, decreasing V/Q ratio towards the base (from 2.1 at apex to 0.3 at base, average 0.8).

Topic: Ventilation/Perfusion

## Q7. Answer: B — 30% of tidal volume

Page 30: Anatomical dead space is 30% of tidal volume or 2 mL/kg or 150 mL.

Topic: Dead Space

## Q8. Answer: C — 26 mm Hg; not affected by anesthetics

Page 31: P50 is the partial pressure of oxygen for 50% saturation, which is 26 mm Hg. P50 is not affected by anesthetics.

Topic: Oxygen Dissociation Curve

## Q9. Answer: B — Expiratory reserve volume + residual volume

Page 32: FRC is the volume of gas in lungs after end expiration. It is ERV + RV and equals 2400–2600 mL.

Topic: Lung Volumes

## Q10. Answer: D — Inversely proportional to pressure

Page 34: Boyle's law states that at a constant temperature, the volume of a gas is inversely proportional to the pressure.

Topic: Physics Related to Anesthesia



**Q11. Answer: A — The subglottis (at the level of cricoid)**

Page 26: The glottis is the narrowest part in adults while subglottis (at the level of cricoid) is the narrowest part in children up to the age of 6 years.

Topic: Airway Anatomy

**Q12. Answer: B — Cadaveric (mid) position**

Page 27: In complete paralysis of both recurrent and superior laryngeal nerves, the cords are held in mid-position (cadaveric position). Cords are also in cadaveric position during anesthesia with muscle relaxants.

Topic: Laryngeal Nerve Paralysis

**Q13. Answer: D — T5**

Page 27: Trachea starts from cricoid ring (C6) and ends at carina (T5). Anteriorly carina corresponds to second costal cartilage at the junction of manubrium with sternal body (angle of Louis).

Topic: Trachea

**Q14. Answer: C — Shorter (2.5 cm), wider, angle 25°**

Page 28, Table 1.1: Right bronchus is shorter (2.5 cm), wider, angle with vertical 25°. Left bronchus is longer (5 cm), narrower, angle 45°.

Topic: Bronchial Tree

**Q15. Answer: C — Ether**

Page 28: Ciliary activity is inhibited by all inhalational agents except ether.

Topic: Respiratory Mucosa

**Q16. Answer: C — Hypoxia**

Page 29: Peripheral chemoreceptors are present in the carotid body and aortic arch. Carotid body receptors are very sensitive to hypoxia.

Topic: Regulation of Respiration

**Q17. Answer: B — Trachea and then main bronchus**

Page 29: Maximum airway resistance to airflow occurs in trachea and then main bronchus and minimum in terminal bronchi. Turbulent flow in trachea and main bronchi makes resistance proportional to square of flow rates.

Topic: Airway Resistance

**Q18. Answer: C — 3–5 mm Hg**

Page 29: V/Q mismatch creates alveolar to arterial oxygen difference  $[(A-a) pO_2]$  which is normally 3–5 mm Hg.

Topic: Ventilation/Perfusion

**Q19. Answer: B — 60–80 mL**

Page 30: Alveolar dead space is 60–80 mL in standing position and zero in lying position (in lying position perfusion is equal in all parts of lung).

Topic: Dead Space

**Q20. Answer: B — 1.34 mL**

Page 30: 1 g of Hb carries 1.34 mL of oxygen. Oxygen content of arterial blood is 20 mL/dL and that of venous blood is 15 mL/dL.

Topic: Oxygen in Blood

**Q21. Answer: D — 90%**

Page 30: Normally Hb is 97% saturated at normal pO<sub>2</sub> of 95–98 mm Hg. At 60 mm Hg saturation is still 90%. After this point there is a sudden drop.

Topic: Oxygen Dissociation Curve

**Q22. Answer: C — Bicarbonate (90%)**

Page 31: Carbon dioxide is transported in blood as bicarbonate (90%), dissolved (5%), and as carbamino compounds.

Topic: Carbon Dioxide

**Q23. Answer: A — 15–20%**

Page 32: During general anesthesia FRC decreases by 15–20%. FRC is the volume of gas in lungs after end expiration (ERV + RV).

Topic: Lung Volumes

**Q24. Answer: A — 1 litre**

Page 33: Closing capacity is the volume at which airway closes, which is normally 1 litre less than FRC. If FRC falls below closing capacity there will be significant hypoventilation and V/Q abnormalities.

Topic: Pulmonary Function Tests

**Q25. Answer: C — Halothane**

Page 32: All inhalational agents blunt hypoxic pulmonary vasoconstriction (HPV). Maximum blunting is with halothane.

Topic: Hypoxic Pulmonary Vasoconstriction

**Q26. Answer: D — Fail-safe valve**

Page 42: The fail-safe valve (oxygen supply failure protection device) proportionately decreases N<sub>2</sub>O flow when oxygen pressure drops, preventing a hypoxic mixture.

Topic: Safety Features

**Q27. Answer: B — 25%**

Page 43: The O<sub>2</sub>–N<sub>2</sub>O interconnected pulley system ensures a minimum FiO<sub>2</sub> of 25%, making singular N<sub>2</sub>O flow impossible.

Topic: Safety Features

**Q28. Answer: D — Type A (Magill circuit)**

Page 46: Magill (Type A) is the circuit of choice for spontaneous ventilation. Fresh gas flow = minute volume prevents rebreathing.

Topic: Mapleson Circuits

**Q29. Answer: A — Type D**

Page 47: Bain incorporated an inner tube into Mapleson D; fresh gases travel through the inner tube minimizing mixing with exhaled gases.

Topic: Mapleson Circuits

**Q30. Answer: D — Sevoflurane**

Page 51: Sevoflurane reacts with soda lime and barium hydroxide to produce compound A (nephrotoxic). Risk is higher with barium hydroxide vs soda lime.

Topic: CO<sub>2</sub> Absorbents

**Q31. Answer: A — Desflurane**

Page 51: CO production with desiccated soda lime in decreasing order: Desflurane >> Isoflurane >> Halothane = Sevoflurane.

Topic: CO ■ Absorbents

**Q32. Answer: D — KOH 1%**

Page 50: Soda lime composition: Ca(OH) ■ 94%, NaOH 5% (acts as catalyst), KOH 1%, color indicator, and silica (to prevent dust formation).

Topic: CO ■ Absorbents

**Q33. Answer: B — Children < 6 years or < 20 kg**

Page 49: Jackson-Rees (Mapleson F) is the most commonly used pediatric semiclosed circuit for children < 6 years or < 20 kg.

Topic: Mapleson Circuits

**Q34. Answer: A — 1917**

Page 38: First anesthesia machine was invented by Edmund Gaskin Boyle in 1917 (that's why anesthesia machine used to be called as Boyle's machine).

Topic: Anesthesia Machine History

**Q35. Answer: A — Black body with white shoulders**

Page 38: The color of oxygen cylinder is black body with white shoulders (top) and pressure is 2,000 psi.

Topic: Gas Cylinders

**Q36. Answer: B — Weight (not by pressure gauge)**

Page 40: N ■ O is filled in liquid form. A small amount of liquid still vaporizing can show full pressure, so contents are measured by weight, not by pressure gauge.

Topic: Gas Cylinders

**Q37. Answer: C — 2000 psi**

Page 40: Entonox is stored in blue color cylinders with white shoulders at a pressure of 2000 psi, mainly used to produce analgesia during labor.

Topic: Gas Cylinders

**Q38. Answer: B — 2, 5**

Page 41, Table 2.2: Pin index position for oxygen is 2, 5. Nitrous oxide is 3, 5. Air is 1, 5.

Topic: Pin Index System

**Q39. Answer: A — Wrong fitting of central supply pipelines to the machine**

Page 42: Like pin index system which prevents wrong fitting of cylinders, DISS prevents wrong fitting of central supply pipes to the machine by using different diameters for each gas.

Topic: Safety Features

**Q40. Answer: B — 35 L/min at 45–60 psi**

Page 42: Oxygen flush bypasses the intermediate and low pressure system, delivering 35 liters of oxygen per minute at 45–60 psi.

Topic: High Pressure System

**Q41. Answer: B — 45–60 psi**

Page 42: High pressure regulator (1st stage) reduces the pressure to 45–60 psi. The 2nd stage further reduces it to 15–35 psi.

*Topic: Pressure System*

**Q42. Answer: D — Thorpe's tubes**

Page 42: Glass variable-orifice flowmeter tubes are called Thorpe's tubes; they are tapered (narrow at base and wider at apex).

*Topic: Flowmeters and Rotameters*

**Q43. Answer: C — More fluted, more prominent and larger in diameter**

Page 42: The oxygen knob is made physically different from other knobs (it is more fluted, more prominent and large in diameter). It is white colored.

*Topic: Flowmeters and Rotameters*

**Q44. Answer: A — Downstream (last in sequence)**

Page 43: Oxygen tube should be most downstream so that at the common manifold junction, leakage affects O<sub>2</sub> at only 2 positions, minimising hypoxic mixture risk.

*Topic: Flowmeters and Rotameters*

**Q45. Answer: A — High specific heat and high thermal conductivity**

Page 44: The vaporizer material should have high specific heat and high thermal conductivity. As Copper possesses these properties, vaporizers are made of copper.

*Topic: Vaporizers*

**Q46. Answer: A — Tec 6**

Page 44: Tec 6 is a heated and pressurized vaporizer specifically designed for desflurane due to its very low boiling point (23.5°C) and high vapor pressure (680 mm Hg).

*Topic: Vaporizers*

**Q47. Answer: D — Red**

Page 45: Color codes for vaporizers: Red = halothane, Purple = isoflurane, Yellow = sevoflurane, Blue = desflurane.

*Topic: Vaporizers*

**Q48. Answer: A — First (nearest to rotameter)**

Page 45: Agent with lower boiling point placed first (nearest rotameter); otherwise condensed particles will be recovered from the second vaporizer.

*Topic: Vaporizers*

**Q49. Answer: D — Schimmelbusch mask**

Page 45: In the open system, inhalational agent is directly poured over patient mouth and nostril using a Schimmelbusch mask. Open systems are now obsolete.

*Topic: Breathing Circuits*

**Q50. Answer: B — 2.5 × minute volume**

Page 47: Bain circuit can be used for spontaneous ventilation but fresh gas flow requirement is higher, i.e. 2.5 times of minute volume.

*Topic: Mapleson Circuits*

**Q51. Answer: A — > 3 × minute volume (unpredictable)**

Page 46: For controlled ventilation, Magill requires very high flows (3 times of MV or > 20 L) and is still unpredictable, therefore should NOT be used for controlled ventilation.

*Topic: Mapleson Circuits*

**Q52. Answer: C — Waters in 1923**

Page 49: In human beings, the closed circuit system was first introduced by Waters in 1923.

*Topic: Circle System*

**Q53. Answer: D — 30 mL**

Page 53: The negative pressure (suction bulb) test is the most sensitive, detecting leaks as small as 30 mL. It is called the universal leak test.

*Topic: Machine Checking*

**Q54. Answer: D — Supraglottic airway device**

Page 58: LMA is a supraglottic airway device; it sits above the glottis in the hypopharynx. Endotracheal tube is an infraglottic device.

*Topic: Supraglottic Airway Devices*

**Q55. Answer: D — Has a drain tube allowing passage of a gastric tube**

Page 59: ProSeal LMA has a larger and posterior cuff providing better seal and a drain tube which can be used to deflate the stomach.

*Topic: Supraglottic Airway Devices*

**Q56. Answer: C — Tip of nose to tragus plus 1 cm**

Page 55: The appropriate length of a Guedel airway is the distance between tip of nose and tragus plus 1 cm.

*Topic: Airway Equipment*

**Q57. Answer: B — 0 (smallest) to 6 (largest)**

Page 55: Facemask is used to ventilate the patient without intubation. They are available in sizes from 0 (smallest) to 6 (largest).

*Topic: Airway Equipment*

**Q58. Answer: B — Artificial Manual Breathing Unit**

Page 55: AMBU stands for Artificial Manual Breathing Unit. Capacity: 1200 mL for adults, 500 mL for children, 250 mL for newborns.

*Topic: Airway Equipment*

**Q59. Answer: A — Macintosh**

Page 56: Macintosh is the most commonly used laryngoscope. It has a curved blade and is available in 4 sizes.

*Topic: Laryngoscopes*

**Q60. Answer: C — Magill straight blade – neonatal epiglottis is large, leafy and more anterior**

Page 56: Magill is a straight blade for neonates. Neonatal epiglottis is large, leafy and more anterior, so it needs to be lifted by a straight blade to visualize glottis.

*Topic: Laryngoscopes*

**Q61. Answer: B — 94–99%**

Page 57: Studies have reported intubation success rates of 94–99% for video laryngoscopy as a rescue modality after failed direct laryngoscopy.

Topic: Laryngoscopes

**Q62. Answer: D — Upper incisors**

Page 58: Dental injury is a complication of laryngoscopy. Most vulnerable are upper incisors.

Topic: Complications of Laryngoscopy

**Q63. Answer: B — Archie Brain**

Page 58: Classical LMA was discovered by Archie Brain, therefore also called as Brain Mask.

Topic: Supraglottic Airway Devices

**Q64. Answer: B — All types of LMA**

Page 59: Contraindications for LMA include full stomach patients, hiatus hernia, pregnancy (where chances of aspiration are high), and oropharyngeal abscess or mass.

Topic: Supraglottic Airway Devices

**Q65. Answer: C — 8 no.**

Page 59: Intubating LMA (also called as LMA Fastrach): Up to 8 no. endotracheal tube can be guided through it.

Topic: Supraglottic Airway Devices

**Q66. Answer: B — Prefilled with gel, avoiding air cuff complications**

Page 59: I-Gel has a cuff prefilled with gel avoiding complications of air-filled cuff such as cuff leakage, damage and puncture. Like Proseal, I-gel also contains a drain tube.

Topic: Supraglottic Airway Devices

**Q67. Answer: A — Low pressure, high volume cuff causing less tracheal injury**

Page 61, Table 3.1: PVC tube cuff is low pressure and high volume producing less tracheal injury. Red rubber has high pressure low volume cuff with more chances of tracheal injury.

Topic: Endotracheal Tubes

**Q68. Answer: C — Age/3 + 3.5**

Page 61: For children 1 year to 6 years, tube size = Age(years)/3 + 3.5. For a 5-year-old:  $5/3 + 3.5 = 5.1$  (means 5 number tube).

Topic: Endotracheal Tubes

**Q69. Answer: D — 30 cm H<sub>2</sub>O**

Page 61: During cuff inflation, ensure cuff pressure to be less than 30 cm H<sub>2</sub>O to prevent tracheal ischemia.

Topic: Endotracheal Tubes

**Q70. Answer: C — Correct position of endotracheal tube**

Page 63: Checking correct position of tube: Capnography (measuring end tidal CO<sub>2</sub>) is the surest sign.

Topic: Endotracheal Tubes

**Q71. Answer: A — 14 days**

Page 64: Maximum permissible time for which an endotracheal tube can be kept is 14 days. Delayed complications like tracheal stenosis usually occur after prolonged intubation.

Topic: Endotracheal Tubes

**Q72. Answer: A — Sore throat (pharyngitis, laryngitis)**

Page 64: Sore throat (pharyngitis, laryngitis) is the most common postoperative complication. It usually subsides in 2–3 days without treatment.

Topic: Complications of Intubation

**Q73. Answer: A — Oral surgery**

Page 63: Contraindications for nasal intubation include basal skull fractures, CSF rhinorrhea, bleeding disorders, nasal polyp, abscess, foreign body, and adenoids. Oral surgery is actually an INDICATION for nasal intubation.

Topic: Nasal Intubation

**Q74. Answer: A — Ethylene oxide (ETO) gas sterilization**

Page 67: Non-metallic items such as endotracheal tubes, facemasks, airways are best sterilized by ethylene oxide (ETO) gas sterilization.

Topic: Sterilization

**Q75. Answer: C — Artificial nose or condenser humidifier**

Page 66: Passive humidifier is also called as artificial nose, condenser humidifier or heat and moisture exchanger (HME) filter.

Topic: Humidification Devices

**Q76. Answer: D — Normal healthy patient**

Page 71: ASA I = Normal healthy patient. ASA II = Mild-to-moderate systemic disease. ASA III = Severe systemic disease. ASA IV = Incapacitating disease.

Topic: Risk Stratification

**Q77. Answer: B — Faucial pillars, soft palate and uvula**

Page 76: Mallampati Class I: Faucial pillars, soft palate and uvula seen. Class II: Base of uvula and soft palate. Class III: Only soft palate. Class IV: Only hard palate.

Topic: Airway Assessment

**Q78. Answer: A — 2 hours**

Page 73: Clear fluids (water and juices without pulp) can be given up to 2 hours before surgery. For solid food, 8 hours of fasting is must.

Topic: Preoperative Instructions

**Q79. Answer: B — Midazolam**

Page 73: In current day surgical practice where majority of patients are getting admitted on the same day, the benzodiazepine of choice is Midazolam.

Topic: Premedication

**Q80. Answer: C — Does not cross the blood-brain barrier, so no central side effects**

Page 74: Glycopyrrolate is preferred because it does not cross the blood-brain barrier therefore is devoid of central side effects.

Topic: Premedication

**Q81. Answer: D — Brain dead patients for organ donation**

Page 71: ASA VI = Brain dead patients (for organ donation).

Topic: Risk Stratification

**Q82. Answer: B — Age, obesity, and risk associated with surgical procedure**

Page 71: The major limitation of ASA classification was that it did not include age, obesity (two major enemies of anesthetist) and risk associated with surgical procedure.

Topic: Risk Stratification

**Q83. Answer: D — 4 weeks**

Page 71: Estrogen increases chances of postoperative DVT and thromboembolism, therefore standard dose estrogen pills should be stopped 4 weeks before. Low dose or progesterone-only pills need not be stopped.

Topic: Preoperative Instructions

**Q84. Answer: D — 4 days; INR < 1.5**

Page 71: Warfarin should be stopped 4 days prior to surgery. INR must be < 1.5 to consider the patient for surgery.

Topic: Preoperative Instructions

**Q85. Answer: B — Continue aspirin as stopping carries risk of ischemia**

Page 72: Although aspirin increases risk of bleeding, stopping aspirin carries risk of ischemia, therefore it is recommended to continue aspirin.

Topic: Preoperative Instructions

**Q86. Answer: D — 6–8 weeks**

Page 72: Normal restoration of effects (particularly recovery of ciliary activity) takes 6–8 weeks, therefore smoking should ideally be stopped 8 weeks before surgery.

Topic: Preoperative Instructions

**Q87. Answer: A — 60 minutes**

Page 74: The timing of antibiotic should be adjusted so that peak blood levels are achieved at the time of skin incision. Antibiotic prophylaxis must be given within 60 minutes before skin incision.

Topic: Premedication

**Q88. Answer: A — 4 hours**

Page 73: For neonates and infants, the recommended fasting period for breast milk is 4 hours while for formula milk and solids a fasting of 6 hours is required.

Topic: Preoperative Instructions

**Q89. Answer: D — Pierre Robin syndrome**

Page 75: Pierre-Robin syndrome causes difficult intubation due to hypoplastic mandible, macroglossia and high-arched palate.

Topic: Causes of Difficult Intubation



**Q90. Answer: D — 6.5 cm or more (> 3 finger breadths)**

Page 76: Thyromental distance: Normal = 6.5 cm or more (> 3 finger breadths). 6.0–6.5 cm = difficult laryngoscopy. < 6.0 cm = laryngoscopy may be impossible.

Topic: Assessment of Difficult Intubation

**Q91. Answer: D — Glottis fully visible**

Page 76: Cormack-Lehane grading: Grade I = Glottis fully visible. Grade II = Only posterior glottis visible. Grade III = Only epiglottis visible. Grade IV = No recognizable structures.

Topic: Assessment of Difficult Intubation

**Q92. Answer: A — Awake fiberoptic intubation**

Page 77: If difficult airway is anticipated then intubate in awake state preferably with fiberoptic bronchoscope. Fiberoptic intubation is considered the gold standard technique.

Topic: Management of Difficult Airway

**Q93. Answer: D — Cricothyroidotomy**

Page 78: In CICV emergency, stepwise management ends with cricothyroidotomy, which should be replaced by definitive tracheostomy as soon as possible.

Topic: Management of Difficult Airway

**Q94. Answer: D — Base of uvula and soft palate**

Page 76: Mallampati Class II: Base of uvula and soft palate is seen. In Class II, oral intubation can be done comfortably.

Topic: Assessment of Difficult Intubation

**Q95. Answer: A — > 5 cm (2 finger breadths)**

Page 76: Mentohyoid distance normal > 5 cm (2 finger breadth).

Topic: Assessment of Difficult Intubation

**Q96. Answer: A — Only the epiglottis**

Page 77: Grade III = Only epiglottis visible. Grade IV = No recognizable structures.

Topic: Assessment of Difficult Intubation

**Q97. Answer: C — Surgery under LMA or intubation via ILMA**

Page 78: Plan B includes surgery under LMA if possible, or intubation with intubating LMA (ILMA). Plan C reverts to facemask. Plan D is for CICV emergencies.

Topic: Management of Difficult Airway

**Q98. Answer: C — Neutral position with neck stabilization**

Page 79: Oral intubation with 'neck stabilization' with neck in 'neutral position' should be tried first and is usually successful.

Topic: Cervical Spine Injury

**Q99. Answer: C — Cervical vertebrae fusion restricting neck movements**

Page 75: Restricted neck movements can be caused by Klippel-Feil syndrome (cervical vertebrae fusion), rheumatoid arthritis, spondylitis, and disc disease.

Topic: Causes of Difficult Intubation

**Q100. Answer: C — Mallampati grading, thyromental distance and neck movements**

Page 76: Mallampati grading, thyromental distance and neck movements are assessed in every case for airway assessment.

*Topic: Assessment of Difficult Intubation*

# Preoperative Assessment and Premedication MCQ – Chapter 4

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 3: Quintessential in Anesthesia (Chapter 4)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. What is the primary purpose of preoperative assessment?**

- A. To delay the surgery
- B. To evaluate the patient's fitness for anesthesia and surgery
- C. To prescribe antibiotics
- D. To schedule follow-up visits

**Ans: B**

**Q2. Which ASA physical status classification denotes a normal healthy patient?**

- A. ASA I
- B. ASA II
- C. ASA III
- D. ASA IV

**Ans: A**

**Q3. ASA II classification refers to a patient with:**

- A. Severe systemic disease that limits activity
- B. Mild systemic disease with no functional limitation
- C. Incapacitating disease that is a constant threat to life
- D. A moribund patient not expected to survive 24 hours

**Ans: B**

**Q4. Which ASA classification is given to a moribund patient not expected to survive 24 hours with or without surgery?**

- A. ASA III
- B. ASA IV
- C. ASA V
- D. ASA VI

**Ans: C**

**Q5. The suffix 'E' in ASA classification denotes:**

- A. Elderly patient
- B. Emergency surgery
- C. Endocrine disorder
- D. Extremity surgery

**Ans: B**

**Q6. During preoperative assessment, Mallampati classification is used to assess:**

- A. Cardiac function
- B. Airway and ease of intubation
- C. Renal function
- D. Respiratory reserve

**Ans: B**

**Q7. Mallampati Class I visualization includes:**

- A. Soft palate only
- B. Soft palate, fauces, uvula, and anterior and posterior tonsillar pillars
- C. Soft palate and base of uvula
- D. Hard palate only

**Ans: B**

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| <p><b>Q8. Mallampati Class IV indicates visualization of:</b></p> <ul style="list-style-type: none"> <li>A. Full uvula and tonsillar pillars</li> <li>B. Soft palate and base of uvula</li> <li>C. Hard palate only (soft palate not visible)</li> <li>D. Soft palate, fauces, and uvula</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q9. Which of the following is NOT a component of the preoperative airway assessment?</b></p> <ul style="list-style-type: none"> <li>A. Thyromental distance</li> <li>B. Mallampati score</li> <li>C. Neck mobility</li> <li>D. Ankle-brachial index</li> </ul>                                | <p><b>Ans: D</b></p> |
| <p><b>Q10. A thyromental distance of less than 6 cm suggests:</b></p> <ul style="list-style-type: none"> <li>A. Easy intubation</li> <li>B. Difficult intubation</li> <li>C. Normal airway</li> <li>D. Need for tracheostomy</li> </ul>                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q11. Preoperative fasting guidelines recommend NPO for clear fluids for how many hours before elective surgery in adults?</b></p> <ul style="list-style-type: none"> <li>A. 1 hour</li> <li>B. 2 hours</li> <li>C. 4 hours</li> <li>D. 6 hours</li> </ul>                                     | <p><b>Ans: B</b></p> |
| <p><b>Q12. The recommended fasting time for solid food before elective surgery in adults is:</b></p> <ul style="list-style-type: none"> <li>A. 2 hours</li> <li>B. 4 hours</li> <li>C. 6 hours</li> <li>D. 8 hours</li> </ul>                                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q13. Which investigation is mandatory in all patients above 40 years of age undergoing surgery?</b></p> <ul style="list-style-type: none"> <li>A. CT scan of brain</li> <li>B. ECG</li> <li>C. Echocardiography</li> <li>D. Pulmonary function test</li> </ul>                                | <p><b>Ans: B</b></p> |
| <p><b>Q14. Which of the following drugs should be continued on the morning of surgery?</b></p> <ul style="list-style-type: none"> <li>A. Oral hypoglycemics</li> <li>B. Antihypertensive medications</li> <li>C. Warfarin</li> <li>D. Aspirin (for minor surgery)</li> </ul>                        | <p><b>Ans: B</b></p> |

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| <p><b>Q15. Preoperative instructions include stopping aspirin how many days before elective surgery?</b></p> <p>A. 1 day<br/>B. 3 days<br/>C. 7 days<br/>D. 14 days</p>                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q16. The primary goal of premedication is:</b></p> <p>A. To induce general anesthesia<br/>B. To allay anxiety and facilitate smooth induction<br/>C. To replace intraoperative anesthetics<br/>D. To provide postoperative analgesia only</p>               | <p><b>Ans: B</b></p> |
| <p><b>Q17. Which of the following is a desired effect of premedication?</b></p> <p>A. Increased salivation<br/>B. Anxiolysis<br/>C. Hypertension<br/>D. Bronchospasm</p>                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q18. Which benzodiazepine is most commonly used for premedication?</b></p> <p>A. Chlordiazepoxide<br/>B. Midazolam<br/>C. Flurazepam<br/>D. Clonazepam</p>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q19. The antisialagogue effect of premedication is achieved by using:</b></p> <p>A. Midazolam<br/>B. Glycopyrrolate<br/>C. Fentanyl<br/>D. Ondansetron</p>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q20. Which anticholinergic drug crosses the blood-brain barrier and can cause central effects?</b></p> <p>A. Glycopyrrolate<br/>B. Atropine<br/>C. Ipratropium<br/>D. Hyoscine butylbromide</p>                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q21. Which of the following is NOT a goal of premedication?</b></p> <p>A. Reduction of gastric volume and acidity<br/>B. Prevention of autonomic reflexes<br/>C. Induction of surgical anesthesia<br/>D. Reduction of postoperative nausea and vomiting</p> | <p><b>Ans: C</b></p> |

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| <p><b>Q22. Ranitidine given as premedication acts by:</b></p> <ul style="list-style-type: none"> <li>A. Blocking muscarinic receptors</li> <li>B. Blocking H2 receptors to reduce gastric acid secretion</li> <li>C. Blocking dopamine receptors</li> <li>D. Enhancing gastric motility</li> </ul>                                      | <p><b>Ans: B</b></p> |
| <p><b>Q23. Metoclopramide given as premedication helps by:</b></p> <ul style="list-style-type: none"> <li>A. Reducing anxiety</li> <li>B. Increasing gastric emptying and increasing lower esophageal sphincter tone</li> <li>C. Providing analgesia</li> <li>D. Reducing bronchial secretions</li> </ul>                               | <p><b>Ans: B</b></p> |
| <p><b>Q24. Sodium citrate given preoperatively is used to:</b></p> <ul style="list-style-type: none"> <li>A. Reduce bronchial secretions</li> <li>B. Neutralize gastric acidity</li> <li>C. Provide sedation</li> <li>D. Prevent bradycardia</li> </ul>                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q25. Which opioid is commonly used as premedication for its analgesic and sedative properties?</b></p> <ul style="list-style-type: none"> <li>A. Naloxone</li> <li>B. Morphine</li> <li>C. Naltrexone</li> <li>D. Flumazenil</li> </ul>                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q26. The ideal time for giving oral premedication before surgery is:</b></p> <ul style="list-style-type: none"> <li>A. 15 minutes</li> <li>B. 30–60 minutes</li> <li>C. 60–90 minutes</li> <li>D. 2–3 hours</li> </ul>                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q27. Which of the following systems must be evaluated during preoperative assessment?</b></p> <ul style="list-style-type: none"> <li>A. Only cardiovascular system</li> <li>B. Cardiovascular, respiratory, hepatic, renal, and nervous systems</li> <li>C. Only respiratory system</li> <li>D. Only the surgical site</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q28. A patient with controlled diabetes mellitus on oral hypoglycemics is classified as:</b></p> <ul style="list-style-type: none"> <li>A. ASA I</li> <li>B. ASA II</li> <li>C. ASA III</li> <li>D. ASA IV</li> </ul>                                                                                                             | <p><b>Ans: B</b></p> |

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| <p><b>Q29. A patient with unstable angina is classified as:</b></p> <p>A. ASA II<br/>B. ASA III<br/>C. ASA IV<br/>D. ASA V</p>                                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q30. Which preoperative investigation is essential before major surgery with expected blood loss?</b></p> <p>A. Echocardiography<br/>B. Blood grouping and cross-matching<br/>C. CT angiography<br/>D. MRI brain</p>                     | <p><b>Ans: B</b></p> |
| <p><b>Q31. Glycopyrrolate differs from atropine in that it:</b></p> <p>A. Crosses the blood-brain barrier<br/>B. Does not cross the blood-brain barrier<br/>C. Causes more tachycardia<br/>D. Has a longer duration of action on the heart</p> | <p><b>Ans: B</b></p> |
| <p><b>Q32. Which drug is used as premedication to prevent aspiration pneumonitis in high-risk patients?</b></p> <p>A. Midazolam<br/>B. Ranitidine + Metoclopramide + Sodium citrate<br/>C. Morphine<br/>D. Diazepam</p>                        | <p><b>Ans: B</b></p> |
| <p><b>Q33. Preoperative instructions regarding smoking cessation recommend stopping at least:</b></p> <p>A. 6 hours before surgery<br/>B. 24 hours before surgery<br/>C. 6–8 weeks before surgery<br/>D. 1 week before surgery</p>             | <p><b>Ans: C</b></p> |
| <p><b>Q34. Which of the following is a contraindication to the use of atropine as premedication?</b></p> <p>A. Bradycardia<br/>B. Glaucoma<br/>C. Excessive secretions<br/>D. Vagal-mediated reflex</p>                                        | <p><b>Ans: B</b></p> |
| <p><b>Q35. Which premedication drug has both antiemetic and anxiolytic properties?</b></p> <p>A. Glycopyrrolate<br/>B. Promethazine<br/>C. Ranitidine<br/>D. Sodium citrate</p>                                                                | <p><b>Ans: B</b></p> |



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| <p><b>Q36. The MET (Metabolic Equivalent of Task) is used in preoperative assessment to evaluate:</b></p> <ul style="list-style-type: none"> <li>A. Airway difficulty</li> <li>B. Functional cardiac capacity</li> <li>C. Hepatic function</li> <li>D. Coagulation status</li> </ul>                                    | <p><b>Ans: B</b></p> |
| <p><b>Q37. A patient who can climb two flights of stairs without symptoms has a functional capacity of at least:</b></p> <ul style="list-style-type: none"> <li>A. 1 MET</li> <li>B. 2 METs</li> <li>C. 4 METs</li> <li>D. 10 METs</li> </ul>                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q38. Preoperative consent for anesthesia should include information about:</b></p> <ul style="list-style-type: none"> <li>A. Only the type of anesthesia</li> <li>B. Risks, benefits, alternatives, and plan of anesthesia</li> <li>C. Only the surgical procedure</li> <li>D. Only postoperative care</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q39. Which of the following oral hypoglycemic drugs should be stopped before surgery?</b></p> <ul style="list-style-type: none"> <li>A. Metformin</li> <li>B. Ranitidine</li> <li>C. Atenolol</li> <li>D. Amlodipine</li> </ul>                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q40. Warfarin is typically stopped how many days before elective surgery?</b></p> <ul style="list-style-type: none"> <li>A. 1 day</li> <li>B. 2 days</li> <li>C. 3–5 days</li> <li>D. 10 days</li> </ul>                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q41. In preoperative assessment, the term 'NPO' stands for:</b></p> <ul style="list-style-type: none"> <li>A. No Pain Observed</li> <li>B. Nil Per Oram (nothing by mouth)</li> <li>C. Normal Preoperative Order</li> <li>D. Non-Prescribed Opioid</li> </ul>                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q42. The dose of midazolam commonly used for oral premedication in adults is:</b></p> <ul style="list-style-type: none"> <li>A. 0.1 mg</li> <li>B. 0.25–0.5 mg/kg</li> <li>C. 7.5–15 mg</li> <li>D. 50 mg</li> </ul>                                                                                              | <p><b>Ans: C</b></p> |

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| <p><b>Q43. Which of the following is an advantage of midazolam over diazepam as premedication?</b></p> <ul style="list-style-type: none"> <li>A. Longer duration of action</li> <li>B. Water soluble, less painful on injection, and produces anterograde amnesia</li> <li>C. Greater respiratory depression</li> <li>D. Cheaper cost</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q44. Preoperative assessment of the cervical spine is important in patients with:</b></p> <ul style="list-style-type: none"> <li>A. Diabetes mellitus</li> <li>B. Rheumatoid arthritis and Down syndrome</li> <li>C. Hypertension</li> <li>D. Asthma</li> </ul>                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q45. Which preoperative blood test is essential for all female patients of reproductive age?</b></p> <ul style="list-style-type: none"> <li>A. Thyroid function test</li> <li>B. Pregnancy test (urine/serum beta-hCG)</li> <li>C. Lipid profile</li> <li>D. Vitamin D levels</li> </ul>                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q46. Preoperative instruction to patients about dentures includes:</b></p> <ul style="list-style-type: none"> <li>A. Keeping them in place during surgery</li> <li>B. Removing all dental prostheses before entering the operating room</li> <li>C. Gluing them in place</li> <li>D. Covering them with gauze</li> </ul>                   | <p><b>Ans: B</b></p> |
| <p><b>Q47. Dexmedetomidine used as premedication acts on:</b></p> <ul style="list-style-type: none"> <li>A. GABA receptors</li> <li>B. Alpha-2 adrenergic receptors</li> <li>C. Opioid receptors</li> <li>D. Dopamine receptors</li> </ul>                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q48. Which of the following is true about premedication in pediatric patients?</b></p> <ul style="list-style-type: none"> <li>A. Anticholinergics are never used</li> <li>B. Midazolam syrup (0.5 mg/kg) is commonly used for anxiolysis</li> <li>C. Opioids are the first choice</li> <li>D. Premedication is not required</li> </ul>     | <p><b>Ans: B</b></p> |
| <p><b>Q49. The fasting guideline for breast milk in infants before surgery is:</b></p> <ul style="list-style-type: none"> <li>A. 2 hours</li> <li>B. 4 hours</li> <li>C. 6 hours</li> <li>D. 8 hours</li> </ul>                                                                                                                                  | <p><b>Ans: B</b></p> |

**Q50. Which of the following investigations is indicated preoperatively in a patient with liver disease?**

- A. ECG only
- B. Coagulation profile (PT, INR, aPTT)
- C. Chest X-ray only
- D. Blood sugar only

**Ans: B**

# Answer Key & Explanations

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## **Q1. Answer: B — To evaluate the patient's fitness for anesthesia and surgery**

The primary purpose of preoperative assessment is to evaluate the patient's fitness for anesthesia and the planned surgical procedure.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q2. Answer: A — ASA I**

ASA I denotes a normal healthy patient with no organic, physiological, or psychiatric disturbance.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q3. Answer: B — Mild systemic disease with no functional limitation**

ASA II refers to a patient with mild systemic disease with no functional limitation (e.g., controlled hypertension, mild diabetes).

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q4. Answer: C — ASA V**

ASA V is assigned to a moribund patient who is not expected to survive 24 hours with or without surgery.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q5. Answer: B — Emergency surgery**

The suffix 'E' is added to denote an emergency surgical procedure.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q6. Answer: B — Airway and ease of intubation**

Mallampati classification assesses the oropharyngeal structures to predict ease of intubation.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q7. Answer: B — Soft palate, fauces, uvula, and anterior and posterior tonsillar pillars**

Mallampati Class I shows soft palate, fauces, uvula, and both tonsillar pillars fully visible.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q8. Answer: C — Hard palate only (soft palate not visible)**

In Mallampati Class IV, only the hard palate is visible, indicating potentially difficult intubation.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q9. Answer: D — Ankle-brachial index**

Ankle-brachial index is used for peripheral vascular disease assessment, not airway evaluation.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

## **Q10. Answer: B — Difficult intubation**

A thyromental distance less than 6 cm (3 finger breadths) suggests potential difficulty in intubation.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

**Q11. Answer: B — 2 hours**

Current guidelines recommend fasting from clear fluids for at least 2 hours before elective surgery.

Topic: Instructions | Ref: Ch 4, Page 48

**Q12. Answer: C — 6 hours**

Patients should fast from solid food for at least 6 hours before elective surgery.

Topic: Instructions | Ref: Ch 4, Page 48

**Q13. Answer: B — ECG**

ECG is recommended as a routine investigation for all patients above 40 years of age.

Topic: Preoperative Assessment | Ref: Ch 4, Page 48

**Q14. Answer: B — Antihypertensive medications**

Antihypertensive medications should generally be continued on the morning of surgery to avoid rebound hypertension.

Topic: Instructions | Ref: Ch 4, Page 48

**Q15. Answer: C — 7 days**

Aspirin should be stopped 7 days before elective surgery as it irreversibly inhibits platelet function.

Topic: Instructions | Ref: Ch 4, Page 48

**Q16. Answer: B — To allay anxiety and facilitate smooth induction**

The primary goal of premedication is to allay anxiety, reduce secretions, and facilitate smooth induction of anesthesia.

Topic: Premedication | Ref: Ch 4, Page 50

**Q17. Answer: B — Anxiolysis**

Anxiolysis (relief of anxiety) is one of the primary desired effects of premedication.

Topic: Premedication | Ref: Ch 4, Page 50

**Q18. Answer: B — Midazolam**

Midazolam is the most commonly used benzodiazepine for premedication due to its rapid onset, short duration, and amnestic properties.

Topic: Premedication | Ref: Ch 4, Page 50

**Q19. Answer: B — Glycopyrrolate**

Glycopyrrolate is an anticholinergic agent used as a premedicant primarily for its antisialagogue (drying) effect.

Topic: Premedication | Ref: Ch 4, Page 50

**Q20. Answer: B — Atropine**

Atropine is a tertiary amine that crosses the blood-brain barrier and can cause central anticholinergic effects.

Topic: Premedication | Ref: Ch 4, Page 50

**Q21. Answer: C — Induction of surgical anesthesia**

Induction of surgical anesthesia is not a goal of premedication; it is achieved by induction agents.

Topic: Premedication | Ref: Ch 4, Page 50

**Q22. Answer: B — Blocking H2 receptors to reduce gastric acid secretion**

Ranitidine is an H2 receptor antagonist that reduces gastric acid volume and increases pH.

Topic: Premedication | Ref: Ch 4, Page 50

**Q23. Answer: B — Increasing gastric emptying and increasing lower esophageal sphincter tone**

Metoclopramide promotes gastric emptying and increases lower esophageal sphincter tone, reducing aspiration risk.

Topic: Premedication | Ref: Ch 4, Page 50

**Q24. Answer: B — Neutralize gastric acidity**

Sodium citrate (0.3 M) is a non-particulate antacid used to neutralize gastric acid and raise gastric pH.

Topic: Premedication | Ref: Ch 4, Page 50

**Q25. Answer: B — Morphine**

Morphine is used as a premedicant for analgesia and sedation, especially in patients with pain.

Topic: Premedication | Ref: Ch 4, Page 50

**Q26. Answer: C — 60–90 minutes**

Oral premedication is generally given 60–90 minutes before surgery for adequate absorption and effect.

Topic: Premedication | Ref: Ch 4, Page 50

**Q27. Answer: B — Cardiovascular, respiratory, hepatic, renal, and nervous systems**

A thorough preoperative assessment evaluates all major organ systems including cardiovascular, respiratory, hepatic, renal, and nervous systems.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

**Q28. Answer: B — ASA II**

A patient with well-controlled systemic disease like mild diabetes is classified as ASA II.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

**Q29. Answer: C — ASA IV**

Unstable angina represents a severe systemic disease that is a constant threat to life, hence ASA IV.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

**Q30. Answer: B — Blood grouping and cross-matching**

Blood grouping and cross-matching is essential before any major surgery where significant blood loss is anticipated.

Topic: Preoperative Assessment | Ref: Ch 4, Page 48

**Q31. Answer: B — Does not cross the blood-brain barrier**

Glycopyrrolate is a quaternary ammonium compound that does not cross the blood-brain barrier, hence no central effects.

Topic: Premedication | Ref: Ch 4, Page 50

**Q32. Answer: B — Ranitidine + Metoclopramide + Sodium citrate**

The combination of H2 blocker, prokinetic, and non-particulate antacid is used to reduce aspiration risk.

Topic: Premedication | Ref: Ch 4, Page 50

**Q33. Answer: C — 6–8 weeks before surgery**

Ideally, smoking should be stopped 6–8 weeks before surgery to allow improvement in mucociliary function and reduction in secretions.

Topic: Instructions | Ref: Ch 4, Page 48

**Q34. Answer: B — Glaucoma**

Atropine is contraindicated in patients with narrow-angle glaucoma as it can precipitate an acute attack.

Topic: Premedication | Ref: Ch 4, Page 50

**Q35. Answer: B — Promethazine**

Promethazine (a phenothiazine) has antiemetic, sedative, and mild anxiolytic properties making it useful as premedication.

Topic: Premedication | Ref: Ch 4, Page 50

**Q36. Answer: B — Functional cardiac capacity**

METs are used to assess a patient's functional cardiac capacity and exercise tolerance preoperatively.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

**Q37. Answer: C — 4 METs**

Climbing two flights of stairs corresponds to approximately 4 METs, which is considered adequate for most surgeries.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

**Q38. Answer: B — Risks, benefits, alternatives, and plan of anesthesia**

Informed consent for anesthesia must cover the anesthetic plan, risks, benefits, and available alternatives.

Topic: Instructions | Ref: Ch 4, Page 48

**Q39. Answer: A — Metformin**

Metformin should be withheld before surgery due to the risk of lactic acidosis, especially if renal function is compromised.

Topic: Instructions | Ref: Ch 4, Page 48

**Q40. Answer: C — 3–5 days**

Warfarin is typically stopped 3–5 days before elective surgery to allow INR to normalize.

Topic: Instructions | Ref: Ch 4, Page 48

**Q41. Answer: B — Nil Per Oram (nothing by mouth)**

NPO stands for Nil Per Oram, meaning nothing by mouth, referring to preoperative fasting.

Topic: Instructions | Ref: Ch 4, Page 48

**Q42. Answer: C — 7.5–15 mg**

Oral midazolam 7.5–15 mg is commonly used for adult premedication.

Topic: Premedication | Ref: Ch 4, Page 50

**Q43. Answer: B — Water soluble, less painful on injection, and produces anterograde amnesia**

Midazolam is water-soluble (less painful on IV injection), has a shorter duration, and provides reliable anterograde amnesia.

Topic: Premedication | Ref: Ch 4, Page 50

**Q44. Answer: B — Rheumatoid arthritis and Down syndrome**

Patients with rheumatoid arthritis and Down syndrome may have atlantoaxial instability requiring cervical spine assessment.

*Topic: Preoperative Assessment | Ref: Ch 4, Page 47*

**Q45. Answer: B — Pregnancy test (urine/serum beta-hCG)**

A pregnancy test should be considered for all female patients of childbearing age before elective surgery.

*Topic: Preoperative Assessment | Ref: Ch 4, Page 48*

**Q46. Answer: B — Removing all dental prostheses before entering the operating room**

All removable dental prostheses should be removed preoperatively to prevent airway obstruction or damage during intubation.

*Topic: Instructions | Ref: Ch 4, Page 48*

**Q47. Answer: B — Alpha-2 adrenergic receptors**

Dexmedetomidine is a selective alpha-2 adrenergic agonist that provides sedation, anxiolysis, and analgesia.

*Topic: Premedication | Ref: Ch 4, Page 50*

**Q48. Answer: B — Midazolam syrup (0.5 mg/kg) is commonly used for anxiolysis**

Oral midazolam (0.5 mg/kg) is the most commonly used premedicant in children for anxiolysis and separation from parents.

*Topic: Premedication | Ref: Ch 4, Page 50*

**Q49. Answer: B — 4 hours**

Breast milk has a fasting time of 4 hours before elective surgery in infants.

*Topic: Instructions | Ref: Ch 4, Page 48*

**Q50. Answer: B — Coagulation profile (PT, INR, aPTT)**

Coagulation profile is essential in liver disease patients as the liver synthesizes most clotting factors.

*Topic: Preoperative Assessment | Ref: Ch 4, Page 48*



# Difficult Airway Management MCQ – Chapter 5

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 3: Quintessential in Anesthesia (Chapter 5)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. What is the definition of a difficult airway?**

- A. Any airway requiring general anesthesia
- B. A clinical situation where a trained anesthesiologist experiences difficulty with face mask ventilation, intubation, or both
- C. An airway requiring fiber-optic intubation
- D. Any airway in a pediatric patient

**Ans: B**

**Q2. Which of the following is a congenital cause of difficult intubation?**

- A. Rheumatoid arthritis
- B. Pierre Robin syndrome
- C. Acromegaly
- D. Ludwig's angina

**Ans: B**

**Q3. Treacher Collins syndrome causes difficult intubation due to:**

- A. Large tongue
- B. Mandibulofacial dysostosis with micrognathia and malar hypoplasia
- C. Cervical spine fusion
- D. Tracheal stenosis

**Ans: B**

**Q4. Which acquired condition can cause difficult airway due to limited mouth opening?**

- A. Marfan syndrome
- B. Temporomandibular joint ankylosis
- C. Down syndrome
- D. Turner syndrome

**Ans: B**

**Q5. Acromegaly causes difficult airway due to:**

- A. Short neck
- B. Macroglossia and subglottic narrowing
- C. Cervical spine instability
- D. Tracheomalacia

**Ans: B**

**Q6. Which of the following infections can cause acute difficult airway?**

- A. Pneumonia
- B. Ludwig's angina
- C. Urinary tract infection
- D. Cellulitis of the leg

**Ans: B**

**Q7. Difficult intubation is defined as requiring more than how many attempts at laryngoscopy?**

- A. 1 attempt
- B. 2 attempts
- C. 3 attempts
- D. 5 attempts

**Ans: C**

|                                                                                                                                                                                                                                                                                     |        |
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| <p><b>Q8. The Mallampati classification is performed with the patient in which position?</b></p> <p>A. Supine with neck extended<br/>B. Sitting upright with mouth fully open and tongue protruded without phonation<br/>C. Left lateral position<br/>D. Trendelenburg position</p> | Ans: B |
| <p><b>Q9. In Mallampati Class III, which structures are visible?</b></p> <p>A. Soft palate, fauces, uvula, and pillars<br/>B. Soft palate, fauces, and uvula<br/>C. Soft palate and base of uvula only<br/>D. Hard palate only</p>                                                  | Ans: C |
| <p><b>Q10. What is the normal thyromental distance in adults?</b></p> <p>A. Less than 4 cm<br/>B. Greater than 6.5 cm (3 finger breadths)<br/>C. 2 cm<br/>D. 10 cm</p>                                                                                                              | Ans: B |
| <p><b>Q11. The sternomental distance predicting difficult intubation is less than:</b></p> <p>A. 10 cm<br/>B. 12.5 cm<br/>C. 15 cm<br/>D. 20 cm</p>                                                                                                                                 | Ans: B |
| <p><b>Q12. Wilson's risk score for difficult intubation includes all of the following EXCEPT:</b></p> <p>A. Weight<br/>B. Head and neck movement<br/>C. Jaw movement<br/>D. Serum albumin level</p>                                                                                 | Ans: D |
| <p><b>Q13. The Cormack-Lehane grading system classifies the view at:</b></p> <p>A. Nasal endoscopy<br/>B. Direct laryngoscopy<br/>C. CT scan<br/>D. Lateral X-ray of neck</p>                                                                                                       | Ans: B |
| <p><b>Q14. Cormack-Lehane Grade I indicates visualization of:</b></p> <p>A. Only the epiglottis<br/>B. Entire glottic opening<br/>C. Posterior commissure only<br/>D. Neither glottis nor epiglottis</p>                                                                            | Ans: B |
| <p><b>Q15. Cormack-Lehane Grade III indicates visualization of:</b></p> <p>A. Full glottis<br/>B. Posterior part of glottis<br/>C. Only the epiglottis<br/>D. Entire vocal cords</p>                                                                                                | Ans: C |

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| <p><b>Q16. Cormack-Lehane Grade IV indicates:</b></p> <ul style="list-style-type: none"> <li>A. Full view of vocal cords</li> <li>B. Partial view of glottis</li> <li>C. Only epiglottis visible</li> <li>D. Neither glottis nor epiglottis visible</li> </ul>                                                    | <p><b>Ans: D</b></p> |
| <p><b>Q17. An inter-incisor distance (mouth opening) of less than how many centimeters suggests difficult intubation?</b></p> <ul style="list-style-type: none"> <li>A. 1 cm</li> <li>B. 2 cm</li> <li>C. 3 cm (2 finger breadths)</li> <li>D. 5 cm</li> </ul>                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q18. The 'sniffing position' for intubation involves:</b></p> <ul style="list-style-type: none"> <li>A. Neck flexion only</li> <li>B. Flexion of the lower cervical spine and extension at the atlanto-occipital joint</li> <li>C. Full neck extension</li> <li>D. Lateral neck rotation</li> </ul>         | <p><b>Ans: B</b></p> |
| <p><b>Q19. The BURP maneuver stands for:</b></p> <ul style="list-style-type: none"> <li>A. Backward, Upward, Rightward Pressure</li> <li>B. Backward, Upward, Rightward Push</li> <li>C. Backward, Upward, Rightward Pressure on the larynx</li> <li>D. Bilateral, Upward, Rotating Pressure</li> </ul>           | <p><b>Ans: C</b></p> |
| <p><b>Q20. Which is the gold standard technique for managing anticipated difficult airway?</b></p> <ul style="list-style-type: none"> <li>A. Blind nasal intubation</li> <li>B. Awake fiber-optic intubation</li> <li>C. Retrograde intubation</li> <li>D. Surgical tracheostomy</li> </ul>                       | <p><b>Ans: B</b></p> |
| <p><b>Q21. The ASA Difficult Airway Algorithm first step in an anticipated difficult airway is:</b></p> <ul style="list-style-type: none"> <li>A. Immediate tracheostomy</li> <li>B. Awake intubation</li> <li>C. Rapid sequence induction</li> <li>D. LMA insertion</li> </ul>                                   | <p><b>Ans: B</b></p> |
| <p><b>Q22. Which supraglottic airway device is used as a rescue device in the 'cannot intubate, cannot ventilate' scenario?</b></p> <ul style="list-style-type: none"> <li>A. Nasopharyngeal airway</li> <li>B. Laryngeal Mask Airway (LMA)</li> <li>C. Oropharyngeal airway</li> <li>D. Nasal cannula</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q23. In a 'cannot intubate, cannot ventilate' (CICV) emergency, the immediate surgical airway of choice is:</b></p> <ul style="list-style-type: none"> <li>A. Tracheostomy</li> <li>B. Cricothyrotomy (cricothyroid membrane puncture)</li> <li>C. Jet ventilation through nasal cannula</li> <li>D. Fiber-optic intubation</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q24. The cricothyroid membrane is located between:</b></p> <ul style="list-style-type: none"> <li>A. Thyroid and cricoid cartilages</li> <li>B. Cricoid cartilage and first tracheal ring</li> <li>C. Hyoid bone and thyroid cartilage</li> <li>D. Two tracheal rings</li> </ul>                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q25. Which bougie is used as an adjunct during difficult intubation?</b></p> <ul style="list-style-type: none"> <li>A. Foley catheter</li> <li>B. Gum elastic bougie (Eschmann introducer)</li> <li>C. Chest drain</li> <li>D. Central venous catheter</li> </ul>                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q26. Obesity contributes to difficult airway management due to all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Reduced functional residual capacity</li> <li>B. Excessive soft tissue in the airway</li> <li>C. Difficult positioning</li> <li>D. Increased thyromental distance</li> </ul>                              | <p><b>Ans: D</b></p> |
| <p><b>Q27. Which congenital syndrome is associated with macroglossia causing difficult airway?</b></p> <ul style="list-style-type: none"> <li>A. Turner syndrome</li> <li>B. Down syndrome (Trisomy 21)</li> <li>C. Klinefelter syndrome</li> <li>D. Marfan syndrome</li> </ul>                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q28. Ankylosing spondylitis causes difficult intubation primarily due to:</b></p> <ul style="list-style-type: none"> <li>A. Macroglossia</li> <li>B. Fixed flexion deformity of the cervical spine with reduced neck mobility</li> <li>C. Trismus</li> <li>D. Laryngeal edema</li> </ul>                                               | <p><b>Ans: B</b></p> |
| <p><b>Q29. The 'upper lip bite test' assesses:</b></p> <ul style="list-style-type: none"> <li>A. Tongue size</li> <li>B. Jaw protrusion and mandibular mobility</li> <li>C. Neck extension</li> <li>D. Nasal patency</li> </ul>                                                                                                              | <p><b>Ans: B</b></p> |

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| <p><b>Q30. A short, thick neck is associated with difficult intubation because it:</b></p> <ul style="list-style-type: none"> <li>A. Increases the thyromental distance</li> <li>B. Limits neck extension and makes alignment of airway axes difficult</li> <li>C. Causes vocal cord paralysis</li> <li>D. Increases mouth opening</li> </ul>                                  | <p><b>Ans: B</b></p> |
| <p><b>Q31. Which type of laryngoscope blade is preferred for difficult intubation?</b></p> <ul style="list-style-type: none"> <li>A. Miller (straight) blade</li> <li>B. McCoy (levering tip) blade</li> <li>C. Standard Macintosh blade only</li> <li>D. No blade can help</li> </ul>                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q32. Retrograde intubation involves passing a guide wire through the:</b></p> <ul style="list-style-type: none"> <li>A. Nose into the trachea</li> <li>B. Cricothyroid membrane upward and out through the mouth</li> <li>C. Mouth directly into the trachea</li> <li>D. Tracheostomy site</li> </ul>                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q33. Which of the following is NOT a predictor of difficult mask ventilation?</b></p> <ul style="list-style-type: none"> <li>A. Beard</li> <li>B. BMI &gt; 26</li> <li>C. Young age</li> <li>D. Edentulous patient</li> </ul>                                                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q34. The mnemonic LEMON for airway assessment stands for:</b></p> <ul style="list-style-type: none"> <li>A. Look, Evaluate, Mallampati, Obstruction, Neck mobility</li> <li>B. Laryngoscopy, Endotracheal, Mask, Oxygen, Nasopharyngeal</li> <li>C. Length, Extension, Mouth, Obstruction, Narrowing</li> <li>D. Look, Epiglottis, Mandible, Oropharynx, Neck</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q35. The 3-3-2 rule evaluates:</b></p> <ul style="list-style-type: none"> <li>A. Number of intubation attempts</li> <li>B. Mouth opening (3 fingers), hyoid-mental distance (3 fingers), and thyroid-to-floor of mouth (2 fingers)</li> <li>C. Three types of laryngoscope blades</li> <li>D. Three airway axes</li> </ul>                                               | <p><b>Ans: B</b></p> |
| <p><b>Q36. Video laryngoscopy is useful in difficult airway because it:</b></p> <ul style="list-style-type: none"> <li>A. Eliminates the need for neck extension</li> <li>B. Provides an indirect view of the glottis without alignment of airway axes</li> <li>C. Can only be used in awake patients</li> <li>D. Replaces the need for any backup plan</li> </ul>             | <p><b>Ans: B</b></p> |

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| <p><b>Q37. For awake fiber-optic intubation, topical anesthesia of the airway is achieved using:</b></p> <ul style="list-style-type: none"> <li>A. Bupivacaine 0.5%</li> <li>B. Lidocaine 4% (spray/nebulization) and nerve blocks</li> <li>C. Thiopentone gargle</li> <li>D. Propofol spray</li> </ul>   | <p><b>Ans: B</b></p> |
| <p><b>Q38. The superior laryngeal nerve block anesthetizes the airway:</b></p> <ul style="list-style-type: none"> <li>A. Below the vocal cords</li> <li>B. From the epiglottis to the vocal cords (supraglottic area)</li> <li>C. Only the nasal cavity</li> <li>D. Only the tongue</li> </ul>            | <p><b>Ans: B</b></p> |
| <p><b>Q39. Transtracheal injection (recurrent laryngeal nerve block) anesthetizes the area:</b></p> <ul style="list-style-type: none"> <li>A. Above the vocal cords</li> <li>B. Below the vocal cords (infraglottic area)</li> <li>C. Nasal cavity</li> <li>D. Oral cavity</li> </ul>                     | <p><b>Ans: B</b></p> |
| <p><b>Q40. The intubating LMA (Fastrach) is designed to:</b></p> <ul style="list-style-type: none"> <li>A. Replace surgical airway</li> <li>B. Facilitate blind or guided tracheal intubation through the LMA</li> <li>C. Provide long-term ventilation</li> <li>D. Measure tidal volume</li> </ul>       | <p><b>Ans: B</b></p> |
| <p><b>Q41. The maximum number of intubation attempts recommended before declaring failed intubation is:</b></p> <ul style="list-style-type: none"> <li>A. 2</li> <li>B. 3</li> <li>C. 5</li> <li>D. Unlimited</li> </ul>                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q42. Which of the following tumors can cause difficult airway?</b></p> <ul style="list-style-type: none"> <li>A. Renal cell carcinoma</li> <li>B. Laryngeal papilloma or carcinoma</li> <li>C. Hepatocellular carcinoma</li> <li>D. Prostatic carcinoma</li> </ul>                                  | <p><b>Ans: B</b></p> |
| <p><b>Q43. Post-radiation therapy to the neck can cause difficult intubation due to:</b></p> <ul style="list-style-type: none"> <li>A. Increased mouth opening</li> <li>B. Tissue fibrosis, edema, and reduced neck mobility</li> <li>C. Enlarged thyroid</li> <li>D. Vocal cord hypermobility</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q44. The difficult airway cart should contain all of the following EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Various sizes of ETT and LMA</li> <li>B. Fiber-optic bronchoscope</li> <li>C. Cricothyrotomy set</li> <li>D. Cardiopulmonary bypass machine</li> </ul>                                                                                                   | <p><b>Ans: D</b></p> |
| <p><b>Q45. Preoxygenation before intubation attempts is important because it:</b></p> <ul style="list-style-type: none"> <li>A. Makes intubation easier</li> <li>B. Provides an oxygen reserve (extends safe apnea time)</li> <li>C. Reduces blood pressure</li> <li>D. Causes bronchodilation</li> </ul>                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q46. In a patient with cervical spine injury, the preferred method of intubation is:</b></p> <ul style="list-style-type: none"> <li>A. Conventional direct laryngoscopy with full neck extension</li> <li>B. Manual in-line stabilization with fiber-optic or video laryngoscopy</li> <li>C. Blind nasal intubation without precautions</li> <li>D. Immediate tracheostomy</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q47. Jet ventilation through a cricothyrotomy cannula uses oxygen at a pressure of:</b></p> <ul style="list-style-type: none"> <li>A. 1–2 psi</li> <li>B. 50 psi (high-pressure source)</li> <li>C. 100 psi</li> <li>D. Normal wall oxygen at 4 L/min</li> </ul>                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q48. Which of the following is a sign of successful tracheal intubation?</b></p> <ul style="list-style-type: none"> <li>A. Gurgling sounds on auscultation</li> <li>B. Presence of end-tidal CO<sub>2</sub> on capnography</li> <li>C. Distension of the abdomen</li> <li>D. Absence of chest rise</li> </ul>                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q49. The lightwand (lighted stylet) technique works on the principle of:</b></p> <ul style="list-style-type: none"> <li>A. Fluoroscopy</li> <li>B. Transillumination of the anterior neck when the tip is in the trachea</li> <li>C. Ultrasound guidance</li> <li>D. Magnetic navigation</li> </ul>                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q50. Which of the following is the most important step after a failed intubation attempt?</b></p> <ul style="list-style-type: none"> <li>A. Immediately attempt again with the same technique</li> <li>B. Ensure adequate oxygenation and ventilation before the next attempt</li> <li>C. Proceed directly to surgical airway</li> <li>D. Extubate the patient</li> </ul>             | <p><b>Ans: B</b></p> |



# Answer Key & Explanations

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**Q1. Answer: B — A clinical situation where a trained anesthesiologist experiences difficulty with face mask ventilation, intubation, or both**

A difficult airway is defined as a clinical situation in which a conventionally trained anesthesiologist experiences difficulty with face mask ventilation, tracheal intubation, or both.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q2. Answer: B — Pierre Robin syndrome**

Pierre Robin syndrome (micrognathia, glossoptosis, cleft palate) is a congenital cause of difficult intubation.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q3. Answer: B — Mandibulofacial dysostosis with micrognathia and malar hypoplasia**

Treacher Collins syndrome involves mandibulofacial dysostosis with micrognathia, making laryngoscopy and intubation difficult.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q4. Answer: B — Temporomandibular joint ankylosis**

TMJ ankylosis causes severe restriction of mouth opening, making conventional laryngoscopy impossible.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q5. Answer: B — Macroglossia and subglottic narrowing**

Acromegaly leads to macroglossia, prognathism, and subglottic narrowing, all contributing to difficult airway management.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q6. Answer: B — Ludwig's angina**

Ludwig's angina causes submandibular space infection with floor of mouth swelling, leading to airway compromise.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q7. Answer: C — 3 attempts**

Difficult intubation is defined as requiring more than 3 attempts at laryngoscopy or attempts lasting longer than 10 minutes.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q8. Answer: B — Sitting upright with mouth fully open and tongue protruded without phonation**

Mallampati assessment is done with patient sitting, mouth wide open, tongue maximally protruded, without phonation.

Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53

**Q9. Answer: C — Soft palate and base of uvula only**

Mallampati Class III shows only the soft palate and base of uvula, suggesting potentially difficult intubation.

Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53

**Q10. Answer: B — Greater than 6.5 cm (3 finger breadths)**

Normal thyromental distance is greater than 6.5 cm (3 finger breadths). Less than 6 cm suggests difficult intubation.

Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53

**Q11. Answer: B — 12.5 cm**

A sternomental distance less than 12.5 cm with the head fully extended and mouth closed is predictive of difficult intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q12. Answer: D — Serum albumin level**

Wilson's risk score includes weight, head/neck movement, jaw movement, receding mandible, and buck teeth. Serum albumin is not part of it.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q13. Answer: B — Direct laryngoscopy**

Cormack-Lehane grading classifies the laryngoscopic view obtained during direct laryngoscopy.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q14. Answer: B — Entire glottic opening**

Grade I shows the entire glottic opening, indicating easy intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q15. Answer: C — Only the epiglottis**

Grade III shows only the epiglottis with no part of the glottis visible, indicating difficult intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q16. Answer: D — Neither glottis nor epiglottis visible**

Grade IV means neither the glottis nor the epiglottis can be visualized – the most difficult scenario.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q17. Answer: C — 3 cm (2 finger breadths)**

A mouth opening less than 3 cm (2 finger breadths) suggests difficulty with laryngoscopy and intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q18. Answer: B — Flexion of the lower cervical spine and extension at the atlanto-occipital joint**

The sniffing position involves flexion of the lower cervical spine (pillow under head) and extension at the atlanto-occipital joint to align the oral, pharyngeal, and laryngeal axes.

*Topic: Management | Ref: Ch 5, Page 54*

**Q19. Answer: C — Backward, Upward, Rightward Pressure on the larynx**

BURP stands for Backward, Upward, and Rightward Pressure on the thyroid cartilage to improve laryngoscopic view.

*Topic: Management | Ref: Ch 5, Page 54*

**Q20. Answer: B — Awake fiber-optic intubation**

Awake fiber-optic intubation is considered the gold standard for managing an anticipated difficult airway.

*Topic: Management | Ref: Ch 5, Page 54*

**Q21. Answer: B — Awake intubation**

The ASA algorithm recommends awake intubation as the primary approach when a difficult airway is anticipated.

*Topic: Management | Ref: Ch 5, Page 54*

**Q22. Answer: B — Laryngeal Mask Airway (LMA)**

The LMA is a key rescue device in failed intubation scenarios and is part of the difficult airway algorithm.

Topic: Management | Ref: Ch 5, Page 54

**Q23. Answer: B — Cricothyrotomy (cricothyroid membrane puncture)**

Emergency cricothyrotomy through the cricothyroid membrane is the life-saving procedure in CICV situations.

Topic: Management | Ref: Ch 5, Page 54

**Q24. Answer: A — Thyroid and cricoid cartilages**

The cricothyroid membrane lies between the inferior border of the thyroid cartilage and the superior border of the cricoid cartilage.

Topic: Management | Ref: Ch 5, Page 54

**Q25. Answer: B — Gum elastic bougie (Eschmann introducer)**

The gum elastic bougie (Eschmann stylet/introducer) is a commonly used adjunct to facilitate tracheal intubation when the glottic view is poor.

Topic: Management | Ref: Ch 5, Page 54

**Q26. Answer: D — Increased thyromental distance**

Obesity does not increase thyromental distance. It causes excessive soft tissue, reduced FRC, and difficult positioning.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q27. Answer: B — Down syndrome (Trisomy 21)**

Down syndrome is associated with macroglossia, short neck, and atlantoaxial instability, all contributing to difficult airway.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q28. Answer: B — Fixed flexion deformity of the cervical spine with reduced neck mobility**

Ankylosing spondylitis causes fusion of cervical vertebrae leading to a fixed, flexed cervical spine with severely limited extension.

Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52

**Q29. Answer: B — Jaw protrusion and mandibular mobility**

The upper lip bite test assesses the ability to protrude the mandible (jaw protrusion), which correlates with ease of laryngoscopy.

Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53

**Q30. Answer: B — Limits neck extension and makes alignment of airway axes difficult**

A short, thick neck limits neck extension and makes alignment of the oral, pharyngeal, and laryngeal axes difficult.

Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53

**Q31. Answer: B — McCoy (levering tip) blade**

The McCoy blade has a hinged tip that can be levered to lift the epiglottis, improving the view in difficult laryngoscopy.

Topic: Management | Ref: Ch 5, Page 54

**Q32. Answer: B — Cricothyroid membrane upward and out through the mouth**

In retrograde intubation, a wire is passed through the cricothyroid membrane in a cephalad direction, emerging from the mouth, then used as a guide for the ETT.

Topic: Management | Ref: Ch 5, Page 54

**Q33. Answer: C — Young age**

Young age is not a predictor of difficult mask ventilation. Beard, obesity, edentulous status, and snoring are predictors.

Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53

**Q34. Answer: A — Look, Evaluate, Mallampati, Obstruction, Neck mobility**

LEMON stands for Look externally, Evaluate 3-3-2 rule, Mallampati, Obstruction, and Neck mobility.

Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53

**Q35. Answer: B — Mouth opening (3 fingers), hyoid-mental distance (3 fingers), and thyroid-to-floor of mouth (2 fingers)**

The 3-3-2 rule assesses inter-incisor distance (3 fingers), hyoid-mental distance (3 fingers), and thyroid notch-to-floor of mouth (2 fingers).

Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53

**Q36. Answer: B — Provides an indirect view of the glottis without alignment of airway axes**

Video laryngoscopy provides an indirect view of the glottis on a screen without requiring alignment of the oral, pharyngeal, and laryngeal axes.

Topic: Management | Ref: Ch 5, Page 54

**Q37. Answer: B — Lidocaine 4% (spray/nebulization) and nerve blocks**

Topical lidocaine 4% via spray or nebulization, supplemented by nerve blocks (superior laryngeal, transtracheal), provides airway anesthesia for awake FOI.

Topic: Management | Ref: Ch 5, Page 54

**Q38. Answer: B — From the epiglottis to the vocal cords (supraglottic area)**

The internal branch of the superior laryngeal nerve supplies sensation from the epiglottis to and including the vocal cords.

Topic: Management | Ref: Ch 5, Page 54

**Q39. Answer: B — Below the vocal cords (infraglottic area)**

Transtracheal injection through the cricothyroid membrane anesthetizes the trachea below the vocal cords (recurrent laryngeal nerve territory).

Topic: Management | Ref: Ch 5, Page 54

**Q40. Answer: B — Facilitate blind or guided tracheal intubation through the LMA**

The intubating LMA (ILMA/Fastrach) is specifically designed to allow passage of an endotracheal tube through the device.

Topic: Management | Ref: Ch 5, Page 54

**Q41. Answer: B — 3**

After 3 failed attempts at intubation by an experienced laryngoscopist, the situation should be declared as failed intubation and the algorithm followed.

Topic: Management | Ref: Ch 5, Page 54

**Q42. Answer: B — Laryngeal papilloma or carcinoma**

Laryngeal tumors can distort airway anatomy and obstruct the glottic opening, making intubation difficult.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q43. Answer: B — Tissue fibrosis, edema, and reduced neck mobility**

Radiation to the neck causes tissue fibrosis, reduced mobility, and edema leading to difficult airway management.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q44. Answer: D — Cardiopulmonary bypass machine**

A difficult airway cart contains various airway devices (ETTs, LMAs, bougies, fiber-optic scope, cricothyrotomy set) but not a bypass machine.

*Topic: Management | Ref: Ch 5, Page 54*

**Q45. Answer: B — Provides an oxygen reserve (extends safe apnea time)**

Preoxygenation with 100% O<sub>2</sub> denitrogenates the FRC, providing an oxygen reserve that extends safe apnea time during intubation attempts.

*Topic: Management | Ref: Ch 5, Page 54*

**Q46. Answer: B — Manual in-line stabilization with fiber-optic or video laryngoscopy**

In cervical spine injury, intubation is performed with manual in-line stabilization (MILS) using fiber-optic or video laryngoscopy to avoid neck movement.

*Topic: Management | Ref: Ch 5, Page 54*

**Q47. Answer: B — 50 psi (high-pressure source)**

Transtacheal jet ventilation uses a high-pressure oxygen source (typically 50 psi) through a narrow cannula in the cricothyroid membrane.

*Topic: Management | Ref: Ch 5, Page 54*

**Q48. Answer: B — Presence of end-tidal CO<sub>2</sub> on capnography**

Capnography (detection of end-tidal CO<sub>2</sub>) is the gold standard for confirming correct tracheal tube placement.

*Topic: Management | Ref: Ch 5, Page 54*

**Q49. Answer: B — Transillumination of the anterior neck when the tip is in the trachea**

The lightwand uses transillumination – a bright glow seen in the midline of the anterior neck confirms the stylet tip is in the trachea.

*Topic: Management | Ref: Ch 5, Page 54*

**Q50. Answer: B — Ensure adequate oxygenation and ventilation before the next attempt**

After a failed attempt, the priority is to ensure adequate oxygenation and ventilation (bag-mask or LMA) before considering the next attempt or alternative plan.

*Topic: Management | Ref: Ch 5, Page 54*

# Monitoring in Anesthesia MCQ – Chapter 6

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 3: Quintessential in Anesthesia (Chapter 6)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Which of the following is a method of clinical monitoring during anesthesia?**

- A. Pulse oximetry
- B. Observation of skin color, pupil size, and respiratory pattern
- C. Invasive arterial pressure
- D. BIS monitoring

**Ans: B**

**Q2. Clinical signs of adequate depth of anesthesia include all EXCEPT:**

- A. Stable heart rate and blood pressure
- B. Absence of sweating and lacrimation
- C. BIS value of 40–60
- D. Central and constricted pupils

**Ans: C**

**Q3. Dilated pupils during anesthesia may indicate:**

- A. Adequate depth of anesthesia
- B. Light anesthesia, hypoxia, or hypercarbia
- C. Excessive muscle relaxation
- D. Hypothermia

**Ans: B**

**Q4. The minimum mandatory monitoring standards during anesthesia include all EXCEPT:**

- A. Pulse oximetry
- B. ECG
- C. Capnography
- D. Cardiac output monitoring

**Ans: D**

**Q5. ECG monitoring during anesthesia primarily detects:**

- A. Cardiac output
- B. Arrhythmias, ischemia, and heart rate
- C. Blood pressure
- D. Venous return

**Ans: B**

**Q6. Which ECG lead is most sensitive for detecting intraoperative myocardial ischemia?**

- A. Lead I
- B. Lead II
- C. Lead V5
- D. Lead aVR

**Ans: C**

**Q7. Lead II on ECG is preferred during anesthesia for monitoring:**

- A. Left ventricular ischemia
- B. P waves and inferior wall ischemia
- C. Right bundle branch block
- D. Pericardial effusion

**Ans: B**

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| <p><b>Q8. Non-invasive blood pressure (NIBP) measurement by oscillometric method measures:</b></p> <ul style="list-style-type: none"> <li>A. Only systolic pressure directly</li> <li>B. Mean arterial pressure directly, then calculates systolic and diastolic</li> <li>C. Only diastolic pressure directly</li> <li>D. Cardiac output</li> </ul>               | <p><b>Ans: B</b></p> |
| <p><b>Q9. Invasive arterial blood pressure monitoring is indicated in all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Major cardiovascular surgery</li> <li>B. Patients requiring frequent arterial blood gas analysis</li> <li>C. Minor ambulatory surgery in healthy patients</li> <li>D. Patients with severe hemodynamic instability</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q10. The most common site for invasive arterial cannulation is the:</b></p> <ul style="list-style-type: none"> <li>A. Brachial artery</li> <li>B. Radial artery</li> <li>C. Femoral artery</li> <li>D. Dorsalis pedis artery</li> </ul>                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q11. Allen's test is performed before radial artery cannulation to assess:</b></p> <ul style="list-style-type: none"> <li>A. Venous drainage of the hand</li> <li>B. Adequacy of collateral circulation from the ulnar artery</li> <li>C. Nerve supply to the hand</li> <li>D. Patency of the brachial artery</li> </ul>                                    | <p><b>Ans: B</b></p> |
| <p><b>Q12. Central venous pressure (CVP) monitoring catheter tip should ideally be positioned at:</b></p> <ul style="list-style-type: none"> <li>A. Right atrium</li> <li>B. Junction of superior vena cava and right atrium</li> <li>C. Right ventricle</li> <li>D. Pulmonary artery</li> </ul>                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q13. Normal CVP value is:</b></p> <ul style="list-style-type: none"> <li>A. 0–2 mmHg</li> <li>B. 2–8 mmHg (3–11 cm H<sub>2</sub>O)</li> <li>C. 15–20 mmHg</li> <li>D. 25–30 mmHg</li> </ul>                                                                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q14. CVP monitoring is useful for assessing:</b></p> <ul style="list-style-type: none"> <li>A. Left ventricular function</li> <li>B. Right heart filling pressure and intravascular volume status</li> <li>C. Pulmonary artery pressure</li> <li>D. Cerebral perfusion pressure</li> </ul>                                                                  | <p><b>Ans: B</b></p> |



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| <p><b>Q15. The pulmonary artery catheter (Swan-Ganz catheter) can measure all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Pulmonary artery pressure</li> <li>B. Cardiac output by thermodilution</li> <li>C. Pulmonary capillary wedge pressure</li> <li>D. Intracranial pressure</li> </ul>                  | <p><b>Ans: D</b></p> |
| <p><b>Q16. Pulmonary capillary wedge pressure (PCWP) is a surrogate for:</b></p> <ul style="list-style-type: none"> <li>A. Right atrial pressure</li> <li>B. Left atrial pressure and left ventricular end-diastolic pressure</li> <li>C. Systemic vascular resistance</li> <li>D. Pulmonary vascular resistance</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q17. Pulse oximetry measures:</b></p> <ul style="list-style-type: none"> <li>A. Partial pressure of oxygen in blood</li> <li>B. Functional oxygen saturation of hemoglobin (SpO<sub>2</sub>)</li> <li>C. Oxygen content of blood</li> <li>D. Dissolved oxygen in plasma</li> </ul>                                    | <p><b>Ans: B</b></p> |
| <p><b>Q18. Pulse oximetry works on the principle of:</b></p> <ul style="list-style-type: none"> <li>A. Fluorescence</li> <li>B. Beer-Lambert law (spectrophotometry with two wavelengths of light)</li> <li>C. Piezoelectric effect</li> <li>D. Wheatstone bridge</li> </ul>                                                | <p><b>Ans: B</b></p> |
| <p><b>Q19. Which of the following causes a falsely low pulse oximetry reading?</b></p> <ul style="list-style-type: none"> <li>A. Carbon monoxide poisoning</li> <li>B. Methylene blue dye injection</li> <li>C. Anemia</li> <li>D. Jaundice</li> </ul>                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q20. Capnography measures:</b></p> <ul style="list-style-type: none"> <li>A. Oxygen saturation</li> <li>B. End-tidal carbon dioxide (EtCO<sub>2</sub>) concentration</li> <li>C. Tidal volume</li> <li>D. Airway pressure</li> </ul>                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q21. Normal end-tidal CO<sub>2</sub> (EtCO<sub>2</sub>) value is approximately:</b></p> <ul style="list-style-type: none"> <li>A. 10–20 mmHg</li> <li>B. 35–45 mmHg</li> <li>C. 60–80 mmHg</li> <li>D. 5–10 mmHg</li> </ul>                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q22. A sudden drop in EtCO<sub>2</sub> to zero during anesthesia most likely indicates:</b></p> <ul style="list-style-type: none"> <li>A. Bronchospasm</li> <li>B. Circuit disconnection or esophageal intubation</li> <li>C. Hyperventilation</li> <li>D. Increased cardiac output</li> </ul>                        | <p><b>Ans: B</b></p> |

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| <p><b>Q23. A gradual rise in EtCO<sub>2</sub> during anesthesia may indicate:</b></p> <ul style="list-style-type: none"> <li>A. Hyperventilation</li> <li>B. Hypoventilation, increased CO<sub>2</sub> production (malignant hyperthermia), or rebreathing</li> <li>C. Pulmonary embolism</li> <li>D. Cardiac arrest</li> </ul>           | <p><b>Ans: B</b></p> |
| <p><b>Q24. Capnography works on the principle of:</b></p> <ul style="list-style-type: none"> <li>A. Mass spectrometry</li> <li>B. Infrared absorption spectroscopy</li> <li>C. Piezoelectric effect</li> <li>D. Paramagnetic analysis</li> </ul>                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q25. Mainstream capnography differs from sidestream in that:</b></p> <ul style="list-style-type: none"> <li>A. It samples gas away from the patient circuit</li> <li>B. The sensor is placed directly in the breathing circuit (at the airway)</li> <li>C. It has a longer response time</li> <li>D. It is less accurate</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q26. Spirometry during anesthesia can measure all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Tidal volume</li> <li>B. Minute ventilation</li> <li>C. Compliance and resistance</li> <li>D. Cardiac output</li> </ul>                                                                                                 | <p><b>Ans: D</b></p> |
| <p><b>Q27. Airway pressure monitoring helps detect all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Circuit disconnection</li> <li>B. Bronchospasm</li> <li>C. Endobronchial intubation</li> <li>D. Myocardial ischemia</li> </ul>                                                                                           | <p><b>Ans: D</b></p> |
| <p><b>Q28. Oxygen analyzers in the anesthetic circuit commonly use which principle?</b></p> <ul style="list-style-type: none"> <li>A. Infrared absorption</li> <li>B. Paramagnetic or galvanic cell (fuel cell) principle</li> <li>C. Mass spectrometry only</li> <li>D. Colorimetric detection</li> </ul>                                | <p><b>Ans: B</b></p> |
| <p><b>Q29. The normal core body temperature range during anesthesia should be maintained at:</b></p> <ul style="list-style-type: none"> <li>A. 34–35°C</li> <li>B. 36–37.5°C</li> <li>C. 38–39°C</li> <li>D. 32–34°C</li> </ul>                                                                                                           | <p><b>Ans: B</b></p> |

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| <p><b>Q30. Sites for core temperature monitoring include all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Nasopharynx</li> <li>B. Distal esophagus</li> <li>C. Tympanic membrane</li> <li>D. Axilla</li> </ul>                                                                                                                   | <p><b>Ans: D</b></p> |
| <p><b>Q31. Perioperative hypothermia is defined as core temperature below:</b></p> <ul style="list-style-type: none"> <li>A. 37°C</li> <li>B. 36°C</li> <li>C. 35°C</li> <li>D. 34°C</li> </ul>                                                                                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q32. Consequences of perioperative hypothermia include all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Impaired coagulation</li> <li>B. Increased wound infection</li> <li>C. Shivering and increased oxygen consumption</li> <li>D. Decreased duration of neuromuscular blockade</li> </ul>                              | <p><b>Ans: D</b></p> |
| <p><b>Q33. Malignant hyperthermia is suspected when temperature monitoring shows:</b></p> <ul style="list-style-type: none"> <li>A. Gradual decrease in temperature</li> <li>B. Rapid and unexplained rise in temperature with rising EtCO<sub>2</sub></li> <li>C. Stable temperature throughout</li> <li>D. Temperature of 36.5°C</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q34. The mechanism of redistribution hypothermia in the first hour of anesthesia is:</b></p> <ul style="list-style-type: none"> <li>A. Excessive fluid administration</li> <li>B. Heat redistribution from core to peripheral compartment due to vasodilation</li> <li>C. Cold operating room only</li> <li>D. Shivering</li> </ul>     | <p><b>Ans: B</b></p> |
| <p><b>Q35. Neuromuscular monitoring is used to assess:</b></p> <ul style="list-style-type: none"> <li>A. Depth of anesthesia</li> <li>B. Degree of neuromuscular blockade and recovery from muscle relaxants</li> <li>C. Blood pressure</li> <li>D. Cerebral activity</li> </ul>                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q36. The most commonly used nerve for peripheral nerve stimulation in neuromuscular monitoring is:</b></p> <ul style="list-style-type: none"> <li>A. Median nerve</li> <li>B. Ulnar nerve (adductor pollicis response)</li> <li>C. Radial nerve</li> <li>D. Femoral nerve</li> </ul>                                                    | <p><b>Ans: B</b></p> |

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| <p><b>Q37. Train-of-four (TOF) stimulation consists of:</b></p> <ul style="list-style-type: none"> <li>A. Single twitch at 1 Hz</li> <li>B. Four supramaximal stimuli at 2 Hz (every 0.5 seconds)</li> <li>C. Tetanic stimulation at 50 Hz</li> <li>D. Double burst stimulation</li> </ul>               | <p><b>Ans: B</b></p> |
| <p><b>Q38. A TOF ratio of greater than 0.9 indicates:</b></p> <ul style="list-style-type: none"> <li>A. Complete neuromuscular block</li> <li>B. Adequate recovery from neuromuscular blockade (safe to extubate)</li> <li>C. Need for more muscle relaxant</li> <li>D. Onset of block</li> </ul>        | <p><b>Ans: B</b></p> |
| <p><b>Q39. TOF count of zero (no twitches visible) indicates:</b></p> <ul style="list-style-type: none"> <li>A. No neuromuscular block</li> <li>B. Intense (deep) neuromuscular block (&gt;95% receptor occupation)</li> <li>C. Partial block only</li> <li>D. Equipment malfunction</li> </ul>          | <p><b>Ans: B</b></p> |
| <p><b>Q40. Post-tetanic count (PTC) is used when:</b></p> <ul style="list-style-type: none"> <li>A. TOF shows 4 twitches</li> <li>B. TOF count is zero (intense block) to assess depth of profound block</li> <li>C. Patient is fully recovered</li> <li>D. No muscle relaxant has been given</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q41. Fade in TOF response is characteristic of:</b></p> <ul style="list-style-type: none"> <li>A. Depolarizing block (Phase I succinylcholine block)</li> <li>B. Non-depolarizing neuromuscular block</li> <li>C. No block at all</li> <li>D. Myasthenia gravis only</li> </ul>                    | <p><b>Ans: B</b></p> |
| <p><b>Q42. The Bispectral Index (BIS) monitor is used to assess:</b></p> <ul style="list-style-type: none"> <li>A. Neuromuscular blockade</li> <li>B. Depth of anesthesia/level of consciousness</li> <li>C. Blood pressure</li> <li>D. Oxygen saturation</li> </ul>                                     | <p><b>Ans: B</b></p> |
| <p><b>Q43. The recommended BIS value range for adequate general anesthesia is:</b></p> <ul style="list-style-type: none"> <li>A. 80–100</li> <li>B. 40–60</li> <li>C. 0–20</li> <li>D. 60–80</li> </ul>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q44. A BIS value of 100 indicates:</b></p> <ul style="list-style-type: none"> <li>A. Deep anesthesia</li> <li>B. Fully awake state</li> <li>C. Brain death</li> <li>D. Burst suppression</li> </ul>                                                                                                | <p><b>Ans: B</b></p> |

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| <p><b>Q45. Evoked potential monitoring during anesthesia is used to assess:</b></p> <ul style="list-style-type: none"> <li>A. Blood pressure changes</li> <li>B. Integrity of neural pathways (spinal cord, brainstem, cortex)</li> <li>C. Muscle relaxation</li> <li>D. Temperature changes</li> </ul>                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q46. Monitoring blood loss during surgery can be estimated by all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Weighing surgical swabs/sponges</li> <li>B. Measuring suction bottle contents</li> <li>C. Monitoring pulse oximetry alone</li> <li>D. Visual estimation and hemodynamic parameters</li> </ul>                                   | <p><b>Ans: C</b></p> |
| <p><b>Q47. The formula for estimated blood loss by swab weighing is:</b></p> <ul style="list-style-type: none"> <li>A. Weight of dry swab minus weight of wet swab</li> <li>B. Weight of wet (blood-soaked) swab minus weight of dry swab (1 gram = 1 mL blood)</li> <li>C. Volume of suction divided by 2</li> <li>D. Hematocrit multiplied by weight</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q48. Maximum allowable blood loss (MABL) depends on:</b></p> <ul style="list-style-type: none"> <li>A. Patient's height only</li> <li>B. Estimated blood volume, starting hematocrit, and minimum acceptable hematocrit</li> <li>C. Type of surgery only</li> <li>D. Duration of surgery</li> </ul>                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q49. Urine output monitoring during surgery is a reliable indicator of:</b></p> <ul style="list-style-type: none"> <li>A. Depth of anesthesia</li> <li>B. Renal perfusion and adequacy of intravascular volume</li> <li>C. Neuromuscular blockade</li> <li>D. Blood glucose level</li> </ul>                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q50. Which of the following is used as a point-of-care test to rapidly assess hemoglobin during surgery?</b></p> <ul style="list-style-type: none"> <li>A. CT scan</li> <li>B. HemoCue or arterial blood gas analyzer</li> <li>C. Pulse oximetry</li> <li>D. Capnography</li> </ul>                                                                         | <p><b>Ans: B</b></p> |

# Answer Key & Explanations

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## **Q1. Answer: B — Observation of skin color, pupil size, and respiratory pattern**

Clinical monitoring involves direct observation by the anesthesiologist including skin color, pupil size, chest movements, and peripheral perfusion without instruments.

*Topic: Clinical Monitoring | Ref: Ch 6, Page 57*

## **Q2. Answer: C — BIS value of 40–60**

BIS monitoring is an instrumental (advanced) monitor, not a clinical sign. Clinical signs include stable vitals, pupil size, and absence of autonomic responses.

*Topic: Clinical Monitoring | Ref: Ch 6, Page 57*

## **Q3. Answer: B — Light anesthesia, hypoxia, or hypercarbia**

Dilated pupils during anesthesia may indicate light anesthesia, hypoxia, hypercarbia, or sympathetic stimulation.

*Topic: Clinical Monitoring | Ref: Ch 6, Page 57*

## **Q4. Answer: D — Cardiac output monitoring**

Minimum monitoring standards include pulse oximetry, ECG, NIBP, capnography, and temperature. Cardiac output monitoring is an advanced monitor used in select cases.

*Topic: Advance Monitoring (Instrumental Monitoring) | Ref: Ch 6, Page 57*

## **Q5. Answer: B — Arrhythmias, ischemia, and heart rate**

ECG monitoring detects cardiac rhythm disturbances, myocardial ischemia (ST changes), and heart rate during anesthesia.

*Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57*

## **Q6. Answer: C — Lead V5**

Lead V5 detects approximately 75% of intraoperative ST-segment changes indicative of left ventricular ischemia.

*Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57*

## **Q7. Answer: B — P waves and inferior wall ischemia**

Lead II provides the best P-wave visualization for rhythm analysis and detects inferior wall ischemia.

*Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57*

## **Q8. Answer: B — Mean arterial pressure directly, then calculates systolic and diastolic**

The oscillometric method directly measures mean arterial pressure (at maximum oscillation amplitude) and derives systolic and diastolic pressures.

*Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57*

## **Q9. Answer: C — Minor ambulatory surgery in healthy patients**

Invasive arterial monitoring is not indicated for minor ambulatory surgery in healthy patients. It is reserved for major surgeries and hemodynamically unstable patients.

*Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57*

**Q10. Answer: B — Radial artery**

The radial artery is the most common site for arterial cannulation due to superficial location, good collateral circulation (ulnar artery), and accessibility.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

**Q11. Answer: B — Adequacy of collateral circulation from the ulnar artery**

Allen's test assesses the adequacy of ulnar artery collateral flow to the hand before radial artery cannulation.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

**Q12. Answer: B — Junction of superior vena cava and right atrium**

The CVP catheter tip should be at the junction of the SVC and right atrium for accurate pressure measurement.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

**Q13. Answer: B — 2–8 mmHg (3–11 cm H<sub>2</sub>O)**

Normal CVP ranges from 2–8 mmHg (approximately 3–11 cm H<sub>2</sub>O).

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

**Q14. Answer: B — Right heart filling pressure and intravascular volume status**

CVP reflects right atrial pressure, which indicates right heart filling pressure and helps guide fluid management.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

**Q15. Answer: D — Intracranial pressure**

The PA catheter measures PA pressures, PCWP, cardiac output, and mixed venous oxygen saturation. ICP requires a separate intracranial monitor.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

**Q16. Answer: B — Left atrial pressure and left ventricular end-diastolic pressure**

PCWP approximates left atrial pressure and hence left ventricular preload (LVEDP) in the absence of mitral valve disease.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

**Q17. Answer: B — Functional oxygen saturation of hemoglobin (SpO<sub>2</sub>)**

Pulse oximetry measures the percentage of oxyhemoglobin (SpO<sub>2</sub>) using spectrophotometric principles at two wavelengths.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q18. Answer: B — Beer-Lambert law (spectrophotometry with two wavelengths of light)**

Pulse oximetry uses two wavelengths (660 nm red and 940 nm infrared) based on the Beer-Lambert law to determine oxygen saturation.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q19. Answer: B — Methylene blue dye injection**

Methylene blue absorbs light at 660 nm, causing falsely low SpO<sub>2</sub> readings. Carbon monoxide causes falsely HIGH readings.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q20. Answer: B — End-tidal carbon dioxide (EtCO<sub>2</sub>) concentration**

Capnography continuously measures and displays the CO<sub>2</sub> concentration in expired gases, with EtCO<sub>2</sub> reflecting alveolar CO<sub>2</sub>.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q21. Answer: B — 35–45 mmHg**

Normal EtCO<sub>2</sub> is 35–45 mmHg, which is slightly lower than arterial PaCO<sub>2</sub> due to alveolar dead space.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q22. Answer: B — Circuit disconnection or esophageal intubation**

A sudden drop of EtCO<sub>2</sub> to zero indicates circuit disconnection, extubation, or esophageal intubation (no alveolar gas being sampled).

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q23. Answer: B — Hypoventilation, increased CO<sub>2</sub> production (malignant hyperthermia), or rebreathing**

Gradual EtCO<sub>2</sub> rise suggests hypoventilation, increased metabolic CO<sub>2</sub> production (e.g., malignant hyperthermia, fever), or CO<sub>2</sub> rebreathing.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q24. Answer: B — Infrared absorption spectroscopy**

Capnographs use infrared absorption – CO<sub>2</sub> absorbs infrared light at a wavelength of 4.26 micrometers.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q25. Answer: B — The sensor is placed directly in the breathing circuit (at the airway)**

Mainstream capnography places the sensor directly in the breathing circuit, providing faster response time but adding weight/dead space.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q26. Answer: D — Cardiac output**

Intraoperative spirometry measures tidal volume, minute ventilation, flow rates, compliance, and resistance – not cardiac output.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q27. Answer: D — Myocardial ischemia**

Airway pressure monitoring detects mechanical problems (disconnection, obstruction, bronchospasm, endobronchial intubation) but not myocardial ischemia.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q28. Answer: B — Paramagnetic or galvanic cell (fuel cell) principle**

Oxygen analyzers use paramagnetic properties of O<sub>2</sub> or electrochemical (galvanic/fuel cell) principles to measure FiO<sub>2</sub>.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

**Q29. Answer: B — 36–37.5°C**

Core body temperature should be maintained between 36–37.5°C during anesthesia to avoid complications of hypothermia or hyperthermia.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

**Q30. Answer: D — Axilla**

The axilla measures peripheral (shell) temperature. Core temperature sites include nasopharynx, distal esophagus, tympanic membrane, and pulmonary artery.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65



**Q31. Answer: B — 36°C**

Perioperative hypothermia is defined as core temperature below 36°C and is common during general anesthesia.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

**Q32. Answer: D — Decreased duration of neuromuscular blockade**

Hypothermia prolongs the duration of neuromuscular blockade (not decreases it). Other complications include coagulopathy, wound infection, and shivering.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

**Q33. Answer: B — Rapid and unexplained rise in temperature with rising EtCO<sub>2</sub>**

Malignant hyperthermia presents with rapidly rising temperature, markedly elevated EtCO<sub>2</sub>, muscle rigidity, and tachycardia.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

**Q34. Answer: B — Heat redistribution from core to peripheral compartment due to vasodilation**

Anesthetic-induced vasodilation causes redistribution of heat from the warm core to the cooler peripheral compartment, dropping core temperature rapidly in the first hour.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

**Q35. Answer: B — Degree of neuromuscular blockade and recovery from muscle relaxants**

Neuromuscular monitoring quantifies the degree of neuromuscular blockade and guides timing of reversal and extubation.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

**Q36. Answer: B — Ulnar nerve (adductor pollicis response)**

The ulnar nerve at the wrist is most commonly stimulated, and the response of the adductor pollicis muscle is observed.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

**Q37. Answer: B — Four supramaximal stimuli at 2 Hz (every 0.5 seconds)**

TOF consists of four supramaximal stimuli delivered at 2 Hz (0.5 second intervals), with the ratio of 4th to 1st twitch indicating block depth.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

**Q38. Answer: B — Adequate recovery from neuromuscular blockade (safe to extubate)**

A TOF ratio  $\geq 0.9$  indicates adequate recovery from neuromuscular blockade and is the threshold for safe extubation.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

**Q39. Answer: B — Intense (deep) neuromuscular block (>95% receptor occupation)**

Zero twitches on TOF indicates intense/deep block with greater than 95% of receptors occupied.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

**Q40. Answer: B — TOF count is zero (intense block) to assess depth of profound block**

PTC is used during intense block (TOF count = 0) to quantify the depth of block by counting post-tetanic facilitation twitches.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

**Q41. Answer: B — Non-depolarizing neuromuscular block**

Fade (progressive decrease in twitch height) is characteristic of non-depolarizing block. Phase I depolarizing block shows uniform decrease without fade.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

**Q42. Answer: B — Depth of anesthesia/level of consciousness**

BIS is a processed EEG parameter that provides a numerical value (0–100) correlating with the depth of anesthesia/consciousness.

Topic: Central Nervous System (CNS) Monitoring | Ref: Ch 6, Page 68

**Q43. Answer: B — 40–60**

BIS values of 40–60 indicate adequate depth of general anesthesia with low probability of awareness.

Topic: Central Nervous System (CNS) Monitoring | Ref: Ch 6, Page 68

**Q44. Answer: B — Fully awake state**

BIS of 100 represents a fully awake state, while 0 represents EEG silence (isoelectric).

Topic: Central Nervous System (CNS) Monitoring | Ref: Ch 6, Page 68

**Q45. Answer: B — Integrity of neural pathways (spinal cord, brainstem, cortex)**

Evoked potentials (SSEP, MEP, BAEP) monitor the functional integrity of neural pathways during surgeries that risk neurological injury.

Topic: Central Nervous System (CNS) Monitoring | Ref: Ch 6, Page 68

**Q46. Answer: C — Monitoring pulse oximetry alone**

Pulse oximetry alone cannot estimate blood loss. Methods include swab weighing, suction measurement, visual estimation, and hemodynamic monitoring.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

**Q47. Answer: B — Weight of wet (blood-soaked) swab minus weight of dry swab (1 gram = 1 mL blood)**

Blood loss = (weight of blood-soaked swab – weight of dry swab) in grams, where 1 gram of blood ≈ 1 mL.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

**Q48. Answer: B — Estimated blood volume, starting hematocrit, and minimum acceptable hematocrit**

$MABL = EBV \times (\text{Starting Hct} - \text{Minimum acceptable Hct}) / \text{Starting Hct}$ . It depends on blood volume and hematocrit values.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

**Q49. Answer: B — Renal perfusion and adequacy of intravascular volume**

Urine output (target ≥0.5 mL/kg/hr) reflects renal perfusion and is a useful guide to the adequacy of fluid resuscitation and cardiac output.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

**Q50. Answer: B — HemoCue or arterial blood gas analyzer**

HemoCue and ABG analyzers provide rapid point-of-care hemoglobin measurement to guide transfusion decisions intraoperatively.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

# Fluids and Blood Transfusion MCQ – Chapter 7

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 3: Quintessential in Anesthesia (Chapter 7)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

**Q1. According to the textbook, fluids are broadly divided into which two categories?**

- A. Isotonic and hypertonic
- B. Crystalloids and colloids
- C. Dextrose and saline
- D. Natural and synthetic

**Ans: B**

**Q2. What is the composition of Plasmalyte solution as per the textbook?**

- A. Na+ 154, Cl- 154 mEq/L
- B. Na+ 140, Cl- 98, K+ 3, Mg2+ 3, Acetate 27, Gluconate 23 mEq/L
- C. Na+ 131, Cl- 111, K+ 5 mEq/L
- D. Na+ 130, Cl- 109, Lactate 28 mEq/L

**Ans: B**

**Q3. Normal saline is preferred over Ringer lactate in all of the following conditions EXCEPT:**

- A. Hypochloremic metabolic alkalosis
- B. Brain injury
- C. Hyponatremia
- D. Hypokalemia

**Ans: D**

**Q4. According to Table 7.1, crystalloids should be replaced in what ratio to blood loss?**

- A. 1:1 ratio
- B. 1.5:1 ratio (150 mL crystalloid for 100 mL blood loss)
- C. 3:1 ratio
- D. 2:1 ratio

**Ans: B**

**Q5. Why should blood NOT be given through the same drip set as Ringer lactate?**

- A. RL is too viscous
- B. RL contains calcium which can cause clotting of blood
- C. RL is hypertonic and can lyse RBCs
- D. RL contains potassium which causes hemolysis

**Ans: B**

**Q6. Which of the following is an isotonic crystalloid solution?**

- A. 5% Dextrose
- B. Normal saline (0.9% NaCl)
- C. 3% Hypertonic saline
- D. 0.45% NaCl

**Ans: B**

**Q7. The sodium content of normal saline (0.9% NaCl) is:**

- A. 100 mEq/L
- B. 130 mEq/L
- C. 154 mEq/L
- D. 170 mEq/L

**Ans: C**

|                                                                                                                                                                                                                                                                                                                                                                                |                      |
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| <p><b>Q8. Ringer's lactate solution contains all of the following EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Na<sup>+</sup> 131 mEq/L</li> <li>B. K<sup>+</sup> 5 mEq/L</li> <li>C. Lactate 29 mEq/L</li> <li>D. Bicarbonate 29 mEq/L</li> </ul>                                                                                                                | <p><b>Ans: D</b></p> |
| <p><b>Q9. The lactate in Ringer's lactate is converted to bicarbonate in the:</b></p> <ul style="list-style-type: none"> <li>A. Kidney</li> <li>B. Liver</li> <li>C. Lung</li> <li>D. Muscle</li> </ul>                                                                                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q10. Excessive administration of normal saline can cause:</b></p> <ul style="list-style-type: none"> <li>A. Metabolic alkalosis</li> <li>B. Hyperchloremic metabolic acidosis</li> <li>C. Respiratory alkalosis</li> <li>D. Hypokalemia</li> </ul>                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q11. 5% Dextrose is considered isotonic in the bag but acts as a hypotonic solution in vivo because:</b></p> <ul style="list-style-type: none"> <li>A. It contains sodium</li> <li>B. Dextrose is rapidly metabolized, leaving free water that distributes across all compartments</li> <li>C. It has high osmolarity</li> <li>D. It contains lactate</li> </ul>         | <p><b>Ans: B</b></p> |
| <p><b>Q12. According to Table 7.1, the intravascular half-life of crystalloids is approximately:</b></p> <ul style="list-style-type: none"> <li>A. 5 minutes</li> <li>B. 30 minutes</li> <li>C. 2–4 hours</li> <li>D. 24 hours</li> </ul>                                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q13. Colloids differ from crystalloids in that they:</b></p> <ul style="list-style-type: none"> <li>A. Are freely filtered across capillary membranes</li> <li>B. Contain large molecular weight substances that maintain oncotic pressure and stay intravascular longer</li> <li>C. Are always synthetic</li> <li>D. Cannot be used for volume resuscitation</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q14. Which of the following is a natural colloid?</b></p> <ul style="list-style-type: none"> <li>A. Hydroxyethyl starch (HES)</li> <li>B. Dextran</li> <li>C. Human albumin</li> <li>D. Gelatin (Haemaccel)</li> </ul>                                                                                                                                                   | <p><b>Ans: C</b></p> |

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| <p><b>Q15. The approximate intravascular half-life of infused colloids compared to crystalloids is:</b></p> <ul style="list-style-type: none"> <li>A. Shorter</li> <li>B. Same</li> <li>C. Longer (several hours)</li> <li>D. Only minutes</li> </ul>                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q16. Dextran 40 is used primarily for:</b></p> <ul style="list-style-type: none"> <li>A. Volume expansion only</li> <li>B. Improving microcirculation and reducing blood viscosity (rheological effects)</li> <li>C. Providing nutrition</li> <li>D. Correcting acidosis</li> </ul>                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q17. A major side effect of dextran infusion is:</b></p> <ul style="list-style-type: none"> <li>A. Hyperkalemia</li> <li>B. Anaphylactic reactions and interference with blood cross-matching</li> <li>C. Metabolic alkalosis</li> <li>D. Hypertension</li> </ul>                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q18. Hydroxyethyl starch (HES) solutions can cause:</b></p> <ul style="list-style-type: none"> <li>A. Improved renal function</li> <li>B. Coagulopathy and renal impairment at high doses</li> <li>C. Hypernatremia</li> <li>D. Metabolic alkalosis</li> </ul>                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q19. Gelatin-based colloids (e.g., Haemaccel) have which disadvantage?</b></p> <ul style="list-style-type: none"> <li>A. Very long intravascular half-life</li> <li>B. Risk of anaphylactoid reactions and relatively short duration of action</li> <li>C. Severe hyperkalemia</li> <li>D. Incompatible with all blood products</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q20. The daily maintenance fluid requirement for a 70 kg adult using the 4-2-1 rule is approximately:</b></p> <ul style="list-style-type: none"> <li>A. 50 mL/hr</li> <li>B. 110 mL/hr</li> <li>C. 150 mL/hr</li> <li>D. 200 mL/hr</li> </ul>                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q21. The 4-2-1 rule for hourly maintenance fluid calculation gives 4 mL/kg/hr for:</b></p> <ul style="list-style-type: none"> <li>A. First 20 kg</li> <li>B. First 10 kg</li> <li>C. Body weight above 20 kg</li> <li>D. All body weight</li> </ul>                                                                                        | <p><b>Ans: B</b></p> |

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| <p><b>Q22. Intraoperative fluid therapy must account for:</b></p> <ul style="list-style-type: none"> <li>A. Maintenance requirements only</li> <li>B. Maintenance, deficit (fasting), and ongoing losses (surgical/third space)</li> <li>C. Blood loss only</li> <li>D. Urine output only</li> </ul>                                              | <p><b>Ans: B</b></p> |
| <p><b>Q23. Third-space fluid loss refers to:</b></p> <ul style="list-style-type: none"> <li>A. Blood loss into the suction canister</li> <li>B. Functional loss of fluid into a non-equilibrating space (e.g., bowel wall, traumatized tissues)</li> <li>C. Urine output</li> <li>D. Sweat loss</li> </ul>                                        | <p><b>Ans: B</b></p> |
| <p><b>Q24. For major abdominal surgery, the estimated third-space fluid loss is approximately:</b></p> <ul style="list-style-type: none"> <li>A. 1–2 mL/kg/hr</li> <li>B. 4–8 mL/kg/hr</li> <li>C. 6–10 mL/kg/hr</li> <li>D. 0.5 mL/kg/hr</li> </ul>                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q25. The preoperative fasting deficit is calculated as:</b></p> <ul style="list-style-type: none"> <li>A. Body weight × hours of fasting</li> <li>B. Hourly maintenance rate × number of hours NPO</li> <li>C. Blood volume × hematocrit</li> <li>D. Urine output × hours</li> </ul>                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q26. Whole blood transfusion contains all components of blood and has a hematocrit of approximately:</b></p> <ul style="list-style-type: none"> <li>A. 20–25%</li> <li>B. 35–40%</li> <li>C. 55–60%</li> <li>D. 70–80%</li> </ul>                                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q27. According to Table 7.1, colloids in high doses can interfere with clotting primarily by:</b></p> <ul style="list-style-type: none"> <li>A. Activating platelets</li> <li>B. Decreasing Factor VIII level and causing renal dysfunction</li> <li>C. Increasing fibrinogen levels</li> <li>D. Stimulating thrombin production</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q28. One unit of packed RBCs raises hemoglobin by approximately:</b></p> <ul style="list-style-type: none"> <li>A. 0.5 g/dL</li> <li>B. 1 g/dL</li> <li>C. 3 g/dL</li> <li>D. 5 g/dL</li> </ul>                                                                                                                                             | <p><b>Ans: B</b></p> |

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| <p><b>Q29. The storage temperature for packed red blood cells is:</b></p> <ul style="list-style-type: none"> <li>A. Room temperature (20–24°C)</li> <li>B. 1–6°C</li> <li>C. -20°C</li> <li>D. -80°C</li> </ul>                                                                                                                                               | Ans: B |
| <p><b>Q30. Fresh frozen plasma (FFP) is indicated for:</b></p> <ul style="list-style-type: none"> <li>A. Anemia</li> <li>B. Replacement of coagulation factors in patients with coagulopathy</li> <li>C. Volume expansion as first choice</li> <li>D. Thrombocytopenia</li> </ul>                                                                             | Ans: B |
| <p><b>Q31. Platelet concentrate transfusion is indicated when platelet count falls below:</b></p> <ul style="list-style-type: none"> <li>A. 150,000/μL</li> <li>B. 100,000/μL</li> <li>C. 50,000/μL (or &lt;75,000 with active bleeding/surgery)</li> <li>D. 200,000/μL</li> </ul>                                                                            | Ans: C |
| <p><b>Q32. Platelets are stored at:</b></p> <ul style="list-style-type: none"> <li>A. 1–6°C</li> <li>B. 20–24°C with continuous agitation</li> <li>C. -30°C</li> <li>D. 37°C</li> </ul>                                                                                                                                                                       | Ans: B |
| <p><b>Q33. The most common cause of fatal transfusion reaction is:</b></p> <ul style="list-style-type: none"> <li>A. Allergic reaction</li> <li>B. ABO-incompatible (hemolytic) transfusion due to clerical error</li> <li>C. Febrile non-hemolytic reaction</li> <li>D. Iron overload</li> </ul>                                                             | Ans: B |
| <p><b>Q34. Signs of acute hemolytic transfusion reaction under general anesthesia include:</b></p> <ul style="list-style-type: none"> <li>A. Hypothermia and bradycardia</li> <li>B. Unexplained hypotension, fever, hemoglobinuria, and DIC</li> <li>C. Hypertension and tachycardia only</li> <li>D. Bronchospasm only</li> </ul>                           | Ans: B |
| <p><b>Q35. The first step in managing suspected hemolytic transfusion reaction is:</b></p> <ul style="list-style-type: none"> <li>A. Continue transfusion at slower rate</li> <li>B. Stop the transfusion immediately and maintain IV access</li> <li>C. Administer more fluids through the same line</li> <li>D. Give antihistamines and continue</li> </ul> | Ans: B |
| <p><b>Q36. Transfusion-related acute lung injury (TRALI) presents with:</b></p> <ul style="list-style-type: none"> <li>A. Delayed hemolysis after 7 days</li> <li>B. Acute respiratory distress, non-cardiogenic pulmonary edema within 6 hours of transfusion</li> <li>C. Chronic iron overload</li> <li>D. Urticaria only</li> </ul>                        | Ans: B |



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| <p><b>Q37. Massive blood transfusion is defined as replacement of:</b></p> <ul style="list-style-type: none"> <li>A. One unit of blood</li> <li>B. More than one blood volume (or &gt;10 units PRBCs) in 24 hours</li> <li>C. Any amount of FFP</li> <li>D. 500 mL of blood</li> </ul>                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q38. Complications of massive blood transfusion include all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Hypothermia</li> <li>B. Hyperkalemia</li> <li>C. Citrate toxicity (hypocalcemia)</li> <li>D. Polycythemia</li> </ul>                                                                                          | <p><b>Ans: D</b></p> |
| <p><b>Q39. Citrate toxicity from massive transfusion manifests as:</b></p> <ul style="list-style-type: none"> <li>A. Hyperkalemia</li> <li>B. Hypocalcemia (ionized calcium reduction) causing cardiac depression</li> <li>C. Hypernatremia</li> <li>D. Metabolic alkalosis</li> </ul>                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q40. Hyperkalemia in stored blood occurs because:</b></p> <ul style="list-style-type: none"> <li>A. Potassium is added as preservative</li> <li>B. Red cell membrane Na-K ATPase pump fails during storage, leaking potassium into plasma</li> <li>C. Citrate releases potassium</li> <li>D. Platelets release potassium</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q41. Febrile non-hemolytic transfusion reaction is caused by:</b></p> <ul style="list-style-type: none"> <li>A. ABO incompatibility</li> <li>B. Antibodies against donor leukocyte antigens or cytokines in stored blood</li> <li>C. Bacterial contamination</li> <li>D. Volume overload</li> </ul>                                 | <p><b>Ans: B</b></p> |
| <p><b>Q42. Transfusion-associated circulatory overload (TACO) is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. Hypotension and bradycardia</li> <li>B. Pulmonary edema, hypertension, dyspnea due to volume overload</li> <li>C. Hemolysis</li> <li>D. Fever only</li> </ul>                                        | <p><b>Ans: B</b></p> |
| <p><b>Q43. Infectious complications of blood transfusion include all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Hepatitis B and C</li> <li>B. HIV</li> <li>C. Malaria</li> <li>D. Tuberculosis (commonly)</li> </ul>                                                                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q44. Autologous blood transfusion refers to:</b></p> <ul style="list-style-type: none"> <li>A. Transfusing blood from a family member</li> <li>B. Transfusing the patient's own previously collected or salvaged blood</li> <li>C. Using synthetic blood substitutes</li> <li>D. Directed donor transfusion</li> </ul>              | <p><b>Ans: B</b></p> |

|                                                                                                                                                                                                                                                                                                                     |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q45. Types of autologous blood transfusion include all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Preoperative autologous donation (PAD)</li> <li>B. Acute normovolemic hemodilution (ANH)</li> <li>C. Intraoperative cell salvage</li> <li>D. Emergency crossmatch from blood bank</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q46. In acute normovolemic hemodilution (ANH), blood is removed preoperatively and replaced with:</b></p> <ul style="list-style-type: none"> <li>A. Nothing</li> <li>B. Crystalloids or colloids to maintain normovolemia</li> <li>C. Whole blood from donors</li> <li>D. Platelets only</li> </ul>           | <p><b>Ans: B</b></p> |
| <p><b>Q47. Intraoperative cell salvage is contraindicated in:</b></p> <ul style="list-style-type: none"> <li>A. Cardiac surgery</li> <li>B. Surgery for malignancy (relative) and infected surgical fields</li> <li>C. Orthopedic surgery</li> <li>D. Vascular surgery</li> </ul>                                   | <p><b>Ans: B</b></p> |
| <p><b>Q48. The trigger for blood transfusion in a healthy patient is generally a hemoglobin of:</b></p> <ul style="list-style-type: none"> <li>A. 10 g/dL</li> <li>B. 7–8 g/dL</li> <li>C. 12 g/dL</li> <li>D. 5 g/dL</li> </ul>                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q49. The anticoagulant used in blood storage bags is:</b></p> <ul style="list-style-type: none"> <li>A. Heparin</li> <li>B. CPDA-1 (Citrate-Phosphate-Dextrose-Adenine)</li> <li>C. Warfarin</li> <li>D. EDTA</li> </ul>                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q50. Blood must be transfused through a filter of pore size:</b></p> <ul style="list-style-type: none"> <li>A. 20 micrometers</li> <li>B. 170–260 micrometers (standard blood filter)</li> <li>C. 0.22 micrometers</li> <li>D. 500 micrometers</li> </ul>                                                     | <p><b>Ans: B</b></p> |

# Answer Key & Explanations

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## Q1. Answer: B — Crystalloids and colloids

Fluids are divided into crystalloids and colloids as stated at the beginning of Chapter 7.

Topic: Fluids | Ref: Ch 7, Page 71

## Q2. Answer: B — Na<sup>+</sup> 140, Cl<sup>-</sup> 98, K<sup>+</sup> 3, Mg<sup>2+</sup> 3, Acetate 27, Gluconate 23 mEq/L

Plasmalyte composition: Na<sup>+</sup> 140, Cl<sup>-</sup> 98, K<sup>+</sup> 3, Mg<sup>2+</sup> 3, Acetate 27, Gluconate 23 mEq/L. Acetate and gluconate are metabolized to generate bicarbonate.

Topic: Crystalloids | Ref: Ch 7, Page 72

## Q3. Answer: D — Hypokalemia

As per the textbook, NS is preferred over RL for treating: hypochloremic metabolic alkalosis, brain injury (Ca<sup>2+</sup> in RL can increase neuronal injury), and hyponatremia.

Topic: Crystalloids | Ref: Ch 7, Page 72

## Q4. Answer: B — 1.5:1 ratio (150 mL crystalloid for 100 mL blood loss)

As per Table 7.1, crystalloids should be replaced in a ratio of 1.5:1. For example, blood loss of 100 mL should be replaced with 150 mL of crystalloid.

Topic: Fluids | Ref: Ch 7, Page 71

## Q5. Answer: B — RL contains calcium which can cause clotting of blood

As RL contains calcium, blood should not be given through the same drip set as calcium can initiate coagulation of stored blood.

Topic: Crystalloids | Ref: Ch 7, Page 72

## Q6. Answer: B — Normal saline (0.9% NaCl)

Normal saline (0.9% NaCl) is an isotonic crystalloid with an osmolarity of 308 mOsm/L, close to plasma osmolarity.

Topic: Crystalloids | Ref: Ch 7, Page 71

## Q7. Answer: C — 154 mEq/L

Normal saline contains 154 mEq/L of sodium and 154 mEq/L of chloride.

Topic: Crystalloids | Ref: Ch 7, Page 71

## Q8. Answer: D — Bicarbonate 29 mEq/L

Ringer's lactate contains Na<sup>+</sup> 131, Cl<sup>-</sup> 111, K<sup>+</sup> 5, Ca<sup>2+</sup> 2, Lactate 29 mEq/L, pH 6.5. It does not contain bicarbonate directly.

Topic: Crystalloids | Ref: Ch 7, Page 71

## Q9. Answer: B — Liver

Lactate is metabolized by the liver to bicarbonate, providing a buffering effect.

Topic: Crystalloids | Ref: Ch 7, Page 71

## Q10. Answer: B — Hyperchloremic metabolic acidosis

Large volumes of NS (high chloride 154 mEq/L) can cause dilutional hyperchloremic metabolic acidosis.

Topic: Crystalloids | Ref: Ch 7, Page 71

**Q11. Answer: B — Dextrose is rapidly metabolized, leaving free water that distributes across all compartments**

After infusion, glucose is rapidly metabolized, leaving free water that distributes evenly across ICF and ECF, effectively acting as a hypotonic solution.

Topic: Crystalloids | Ref: Ch 7, Page 71

**Q12. Answer: B — 30 minutes**

As per Table 7.1, crystalloids have an intravascular half-life of 30 minutes, so they expand plasma volume for less time compared to colloids.

Topic: Crystalloids | Ref: Ch 7, Page 71 (Table 7.1)

**Q13. Answer: B — Contain large molecular weight substances that maintain oncotic pressure and stay intravascular longer**

Colloids contain large molecules (proteins/starches) that generate oncotic pressure, keeping fluid in the intravascular space longer than crystalloids.

Topic: Colloids | Ref: Ch 7, Page 72

**Q14. Answer: C — Human albumin**

Human albumin is the only natural colloid. HES, dextrans, and gelatins are synthetic colloids.

Topic: Colloids | Ref: Ch 7, Page 72

**Q15. Answer: C — Longer (several hours)**

Colloids have a longer intravascular half-life (hours) compared to crystalloids because large molecules are retained within the vascular space.

Topic: Colloids | Ref: Ch 7, Page 72

**Q16. Answer: B — Improving microcirculation and reducing blood viscosity (rheological effects)**

Dextran 40 (low molecular weight) improves microcirculation by reducing blood viscosity and preventing rouleaux formation.

Topic: Colloids | Ref: Ch 7, Page 72

**Q17. Answer: B — Anaphylactic reactions and interference with blood cross-matching**

Dextrans can cause anaphylaxis, interfere with blood cross-matching (rouleaux formation), and impair coagulation at high doses.

Topic: Colloids | Ref: Ch 7, Page 72

**Q18. Answer: B — Coagulopathy and renal impairment at high doses**

HES can cause dose-dependent coagulopathy (reduced Factor VIII, von Willebrand factor) and nephrotoxicity, especially in critically ill patients.

Topic: Colloids | Ref: Ch 7, Page 72

**Q19. Answer: B — Risk of anaphylactoid reactions and relatively short duration of action**

Gelatins have a shorter duration than other colloids and carry a risk of anaphylactoid/allergic reactions.

Topic: Colloids | Ref: Ch 7, Page 72

**Q20. Answer: B — 110 mL/hr**

4-2-1 rule: 4 mL/kg/hr for first 10 kg (40) + 2 mL/kg/hr for next 10 kg (20) + 1 mL/kg/hr for remaining 50 kg (50) = 110 mL/hr.

Topic: Fluid Management | Ref: Ch 7, Page 73

**Q21. Answer: B — First 10 kg**

The 4-2-1 rule: 4 mL/kg/hr for the first 10 kg, 2 mL/kg/hr for the next 10 kg, and 1 mL/kg/hr for each kg above 20 kg.

Topic: Fluid Management | Ref: Ch 7, Page 73

**Q22. Answer: B — Maintenance, deficit (fasting), and ongoing losses (surgical/third space)**

Intraoperative fluids must replace maintenance needs, preoperative fasting deficit, and ongoing surgical/third-space losses.

Topic: Fluid Management | Ref: Ch 7, Page 73

**Q23. Answer: B — Functional loss of fluid into a non-equilibrating space (e.g., bowel wall, traumatized tissues)**

Third-space loss is the redistribution of fluid into a non-functional compartment (traumatized tissue, bowel wall, peritoneal cavity) that does not participate in normal fluid exchange.

Topic: Fluid Management | Ref: Ch 7, Page 73

**Q24. Answer: C — 6–10 mL/kg/hr**

Third-space losses vary: 1–2 mL/kg/hr for minimal trauma, 4–6 for moderate, and 6–10 mL/kg/hr for major abdominal surgery.

Topic: Fluid Management | Ref: Ch 7, Page 73

**Q25. Answer: B — Hourly maintenance rate × number of hours NPO**

Fasting deficit = maintenance fluid rate (by 4-2-1 rule) × hours of fasting (NPO period).

Topic: Fluid Management | Ref: Ch 7, Page 73

**Q26. Answer: B — 35–40%**

Whole blood has a hematocrit of approximately 35–40% and contains RBCs, plasma, platelets, and clotting factors.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

**Q27. Answer: B — Decreasing Factor VIII level and causing renal dysfunction**

As per Table 7.1, colloids in high doses can interfere with clotting by primarily decreasing Factor VIII level and can also cause renal dysfunction.

Topic: Colloids | Ref: Ch 7, Page 71 (Table 7.1)

**Q28. Answer: B — 1 g/dL**

One unit of PRBCs raises hemoglobin by approximately 1 g/dL (or hematocrit by 3%) in an average adult.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

**Q29. Answer: B — 1–6°C**

PRBCs are stored at 1–6°C with a shelf life of 35–42 days depending on the anticoagulant/preservative used.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

**Q30. Answer: B — Replacement of coagulation factors in patients with coagulopathy**

FFP contains all coagulation factors and is indicated for coagulopathy, DIC, massive transfusion, and warfarin reversal.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

**Q31. Answer: C — 50,000/ $\mu$ L (or <75,000 with active bleeding/surgery)**

Platelet transfusion is generally indicated when count is <50,000/ $\mu$ L or <75,000–100,000/ $\mu$ L with active bleeding or before surgery.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

**Q32. Answer: B — 20–24°C with continuous agitation**

Platelets are stored at room temperature (20–24°C) with continuous gentle agitation, with a shelf life of 5 days.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

**Q33. Answer: B — ABO-incompatible (hemolytic) transfusion due to clerical error**

ABO-incompatible acute hemolytic transfusion reaction, usually due to clerical/identification error, is the most common cause of fatal transfusion reactions.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q34. Answer: B — Unexplained hypotension, fever, hemoglobinuria, and DIC**

Under GA, signs include unexplained hypotension, tachycardia, fever, hemoglobinuria (dark urine), diffuse bleeding (DIC), and bronchospasm.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q35. Answer: B — Stop the transfusion immediately and maintain IV access**

The immediate action is to STOP the transfusion, maintain IV access with new tubing, and send the blood bag and patient sample for verification.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q36. Answer: B — Acute respiratory distress, non-cardiogenic pulmonary edema within 6 hours of transfusion**

TRALI presents with acute hypoxemia, bilateral pulmonary infiltrates, and respiratory distress within 6 hours of transfusion, without circulatory overload.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q37. Answer: B — More than one blood volume (or >10 units PRBCs) in 24 hours**

Massive transfusion is defined as replacement of more than one blood volume (approximately 70 mL/kg or >10 units PRBCs) within 24 hours.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q38. Answer: D — Polycythemia**

Massive transfusion complications include hypothermia, hyperkalemia, citrate toxicity (hypocalcemia), coagulopathy, and acid-base disturbance – not polycythemia.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q39. Answer: B — Hypocalcemia (ionized calcium reduction) causing cardiac depression**

Citrate (anticoagulant in stored blood) chelates ionized calcium, causing hypocalcemia which can lead to cardiac depression and coagulopathy.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q40. Answer: B — Red cell membrane Na-K ATPase pump fails during storage, leaking potassium into plasma**

During storage, the RBC membrane Na-K ATPase pump becomes inactive, causing potassium to leak from cells into the plasma of stored blood.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q41. Answer: B — Antibodies against donor leukocyte antigens or cytokines in stored blood**

Febrile non-hemolytic reactions are caused by recipient antibodies against donor WBC antigens or accumulated cytokines in stored blood components.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q42. Answer: B — Pulmonary edema, hypertension, dyspnea due to volume overload**

TACO presents with signs of circulatory overload: pulmonary edema, raised JVP, hypertension, and dyspnea, especially in elderly and cardiac patients.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q43. Answer: D — Tuberculosis (commonly)**

Transfusion-transmitted infections include hepatitis B, C, HIV, malaria, syphilis, and CMV. TB is not a common transfusion-transmitted infection.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

**Q44. Answer: B — Transfusing the patient's own previously collected or salvaged blood**

Autologous transfusion involves collecting, storing, and re-infusing the patient's own blood, eliminating risks of alloimmunization and transfusion-transmitted infections.

Topic: Autologous Blood Transfusion | Ref: Ch 7, Page 79

**Q45. Answer: D — Emergency crossmatch from blood bank**

Autologous methods include preoperative donation, acute normovolemic hemodilution, and intraoperative cell salvage. Emergency crossmatch uses donor (allogeneic) blood.

Topic: Autologous Blood Transfusion | Ref: Ch 7, Page 79

**Q46. Answer: B — Crystalloids or colloids to maintain normovolemia**

In ANH, blood is withdrawn immediately before surgery and replaced with crystalloid/colloid to maintain intravascular volume. The blood is re-infused after surgical bleeding.

Topic: Autologous Blood Transfusion | Ref: Ch 7, Page 79

**Q47. Answer: B — Surgery for malignancy (relative) and infected surgical fields**

Cell salvage is relatively contraindicated in malignancy (risk of tumor cell dissemination) and contaminated/infected surgical fields.

Topic: Autologous Blood Transfusion | Ref: Ch 7, Page 79

**Q48. Answer: B — 7–8 g/dL**

In otherwise healthy patients, transfusion is generally triggered at hemoglobin 7–8 g/dL (restrictive strategy). Higher thresholds apply for cardiac disease.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

**Q49. Answer: B — CPDA-1 (Citrate-Phosphate-Dextrose-Adenine)**

CPDA-1 is the standard anticoagulant-preservative used in blood bags. Citrate prevents coagulation, phosphate and dextrose support RBC metabolism, and adenine maintains ATP levels.

*Topic: Blood Transfusion | Ref: Ch 7, Page 75*

**Q50. Answer: B — 170–260 micrometers (standard blood filter)**

Standard blood transfusion sets have a 170–260 micrometer filter to remove clots and debris. Microaggregate filters (20–40 µm) are used for special indications.

*Topic: Blood Transfusion | Ref: Ch 7, Page 75*



# History of Anesthesia MCQ – Chapter 8

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 4: Introduction to Anesthesia (Chapter 8)

**25 Questions • 20 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Who used the term 'anesthesia' in 1st AD to describe the narcotic effects of the plant Mandragora?**

- A. Dioscorides
- B. Oliver Wendell Holmes
- C. John Lundy
- D. Ralph Waters

**Ans: A**

**Q2. The word 'anesthesia', which means no senses, was coined by:**

- A. Dioscorides
- B. Oliver Wendell Holmes
- C. Humphry Davy
- D. Horace Wells

**Ans: B**

**Q3. The term 'balanced anesthesia' was coined by:**

- A. Priestley and Davy
- B. Morton and Snow
- C. John Lundy and Ralph Waters
- D. Carl Koller and August Bier

**Ans: C**

**Q4. Which was the first anesthetic gas, and who synthesized it in 1774?**

- A. Ether, synthesized by Valerius Cordus
- B. Nitrous oxide, synthesized by Priestley
- C. Chloroform, synthesized by John Snow
- D. Cyclopropane, synthesized by Humphry Davy

**Ans: B**

**Q5. Who first used nitrous oxide on himself for a toothache and called it 'laughing gas'?**

- A. Joseph Priestley
- B. Horace Wells
- C. Humphry Davy
- D. W.T.G. Morton

**Ans: C**

**Q6. The first public clinical demonstration of nitrous oxide (1845) was given by Horace Wells for what procedure?**

- A. Tooth extraction
- B. Removal of jaw tumor
- C. Childbirth
- D. Limb amputation

**Ans: A**

|                                                                                                                                                                                                                                                          |                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. Horace Wells became addicted to which agent and later committed suicide by cutting his femoral artery?</b></p> <p>A. Nitrous oxide<br/>B. Ether<br/>C. Cocaine<br/>D. Chloroform</p>                                                           | <p><b>Ans: D</b></p> |
| <p><b>Q8. Ether was prepared by Valerius Cordus in 1540 and was known by what name?</b></p> <p>A. Laughing gas<br/>B. Sweet oil of vitriol<br/>C. Dephlogisticated air<br/>D. Spirit of salt</p>                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q9. Who gave the first public demonstration of ether anesthesia on 16th October 1846?</b></p> <p>A. WTG Morton<br/>B. John Snow<br/>C. Horace Wells<br/>D. Carl Koller</p>                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q10. Which date is celebrated as World Anesthesia Day?</b></p> <p>A. 14th February<br/>B. 1st AD<br/>C. 16th October<br/>D. 12th March</p>                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q11. The first successful demonstration of ether anesthesia by Morton took place at which hospital (Fig. 8.1)?</b></p> <p>A. Johns Hopkins Hospital<br/>B. Massachusetts General Hospital<br/>C. St. Thomas Hospital<br/>D. Hindu Rao Hospital</p> | <p><b>Ans: B</b></p> |
| <p><b>Q12. Who popularized chloroform and used it on Queen Victoria for the birth of her 8th child?</b></p> <p>A. WTG Morton<br/>B. Humphry Davy<br/>C. John Snow<br/>D. Valerius Cordus</p>                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q13. In how many patients did John Snow use chloroform successfully?</b></p> <p>A. 4,000 patients<br/>B. 1,000 patients<br/>C. 10,000 patients<br/>D. 100 patients</p>                                                                             | <p><b>Ans: A</b></p> |

|                                                                                                                                                                                                                                                                                                         |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q14. The first intravenous anesthetic used was thiopentone. Who used it and in which year?</b></p> <p>A. Carl Koller, 1884<br/>B. John Lundy, 1934<br/>C. Harold Griffith, 1942<br/>D. August Bier, 1898</p>                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q15. The first local anesthetic used was cocaine. Who used it and for what purpose in 1884?</b></p> <p>A. Carl Koller, for anesthetizing the cornea<br/>B. J Leonard Corning, for spinal anesthesia<br/>C. August Bier, for epidural anesthesia<br/>D. John Lundy, for intravenous anesthesia</p> | <p><b>Ans: A</b></p> |
| <p><b>Q16. Who first gave spinal anesthesia in dogs?</b></p> <p>A. August Bier<br/>B. Carl Koller<br/>C. J Leonard Corning<br/>D. Cathelin and Sicard</p>                                                                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q17. The first spinal anesthesia in human beings was given by August Bier in which year?</b></p> <p>A. 1846<br/>B. 1884<br/>C. 1898<br/>D. 1901</p>                                                                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q18. The first epidural anesthesia (1901) was given by:</b></p> <p>A. August Bier<br/>B. Cathelin and Sicard<br/>C. J Leonard Corning<br/>D. Carl Koller</p>                                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q19. Who was the first to use 'curare' for muscle relaxation in 1942?</b></p> <p>A. Harold Griffith<br/>B. Bovet<br/>C. John Lundy<br/>D. Ivan Magill</p>                                                                                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q20. Bovet received the Nobel Prize (1949) for using which muscle relaxant in practice?</b></p> <p>A. Curare<br/>B. Pancuronium<br/>C. Suxamethonium (Succinylcholine)<br/>D. Atracurium</p>                                                                                                      | <p><b>Ans: C</b></p> |

|                                                                                                                                                                                                                                      |                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q21. Who received the Nobel Prize in 1949 in connection with muscle relaxants?</b></p> <p>A. Harold Griffith<br/>B. Bovet<br/>C. John Lundy<br/>D. Edmund Gaskin Boyle</p>                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q22. The first Boyle's machine was invented in 1917 by:</b></p> <p>A. Ivan Magill<br/>B. Edmund Gaskin Boyle<br/>C. Robert Boyle<br/>D. John Lundy</p>                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q23. The first endotracheal intubation was done by:</b></p> <p>A. Edmund Gaskin Boyle<br/>B. WTG Morton<br/>C. Ivan Magill<br/>D. Carl Koller</p>                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q24. Apart from synthesizing nitrous oxide, Priestley is also credited with synthesizing which other gas, which he called 'Dephlogisticated air'?</b></p> <p>A. Carbon dioxide<br/>B. Oxygen<br/>C. Nitrogen<br/>D. Helium</p> | <p><b>Ans: B</b></p> |
| <p><b>Q25. Ether anesthesia in Morton's first demonstration was used for removal of what?</b></p> <p>A. Jaw tumor<br/>B. A tooth<br/>C. An appendix<br/>D. A limb</p>                                                                | <p><b>Ans: A</b></p> |

# Answer Key & Explanations

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## Q1. Answer: A — Dioscorides

Page 83: History of anesthesia dates back to 1st AD when Dioscorides used the term anesthesia to describe the narcotic effects of the plant Mandragora.

Topic: Introduction

## Q2. Answer: B — Oliver Wendell Holmes

Page 83: The word anesthesia, which means no senses, was coined by Oliver Wendell Holmes.

Topic: Introduction

## Q3. Answer: C — John Lundy and Ralph Waters

Page 83: The term 'balanced anesthesia' was coined by John Lundy and Ralph Waters.

Topic: Introduction

## Q4. Answer: B — Nitrous oxide, synthesized by Priestley

Page 83: The first anesthetic gas, i.e. nitrous oxide, was synthesized by Priestley in 1774 (Priestley also synthesized oxygen, which he called Dephlogisticated air).

Topic: Nitrous Oxide

## Q5. Answer: C — Humphry Davy

Page 83: The first clinical use of nitrous oxide was done by Humphry Davy, who used it on himself for toothache and called nitrous oxide as 'laughing gas'.

Topic: Nitrous Oxide

## Q6. Answer: A — Tooth extraction

Page 83: The first public clinical demonstration of nitrous oxide was given by Horace Wells for tooth extraction (1845), but unfortunately the patient cried in pain.

Topic: Nitrous Oxide

## Q7. Answer: D — Chloroform

Page 83: Horace Wells became so frustrated that he became a chloroform addict and committed suicide by cutting his femoral artery.

Topic: Nitrous Oxide

## Q8. Answer: B — Sweet oil of vitriol

Page 83: Ether was prepared by Valerius Cordus in 1540 when it was known as 'sweet oil of vitriol'.

Topic: Ether

## Q9. Answer: A — WTG Morton

Page 83: The first public demonstration of ether was given by WTG Morton on 16th October 1846, and that is why 16th October is celebrated as World Anesthesia Day.

Topic: Ether

**Q10. Answer: C — 16th October**

Page 83: The first public demonstration of ether was given by WTG Morton on 16th October 1846, and that is why 16th October is celebrated as World Anesthesia Day.

Topic: Ether

**Q11. Answer: B — Massachusetts General Hospital**

Page 83, Fig. 8.1: The first successful demonstration of ether anesthesia was by Morton in Massachusetts General Hospital.

Topic: Ether

**Q12. Answer: C — John Snow**

Page 84: It was John Snow who popularized chloroform, using it successfully in 4,000 patients. He used chloroform on Queen Victoria for the birth of her 8th child.

Topic: Chloroform

**Q13. Answer: A — 4,000 patients**

Page 84: It was John Snow who popularized chloroform, using it successfully in 4,000 patients.

Topic: Chloroform

**Q14. Answer: B — John Lundy, 1934**

Page 84: The first intravenous anesthetic used was thiopentone by John Lundy in 1934.

Topic: Intravenous Anesthetics

**Q15. Answer: A — Carl Koller, for anesthetizing the cornea**

Page 84: The first local anesthetic used was cocaine by Carl Koller for anesthetizing the cornea (1884).

Topic: Local Anesthesia

**Q16. Answer: C — J Leonard Corning**

Page 84: The first spinal anesthesia in dogs was given by J Leonard Corning, while the first spinal anesthesia in human beings was given by August Bier in 1898.

Topic: Local Anesthesia

**Q17. Answer: C — 1898**

Page 84: The first spinal anesthesia in human beings was given by August Beir (Bier) in 1898.

Topic: Local Anesthesia

**Q18. Answer: B — Cathelin and Sicard**

Page 84: The first epidural anesthesia was given by Cathelin and Sicard (1901).

Topic: Local Anesthesia

**Q19. Answer: A — Harold Griffith**

Page 84: Harold Griffith was the first to use 'curare' for muscle relaxation (1942).

Topic: Muscle Relaxants

**Q20. Answer: C — Suxamethonium (Succinylcholine)**

Page 84: Bovet got the Nobel Prize (1949) for using Suxamethonium (Succinylcholine) in practice.

Topic: Muscle Relaxants

**Q21. Answer: B — Bovet**

Page 84: Bovet got the Nobel Prize (1949) for using Suxamethonium (Succinylcholine) in practice.

*Topic: Muscle Relaxants*

**Q22. Answer: B — Edmund Gaskin Boyle**

Page 84: The first Boyle's machine was invented by Edmund Gaskin Boyle in 1917.

*Topic: Instruments*

**Q23. Answer: C — Ivan Magill**

Page 84: The first endotracheal intubation was done by Ivan Magill.

*Topic: Instruments*

**Q24. Answer: B — Oxygen**

Page 83: Priestley also synthesized oxygen, which he called Dephlogisticated air.

*Topic: Nitrous Oxide*

**Q25. Answer: A — Jaw tumor**

Page 100: First public demonstration of ether was given by WTG Morton on 16th October 1846 for the removal of jaw tumor.

*Topic: Ether*



# Introduction to General Anesthesia MCQ – Chapter 9

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 5: General Anesthesia (Chapter 9)

**25 Questions • 20 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. What are the three basic components (TRIAD) of general anesthesia?**

- A. Narcosis, analgesia and muscle relaxation
- B. Amnesia, analgesia and hypnosis
- C. Sedation, paralysis and reflex abolition
- D. Narcosis, amnesia and homeostasis

**Ans: A**

**Q2. The concept of 'balanced anesthesia' was laid in 1926 by:**

- A. Oliver Wendell Holmes
- B. John Lundy
- C. Guedel
- D. Ralph Waters

**Ans: B**

**Q3. Which of the following is NOT one of the listed components of balanced anesthesia?**

- A. Amnesia
- B. Abolition of reflexes
- C. Hypothermia
- D. Maintenance of normal hemodynamics and physiologic homeostasis

**Ans: C**

**Q4. Which drugs are used to premedicate the patient the night before surgery to induce good sleep and produce amnesia?**

- A. Opioids
- B. Benzodiazepines (lorazepam or diazepam)
- C. Beta-blockers
- D. Anticholinergics

**Ans: B**

**Q5. Preoxygenation with 100% oxygen for 3 minutes replaces 99% of the nitrogen in the lungs. What is this process called?**

- A. Denitrogenation
- B. Diffusion hypoxia
- C. Concentration effect
- D. Second gas effect

**Ans: A**

**Q6. Why are intravenous agents such as propofol preferred for induction?**

- A. Because of their long duration of action
- B. Because of their ultrafast action (within 15 seconds)
- C. Because they provide muscle relaxation
- D. Because they prevent aspiration

**Ans: B**

|                                                                                                                                                                                                                                                                             |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. Which muscle relaxant is the ideal agent for intubation due to its rapid onset, but whose use is restricted to rapid sequence or difficult intubation?</b></p> <p>A. Vecuronium<br/>B. Atracurium<br/>C. Suxamethonium (succinylcholine)<br/>D. Rocuronium</p>    | <p><b>Ans: C</b></p> |
| <p><b>Q8. After intubation, the position of the cuffed endotracheal tube is verified by:</b></p> <p>A. Chest X-ray only<br/>B. Auscultation and capnography<br/>C. Pulse oximetry only<br/>D. Arterial blood gas</p>                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q9. In a typical maintenance gas mixture, what proportions of oxygen and nitrous oxide are used?</b></p> <p>A. Oxygen 66% + nitrous oxide 33%<br/>B. Oxygen 50% + nitrous oxide 50%<br/>C. Oxygen 33% + nitrous oxide 66%<br/>D. Oxygen 21% + nitrous oxide 79%</p>   | <p><b>Ans: C</b></p> |
| <p><b>Q10. Why do the majority of anesthesiologists in current practice use air instead of nitrous oxide?</b></p> <p>A. Nitrous oxide is too expensive<br/>B. Nitrous oxide depletes ozone<br/>C. Nitrous oxide causes hypotension<br/>D. Air provides better analgesia</p> | <p><b>Ans: B</b></p> |
| <p><b>Q11. At the end of surgery, the muscle blockade is reversed with which combination?</b></p> <p>A. Neostigmine + glycopyrrolate<br/>B. Atropine + suxamethonium<br/>C. Propofol + fentanyl<br/>D. Naloxone + flumazenil</p>                                            | <p><b>Ans: A</b></p> |
| <p><b>Q12. The stages of anesthesia were described by Guedel using which agent?</b></p> <p>A. Halothane<br/>B. Ether<br/>C. Nitrous oxide<br/>D. Propofol</p>                                                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q13. Which stage of anesthesia is the stage of analgesia?</b></p> <p>A. Stage 1<br/>B. Stage 2<br/>C. Stage 3<br/>D. Stage 4</p>                                                                                                                                      | <p><b>Ans: A</b></p> |

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| <p><b>Q14. Stage 2 of anesthesia is known as the stage of:</b></p> <ul style="list-style-type: none"> <li>A. Analgesia</li> <li>B. Excitement</li> <li>C. Surgical anesthesia</li> <li>D. Medullary paralysis</li> </ul>                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q15. Surgical anesthesia (Stage 3) is divided into how many planes?</b></p> <ul style="list-style-type: none"> <li>A. Two planes</li> <li>B. Three planes</li> <li>C. Four planes</li> <li>D. Five planes</li> </ul>                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q16. The ideal stage for surgical anesthesia / intubation is:</b></p> <ul style="list-style-type: none"> <li>A. Stage III, plane 1</li> <li>B. Stage III, plane 3</li> <li>C. Stage II</li> <li>D. Stage IV</li> </ul>                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q17. The onset of surgical anesthesia (stage III) is best indicated by:</b></p> <ul style="list-style-type: none"> <li>A. Onset of regular respiration</li> <li>B. Pupillary constriction</li> <li>C. Loss of eyelash reflex</li> <li>D. Diaphragmatic paralysis</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q18. Which is the first reflex to go during induction, used to test loss of consciousness?</b></p> <ul style="list-style-type: none"> <li>A. Carinal reflex</li> <li>B. Corneal reflex</li> <li>C. Eyelash reflex</li> <li>D. Laryngeal reflex</li> </ul>                   | <p><b>Ans: C</b></p> |
| <p><b>Q19. Which is the LAST reflex to be abolished during deepening anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Eyelash reflex</li> <li>B. Carinal reflex</li> <li>C. Conjunctival reflex</li> <li>D. Corneal reflex</li> </ul>                                | <p><b>Ans: B</b></p> |
| <p><b>Q20. The initial pupillary dilatation seen in Stage 2 is because of:</b></p> <ul style="list-style-type: none"> <li>A. Parasympathetic stimulation</li> <li>B. Sympathetic stimulation</li> <li>C. Direct drug effect on the iris</li> <li>D. Hypoxia</li> </ul>            | <p><b>Ans: B</b></p> |

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| <p><b>Q21. The return of reflexes during recovery occurs in what sequence relative to their abolition?</b></p> <p>A. In the same sequence<br/>B. In the opposite sequence<br/>C. Randomly<br/>D. All return simultaneously</p>                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q22. In Stage 3, Plane 4 (diaphragmatic paralysis), the respiration is described as:</b></p> <p>A. Regular<br/>B. Irregular<br/>C. Jerky<br/>D. Absent</p>                                                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q23. The term 'anesthesia' (as the basic concept of general anesthesia) was given by:</b></p> <p>A. John Lundy<br/>B. Guedel<br/>C. Oliver Wendell Holmes<br/>D. Ralph Waters</p>                                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q24. During preoxygenation, the patient is given 100% oxygen for how long?</b></p> <p>A. 30 seconds<br/>B. 1 minute<br/>C. 3 minutes<br/>D. 10 minutes</p>                                                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q25. In Stage 3 Plane 2, which reflex is abolished and what is the eye position?</b></p> <p>A. Corneal reflex abolished, eyes fixed centrally<br/>B. Laryngeal reflex abolished, eyes roving<br/>C. Eyelash reflex abolished, eyes divergent<br/>D. Carinal reflex abolished, eyes fully dilated</p> | <p><b>Ans: A</b></p> |

# Answer Key & Explanations

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## **Q1. Answer: A — Narcosis, analgesia and muscle relaxation**

Page 87: The basic components (TRIAD) of general anesthesia are narcosis, analgesia and muscle relaxation.

*Topic: Components of General Anesthesia*

## **Q2. Answer: B — John Lundy**

Page 87: The anesthesia in present day should be balanced anesthesia, a concept laid by John Lundy in 1926.

*Topic: Components of General Anesthesia*

## **Q3. Answer: C — Hypothermia**

Page 87: The components of balanced anesthesia are narcosis, analgesia, muscle relaxation, amnesia, abolition of reflexes, and maintenance of normal hemodynamics and physiologic homeostasis.

*Topic: Components of General Anesthesia*

## **Q4. Answer: B — Benzodiazepines (lorazepam or diazepam)**

Page 87: Selected indoor patients may be premedicated with lorazepam or diazepam night before surgery. Benzodiazepines are used to induce good sleep and produce amnesia.

*Topic: General Anesthesia Protocol*

## **Q5. Answer: A — Denitrogenation**

Page 87: As 99% of the nitrogen in lungs can be replaced with oxygen (called as Denitrogenation) in 3 minutes, preoxygenation increases the oxygen reserve enabling the patient to sustain hypoxia for longer periods.

*Topic: General Anesthesia Protocol*

## **Q6. Answer: B — Because of their ultrafast action (within 15 seconds)**

Page 87: Intravenous agents are preferred for induction because of their ultrafast action (within 15 seconds). After the patient becomes unconscious, the aim is to secure the airway as early as possible.

*Topic: General Anesthesia Protocol*

## **Q7. Answer: C — Suxamethonium (succinylcholine)**

Page 87: Due to its rapid onset, suxamethonium (succinylcholine) is the ideal muscle relaxant for intubation; however, its use is only restricted to rapid sequence or difficult intubation. In normal circumstances, patients are generally intubated with non-depolarizers.

*Topic: General Anesthesia Protocol*

## **Q8. Answer: B — Auscultation and capnography**

Page 87: After the patient is intubated with a cuffed endotracheal tube, the position of the tube is verified by auscultation and capnography. The tube is fixed with adhesive and positive pressure ventilation (IPPV) is started.

*Topic: General Anesthesia Protocol*

## **Q9. Answer: C — Oxygen 33% + nitrous oxide 66%**

Page 87: Anesthesia is maintained with oxygen (33%) + nitrous oxide (66%) + inhalational agent. As nitrous oxide depletes ozone, the majority of anesthesiologists in current day practice use air instead of nitrous oxide.

*Topic: General Anesthesia Protocol*

**Q10. Answer: B — Nitrous oxide depletes ozone**

Page 87: As nitrous oxide depletes ozone, the majority of anesthesiologists in current day practice use air instead of nitrous oxide.

*Topic: General Anesthesia Protocol*

**Q11. Answer: A — Neostigmine + glycopyrrolate**

Page 87: At the end of the surgery, the muscle blockade effect of non-depolarizers is reversed with neostigmine + glycopyrrolate, and then the patient is extubated after thorough suctioning of oral cavity.

*Topic: General Anesthesia Protocol*

**Q12. Answer: B — Ether**

Page 88: Stages were described by Guedel with ether (Table 9.1). In current anesthesia, newer inhalational agents are so fast acting that these stages do not exist.

*Topic: Stages of Anesthesia*

**Q13. Answer: A — Stage 1**

Page 88, Table 9.1: Stage-1 is the stage of analgesia (from analgesia to loss of consciousness). Respiration is irregular, tidal volume small, pupils constricted.

*Topic: Stages of Anesthesia*

**Q14. Answer: B — Excitement**

Page 88, Table 9.1: Stage-2 is the stage of excitement (from loss of consciousness to rhythmic respiration). Pupils are dilated due to sympathetic stimulation and the eyelash reflex is abolished.

*Topic: Stages of Anesthesia*

**Q15. Answer: C — Four planes**

Page 88, Table 9.1: Stage-3 (stage of surgical anesthesia) is divided into four planes.

*Topic: Stages of Anesthesia*

**Q16. Answer: B — Stage III, plane 3**

Page 88: The stage of surgical anesthesia is stage III plane 3 because laryngeal reflexes go in this stage and intubation can be performed.

*Topic: Stages of Anesthesia*

**Q17. Answer: A — Onset of regular respiration**

Page 88: Onset of surgical anesthesia (stage III) is best indicated by onset of regular respiration.

*Topic: Stages of Anesthesia*

**Q18. Answer: C — Eyelash reflex**

Page 88: The first reflex to go is eyelash, therefore loss of consciousness is tested by absence of eyelash reflex, while the last reflex to go is the carinal reflex (tested by touching the carina with a suction catheter, which should initiate cough).

*Topic: Stages of Anesthesia*

**Q19. Answer: B — Carinal reflex**

Page 88: The last reflex to go is the carinal reflex, tested by touching the carina with a suction catheter, which should initiate cough.

*Topic: Stages of Anesthesia*

**Q20. Answer: B — Sympathetic stimulation**

Page 88: Initial pupillary dilatation seen in stage II is because of sympathetic stimulation.

*Topic: Stages of Anesthesia*

**Q21. Answer: B — In the opposite sequence**

Page 88: The return of reflexes is in the opposite sequence, i.e. the carinal reflex returns first to come and the eyelash reflex last.

*Topic: Stages of Anesthesia*

**Q22. Answer: C — Jerky**

Page 88, Table 9.1: In Plane 4 (diaphragmatic paralysis) respiration is jerky, tidal volume small, pupils fully dilated, and the carinal and anal sphincter reflexes are abolished.

*Topic: Stages of Anesthesia*

**Q23. Answer: C — Oliver Wendell Holmes**

Page 87: The term anesthesia was given by Oliver Wendell Holmes.

*Topic: Components of General Anesthesia*

**Q24. Answer: C — 3 minutes**

Page 87: The anesthesiologist preoxygenates the patient with 100% oxygen for 3 minutes, replacing 99% of the nitrogen in the lungs (denitrogenation).

*Topic: General Anesthesia Protocol*

**Q25. Answer: A — Corneal reflex abolished, eyes fixed centrally**

Page 88, Table 9.1: In Plane 2 (cessation of eye movement to respiratory paresis), respiration is regular, tidal volume medium, pupils  $\frac{1}{2}$  dilated, eyes fixed centrally, and the corneal reflex is abolished.

*Topic: Stages of Anesthesia*



# Inhalational Agents MCQ – Chapter 10

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 5: General Anesthesia (Chapter 10)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Inhalational agents are mainly used in anesthesia for:**

- A. Induction only
- B. Maintenance of anesthesia
- C. Premedication
- D. Postoperative analgesia

**Ans: B**

**Q2. According to the Meyer Overton rule, the anesthetic potency of an inhalational agent is directly proportional to its:**

- A. Blood gas partition coefficient
- B. Molecular weight
- C. Lipid solubility (oil gas partition coefficient)
- D. Vapor pressure

**Ans: C**

**Q3. The most acceptable mechanism of action of inhalational agents involves shifting of the mechanism from lipid sites to:**

- A. Protein sites — direct binding to cellular proteins altering their enzymes
- B. Sodium channels exclusively
- C. Potassium channels
- D. The lipid bilayer permanently

**Ans: A**

**Q4. Which inhalational agents act principally by inhibiting N-methyl-D-aspartate (NMDA) receptors rather than via GABA?**

- A. Halothane and isoflurane
- B. Sevoflurane and desflurane
- C. Nitrous oxide and xenon
- D. Enflurane and ether

**Ans: C**

**Q5. Potency of an inhalational agent is best determined by which parameter?**

- A. Blood gas partition coefficient
- B. Minimum alveolar concentration (MAC)
- C. Vapor pressure
- D. Boiling point

**Ans: B**

**Q6. According to Table 10.1, which agent is the MOST potent and which is the LEAST potent?**

- A. Most potent: Nitrous oxide; Least potent: Halothane
- B. Most potent: Halothane (MAC 0.74%); Least potent: Nitrous oxide (MAC 104%)
- C. Most potent: Desflurane; Least potent: Sevoflurane
- D. Most potent: Isoflurane; Least potent: Desflurane

**Ans: B**

|                                                                                                                                                                                                                                                                                                                                    |        |
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| <p><b>Q7. At what age is the maximum MAC seen in human beings?</b></p> <p>A. At birth<br/>B. 6 months<br/>C. 1 year<br/>D. Puberty</p>                                                                                                                                                                                             | Ans: B |
| <p><b>Q8. Which of the following INCREASES the MAC?</b></p> <p>A. Hyponatremia<br/>B. Pregnancy<br/>C. Hypernatremia<br/>D. Hypothermia</p>                                                                                                                                                                                        | Ans: C |
| <p><b>Q9. Which of the following has NO effect on MAC?</b></p> <p>A. Obesity<br/>B. Anemia (severe)<br/>C. Hypoxia<br/>D. Hypercarbia</p>                                                                                                                                                                                          | Ans: A |
| <p><b>Q10. Increasing barometric pressure affects the MAC by which mechanism?</b></p> <p>A. Increases the MAC (pressure reversal theory of anesthesia)<br/>B. Decreases the MAC<br/>C. Has no effect on MAC<br/>D. Abolishes the MAC</p>                                                                                           | Ans: A |
| <p><b>Q11. The most important factor determining the speed of induction and recovery of an inhalational agent is the:</b></p> <p>A. Oil gas partition coefficient<br/>B. Blood gas (B/G) partition coefficient<br/>C. MAC value<br/>D. Molecular weight</p>                                                                        | Ans: B |
| <p><b>Q12. According to Table 10.2, which agent has the LOWEST blood gas partition coefficient (0.14), giving the fastest induction?</b></p> <p>A. Nitrous oxide<br/>B. Desflurane<br/>C. Xenon<br/>D. Halothane</p>                                                                                                               | Ans: C |
| <p><b>Q13. A low blood gas partition coefficient agent (e.g. nitrous oxide, desflurane) is relatively SAFE in shock because:</b></p> <p>A. It increases cardiac output<br/>B. Its alveolar concentration will not be so high even when cardiac output is decreased<br/>C. It is more potent<br/>D. It is more soluble in blood</p> | Ans: B |

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| <p><b>Q14. What is the augmented inflow effect of nitrous oxide?</b></p> <ul style="list-style-type: none"> <li>A. Large uptake of nitrous oxide creates a void in the alveoli, drawing in more nitrous oxide from the machine</li> <li>B. Nitrous oxide expands gas-filled spaces</li> <li>C. Nitrous oxide increases the concentration of a second gas</li> <li>D. Nitrous oxide is excreted through the lungs after surgery</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q15. The second gas effect describes how nitrous oxide increases the concentration of another agent (e.g. halothane or isoflurane) given along with it. What is the term for nitrous oxide increasing its OWN concentration?</b></p> <ul style="list-style-type: none"> <li>A. Second gas effect</li> <li>B. Concentration effect</li> <li>C. Diffusion hypoxia</li> <li>D. Pressure effect</li> </ul>                              | <p><b>Ans: B</b></p> |
| <p><b>Q16. Diffusion hypoxia (Fink effect) at the end of surgery is prevented by:</b></p> <ul style="list-style-type: none"> <li>A. Giving 100% oxygen for 5–10 minutes after discontinuing nitrous oxide</li> <li>B. Continuing nitrous oxide for longer</li> <li>C. Hyperventilating the patient</li> <li>D. Giving a second gas</li> </ul>                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q17. Among the inhalational agents, the earliest recovery is seen with which agent, and the most delayed with which?</b></p> <ul style="list-style-type: none"> <li>A. Earliest: Halothane; Most delayed: Desflurane</li> <li>B. Earliest: Desflurane (B/G 0.42); Most delayed: Halothane (B/G 2.4)</li> <li>C. Earliest: Isoflurane; Most delayed: Sevoflurane</li> <li>D. Earliest: Nitrous oxide; Most delayed: Xenon</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q18. Maximum metabolism among current inhalational agents is seen with which agent?</b></p> <ul style="list-style-type: none"> <li>A. Desflurane (&lt;0.02%)</li> <li>B. Sevoflurane</li> <li>C. Halothane (20%)</li> <li>D. Isoflurane</li> </ul>                                                                                                                                                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q19. Ventilatory responses to increased CO<sub>2</sub> and hypoxia are blunted maximally by which agent?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Sevoflurane</li> <li>D. Nitrous oxide</li> </ul>                                                                                                                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q20. All inhalational agents blunt hypoxic pulmonary vasoconstriction (HPV) EXCEPT which one (which is itself a pulmonary vasoconstrictor)?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Nitrous oxide</li> <li>C. Sevoflurane</li> <li>D. Desflurane</li> </ul>                                                                                                                                         | <p><b>Ans: B</b></p> |

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| <p><b>Q21. The minimum decrease in cardiac output among inhalational agents is seen with:</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Enflurane</li> <li>D. Sevoflurane</li> </ul>                                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q22. Which inhalational agent is the agent of choice for cardiac patients (except in myocardial ischemia)?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Sevoflurane</li> <li>C. Isoflurane</li> <li>D. Desflurane</li> </ul>                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q23. Which agent does NOT sensitize the myocardium to adrenaline?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Enflurane</li> <li>D. Sevoflurane</li> </ul>                                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q24. Increase in cerebral blood flow and intracranial tension (ICT) caused by inhalational agents is maximum with which agent?</b></p> <ul style="list-style-type: none"> <li>A. Sevoflurane</li> <li>B. Isoflurane</li> <li>C. Desflurane</li> <li>D. Halothane</li> </ul>                                                | <p><b>Ans: D</b></p> |
| <p><b>Q25. Which inhalational agent is preferred as the agent of choice for neurosurgery in current practice?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Sevoflurane</li> <li>C. Desflurane</li> <li>D. Nitrous oxide</li> </ul>                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q26. Direct hepatocellular damage is classically seen with which agent but NOT with sevoflurane?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Nitrous oxide</li> <li>C. Xenon</li> <li>D. Ether</li> </ul>                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q27. Nephrotoxicity produced by fluorinated inhalational agents is characterized as:</b></p> <ul style="list-style-type: none"> <li>A. Oliguric renal failure with hyperkalemia</li> <li>B. Vasopressin-resistant polyuric renal failure</li> <li>C. Acute glomerulonephritis</li> <li>D. Renal artery stenosis</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q28. Although sevoflurane produces high fluoride levels, it does NOT cause nephrotoxicity because:</b></p> <ul style="list-style-type: none"> <li>A. Its low blood gas coefficient (0.69) allows rapid elimination from the body</li> <li>B. It does not undergo metabolism</li> <li>C. It is not fluorinated</li> <li>D. It increases renal blood flow</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q29. Nitrous oxide can cause megaloblastic and aplastic anemia if used for more than how long, due to interaction with vitamin B12?</b></p> <ul style="list-style-type: none"> <li>A. More than 12 hours</li> <li>B. More than 1 hour</li> <li>C. More than 24 hours</li> <li>D. More than 1 week</li> </ul>                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q30. Prolonged nitrous oxide exposure can impair myelin formation leading to which neurological complication?</b></p> <ul style="list-style-type: none"> <li>A. Subacute degeneration of spinal cord</li> <li>B. Guillain-Barré syndrome</li> <li>C. Multiple sclerosis</li> <li>D. Parkinson's disease</li> </ul>                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q31. Maximum muscle relaxation among current inhalational agents is seen with:</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Nitrous oxide</li> <li>C. Desflurane</li> <li>D. Isoflurane</li> </ul>                                                                                                                                      | <p><b>Ans: C</b></p> |
| <p><b>Q32. Trilene (trichloroethylene) reacting with soda lime can produce dichloroacetylene and which other gas that can cause ARDS?</b></p> <ul style="list-style-type: none"> <li>A. Carbon monoxide</li> <li>B. Compound A</li> <li>C. Phosgene</li> <li>D. Hydrogen fluoride</li> </ul>                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q33. Which inhalational agent produces Compound A on reaction with soda lime and barium?</b></p> <ul style="list-style-type: none"> <li>A. Desflurane</li> <li>B. Sevoflurane</li> <li>C. Halothane</li> <li>D. Isoflurane</li> </ul>                                                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q34. Desflurane (and other agents) reacting with DESICCATED (dried) soda lime can produce which toxic gas?</b></p> <ul style="list-style-type: none"> <li>A. Phosgene</li> <li>B. Compound A</li> <li>C. Carbon monoxide</li> <li>D. Nitric oxide</li> </ul>                                                                                                       | <p><b>Ans: C</b></p> |

|                                                                                                                                                                                                                                                                                                                    |                      |
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| <p><b>Q35. Which agents are responsible for ozone depletion and global warming (in order)?</b></p> <p>A. Halothane followed by isoflurane<br/> B. N<sub>2</sub>O followed by desflurane<br/> C. Sevoflurane followed by xenon<br/> D. Ether followed by cyclopropane</p>                                           | <p><b>Ans: B</b></p> |
| <p><b>Q36. Which inhalational agents are good amnesic agents EXCEPT which one?</b></p> <p>A. Halothane<br/> B. Nitrous oxide<br/> C. Sevoflurane<br/> D. Isoflurane</p>                                                                                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q37. Nitrous oxide is prepared by heating ammonium nitrate at what temperature?</b></p> <p>A. 100–150°C<br/> B. 245–270°C<br/> C. 350–400°C<br/> D. 50–60°C</p>                                                                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q38. Nitrous oxide is stored in cylinders of which color, and is how many times more soluble than nitrogen?</b></p> <p>A. Black cylinders; 1.5 times<br/> B. Blue cylinders; 35 times more soluble than nitrogen<br/> C. Green cylinders; 10 times<br/> D. White cylinders; 5 times</p>                      | <p><b>Ans: B</b></p> |
| <p><b>Q39. Nitrous oxide is absolutely contraindicated in pneumothorax because it doubles the pneumothorax volume in how long?</b></p> <p>A. 10 minutes (and triples in 30 minutes)<br/> B. 1 minute<br/> C. 1 hour<br/> D. 5 minutes</p>                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q40. Entonox is a mixture of:</b></p> <p>A. 50% oxygen and 50% nitrous oxide, used commonly for labor analgesia<br/> B. 33% oxygen and 66% nitrous oxide<br/> C. 21% oxygen and 79% nitrogen<br/> D. 100% nitrous oxide</p>                                                                                  | <p><b>Ans: A</b></p> |
| <p><b>Q41. Why should Entonox be stored above 10°C?</b></p> <p>A. Otherwise the cylinder explodes<br/> B. Otherwise low temperature can cause separation of oxygen and nitrous oxide, delivering high concentration of nitrous oxide<br/> C. Otherwise it loses potency<br/> D. Otherwise it becomes flammable</p> | <p><b>Ans: B</b></p> |

|                                                                                                                                                                                                                                                                                                                                                  |                      |
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| <p><b>Q42. Which noble gas, having the lowest blood gas coefficient (0.14), provides the fastest induction and recovery (almost 3 times faster than N<sub>2</sub>O)?</b></p> <p>A. Helium<br/>B. Argon<br/>C. Xenon<br/>D. Krypton</p>                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q43. Why is xenon unsuitable for asthma patients?</b></p> <p>A. Because of its high density (almost 6 times of air), which increases airway resistance<br/>B. Because it causes bronchoconstriction directly<br/>C. Because it depresses respiration<br/>D. Because it is flammable</p>                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q44. What is the most important risk factor for halothane hepatitis?</b></p> <p>A. Hypoxia<br/>B. Multiple and frequent exposures<br/>C. Obesity<br/>D. Female gender</p>                                                                                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q45. Halothane is contraindicated in pheochromocytoma because it:</b></p> <p>A. Sensitizes the myocardium to adrenaline, producing ventricular arrhythmias<br/>B. Causes severe hypertension<br/>C. Increases catecholamine synthesis<br/>D. Causes bronchospasm</p>                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q46. Which agent is the inhalational agent of choice for a known case of halothane hepatitis?</b></p> <p>A. Isoflurane<br/>B. Desflurane<br/>C. Sevoflurane<br/>D. Enflurane</p>                                                                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q47. Hypotension is maximum with which agent, making it the inhalational agent of choice for controlled hypotension?</b></p> <p>A. Halothane<br/>B. Isoflurane<br/>C. Sevoflurane<br/>D. Nitrous oxide</p>                                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q48. Desflurane requires a special vaporizer (TEC 6) which is temperature and pressure compensated because:</b></p> <p>A. Its vapor pressure is very high (681 mm Hg) and boiling point is less than 23°C, so it can boil at room temperature<br/>B. It is highly flammable<br/>C. It is unstable in light<br/>D. It reacts with metal</p> | <p><b>Ans: A</b></p> |



**Q49. Sevoflurane is the inhalational agent of choice for which patients (due to maintained hepatic blood flow and no trifluoroacetic acid)?**

- A. Cardiac patients
- B. Renal patients
- C. Hepatic patients
- D. Obese patients

**Ans: C**

**Q50. Why is ether considered the only 'complete anesthetic' and the safest inhalational agent for remote locations?**

- A. It is the most potent agent
- B. It has all three basic properties of anesthesia (narcosis, analgesia, muscle relaxation), does not cause cardiac/respiratory depression, preserves ciliary activity, can be given by open drop method, and is very cheap
- C. It has the fastest induction and recovery
- D. It is non-inflammable

**Ans: B**

# Answer Key & Explanations

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## Q1. Answer: B — Maintenance of anesthesia

Page 89: Inhalational agents are mainly used in anesthesia for maintenance of anesthesia; however, they can also be used for induction, especially in children.

Topic: Classification

## Q2. Answer: C — Lipid solubility (oil gas partition coefficient)

Page 89: Anesthetic potency of an inhalational agent is directly proportional to its lipid solubility (oil gas partition coefficient). This is called the Meyer Overton rule, which led to the unitary theory of narcosis.

Topic: Mechanism of Action

## Q3. Answer: A — Protein sites — direct binding to cellular proteins altering their enzymes

Page 89: Exceptions to the Meyer Overton rule lead to shifting of mechanism from lipid sites to protein sites; the most acceptable mechanism is that they directly bind to cellular proteins, altering their enzymes.

Topic: Mechanism of Action

## Q4. Answer: C — Nitrous oxide and xenon

Page 89: Modulation of GABA receptors is the principal mechanism for all volatile agents except the gases — nitrous oxide and xenon, which act by inhibiting N-methyl-D-aspartate (NMDA) receptors.

Topic: Mechanism of Action

## Q5. Answer: B — Minimum alveolar concentration (MAC)

Page 90: The best estimate for determining the potency of an inhalational anesthetic is minimum alveolar concentration (MAC), defined as the minimum concentration of agent to produce immobility in 50% of subjects given noxious stimuli.

Topic: Potency / MAC

## Q6. Answer: B — Most potent: Halothane (MAC 0.74%); Least potent: Nitrous oxide (MAC 104%)

Page 90, Table 10.1: Halothane with MAC of 0.74% is the most potent, while nitrous oxide with a very high MAC of 104% is the least potent among the agents used nowadays.

Topic: Potency / MAC

## Q7. Answer: B — 6 months

Page 90: Age — maximum MAC in human beings is at the age of 6 months, thereafter decreasing steadily throughout life (except a slight increase at puberty).

Topic: Factors Affecting MAC

## Q8. Answer: C — Hypernatremia

Page 90: Factors increasing MAC include hyperthermia  $>42^{\circ}\text{C}$ , chronic intoxication of alcohol and amphetamines, cocaine, ephedrine, increasing barometric pressure, and hypernatremia. Hyponatremia, pregnancy and hypothermia decrease MAC.

Topic: Factors Affecting MAC

## Q9. Answer: A — Obesity

Page 90: Factors having no effect on MAC are thyroid disorders, sex (MAC is the same in males and females), and obesity (MAC does not increase in obese patients).

Topic: Factors Affecting MAC

**Q10. Answer: A — Increases the MAC (pressure reversal theory of anesthesia)**

Page 90: Barometric pressure — increasing pressure increases the MAC (pressure reversal theory of anesthesia).

Topic: Factors Affecting MAC

**Q11. Answer: B — Blood gas (B/G) partition coefficient**

Page 91: The blood gas partition coefficient (B/G coefficient) is the most important factor determining the uptake of an agent and so the speed of induction and recovery. Agents with low B/G coefficient will have high alveolar concentration (fast induction).

Topic: Uptake and Distribution

**Q12. Answer: C — Xenon**

Page 91, Table 10.2: Xenon has the lowest blood gas partition coefficient (0.14), so the fastest induction is seen with xenon. Among the agents used now-a-days, induction is fastest with desflurane (0.42) and slowest with halothane (2.4).

Topic: Uptake and Distribution

**Q13. Answer: B — Its alveolar concentration will not be so high even when cardiac output is decreased**

Page 91: In low cardiac output states such as shock, agents with high B/G coefficient can achieve dangerously high concentrations. That is why agents with low B/G coefficient such as nitrous oxide, desflurane or sevoflurane are relatively safe in shock.

Topic: Uptake and Distribution

**Q14. Answer: A — Large uptake of nitrous oxide creates a void in the alveoli, drawing in more nitrous oxide from the machine**

Page 92: When nitrous oxide is given in high concentration, the large uptake of nitrous oxide creates a void to in-draw more nitrous oxide from the machine — this is called augmented inflow effect (nitrous oxide has augmented its inflow from the machine).

Topic: Uptake and Distribution

**Q15. Answer: B — Concentration effect**

Page 92: The concentration effect is where nitrous oxide increases its own concentration. The second gas effect is when another agent given along with nitrous oxide also has its concentration increased by the same mechanism.

Topic: Uptake and Distribution

**Q16. Answer: A — Giving 100% oxygen for 5–10 minutes after discontinuing nitrous oxide**

Page 92: Towards the end of surgery when nitrous oxide delivery is stopped, the gradient reverses and nitrous oxide gushes into alveoli replacing oxygen, causing hypoxia (diffusion hypoxia). To avoid this, 100% oxygen should be given for 5–10 minutes after discontinuing nitrous oxide.

Topic: Uptake and Distribution

**Q17. Answer: B — Earliest: Desflurane (B/G 0.42); Most delayed: Halothane (B/G 2.4)**

Page 92: Recovery is rapid with agents with low B/G coefficient. Therefore, earliest recovery will be with desflurane (B/G coefficient 0.42) and most delayed with halothane (B/G coefficient 2.4).

Topic: Recovery

**Q18. Answer: C — Halothane (20%)**

Page 92: All inhalational agents undergo oxidation by cytochrome P-450 enzymes. Among current agents, maximum metabolism is seen with halothane (20%). Desflurane is negligibly metabolized (<0.02%) and nitrous oxide does not undergo any metabolism in the human body.

Topic: Metabolism

**Q19. Answer: A — Halothane**

Page 92: Initially all agents decrease the tidal volume and increase the frequency, finally decreasing the minute volume. Maximum inhibition to ventilatory response is seen with halothane.

*Topic: Systemic Effects – Respiratory*

**Q20. Answer: B — Nitrous oxide**

Page 92: All inhalational agents are pulmonary vasodilators (except nitrous oxide which is a pulmonary vasoconstrictor) thereby blunting HPV response and worsening V/Q mismatch.

*Topic: Systemic Effects – Respiratory*

**Q21. Answer: B — Isoflurane**

Page 92–93: By causing direct myocardial depression, all new inhalational agents decrease the cardiac output. The minimum decrease in cardiac output is seen with isoflurane.

*Topic: Systemic Effects – Cardiovascular*

**Q22. Answer: C — Isoflurane**

Page 93: Among all inhalational agents, isoflurane minimally depresses myocardium, does not cause bradycardia, and baroreceptor reflex is preserved, so cardiac output is best maintained — making isoflurane the inhalational agent of choice for cardiac patients (except myocardial ischemia, where coronary steal is a theoretical concern).

*Topic: Systemic Effects – Cardiovascular*

**Q23. Answer: B — Isoflurane**

Page 93: Halothane (and theoretically other agents too, except isoflurane) can sensitize the myocardium to adrenaline. Isoflurane does not sensitize the myocardium to adrenaline.

*Topic: Systemic Effects – Cardiovascular*

**Q24. Answer: D — Halothane**

Page 93: All inhalational agents (including nitrous oxide) increase cerebral blood flow and hence ICT. The order is halothane >> desflurane ≈ isoflurane > sevoflurane.

*Topic: Systemic Effects – CNS*

**Q25. Answer: B — Sevoflurane**

Page 93: Due to more favorable effects on cerebral hemodynamics, better preservation of autoregulation, equivalent reduction in cerebral metabolic rate, and smoother recovery, sevoflurane is preferred over isoflurane as the inhalational agent of choice for neurosurgery in current day practice.

*Topic: Systemic Effects – CNS*

**Q26. Answer: A — Halothane**

Page 93: Direct hepatocellular damage is seen with halothane (and theoretically with isoflurane and desflurane) but not with sevoflurane. Maximal decrease in blood supply to liver is seen with halothane and least with sevoflurane.

*Topic: Systemic Effects – Liver*

**Q27. Answer: B — Vasopressin-resistant polyuric renal failure**

Page 93: All inhalational agents depress renal function by decreasing renal blood flow. Direct renal toxicity is due to inorganic fluoride. The nephrotoxicity produced by fluorinated agents is vasopressin-resistant polyuric renal failure. Fluoride is toxic at 50 µM.

*Topic: Systemic Effects – Renal*

**Q28. Answer: A — Its low blood gas coefficient (0.69) allows rapid elimination from the body**

Page 93, Table 10.3 note: Sevoflurane, although it produces high fluoride, still does not cause nephrotoxicity even after long exposures because its low blood gas coefficient (0.69) allows its rapid elimination from the body.

*Topic: Systemic Effects – Renal*

**Q29. Answer: A — More than 12 hours**

Page 93–94: Nitrous oxide interacts with vitamin B12 and inhibits methionine synthetase, leading to bone marrow depression causing aplastic and megaloblastic anemia. Vitamin B12 inactivation is seen if nitrous oxide is used for more than 12 hours.

*Topic: Systemic Effects – Hematopoietic*

**Q30. Answer: A — Subacute degeneration of spinal cord**

Page 94: Nitrous oxide, by inhibiting the production of thymidate, can impair myelin formation leading to subacute degeneration of spinal cord.

*Topic: Systemic Effects – Spinal Cord*

**Q31. Answer: C — Desflurane**

Page 94: All inhalational agents are centrally acting muscle relaxants (except nitrous oxide). Among the agents used now-a-days, maximum relaxation is seen with desflurane.

*Topic: Systemic Effects – Neuromuscular*

**Q32. Answer: C — Phosgene**

Page 94: Trilene can produce dichloroacetylene (which is neurotoxic) and phosgene gas (which can cause ARDS) on reaction with soda lime.

*Topic: Reaction with Soda Lime*

**Q33. Answer: B — Sevoflurane**

Page 94: Sevoflurane can produce a toxic compound called compound A with soda lime and barium hydroxide. Compound A can cause nephrotoxicity.

*Topic: Reaction with Soda Lime*

**Q34. Answer: C — Carbon monoxide**

Page 94: Desflurane (and other agents too) can produce carbon monoxide by reacting with desiccated soda lime.

*Topic: Reaction with Soda Lime*

**Q35. Answer: B — N<sub>2</sub>O followed by desflurane**

Page 94: Inhalational agents are responsible for ozone depletion and global warming, with N<sub>2</sub>O followed by desflurane. The best way to reduce this environmental impact is to use closed-circuit anesthesia and avoid N<sub>2</sub>O as much as possible.

*Topic: Environmental Effects*

**Q36. Answer: B — Nitrous oxide**

Page 94: All inhalational agents are good amnesic agents except nitrous oxide.

*Topic: Systemic Effects*

**Q37. Answer: B — 245–270°C**

Page 94: Nitrous oxide is prepared by heating ammonium nitrate between 245 and 270°C.

*Topic: Nitrous Oxide*

**Q38. Answer: B — Blue cylinders; 35 times more soluble than nitrogen**

Page 94: Nitrous oxide is stored in blue color cylinders at a pressure of 760 psi. It is 1.5 times heavier than air and 35 times more soluble than nitrogen.

Topic: Nitrous Oxide

**Q39. Answer: A — 10 minutes (and triples in 30 minutes)**

Page 95: Nitrous oxide doubles the pneumothorax volume in 10 minutes and triples in 30 minutes, producing severe cardiorespiratory compromise. The 35 times more solubility of nitrous oxide than nitrogen makes it contraindicated in closed gas spaces.

Topic: Nitrous Oxide

**Q40. Answer: A — 50% oxygen and 50% nitrous oxide, used commonly for labor analgesia**

Page 96: Entonox is a mixture of 50% oxygen and 50% nitrous oxide, used commonly for labor analgesia. It is stored in blue colored cylinders with blue and white shoulders at a pressure of 2,000 psi.

Topic: Entonox

**Q41. Answer: B — Otherwise low temperature can cause separation of oxygen and nitrous oxide, delivering high concentration of nitrous oxide**

Page 96: Entonox should be stored above 10°C otherwise low temperature can cause separation of oxygen and nitrous oxide, and high concentration of nitrous oxide can be delivered. To prevent this, the cylinder should be inverted 2–3 times to ensure good mixing.

Topic: Entonox

**Q42. Answer: C — Xenon**

Page 96: Xenon has the lowest blood gas coefficient (0.14), therefore induction and recovery is fastest (almost 3 times faster than N<sub>2</sub>O which has blood gas coefficient of 0.47). It is more potent than nitrous oxide (MAC 70%).

Topic: Xenon

**Q43. Answer: A — Because of its high density (almost 6 times of air), which increases airway resistance**

Page 96: Because of high density (almost 6 times of air), xenon increases airway resistance, making it unsuitable for asthma patients.

Topic: Xenon

**Q44. Answer: B — Multiple and frequent exposures**

Page 97: The most important risk factor for halothane hepatitis is multiple and frequent exposures. Other risk factors are hypoxia, middle age, obesity, female gender, and patients suffering from autoimmune diseases. The pathologic lesion is centrilobular necrosis.

Topic: Halothane

**Q45. Answer: A — Sensitizes the myocardium to adrenaline, producing ventricular arrhythmias**

Page 97: Halothane sensitizes the heart to adrenaline (exogenous to heart) producing ventricular arrhythmias; therefore the maximum permissible dose of adrenaline with halothane for local ischemia is limited, and for the same reason it is contraindicated in pheochromocytoma.

Topic: Halothane

**Q46. Answer: C — Sevoflurane**

Page 98: Sevoflurane is the inhalational agent of choice for a known case of halothane hepatitis because sevoflurane does not produce trifluoroacetic acid, avoiding the possibility of hepatitis in such a patient.

Topic: Halothane

**Q47. Answer: B — Isoflurane**

Page 98: Hypotension is maximum with isoflurane (it is the inhalational agent of choice for controlled hypotension). At the same time, it minimally depresses myocardium and does not cause bradycardia.

*Topic: Isoflurane*

**Q48. Answer: A — Its vapor pressure is very high (681 mm Hg) and boiling point is less than 23°C, so it can boil at room temperature**

Page 98–99: Desflurane has a very high vapor pressure (681 mm Hg) and boiling point less than 23°C, therefore it can boil at room temperature; that is why a special vaporizer (TEC 6) which is temperature and pressure compensated is required for its delivery.

*Topic: Desflurane*

**Q49. Answer: C — Hepatic patients**

Page 99: Sevoflurane decreases portal blood flow but increases hepatic artery blood flow, so hepatic blood flow is best maintained. It does not produce trifluoroacetic acid and cannot cause immunologic hepatitis. These properties make sevoflurane the inhalational agent of choice for hepatic patients.

*Topic: Sevoflurane*

**Q50. Answer: B — It has all three basic properties of anesthesia (narcosis, analgesia, muscle relaxation), does not cause cardiac/respiratory depression, preserves ciliary activity, can be given by open drop method, and is very cheap**

Page 100: Ether is the only anesthetic till date which can be considered complete, i.e. it has all three basic properties of anesthesia (narcosis, analgesia and muscle relaxation). It is the safest inhalational anesthetic (no cardiac or respiratory depression), the only agent which preserves ciliary activity, can be given by open drop method, and is very cheap — making it the agent of choice for remote locations. Its main disadvantage is that it is highly inflammable and explosive.

*Topic: Ether*

# Gases Used in Anesthesia MCQ – Chapter 11

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 5: General Anesthesia (Chapter 11)

**25 Questions • 20 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

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**Q1. Medical oxygen is prepared by which process?**

- A. Fractional distillation of air
- B. Heating ammonium nitrate
- C. Electrolysis of water
- D. Reaction of zeolites

**Ans: A**

**Q2. Oxygen concentrators produce oxygen from ambient air by selective absorption of nitrogen on:**

- A. Activated charcoal
- B. Sodalime
- C. Zeolites
- D. Silica gel

**Ans: C**

**Q3. Medical oxygen is stored in cylinders with which color coding, at what pressure?**

- A. Black body and white shoulders at 2,000 psi
- B. Blue body at 760 psi
- C. Grey body at 750 psi
- D. Brown body at 2,000 psi

**Ans: A**

**Q4. What is the critical temperature of oxygen?**

- A. 36.5°C
- B. -119°C
- C. 50°C
- D. -183°C

**Ans: B**

**Q5. Which is the most common type of hypoxia seen during anesthesia?**

- A. Anemic hypoxia
- B. Stagnant hypoxia
- C. Hypoxic hypoxia
- D. Histotoxic hypoxia

**Ans: C**

**Q6. Among the body stores of oxygen, which is practically the only store, lasting only 2–3 minutes?**

- A. Oxygen bound to hemoglobin
- B. Dissolved fraction in blood
- C. Functional residual capacity (FRC) in lungs
- D. Oxygen in tissues

**Ans: C**

**Q7. At an oxygen saturation (SpO<sub>2</sub>) of less than 60%, hypoxia produces what cardiovascular effect?**

- A. Sympathetic stimulation with tachycardia and hypertension
- B. Vasodilatation with reflex tachycardia and maintained cardiac output
- C. Direct cardiac depression with bradyarrhythmia or even cardiac arrest
- D. No cardiovascular effect

**Ans: C**

|                                                                                                                                                                                                                                                                                                                                            |                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q8. Clinical cyanosis is produced when the reduced hemoglobin level is more than:</b></p> <ul style="list-style-type: none"> <li>A. 5 g/dL</li> <li>B. 1.5 g/dL</li> <li>C. 0.5 g/dL</li> <li>D. 10 g/dL</li> </ul>                                                                                                                  | <p><b>Ans: A</b></p> |
| <p><b>Q9. Which type of hypoxia responds best to oxygen therapy?</b></p> <ul style="list-style-type: none"> <li>A. Histotoxic hypoxia</li> <li>B. Hypoxic hypoxia</li> <li>C. Right to left shunts</li> <li>D. Anemic hypoxia</li> </ul>                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q10. Prolonged 100% oxygen is usually considered safe for up to how long in normal adults?</b></p> <ul style="list-style-type: none"> <li>A. 8–12 hours</li> <li>B. 2–3 hours</li> <li>C. 24 hours</li> <li>D. 48 hours</li> </ul>                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q11. In neonates, pulmonary oxygen toxicity manifests as:</b></p> <ul style="list-style-type: none"> <li>A. Bronchopulmonary dysplasia</li> <li>B. Retrolental fibroplasia</li> <li>C. Cerebral convulsions</li> <li>D. Avascular necrosis</li> </ul>                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q12. Retrolental fibroplasia from oxygen toxicity is dependent on arterial oxygen concentration. Which group is at greatest risk?</b></p> <ul style="list-style-type: none"> <li>A. Adults on prolonged oxygen</li> <li>B. Premature neonates (&lt; 36 weeks)</li> <li>C. Full-term neonates</li> <li>D. Elderly patients</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q13. Hyperbaric oxygen therapy is practically restricted to which two conditions?</b></p> <ul style="list-style-type: none"> <li>A. Asthma and COPD</li> <li>B. Decompression sickness and severe cases of carbon monoxide poisoning</li> <li>C. Pneumonia and ARDS</li> <li>D. Anemia and shock</li> </ul>                          | <p><b>Ans: B</b></p> |
| <p><b>Q14. The half-life of carbon monoxide is 214 minutes at 1 atm. At 2.5 atm of hyperbaric oxygen, it is reduced to:</b></p> <ul style="list-style-type: none"> <li>A. 100 minutes</li> <li>B. 50 minutes</li> <li>C. 19 minutes</li> <li>D. 5 minutes</li> </ul>                                                                       | <p><b>Ans: C</b></p> |

|                                                                                                                                                                                                                                                                                                                                                                                                  |                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q15. CNS toxicity of hyperbaric oxygen (the Paul Bert effect, with seizures) is usually seen if the pressure used is:</b></p> <ul style="list-style-type: none"> <li>A. More than 3 atm</li> <li>B. More than 1 atm</li> <li>C. More than 0.5 atm</li> <li>D. Less than 1 atm</li> </ul>                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q16. In the past, carbon dioxide (5%) was used to hasten recovery at emergence. What is its current use in anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Treatment of postspinal headache</li> <li>B. Gas insufflation for laparoscopic surgeries and cryosurgeries</li> <li>C. Opening the glottis for intubation</li> <li>D. It is no longer used in any form</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q17. Heliox is a mixture of which gases?</b></p> <ul style="list-style-type: none"> <li>A. 79% nitrogen + 21% oxygen</li> <li>B. 50% helium + 50% oxygen</li> <li>C. 79% helium + 21% oxygen</li> <li>D. 66% nitrous oxide + 33% oxygen</li> </ul>                                                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q18. The most important characteristic of helium is its:</b></p> <ul style="list-style-type: none"> <li>A. High density</li> <li>B. Low density</li> <li>C. High solubility</li> <li>D. Anesthetic potency</li> </ul>                                                                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q19. Heliox decreases turbulence and airway resistance, enabling oxygen to pass through narrow orifices. It is therefore used for:</b></p> <ul style="list-style-type: none"> <li>A. Upper airway obstruction (tracheal/bronchial stenosis or foreign body)</li> <li>B. Treatment of pneumothorax</li> <li>C. Labor analgesia</li> <li>D. Cardiac arrest resuscitation</li> </ul>          | <p><b>Ans: A</b></p> |
| <p><b>Q20. Nitric oxide is the major vasodilatory compound produced in endothelium during the conversion of:</b></p> <ul style="list-style-type: none"> <li>A. L-arginine to L-citrulline by nitric oxide synthase</li> <li>B. Glucose to pyruvate</li> <li>C. Ammonium nitrate to nitrous oxide</li> <li>D. Bicarbonate to carbon dioxide</li> </ul>                                            | <p><b>Ans: A</b></p> |
| <p><b>Q21. Inhaled nitric oxide is used for diagnosis and treatment of:</b></p> <ul style="list-style-type: none"> <li>A. Asthma</li> <li>B. Primary pulmonary hypertension or secondary pulmonary hypertension associated with ARDS</li> <li>C. Systemic hypertension</li> <li>D. Carbon monoxide poisoning</li> </ul>                                                                          | <p><b>Ans: B</b></p> |

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| <p><b>Q22. Medical air supplied through the central supply is at what pressure?</b></p> <ul style="list-style-type: none"> <li>A. 60 psi</li> <li>B. 750 psi</li> <li>C. 2,000 psi</li> <li>D. 760 psi</li> </ul>                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q23. Medical air can be used as a respired gas (replacement to nitrous oxide) or as a gas to:</b></p> <ul style="list-style-type: none"> <li>A. Drive pneumatic devices like ventilators</li> <li>B. Treat pulmonary hypertension</li> <li>C. Insufflate the abdomen in laparoscopy</li> <li>D. Hasten recovery at emergence</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q24. Excessive (100%) oxygen toxicity occurs because of toxic radicals. Which of the following is one such radical?</b></p> <ul style="list-style-type: none"> <li>A. Bicarbonate</li> <li>B. Superoxide</li> <li>C. Carbamino compound</li> <li>D. Methemoglobin</li> </ul>                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q25. For decompression sickness, hyperbaric therapy is given as 100% oxygen at what pressure, followed by 1.9 atm?</b></p> <ul style="list-style-type: none"> <li>A. 2.8 atm</li> <li>B. 6 atm</li> <li>C. 1 atm</li> <li>D. 3 atm</li> </ul>                                                                                           | <p><b>Ans: A</b></p> |

# Answer Key & Explanations

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## Q1. Answer: A — Fractional distillation of air

Page 103: Medical oxygen is prepared by fractional distillation of air. Oxygen concentrators are portable devices which produce oxygen from ambient air by selective absorption of nitrogen on zeolites.

Topic: Oxygen

## Q2. Answer: C — Zeolites

Page 103: Oxygen concentrators are portable devices which can produce oxygen from ambient air by selective absorption of nitrogen on zeolites. Oxygen concentrators are suitable for use in hospitals, home or at remote locations.

Topic: Oxygen

## Q3. Answer: A — Black body and white shoulders at 2,000 psi

Page 103: Medical oxygen is stored in cylinders with black body and white shoulders at a pressure of 2,000 psi (pounds per square inch). It is also stored as liquid oxygen.

Topic: Oxygen

## Q4. Answer: B — -119°C

Page 103: The critical temperature of oxygen is -119°C, therefore liquid oxygen should be stored below -119°C (to keep any gas in liquefied state it has to be kept below its critical temperature).

Topic: Oxygen

## Q5. Answer: C — Hypoxic hypoxia

Page 103: Hypoxic hypoxia is the most common type of hypoxia seen during anesthesia.

Topic: Hypoxia

## Q6. Answer: C — Functional residual capacity (FRC) in lungs

Page 103: Body stores of oxygen are theoretically 1,500 mL (bound to hemoglobin, dissolved fraction, and in lungs as functional residual capacity). Among these, only the available oxygen in lungs (FRC) practically is the only store in body, which lasts only for 2–3 minutes.

Topic: Hypoxia

## Q7. Answer: C — Direct cardiac depression with bradyarrhythmia or even cardiac arrest

Page 103: At SpO<sub>2</sub> <60%, hypoxia is a direct cardiac depressant and can produce bradyarrhythmia or even cardiac arrest. At SpO<sub>2</sub> >80% it stimulates the sympathetic system (tachycardia, hypertension), and at SpO<sub>2</sub> 60–80% it causes vasodilatation with reflex tachycardia where cardiac output is maintained.

Topic: Hypoxia

## Q8. Answer: A — 5 g/dL

Page 103: The clinical presentation of hypoxia is cyanosis, which is produced if reduced Hb level is more than 5 g/dL (roughly corresponds to O<sub>2</sub> saturation <85% or pO<sub>2</sub> <50 mm Hg), methemoglobin >1.5 g/dL, and sulfhemoglobin >0.5 g/dL.

Topic: Hypoxia

## Q9. Answer: B — Hypoxic hypoxia

Page 103–104: Hypoxic hypoxia responds best to oxygen therapy, while there is no benefit to oxygen therapy in histotoxic hypoxia and right to left shunts.

Topic: Hypoxia

**Q10. Answer: A — 8–12 hours**

Page 104: 100% oxygen is usually considered safe up to 8–12 hours in normal adults, but in infants and neonates 100% oxygen for more than 2–3 hours can cause pulmonary toxicity. Pulmonary toxicity depends on alveolar concentration (not on arterial concentration).

Topic: Oxygen Toxicity

**Q11. Answer: A — Bronchopulmonary dysplasia**

Page 104: Oxygen toxicity occurs because of toxic radicals (superoxide, hydroxyl ions, singlet oxygen, hydrogen peroxide) which damage the capillary membrane, producing an ARDS-like picture. In neonates pulmonary toxicity manifests as bronchopulmonary dysplasia.

Topic: Oxygen Toxicity

**Q12. Answer: B — Premature neonates (< 36 weeks)**

Page 104: Retrolental fibroplasia is dependent on arterial oxygen concentration. Premature neonates (< 36 weeks) are at greatest risk.

Topic: Oxygen Toxicity

**Q13. Answer: B — Decompression sickness and severe cases of carbon monoxide poisoning**

Page 104: Hyperbaric oxygen means delivering oxygen above atmospheric pressure (>760 mm Hg or >1 atm). Although there are many indications, practically its use is only restricted to decompression sickness and severe cases of carbon monoxide poisoning.

Topic: Hyperbaric Oxygen

**Q14. Answer: C — 19 minutes**

Page 104: In carbon monoxide poisoning the half-life of CO at 1 atm is 214 minutes, which can be reduced to 19 minutes at 2.5 atm. In current practice its use is reserved only for severe cases; mild-to-moderate cases are managed by 100% normobaric oxygen.

Topic: Hyperbaric Oxygen

**Q15. Answer: A — More than 3 atm**

Page 104: CNS toxicity of hyperbaric oxygen manifests as seizures preceded by facial numbness, twitching and olfactory/gustatory sensations (Paul Bert effect). The Paul Bert effect is usually seen if the pressure used is > 3 atm.

Topic: Hyperbaric Oxygen

**Q16. Answer: B — Gas insufflation for laparoscopic surgeries and cryosurgeries**

Page 105: In the past CO<sub>2</sub> (5%) was used to hasten recovery at emergence, for glottis opening for blind intubation, and for treatment of postspinal headache; however, it is not used now in anesthesia. Currently it is used only on the surgical side for gas insufflation for laparoscopic surgeries and cryosurgeries. It is stored in grey color cylinders at 750 psi.

Topic: Carbon Dioxide

**Q17. Answer: C — 79% helium + 21% oxygen**

Page 105: Helium, mixed with oxygen (79% helium + 21% oxygen) is available as Heliox. Heliox is stored in black color cylinders (representing oxygen) with brown and white shoulders (brown representing helium). Helium is stored in brown cylinders at 2,000 psi.

Topic: Helium/Heliox

**Q18. Answer: B — Low density**

Page 105: The most important characteristic of helium is its low density (specific gravity of helium is 341 while that of air is 1,000). This allows it to reduce turbulence and airway resistance.

Topic: Helium/Heliox

**Q19. Answer: A — Upper airway obstruction (tracheal/bronchial stenosis or foreign body)**

Page 105: When oxygen is mixed with helium (Heliox), the density of the gas mixture decreases, reducing turbulence and resistance, enabling oxygen to pass even through narrow orifices. Therefore, Heliox is used for upper airway obstruction (tracheal or bronchial stenosis or foreign body) and microlaryngeal surgeries (where small size endotracheal tubes are used).

Topic: Helium/Heliox

**Q20. Answer: A — L-arginine to L-citrulline by nitric oxide synthase**

Page 105: Nitric oxide is the major vasodilatory compound produced in pulmonary and systemic endothelium during the conversion of L-arginine to L-citrulline by nitric oxide synthase. It produces vasodilatation through cGMP.

Topic: Nitric Oxide

**Q21. Answer: B — Primary pulmonary hypertension or secondary pulmonary hypertension associated with ARDS**

Page 105: Because of its pulmonary vasodilatory property, inhaled nitric oxide is used for diagnosis and treatment of primary pulmonary hypertension or secondary pulmonary hypertension associated with ARDS.

Topic: Nitric Oxide

**Q22. Answer: A — 60 psi**

Page 105: Medical air is usually supplied through central supply at a pressure of 60 psi; however, it is also available in grey cylinders with black and white shoulders at a pressure of 2,000 psi.

Topic: Air

**Q23. Answer: A — Drive pneumatic devices like ventilators**

Page 105: Medical air can be used as a respired gas (as a replacement to nitrous oxide) or as a gas to drive pneumatic devices like ventilators.

Topic: Air

**Q24. Answer: B — Superoxide**

Page 104: Oxygen toxicity occurs because of toxic radicals of oxygen like superoxide and hydroxyl ions, singlet oxygen, and hydrogen peroxide. These radicals damage the capillary membrane increasing capillary permeability and an ARDS-like picture.

Topic: Oxygen Toxicity

**Q25. Answer: A — 2.8 atm**

Page 104–105: Patients with decompression sickness are given 100% oxygen at 2.8 atm followed by 1.9 atm, interspersed by periods of 5–15 minutes of air breathing. The exact safe levels are not defined; however, not more than 2.8 atm should be used with 100% oxygen in chamber.

Topic: Hyperbaric Oxygen

# Intravenous Anesthetics MCQ – Chapter 12

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 5: General Anesthesia (Chapter 12)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

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**Q1. Which intravenous anesthetic is the most commonly used agent for induction in present day anesthesia?**

- A. Thiopentone
- B. Ketamine
- C. Propofol
- D. Etomidate

**Ans: C**

**Q2. Thiopentone was the first intravenous anesthetic used in clinical practice. By whom and in which year?**

- A. Water and Lundy, 1934
- B. Stevens, 1962
- C. Domino and Corssen, 1965
- D. Carl Koller, 1884

**Ans: A**

**Q3. Why should thiopentone NOT be prepared with Ringer lactate?**

- A. It will get precipitated due to acidic pH of Ringer lactate
- B. It becomes inactivated
- C. It causes hemolysis
- D. It increases its potency dangerously

**Ans: A**

**Q4. Although the elimination half-life of thiopentone is 10.4 hours, consciousness is regained after 15–20 minutes because of:**

- A. Rapid metabolism
- B. Redistribution
- C. Rapid renal excretion
- D. Protein binding

**Ans: B**

**Q5. Thiopentone is utilized for cerebral protection in which type of ischemia?**

- A. Global ischemia seen in cardiac arrest
- B. Focal ischemia
- C. Both global and focal equally
- D. It offers no cerebral protection

**Ans: B**

**Q6. At subanesthetic doses, thiopentone acts as a/an:**

- A. Potent analgesic
- B. Antanalgesic
- C. Muscle relaxant
- D. Bronchodilator

**Ans: B**

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| <p><b>Q7. Repeated higher doses of thiopentone can cause prolonged apnea, which is therefore termed:</b></p> <p>A. Transient apnea<br/>B. Double apnea<br/>C. Central apnea<br/>D. Obstructive apnea</p>                                                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q8. The high alkalinity of thiopentone is responsible for which local complication after intra-arterial injection?</b></p> <p>A. Tissue necrosis only<br/>B. Vasospasm and thrombus formation leading to gangrene of the limb<br/>C. Hyperpigmentation<br/>D. Allergic urticaria</p>                                                | <p><b>Ans: B</b></p> |
| <p><b>Q9. Thiopentone is absolutely contraindicated in which condition?</b></p> <p>A. Acute intermittent and variegate porphyria<br/>B. Porphyria cutanea tarda<br/>C. Asthma<br/>D. Renal disease</p>                                                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q10. What is the dose of thiopentone?</b></p> <p>A. 2 mg/kg<br/>B. 5 mg/kg<br/>C. 0.1 mg/kg<br/>D. 10 mg/kg</p>                                                                                                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q11. Methohexitone differs from thiopentone in that it is epileptogenic, making it the agent of choice for:</b></p> <p>A. Electroconvulsive therapy (ECT)<br/>B. Status epilepticus<br/>C. Cerebral protection<br/>D. Day care surgery</p>                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q12. What is the chemical name of propofol?</b></p> <p>A. Sodium ethyl thiobarbiturate<br/>B. 2,6 di-isopropylphenol<br/>C. Phencyclidine derivative<br/>D. Imidazole derivative</p>                                                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q13. Why is it mandatory to discard the propofol vial within 6 hours of opening?</b></p> <p>A. Because it loses potency<br/>B. Because it contains egg lecithin and glycerol, which are a good media for bacterial growth, with high chances of contamination<br/>C. Because it precipitates<br/>D. Because it becomes alkaline</p> | <p><b>Ans: B</b></p> |

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| <p><b>Q14. Propofol decreases central serotonin release and therefore acts as a potent:</b></p> <ul style="list-style-type: none"> <li>A. Analgesic</li> <li>B. Muscle relaxant</li> <li>C. Antiemetic (even more effective than ondansetron)</li> <li>D. Anticonvulsant only</li> </ul>                                                                                                                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q15. Propofol is the intravenous agent of choice for which type of surgery, owing to its early smooth recovery and inactive metabolites?</b></p> <ul style="list-style-type: none"> <li>A. Cardiac surgery</li> <li>B. Aneurysm surgery</li> <li>C. Day care surgery</li> <li>D. Neurosurgery</li> </ul>                                                                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q16. Propofol is the agent of choice for induction in patients susceptible to:</b></p> <ul style="list-style-type: none"> <li>A. Malignant hyperthermia</li> <li>B. Porphyria</li> <li>C. Asthma</li> <li>D. Egg allergy</li> </ul>                                                                                                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q17. Propofol infusion syndrome occurs because of the failure of free fatty acid metabolism. It is associated with severe:</b></p> <ul style="list-style-type: none"> <li>A. Respiratory alkalosis</li> <li>B. Metabolic acidosis (lactic acidosis), cardiac failure and cardiomyopathy</li> <li>C. Hyperthermia</li> <li>D. Hypokalemia</li> </ul>                                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q18. Regarding egg allergy and propofol, which statement is correct?</b></p> <ul style="list-style-type: none"> <li>A. Egg allergy is an absolute contraindication</li> <li>B. History of egg allergy is not a contraindication (as egg allergy is to egg white/albumin, not lecithin); but avoid in patients with history of anaphylaxis to egg</li> <li>C. Propofol cannot be used in any allergic patient</li> <li>D. Soy allergy is an absolute contraindication</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q19. Which intravenous agent is the most cardiovascular stable among all IV agents?</b></p> <ul style="list-style-type: none"> <li>A. Thiopentone</li> <li>B. Propofol</li> <li>C. Ketamine</li> <li>D. Etomidate</li> </ul>                                                                                                                                                                                                                                                    | <p><b>Ans: D</b></p> |
| <p><b>Q20. The major side effect of etomidate (especially on long-term infusion) is:</b></p> <ul style="list-style-type: none"> <li>A. Adrenocortical suppression</li> <li>B. Bronchospasm</li> <li>C. Hypotension</li> <li>D. Hyperthermia</li> </ul>                                                                                                                                                                                                                                | <p><b>Ans: A</b></p> |

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| <p><b>Q21. Etomidate is the intravenous anesthetic of choice for which patients?</b></p> <ul style="list-style-type: none"> <li>A. Aneurysm surgery and patients with cardiac disease</li> <li>B. Asthma patients</li> <li>C. Day care surgery</li> <li>D. Patients with porphyria</li> </ul>              | <p><b>Ans: A</b></p> |
| <p><b>Q22. Benzodiazepines mediate their action through which receptor?</b></p> <ul style="list-style-type: none"> <li>A. NMDA receptor</li> <li>B. GABA-A receptor (increasing chloride conductance)</li> <li>C. Opioid mu receptor</li> <li>D. Alpha 2 adrenergic receptor</li> </ul>                    | <p><b>Ans: B</b></p> |
| <p><b>Q23. Which benzodiazepine is the only one used in the intraoperative period, being 3 times more potent than diazepam?</b></p> <ul style="list-style-type: none"> <li>A. Diazepam</li> <li>B. Lorazepam</li> <li>C. Midazolam</li> <li>D. Nitrazepam</li> </ul>                                       | <p><b>Ans: C</b></p> |
| <p><b>Q24. According to Table 12.1, what is the elimination half-life of midazolam?</b></p> <ul style="list-style-type: none"> <li>A. 2–3 hours</li> <li>B. 15 hours</li> <li>C. 30–60 hours</li> <li>D. 10 hours</li> </ul>                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q25. Flumazenil is a competitive antagonist at benzodiazepine receptors. Which effect is minimally reversed by it?</b></p> <ul style="list-style-type: none"> <li>A. Hypnosis</li> <li>B. Respiratory depression</li> <li>C. Sedation</li> <li>D. Amnesia</li> </ul>                                 | <p><b>Ans: D</b></p> |
| <p><b>Q26. Ketamine produces which type of anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Balanced anesthesia</li> <li>B. Dissociative anesthesia</li> <li>C. Neuroleptanesthesia</li> <li>D. Spinal anesthesia</li> </ul>                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q27. The main mechanism by which ketamine exerts its effect is by:</b></p> <ul style="list-style-type: none"> <li>A. Increasing GABA activity</li> <li>B. Inhibiting N-methyl-D-aspartate (NMDA) receptors</li> <li>C. Blocking sodium channels</li> <li>D. Stimulating alpha 2 receptors</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q28. What is the most common side effect of ketamine, occurring in 30–40% of patients?</b></p> <ul style="list-style-type: none"> <li>A. Bradycardia</li> <li>B. Hallucinations (mainly auditory)</li> <li>C. Hypotension</li> <li>D. Respiratory depression</li> </ul>                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q29. Ketamine stimulates the sympathetic system causing tachycardia and hypertension, making it the intravenous anesthetic of choice for:</b></p> <ul style="list-style-type: none"> <li>A. Hypertensive patients</li> <li>B. Shock</li> <li>C. Ischemic heart disease</li> <li>D. Pheochromocytoma</li> </ul>                                      | <p><b>Ans: B</b></p> |
| <p><b>Q30. Ketamine is a potent bronchodilator and can be used in the treatment of:</b></p> <ul style="list-style-type: none"> <li>A. Refractory status asthmaticus unresponsive to conventional therapy</li> <li>B. Pulmonary edema</li> <li>C. Pneumothorax</li> <li>D. COPD exacerbation only</li> </ul>                                               | <p><b>Ans: A</b></p> |
| <p><b>Q31. Why is the use of atropine or glycopyrrolate necessary with ketamine?</b></p> <ul style="list-style-type: none"> <li>A. To prevent hypotension</li> <li>B. Because tracheobronchial and salivary secretions are increased, producing laryngospasm</li> <li>C. To reduce hallucinations</li> <li>D. To increase intraocular pressure</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q32. What is the intravenous and intramuscular dose of ketamine?</b></p> <ul style="list-style-type: none"> <li>A. IV 2 mg/kg, IM 5–10 mg/kg</li> <li>B. IV 5 mg/kg, IM 2 mg/kg</li> <li>C. IV 0.2 mg/kg, IM 1 mg/kg</li> <li>D. IV 10 mg/kg, IM 20 mg/kg</li> </ul>                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q33. Ketamine is contraindicated in head injury because it:</b></p> <ul style="list-style-type: none"> <li>A. Increases intracranial tension (though it can be used in mechanically-ventilated patients)</li> <li>B. Causes severe hypotension</li> <li>C. Causes bradycardia</li> <li>D. Depresses respiration</li> </ul>                          | <p><b>Ans: A</b></p> |
| <p><b>Q34. Which are the most important receptors for the action of opioids?</b></p> <ul style="list-style-type: none"> <li>A. Kappa (<math>\kappa</math>) receptors</li> <li>B. Delta (<math>\delta</math>) receptors</li> <li>C. Mu (<math>\mu</math>) receptors</li> <li>D. Nociceptin receptors</li> </ul>                                            | <p><b>Ans: C</b></p> |

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| <p><b>Q35. The most important characteristic of agonist-antagonist opioids is:</b></p> <ul style="list-style-type: none"> <li>A. Ceiling effect on respiratory depression</li> <li>B. Highest analgesic potency</li> <li>C. Complete reversal of all opioid effects</li> <li>D. Absence of any analgesia</li> </ul>                           | <p><b>Ans: A</b></p> |
| <p><b>Q36. What is the most common cause of death in morphine poisoning?</b></p> <ul style="list-style-type: none"> <li>A. Cardiac arrest</li> <li>B. Respiratory depression</li> <li>C. Convulsions</li> <li>D. Hypotension</li> </ul>                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q37. Muscle rigidity (wooden chest or stiff chest syndrome) caused by opioids is most commonly seen with which opioid?</b></p> <ul style="list-style-type: none"> <li>A. Morphine</li> <li>B. Pethidine</li> <li>C. Alfentanil</li> <li>D. Codeine</li> </ul>                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q38. Tolerance is seen to all actions of opioids EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Constipation and miosis</li> <li>B. Analgesia</li> <li>C. Respiratory depression</li> <li>D. Sedation</li> </ul>                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q39. Pethidine (meperidine) is a chemical congener of atropine and therefore has which additional side effects?</b></p> <ul style="list-style-type: none"> <li>A. Blurred vision, dry mouth and tachycardia</li> <li>B. Bradycardia and miosis</li> <li>C. Hypertension and hyperthermia</li> <li>D. Diarrhea and salivation</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q40. Pethidine is the drug of choice for which condition, as it abolishes shivering?</b></p> <ul style="list-style-type: none"> <li>A. Postoperative shivering</li> <li>B. Postoperative nausea</li> <li>C. Pulmonary edema</li> <li>D. Biliary colic</li> </ul>                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q41. Pethidine is absolutely contraindicated in patients on which drugs?</b></p> <ul style="list-style-type: none"> <li>A. Monoamine oxidase inhibitors (MAO)</li> <li>B. Beta-blockers</li> <li>C. Calcium channel blockers</li> <li>D. Diuretics</li> </ul>                                                                           | <p><b>Ans: A</b></p> |

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| <p><b>Q42. Which opioid is the most common used opioid in anesthesia and is the most cardiac stable opioid?</b></p> <p>A. Morphine<br/>B. Fentanyl<br/>C. Pethidine<br/>D. Remifentanyl</p>                                                                                                                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q43. Which is the most potent human opioid (500 times more potent than morphine) available in clinical practice?</b></p> <p>A. Alfentanil<br/>B. Sufentanil<br/>C. Remifentanyl<br/>D. Fentanyl</p>                                                                                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q44. Remifentanyl is the opioid of choice for day care surgery because of its ultrashort action. It is metabolized rapidly by:</b></p> <p>A. Liver enzymes<br/>B. Esterases in red cells and tissues<br/>C. Renal excretion<br/>D. Lungs</p>                                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q45. Remifentanyl contains glycine, which makes it UNSUITABLE for:</b></p> <p>A. Day care surgery<br/>B. Postoperative analgesia and painless labor<br/>C. Cardiac surgery<br/>D. Intubation</p>                                                                                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q46. Which opioid has the highest addiction potential?</b></p> <p>A. Codeine<br/>B. Heroin<br/>C. Methadone<br/>D. Tramadol</p>                                                                                                                                                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q47. Naloxone is a pure opioid antagonist with a short duration of action (30–60 min), thereby increasing the chances of:</b></p> <p>A. Renarcotization with long acting agents like morphine<br/>B. Respiratory arrest<br/>C. Hypotension<br/>D. Seizures</p>                                                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q48. Peripheral opioid antagonists like methylnaltrexone and alvimopan are quaternary ammonium compounds not crossing the blood brain barrier. They are useful for:</b></p> <p>A. Reversing respiratory depression<br/>B. Reversing peripheral side effects (particularly constipation) without reversing analgesia<br/>C. Treating opioid addiction<br/>D. Reversing sedation</p> | <p><b>Ans: B</b></p> |

**Q49.** Dexmedetomidine is a new alpha 2 agonist. When used as an adjuvant, the MAC of inhalational agents is decreased and narcotic requirement is reduced by approximately:

- A. 10%
- B. 50%
- C. 25%
- D. 90%

**Ans: B**

**Q50.** What is the most common side effect of dexmedetomidine, seen in almost one-third of patients?

- A. Hypertension
- B. Hypotension
- C. Bradycardia
- D. Respiratory depression

**Ans: B**



# Answer Key & Explanations

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## Q1. Answer: C — Propofol

Page 106: Induction — the most commonly used intravenous anesthetic for induction is propofol. Indications for induction with thiopentone are very limited; ketamine is used only in specific situations.

Topic: Classification and Uses

## Q2. Answer: A — Water and Lundy, 1934

Page 106: Thiopentone was the first intravenous anesthetic used in clinical practice by Water and Lundy in 1934. It is an ultra short-acting barbiturate. Chemically it is sodium ethyl thiobarbiturate.

Topic: Thiopentone

## Q3. Answer: A — It will get precipitated due to acidic pH of Ringer lactate

Page 106: Thiopentone should not be prepared with Ringer lactate, otherwise it will get precipitated due to the acidic pH of Ringer lactate solution. The pH of sodium thiopentone (2.5%) solution is 10.4 (highly alkaline).

Topic: Thiopentone

## Q4. Answer: B — Redistribution

Page 106–107: The elimination half-life of thiopentone is 10.4 hours, but consciousness is regained after 15–20 minutes because of redistribution, which means the drug is redistributed from brain to tissues with less vascularity like muscle or fat.

Topic: Thiopentone

## Q5. Answer: B — Focal ischemia

Page 107: Thiopentone decreases cerebral oxygen consumption (CMRO<sub>2</sub>), cerebral metabolic rate (CMR) and intracranial tension; therefore, it can be utilized for cerebral protection in case of focal ischemia. It does not offer any protection in global ischemia seen in cardiac arrest.

Topic: Thiopentone

## Q6. Answer: B — Antanalgesic

Page 107: Antanalgesic action — at subanesthetic doses, thiopentone acts as antanalgesic. Because of the phenyl group, thiopentone is also a very potent anticonvulsant.

Topic: Thiopentone

## Q7. Answer: B — Double apnea

Page 107: Transient apnea is common after thiopentone. At higher dosage, thiopentone depresses respiration after brief recovery from transient apnea, therefore also termed as double apnea.

Topic: Thiopentone

## Q8. Answer: B — Vasospasm and thrombus formation leading to gangrene of the limb

Page 108: Because of its high alkalinity, in acidic pH of blood, thiopentone gets precipitated forming crystals, which can cause endothelial damage precipitating vasospasm and thrombus formation blocking microcirculation. Both these processes can lead to ischemia and gangrene of the limb.

Topic: Thiopentone

**Q9. Answer: A — Acute intermittent and variegate porphyria**

Page 108: Thiopentone induces the enzyme aminolevulinic acid synthetase, which stimulates the formation of porphyrin in susceptible individuals. Therefore, it is absolutely contraindicated in acute intermittent and variegate porphyria (but can be used safely in porphyria cutanea tarda).

Topic: Thiopentone

**Q10. Answer: B — 5 mg/kg**

Page 107: The dose of thiopentone is 5 mg/kg body weight. Males require more than females, obese more than thin, and young more than old.

Topic: Thiopentone

**Q11. Answer: A — Electroconvulsive therapy (ECT)**

Page 108: The most striking difference between methohexitone and thiopentone is that methohexitone is epileptogenic while thiopentone is anticonvulsant, making it the agent of choice for electroconvulsive therapy (ECT). Histamine release is less compared to thiopentone, therefore it is the preferred barbiturate for asthma patients.

Topic: Methohexitone

**Q12. Answer: B — 2,6 di-isopropylphenol**

Page 109: Propofol consists of a phenol ring with isopropyl group attached, its chemical name is 2,6 di-isopropylphenol. It is available as 1% and 2% milky white solution prepared in soyabean oil.

Topic: Propofol

**Q13. Answer: B — Because it contains egg lecithin and glycerol, which are a good media for bacterial growth, with high chances of contamination**

Page 109: Propofol contains egg lecithin and glycerol. As egg is a good media for bacterial growth, the chances of contamination of opened vial are very high. Although the recent formulations contain antimicrobial agents, they do not guarantee immunity against contamination; therefore, it is mandatory to discard the propofol vial within 6 hours.

Topic: Propofol

**Q14. Answer: C — Antiemetic (even more effective than ondansetron)**

Page 109: By decreasing the central serotonin release, propofol acts as a potent antiemetic agent (even more effective than ondansetron for postoperative nausea and vomiting).

Topic: Propofol

**Q15. Answer: C — Day care surgery**

Page 109: Because of its early and smooth recovery and inactive metabolites and antiemetic effects, propofol is the IV agent of choice for day care surgery. Along with opioids (remifentanyl), propofol is the agent of choice for total intravenous anesthesia (TIVA).

Topic: Propofol

**Q16. Answer: A — Malignant hyperthermia**

Page 109: Propofol is the agent of choice for induction in susceptible individuals for malignant hyperthermia. Propofol infusion can also be used to produce sedation in ICU.

Topic: Propofol

**Q17. Answer: B — Metabolic acidosis (lactic acidosis), cardiac failure and cardiomyopathy**

Page 110: Propofol infusion syndrome is rare but a lethal complication, usually seen if propofol infusion is continued for more than 48 hours, and more common in children. It occurs because of the failure of free fatty acid metabolism and is associated with severe metabolic acidosis (lactic acidosis), acute cardiac failure, cardiomyopathy, skeletal myopathy, hyperkalemia, lipemia and hepatomegaly.

Topic: Propofol

**Q18. Answer: B — History of egg allergy is not a contraindication (as egg allergy is to egg white/albumin, not lecithin); but avoid in patients with history of anaphylaxis to egg**

Page 110: Patients with egg allergy — since egg allergy is almost always from egg white (albumin) and not from lecithin (which is prepared from yoke), history of egg allergy is not a contraindication for propofol. However, it is advisable to not use propofol in patients with history of anaphylaxis to egg. Patients with soy allergy can also safely receive propofol.

Topic: Propofol

**Q19. Answer: D — Etomidate**

Page 110: Etomidate is most cardiovascular stable among all IV agents. It has minimal respiratory depression, no histamine release, and decreases ICT by almost 50% in patients with elevated ICT.

Topic: Etomidate

**Q20. Answer: A — Adrenocortical suppression**

Page 110: The major side effect of etomidate is adrenocortical suppression on long-term infusion. Single use only causes temporary adrenocortical suppression, which recovers with vitamin C supplementation. Nausea and vomiting incidence (40%) is highest among all intravenous anesthetics.

Topic: Etomidate

**Q21. Answer: A — Aneurysm surgery and patients with cardiac disease**

Page 110: Etomidate is the intravenous anesthetic of choice for aneurysm surgery and patients with cardiac disease. It is only porphyrinogenic in rats, not in humans, and therefore can be safely used in porphyria in human beings.

Topic: Etomidate

**Q22. Answer: B — GABA-A receptor (increasing chloride conductance)**

Page 111: Benzodiazepines mediate their action through the GABA receptor (GABA-A subtype) increasing the membrane conductance to chloride ion, causing hyperpolarization of membrane (GABA facilitatory).

Topic: Benzodiazepines

**Q23. Answer: C — Midazolam**

Page 111: Midazolam is 3 times more potent than diazepam. In current day practice, midazolam is the only BZ used in the intraoperative period. Diazepam and lorazepam are hardly used intraoperatively because they are oil based (injection is painful), sleep is not smooth, and elimination half-lives are prolonged.

Topic: Benzodiazepines

**Q24. Answer: A — 2–3 hours**

Page 111, Table 12.1: Elimination half-life — diazepam 30–60 hours, midazolam 2–3 hours, lorazepam 15 hours. Midazolam is water-based (painless injection), while diazepam and lorazepam are oil-based (painful injection).

Topic: Benzodiazepines

**Q25. Answer: D — Amnesia**

Page 111–112: Flumazenil antagonizes hypnosis, respiratory depression and sedation, but amnesia is minimally reversed (particularly at low doses of flumazenil). The disadvantage of flumazenil is its short half-life (1–2 hours), therefore there are chances of re-sedation with long-acting agents like diazepam and lorazepam.

Topic: Benzodiazepines

**Q26. Answer: B — Dissociative anesthesia**

Page 112: Ketamine produces dissociative anesthesia, a state which dissociates the individual from surroundings and himself, i.e. the individual is in a cataleptic state. It is a phencyclidine derivative, synthesized by Stevens in 1962 and first used in humans by Domino and Corssen in 1965.

Topic: Ketamine

**Q27. Answer: B — Inhibiting N-methyl-D-aspartate (NMDA) receptors**

Page 112: The main mechanism by which ketamine exerts its effect is by inhibiting N-methyl-D-aspartate (NMDA) receptors. Interaction through opioid ( $\mu$ ) receptors has also been postulated. Its primary site of action is the thalamo-neocortical projection.

Topic: Ketamine

**Q28. Answer: B — Hallucinations (mainly auditory)**

Page 112: Ketamine produces both auditory and visual hallucinations (mainly auditory). The incidence is 30–40%. Hallucination is the most common side effect of ketamine. Hallucinations and emergence reactions can be decreased by giving benzodiazepines along with ketamine.

Topic: Ketamine

**Q29. Answer: B — Shock**

Page 112: Ketamine stimulates the sympathetic system causing tachycardia and hypertension, therefore it is the intravenous anesthetic of choice for shock. However, in debilitated patients in whom catecholamines have been depleted, ketamine can cause direct myocardial depression.

Topic: Ketamine

**Q30. Answer: A — Refractory status asthmaticus unresponsive to conventional therapy**

Page 113: Ketamine is such a potent bronchodilator that it can be used in the treatment of refractory status asthmaticus unresponsive to conventional therapy. Pharyngeal and laryngeal reflexes are preserved (but silent aspiration can still occur).

Topic: Ketamine

**Q31. Answer: B — Because tracheobronchial and salivary secretions are increased, producing laryngospasm**

Page 113: Tracheobronchial and salivary secretions are increased with ketamine, producing laryngospasm. Therefore, the use of atropine/glycopyrrolate is necessary with ketamine.

Topic: Ketamine

**Q32. Answer: A — IV 2 mg/kg, IM 5–10 mg/kg**

Page 113: Ketamine can be given IV, IM, oral, nasal, per rectal and intrathecally. The dose is: Intravenous 2 mg/kg, Intramuscular 5–10 mg/kg.

Topic: Ketamine

**Q33. Answer: A — Increases intracranial tension (though it can be used in mechanically-ventilated patients)**

Page 113: Ketamine is contraindicated in head injury as it increases intracranial tension (it can be used in mechanically-ventilated patients). It is also contraindicated in ocular surgeries and glaucoma (increases IOP), ischemic heart disease (increases myocardial oxygen demand), vascular aneurysm (hypertension), and psychiatric diseases.

Topic: Ketamine

**Q34. Answer: C — Mu ( $\mu$ ) receptors**

Page 114:  $\mu$  (mu) receptors are the most important receptors for the action of opioids. They are further divided into  $\mu_1$  and  $\mu_2$ .  $\mu_1$  mediates analgesia (mainly supraspinal), sedation, bradycardia, miosis, urinary retention, and muscle rigidity;  $\mu_2$  mediates respiratory depression, constipation, and physical dependence.

Topic: Opioids

**Q35. Answer: A — Ceiling effect on respiratory depression**

Page 115: As agonist-antagonists exhibit agonistic action only up to a certain dose, the analgesia and respiratory depression achieve a plateau after this dose. This effect is called the ceiling effect, which is the most important characteristic of agonist-antagonists.

Topic: Opioids

**Q36. Answer: B — Respiratory depression**

Page 115: Respiratory depression is the most common cause of death in morphine poisoning. Children and old patients are more prone for respiratory depression. Opioids inhibit the ventilatory response to hypoxia and hypercarbia.

Topic: Opioids

**Q37. Answer: C — Alfentanil**

Page 116: Opioids can cause muscle rigidity which sometimes becomes so severe in thoracic muscles that it causes hypoxia. This is called wooden chest syndrome or stiff chest syndrome. Muscle rigidity is most commonly seen with alfentanil. Treatment includes naloxone, benzodiazepines, muscle relaxants, and elective ventilation.

Topic: Opioids

**Q38. Answer: A — Constipation and miosis**

Page 116: Tolerance has been seen to all actions of opioids except constipation and miosis. Tolerance is mainly pharmacodynamic. The chronic use of opioids not only produces tolerance to pain but can produce hyperalgesia.

Topic: Opioids

**Q39. Answer: A — Blurred vision, dry mouth and tachycardia**

Page 116: Pethidine is an atropine congener, therefore it has additional side effects like atropine, i.e. blurred vision, dry mouth and tachycardia. It is one tenth as potent as morphine.

Topic: Opioids

**Q40. Answer: A — Postoperative shivering**

Page 116: Pethidine abolishes shivering, therefore it is the drug of choice for shivering. It is also a directly acting myocardial depressant and has local analgesic properties.

Topic: Opioids

**Q41. Answer: A — Monoamine oxidase inhibitors (MAO)**

Page 116: Pethidine is absolutely contraindicated in patients on monoamine oxidase inhibitors (MAO) therapy. A patient on MAO inhibitors undergoes abnormal metabolism of pethidine to produce Nor-Pethidine, which can cause restlessness, hypertension, convulsion, coma or even death.

Topic: Opioids

**Q42. Answer: B — Fentanyl**

Page 117: Fentanyl is the most common used opioid in anesthesia. It is 100 times more potent than morphine. Due to high lipid solubility it has rapid onset (2 to 5 min) and rapid recovery (1–2 hours). It is the most cardiac stable opioid.

Topic: Opioids

**Q43. Answer: B — Sufentanil**

Page 117: Sufentanil is the most potent human opioid (500 times of morphine) available in clinical practice. As sufentanil maximally inhibits airway reflexes, it is the agent of choice for inhibiting stress response to laryngoscopy and intubation.

Topic: Opioids

**Q44. Answer: B — Esterases in red cells and tissues**

Page 117: Remifentanyl is an ultrashort acting opioid metabolized rapidly by esterases in red cells and tissues. Onset is fastest (1.1 minutes) and recovery is also most rapid among opioids (5–10 minutes), making remifentanyl the opioid of choice for day care surgery.

Topic: Opioids

**Q45. Answer: B — Postoperative analgesia and painless labor**

Page 117: Remifentanyl contains glycine which can cause motor weakness, making it unsuitable for postoperative analgesia and painless labor. It also causes significant hypotension.

Topic: Opioids

**Q46. Answer: B — Heroin**

Page 117: Among other opioid agonists, codeine is used for suppressing cough, heroin has got the highest addiction potential, and methadone is mainly used in prevention of opioid withdrawal symptoms.

Topic: Opioids

**Q47. Answer: A — Renarcotization with long acting agents like morphine**

Page 118: Naloxone is a pure opioid antagonist acting on all opioid receptors with maximum propensity for mu receptors. The duration of action is 30–60 min, thereby increasing the chances of renarcotization with long acting agents like morphine. Naloxone can be given intra-tracheally.

Topic: Opioid Antagonists

**Q48. Answer: B — Reversing peripheral side effects (particularly constipation) without reversing analgesia**

Page 118: Peripheral antagonists — methylnaltrexone and alvimopan are quaternary ammonium compounds not crossing the blood brain barrier, therefore they hold a very promising role in reversing the peripheral side effects (particularly constipation) without reversing the analgesia.

Topic: Opioid Antagonists

**Q49. Answer: B — 50%**

Page 118–119: Dexmedetomidine is a new  $\alpha_2$  agonist which is more selective with lesser side effects compared to clonidine. As an adjuvant, the MAC of inhalational agents is decreased and narcotic requirement has been found to be reduced by 50% in patients who receive dexmedetomidine infusion.

*Topic: Alpha 2 Adrenergic Agonists*

**Q50. Answer: B — Hypotension**

Page 119: Side effects of dexmedetomidine include initial hypertension; hypotension is the most common side effect, seen in almost one-third of the patients. The effect of dexmedetomidine can be reversed with atipamezole.

*Topic: Alpha 2 Adrenergic Agonists*

# Muscle Relaxants MCQ – Chapter 13

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 5: General Anesthesia (Chapter 13)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

**Q1. Acetylcholine is synthesized in the nerve cytoplasm from choline and acetyl coenzyme A by which enzyme?**

- A. Acetylcholinesterase
- B. Choline acetyltransferase
- C. Pseudocholinesterase
- D. Monoamine oxidase

**Ans: B**

**Q2. Acetylcholine is destroyed in less than 1 ms at the neuromuscular junction by which enzyme?**

- A. Pseudocholinesterase (butyrylcholinesterase)
- B. Choline acetyltransferase
- C. Acetylcholinesterase (true cholinesterase)
- D. Catechol-O-methyltransferase

**Ans: C**

**Q3. Which subunit of the acetylcholine receptor is the important site where ACh, succinylcholine and competitive antagonists bind?**

- A. Beta ( $\beta$ ) subunit
- B. Alpha ( $\alpha$ ) subunit
- C. Gamma ( $\gamma$ ) subunit
- D. Delta ( $\delta$ ) subunit

**Ans: B**

**Q4. Which muscle is considered the most resistant to muscle relaxants?**

- A. Adductor pollicis
- B. Diaphragm
- C. Orbicularis oculi
- D. Laryngeal muscles

**Ans: B**

**Q5. In clinical practice, central muscles are blocked earlier than peripheral muscles (limb muscles). This is because:**

- A. Central muscles are smaller
- B. Of early and more blood supply delivering more relaxant to central muscles
- C. Central muscles have fewer receptors
- D. Peripheral muscles are more sensitive

**Ans: B**

**Q6. Which muscle is most commonly monitored for neuromuscular blockade?**

- A. Diaphragm
- B. Adductor pollicis (supplied by ulnar nerve)
- C. Orbicularis oculi
- D. Masseter

**Ans: B**

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. Succinylcholine is metabolized by which enzyme?</b></p> <ul style="list-style-type: none"> <li>A. Acetylcholinesterase</li> <li>B. Plasma pseudocholinesterase (butyrylcholinesterase)</li> <li>C. Choline acetyltransferase</li> <li>D. Hoffman degradation</li> </ul>                                                                                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q8. The excessive repeated depolarization and contractions produced by succinylcholine can be directly visualized as:</b></p> <ul style="list-style-type: none"> <li>A. Fasciculations</li> <li>B. Tetany</li> <li>C. Tremor</li> <li>D. Fading</li> </ul>                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q9. According to Table 13.1, which feature distinguishes a depolarizing block from a nondepolarizing block?</b></p> <ul style="list-style-type: none"> <li>A. It shows fading on neuromuscular monitoring</li> <li>B. It is preceded by muscle fasciculations</li> <li>C. It exhibits post-tetanic facilitation</li> <li>D. It is reversed by neostigmine</li> </ul>                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q10. What is the most prominent effect of suxamethonium?</b></p> <ul style="list-style-type: none"> <li>A. Bradycardia</li> <li>B. Hyperkalemia</li> <li>C. Increased intraocular tension</li> <li>D. Increased intragastric pressure</li> </ul>                                                                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q11. Postoperative myalgia (muscle pains) from succinylcholine can be reduced by precurarization, which involves:</b></p> <ul style="list-style-type: none"> <li>A. Giving one-tenth dose of nondepolarizing relaxant 3 minutes prior to succinylcholine</li> <li>B. Giving a double dose of succinylcholine</li> <li>C. Giving neostigmine before succinylcholine</li> <li>D. Hyperventilating the patient</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q12. Succinylcholine is the most commonly implicated drug for which life-threatening condition?</b></p> <ul style="list-style-type: none"> <li>A. Anaphylaxis</li> <li>B. Malignant hyperthermia</li> <li>C. Masseter spasm</li> <li>D. Pulmonary edema</li> </ul>                                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q13. Which serum potassium level is an absolute contraindication for using suxamethonium?</b></p> <ul style="list-style-type: none"> <li>A. Serum K+ &gt; 5.5 mEq/L</li> <li>B. Serum K+ &gt; 3.5 mEq/L</li> <li>C. Serum K+ &lt; 3.0 mEq/L</li> <li>D. Serum K+ &gt; 4.0 mEq/L</li> </ul>                                                                                                                             | <p><b>Ans: A</b></p> |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q14. Prolonged apnea after succinylcholine can be due to all of the following EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Low pseudocholinesterase</li> <li>B. Atypical pseudocholinesterase</li> <li>C. Phase II block</li> <li>D. Hyperkalemia</li> </ul>                                                                                                                 | <p><b>Ans: D</b></p> |
| <p><b>Q15. Atypical pseudocholinesterase is a genetic disorder diagnosed by which test?</b></p> <ul style="list-style-type: none"> <li>A. Fluoride number</li> <li>B. Dibucaine number</li> <li>C. Train of four ratio</li> <li>D. Tensilon test</li> </ul>                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q16. Phase II block (dual block) is produced by an excessive dose of suxamethonium (&gt; 5 mg/kg or &gt; 500 mg). Which finding is pathognomonic of phase II block?</b></p> <ul style="list-style-type: none"> <li>A. Fasciculations</li> <li>B. Succinylcholine showing fading on neuromuscular monitoring</li> <li>C. Post-tetanic depression</li> <li>D. Hyperkalemia</li> </ul>       | <p><b>Ans: B</b></p> |
| <p><b>Q17. The onset of action of a nondepolarizer is inversely proportional to its:</b></p> <ul style="list-style-type: none"> <li>A. Potency</li> <li>B. Molecular weight</li> <li>C. Duration</li> <li>D. Lipid solubility</li> </ul>                                                                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q18. The basic difference between steroidal and benzyliisoquinoline (BZIQ) compounds is that:</b></p> <ul style="list-style-type: none"> <li>A. Histamine release is seen with BZIQ compounds and vagolytic property with steroidal compounds</li> <li>B. Steroidal compounds release histamine</li> <li>C. BZIQ compounds are vagolytic</li> <li>D. Both groups are identical</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q19. The main reason for not preferring pancuronium in current day practice is its:</b></p> <ul style="list-style-type: none"> <li>A. Cardiac instability (tachycardia and hypertension)</li> <li>B. Histamine release</li> <li>C. Short duration</li> <li>D. Hepatic metabolism</li> </ul>                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q20. Which nondepolarizing muscle relaxant is the most cardiovascular stable, making it the muscle relaxant of choice for cardiac patients?</b></p> <ul style="list-style-type: none"> <li>A. Pancuronium</li> <li>B. Vecuronium</li> <li>C. Atracurium</li> <li>D. Gallamine</li> </ul>                                                                                                  | <p><b>Ans: B</b></p> |

|                                                                                                                                                                                                                                                                                 |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q21. Which nondepolarizing muscle relaxant is the agent of choice for intubation and precurarization due to its early onset (60–90 seconds)?</b></p> <p>A. Vecuronium<br/>B. Pancuronium<br/>C. Rocuronium<br/>D. Atracurium</p>                                          | <p><b>Ans: C</b></p> |
| <p><b>Q22. Rapacuronium was withdrawn from the market because it produced:</b></p> <p>A. Intense bronchospasm in a significant number of patients (&gt; 9%)<br/>B. Severe hyperkalemia<br/>C. Malignant hyperthermia<br/>D. Prolonged apnea</p>                                 | <p><b>Ans: A</b></p> |
| <p><b>Q23. D-tubocurare is no more used in current practice because of its:</b></p> <p>A. Highest propensity for histamine release and ganglion blockade (and hence severe hypotension)<br/>B. Cardiac instability<br/>C. Renal toxicity<br/>D. Short duration of action</p>    | <p><b>Ans: A</b></p> |
| <p><b>Q24. Atracurium undergoes a unique method of degradation which is pH and temperature dependent. What is this called?</b></p> <p>A. Ester hydrolysis<br/>B. Hoffman degradation<br/>C. Pseudocholinesterase metabolism<br/>D. Renal elimination</p>                        | <p><b>Ans: B</b></p> |
| <p><b>Q25. Because its metabolism is not dependent on hepatic or renal functions, atracurium is the relaxant of choice for:</b></p> <p>A. Cardiac patients only<br/>B. Hepatic failure, renal failure, newborn and old age<br/>C. Day care surgery<br/>D. Intubation</p>        | <p><b>Ans: B</b></p> |
| <p><b>Q26. Atracurium's metabolic product laudanosine can cross the blood brain barrier and produce:</b></p> <p>A. Convulsions<br/>B. Bronchospasm<br/>C. Hypotension<br/>D. Bradycardia</p>                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q27. The chief advantage of cisatracurium over atracurium is that:</b></p> <p>A. It does not release histamine and laudanosine production is 5 times less<br/>B. It has a faster onset<br/>C. It is eliminated by kidneys<br/>D. It is more potent and cardiac stable</p> | <p><b>Ans: A</b></p> |

|                                                                                                                                                                                                                                                                                                |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q28. Which nondepolarizing muscle relaxant is metabolized by pseudocholinesterase and is the muscle relaxant of choice for day care surgery?</b></p> <p>A. Mivacurium<br/>B. Doxacurium<br/>C. Pancuronium<br/>D. Pipecuronium</p>                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q29. Gallamine is the only nondepolarizer which can cross the placenta, therefore it is absolutely contraindicated in:</b></p> <p>A. Renal disease<br/>B. Pregnancy<br/>C. Hepatic disease<br/>D. Cardiac disease</p>                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q30. Gantacurium, an onium chlorofumarate under research, is metabolized by forming a cysteine adduct followed by ester hydrolysis. It may emerge as an alternative to which drug?</b></p> <p>A. Atracurium<br/>B. Suxamethonium for intubation<br/>C. Vecuronium<br/>D. Pancuronium</p> | <p><b>Ans: B</b></p> |
| <p><b>Q31. When two muscle relaxants of the SAME group (steroid + steroid or BZIQ + BZIQ) are added, the effect produced is:</b></p> <p>A. Additive<br/>B. Synergistic<br/>C. Antagonistic<br/>D. No effect</p>                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q32. Inhalational agents prolong the block of muscle relaxants and decrease the requirement of relaxant by:</b></p> <p>A. 20–30%<br/>B. 50–60%<br/>C. 5–10%<br/>D. 70–80%</p>                                                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q33. Which class of antibiotics prolongs the effect of both depolarizers and NDMR, and is reversed by 4-aminopyridine?</b></p> <p>A. Penicillins<br/>B. Aminoglycosides<br/>C. Cephalosporins<br/>D. Macrolides</p>                                                                      | <p><b>Ans: B</b></p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q34. According to Table 13.2, in conditions with extrajunctional receptors (burns, hemiplegia, paraplegia, denervation), the response to depolarizers and nondepolarizers is:</b></p> <ul style="list-style-type: none"> <li>A. Increased sensitivity/hyperkalemia to depolarizers and resistance to nondepolarizers</li> <li>B. Resistance to depolarizers and increased sensitivity to nondepolarizers</li> <li>C. Increased sensitivity to both</li> <li>D. Resistance to both</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q35. In myasthenia gravis, the response to muscle relaxants is:</b></p> <ul style="list-style-type: none"> <li>A. Increased sensitivity to depolarizers and resistance to nondepolarizers</li> <li>B. Resistance to depolarizers and increased sensitivity to nondepolarizers</li> <li>C. Increased sensitivity to both</li> <li>D. Resistance to both</li> </ul>                                                                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q36. According to the rules of reversal, a reversal agent should be given only after:</b></p> <ul style="list-style-type: none"> <li>A. Immediately after the last dose of relaxant</li> <li>B. Some evidence of spontaneous recovery has appeared (e.g. spontaneous respiration, TO4 shows at least 1 twitch)</li> <li>C. The patient is fully awake</li> <li>D. 30 minutes regardless of recovery</li> </ul>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q37. Which anticholinesterase, being NOT a quaternary ammonium compound, can cross the blood brain barrier?</b></p> <ul style="list-style-type: none"> <li>A. Neostigmine</li> <li>B. Pyridostigmine</li> <li>C. Edrophonium</li> <li>D. Physostigmine</li> </ul>                                                                                                                                                                                                                            | <p><b>Ans: D</b></p> |
| <p><b>Q38. Why is glycopyrrolate preferred over atropine when given with cholinesterase inhibitors during reversal?</b></p> <ul style="list-style-type: none"> <li>A. Because glycopyrrolate is devoid of central side effects</li> <li>B. Because glycopyrrolate is cheaper</li> <li>C. Because glycopyrrolate has a faster onset</li> <li>D. Because glycopyrrolate increases heart rate more</li> </ul>                                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q39. Which is the most commonly used reversal agent, and what is its dose?</b></p> <ul style="list-style-type: none"> <li>A. Neostigmine, 0.04–0.08 mg/kg</li> <li>B. Edrophonium, 0.5 mg/kg</li> <li>C. Pyridostigmine, 0.2 mg/kg</li> <li>D. Physostigmine, 0.1 mg/kg</li> </ul>                                                                                                                                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q40. Which is the reversal agent of choice for mivacurium?</b></p> <ul style="list-style-type: none"> <li>A. Neostigmine</li> <li>B. Pyridostigmine</li> <li>C. Edrophonium</li> <li>D. Sugammadex</li> </ul>                                                                                                                                                                                                                                                                                | <p><b>Ans: C</b></p> |

|                                                                                                                                                                                                                                                                                                                                                                                    |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q41. Sugammadex (gamma cyclodextrin) directly binds which type of muscle relaxants to form a complex eliminated unchanged through the kidney?</b></p> <p>A. Benzylisoquinoline type<br/>B. Steroidal type of nondepolarizing muscle relaxants<br/>C. Depolarizing type<br/>D. Centrally acting relaxants</p>                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q42. A major advantage of sugammadex over anticholinesterases is that:</b></p> <p>A. It is effective in reversing profound block, while anticholinesterases cannot reverse deeper blocks<br/>B. It is cheaper<br/>C. It can reverse benzylisoquinoline relaxants<br/>D. It can be used in severe renal insufficiency</p>                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q43. Which clinical sign is considered the BEST clinical sign of adequate reversal, though difficult to perform, making head lift the most widely used test?</b></p> <p>A. Spontaneous eye opening<br/>B. Sustained jaw clench over a tongue blade<br/>C. Recovery of cough reflex<br/>D. Spontaneous limb movements</p>                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q44. A train of four (TO4) ratio of what value indicates adequate recovery, with a ratio above 0.9 guaranteeing recovery?</b></p> <p>A. &gt; 0.7 (70%)<br/>B. &gt; 0.3 (30%)<br/>C. &gt; 0.5 (50%)<br/>D. &gt; 0.1 (10%)</p>                                                                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q45. Respiratory acidosis and metabolic alkalosis can double the dose of neostigmine required, and it becomes almost impossible to reverse a patient with pCO<sub>2</sub> more than:</b></p> <p>A. 50 mm Hg<br/>B. 40 mm Hg<br/>C. 60 mm Hg<br/>D. 30 mm Hg</p>                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q46. Centrally acting muscle relaxants produce muscle relaxation by which mechanism?</b></p> <p>A. By acting at the neuromuscular junction<br/>B. Through a central mechanism (both supraspinal and spinal level), inhibiting polysynaptic reflexes; they have no action on the NMJ<br/>C. By blocking acetylcholine receptors<br/>D. By inhibiting pseudocholinesterase</p> | <p><b>Ans: B</b></p> |

**Q47. For the treatment of tetanus, which centrally acting muscle relaxant is most effective?**

- A. Intravenous diazepam
- B. Baclofen
- C. Tizanidine
- D. Chlorzoxazone

**Ans: A**

**Q48. Suxamethonium increases which pressures in the body?**

- A. Only intraocular pressure
- B. Intraocular, intragastric and intracranial pressures
- C. Only intracranial pressure
- D. Only intragastric pressure

**Ans: B**

**Q49. Which nondepolarizing muscle relaxant has the longest duration of action?**

- A. Vecuronium
- B. Mivacurium
- C. Doxacurium
- D. Rocuronium

**Ans: C**

**Q50. Which nondepolarizing muscle relaxants are SAFE in renal failure?**

- A. Gallamine, metocurine, doxacurium, pancuronium
- B. Atracurium, cisatracurium, rocuronium, mivacurium
- C. Vecuronium and pancuronium only
- D. Tubocurare and gallamine

**Ans: B**



# Answer Key & Explanations

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## **Q1. Answer: B — Choline acetyltransferase**

Page 122: Acetylcholine is synthesized in the nerve cytoplasm by combination of choline and acetyl coenzyme A. This reaction is mediated by the enzyme choline acetyltransferase. The synthesized ACh is stored in vesicles.

*Topic: Physiology of NMJ*

## **Q2. Answer: C — Acetylcholinesterase (true cholinesterase)**

Page 123: The acetylcholine is destroyed in less than 1 ms by acetylcholinesterase (true cholinesterase) which is synthesized by the underlying muscle.

*Topic: Physiology of NMJ*

## **Q3. Answer: B — Alpha ( $\alpha$ ) subunit**

Page 123: The alpha ( $\alpha$ ) subunit is important because ACh, succinylcholine and competitive antagonists (nondepolarizing relaxants) bind to this site. The receptor has two gates — an upper voltage-dependent and a lower time-dependent gate.

*Topic: Acetylcholine Receptor*

## **Q4. Answer: B — Diaphragm**

Page 123: Laryngeal muscles and diaphragm may require more than double the doses required to block adductor pollicis. In fact, the diaphragm is considered as the most resistant muscle.

*Topic: Sequence of Muscle Blockade*

## **Q5. Answer: B — Of early and more blood supply delivering more relaxant to central muscles**

Page 123: Central muscles are blocked earlier than peripheral muscles simply because of early and more blood supply delivering more quantity of muscle relaxants to central muscles as compared to peripheral muscles. The sequence of recovery is also the same — central muscles recover first and limb muscles last.

*Topic: Sequence of Muscle Blockade*

## **Q6. Answer: B — Adductor pollicis (supplied by ulnar nerve)**

Page 123: The most commonly chosen muscle is adductor pollicis, supplied by the ulnar nerve. Relaxation of adductor pollicis means laryngeal muscles have already been blocked and intubation can be performed.

*Topic: Neuromuscular Monitoring*

## **Q7. Answer: B — Plasma pseudocholinesterase (butyrylcholinesterase)**

Page 124: Succinylcholine is immediately metabolized in plasma by pseudocholinesterase (butyrylcholinesterase), which is synthesized in liver and present in plasma in abundant quantities. To prevent its metabolism, it should be given at a faster rate.

*Topic: Suxamethonium*

## **Q8. Answer: A — Fasciculations**

Page 124: Excessive succinylcholine produces repeated depolarization and contractions which can be directly visualized as fasciculations. Persistent depolarization makes the membrane refractory, producing what is termed phase I block.

*Topic: Suxamethonium*

**Q9. Answer: B — It is preceded by muscle fasciculations**

Page 124, Table 13.1: A depolarizing block (phase I block) is preceded by muscle fasciculations, shows no fading, no post-tetanic facilitation, and is not reversed by cholinesterase inhibitors. A nondepolarizing block shows no fasciculations, fading, post-tetanic facilitation, and is reversed by neostigmine.

Topic: Suxamethonium

**Q10. Answer: B — Hyperkalemia**

Page 125: Hyperkalemia is the most prominent effect of suxamethonium, which occurs due to excessive muscle fasciculations. In normal circumstances serum potassium increases by 0.5 mEq/L. Hyperkalemia increases the risk of ventricular arrhythmias.

Topic: Suxamethonium

**Q11. Answer: A — Giving one-tenth dose of nondepolarizing relaxant 3 minutes prior to succinylcholine**

Page 125: Precurarization is the technique in which one-tenth dose of nondepolarizing muscle relaxant is given 3 minutes prior to succinylcholine. This small dose decreases fasciculations and hence myalgia. Incidence of muscle pains is 40–50%.

Topic: Suxamethonium

**Q12. Answer: B — Malignant hyperthermia**

Page 125: Malignant hyperthermia — succinylcholine is the most commonly implicated drug. Masseter spasm with succinylcholine, especially in children, may be a precursor sign for malignant hyperthermia.

Topic: Suxamethonium

**Q13. Answer: A — Serum K<sup>+</sup> > 5.5 mEq/L**

Page 125: Hyperkalemia — serum K<sup>+</sup> > 5.5 mEq/L is an absolute contraindication for using suxamethonium, because suxamethonium further increases serum potassium.

Topic: Suxamethonium

**Q14. Answer: D — Hyperkalemia**

Page 126: Prolonged apnea after succinylcholine can be because of low pseudocholinesterase, atypical pseudocholinesterase, or phase II block.

Topic: Suxamethonium

**Q15. Answer: B — Dibucaine number**

Page 126: Atypical (abnormal) pseudocholinesterase is a genetic disease in which the patient has abnormal enzyme not able to metabolize suxamethonium. Incidence is 1 in 3,000. This is diagnosed by the dibucaine number. Dibucaine inhibits 80% of normal enzyme and 20% of abnormal enzyme.

Topic: Suxamethonium

**Q16. Answer: B — Succinylcholine showing fading on neuromuscular monitoring**

Page 127: Phase II block (also called dual block) is prolonged block seen after excessive dose of suxamethonium (> 5 mg/kg or > 500 mg). Succinylcholine showing fading on neuromuscular monitoring is pathognomonic of phase II block.

Topic: Suxamethonium

**Q17. Answer: A — Potency**

Page 127–128: Onset of a nondepolarizer is inversely proportional to potency. More the potency, lesser the dose required, means less molecules delivered at the neuromuscular junction to compete with acetylcholine, delaying the onset of action.

Topic: Nondepolarizing Muscle Relaxants

**Q18. Answer: A — Histamine release is seen with BZIQ compounds and vagolytic property with steroidal compounds**

Page 127: The basic difference between steroidal and benzyloquinoline (BZIQ) compounds is that histamine release is seen with BZIQ compounds and vagolytic property with steroidal compounds.

*Topic: Nondepolarizing Muscle Relaxants*

**Q19. Answer: A — Cardiac instability (tachycardia and hypertension)**

Page 128: The main reason for not preferring pancuronium is its cardiac instability. It releases nor-adrenaline and causes vagal blockade, leading to tachycardia and hypertension. By releasing nor-adrenaline it increases the chances of arrhythmias with halothane.

*Topic: Steroidal Compounds*

**Q20. Answer: B — Vecuronium**

Page 128: As vecuronium is the most cardiovascular stable, it is the muscle relaxant of choice for cardiac patients. Long-term use in ICU has resulted in polyneuropathy due to accumulation of its metabolite, 3-hydroxy metabolite.

*Topic: Steroidal Compounds*

**Q21. Answer: C — Rocuronium**

Page 128: The advantage of rocuronium over other nondepolarizing muscle relaxants is its early onset (within 60–90 seconds compared to 3–4 minutes for other relaxants). As its onset is comparable to succinylcholine, it is the nondepolarizer of choice for intubation and precurarization. It is also the only nondepolarizing muscle relaxant which can be given by intramuscular route.

*Topic: Steroidal Compounds*

**Q22. Answer: A — Intense bronchospasm in a significant number of patients (> 9%)**

Page 128: Rapacuronium is the analog of vecuronium with less potency. It has been withdrawn from the market because it produced intense bronchospasm in a significant number of patients (> 9%).

*Topic: Steroidal Compounds*

**Q23. Answer: A — Highest propensity for histamine release and ganglion blockade (and hence severe hypotension)**

Page 128: D-tubocurarine is not used in present day practice because of its highest propensity for histamine release and ganglion blockade (and hence severe hypotension). It was named so because it was carried in bamboo tubes and used as arrow poison by Amazon people.

*Topic: Benzyloquinoline Compounds*

**Q24. Answer: B — Hoffman degradation**

Page 129: The majority of atracurium undergoes spontaneous degradation in plasma called Hoffman degradation. Hoffman degradation is a pH and temperature-dependent reaction; higher pH and temperature favor more degradation.

*Topic: Benzyloquinoline Compounds*

**Q25. Answer: B — Hepatic failure, renal failure, newborn and old age**

Page 129: As atracurium metabolism is not dependent on hepatic or renal functions, it is the relaxant of choice for hepatic failure, renal failure, newborn (immature hepatic/renal function), old age, and when reversal agent is contraindicated.

*Topic: Benzyloquinoline Compounds*

**Q26. Answer: A — Convulsions**

Page 129: At higher doses, atracurium's metabolic product laudanosine can cross the blood brain barrier and can produce convulsions. As it is a benzylisoquinoline compound, it can release histamine producing allergic reactions ranging from pruritic rash to angioneurotic edema, bronchospasm or hypotension.

Topic: Benzylisoquinoline Compounds

**Q27. Answer: A — It does not release histamine and laudanosine production is 5 times less**

Page 129: Cisatracurium is an isomer of atracurium with 4 times more potency. The chief advantage of cisatracurium over atracurium is that it does not release histamine and laudanosine production is 5 times less than atracurium, therefore always preferred over atracurium. It is only metabolized by Hoffman degradation.

Topic: Benzylisoquinoline Compounds

**Q28. Answer: A — Mivacurium**

Page 129: Like succinylcholine, mivacurium is metabolized in plasma by pseudocholinesterase, therefore prolonged block is expected in conditions causing low pseudocholinesterase. Its short duration of action (5–10 minutes) makes it a muscle relaxant of choice for day care surgery. The major side effect is histamine release.

Topic: Benzylisoquinoline Compounds

**Q29. Answer: B — Pregnancy**

Page 129: Gallamine is the only nondepolarizer which can cross the placenta and cause fetal death, therefore absolutely contraindicated in pregnancy. More than 80% is excreted by kidneys, therefore it is also contraindicated in renal disease. It has maximum propensity for vagal blockade (vagolytic), causing severe tachycardia.

Topic: Benzylisoquinoline Compounds

**Q30. Answer: B — Suxamethonium for intubation**

Page 130: Gantacurium (previously AV430A) is a mixed onium chlorofumarate having fast onset (< 90 seconds) and ultra short duration (shortest among nondepolarizers). It may emerge as an alternative to suxamethonium for intubation. It is rapidly metabolized by forming a cysteine adduct followed by ester hydrolysis.

Topic: Compounds Under Research

**Q31. Answer: A — Additive**

Page 130: Two muscle relaxants of the same group (steroid + steroid or BZIQ + BZIQ) produce an additive effect, while addition of different groups (steroid + BZIQ) produces a synergistic effect.

Topic: Drug Interactions

**Q32. Answer: A — 20–30%**

Page 130: Inhalational agents prolong the block by both depolarizers and NDMR and decrease the requirement of relaxant by 20–30%. Maximum relaxation among inhalational agents is produced by desflurane (Desflurane > Sevoflurane > Isoflurane > Halothane).

Topic: Drug Interactions

**Q33. Answer: B — Aminoglycosides**

Page 130: Antibiotics prolong the effect of both depolarizers and NDMR. Aminoglycosides (neomycin, streptomycin, gentamicin, kanamycin, polymyxin, tobramycin), lincomycin and clindamycin are implicated. To reverse the effect of block by aminoglycosides, the drug used is 4-aminopyridine.

Topic: Drug Interactions

**Q34. Answer: A — Increased sensitivity/hyperkalemia to depolarizers and resistance to nondepolarizers**

Page 131, Table 13.2: In conditions with extrajunctional receptors (burns, hemiplegia, paraplegia, denervation muscle injuries, tetanus, cerebral palsy), there is increased sensitivity and hyperkalemia to depolarizers but resistance to nondepolarizers, because extrajunctional receptors are sensitive to depolarizing agents and resistant to nondepolarizers.

Topic: Muscular Diseases

**Q35. Answer: B — Resistance to depolarizers and increased sensitivity to nondepolarizers**

Page 131, Table 13.2: In myasthenia gravis there is resistance to depolarizers and increased sensitivity to nondepolarizers. This autoimmune disease destroys acetylcholine receptors, leaving fewer receptors for succinylcholine (increasing resistance) but less competition for nondepolarizers (increasing their sensitivity).

Topic: Muscular Diseases

**Q36. Answer: B — Some evidence of spontaneous recovery has appeared (e.g. spontaneous respiration, TO4 shows at least 1 twitch)**

Page 131: Reversal should be given only after some evidence of spontaneous recovery has appeared, like the patient has started spontaneous respiration and TO4 shows at least 1 twitch; otherwise reversal agents may enter the open channels and themselves produce block.

Topic: Reversal of Block

**Q37. Answer: D — Physostigmine**

Page 131: Anticholinesterases used for reversal are neostigmine, pyridostigmine, edrophonium and physostigmine. All except physostigmine are quaternary ammonium compounds and therefore do not cross the blood brain barrier (physostigmine does).

Topic: Reversal of Block

**Q38. Answer: A — Because glycopyrrolate is devoid of central side effects**

Page 132: To prevent muscarinic side effects (bradycardia, bronchospasm, secretions), atropine or glycopyrrolate has to be given with cholinesterase inhibitors. Glycopyrrolate, being devoid of central side effects, is always preferred over atropine.

Topic: Reversal of Block

**Q39. Answer: A — Neostigmine, 0.04–0.08 mg/kg**

Page 132: Neostigmine is the most commonly used reversal agent. Dose is 0.04–0.08 mg/kg. The effect begins in 5–10 minutes and peaks in half hour and lasts for one hour. The anticholinergic preferred with neostigmine is glycopyrrolate.

Topic: Reversal of Block

**Q40. Answer: C — Edrophonium**

Page 132: As neostigmine and pyridostigmine inhibit pseudocholinesterase, they cannot be used to reverse mivacurium (which is metabolized by pseudocholinesterase). Edrophonium does not inhibit pseudocholinesterase, therefore it is the reversal agent of choice for mivacurium.

Topic: Reversal of Block

**Q41. Answer: B — Steroidal type of nondepolarizing muscle relaxants**

Page 132: Sugammadex directly binds to the steroidal type of nondepolarizing muscle relaxants to form a complex which gets eliminated unchanged through the kidney. Its major disadvantage is that it cannot reverse benzylisoquinoline type of nondepolarizers.

Topic: Reversal of Block

**Q42. Answer: A — It is effective in reversing profound block, while anticholinesterases cannot reverse deeper blocks**

Page 132: Anticholinesterases are unable to reverse deeper blocks, while sugammadex is effective in reversing profound blockade. The reversal of block with cholinesterase inhibitors is slow (10–15 min) while reversal with sugammadex is very fast (2–3 minutes). It also has no muscarinic side effects, so there is no need to use glycopyrrolate or atropine.

Topic: Reversal of Block

**Q43. Answer: B — Sustained jaw clench over a tongue blade**

Page 132: Sustained jaw clench over a tongue blade is considered the best clinical sign of adequate reversal; however, it is difficult to perform by the patient, therefore head lift (ability to lift head for more than 5 seconds) is still the most widely used clinical test.

Topic: Signs of Adequate Reversal

**Q44. Answer: A — > 0.7 (70%)**

Page 132: Train of four (TO4) ratio > 0.7 (70%) indicates adequate recovery, but a ratio > 0.9 guarantees recovery. Most clinicians believe that in addition to TO4 ratio > 0.9, patients must not exhibit any clinical symptom/sign of impaired neuromuscular recovery. General weakness is considered as the most sensitive symptom of inadequate reversal.

Topic: Signs of Adequate Reversal

**Q45. Answer: A — 50 mm Hg**

Page 133: Acid base abnormalities — respiratory acidosis and metabolic alkalosis double the dose of neostigmine. It becomes almost impossible to reverse a patient with pCO<sub>2</sub> more than 50 mm Hg.

Topic: Common Causes of Inadequate Reversal

**Q46. Answer: B — Through a central mechanism (both supraspinal and spinal level), inhibiting polysynaptic reflexes; they have no action on the NMJ**

Page 133: Centrally acting muscle relaxants produce muscle relaxation through a central mechanism both at supraspinal and spinal level. Polysynaptic reflexes involved in maintenance of muscle tone are inhibited at both spinal and supraspinal level. These drugs have no action on the neuromuscular junction.

Topic: Centrally Acting Muscle Relaxants

**Q47. Answer: A — Intravenous diazepam**

Page 133: Among centrally acting muscle relaxants, for tetanus, intravenous diazepam is most effective. They are also used for muscle spasms, spastic neurological diseases (cerebral palsy, spinal injuries), and closed reductions/dislocations in orthopedics.

Topic: Centrally Acting Muscle Relaxants

**Q48. Answer: B — Intraocular, intragastric and intracranial pressures**

Page 125: Intraocular, intragastric and intracranial pressures are all increased by suxamethonium. Ocular muscles undergo tonic contraction increasing intraocular tension; neck muscle contraction blocks jugular venous outflow increasing intracranial tension; and contraction of abdominal muscles increases intragastric pressure.

Topic: Suxamethonium

**Q49. Answer: C — Doxacurium**

Page 128–129: Doxacurium has a long duration of action (60 minutes), the longest acting nondepolarizer, with high potency and almost complete elimination as unchanged drug by kidneys, making it a least preferred drug by clinicians.

Topic: Nondepolarizing Muscle Relaxants

**Q50. Answer: B — Atracurium, cisatracurium, rocuronium, mivacurium**

Page 134: NDMR safe in renal failure are atracurium, cisatracurium, rocuronium and mivacurium. NDMR absolutely contraindicated in renal failure are gallamine, metocurine, doxacurium, pancuronium and tubocurare.

*Topic: Nondepolarizing Muscle Relaxants*

# Perioperative Complications of General Anesthesia MCQ – Chapter 14

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 5: General Anesthesia (Chapter 14)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

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**Q1. Although the more turbulent periods in anesthesia are induction and emergence, maximum complications are reported in which period?**

- A. Maintenance period
- B. Induction period
- C. Emergence period
- D. Recovery period

**Ans: A**

**Q2. Which has been cited as the most common cause of anesthesia-related death in the perioperative period?**

- A. Cardiac arrest
- B. Postoperative respiratory depression
- C. Anaphylaxis
- D. Aspiration

**Ans: B**

**Q3. What is the most important risk factor for aspiration (single most important predisposing factor)?**

- A. Pregnancy
- B. Full stomach
- C. Hiatus hernia
- D. Diabetes

**Ans: B**

**Q4. Aspiration of late pregnancy was described by Mendelson and is called:**

- A. Mendelson syndrome
- B. Breuer Lockhart reflex
- C. Sellick's syndrome
- D. Fink effect

**Ans: A**

**Q5. What is the most important risk factor of aspirate that produces the most fulminant reaction?**

- A. Volume > 25 mL
- B. pH < 2.5
- C. Solid particles
- D. Temperature of aspirate

**Ans: A**

**Q6. During the cricoid pressure (Sellick's maneuver), backward and downward pressure of how much force is applied on the cricoid cartilage?**

- A. 30–40 Newtons (8–9 pounds)
- B. 10–20 Newtons
- C. 50–60 Newtons
- D. 5–8 Newtons

**Ans: A**

|                                                                                                                                                                                                                                                                                                |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. Which is the most common postoperative pulmonary complication, and its most common cause?</b></p> <p>A. Pneumothorax; caused by barotrauma<br/> B. Atelectasis; caused by secretions<br/> C. Pulmonary embolism; caused by DVT<br/> D. Pulmonary edema; caused by fluid overload</p> | <p><b>Ans: B</b></p> |
| <p><b>Q8. What is the most common cause of laryngospasm in anesthesia?</b></p> <p>A. Tongue fall<br/> B. Secretions<br/> C. Light anesthesia<br/> D. Endotracheal tube kinking</p>                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q9. Negative pressure pulmonary edema occurs due to upper airway obstruction, particularly from:</b></p> <p>A. Laryngospasm<br/> B. Bronchospasm<br/> C. Tongue fall<br/> D. Secretions</p>                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q10. Pulmonary embolism usually presents in which postoperative period?</b></p> <p>A. Second week (but can present with sudden onset from 2nd to 4th postoperative day)<br/> B. First 24 hours only<br/> C. After 1 month<br/> D. Only during induction</p>                              | <p><b>Ans: A</b></p> |
| <p><b>Q11. Postoperative hypertension and tachycardia are more responsible for unplanned ICU admission and carry higher morbidity and mortality than:</b></p> <p>A. Hypotension and bradycardia<br/> B. Aspiration<br/> C. Hypoxia<br/> D. Arrhythmias</p>                                     | <p><b>Ans: A</b></p> |
| <p><b>Q12. What is the most common arrhythmia seen in the perioperative period?</b></p> <p>A. Ventricular fibrillation<br/> B. Bradycardia<br/> C. Tachycardia<br/> D. Atrial fibrillation</p>                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q13. What is the most common cause of hypotension in the perioperative period?</b></p> <p>A. Anesthetic drugs<br/> B. Intravascular volume depletion<br/> C. Cardiac arrhythmias<br/> D. Anaphylaxis</p>                                                                                 | <p><b>Ans: B</b></p> |

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| <p><b>Q14. Which is the most commonly used agent to produce controlled hypotension as a continuous infusion?</b></p> <p>A. Sodium nitroprusside<br/>B. Nitroglycerin<br/>C. Labetalol<br/>D. Dexmedetomidine</p>                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q15. In controlled hypotension, in a person with normal BP, blood pressure is reduced to a mean arterial pressure of:</b></p> <p>A. 50–65 mm Hg<br/>B. 30–40 mm Hg<br/>C. 70–80 mm Hg<br/>D. 90–100 mm Hg</p>                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q16. More than 80% of cardiac arrests occur at which time?</b></p> <p>A. Time of induction<br/>B. Maintenance period<br/>C. Emergence<br/>D. Recovery in PACU</p>                                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q17. Which is the most common CNS complication of anesthesia?</b></p> <p>A. Awareness<br/>B. Convulsions<br/>C. Delayed recovery<br/>D. Permanent brain damage</p>                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q18. Brachial plexus injuries are the most common postoperative nerve injury associated with general anesthesia, followed by which nerve?</b></p> <p>A. Ulnar nerve<br/>B. Common peroneal nerve<br/>C. Radial nerve<br/>D. Femoral nerve</p>                    | <p><b>Ans: A</b></p> |
| <p><b>Q19. For tourniquet use, the pressure should not exceed more than how much above systolic pressure?</b></p> <p>A. 100 mm Hg (maximum 250 mm Hg)<br/>B. 50 mm Hg (maximum 150 mm Hg)<br/>C. 200 mm Hg (maximum 350 mm Hg)<br/>D. 20 mm Hg (maximum 100 mm Hg)</p> | <p><b>Ans: A</b></p> |
| <p><b>Q20. The incidence of awareness (inadequate amnesia) is less than 1%. What is the most common type?</b></p> <p>A. Visual<br/>B. Auditory<br/>C. Tactile<br/>D. Painful</p>                                                                                       | <p><b>Ans: B</b></p> |

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| <p><b>Q21. What is the incidence of postoperative cognitive dysfunction (POCD) at 1 year after cardiac surgery?</b></p> <p>A. 20% to 40%<br/>B. 4–10%<br/>C. 50–60%<br/>D. 1–2%</p>                                    | <p><b>Ans: A</b></p> |
| <p><b>Q22. What is the second most common complaint after pain in the postoperative period?</b></p> <p>A. Postoperative nausea and vomiting (PONV)<br/>B. Shivering<br/>C. Sore throat<br/>D. Headache</p>             | <p><b>Ans: A</b></p> |
| <p><b>Q23. Which are the drugs of choice for treatment as well as prophylaxis of PONV?</b></p> <p>A. Serotonin antagonists (ondansetron, granisetron)<br/>B. Metoclopramide<br/>C. Dexamethasone<br/>D. Droperidol</p> | <p><b>Ans: A</b></p> |
| <p><b>Q24. Regarding PONV risk factors, female gender increases vulnerability by how much compared to males?</b></p> <p>A. Three times<br/>B. Two times<br/>C. Four times<br/>D. 1.8 times</p>                         | <p><b>Ans: A</b></p> |
| <p><b>Q25. What is the most common thermal perturbation seen in anesthesia?</b></p> <p>A. Hyperthermia<br/>B. Hypothermia<br/>C. Malignant hyperthermia<br/>D. Fever</p>                                               | <p><b>Ans: B</b></p> |
| <p><b>Q26. Which is the drug of choice for postoperative shivering?</b></p> <p>A. Tramadol<br/>B. Pethidine<br/>C. Clonidine<br/>D. Propofol</p>                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q27. Malignant hyperthermia is due to an abnormality of which receptor?</b></p> <p>A. Type 1 Ryanodine receptor (RYR1)<br/>B. NMDA receptor<br/>C. GABA-A receptor<br/>D. Acetylcholine receptor</p>             | <p><b>Ans: A</b></p> |

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| <p><b>Q28. Which muscle relaxant is the most commonly implicated drug in malignant hyperthermia?</b></p> <p>A. Vecuronium<br/>B. Atracurium<br/>C. Succinylcholine<br/>D. Rocuronium</p>                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q29. What is the most sensitive early sign of malignant hyperthermia?</b></p> <p>A. Hyperthermia<br/>B. Rapid increase in end tidal CO<sub>2</sub> (ETCO<sub>2</sub>)<br/>C. Masseter spasm<br/>D. Hyperkalemia</p>                                          | <p><b>Ans: B</b></p> |
| <p><b>Q30. What is the most common cause of death in malignant hyperthermia?</b></p> <p>A. Renal failure<br/>B. Hyperkalemia induced ventricular fibrillation<br/>C. Pulmonary edema<br/>D. DIC</p>                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q31. What is the specific treatment of malignant hyperthermia?</b></p> <p>A. Dantrolene, 2 mg/kg repeated every 5 minutes (max 10 mg/kg)<br/>B. Adrenaline<br/>C. Sodium bicarbonate<br/>D. Insulin and dextrose</p>                                         | <p><b>Ans: A</b></p> |
| <p><b>Q32. What is considered the gold standard test for diagnosing susceptibility to malignant hyperthermia?</b></p> <p>A. Blood creatinine kinase (CK) levels<br/>B. Muscle contracture test (with halothane and caffeine)<br/>C. Genetic testing<br/>D. ECG</p> | <p><b>Ans: B</b></p> |
| <p><b>Q33. Which agents can DELAY or PREVENT malignant hyperthermia?</b></p> <p>A. Halothane and succinylcholine<br/>B. Propofol and nondepolarizing muscle relaxants<br/>C. Sevoflurane and atropine<br/>D. Isoflurane and ketamine</p>                           | <p><b>Ans: B</b></p> |
| <p><b>Q34. What is the most common cause of anaphylaxis in the perioperative period?</b></p> <p>A. Latex allergy<br/>B. Muscle relaxants (60–70%)<br/>C. Antibiotics<br/>D. Inhalational agents</p>                                                                | <p><b>Ans: B</b></p> |

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| <p><b>Q35. What is the mainstay of treatment of anaphylaxis?</b></p> <p>A. Steroids (hydrocortisone)<br/> B. Adrenaline<br/> C. Antihistamine (diphenhydramine)<br/> D. Intravenous fluids</p>                                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q36. In the lithotomy position, which nerve injury is the most common neuropathy of the lower limb?</b></p> <p>A. Saphenous nerve<br/> B. Common peroneal nerve (compressed between head of fibula and bar)<br/> C. Femoral nerve<br/> D. Obturator nerve</p>                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q37. Which ocular complication is a common complication of general anesthesia if the eyes remain open?</b></p> <p>A. Exposure keratitis (corneal abrasion)<br/> B. Glaucoma<br/> C. Retinal detachment<br/> D. Cataract</p>                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q38. Fire requires three prerequisites (the triad of fire). What are they?</b></p> <p>A. Ignition source, fuel and an oxidizer (gas supporting combustion)<br/> B. Heat, light and oxygen<br/> C. Spark, electricity and nitrous oxide<br/> D. Cautery, laser and air</p>                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q39. Cautery is responsible for what percentage of fires seen in the operation theater?</b></p> <p>A. 68%<br/> B. 25%<br/> C. 50%<br/> D. 90%</p>                                                                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q40. The hydrodynamics of regurgitation: a pressure of more than how much (cm H<sub>2</sub>O) is required to overcome the competency of the lower esophageal sphincter (LES)?</b></p> <p>A. &gt; 20 cm H<sub>2</sub>O<br/> B. &gt; 5 cm H<sub>2</sub>O<br/> C. &gt; 50 cm H<sub>2</sub>O<br/> D. &gt; 10 cm H<sub>2</sub>O</p> | <p><b>Ans: A</b></p> |
| <p><b>Q41. Which drug is preferred for prevention of nausea and vomiting in conditions associated with high incidence of aspiration, as it fastens gastric emptying and increases LES tone?</b></p> <p>A. Ondansetron<br/> B. Metoclopramide<br/> C. Dexamethasone<br/> D. Droperidol</p>                                            | <p><b>Ans: B</b></p> |

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| <p><b>Q42. Which drugs can cause convulsions as a perioperative complication?</b></p> <p>A. Local anesthetics, methohexitone, enflurane, sevoflurane and atracurium/cisatracurium<br/> B. Propofol, thiopentone and morphine<br/> C. Isoflurane, halothane and nitrous oxide<br/> D. Vecuronium and pancuronium</p>                                                                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q43. What is the lifetime risk of re-infarction in a known patient of myocardial infarction undergoing surgery?</b></p> <p>A. 6%<br/> B. 20%<br/> C. 50%<br/> D. 1%</p>                                                                                                                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q44. What is the most popular recovery score used for discharge from the postanesthesia care unit (PACU)?</b></p> <p>A. Glasgow Coma Scale<br/> B. ALDRETE score (color, respiration, consciousness level, circulation and activity)<br/> C. APGAR score<br/> D. Ramsay score</p>                                                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q45. The Breuer Lockhart reflex refers to which phenomenon?</b></p> <p>A. Light anesthesia during Lord's (anal) stretching and cervical dilatation initiating parasympathetic overactivity causing laryngospasm, bronchospasm, bradycardia and even cardiac arrest<br/> B. Cricoid pressure preventing aspiration<br/> C. Pupillary dilatation during surgery<br/> D. Diffusion hypoxia after nitrous oxide</p>                     | <p><b>Ans: A</b></p> |
| <p><b>Q46. Which non-anesthetic causes of hyperthermia in anesthesia is correctly matched?</b></p> <p>A. Neuroleptic malignant syndrome — phenothiazines<br/> B. Malignant hyperthermia — propofol<br/> C. Hypermetabolic state — hypothyroidism<br/> D. Hyperthermia — beta-blockers</p>                                                                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q47. In the differential diagnosis between malignant hyperthermia (MH) and neuroleptic malignant syndrome (NMS), which feature points to NMS?</b></p> <p>A. Rapid and disproportionate CO<sub>2</sub> rise<br/> B. History of intake of antidopaminergic drugs (phenothiazines, metoclopramide) or missing levodopa in Parkinson patient<br/> C. Inability to reverse muscle rigidity with nondepolarizers<br/> D. Hyperkalemia</p> | <p><b>Ans: B</b></p> |
| <p><b>Q48. Which position can cause spinal cord ischemia and quadriplegia due to extreme flexion of the neck?</b></p> <p>A. Lithotomy<br/> B. Trendelenburg<br/> C. Sitting position<br/> D. Supine</p>                                                                                                                                                                                                                                   | <p><b>Ans: C</b></p> |

**Q49. In the perioperative period, what is the most common cause of tachycardia in the intraoperative period?**

- A. Inadequate depth of anesthesia
- B. Pain and anxiety
- C. Hypovolemia
- D. Hyperthermia

**Ans: A**

**Q50. In the PACU, vitals should be monitored at what interval for the first 15 minutes?**

- A. Every 5 minutes
- B. Every 1 minute
- C. Every 15 minutes
- D. Every 30 minutes

**Ans: A**



# Answer Key & Explanations

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## Q1. Answer: A — Maintenance period

Page 136: Although the more turbulent periods are induction and emergence, maximum complications are reported in the maintenance period, necessitating the same vigilance standards in maintenance period as in induction and recovery phase.

Topic: Introduction

## Q2. Answer: B — Postoperative respiratory depression

Page 136: Postoperative respiratory depression has been cited as the most common cause of anesthesia related death in perioperative period, and the usual cause is transferring the patient to PACU without oxygen and reluctance to give oxygen in the postoperative period.

Topic: Mortality

## Q3. Answer: B — Full stomach

Page 137: Among predisposing factors for regurgitation, full stomach is the single most important factor. Other factors include depressed level of consciousness and conditions decreasing the tone of LES.

Topic: Pulmonary Aspiration

## Q4. Answer: A — Mendelson syndrome

Page 137: Pregnancy (acid aspiration in late pregnancy) was described by Mendelson and is called as Mendelson syndrome. The gravid uterus compresses the stomach leading to distortion of gastro-esophageal angle making LES to become incompetent.

Topic: Pulmonary Aspiration

## Q5. Answer: A — Volume > 25 mL

Page 137: Risk factors of aspiration are volume > 25 mL (the most important risk factor), pH < 2.5, and solid particles. Therefore, acidic solid particles in large quantity (> 25 mL) will produce the most fulminant reaction.

Topic: Pulmonary Aspiration

## Q6. Answer: A — 30–40 Newtons (8–9 pounds)

Page 138: As soon as the patient becomes unconscious, the assistant applies the backward and downward pressure of 30–40 Newtons (equivalent to 8–9 pounds) on cricoid cartilage (called as Sellick's maneuver) which compresses esophagus between assistant finger and vertebral column preventing aspiration. Cricoid pressure should be released only once the cuff is inflated and tube position is confirmed.

Topic: Rapid Sequence Induction

## Q7. Answer: B — Atelectasis; caused by secretions

Page 139: Atelectasis is the most common postoperative pulmonary complication. The most common cause of atelectasis is the secretions. V/Q mismatch caused by atelectasis is the most common cause of postoperative hypoxia.

Topic: V/Q Mismatch

## Q8. Answer: B — Secretions

Page 139: Blood and mucous can irritate the larynx producing laryngospasm (glottic closure) and desaturation. Secretions are the most common cause of laryngospasm in anesthesia. Treatment includes removing secretions by suction, IPPV with bag and mask, and small doses of succinylcholine (25–50 mg IV) if not relieved.

Topic: Upper Airway Obstruction

**Q9. Answer: A — Laryngospasm**

Page 139: The causes of pulmonary edema in perioperative period include cardiac failure, fluid overload, transfusion related, aspiration, sepsis, or negative pressure pulmonary edema due to upper airway obstruction (particularly laryngospasm).

Topic: Pulmonary Edema

**Q10. Answer: A — Second week (but can present with sudden onset from 2nd to 4th postoperative day)**

Page 139: Pulmonary embolism usually presents in second week but can present with sudden onset from 2nd to 4th postoperative day. Signs include dyspnea, substernal/chest pain, pleural pain, hemoptysis, tachycardia, increased CVP, hypotension, and gallop rhythm.

Topic: Pulmonary Embolism

**Q11. Answer: A — Hypotension and bradycardia**

Page 140: It has been observed that postoperative hypertension and tachycardia are more responsible for unplanned ICU admission and carry higher morbidity and mortality than hypotension and bradycardia.

Topic: Cardiovascular Complications

**Q12. Answer: C — Tachycardia**

Page 140: Although any arrhythmia may occur, the most common arrhythmia seen in perioperative period is tachycardia. The most common cause of tachycardia in intraoperative period is inadequate depth of anesthesia, and in postoperative period is pain and anxiety.

Topic: Cardiac Arrhythmias

**Q13. Answer: B — Intravascular volume depletion**

Page 140: The most common cause of hypotension in perioperative period is intravascular volume depletion, which could be because of excessive blood and fluid loss (particularly in major GI surgeries) or inadequate volume replacement.

Topic: Hypotension

**Q14. Answer: B — Nitroglycerin**

Page 141: To produce controlled hypotension, short acting vasodilator especially nitroglycerin as continuous infusion is the most commonly used agent. The concern of methemoglobinemia by nitroglycerine is very rare and treated with methylene blue. Sodium nitroprusside is now not preferred due to possibility of cyanide toxicity.

Topic: Hypotension / Controlled Hypotension

**Q15. Answer: A — 50–65 mm Hg**

Page 141: In a person with normal BP, no organ dysfunction occurs if blood pressure is maintained within the limits of autoregulation, i.e. if mean arterial pressure is maintained between 50–65 mm Hg. For a patient with normal BP, controlled hypotension means reduction of mean arterial pressure to 50–65 mm Hg.

Topic: Hypotension / Controlled Hypotension

**Q16. Answer: A — Time of induction**

Page 141: More than 80% of cardiac arrest occurs at the time of induction. The common causes of cardiac arrest are inability to ventilate, side effects of anesthetic agents like severe hypotension with inhalational agents or bradycardia proceeding to asystole after succinylcholine, and anaphylaxis.

Topic: Cardiac Arrest

**Q17. Answer: C — Delayed recovery**

Page 141: Delayed recovery (which means patient not regaining consciousness within 30–60 minutes after discontinuing anesthesia) is the most common CNS complication and is usually due to the residual effect of anesthetic agents.

Topic: Neurological Complications

**Q18. Answer: A — Ulnar nerve**

Page 142: Brachial plexus injuries are the most common postoperative nerve injury associated with general anesthesia, followed by ulnar nerve. However, after regional anesthesia, lumbosacral nerve roots are most commonly involved nerves. When combined (GA+RA), spinal cord injury becomes the most common injury, followed by brachial plexus.

Topic: Peripheral Neuropathies

**Q19. Answer: A — 100 mm Hg (maximum 250 mm Hg)**

Page 142: Tourniquet palsies — pressure in tourniquet should not exceed more than 100 mm Hg above systolic pressure (maximum 250 mm Hg) and duration should not exceed more than 2 hours for both upper and lower limb.

Topic: Peripheral Neuropathies

**Q20. Answer: B — Auditory**

Page 142: Incidence of awareness is < 1% however very important from medico legal point of view. Most common type is auditory. Awareness usually occurs when oxygen, nitrous oxide and opioids are used for maintenance of anesthesia.

Topic: Awareness

**Q21. Answer: A — 20% to 40%**

Page 142: The incidence of postoperative cognitive dysfunction (POCD) at 1 year after cardiac surgery is 20% to 40% and 4–10% after noncardiac surgery. It is more common in old age.

Topic: Postoperative Cognitive Dysfunction

**Q22. Answer: A — Postoperative nausea and vomiting (PONV)**

Page 142–143: The incidence of postoperative nausea and vomiting (PONV) is 20% to 30% making it the second most common complaint after pain in postoperative period.

Topic: Postoperative Nausea and Vomiting

**Q23. Answer: A — Serotonin antagonists (ondansetron, granisetron)**

Page 143: Serotonin antagonists (ondansetron, granisetron) remain the drugs of choice for treatment as well as prophylaxis for PONV. Interestingly all are equally effective at equipotent doses. Metoclopramide is the preferred antiemetic in conditions associated with high incidence of aspiration.

Topic: Postoperative Nausea and Vomiting

**Q24. Answer: A — Three times**

Page 143: Female gender is three times more vulnerable than males. General anesthesia has 4 times more incidence of PONV than regional anesthesia (most common reason being opioids). Nonsmokers are 1.8 times more vulnerable than smokers.

Topic: Postoperative Nausea and Vomiting

**Q25. Answer: B — Hypothermia**

Page 143: Hypothermia is the most common thermal perturbation seen in anesthesia. Shivering occurs as a protective mechanism to hypothermia. O<sub>2</sub> consumption may increase to 4 times (400%) during shivering, therefore oxygen supplementation during shivering is mandatory.

Topic: Thermal Perturbations

**Q26. Answer: B — Pethidine**

Page 143: Pethidine is the drug of choice for shivering. Tramadol is the 2nd drug of choice after pethidine. Other drugs include alpha 2 agonists (clonidine, dexmedetomidine), butorphanol, chlorpromazine, physostigmine, magnesium sulfate, ketanserin and propofol.

Topic: Shivering

**Q27. Answer: A — Type 1 Ryanodine receptor (RYR1)**

Page 144: Malignant hyperthermia is due to an abnormality of type 1 Ryanodine receptor (RYR1) which is the calcium releasing channel of sarcoplasmic reticulum. The abnormality leads to excessive accumulation of calcium in sarcoplasmic reticulum causing sustained contraction of muscle. It is a genetic disease, usually autosomal dominant.

Topic: Malignant Hyperthermia

**Q28. Answer: C — Succinylcholine**

Page 144: Succinylcholine is the most commonly implicated drug (muscle relaxant) in malignant hyperthermia. Among inhalational agents, halothane is the most commonly implicated inhalational agent.

Topic: Malignant Hyperthermia

**Q29. Answer: B — Rapid increase in end tidal CO<sub>2</sub> (ETCO<sub>2</sub>)**

Page 144: Rapid increase in end tidal CO<sub>2</sub> (ETCO<sub>2</sub>) is the most sensitive early sign of malignant hyperthermia (may rise to more than 100 mm Hg; normal 32–42 mm Hg). Hyperthermia is the most important sign in diagnosing MH but unfortunately it is a late sign.

Topic: Malignant Hyperthermia

**Q30. Answer: B — Hyperkalemia induced ventricular fibrillation**

Page 145: Hyperkalemia induced ventricular fibrillation is the most common cause of death in malignant hyperthermia.

Topic: Malignant Hyperthermia

**Q31. Answer: A — Dantrolene, 2 mg/kg repeated every 5 minutes (max 10 mg/kg)**

Page 145: The specific treatment is dantrolene, 2 mg/kg to be repeated every 5 minutes (maximum 10 mg/kg). Dantrolene should be continued 1 mg/kg every 6 hourly for next 24 hours as chances of recurrence is 50%. Dantrolene directly binds to ryanodine receptor inhibiting calcium release.

Topic: Malignant Hyperthermia

**Q32. Answer: B — Muscle contracture test (with halothane and caffeine)**

Page 145: Patients with normal CK levels should undergo muscle contracture test. In this test a muscle biopsy is taken and subjected to halothane and caffeine to see the pathological changes of malignant hyperthermia. Muscle contracture test is considered as gold standard test for diagnosing susceptibility to malignant hyperthermia.

Topic: Malignant Hyperthermia

**Q33. Answer: B — Propofol and nondepolarizing muscle relaxants**

Page 145: Propofol (can delay or prevent malignant hyperthermia) and nondepolarizing muscle relaxants (can delay or prevent malignant hyperthermia) are safe drugs for general anesthesia in susceptible patients. Other safe drugs are barbiturates, narcotics, benzodiazepines and nitrous oxide.

Topic: Malignant Hyperthermia

**Q34. Answer: B — Muscle relaxants (60–70%)**

Page 146: The most common cause of anaphylaxis is muscle relaxants (60–70%) followed by latex allergy (12%) and then antibiotics (8–10%). Among muscle relaxants, maximum incidence is with rocuronium to be followed by suxamethonium.

Topic: Anaphylactic Reactions

**Q35. Answer: B — Adrenaline**

Page 146: Adrenaline is the mainstay of treatment of anaphylaxis. Doses: 0.5–1 mg of 1:1000 intramuscularly or subcutaneously. In life-threatening situations, 1:10,000 adrenaline can be given intravenously. Maintenance of airway, steroids (hydrocortisone) and histamine antagonist (diphenhydramine) are also given.

Topic: Anaphylactic Reactions

**Q36. Answer: B — Common peroneal nerve (compressed between head of fibula and bar)**

Page 146: In the lithotomy position, common peroneal nerve (compressed between head of fibula and bar) is the most common neuropathy of the lower limb. Functional residual capacity and vital capacity decrease by approx 20%, increasing the V/Q mismatch.

*Topic: Complications of Different Positions*

**Q37. Answer: A — Exposure keratitis (corneal abrasion)**

Page 146: Exposure keratitis (corneal abrasion) is a common complication of general anesthesia if eyes remain open (because of loss of blinking reflex). To prevent keratitis, artificial tears/eye ointment should be instilled and eyes should be covered with eye pads.

*Topic: Ocular Complications*

**Q38. Answer: A — Ignition source, fuel and an oxidizer (gas supporting combustion)**

Page 147: There are three prerequisites for a fire (the triad of fire): ignition source, fuel and an oxidizer (gas supporting combustion). The ignition source may be cautery (responsible for 68% of fires), laser, electric sparks, or static electricity.

*Topic: Fires and Electric Hazards*

**Q39. Answer: A — 68%**

Page 147: Cautery is responsible for 68% of fires seen in OT. The maximum leakage allowed in operation theater is 10  $\mu$ A (electric sparks and short circuits). Among fuels, silicone is most resistant to fire followed by PVC, while red rubber tubes are most vulnerable to catch fire.

*Topic: Fires and Electric Hazards*

**Q40. Answer: A — > 20 cm H<sub>2</sub>O**

Page 137: Normal intragastric pressure is 5–7 cm H<sub>2</sub>O and regurgitation is prevented by the tone of LES. A pressure of > 20 cm H<sub>2</sub>O is required to overcome the competency of LES. In conditions like pregnancy and hiatus hernia, a pressure > 15 cm H<sub>2</sub>O can cause regurgitation.

*Topic: Pulmonary Aspiration*

**Q41. Answer: B — Metoclopramide**

Page 138, 143: Metoclopramide fastens gastric emptying and increases the tone of LES, therefore it is the preferred antiemetic for conditions associated with high incidence of aspiration.

*Topic: Postoperative Nausea and Vomiting*

**Q42. Answer: A — Local anesthetics, methohexitone, enflurane, sevoflurane and atracurium/cisatracurium**

Page 141: Convulsions may occur due to hypoxia, drugs like local anesthetics, methohexitone, enflurane, sevoflurane and atracurium/cisatracurium, and cerebrovascular accidents.

*Topic: Convulsions*

**Q43. Answer: A — 6%**

Page 141: Surgery and anesthesia are stressful conditions which can precipitate MI in susceptible individuals. The lifetime risk of re-infarction in a known patient of MI is 6%. Treatment includes oxygen, morphine, nitroglycerine, inotropes, thrombolytic therapy and intra-aortic balloon counterpulsations.

*Topic: Myocardial Ischemia*

**Q44. Answer: B — ALDRETE score (color, respiration, consciousness level, circulation and activity)**

Page 148: The criteria for discharge from PACU vary depending on whether patient is discharged to ICU, ward or home. Although there are many recovery scores, the most popular are ALDRETE score (which includes color, respiration, consciousness level, circulation and activity level) and postanesthesia discharge scoring system (PADS).

Topic: Discharge Criteria from PACU

**Q45. Answer: A — Light anesthesia during Lord's (anal) stretching and cervical dilatation initiating parasympathetic overactivity causing laryngospasm, bronchospasm, bradycardia and even cardiac arrest**

Page 139: Light anesthesia during Lord's (anal) stretching and cervical dilatation can initiate parasympathetic over activity causing laryngospasm, bronchospasm, bradycardia and even cardiac arrest. This reflex is called as Breuer Lockhart reflex.

Topic: Bronchospasm

**Q46. Answer: A — Neuroleptic malignant syndrome — phenothiazines**

Page 144: Causes of hyperthermia in anesthesia include hypermetabolic states (thyrotoxicosis, pheochromocytoma), neuroleptic malignant syndrome due to phenothiazines, drug induced (atropine, pethidine, MAO inhibitors, tricyclic antidepressants, amphetamines), injury to hypothalamic temperature regulatory centers, and malignant hyperthermia.

Topic: Hyperthermia

**Q47. Answer: B — History of intake of antidopaminergic drugs (phenothiazines, metoclopramide) or missing levodopa in Parkinson patient**

Page 145: In NMS, there is a history of intake of antidopaminergic drugs (like phenothiazines or metoclopramide) or missing the intake of levodopa in Parkinson patient. The rise in CO<sub>2</sub> and muscle rigidity are slow and in proportion to rise in temperature. As the neuromuscular junction is normal in NMS, muscle rigidity can be reversed with nondepolarizing muscle relaxants (while in MH, it is not possible to reverse rigidity by nondepolarizers).

Topic: Malignant Hyperthermia

**Q48. Answer: C — Sitting position**

Page 146: In the sitting position, spinal cord ischemia and quadriplegia can occur due to extreme flexion of neck. Brachial plexus injury (weight of arm stretching the brachial plexus), femoral and obturator nerve injury, and venous air embolism can also occur.

Topic: Complications of Different Positions

**Q49. Answer: A — Inadequate depth of anesthesia**

Page 140: The most common cause of tachycardia in intraoperative period is inadequate depth of anesthesia, and in postoperative period it is pain and anxiety. Treatment includes increasing the depth or treating the pain; if not due to these causes, it should be treated with esmolol.

Topic: Cardiac Arrhythmias

**Q50. Answer: A — Every 5 minutes**

Page 136: To prevent complications in PACU, vitals should be monitored every 5 minutes for the first 15 minutes and thereafter every 15 minutes in stable patients. All patients in PACU must receive supplemental oxygen and continuous monitoring of oxygen saturation.

Topic: Mortality / PACU

# Local Anesthetics MCQ – Chapter 15

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 6: Regional Anesthesia (Chapter 15)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Chemically, local anesthetics consist of a benzene ring separated from a tertiary amide by either an ester or amide linkage. They are accordingly classified as:**

- A. Aminoesters and aminoamides
- B. Weak acids and weak bases
- C. Short and long acting
- D. Ionized and nonionized

**Ans: A**

**Q2. Aminoester local anesthetics are metabolized by pseudocholinesterase and have a high incidence of allergic reactions due to:**

- A. Methylparaben
- B. Para aminobenzoic acid
- C. Orthotoluidine
- D. Ecgonine

**Ans: B**

**Q3. Which is the shortest acting local anesthetic?**

- A. Procaine
- B. Lignocaine
- C. Chlorprocaine
- D. Dibucaine

**Ans: C**

**Q4. Which is the longest acting local anesthetic?**

- A. Bupivacaine
- B. Ropivacaine
- C. Tetracaine
- D. Dibucaine

**Ans: D**

**Q5. Local anesthetics penetrate the axonal membrane in their undissociated (nonionized) form, then inside the cell they bind to which structure in dissociated (ionized) form?**

- A. Sodium channel (alpha subunit) from the inner side
- B. Potassium channel from outside
- C. GABA receptor
- D. Calcium channel from outside

**Ans: A**

**Q6. Recent studies have proven that local anesthetics not only act on sodium channels but also block which channels?**

- A. Potassium and calcium channels
- B. Chloride channels
- C. Magnesium channels
- D. No other channels

**Ans: A**



|                                                                                                                                                                                                                                                                                                                                                                   |                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. According to current understanding, which nerve fibers are the MOST sensitive to local anesthetic block?</b></p> <ul style="list-style-type: none"> <li>A. Large myelinated A alpha and A beta fibers</li> <li>B. Small myelinated A gamma and A delta fibers</li> <li>C. Nonmyelinated C fibers</li> <li>D. B fibers</li> </ul>                        | <p><b>Ans: B</b></p> |
| <p><b>Q8. In peripheral nerve blocks, what is the functional sequence of blockade?</b></p> <ul style="list-style-type: none"> <li>A. Autonomic &gt; sensory &gt; motor</li> <li>B. Motor &gt; sensory &gt; autonomic</li> <li>C. Sensory &gt; motor &gt; autonomic</li> <li>D. All blocked simultaneously</li> </ul>                                              | <p><b>Ans: B</b></p> |
| <p><b>Q9. Lignocaine has a faster onset than bupivacaine because:</b></p> <ul style="list-style-type: none"> <li>A. It has a lower pKa (7.8) closer to physiologic pH, so more is in nonionized form to cross the membrane</li> <li>B. It has a higher molecular weight</li> <li>C. It is more lipid soluble</li> <li>D. It has higher protein binding</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q10. Why is the onset of local anesthetic action delayed in ischemic or infected (abscess) tissue?</b></p> <ul style="list-style-type: none"> <li>A. Acidic pH ionizes the drug, delaying onset</li> <li>B. Alkaline pH ionizes the drug</li> <li>C. The drug is metabolized faster</li> <li>D. There is no nerve to block</li> </ul>                       | <p><b>Ans: A</b></p> |
| <p><b>Q11. A stimulated nerve is blocked earlier than a nonstimulated nerve. This kind of block is called:</b></p> <ul style="list-style-type: none"> <li>A. Differential block</li> <li>B. Use-dependent block</li> <li>C. Phase II block</li> <li>D. Refractory block</li> </ul>                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q12. The most commonly used vasoconstrictor added to local anesthetics is adrenaline, in a concentration of:</b></p> <ul style="list-style-type: none"> <li>A. 1 in 2,00,000</li> <li>B. 1 in 1,000</li> <li>C. 1 in 10,000</li> <li>D. 1 in 50,000</li> </ul>                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q13. The maximum systemic absorption of local anesthetic is seen after which block?</b></p> <ul style="list-style-type: none"> <li>A. Spinal block</li> <li>B. Intercostal nerve block</li> <li>C. Axillary block</li> <li>D. Infiltration</li> </ul>                                                                                                       | <p><b>Ans: B</b></p> |

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| <p><b>Q14. The potency of a local anesthetic directly correlates with its:</b></p> <ul style="list-style-type: none"> <li>A. Lipid solubility</li> <li>B. pKa</li> <li>C. Protein binding</li> <li>D. Molecular weight</li> </ul>                                                                                                                                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q15. The CNS:CVS dose ratio for lignocaine is 1:7 and for bupivacaine is 1:3. What does this indicate?</b></p> <ul style="list-style-type: none"> <li>A. With bupivacaine, CNS toxicity is involved at much lower doses compared to CVS toxicity (less safety margin)</li> <li>B. Bupivacaine is safer than lignocaine</li> <li>C. Lignocaine causes more cardiac toxicity</li> <li>D. Both have equal toxicity</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q16. What is often the first presentation of local anesthetic toxicity?</b></p> <ul style="list-style-type: none"> <li>A. Cardiac arrest</li> <li>B. Convulsions</li> <li>C. Hypotension</li> <li>D. Bradycardia</li> </ul>                                                                                                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q17. At clinically used doses, all local anesthetics are vasodilators EXCEPT which agents?</b></p> <ul style="list-style-type: none"> <li>A. Cocaine, levobupivacaine and ropivacaine</li> <li>B. Lignocaine and bupivacaine</li> <li>C. Procaine and tetracaine</li> <li>D. Prilocaine and mepivacaine</li> </ul>                                                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q18. Allergic reactions to local anesthetics are common with esters but rare with amides. The reaction with amides is because of which preservative?</b></p> <ul style="list-style-type: none"> <li>A. Methylparaben</li> <li>B. Sodium bisulfite</li> <li>C. Para aminobenzoic acid</li> <li>D. Benzethonium chloride</li> </ul>                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q19. Methemoglobinemia is usually seen with which local anesthetics?</b></p> <ul style="list-style-type: none"> <li>A. Prilocaine and benzocaine</li> <li>B. Lignocaine and bupivacaine</li> <li>C. Cocaine and procaine</li> <li>D. Tetracaine and dibucaine</li> </ul>                                                                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q20. Local anesthetics with adrenaline can cause necrosis and gangrene if used for ring block of which structures (because these have end arteries)?</b></p> <ul style="list-style-type: none"> <li>A. Fingers, toes, penis or pinna</li> <li>B. Forearm and leg</li> <li>C. Abdomen and back</li> <li>D. Scalp and forehead</li> </ul>                                                                                    | <p><b>Ans: A</b></p> |

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| <p><b>Q21. Which was the first local anesthetic used, and what is its source?</b></p> <p>A. Procaine, from coca leaves<br/> B. Cocaine, extracted from the leaves of <i>Erythroxylum coca</i><br/> C. Lignocaine, synthesized chemically<br/> D. Tetracaine, from cinchona bark</p>                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q22. Which local anesthetic is the agent of choice in patients with a history of malignant hyperthermia?</b></p> <p>A. Lignocaine<br/> B. Bupivacaine<br/> C. Procaine<br/> D. Cocaine</p>                                                                                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q23. Which is the most commonly used local anesthetic?</b></p> <p>A. Bupivacaine<br/> B. Lignocaine (xylocaine, lidocaine)<br/> C. Procaine<br/> D. Ropivacaine</p>                                                                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q24. What is the maximum safe dose of lignocaine without and with adrenaline?</b></p> <p>A. 4.5 mg/kg (max 300 mg) without adrenaline; 7 mg/kg (max 500 mg) with adrenaline<br/> B. 2.5 mg/kg without; 3 mg/kg with adrenaline<br/> C. 7 mg/kg without; 4.5 mg/kg with adrenaline<br/> D. 3 mg/kg without; 5 mg/kg with adrenaline</p> | <p><b>Ans: A</b></p> |
| <p><b>Q25. Lignocaine is used intravenously for the treatment of ventricular tachycardia. The preparation used is xylocard 2%, in a dose of:</b></p> <p>A. 2 mg/kg<br/> B. 5 mg/kg<br/> C. 0.5 mg/kg<br/> D. 10 mg/kg</p>                                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q26. Which local anesthetic is the safest, owing to its extrahepatic metabolism (in liver, kidneys, lungs by amidase)?</b></p> <p>A. Lignocaine<br/> B. Prilocaine<br/> C. Bupivacaine<br/> D. Mepivacaine</p>                                                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q27. Bupivacaine is the local anesthetic of choice for postoperative pain relief and painless labor because of its:</b></p> <p>A. Wide differential sensory and motor blockade (motor block only at high concentrations)<br/> B. Short duration of action<br/> C. Rapid onset<br/> D. Low protein binding</p>                          | <p><b>Ans: A</b></p> |

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| <p><b>Q28. What is the maximum safe dose of bupivacaine without and with adrenaline?</b></p> <p>A. 2.5 mg/kg (max 175 mg) without; 3 mg/kg (max 225 mg) with adrenaline<br/> B. 4.5 mg/kg without; 7 mg/kg with adrenaline<br/> C. 3 mg/kg without; 4 mg/kg with adrenaline<br/> D. 1 mg/kg without; 2 mg/kg with adrenaline</p> | <p><b>Ans: A</b></p> |
| <p><b>Q29. Bupivacaine is absolutely contraindicated for which block due to its high cardiotoxicity?</b></p> <p>A. Spinal block<br/> B. Bier's block (IVRA)<br/> C. Epidural block<br/> D. Nerve block</p>                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q30. Which drug is now considered the drug of choice for the treatment of bupivacaine (and other local anesthetic) induced ventricular tachycardia?</b></p> <p>A. Bretylium<br/> B. Amiodarone<br/> C. Lignocaine<br/> D. Procainamide</p>                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q31. Levobupivacaine is the S isomer of bupivacaine. Why is it less cardiotoxic than bupivacaine?</b></p> <p>A. Because it does not contain the R isomer<br/> B. Because it is more lipid soluble<br/> C. Because it is metabolized faster<br/> D. Because it has lower protein binding</p>                                | <p><b>Ans: A</b></p> |
| <p><b>Q32. Ropivacaine cardiotoxicity compared to bupivacaine and levobupivacaine is:</b></p> <p>A. Less than both bupivacaine and levobupivacaine (but still higher than lignocaine)<br/> B. More than both<br/> C. Equal to bupivacaine<br/> D. More than lignocaine but less than bupivacaine only</p>                        | <p><b>Ans: A</b></p> |
| <p><b>Q33. Why is ropivacaine always preferred over bupivacaine for postoperative analgesia and painless labor?</b></p> <p>A. It is less potent and produces less motor block, while sensory analgesia is comparable<br/> B. It has a faster onset<br/> C. It is cheaper<br/> D. It has a longer duration</p>                    | <p><b>Ans: A</b></p> |
| <p><b>Q34. EMLA cream is a eutectic mixture of which two local anesthetics?</b></p> <p>A. 5% prilocaine and 5% lignocaine in equal amount (1:1 ratio)<br/> B. Tetracaine and adrenaline<br/> C. Bupivacaine and ropivacaine<br/> D. Cocaine and tetracaine</p>                                                                   | <p><b>Ans: A</b></p> |

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| <p><b>Q35. Which local anesthetics CANNOT be used for surface (topical) anesthesia as they do not get absorbed from skin or mucous membranes?</b></p> <p>A. Mepivacaine, bupivacaine, levobupivacaine and ropivacaine<br/> B. Lignocaine and prilocaine<br/> C. Cocaine and tetracaine<br/> D. Benzocaine and amethocaine</p>                                   | <p><b>Ans: A</b></p> |
| <p><b>Q36. Which local anesthetic concentration is used for intravenous regional analgesia (Bier's block)?</b></p> <p>A. 5% lignocaine<br/> B. 0.5% lignocaine<br/> C. 2% lignocaine<br/> D. 0.125% bupivacaine</p>                                                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q37. Cardiotoxicity of bupivacaine increases in which conditions?</b></p> <p>A. Pregnancy, hypoxia and acidosis<br/> B. Hyperventilation and alkalosis<br/> C. Hypothermia only<br/> D. Hyperkalemia only</p>                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q38. Local anesthetic induced neurotoxicity classically manifests as cauda equina syndrome, seen with which agents through small bore catheters?</b></p> <p>A. Lignocaine and tetracaine<br/> B. Bupivacaine and ropivacaine<br/> C. Procaine and cocaine<br/> D. Mepivacaine and prilocaine</p>                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q39. For spinal and epidural anesthesia, which type of local anesthetic preparation should be used?</b></p> <p>A. Preservative free preparations<br/> B. Preparations with methylparaben<br/> C. Preparations with adrenaline<br/> D. Acidic preparations (pH 4)</p>                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q40. Tumescent anesthesia is a technique used by plastic surgeons during liposuction. What is a concern with this technique?</b></p> <p>A. Peak concentration may occur in postoperative period producing toxicity<br/> B. It causes immediate methemoglobinemia<br/> C. It is contraindicated in all patients<br/> D. It requires general anesthesia</p> | <p><b>Ans: A</b></p> |
| <p><b>Q41. Which agent is used for refrigeration analgesia, where nerve conduction is blocked by cooling the nerves to -5 to -20°C?</b></p> <p>A. Cryoprobes delivering CO2 or nitrous oxide<br/> B. Liquid nitrogen<br/> C. Ice packs<br/> D. Ethyl chloride spray</p>                                                                                         | <p><b>Ans: A</b></p> |

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| <p><b>Q42. When two local anesthetics are combined (e.g. lignocaine for rapid onset and bupivacaine for long duration), the toxicity is:</b></p> <ul style="list-style-type: none"> <li>A. Additive (not independent), therefore maximum safe doses should be reduced accordingly</li> <li>B. Independent of each other</li> <li>C. Reduced</li> <li>D. Neutralized</li> </ul>                                       | <p><b>Ans: A</b></p> |
| <p><b>Q43. Differential block of local anesthetics depends on concentration. A drug at lower concentration will:</b></p> <ul style="list-style-type: none"> <li>A. Produce only sensory block, while at higher concentration produce motor block</li> <li>B. Produce only motor block</li> <li>C. Produce no block</li> <li>D. Block autonomic fibers only</li> </ul>                                                | <p><b>Ans: A</b></p> |
| <p><b>Q44. Which agent has the property of being the longest acting, most potent, and most toxic local anesthetic?</b></p> <ul style="list-style-type: none"> <li>A. Bupivacaine</li> <li>B. Tetracaine</li> <li>C. Dibucaine (cinchocaine)</li> <li>D. Ropivacaine</li> </ul>                                                                                                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q45. Procaine and chlorprocaine are metabolized by pseudocholinesterase. Cocaine differs in that it is metabolized in:</b></p> <ul style="list-style-type: none"> <li>A. Liver</li> <li>B. Plasma</li> <li>C. Lungs</li> <li>D. Kidneys</li> </ul>                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q46. Concentration minimum (Cm) — the minimum concentration of local anesthetic to block nerve impulse conduction — is increased by:</b></p> <ul style="list-style-type: none"> <li>A. Acidic pH (causing ionization)</li> <li>B. Alkaline pH</li> <li>C. Nerve stimulation</li> <li>D. Adjuvant opioids</li> </ul>                                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q47. In central neuraxial blocks, the sensory sequence of blockade differs from peripheral nerve blocks regarding temperature and pain. In central neuraxial blocks:</b></p> <ul style="list-style-type: none"> <li>A. Temperature is blocked earlier than pain</li> <li>B. Pain is blocked earlier than temperature</li> <li>C. Touch is blocked first</li> <li>D. Proprioception is blocked first</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q48. What is the most common cause of failure of local anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Inappropriate technique, inadequate volume or concentration</li> <li>B. Drug allergy</li> <li>C. Genetic variation</li> <li>D. Tachyphylaxis</li> </ul>                                                                                                                                   | <p><b>Ans: A</b></p> |

**Q49. Adrenaline added to commercial local anesthetic preparations makes them even more acidic (pH around 4). Why?**

- A. Because adrenaline can become unstable at alkaline pH
- B. To increase potency
- C. To reduce allergic reactions
- D. To improve shelf life

**Ans: A**

**Q50. Which opioid is frequently used to supplement the effect of local anesthetics for nerve blocks because of its local anesthetic properties?**

- A. Morphine
- B. Buprenorphine
- C. Fentanyl
- D. Remifentanyl

**Ans: B**

# Answer Key & Explanations

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## Q1. Answer: A — Aminoesters and aminoamides

Page 151: Local anesthetics consist of a benzene ring separated from a tertiary amide either by ester or amide linkage. Based on this intermediate chain they are classified as aminoesters and aminoamides.

Topic: Classification

## Q2. Answer: B — Para aminobenzoic acid

Page 151, Table: Aminoesters are metabolized by pseudocholinesterase (except cocaine which is metabolized in liver) and have a high incidence of allergic reactions due to para aminobenzoic acid. Amides are metabolized primarily in liver and have a low incidence of allergic reactions.

Topic: Classification

## Q3. Answer: C — Chlorprocaine

Page 151: Based on duration of action, short duration (15–30 minutes) agents are chlorprocaine (shortest acting local anesthetic) and procaine. Dibucaine is the longest acting local anesthetic.

Topic: Classification

## Q4. Answer: D — Dibucaine

Page 151: Long duration (2–3 hours) agents include bupivacaine, levobupivacaine, ropivacaine, tetracaine, etidocaine and dibucaine. Dibucaine is the longest acting local anesthetic.

Topic: Classification

## Q5. Answer: A — Sodium channel (alpha subunit) from the inner side

Page 152: Local anesthetics in undissociated (nonionized) form penetrate the axonal membrane. Once inside, the dissociated (ionized, protonated, cationic) form binds to the sodium channel (alpha subunit) from the inner side, blocking the channel and preventing depolarization and action potential.

Topic: Mechanism of Action

## Q6. Answer: A — Potassium and calcium channels

Page 152: Recent studies have proven that local anesthetics not only act on sodium channels, but also block potassium and calcium channels in a manner that hypokalemia and hypercalcemia can antagonize the effect of local anesthetics. They are also found to block N-methyl-D-aspartate (NMDA) receptors.

Topic: Mechanism of Action

## Q7. Answer: B — Small myelinated A gamma and A delta fibers

Page 152: Small myelinated fibers (A gamma and A delta) are the most sensitive to local anesthetic block, followed by large myelinated (A alpha and A beta), then B (mixed), with nonmyelinated C fibers being most resistant. The sequence of blockade is A > B > C. The traditional notion that thin fibers are most sensitive does not hold true in present day practice.

Topic: Sensitivity of Nerve Fiber

## Q8. Answer: B — Motor > sensory > autonomic

Page 153: In peripheral nerve blocks, motor function is blocked earliest followed by sensory and then autonomic (motor > sensory > autonomic). However, in central neuraxial blocks the sequence is autonomic > sensory > motor. Sequence of recovery is in reverse order of blockade.

Topic: Sensitivity of Nerve Fiber



**Q9. Answer: A — It has a lower pKa (7.8) closer to physiologic pH, so more is in nonionized form to cross the membrane**

Page 153: pKa is the pH at which a local anesthetic is 50% ionized and 50% nonionized. Agents with pKa closer to physiologic pH have more nonionized form which diffuses through the axonal membrane, enhancing onset. That is why lignocaine with lower pKa of 7.8 has fast onset compared to bupivacaine with higher pKa of 8.1.

*Topic: Onset of Action*

**Q10. Answer: A — Acidic pH ionizes the drug, delaying onset**

Page 153: Adding sodabarbonate makes more drug available in nonionized form to cross the axonal membrane (faster onset). On the other hand, in ischemic tissue (like abscess) acidic pH ionizes the drug, delaying the onset of action.

*Topic: Onset of Action*

**Q11. Answer: B — Use-dependent block**

Page 153: Since activated channels are blocked more easily, a stimulated nerve will be blocked earlier than a nonstimulated nerve. This kind of block by local anesthetics is called use-dependent block.

*Topic: Onset of Action*

**Q12. Answer: A — 1 in 2,00,000**

Page 153: Vasoconstrictors decrease the systemic absorption increasing the duration of action. The most commonly used vasoconstrictor is adrenaline in a concentration of 1 in 2,00,000. Adrenaline when added to lignocaine increases the duration of both sensory and motor blockade, while with bupivacaine only sensory block is prolonged.

*Topic: Duration of Action*

**Q13. Answer: B — Intercostal nerve block**

Page 153: Systemic absorption depends on the site of injection, proportional to the vascularity of the site. Maximum systemic absorption has been seen after intercostal nerve block. Addition of vasoconstrictors decreases the systemic absorption.

*Topic: Systemic Absorption*

**Q14. Answer: A — Lipid solubility**

Page 153: Potency directly correlates with lipid solubility. Potency (lipid solubility) is also directly proportional to duration; more the potency, longer is the duration (with the exception of chlorprocaine which despite having intermediate potency is shortest acting).

*Topic: Potency*

**Q15. Answer: A — With bupivacaine, CNS toxicity is involved at much lower doses compared to CVS toxicity (less safety margin)**

Page 154: The CNS:CVS dose ratio for lignocaine is 1:7 and for bupivacaine is 1:3, meaning that with bupivacaine, CNS toxicity is involved at much lower doses as compared to CVS — indicating a narrower safety margin. CNS is the first system involved in local anesthetic toxicity.

*Topic: Systemic Toxicity*

**Q16. Answer: B — Convulsions**

Page 154: As the CNS is the first system involved in local anesthetic toxicity, typical sequence is excitation followed by depression of cerebral tissue. Signs include circumoral numbness, dizziness, tongue paresthesia, visual and auditory disturbances, muscle twitching, tremors, convulsions followed by coma and death. Convulsions as first presentation are quite common.

*Topic: Systemic Toxicity*

**Q17. Answer: A — Cocaine, levobupivacaine and ropivacaine**

Page 154: At lower doses local anesthetics may act as vasoconstrictors by directly inhibiting nitric oxide, but at clinically used doses all local anesthetics are vasodilators except cocaine, levobupivacaine and ropivacaine which are vasoconstrictors at all doses.

Topic: Cardiovascular System

**Q18. Answer: A — Methylparaben**

Page 154–155: Allergic reactions are common with esters but rare with amides. The reaction with amides is because of the preservative (methyl paraben) it contains. Cross sensitivity does not exist between classes (i.e. esters and amides) but exists between agents of the same class.

Topic: Immunologic

**Q19. Answer: A — Prilocaine and benzocaine**

Page 155: Methemoglobinemia is usually seen with prilocaine; however, benzocaine can also cause methemoglobinemia. Treatment is IV methylene blue.

Topic: Systemic Effects

**Q20. Answer: A — Fingers, toes, penis or pinna**

Page 155: Local anesthetics with adrenaline can cause necrosis and gangrene if used for ring block of fingers, toes, penis or pinna because these structures have end arteries.

Topic: Local Toxicity

**Q21. Answer: B — Cocaine, extracted from the leaves of Erythroxylum coca**

Page 155: Cocaine was the first local anesthetic used by Carl Koller for anesthetizing the cornea. It is extracted from the leaves of Erythroxylum coca. It is a very potent vasoconstrictor and is the only ester not metabolized by pseudocholinesterase (it is metabolized in liver).

Topic: Individual Agents

**Q22. Answer: C — Procaine**

Page 155: Procaine, like other esters, is metabolized by pseudocholinesterase. It is the agent of choice in patients with history of malignant hyperthermia. In fact, historically it had been used for the treatment of malignant hyperthermia.

Topic: Individual Agents

**Q23. Answer: B — Lignocaine (xylocaine, lidocaine)**

Page 155: Lignocaine (xylocaine, lidocaine) is the most commonly used local anesthetic. It was first synthesized by Lofgren and first used by Gordh. Its solution is very stable, not even decomposed by boiling.

Topic: Individual Agents

**Q24. Answer: A — 4.5 mg/kg (max 300 mg) without adrenaline; 7 mg/kg (max 500 mg) with adrenaline**

Page 156: The maximum safe dose of lignocaine without adrenaline is 4.5 mg/kg (maximum 300 mg) and with adrenaline is 7 mg/kg (maximum 500 mg). Duration of effect without adrenaline is 45–60 minutes and with adrenaline is 2–3 hours.

Topic: Lignocaine

**Q25. Answer: A — 2 mg/kg**

Page 156: For treating ventricular tachycardia, preservative free lignocaine (available as xylocard 2%) is used intravenously in a dose of 2 mg/kg. Intravenous xylocard is also used for blunting the cardiovascular response to laryngoscopy and intubation, and lignocaine infusion is used for neuropathic pain.

Topic: Lignocaine

**Q26. Answer: B — Prilocaine**

Page 156: Prilocaine, other than liver, also has extrahepatic metabolism in kidneys, lungs and by amidase, making it the safest local anesthetic. Methemoglobinemia occurs with prilocaine at high doses, because of accumulation of its metabolite orthotoluidine which converts hemoglobin to methHb.

Topic: Prilocaine

**Q27. Answer: A — Wide differential sensory and motor blockade (motor block only at high concentrations)**

Page 156: The very unique property of bupivacaine is wide differential sensory and motor blockade, which means motor block will occur only at high concentrations, making it a local anesthetic of choice for postoperative pain relief and painless labor. It is 4 times more potent than lignocaine.

Topic: Bupivacaine

**Q28. Answer: A — 2.5 mg/kg (max 175 mg) without; 3 mg/kg (max 225 mg) with adrenaline**

Page 156: The maximum safe dose of bupivacaine without adrenaline is 2.5 mg/kg (maximum 175 mg) and with adrenaline is 3 mg/kg (maximum 225 mg). Addition of adrenaline only prolongs the duration of sensory block.

Topic: Bupivacaine

**Q29. Answer: B — Bier's block (IVRA)**

Page 157: Bupivacaine has high tissue binding (slow reversal of sodium channels) and high degree of protein binding which makes resuscitation after cardiac arrest prolonged and very difficult. Therefore, it is absolutely contraindicated for Bier's block.

Topic: Bupivacaine

**Q30. Answer: B — Amiodarone**

Page 157: Contrary to the previous recommendation of bretylium as the drug of choice, amiodarone nowadays is considered as the drug of choice for treatment of bupivacaine (and other local anesthetic) induced ventricular tachycardia. Management of cardiac arrest includes CPR along with rapid bolus of Intralipid 20%, 1.5 mL/kg.

Topic: Bupivacaine

**Q31. Answer: A — Because it does not contain the R isomer**

Page 157: Levobupivacaine is the S isomer of bupivacaine. As it does not contain the R isomer (which is responsible for cardiotoxicity), the cardiotoxicity of levobupivacaine is less than bupivacaine. Maximum safe dose of levobupivacaine is 2.5 mg/kg (maximum 175 mg).

Topic: Levobupivacaine

**Q32. Answer: A — Less than both bupivacaine and levobupivacaine (but still higher than lignocaine)**

Page 157: Cardiotoxicity and CNS toxicity of ropivacaine is less as compared to bupivacaine and even lesser than levobupivacaine (but still higher than lignocaine). Maximum safe dose is 3 mg/kg (maximum 225 mg). Since its cardiotoxicity is less, it can be given in higher doses.

Topic: Ropivacaine

**Q33. Answer: A — It is less potent and produces less motor block, while sensory analgesia is comparable**

Page 157: Ropivacaine is less potent than bupivacaine, therefore its duration and intensity of motor block will also be proportionately less, while sensory analgesia is comparable. Therefore, ropivacaine is always preferred over bupivacaine for postoperative analgesia and painless labor (also being less cardiotoxic).

Topic: Ropivacaine

**Q34. Answer: A — 5% prilocaine and 5% lignocaine in equal amount (1:1 ratio)**

Page 157: EMLA stands for eutectic (easily melted) mixture of local anesthetics. It is a combination of 5% prilocaine and 5% lignocaine in equal amount (1:1 ratio). It is used for intravenous cannulation in children or sensitive individuals, small skin procedures and removal of mole. The major limitation is its slow onset (30–60 min).

Topic: Methods of Local Anesthesia

**Q35. Answer: A — Mepivacaine, bupivacaine, levobupivacaine and ropivacaine**

Page 157: Mepivacaine, bupivacaine, levobupivacaine and ropivacaine do not have any absorption from skin or mucous membranes therefore cannot be used for topical anesthesia. Also, the absorption of procaine and chlorprocaine is so poor that they should not be used for surface anesthesia.

Topic: Methods of Local Anesthesia

**Q36. Answer: B — 0.5% lignocaine**

Page 156, 158: 0.5% lignocaine is used for Bier's block (intravenous regional analgesia). Spinal uses 5% heavy lignocaine, nerve blocks 1–2%, surface (topical) 4% and 10%, and infiltration 1–2%.

Topic: Methods of Local Anesthesia

**Q37. Answer: A — Pregnancy, hypoxia and acidosis**

Page 156: Cardiotoxicity of bupivacaine increases in pregnancy, hypoxia and acidosis. It is far more cardiotoxic than lignocaine, with the R component contributing more than the S component in causing cardiotoxicity.

Topic: Bupivacaine

**Q38. Answer: A — Lignocaine and tetracaine**

Page 155: The classical example of local anesthetic induced local neurotoxicity is cauda equina syndrome seen with lignocaine and tetracaine if used through small bore spinal catheters. Transient neurologic symptoms (dysesthesia, radicular pain in lower extremities) are seen with lignocaine and mepivacaine. Maximum local neurotoxicity has been reported with chlorprocaine.

Topic: Local Toxicity

**Q39. Answer: A — Preservative free preparations**

Page 155: Antimicrobial preservative (methylparaben) is added to multidose vials. For spinal and epidural anesthesia, preservative free preparations should be used.

Topic: Commercial Preparations

**Q40. Answer: A — Peak concentration may occur in postoperative period producing toxicity**

Page 158: Tumescence anesthesia is used by plastic surgeons during liposuction, where dilute combination of lignocaine with adrenaline is injected subcutaneously in large quantities. Peak concentration may even occur in postoperative period producing toxicity.

Topic: Methods of Local Anesthesia

**Q41. Answer: A — Cryoprobes delivering CO<sub>2</sub> or nitrous oxide**

Page 158: In refrigeration analgesia, nerve conduction is blocked by cooling the nerves using cryoprobes which deliver CO<sub>2</sub> or nitrous oxide at –5 to –20°C.

Topic: Methods of Local Anesthesia

**Q42. Answer: A — Additive (not independent), therefore maximum safe doses should be reduced accordingly**

Page 155: The use of two local anesthetics is quite common in clinical practice (usually lignocaine for rapid onset and bupivacaine/ropivacaine for long duration). The toxicity of two local anesthetics should be considered additive, not independent, and therefore maximum safe doses should be reduced accordingly.

Topic: Toxicity

**Q43. Answer: A — Produce only sensory block, while at higher concentration produce motor block**

Page 154: Differential block depends on concentration. A drug at lower concentration will produce only sensory block while at higher concentration can produce motor block.

Topic: Differential Block

**Q44. Answer: C — Dibucaine (cinchocaine)**

Page 157, Table 15.2: Dibucaine (cinchocaine) is longest acting, most potent and most toxic local anesthetic.

Topic: Individual Agents

**Q45. Answer: A — Liver**

Page 154–155: Esters are metabolized by plasma pseudocholinesterase except cocaine which is metabolized in liver. Cocaine is the only ester not metabolized by pseudocholinesterase. Its metabolite ecgonine is also a CNS stimulant.

Topic: Metabolism

**Q46. Answer: A — Acidic pH (causing ionization)**

Page 154: Concentration minimum (Cm) is the minimum concentration of local anesthetic that will block nerve impulse conduction. Acidic pH increases the Cm by causing ionization. Stimulated nerve will need less Cm. Hypokalemia and hypercalcemia antagonize the block.

Topic: Concentration Minimum

**Q47. Answer: A — Temperature is blocked earlier than pain**

Page 153: The sequence of sensory blockade is pain > temperature (cold before hot) > touch > deep pressure > proprioception, EXCEPT for central neuraxial blocks where temperature is blocked earlier than pain.

Topic: Sensitivity of Nerve Fiber

**Q48. Answer: A — Inappropriate technique, inadequate volume or concentration**

Page 158: The usual cause of failure of local anesthesia is inappropriate technique, inadequate volume or concentration of local anesthetic. However, certain factors like infection (acidosis), genetic variation, or drug tolerance and tachyphylaxis can also contribute to failure.

Topic: Causes of Failure

**Q49. Answer: A — Because adrenaline can become unstable at alkaline pH**

Page 155: Local anesthetics are made acidic (pH around 6) to enhance chemical stability. Local anesthetics containing adrenaline are made even more acidic (pH around 4) because adrenaline can become unstable at alkaline pH.

Topic: Commercial Preparations

**Q50. Answer: B — Buprenorphine**

Page 151: Among other drugs with local anesthetic properties, buprenorphine — because of its local anesthetic properties — is frequently used to supplement the effect of local anesthetics for nerve blocks. Pethidine and tramadol also have local anesthetic properties.

Topic: Other Drugs with LA Properties

# Peripheral Nerve Blocks MCQ – Chapter 16

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 6: Regional Anesthesia (Chapter 16)

**25 Questions • 20 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

**Q1. What has revolutionized the practice of nerve block in modern day anesthetic practice (compared to the conventional method of eliciting paresthesia or using a nerve stimulator)?**

- A. Use of ultrasound
- B. Use of fluoroscopy
- C. Use of CT guidance
- D. Use of MRI

**Ans: A**

**Q2. Brachial plexus block is the second most commonly performed block. Which is the most commonly performed block?**

- A. Central neuraxial block
- B. Intercostal nerve block
- C. Axillary block
- D. Ankle block

**Ans: A**

**Q3. The interscalene approach is unsuitable for hand surgeries because which nerve is usually spared?**

- A. Radial nerve
- B. Ulnar nerve
- C. Median nerve
- D. Musculocutaneous nerve

**Ans: B**

**Q4. Phrenic nerve block occurs in almost all patients with the interscalene approach. Why does unilateral phrenic nerve block usually have no clinically significant consequences?**

- A. Because it is compensated by the other side
- B. Because pulmonary functions are only reduced by 20–25% in a normal individual
- C. Because the diaphragm is not affected
- D. Because phrenic nerve is not blocked

**Ans: B**

**Q5. The supraclavicular approach to brachial plexus block is performed with the needle inserted in which relationship to the subclavian artery (Fig. 16.1)?**

- A. Lateral to the subclavian artery
- B. Medial to the subclavian artery
- C. Directly over the artery
- D. Below the first rib

**Ans: A**

**Q6. What is the most bothersome complication of the supraclavicular block, with an incidence of 1–6%?**

- A. Pneumothorax
- B. Horner syndrome
- C. Phrenic nerve block
- D. Hematoma

**Ans: A**

|                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. The advantage of the axillary approach over the supraclavicular approach is that:</b></p> <ul style="list-style-type: none"> <li>A. Complications like pneumothorax, phrenic nerve block and Horner syndrome can be avoided</li> <li>B. It blocks all nerves including musculocutaneous</li> <li>C. It is suitable for shoulder surgery</li> <li>D. It has no chance of intravascular injection</li> </ul>     | <p><b>Ans: A</b></p> |
| <p><b>Q8. A wrist block is used for hand surgeries. Where is the ulnar nerve blocked in a wrist block?</b></p> <ul style="list-style-type: none"> <li>A. Just lateral to flexor carpi ulnaris tendon</li> <li>B. Between Palmaris longus and flexor carpi radialis</li> <li>C. Posterior to extensor carpi radialis tendon</li> <li>D. At the cubital fossa</li> </ul>                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q9. Intravenous regional block (Bier block) was first given by August Bier. Which local anesthetic is absolutely contraindicated for it?</b></p> <ul style="list-style-type: none"> <li>A. Lignocaine</li> <li>B. Prilocaine</li> <li>C. Bupivacaine</li> <li>D. Procaine</li> </ul>                                                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q10. Which is an advantage of the Bier block?</b></p> <ul style="list-style-type: none"> <li>A. Long duration of action</li> <li>B. Rapid onset (within 5 minutes), easy procedure, almost no chance of failure</li> <li>C. No tourniquet required</li> <li>D. Provides postoperative analgesia</li> </ul>                                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q11. In the Bier block, the tourniquet cannot be released before how long, even if the surgery finishes before it?</b></p> <ul style="list-style-type: none"> <li>A. 30 minutes</li> <li>B. 10 minutes</li> <li>C. 1 hour</li> <li>D. 5 minutes</li> </ul>                                                                                                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q12. Blocks of the lower limb are rarely used for providing anesthesia. The perivascular block (3 in 1 block) blocks which three nerves simultaneously?</b></p> <ul style="list-style-type: none"> <li>A. Femoral, sciatic and obturator nerves</li> <li>B. Radial, ulnar and median nerves</li> <li>C. Tibial, peroneal and sural nerves</li> <li>D. Pudendal, ilioinguinal and iliohypogastric nerves</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q13. An ankle block is performed for foot surgeries. How many nerves are blocked in an ankle block?</b></p> <ul style="list-style-type: none"> <li>A. Three nerves</li> <li>B. Four nerves</li> <li>C. Five nerves (deep peroneal, superficial peroneal, saphenous, posterior tibial, sural)</li> <li>D. Two nerves</li> </ul>                                                                                     | <p><b>Ans: C</b></p> |



|                                                                                                                                                                                                                                                                                                                 |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q14. Cervical plexus block is best utilized for which surgery, where the patient's awake state is the best monitor to assess neurologic status?</b></p> <p>A. Carotid endarterectomy<br/>B. Thyroidectomy only<br/>C. Hernia repair<br/>D. Breast surgery</p>                                             | <p><b>Ans: A</b></p> |
| <p><b>Q15. The phrenic nerve block (3 cm above the clavicle just lateral to the posterior border of sternocleidomastoid) is sometimes performed for:</b></p> <p>A. Intractable hiccups<br/>B. Hernia repair<br/>C. Carotid endarterectomy<br/>D. Penile surgery</p>                                             | <p><b>Ans: A</b></p> |
| <p><b>Q16. Ilioinguinal and iliohypogastric nerve blocks are performed for hernia repair. Where are they blocked?</b></p> <p>A. 3 cm medial to anterior superior iliac spine<br/>B. At the base of the penis<br/>C. Lateral to subclavian artery<br/>D. At the posterior angle of rib</p>                       | <p><b>Ans: A</b></p> |
| <p><b>Q17. For a penile block, the pudendal nerve is blocked by injection of 10–15 mL of local anesthetic at the base of penis. Which preparation is contraindicated?</b></p> <p>A. Xylocaine with adrenaline<br/>B. Plain lignocaine<br/>C. Bupivacaine<br/>D. Prilocaine</p>                                  | <p><b>Ans: A</b></p> |
| <p><b>Q18. The intercostal nerve block is most often performed for which indication?</b></p> <p>A. Pain relief for rib fractures and post-herpetic neuropathies<br/>B. Hand surgery<br/>C. Foot surgery<br/>D. Carotid endarterectomy</p>                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q19. The most bothersome complications of the intercostal nerve block are pneumothorax and:</b></p> <p>A. Risk of systemic local anesthetic toxicity (maximum systemic absorption occurs after intercostal nerve block)<br/>B. Horner syndrome<br/>C. Phrenic nerve block<br/>D. Compartment syndrome</p> | <p><b>Ans: A</b></p> |
| <p><b>Q20. Which is an absolute contraindication for peripheral nerve blocks?</b></p> <p>A. Coagulopathy or being on anticoagulants (except aspirin) and infection at the site of needle placement<br/>B. Hypertension<br/>C. Diabetes<br/>D. Old age</p>                                                       | <p><b>Ans: A</b></p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                   |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q21. In the axillary approach, which nerves are spared, requiring them to be blocked separately?</b></p> <ul style="list-style-type: none"> <li>A. Musculocutaneous and intercostobrachial nerves</li> <li>B. Ulnar and median nerves</li> <li>C. Radial and ulnar nerves</li> <li>D. Phrenic and vagus nerves</li> </ul>                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q22. Why is the infraclavicular route for blocking the brachial plexus not utilized routinely?</b></p> <ul style="list-style-type: none"> <li>A. Due to high failure rate in coracoid approach and increased incidence of pneumothorax and hematoma in classical approach</li> <li>B. Because it cannot block any nerve</li> <li>C. Because it requires general anesthesia</li> <li>D. Because it is too painful</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q23. The paravertebral block, if large volumes are used, can spread to which spaces?</b></p> <ul style="list-style-type: none"> <li>A. Superior and inferior paravertebral spaces, intercostal spaces, and epidural space</li> <li>B. Only the intercostal space</li> <li>C. Only the subarachnoid space</li> <li>D. Only the pleural space</li> </ul>                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q24. The supraclavicular approach is usually employed for surgeries on which area?</b></p> <ul style="list-style-type: none"> <li>A. Lower arm, elbow, forearm and hand</li> <li>B. Shoulder only</li> <li>C. Upper arm only</li> <li>D. Neck only</li> </ul>                                                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q25. The fascia iliaca block (modified femoral block) is usually performed under ultrasound guidance, but can be performed blindly with which technique?</b></p> <ul style="list-style-type: none"> <li>A. Double pop technique (first pop of fascia lata followed by second pop of fascia iliaca)</li> <li>B. Single pop technique</li> <li>C. Loss of resistance to air</li> <li>D. Hanging drop technique</li> </ul>     | <p><b>Ans: A</b></p> |

# Answer Key & Explanations

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## **Q1. Answer: A — Use of ultrasound**

Page 160: The conventional method of giving a nerve block is by eliciting paresthesia and/or using the nerve stimulator. However, the use of ultrasound has revolutionized the practice of nerve block in modern day anesthetic practice, making it technically easy and improving precision.

*Topic: Technique*

## **Q2. Answer: A — Central neuraxial block**

Page 160: Brachial plexus block is the second most commonly performed block (after central neuraxial block). It is used for surgery of upper limb and shoulder, and can be blocked by 4 approaches — interscalene, supraclavicular, infraclavicular and axillary.

*Topic: Brachial Plexus Block*

## **Q3. Answer: B — Ulnar nerve**

Page 160: The interscalene approach blocks the brachial plexus between anterior and middle scalene at the level of cricoid. Blockade occurs at the level of superior and middle trunks. Ulnar nerve is usually spared with this approach, making it unsuitable for hand surgeries.

*Topic: Brachial Plexus Block*

## **Q4. Answer: B — Because pulmonary functions are only reduced by 20–25% in a normal individual**

Page 160: Phrenic nerve block occurs in almost all patients with interscalene approach, but unilateral phrenic nerve block has no clinically significant consequences in a normal individual (pulmonary functions are only reduced by 20–25%).

*Topic: Brachial Plexus Block*

## **Q5. Answer: A — Lateral to the subclavian artery**

Page 160–161, Fig. 16.1: In the supraclavicular approach, the needle is inserted lateral to subclavian vessels. As the dome of pleura is on the medial side, keeping the direction of needle laterally is the best way to avoid pneumothorax.

*Topic: Brachial Plexus Block*

## **Q6. Answer: A — Pneumothorax**

Page 161: Pneumothorax is the most bothersome complication of supraclavicular block. Incidence is 1–6%, but fortunately in more than 98% of the cases pneumothorax is small enough to require any intervention. Keeping the direction of needle lateral to the subclavian artery is the best way to avoid pneumothorax.

*Topic: Brachial Plexus Block*

## **Q7. Answer: A — Complications like pneumothorax, phrenic nerve block and Horner syndrome can be avoided**

Page 161: The advantage of axillary approach over supraclavicular is that complications like pneumothorax, phrenic nerve block and Horner syndrome can be avoided. However, musculocutaneous and intercostobrachial nerves are spared, making it not suitable for arm surgery. Intravascular injection and hematoma formation are high.

*Topic: Brachial Plexus Block*

## **Q8. Answer: A — Just lateral to flexor carpi ulnaris tendon**

Page 161: Wrist block is used for hand surgeries where ulnar nerve is blocked just lateral to flexor carpi ulnaris tendon, median nerve between tendons of Palmaris longus and flexor carpi radialis, and radial nerve by 8–10 mL subcutaneous injection extending from radial artery anteriorly to extensor carpi radialis tendon posteriorly.

*Topic: Individual Nerve Blocks*

**Q9. Answer: C — Bupivacaine**

Page 161: In Bier block, accidental deflation or leak of tourniquet can cause drug toxicity and death; therefore, a toxic drug like bupivacaine is absolutely contraindicated for Bier block. 30–40 mL of 0.5% lignocaine (xylocard) or prilocaine is used.

Topic: Intravenous Regional Block (Bier Block)

**Q10. Answer: B — Rapid onset (within 5 minutes), easy procedure, almost no chance of failure**

Page 161: Advantages of Bier block include easy procedure, almost no chances of failure, rapid onset (within 5 minutes), and good muscle relaxation. Disadvantages include tourniquet discomfort or compartment syndrome, short duration, and the tourniquet cannot be released before 30 minutes (even if surgery finishes before it).

Topic: Intravenous Regional Block (Bier Block)

**Q11. Answer: A — 30 minutes**

Page 162: In Bier block, the tourniquet cannot be released before 30 minutes (even if the surgery finishes before it), because accidental deflation or leak of tourniquet can cause drug toxicity and even death.

Topic: Intravenous Regional Block (Bier Block)

**Q12. Answer: A — Femoral, sciatic and obturator nerves**

Page 162: The perivascular block (3 in 1 block) involves drug injected in femoral canal while maintaining distal pressure, resulting in upward spread of drug, blocking femoral, sciatic and obturator nerves simultaneously (that is why called as 3 in 1 block).

Topic: Blocks of Lower Limb

**Q13. Answer: C — Five nerves (deep peroneal, superficial peroneal, saphenous, posterior tibial, sural)**

Page 162: Ankle block is performed for foot surgeries, where five nerves are blocked — deep peroneal, superficial peroneal and saphenous nerve with subcutaneous infiltration at the dorsum of foot, posterior tibial posterior to medial malleolus, and sural laterally between lateral malleolus and achilles tendon. As it requires multiple injections, ankle block can be painful. Lignocaine with adrenaline should not be used for ankle block.

Topic: Blocks of Lower Limb

**Q14. Answer: A — Carotid endarterectomy**

Page 162: As patient's awake state is the best monitor to assess the neurologic status, cervical plexus block is best utilized for carotid endarterectomy. In selected cases, it is possible to do tracheostomy and thyroidectomy with bilateral cervical plexus block.

Topic: Blocks of Head and Neck

**Q15. Answer: A — Intractable hiccups**

Page 162: Phrenic nerve block is sometimes performed for intractable hiccups; the nerve is blocked 3 cm above the clavicle just lateral to the posterior border of sternocleidomastoid.

Topic: Blocks of Head and Neck

**Q16. Answer: A — 3 cm medial to anterior superior iliac spine**

Page 162: Ilioinguinal and iliohypogastric nerve blocks are performed for hernia repair; these are blocked at a point 3 cm medial to anterior superior iliac spine.

Topic: Blocks of Head and Neck

**Q17. Answer: A — Xylocaine with adrenaline**

Page 162: Penile block — pudendal nerve is blocked by injection of 10–15 mL of local anesthetic at the base of penis. Xylocaine with adrenaline is contraindicated for penile block (because the penis has end arteries).

Topic: Blocks of Head and Neck

**Q18. Answer: A — Pain relief for rib fractures and post-herpetic neuropathies**

Page 163: Intercostal nerve block is most often performed for pain relief for rib fractures and post-herpetic neuropathies; however, it can be used for chest tube insertion or doing thoracoscopies. Block is given at the lower border of rib, usually at the posterior angle of rib.

Topic: Intercostal Nerve Block

**Q19. Answer: A — Risk of systemic local anesthetic toxicity (maximum systemic absorption occurs after intercostal nerve block)**

Page 163: The most bothersome complications of intercostal nerve block may be pneumothorax and risk of systemic local anesthetic toxicity (maximum systemic absorption occurs after intercostal nerve block).

Topic: Intercostal Nerve Block

**Q20. Answer: A — Coagulopathy or being on anticoagulants (except aspirin) and infection at the site of needle placement**

Page 163: Contraindications for peripheral nerve blocks include patient suffering from coagulopathy or on anticoagulants (except aspirin), infection at the site of needle placement, known case of drug allergy to local anesthetics, and pre-existing neuropathy (avoid regional anesthesia to avoid medicolegal issues).

Topic: Contraindications

**Q21. Answer: A — Musculocutaneous and intercostobrachial nerves**

Page 161: In the axillary approach, musculocutaneous and intercostobrachial nerves are spared, therefore axillary block is not suitable for arm surgery. Musculocutaneous and intercostobrachial nerves have to be blocked separately.

Topic: Brachial Plexus Block

**Q22. Answer: A — Due to high failure rate in coracoid approach and increased incidence of pneumothorax and hematoma in classical approach**

Page 161: In the infraclavicular approach, the theoretical advantage is that musculocutaneous and axillary nerve can be blocked. However, due to high failure rate in coracoid approach and increased incidence of pneumothorax and hematoma in classical approach, the infraclavicular route is not utilized routinely.

Topic: Brachial Plexus Block

**Q23. Answer: A — Superior and inferior paravertebral spaces, intercostal spaces, and epidural space**

Page 162–163: In paravertebral block, each spinal nerve is blocked just after it exits from intervertebral foramen. It can be utilized for breast, thoracic or abdominal surgeries. If large volumes are used, the drug may spread laterally into superior and inferior paravertebral spaces, superior and inferior intercostal spaces, and medially along with the nerve sheath into epidural space.

Topic: Blocks of Head and Neck

**Q24. Answer: A — Lower arm, elbow, forearm and hand**

Page 160: The supraclavicular approach is very commonly used and is usually employed for surgeries on lower arm, elbow, forearm and hand. Distal trunk and proximal division of brachial plexus is blocked by this approach.

Topic: Brachial Plexus Block

**Q25. Answer: A — Double pop technique (first pop of fascia lata followed by second pop of fascia iliaca)**

Page 162: Fascia iliaca nerve (modified femoral) block — a large volume of drug is injected just below the fascia iliaca to block femoral nerve as it traverses the fascia iliaca. Usually performed under ultrasound guidance; however, it can be performed blindly with double pop technique (first pop of fascia lata followed by second pop of fascia iliaca).

Topic: Blocks of Lower Limb

# Central Neuraxial Blocks (Spinal and Epidural Anesthesia) MCQ – Chapter 17

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 6: Regional Anesthesia (Chapter 17)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

**Q1. The vertebral column consists of how many vertebrae?**

- A. 33 (7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral, 4 fused coccygeal)
- B. 31
- C. 30
- D. 26

**Ans: A**

**Q2. Which is the correct order of structures encountered (posterior to anterior) while giving spinal anesthesia?**

- A. Skin → subcutaneous → supraspinous ligament → interspinous ligament → ligamentum flavum → dura → arachnoid
- B. Skin → dura → ligamentum flavum → arachnoid
- C. Skin → ligamentum flavum → supraspinous ligament → dura
- D. Skin → arachnoid → dura → ligamentum flavum

**Ans: A**

**Q3. The line joining the highest point on the iliac crest (intercristal line / Tuffier's line) corresponds to which level?**

- A. L4 and L5 interspace or L4 spine
- B. L2 and L3 interspace
- C. T12 and L1 interspace
- D. S1 spine

**Ans: A**

**Q4. The epidural veins form a plexus called the plexus of Batson. When engorged (e.g. in pregnancy), they push the dura medially, which:**

- A. Reduces the subarachnoid space, leading to higher spread of local anesthetic
- B. Increases the subarachnoid space
- C. Has no effect on spread
- D. Blocks CSF circulation

**Ans: A**

**Q5. The spinal cord extends from the medulla oblongata to which level in adults and infants?**

- A. Lower border of L1 in adults and L3 in infants
- B. L3 in adults and L5 in infants
- C. T12 in adults and L1 in infants
- D. L5 in adults and S1 in infants

**Ans: A**

**Q6. The anterior spinal artery is reinforced by which important artery?**

- A. Artery of Adamkiewicz (arteria radicularis magna)
- B. Posterior inferior cerebellar artery
- C. Vertebral artery
- D. Spinal branch of intercostal artery

**Ans: A**

|                                                                                                                                                                                                                                                                              |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. What is the volume of CSF at one time, and how is it distributed?</b></p> <p>A. 140 mL; half in cranium and half in spinal canal<br/> B. 500 mL; all in spinal canal<br/> C. 50 mL; all in cranium<br/> D. 250 mL; entirely in ventricles</p>                      | <p><b>Ans: A</b></p> |
| <p><b>Q8. Several meta-analyses have shown a reduced risk in overall mortality and morbidity by how much in patients undergoing surgeries under central neuraxial blocks compared to general anesthesia?</b></p> <p>A. As high as 30%<br/> B. 5%<br/> C. 50%<br/> D. 10%</p> | <p><b>Ans: A</b></p> |
| <p><b>Q9. What is the most prominent cardiovascular effect of central neuraxial blocks?</b></p> <p>A. Hypertension<br/> B. Hypotension<br/> C. Tachycardia<br/> D. Increased cardiac output</p>                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q10. What is the sequence of blockade in central neuraxial blocks?</b></p> <p>A. Autonomic &gt; sensory &gt; motor<br/> B. Motor &gt; sensory &gt; autonomic<br/> C. Sensory &gt; motor &gt; autonomic<br/> D. Motor &gt; autonomic &gt; sensory</p>                   | <p><b>Ans: A</b></p> |
| <p><b>Q11. How is the autonomic level tested in central neuraxial blocks?</b></p> <p>A. By temperature (cold)<br/> B. By pin prick<br/> C. By toe movement<br/> D. By touch</p>                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q12. What is the most common postoperative complication of central neuraxial blocks in present day practice?</b></p> <p>A. Post-spinal headache<br/> B. Urinary retention<br/> C. Paraplegia<br/> D. Meningitis</p>                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q13. Spinal anesthesia is usually given in which space in adults?</b></p> <p>A. L3-4 (or L4-5)<br/> B. T12-L1<br/> C. L1-2<br/> D. S1-2</p>                                                                                                                            | <p><b>Ans: A</b></p> |



|                                                                                                                                                                                                                                                                                                                                                                    |                      |
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| <p><b>Q14. Which is the most commonly used drug for spinal anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Lignocaine 5%</li> <li>B. Hyperbaric bupivacaine 0.5% (made hyperbaric by 8% dextrose)</li> <li>C. Tetracaine 1%</li> <li>D. Ropivacaine 0.5%</li> </ul>                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q15. Which two factors significantly affect the level (height) of spinal block?</b></p> <ul style="list-style-type: none"> <li>A. Dose and baricity</li> <li>B. Speed of injection and sex</li> <li>C. CSF pressure and addition of vasoconstrictor</li> <li>D. Barbotage and coughing</li> </ul>                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q16. CSF volume has what kind of relation with the level of block?</b></p> <ul style="list-style-type: none"> <li>A. Inverse relation (decreased CSF volume leads to higher spread)</li> <li>B. Direct relation</li> <li>C. No relation</li> <li>D. Exponential relation</li> </ul>                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q17. Pencil tip point end spinal needles (e.g. Whitacre, Sprotte) are dura separating. What is their advantage?</b></p> <ul style="list-style-type: none"> <li>A. Less damage to dural fibers and hence less incidence of postspinal headache</li> <li>B. They are cheaper</li> <li>C. Faster onset of block</li> <li>D. Longer duration of block</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q18. What is the most common complication of spinal anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Hypotension</li> <li>B. Bradycardia</li> <li>C. Post-spinal headache</li> <li>D. Cardiac arrest</li> </ul>                                                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q19. Which is the most preferred vasopressor for the treatment of spinal hypotension?</b></p> <ul style="list-style-type: none"> <li>A. Ephedrine</li> <li>B. Phenylephrine</li> <li>C. Methoxamine</li> <li>D. Dopamine</li> </ul>                                                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q20. Total spinal anesthesia refers to:</b></p> <ul style="list-style-type: none"> <li>A. When the drug reaches up to the cranium and involves the cranial nerves</li> <li>B. A block extending up to T4</li> <li>C. Bilateral motor block of the legs</li> <li>D. A failed spinal block</li> </ul>                                                          | <p><b>Ans: A</b></p> |

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| <p><b>Q21. Which is the drug of choice for treatment of shivering after spinal anesthesia?</b></p> <p>A. Tramadol<br/>B. Pethidine<br/>C. Clonidine<br/>D. Ondansetron</p>                                                                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q22. Post-spinal headache is a low pressure headache due to seepage of CSF. What is the typical postural feature?</b></p> <p>A. Pain in sitting/standing position which gets relieved in lying down<br/>B. Pain present in all positions<br/>C. Pain only in lying down position<br/>D. Pain unrelated to posture</p> | <p><b>Ans: A</b></p> |
| <p><b>Q23. What is the single most important factor in the etiology of post-spinal headache?</b></p> <p>A. Needle size<br/>B. Altitude<br/>C. Hydration status<br/>D. Age of patient</p>                                                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q24. What is the definitive (3rd line) method of treatment of post-spinal headache?</b></p> <p>A. Autologous epidural blood patch<br/>B. Caffeine in Ringer lactate<br/>C. Intravenous fluids<br/>D. Abdominal binders</p>                                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q25. Which cranial nerve is most commonly (90%) involved after spinal anesthesia (due to the longest intracranial course)?</b></p> <p>A. 3rd cranial nerve<br/>B. 6th cranial nerve<br/>C. 7th cranial nerve<br/>D. 8th cranial nerve</p>                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q26. Which organism has been cited as the most common cause of meningitis after spinal anesthesia?</b></p> <p>A. Staphylococcus epidermidis<br/>B. Streptococcus viridans (an oral flora)<br/>C. Escherichia coli<br/>D. Pseudomonas</p>                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q27. Cauda equina syndrome (CES), a classical example of local anesthetic neurotoxicity, is usually seen with which agent given as continuous spinal with small bore catheters?</b></p> <p>A. Bupivacaine<br/>B. Lignocaine<br/>C. Ropivacaine<br/>D. Tetracaine only</p>                                             | <p><b>Ans: B</b></p> |

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| <p><b>Q28. Transient neurological symptoms (TNS) can occur with any local anesthetic but most often with lignocaine. What is the incidence with lignocaine?</b></p> <p>A. 10–11%<br/>B. 1–2%<br/>C. 30–40%<br/>D. 50%</p>                                                                                                                                                                                      | Ans: A |
| <p><b>Q29. Epidural hematoma leading to paraplegia is rare (1:200,000) but occurs in patients on anticoagulants. What is the diagnosis and management?</b></p> <p>A. Confirmed by MRI; immediate neurosurgical intervention to drain the hematoma<br/>B. Confirmed by X-ray; conservative management<br/>C. Confirmed by CSF analysis; antibiotics<br/>D. No treatment needed as it resolves spontaneously</p> | Ans: A |
| <p><b>Q30. Direct trauma to the spinal cord is avoided by giving spinal below which level in adults?</b></p> <p>A. Below L1 (and in children up to 2 years, below L3)<br/>B. Below T12<br/>C. Below L5<br/>D. Below S1</p>                                                                                                                                                                                     | Ans: A |
| <p><b>Q31. What is the most common side effect of intrathecal opioids?</b></p> <p>A. Pruritus<br/>B. Respiratory depression<br/>C. Nausea<br/>D. Urinary retention</p>                                                                                                                                                                                                                                         | Ans: A |
| <p><b>Q32. A saddle block is given in which position and is utilized for which surgeries?</b></p> <p>A. Sitting position (patient seated 5–10 minutes till drug gets fixed); perianal surgeries<br/>B. Lateral position; lower limb surgeries<br/>C. Prone position; spine surgeries<br/>D. Trendelenburg position; abdominal surgeries</p>                                                                    | Ans: A |
| <p><b>Q33. Which is the most commonly used epidural needle?</b></p> <p>A. Tuohy needle (blunt bevel with curve of 15 to 30 degrees at tip, called Huber tip)<br/>B. Quincke-Babcock needle<br/>C. Whitacre needle<br/>D. Crawford needle</p>                                                                                                                                                                   | Ans: A |
| <p><b>Q34. What is the most commonly used method to locate the epidural space?</b></p> <p>A. Loss of resistance technique<br/>B. Hanging drop technique (Gutierrez's sign)<br/>C. Free flow of CSF<br/>D. Nerve stimulation</p>                                                                                                                                                                                | Ans: A |

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| <p><b>Q35. How much local anesthetic is required for blocking 1 segment in epidural anesthesia?</b></p> <p>A. 2–3 mL<br/>B. 0.5 mL<br/>C. 5–6 mL<br/>D. 10 mL</p>                                                                                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q36. What is the most commonly used combination for postoperative analgesia and painless labor through an epidural catheter?</b></p> <p>A. Fentanyl + bupivacaine or ropivacaine (ropivacaine preferred) as continuous infusion<br/>B. Morphine alone<br/>C. Lignocaine with adrenaline<br/>D. Bupivacaine alone</p> | <p><b>Ans: A</b></p> |
| <p><b>Q37. Which local anesthetic is the most preferred for epidural anesthesia because of its high safety profile?</b></p> <p>A. Ropivacaine<br/>B. Bupivacaine<br/>C. Lignocaine<br/>D. Levobupivacaine</p>                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q38. What is the most important factor affecting the spread (level) of block in epidural anesthesia?</b></p> <p>A. Volume of the drug<br/>B. Concentration of the drug<br/>C. Speed of injection<br/>D. Baricity</p>                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q39. What is a key advantage of epidural over spinal anesthesia?</b></p> <p>A. Less hypotension (slow onset gives body time to compensate), making epidural a better choice for cardiac compromised patients<br/>B. Easier technically<br/>C. Denser block<br/>D. Less expensive</p>                                 | <p><b>Ans: A</b></p> |
| <p><b>Q40. The chance of epidural hematoma formation compared to spinal is:</b></p> <p>A. 10 fold higher than spinal (as the veins are denser laterally, epidural should be given in midline)<br/>B. Equal to spinal<br/>C. Less than spinal<br/>D. Half of spinal</p>                                                     | <p><b>Ans: A</b></p> |

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| <p><b>Q41. In combined spinal epidural (CSE) anesthesia, a special spinal needle (120 mm) is passed through the epidural needle. What is its purpose?</b></p> <ul style="list-style-type: none"> <li>A. To allow spinal anesthesia for surgery and the epidural catheter for prolonging/extending the block and postoperative analgesia</li> <li>B. To increase the level of block</li> <li>C. To prevent hypotension</li> <li>D. To avoid post-spinal headache</li> </ul>                                     | <p><b>Ans: A</b></p> |
| <p><b>Q42. Caudal block (epidural sacral block) is commonly utilized in children, with drug injected in the sacral hiatus. For which surgeries is it most often used?</b></p> <ul style="list-style-type: none"> <li>A. Perianal, genital (circumcision), urethral (urethroplasty) and lower abdominal (herniotomy) surgeries</li> <li>B. Upper abdominal surgeries</li> <li>C. Thoracic surgeries</li> <li>D. Neck surgeries</li> </ul>                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q43. Which is an ABSOLUTE contraindication for central neuraxial blocks?</b></p> <ul style="list-style-type: none"> <li>A. Raised intracranial tension (medullary coning can cause death)</li> <li>B. Mild hypotension</li> <li>C. Chronic backache</li> <li>D. Mild fixed cardiac output lesion</li> </ul>                                                                                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q44. What platelet count allows central neuraxial blocks to be safely given, and what count is an absolute contraindication?</b></p> <ul style="list-style-type: none"> <li>A. Safe if &gt; 80,000/cubic mm; absolute contraindication if &lt; 50,000/cubic mm</li> <li>B. Safe if &gt; 50,000; absolute contraindication if &lt; 20,000</li> <li>C. Safe if &gt; 100,000; absolute contraindication if &lt; 80,000</li> <li>D. Safe if &gt; 40,000; absolute contraindication if &lt; 10,000</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q45. Regarding low molecular weight heparin (LMWH) and central neuraxial blocks, it is considered safe to give the block:</b></p> <ul style="list-style-type: none"> <li>A. 12 hours after the last dose of LMWH if prophylactic (and 24 hours if therapeutic dose)</li> <li>B. Immediately after LMWH</li> <li>C. 1 hour after LMWH</li> <li>D. 48 hours after LMWH always</li> </ul>                                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q46. Patients on which antiplatelet agents can be safely given central neuraxial blocks without delay?</b></p> <ul style="list-style-type: none"> <li>A. Aspirin and NSAIDs</li> <li>B. Clopidogrel</li> <li>C. Prasugrel</li> <li>D. Ticlopidine</li> </ul>                                                                                                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q47. What level of block is required for a cesarean section?</b></p> <ul style="list-style-type: none"> <li>A. Up to T4</li> <li>B. Up to T10</li> <li>C. Up to T6</li> <li>D. Sacral segments only</li> </ul>                                                                                                                                                                                                                                                                                           | <p><b>Ans: A</b></p> |

**Q48. According to Table 17.3, what is the quality of block in spinal versus epidural anesthesia?**

- A. Spinal: more dense; Epidural: patchy block is common
- B. Spinal: patchy; Epidural: more dense
- C. Both produce equally dense blocks
- D. Both produce patchy blocks

**Ans: A**

**Q49. Apnea after spinal anesthesia is usually due to:**

- A. Severe hypotension causing medullary ischemia
- B. Direct phrenic nerve block
- C. Local anesthetic toxicity
- D. Bronchospasm

**Ans: A**

**Q50. Why is bradycardia common after spinal anesthesia?**

- A. Due to decreased atrial pressure (Bainbridge reflex), parasympathetic over dominance, and direct inhibition of cardioaccelerator fibers (T1 to T4)
- B. Due to direct stimulation of the SA node
- C. Due to hypertension
- D. Due to increased venous return

**Ans: A**

# Answer Key & Explanations

## Q1. Answer: A — 33 (7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral, 4 fused coccygeal)

Page 164: Vertebral column consists of 33 vertebrae: 7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral and 4 fused coccygeal. It has 4 curves: thoracic and sacral spine are convex posteriorly (kyphotic) while cervical and lumbar spine are convex anteriorly (lordotic).

Topic: Applied Anatomy

## Q2. Answer: A — Skin → subcutaneous → supraspinous ligament → interspinous ligament → ligamentum flavum → dura → arachnoid

Page 164, Fig. 17.1: Structures encountered while giving spinal anesthesia from posterior to anterior are skin, subcutaneous tissue, supraspinous ligament, interspinous ligament, ligamentum flavum (yellow ligament), dura, and arachnoid.

Topic: Applied Anatomy

## Q3. Answer: A — L4 and L5 interspace or L4 spine

Page 165: The line joining the highest point on iliac crest (intercristal line, also called as Tuffier's line) corresponds to L4 and L5 interspace or L4 spine. It is very important to identify this intervertebral space while choosing the space for giving spinal or epidural anesthesia.

Topic: Applied Anatomy

## Q4. Answer: A — Reduces the subarachnoid space, leading to higher spread of local anesthetic

Page 165: The epidural veins form a plexus called the plexus of Batson, which drain into inferior vena cava. Any pressure on inferior vena cava (pregnancy, abdominal tumors, ascites) leads these veins to become engorged, pushing the dura medially, which reduces the subarachnoid space leading to higher spread of local anesthetic.

Topic: Epidural Space

## Q5. Answer: A — Lower border of L1 in adults and L3 in infants

Page 165: Spinal cord extends from medulla oblongata to lower border of L1 in adults and L3 in infants (adult level is achieved by the age of 2 years). Therefore, in infancy spinal anesthesia should be given in L4-5 space while in adults it can be given in L3-4 or L4-5 space.

Topic: Anatomy of Spinal Cord

## Q6. Answer: A — Artery of Adamkiewicz (arteria radicularis magna)

Page 165: The spinal cord is supplied by 2 posterior spinal arteries and 1 anterior spinal artery. The anterior spinal artery is reinforced by the artery of Adamkiewicz (arteria radicularis magna), which usually enters between T5 and L2 and usually (75%) enters from the left side. Damage/vasoconstriction can lead to cord ischemia and paraplegia.

Topic: Blood Supply of Spinal Cord

## Q7. Answer: A — 140 mL; half in cranium and half in spinal canal

Page 165: The CSF is secreted by choroid plexus of third, fourth and lateral ventricles and absorbed into venous sinuses via arachnoid villi. 500 mL is secreted in 24 hours. Volume of CSF at one time is 140 mL, half of which is present in cranium and half in spinal canal. Specific gravity is 1.003–1.009 and pH is 7.35.

Topic: Cerebrospinal Fluid

## Q8. Answer: A — As high as 30%

Page 166: Several meta-analyses have shown a reduced risk in overall mortality and morbidity as high as 30% in patients who underwent surgeries under CNB as compared to general anesthesia (GA).

Topic: Advantages of CNB

**Q9. Answer: B — Hypotension**

Page 166: The most prominent effect of central neuraxial blocks on the cardiovascular system is hypotension, which occurs due to sympathetic block (venodilatation, venous pooling, decreased venous return, decreased cardiac output), bradycardia, blockade of nerve supply to adrenal glands, direct absorption of drug, and compression of IVC/aorta by pregnant uterus.

*Topic: Systemic Effects – Cardiovascular*

**Q10. Answer: A — Autonomic > sensory > motor**

Page 166: In spinal nerve, autonomic fibers are on the most outer (posterior) side while motor on most inner (anterior) side, therefore the sequence of blockade is autonomic > sensory > motor. The autonomic level is 2 segments higher than sensory which is 2 segments higher than motor (zones of differential blockade).

*Topic: Systemic Effects – Nervous System*

**Q11. Answer: A — By temperature (cold)**

Page 166: Autonomic level is tested by temperature (cold), sensory by pin prick (by non-pricking needles) and motor by toe movement.

*Topic: Systemic Effects – Nervous System*

**Q12. Answer: B — Urinary retention**

Page 167: Urinary retention is the most common postoperative complication of central neuraxial blocks in present day practice. It is due to blockade of sacral parasympathetic fibers (S2, 3, 4). Flaccid paralysis of nervi erigentes and engorged penis is one of the signs of successful block.

*Topic: Systemic Effects – Genitourinary*

**Q13. Answer: A — L3-4 (or L4-5)**

Page 165, 167: Spinal anesthesia is usually given in L3-4 (or L4-5) space in adults because the spinal cord ends at L1, avoiding cord injury. The Taylor (lumbosacral) approach has its point of insertion 1 cm medial and 1 cm caudal to the posterior superior iliac spine.

*Topic: Spinal Anesthesia*

**Q14. Answer: B — Hyperbaric bupivacaine 0.5% (made hyperbaric by 8% dextrose)**

Page 167, Table 17.1: Bupivacaine is the most commonly used drug for spinal. The concentration used is 0.5% and it is made hyperbaric by addition of 8% dextrose. Lignocaine 5% (hyperbaric by 7.5% dextrose) is rarely used in current practice due to TNS and cauda equina syndrome.

*Topic: Drugs for Spinal Anesthesia*

**Q15. Answer: A — Dose and baricity**

Page 168–169: Dose and baricity are two very important determinants of level of block. Greater the dose, higher the level. Baricity is the ratio of specific gravity of agent to CSF; hyperbaric solutions settle downwards governed by patient position. Sex, speed of injection, increased CSF pressure (Barbotage), and addition of vasoconstrictor do NOT affect the level of block.

*Topic: Factors Affecting Level of Block*

**Q16. Answer: A — Inverse relation (decreased CSF volume leads to higher spread)**

Page 168–169: CSF volume has an inverse relation with the level of block; decreased CSF volume will lead to higher spread of drug while increased CSF volume will lead to attainment of lower levels.

*Topic: Factors Affecting Level of Block*



**Q17. Answer: A — Less damage to dural fibers and hence less incidence of postspinal headache**

Page 168: Dura separating needles have pencil tip point end (e.g. Whitacre and Sprotte). As these needles only separate the dural fibers, damage to dural fibers and hence incidence of postspinal headache is less, but these needles are expensive. Dura cutting needles (Quincke-Babcock, Greene, Pitkin) cause more damage and high PDPH but are cheaper.

Topic: Spinal Needles

**Q18. Answer: A — Hypotension**

Page 169: Hypotension is the most common complication of spinal anesthesia. Mild hypotension occurs in almost all patients and significant hypotension (systolic BP < 90 mm Hg) may be seen in up to 1/3rd of patients. Mild hypotension is beneficial in reducing blood loss.

Topic: Complications of Spinal Anesthesia

**Q19. Answer: A — Ephedrine**

Page 170: Ephedrine not only produces vasoconstriction ( $\alpha$  agonist) but also increases the cardiac output and heart rate by its  $\beta$  adrenergic properties, therefore it is the most preferred vasopressor for the treatment of spinal hypotension. However, mephentermine is more frequently used in India.

Topic: Complications of Spinal Anesthesia

**Q20. Answer: A — When the drug reaches up to the cranium and involves the cranial nerves**

Page 170: High spinal refers to a level high enough to block phrenic nerve causing diaphragmatic paralysis and apnea. If the drug reaches cranium and involves cranial nerves it is called total spinal. It usually happens due to accidental intrathecal injection of large volume of drug during epidural anesthesia.

Topic: Complications of Spinal Anesthesia

**Q21. Answer: B — Pethidine**

Page 170: Shivering is one of the very common complications of spinal anesthesia, seen in almost half of the patients. Oxygen consumption can increase 4–5 times. The drug of choice for treatment of shivering is pethidine followed by tramadol.

Topic: Complications of Spinal Anesthesia

**Q22. Answer: A — Pain in sitting/standing position which gets relieved in lying down**

Page 171: The typical symptomatology of post spinal headache is that patient will complain of pain in sitting and standing position while pain gets relieved in lying down, whereas headache due to meningitis will remain same in all positions. The pain is usually occipital. CSF loss around 10 mL/hr is considered significant.

Topic: Post-Spinal Headache

**Q23. Answer: A — Needle size**

Page 171: Needle size is the single most important factor; incidence with 16G needle may be 75% while with 26G needle it is 1 to 3%. Use of small gauge and dura separating needles are the most important preventive measures.

Topic: Post-Spinal Headache

**Q24. Answer: A — Autologous epidural blood patch**

Page 172: The third line of management, although interventional, is the definitive method of treatment — autologous epidural blood patch. 10–20 mL of the patient's own blood is given in the same space; blood will clot and seal the rent. 1st blood patch is effective in 95% patients and 2nd in 99%.

Topic: Post-Spinal Headache

**Q25. Answer: B — 6th cranial nerve**

Page 172: All cranial nerves except 1st, 9th and 10th can be involved after spinal anesthesia. Most commonly (90%) the 6th nerve is involved because of the longest intracranial course of the 6th nerve. As a result of low CSF pressure, descent of pons and medulla causes stretching of the 6th nerve at the apex of petrous temporal bone.

*Topic: Neurological Complications*

**Q26. Answer: B — Streptococcus viridans (an oral flora)**

Page 172: Streptococcus viridans (an oral flora) has been cited as the most common organism responsible for meningitis after spinal, emphasizing the mandatory requirement of facemask to be worn during the procedure. However, after epidural, meningitis most often occurs because of Staphylococcus epidermidis which can contaminate the epidural catheter.

*Topic: Neurological Complications*

**Q27. Answer: B — Lignocaine**

Page 172–173: Cauda equina syndrome (CES) is one of the classical examples of neurotoxicity caused by local anesthetic. It is usually seen with lignocaine when given as continuous spinal with small bore catheters. Clinical features include retention of urine, incontinence of feces, loss of sexual function and loss of sensation in perineal region.

*Topic: Neurological Complications*

**Q28. Answer: A — 10–11%**

Page 172: Transient neurological symptoms (TNS) can occur with any local anesthetic but most often with lignocaine. The incidence with lignocaine can be as high as 10–11% (that is why lignocaine is hardly used for spinal). Patients complain of pain in buttocks radiating to legs without neurological deficit, resolving in a few days (< 1 week).

*Topic: Neurological Complications*

**Q29. Answer: A — Confirmed by MRI; immediate neurosurgical intervention to drain the hematoma**

Page 173: Epidural hematoma is rare (1:200,000) but large enough to cause paraplegia in patients on anticoagulants therapy or suffering from coagulopathies. The presentation is sharp back pain with motor weakness and/or bladder/bowel dysfunction. The diagnosis should be confirmed by MRI and management includes immediate neurosurgical intervention to drain the hematoma.

*Topic: Neurological Complications*

**Q30. Answer: A — Below L1 (and in children up to 2 years, below L3)**

Page 173: To avoid direct trauma to spinal cord, spinal in adults should be given below L1 and in children up to 2 years below L3.

*Topic: Neurological Complications*

**Q31. Answer: A — Pruritus**

Page 173: Pruritus is the most common side effect of intrathecal opioids. It can be treated with opioid antagonists.

*Topic: Complications of Spinal Anesthesia*

**Q32. Answer: A — Sitting position (patient seated 5–10 minutes till drug gets fixed); perianal surgeries**

Page 173: A saddle block is the spinal given in sitting position and the patient remains seated for 5–10 minutes till drug gets fixed. As only sacral segments are blocked, it can be utilized only for perianal surgeries. The advantage is that hemodynamic fluctuations are minimal and there is no possibility of high spinal.

*Topic: Saddle Block*

**Q33. Answer: A — Tuohy needle (blunt bevel with curve of 15 to 30 degrees at tip, called Huber tip)**

Page 174, Fig. 17.2: The most commonly used needle for epidural is the Tuohy needle. It has blunt bevel with curve of 15 to 30 degrees at tip (called as Huber tip). Other needles include Weiss (winged) and Crawford (straight blunt bevel with no curve).

Topic: Epidural Anesthesia

**Q34. Answer: A — Loss of resistance technique**

Page 174: The most commonly used method to locate epidural space is by loss of resistance technique. Once the needle pierces the ligamentum flavum, there is sudden loss of resistance and the syringe filled with air or saline will be felt literally sucked in epidural space. The hanging drop technique (Gutierrez's sign) is hardly used in current practice.

Topic: Epidural Anesthesia

**Q35. Answer: A — 2–3 mL**

Page 174: Usually 2–3 mL of local anesthetic is required for blocking 1 segment; therefore normally 15 to 20 mL of drug is required for epidural anesthesia.

Topic: Epidural Anesthesia

**Q36. Answer: A — Fentanyl + bupivacaine or ropivacaine (ropivacaine preferred) as continuous infusion**

Page 174–175: Fentanyl (2–4 µg/mL) + bupivacaine (0.125%) or ropivacaine (0.2%) (ropivacaine preferred) as continuous infusion by syringe pump through epidural catheter is the most commonly used combination for postoperative analgesia and painless labor.

Topic: Epidural Anesthesia

**Q37. Answer: A — Ropivacaine**

Page 174: Ropivacaine, because of its high safety profile, is most preferred for epidural anesthesia. Concentrations used (Table 17.2): Lignocaine 1–2%, Levobupivacaine 0.125–0.75%, Ropivacaine 0.1–1%, Bupivacaine 0.625–0.5%.

Topic: Epidural Anesthesia

**Q38. Answer: A — Volume of the drug**

Page 175: Volume of the drug is the most important factor affecting the spread (level) of block in epidural anesthesia. Old age requires less dose because the volume of epidural space is less. High concentration produces higher spread.

Topic: Epidural Anesthesia

**Q39. Answer: A — Less hypotension (slow onset gives body time to compensate), making epidural a better choice for cardiac compromised patients**

Page 175: Less hypotension — as the onset of action of epidural is slow, body gets sufficient time to compensate for hypotension, making epidural a better choice than spinal for cardiac compromised patients. Other advantages include no postspinal headache (unless dura punctured) and ability to change level and duration via catheter.

Topic: Advantages of Epidural over Spinal

**Q40. Answer: A — 10 fold higher than spinal (as the veins are denser laterally, epidural should be given in midline)**

Page 176: The possibility of hematoma formation can be 10 fold higher than spinal. As the veins are denser laterally, the epidural should be given in midline. To prevent intravascular injection, inject the drug only after confirming negative aspiration of blood.

Topic: Disadvantages of Epidural over Spinal

**Q41. Answer: A — To allow spinal anesthesia for surgery and the epidural catheter for prolonging/extending the block and postoperative analgesia**

Page 176: Combined spinal epidural (CSE) is now a very commonly performed technique where spinal is used for surgery and epidural is utilized if surgery gets prolonged or the surgeon wants to extend the incision. Later, the epidural catheter can be used for postoperative analgesia. A special spinal needle (120 mm) is passed through the epidural needle (90 mm normal spinal needle).

*Topic: Combined Spinal Epidural*

**Q42. Answer: A — Perianal, genital (circumcision), urethral (urethroplasty) and lower abdominal (herniotomy) surgeries**

Page 176: Caudal block is a type of epidural block commonly utilized in children where drug is injected in sacral hiatus after piercing sacrococcygeal membrane. Most often utilized to produce analgesia for perianal surgeries, genital surgeries like circumcision, urethral surgeries like urethroplasties, or lower abdominal surgeries like herniotomies. In children it is easy to perform because sacral hiatus can be easily identified.

*Topic: Caudal Block*

**Q43. Answer: A — Raised intracranial tension (medullary coning can cause death)**

Page 177: Raised intracranial tension is an absolute contraindication — medullary coning can cause death if spinal/epidural is given to patients with raised ICT (due to possibility of accidental dural puncture and increased CSF pressure). Other absolute contraindications include patient refusal, severe hypovolemia/hypotension, patients on anticoagulants, bleeding disorders, infection at local site, and severe fixed cardiac output lesions.

*Topic: Contraindications*

**Q44. Answer: A — Safe if > 80,000/cubic mm; absolute contraindication if < 50,000/cubic mm**

Page 178: Patients with platelet count more than 80,000/cubic mm can be safely given central neuraxial blocks. A platelet count between 50,000–80,000/cubic mm is a relative contraindication while count less than 50,000 becomes an absolute contraindication.

*Topic: Contraindications*

**Q45. Answer: A — 12 hours after the last dose of LMWH if prophylactic (and 24 hours if therapeutic dose)**

Page 177: It is considered safe to give central neuraxial block 12 hours after last dose of LMWH if the dose is prophylactic. However, if doses are therapeutic then CNB should be given only after 24 hours. LMWH can be safely given 6 hours after central neuraxial blocks.

*Topic: Contraindications*

**Q46. Answer: A — Aspirin and NSAIDs**

Page 177–178: Patients on aspirin and NSAIDs can be safely given central neuraxial blocks. For other antiplatelets, CNB should be delayed for 7 days after clopidogrel, 14 days after prasugrel and ticlopidine, 2 days after abciximab, and 8 hours after eptifibatide.

*Topic: Contraindications*

**Q47. Answer: A — Up to T4**

Page 179: For cesarean section, the block level required is up to T4. Although skin incision level for cesarean section is at T11 level, since the intestine and peritoneum are also handled, the block required is up to T4. Other levels: prostate T10, hernia T10, appendix T8, hysterectomy T6, perianal surgeries sacral segments.

*Topic: Level of Block for Surgeries*

**Q48. Answer: A — Spinal: more dense; Epidural: patchy block is common**

Page 178, Table 17.3: In spinal anesthesia the quality of block is more dense, while in epidural patchy block is common. Spinal is cheaper, fast onset, technically easy, with less block failure rate; epidural is expensive, slow onset (15–20 minutes), technically difficult, with higher block failure rate.

*Topic: Spinal vs Epidural Comparison*

**Q49. Answer: A — Severe hypotension causing medullary ischemia**

Page 166, 170: Apnea after spinal anesthesia is usually due to severe hypotension causing medullary ischemia. Other causes of apnea include high spinal (high enough to block phrenic nerve C3, 4, 5 which is rare) and total spinal (when the drug reaches the cranium).

*Topic: Systemic Effects – Respiratory*

**Q50. Answer: A — Due to decreased atrial pressure (Bainbridge reflex), parasympathetic over dominance, and direct inhibition of cardioaccelerator fibers (T1 to T4)**

Page 166: Bradycardia can further decrease cardiac output. The reasons for bradycardia are: decreased atrial pressure because of decreased venous return (Bainbridge reflex), parasympathetic over dominance due to sympathetic block, and direct inhibition of cardioaccelerator fibres (T1 to T4).

*Topic: Systemic Effects – Cardiovascular*

# Anesthesia for Cardiovascular Diseases MCQ – Chapter 18

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 18)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. What level of effort tolerance (in METS) is generally associated with lesser perioperative complications in patients with ischemic heart disease?**

- A. > 2 METS
- B. > 4 METS
- C. > 6 METS
- D. > 8 METS

**Ans: B**

**Q2. How many parameters are included in the Lee revised cardiac risk index?**

- A. 4
- B. 5
- C. 6
- D. 8

**Ans: C**

**Q3. Regarding beta-blockers in IHD patients presenting for surgery, which statement is correct?**

- A. Beta-blockers should be stopped 5 days before surgery
- B. Beta-blockers must not be stopped at any cost if the patient is already on them
- C. Beta-blockers should be freshly started on the day before surgery
- D. Beta-blockers should be replaced with calcium channel blockers preoperatively

**Ans: B**

**Q4. How many days before surgery should clopidogrel be stopped in an IHD patient?**

- A. 3 days
- B. 5 days
- C. 7 days
- D. 10 days

**Ans: B**

**Q5. For how long should elective surgery be deferred after placement of a conventional drug eluting stent?**

- A. 6 weeks
- B. 3 months
- C. 6 months
- D. 1 year

**Ans: D**

**Q6. Which combination of ECG leads can detect 94–95% of intraoperative ischemic events?**

- A. Lead I and aVL
- B. Lead II and V5
- C. Lead III and V1
- D. Lead aVR and V6

**Ans: B**

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| <p><b>Q7. Which is the earliest and most sensitive monitor to detect intraoperative myocardial infarction?</b></p> <p>A. ECG with ST analysis<br/>B. Pulmonary artery catheter<br/>C. Transesophageal echocardiography (TEE)<br/>D. Central venous pressure monitoring</p>                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q8. A patient with IHD and normal LV function is undergoing surgery. Which of the following should be AVOIDED during maintenance of anesthesia?</b></p> <p>A. Isoflurane<br/>B. Sevoflurane<br/>C. Nitrous oxide<br/>D. Opioids</p>                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q9. Mitral stenosis is considered severe when the transvalvular pressure gradient exceeds:</b></p> <p>A. 5 mm Hg<br/>B. 10 mm Hg<br/>C. 15 mm Hg<br/>D. 20 mm Hg</p>                                                                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q10. Why should tachycardia be avoided in patients with mitral stenosis?</b></p> <p>A. It increases the afterload on the left ventricle<br/>B. It causes coronary vasoconstriction<br/>C. It decreases diastole, reducing left ventricular filling and hence cardiac output<br/>D. It increases the degree of mitral regurgitation</p> | <p><b>Ans: C</b></p> |
| <p><b>Q11. Approximately what fraction of mitral stenosis patients have atrial fibrillation?</b></p> <p>A. One-tenth<br/>B. One-fourth<br/>C. One-third<br/>D. One-half</p>                                                                                                                                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q12. Mitral regurgitation is considered severe when the regurgitant fraction exceeds:</b></p> <p>A. 0.3<br/>B. 0.4<br/>C. 0.5<br/>D. 0.6</p>                                                                                                                                                                                           | <p><b>Ans: D</b></p> |
| <p><b>Q13. In mitral regurgitation, which hemodynamic change should be avoided?</b></p> <p>A. Mild tachycardia<br/>B. Decrease in afterload<br/>C. Bradycardia<br/>D. Mild hypotension</p>                                                                                                                                                   | <p><b>Ans: C</b></p> |



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| <p><b>Q14. Which antihypertensives should be stopped on the day of surgery in hypertensive patients?</b></p> <ul style="list-style-type: none"> <li>A. Beta-blockers and calcium channel blockers</li> <li>B. ACE inhibitors and angiotensin II receptor antagonists</li> <li>C. Diuretics and alpha blockers</li> <li>D. All antihypertensives should be continued</li> </ul>                        | <p><b>Ans: B</b></p> |
| <p><b>Q15. In mitral valve prolapse (MVP), which assessment is MOST important preoperatively?</b></p> <ul style="list-style-type: none"> <li>A. Degree of left atrial enlargement</li> <li>B. Whether MVP is associated with mitral regurgitation or not</li> <li>C. Presence of atrial fibrillation</li> <li>D. Left ventricular ejection fraction</li> </ul>                                        | <p><b>Ans: B</b></p> |
| <p><b>Q16. Which induction agent is the agent of choice for shock patients?</b></p> <ul style="list-style-type: none"> <li>A. Propofol</li> <li>B. Thiopentone</li> <li>C. Etomidate</li> <li>D. Ketamine</li> </ul>                                                                                                                                                                                  | <p><b>Ans: D</b></p> |
| <p><b>Q17. Among valvular lesions, which has the most devastating complications including sudden death?</b></p> <ul style="list-style-type: none"> <li>A. Mitral stenosis</li> <li>B. Mitral regurgitation</li> <li>C. Aortic stenosis</li> <li>D. Aortic regurgitation</li> </ul>                                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q18. Aortic stenosis is considered severe when the aortic valve orifice is less than:</b></p> <ul style="list-style-type: none"> <li>A. 1.5 cm<sup>2</sup></li> <li>B. 1.0 cm<sup>2</sup></li> <li>C. 0.8 cm<sup>2</sup></li> <li>D. 0.5 cm<sup>2</sup></li> </ul>                                                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q19. Which vasopressor is the agent of choice for patients with aortic stenosis, and why?</b></p> <ul style="list-style-type: none"> <li>A. Ephedrine, because it increases cardiac output</li> <li>B. Noradrenaline, because it has strong alpha effect</li> <li>C. Phenylephrine, because it does not cause tachycardia</li> <li>D. Dopamine, because it improves renal blood flow</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q20. In aortic regurgitation, which hemodynamic changes should be avoided?</b></p> <ul style="list-style-type: none"> <li>A. Tachycardia and hypotension</li> <li>B. Bradycardia and hypertension</li> <li>C. Tachycardia and hypertension</li> <li>D. Bradycardia and hypotension</li> </ul>                                                                                                   | <p><b>Ans: B</b></p> |

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| <p><b>Q21. Which type of prosthetic heart valve is more prone to thromboembolism but has a longer life?</b></p> <p>A. Bioprosthetic valve (10–15 years)<br/> B. Mechanical valve (20–30 years)<br/> C. Homograft valve<br/> D. Autograft valve</p>                                                                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q22. How many days before surgery should warfarin be stopped in patients with prosthetic heart valves?</b></p> <p>A. 2 days<br/> B. 3 days<br/> C. 5 days<br/> D. 7 days</p>                                                                                                                                                                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q23. According to new guidelines, antibiotic prophylaxis for patients with prosthetic heart valves is recommended for which procedures?</b></p> <p>A. All surgical procedures<br/> B. All procedures involving body cavities<br/> C. Dental procedures involving gingiva/oral mucosa/periapical region and invasive procedures involving incision on infective tissue<br/> D. Only cardiac surgeries</p> | <p><b>Ans: C</b></p> |
| <p><b>Q24. What is the basic aim of anesthetic management in left to right shunts (VSD, ASD, PDA)?</b></p> <p>A. Maintain high SVR and low PVR<br/> B. Maintain low SVR and high PVR<br/> C. Maintain high SVR and high PVR<br/> D. Maintain low SVR and low PVR</p>                                                                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q25. For which congenital defect is antibiotic prophylaxis for infective endocarditis recommended ONLY if there is a concomitant valvular abnormality?</b></p> <p>A. VSD<br/> B. ASD<br/> C. PDA<br/> D. Tetralogy of Fallot</p>                                                                                                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q26. In left to right shunts, what effect does increased pulmonary blood flow have on IV induction agents?</b></p> <p>A. Significantly delays onset<br/> B. Causes dilution but has no clinical implication<br/> C. Significantly hastens onset<br/> D. Causes unpredictable effects requiring dose reduction</p>                                                                                        | <p><b>Ans: B</b></p> |

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| <p><b>Q27. A patient in shock must be resuscitated to at least what minimum blood pressure before considering anesthesia?</b></p> <p>A. MAP &gt; 50 mm Hg or SBP &gt; 80 mm Hg<br/> B. MAP &gt; 65 mm Hg or SBP &gt; 90 mm Hg<br/> C. MAP &gt; 75 mm Hg or SBP &gt; 100 mm Hg<br/> D. MAP &gt; 80 mm Hg or SBP &gt; 110 mm Hg</p> | <p><b>Ans: B</b></p> |
| <p><b>Q28. Which is the induction agent of choice for right to left shunts (cyanotic heart diseases)?</b></p> <p>A. Propofol<br/> B. Thiopentone<br/> C. Etomidate<br/> D. Ketamine</p>                                                                                                                                           | <p><b>Ans: D</b></p> |
| <p><b>Q29. What happens to the onset of IV induction agents in right to left shunts?</b></p> <p>A. Onset is delayed<br/> B. Onset is rapid as pulmonary circulation is bypassed<br/> C. Onset is unchanged<br/> D. Onset is slower due to polycythemia</p>                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q30. Which muscle relaxant is the agent of choice in right to left shunts, and why?</b></p> <p>A. Vecuronium, because it is cardiac stable<br/> B. Atracurium, because it does not affect SVR<br/> C. Pancuronium, because it causes hypertension<br/> D. Succinylcholine, because of its rapid onset</p>                   | <p><b>Ans: C</b></p> |
| <p><b>Q31. In right to left shunts, the FiO<sub>2</sub> used is 50% instead of 33% because:</b></p> <p>A. Higher oxygen improves cerebral perfusion<br/> B. High FiO<sub>2</sub> decreases PVR, which is beneficial<br/> C. It compensates for polycythemia<br/> D. N<sub>2</sub>O at higher concentrations is toxic</p>          | <p><b>Ans: B</b></p> |
| <p><b>Q32. In coarctation of aorta, what is the minimum mean arterial pressure that should be maintained in the lower limb?</b></p> <p>A. &gt; 30 mm Hg<br/> B. &gt; 40 mm Hg<br/> C. &gt; 50 mm Hg<br/> D. &gt; 60 mm Hg</p>                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q33. Regarding heart failure patients, which statement is correct?</b></p> <p>A. Elective surgery can proceed if LVEF &gt; 30%<br/> B. Elective surgery is contraindicated in heart failure patients<br/> C. Only regional anesthesia is suitable<br/> D. Beta-blockers must be stopped before surgery</p>                  | <p><b>Ans: B</b></p> |

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| <p><b>Q34. What is the most common cause of right heart failure?</b></p> <ul style="list-style-type: none"> <li>A. Pulmonary embolism</li> <li>B. Chronic lung disease</li> <li>C. Left heart failure</li> <li>D. Primary pulmonary hypertension</li> </ul>                                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q35. Which drugs remain the primary treatment for chronic heart failure?</b></p> <ul style="list-style-type: none"> <li>A. Beta-blockers and calcium channel blockers</li> <li>B. Digitalis and diuretics</li> <li>C. ACE inhibitors, ARBs and aldosterone antagonists (RAAS inhibitors)</li> <li>D. Nitrates and hydralazine</li> </ul>                                        | <p><b>Ans: C</b></p> |
| <p><b>Q36. Which preoperative electrolyte is particularly important to check in heart failure patients on diuretics and digitalis?</b></p> <ul style="list-style-type: none"> <li>A. Sodium</li> <li>B. Calcium</li> <li>C. Potassium</li> <li>D. Magnesium</li> </ul>                                                                                                                | <p><b>Ans: C</b></p> |
| <p><b>Q37. Why is positive pressure ventilation beneficial in heart failure patients?</b></p> <ul style="list-style-type: none"> <li>A. It increases myocardial contractility</li> <li>B. It decreases venous return (preload)</li> <li>C. It increases systemic vascular resistance</li> <li>D. It improves coronary blood flow</li> </ul>                                           | <p><b>Ans: B</b></p> |
| <p><b>Q38. What is the most common genetic cardiovascular disease?</b></p> <ul style="list-style-type: none"> <li>A. Dilated cardiomyopathy</li> <li>B. Hypertrophic cardiomyopathy</li> <li>C. Restrictive cardiomyopathy</li> <li>D. Arrhythmogenic right ventricular cardiomyopathy</li> </ul>                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q39. What is the most important feature of hypertrophic cardiomyopathy (HCM)?</b></p> <ul style="list-style-type: none"> <li>A. Diastolic dysfunction</li> <li>B. Dynamic left ventricular outflow tract (LVOT) obstruction</li> <li>C. Mitral regurgitation</li> <li>D. Systolic anterior motion of the mitral valve</li> </ul>                                                | <p><b>Ans: B</b></p> |
| <p><b>Q40. Pulmonary edema in HCM patients CANNOT be treated by diuretics, nitrates, and digitalis. What is the recommended treatment?</b></p> <ul style="list-style-type: none"> <li>A. Dobutamine and milrinone</li> <li>B. Phenylephrine (vasopressor) and esmolol (<math>\beta</math>-blocker)</li> <li>C. Adrenaline and atropine</li> <li>D. Furosemide and morphine</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q41. Why is pancuronium contraindicated in HCM?</b></p> <ul style="list-style-type: none"> <li>A. It causes bradycardia</li> <li>B. It can cause sympathetic stimulation which worsens LVOT obstruction</li> <li>C. It causes histamine release</li> <li>D. It has a prolonged duration of action</li> </ul>                                                                                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q42. Which is the most common type of cardiomyopathy?</b></p> <ul style="list-style-type: none"> <li>A. Hypertrophic cardiomyopathy</li> <li>B. Dilated cardiomyopathy</li> <li>C. Restrictive cardiomyopathy</li> <li>D. Obliterative cardiomyopathy</li> </ul>                                                                                                                                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q43. In patients with pericardial diseases (constrictive pericarditis/cardiac tamponade), which induction agent is preferred?</b></p> <ul style="list-style-type: none"> <li>A. Propofol</li> <li>B. Thiopentone</li> <li>C. Etomidate</li> <li>D. Ketamine</li> </ul>                                                                                                                                                                                                                          | <p><b>Ans: D</b></p> |
| <p><b>Q44. In pericardial diseases, which of the following statements is correct regarding anesthetic technique?</b></p> <ul style="list-style-type: none"> <li>A. Central neuraxial blocks are the technique of choice</li> <li>B. Pericardiocentesis should be done under GA before the procedure</li> <li>C. CNB (spinal/epidural) are contraindicated and pericardiocentesis should be done under local anesthesia before GA</li> <li>D. Only monitored anesthesia care is appropriate</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q45. Which muscle relaxant is preferred for patients with pericardial diseases, and why?</b></p> <ul style="list-style-type: none"> <li>A. Vecuronium, because it is cardiac stable</li> <li>B. Pancuronium, because it stimulates sympathetic system and increases cardiac output</li> <li>C. Atracurium, because it causes histamine release</li> <li>D. Rocuronium, because of rapid onset</li> </ul>                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q46. What is the most common indication for cardiac implanted electronic devices (CIED)?</b></p> <ul style="list-style-type: none"> <li>A. Atrial fibrillation</li> <li>B. Ventricular tachycardia</li> <li>C. Sick sinus syndrome</li> <li>D. Complete heart block</li> </ul>                                                                                                                                                                                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q47. What is the most common source of electromagnetic interference (EMI) that can cause CIED malfunction during surgery?</b></p> <ul style="list-style-type: none"> <li>A. MRI</li> <li>B. Radiofrequency devices</li> <li>C. Cautery (electrocautery)</li> <li>D. Ultrasound devices</li> </ul>                                                                                                                                                                                               | <p><b>Ans: C</b></p> |

**Q48.** Before surgery, CIED should be programmed to asynchronous mode. In this mode, the pacemaker delivers a fixed rate of approximately:

- A. 50–55 beats/minute
- B. 60–65 beats/minute
- C. 70–72 beats/minute
- D. 80–85 beats/minute

**Ans: C**

**Q49.** Hypertension is defined as blood pressure exceeding which value on at least 2 occasions?

- A. > 120/80 mm Hg
- B. > 130/85 mm Hg
- C. > 140/90 mm Hg
- D. > 150/95 mm Hg

**Ans: C**

**Q50.** What is the general consensus for the diastolic BP threshold to delay elective surgery in hypertensive patients?

- A. > 90 mm Hg
- B. > 100 mm Hg
- C. > 110 mm Hg
- D. > 120 mm Hg

**Ans: C**

# Answer Key & Explanations

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## Q1. Answer: B — > 4 METS

As per the text, effort tolerance > 4 METS is generally associated with lesser complications in IHD patients undergoing surgery.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

## Q2. Answer: C — 6

The Lee revised cardiac risk index includes 6 parameters: high risk surgery, history of IHD, cerebrovascular disease, congestive heart failure, insulin dependent diabetes mellitus, and preoperative serum creatinine > 2 mg/dL.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

## Q3. Answer: B — Beta-blockers must not be stopped at any cost if the patient is already on them

Beta-blockers must not be stopped at any cost if the patient is on them, as stopping can significantly increase morbidity and mortality. They should not be started afresh unless they can be started at least 1 week before surgery.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

## Q4. Answer: B — 5 days

Clopidogrel should be stopped 5 days before surgery. However, central neuraxial block can be given only after 7 days of stopping clopidogrel.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

## Q5. Answer: D — 1 year

Elective surgery must be deferred for 6 months with new generation drug eluting stents and 1 year with conventional drug eluting stents.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

## Q6. Answer: B — Lead II and V5

Lead V5 is best for detecting left ventricle infarction and lead II for arrhythmia (and inferior wall MI). A combination of lead II and V5 can detect 94–95% of intraoperative ischemic events.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

## Q7. Answer: C — Transesophageal echocardiography (TEE)

TEE is the earliest and most sensitive monitor to detect intraoperative infarction, capable of detecting more than 99% of ischemic events.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 184

## Q8. Answer: C — Nitrous oxide

Nitrous oxide should be avoided in IHD patients for two reasons: it may cause mild sympathetic stimulation and it is a pulmonary vasoconstrictor. Inhalational agents can be safely used if LV function is normal.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 184

## Q9. Answer: B — 10 mm Hg

Mitral stenosis is considered severe if the transvalvular pressure gradient is > 10 mm Hg (normal < 5 mm Hg).

Topic: Valvular Diseases – Mitral Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 185

**Q10. Answer: C — It decreases diastole, reducing left ventricular filling and hence cardiac output**

In mitral stenosis, tachycardia decreases diastole which reduces left ventricular filling and hence stroke volume and cardiac output.

Topic: Valvular Diseases – Mitral Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 185

**Q11. Answer: C — One-third**

Atrial fibrillation is seen in one-third of the patients with mitral stenosis, and a significant number will be on oral anticoagulants contraindicating the use of CNB.

Topic: Valvular Diseases – Mitral Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 185

**Q12. Answer: D — 0.6**

Mitral regurgitation is said to be severe if the regurgitant fraction is > 0.6.

Topic: Valvular Diseases – Mitral Regurgitation | Ref: Ch 18, Section: Valvular Diseases, Page 185

**Q13. Answer: C — Bradycardia**

In MR, bradycardia should be avoided because it increases diastole which increases the regurgitation. Hypertension should also be avoided as it increases afterload.

Topic: Valvular Diseases – Mitral Regurgitation | Ref: Ch 18, Section: Valvular Diseases, Page 185

**Q14. Answer: B — ACE inhibitors and angiotensin II receptor antagonists**

All antihypertensives should be continued except ACE inhibitors and angiotensin II receptor antagonists, which are stopped on the day of surgery as they can cause profound intraoperative hypotension.

Topic: Hypertension | Ref: Ch 18, Section: Hypertension, Page 190

**Q15. Answer: B — Whether MVP is associated with mitral regurgitation or not**

The hemodynamic management principles of MVP without MR and with MR are different, so it is very important to assess preoperatively whether MVP is associated with MR or not.

Topic: Valvular Diseases – Mitral Valve Prolapse | Ref: Ch 18, Section: Valvular Diseases, Page 185-186

**Q16. Answer: D — Ketamine**

Ketamine is the induction agent of choice for shock patients due to its sympathomimetic effects. However, in chronic/severe shock with sympathetic depletion, its direct myocardial depressant effect may predominate.

Topic: Hypotension/Shock | Ref: Ch 18, Section: Hypotension/Shock, Page 190-191

**Q17. Answer: C — Aortic stenosis**

Among the valvular lesions, aortic stenosis has the most devastating complications like infarction, arrhythmia or even sudden death.

Topic: Valvular Diseases – Aortic Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 186

**Q18. Answer: C — 0.8 cm<sup>2</sup>**

Severe AS is defined as transvalvular gradient > 50 mm Hg or aortic valve orifice less than 0.8 cm<sup>2</sup>.

Topic: Valvular Diseases – Aortic Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 186

**Q19. Answer: C — Phenylephrine, because it does not cause tachycardia**

As phenylephrine does not cause tachycardia, it is the vasopressor of choice for patients with aortic stenosis.

Topic: Valvular Diseases – Aortic Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 186



**Q20. Answer: B — Bradycardia and hypertension**

In AR, bradycardia by increasing diastole and hypertension by increasing afterload can increase the regurgitant fraction, so both should be avoided.

Topic: Valvular Diseases – Aortic Regurgitation | Ref: Ch 18, Section: Valvular Diseases, Page 186

**Q21. Answer: B — Mechanical valve (20–30 years)**

Mechanical prosthetic heart valves are more prone for thromboembolism but have a long life of 20–30 years, while bioprosthetic valves are less prone for thromboembolism but have shorter life of 10–15 years.

Topic: Prosthetic Heart Valves | Ref: Ch 18, Section: Prosthetic Heart Valves, Page 186

**Q22. Answer: C — 5 days**

Warfarin should be stopped 5 days before surgery and heparin should be started, which is then stopped 12–24 hours before surgery.

Topic: Prosthetic Heart Valves | Ref: Ch 18, Section: Prosthetic Heart Valves, Page 186

**Q23. Answer: C — Dental procedures involving gingiva/oral mucosa/periapical region and invasive procedures involving incision on infective tissue**

The new guidelines recommend antibiotic prophylaxis only for dental procedures involving gingiva, oral mucosa and periapical region of teeth, and invasive procedures involving incision on infective tissue.

Topic: Prosthetic Heart Valves | Ref: Ch 18, Section: Prosthetic Heart Valves, Page 186

**Q24. Answer: B — Maintain low SVR and high PVR**

In left to right shunts, increase in SVR or decrease in PVR increases the shunt fraction. The aim is to maintain low systemic vascular resistance and high pulmonary vascular resistance.

Topic: Congenital Heart Diseases – Left to Right Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

**Q25. Answer: B — ASD**

Antibiotic prophylaxis is recommended for VSD and PDA, but for ASD only if there is a concomitant valvular abnormality (mitral valve prolapse may be associated with ASD).

Topic: Congenital Heart Diseases – Left to Right Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

**Q26. Answer: B — Causes dilution but has no clinical implication**

Although increase in pulmonary blood flow causes dilution of IV agents in left to right shunts, this has no clinical implication.

Topic: Congenital Heart Diseases – Left to Right Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

**Q27. Answer: B — MAP > 65 mm Hg or SBP > 90 mm Hg**

Patient must be resuscitated to at least mean arterial pressure > 65 mm Hg or systolic pressure > 90 mm Hg before considering anesthesia.

Topic: Hypotension/Shock | Ref: Ch 18, Section: Hypotension/Shock, Page 190-191

**Q28. Answer: D — Ketamine**

As ketamine increases the systemic vascular resistance, it is the agent of choice for induction in right to left shunts.

Topic: Congenital Heart Diseases – Right to Left Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

**Q29. Answer: B — Onset is rapid as pulmonary circulation is bypassed**

In right to left shunts, as pulmonary circulation is bypassed, the induction with IV agents will be rapid.

Topic: Congenital Heart Diseases – Right to Left Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

**Q30. Answer: C — Pancuronium, because it causes hypertension**

Pancuronium causes hypertension (increases SVR), which is desirable in right to left shunts to reduce shunt fraction, making it the muscle relaxant of choice.

Topic: Congenital Heart Diseases – Right to Left Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

**Q31. Answer: B — High FiO<sub>2</sub> decreases PVR, which is beneficial**

High FiO<sub>2</sub> decreases PVR, which is beneficial in right to left shunts (aim is to decrease PVR), so oxygen concentration used is 50%.

Topic: Congenital Heart Diseases – Right to Left Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

**Q32. Answer: B — > 40 mm Hg**

In coarctation of aorta, blood pressure monitoring in lower limbs is mandatory and mean arterial pressure of the lower limb should remain at least > 40 mm Hg.

Topic: Congenital Heart Diseases – Coarctation of Aorta | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

**Q33. Answer: B — Elective surgery is contraindicated in heart failure patients**

The patient in heart failure carries very high risk of perioperative morbidity and mortality, therefore elective surgery is contraindicated.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

**Q34. Answer: C — Left heart failure**

The most common cause of right heart failure is left heart failure.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

**Q35. Answer: C — ACE inhibitors, ARBs and aldosterone antagonists (RAAS inhibitors)**

The RAAS system plays an important role in heart failure pathophysiology. ACE inhibitors, ARBs and aldosterone antagonists remain the primary drugs for treatment of chronic heart failure.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

**Q36. Answer: C — Potassium**

As heart failure patients are on diuretics and digitalis, checking preoperative electrolytes, especially potassium, is a must.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

**Q37. Answer: B — It decreases venous return (preload)**

Positive pressure ventilation is beneficial in heart failure by decreasing the venous return (preload), which is one of the basic principles of heart failure management.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

**Q38. Answer: B — Hypertrophic cardiomyopathy**

Hypertrophic cardiomyopathy (HCM), also called subaortic stenosis, is the most common genetic cardiovascular disease.

Topic: Cardiomyopathies – HCM | Ref: Ch 18, Section: Cardiomyopathies, Page 188

**Q39. Answer: B — Dynamic left ventricular outflow tract (LVOT) obstruction**

Dynamic left ventricular outflow obstruction (LVOT) is the most important feature of HCM.

Topic: Cardiomyopathies – HCM | Ref: Ch 18, Section: Cardiomyopathies, Page 188

**Q40. Answer: B — Phenylephrine (vasopressor) and esmolol (β-blocker)**

In HCM, pulmonary edema cannot be treated by diuretics (decrease preload), nitrates (decrease afterload), or digitalis (increase contractility) as all worsen LVOT obstruction. It is treated by vasopressors (phenylephrine) and β-blockers (esmolol).

Topic: Cardiomyopathies – HCM | Ref: Ch 18, Section: Cardiomyopathies, Page 188-189

**Q41. Answer: B — It can cause sympathetic stimulation which worsens LVOT obstruction**

Pancuronium is contraindicated in HCM because it can cause sympathetic stimulation, which increases myocardial contractility and worsens LVOT obstruction.

Topic: Cardiomyopathies – HCM | Ref: Ch 18, Section: Cardiomyopathies, Page 188

**Q42. Answer: B — Dilated cardiomyopathy**

Dilated cardiomyopathy is the most common cardiomyopathy. It may be primary (idiopathic) or secondary like alcoholic or peripartum cardiomyopathy.

Topic: Cardiomyopathies – DCM | Ref: Ch 18, Section: Cardiomyopathies, Page 189

**Q43. Answer: D — Ketamine**

In pericardial diseases, ketamine is preferred for induction as it stimulates the sympathetic system and increases cardiac output in these low cardiac output states.

Topic: Pericardial Diseases | Ref: Ch 18, Section: Pericardial Diseases, Page 189

**Q44. Answer: C — CNB (spinal/epidural) are contraindicated and pericardiocentesis should be done under local anesthesia before GA**

Patients with pericardial diseases are in low cardiac output state, so CNB are contraindicated. Pericardiocentesis should be done under local anesthesia before proceeding to GA.

Topic: Pericardial Diseases | Ref: Ch 18, Section: Pericardial Diseases, Page 189

**Q45. Answer: B — Pancuronium, because it stimulates sympathetic system and increases cardiac output**

Pancuronium is preferred for pericardial diseases because it stimulates the sympathetic system and increases cardiac output, which is needed in the low cardiac output state.

Topic: Pericardial Diseases | Ref: Ch 18, Section: Pericardial Diseases, Page 189

**Q46. Answer: C — Sick sinus syndrome**

CIED are used for treating bradyarrhythmias, with sick sinus syndrome being the most common indication.

Topic: Cardiac Implanted Electronic Devices | Ref: Ch 18, Section: CIED/Pacemakers, Page 189

**Q47. Answer: C — Cautery (electrocautery)**

The most common source of EMI is cautery, followed by radiofrequency devices and MRI. MRI is contraindicated for patients with CIED.

Topic: Cardiac Implanted Electronic Devices | Ref: Ch 18, Section: CIED/Pacemakers, Page 189

**Q48. Answer: C — 70–72 beats/minute**

In asynchronous mode, pacemakers deliver a fixed rate of 70–72 beats/minute irrespective of the patient's own rate, avoiding the risk of CIED being affected by interference.

Topic: Cardiac Implanted Electronic Devices | Ref: Ch 18, Section: CIED/Pacemakers, Page 189

**Q49. Answer: C — > 140/90 mm Hg**

Hypertension is defined as blood pressure > 140/90 on at least 2 occasions 1–2 weeks apart, while hypertensive crisis is defined as BP > 180/120.

*Topic: Hypertension | Ref: Ch 18, Section: Hypertension, Page 190*

**Q50. Answer: C — > 110 mm Hg**

Although there is no universal standard cut-off, the general consensus is to delay elective surgery if diastolic blood pressure is > 110 mm Hg.

*Topic: Hypertension | Ref: Ch 18, Section: Hypertension, Page 190*

# Anesthesia for Respiratory Diseases MCQ – Chapter 19

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 19)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Which two factors are the most important predictors of postoperative pulmonary complications?**

- A. Age > 60 years and ASA grade > 2
- B. Dyspnea at minimal activity and site of surgery
- C. Heavy smoking and congestive heart failure
- D. Serum albumin < 3.5 g/dL and severity of pulmonary disease

**Ans: B**

**Q2. What is the ideal duration for stopping smoking before surgery?**

- A. 2 weeks
- B. 4 weeks
- C. 8 weeks
- D. 12 weeks

**Ans: C**

**Q3. Stopping smoking 12 hours before surgery is beneficial primarily because it reduces:**

- A. Nicotine levels
- B. Carboxyhemoglobin levels
- C. Airway reactivity
- D. Mucus production

**Ans: B**

**Q4. Which of the following is NOT listed as a major risk factor for postoperative pulmonary complications?**

- A. Age > 60 years
- B. Serum albumin < 3.5 g/dL
- C. BMI > 30
- D. ASA grade > 2

**Ans: C**

**Q5. What surgical duration is cited as a risk factor for postoperative pulmonary complications?**

- A. > 1 hour
- B. > 2 hours
- C. > 3 hours
- D. > 4 hours

**Ans: C**

**Q6. Regarding pulmonary function tests (PFT) in patients with pulmonary disease, the text states that:**

- A. PFT should be done routinely in all patients
- B. PFT determines severity but routine use is not recommended
- C. PFT is the best predictor of postoperative complications
- D. PFT has no role in assessing pulmonary disease

**Ans: B**

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| <p><b>Q7. Why should premedication with H2 blockers (Ranitidine) be avoided in patients with pulmonary disease?</b></p> <ul style="list-style-type: none"> <li>A. It causes excessive sedation</li> <li>B. H2 blockade leads to unopposed H1 activation and bronchospasm</li> <li>C. It increases tracheobronchial secretions</li> <li>D. It causes respiratory depression</li> </ul>                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q8. Why is premedication with benzodiazepines avoided in patients with pulmonary disease?</b></p> <ul style="list-style-type: none"> <li>A. They cause bronchospasm</li> <li>B. They increase tracheobronchial secretions</li> <li>C. These patients are more sensitive to respiratory depression</li> <li>D. They cause histamine release</li> </ul>                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q9. Regarding anticholinergic premedication in pulmonary disease patients, the text states:</b></p> <ul style="list-style-type: none"> <li>A. It is contraindicated due to excessive drying of secretions</li> <li>B. It is theoretically beneficial by causing bronchodilatation but clinical advantage has not been seen</li> <li>C. It is the premedication of choice</li> <li>D. It should always be given with bronchodilators</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q10. What is the role of prophylactic nebulization in an asymptomatic patient with pulmonary disease?</b></p> <ul style="list-style-type: none"> <li>A. It is the standard of care preoperatively</li> <li>B. It should be given 30 minutes before surgery</li> <li>C. There is no role of prophylactic nebulization in asymptomatic patients</li> <li>D. It is recommended only for COPD patients</li> </ul>                                  | <p><b>Ans: C</b></p> |
| <p><b>Q11. For patients with pulmonary disease, which type of anesthesia is always preferred?</b></p> <ul style="list-style-type: none"> <li>A. General anesthesia with controlled ventilation</li> <li>B. Regional anesthesia</li> <li>C. Total intravenous anesthesia</li> <li>D. General anesthesia with spontaneous ventilation</li> </ul>                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q12. Central neuraxial block above which level can cause significant distress in COPD patients?</b></p> <ul style="list-style-type: none"> <li>A. T4</li> <li>B. T6</li> <li>C. T8</li> <li>D. T10</li> </ul>                                                                                                                                                                                                                                  | <p><b>Ans: B</b></p> |

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| <p><b>Q13. Why does a neuraxial block above T6 cause particular distress in COPD patients?</b></p> <p>A. It causes cardiac depression<br/> B. It leads to bronchospasm from parasympathetic stimulation<br/> C. These patients have active expiration dependent on abdominal muscles which get paralyzed<br/> D. It causes severe hypotension</p> | <p><b>Ans: C</b></p> |
| <p><b>Q14. Dry gases during general anesthesia can cause atelectasis by which mechanism?</b></p> <p>A. Causing bronchospasm<br/> B. Inhibiting ciliary function<br/> C. Increasing dead space<br/> D. Causing V/Q mismatch</p>                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q15. Which is the ONLY inhalational agent that does NOT inhibit ciliary activity?</b></p> <p>A. Sevoflurane<br/> B. Halothane<br/> C. Ether<br/> D. Isoflurane</p>                                                                                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q16. What is the recommended tidal volume for ventilating patients with pulmonary disease to avoid barotrauma?</b></p> <p>A. 4–6 mL/kg<br/> B. 6–8 mL/kg<br/> C. 8–10 mL/kg<br/> D. 10–12 mL/kg</p>                                                                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q17. What respiratory rate is recommended for ventilating patients with pulmonary disease?</b></p> <p>A. 4–6 breaths/min<br/> B. 6–10 breaths/min<br/> C. 10–14 breaths/min<br/> D. 14–18 breaths/min</p>                                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q18. Which of the following is NOT part of the ventilation principles for pulmonary disease patients?</b></p> <p>A. Low tidal volume (6–8 mL/kg)<br/> B. Low I:E ratio<br/> C. High respiratory rate (16–20 breaths/min)<br/> D. Slow respiratory rate (6–10 breaths/min)</p>                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q19. Improvement in pulmonary functions postoperatively should ideally be guided by:</b></p> <p>A. Peak expiratory flow rate (PEFR)<br/> B. FEV1<br/> C. Functional residual capacity (FRC)<br/> D. Vital capacity</p>                                                                                                                      | <p><b>Ans: C</b></p> |



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| <p><b>Q20. What is the most preferred approach for pain management in thoracic and upper abdominal surgeries in respiratory compromised patients?</b></p> <ul style="list-style-type: none"> <li>A. Parenteral opioids</li> <li>B. Intercostal nerve block</li> <li>C. Thoracic epidural</li> <li>D. Intrapleural analgesia</li> </ul>                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q21. Why is cryoanalgesia not commonly used for postoperative pain relief despite producing effective pain relief?</b></p> <ul style="list-style-type: none"> <li>A. Risk of pneumothorax and nerve damage</li> <li>B. Delayed onset, nonavailability of probes, and very long-lasting effect (nerve recovery 1–6 months)</li> <li>C. Risk of respiratory depression</li> <li>D. Requires repeated applications</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q22. Why should parenteral opioids be avoided in respiratory compromised patients for postoperative pain management?</b></p> <ul style="list-style-type: none"> <li>A. They cause bronchospasm</li> <li>B. They cause histamine release</li> <li>C. They cause respiratory depression</li> <li>D. They have poor analgesic efficacy</li> </ul>                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q23. In local infiltration for thoracotomy pain, how many levels above and below the incision site should be infiltrated?</b></p> <ul style="list-style-type: none"> <li>A. 1 level above and below</li> <li>B. 2 levels above and below</li> <li>C. 3 levels above and below</li> <li>D. 4 levels above and below</li> </ul>                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q24. For elective surgery in an asthma patient, what percentage of predicted peak expiratory flow rate (PEFR) should ideally be achieved?</b></p> <ul style="list-style-type: none"> <li>A. 60%</li> <li>B. 70%</li> <li>C. 80%</li> <li>D. 90%</li> </ul>                                                                                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q25. Why is ketamine generally avoided in asthma patients despite being the best bronchodilator?</b></p> <ul style="list-style-type: none"> <li>A. It causes cardiac depression</li> <li>B. Its side effects (especially vivid reactions) and propensity to increase tracheobronchial secretions</li> <li>C. It causes histamine release</li> <li>D. It has no IV formulation available</li> </ul>                         | <p><b>Ans: B</b></p> |

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| <p><b>Q26. In which specific clinical scenario is the use of ketamine indicated in asthma patients?</b></p> <ul style="list-style-type: none"> <li>A. All elective surgeries in asthma patients</li> <li>B. Patients in active asthma (wheezing) undergoing life-threatening emergency surgeries</li> <li>C. Patients with mild intermittent asthma</li> <li>D. As premedication for asthma patients</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q27. Ketamine in a patient taking theophylline can precipitate:</b></p> <ul style="list-style-type: none"> <li>A. Cardiac arrhythmias</li> <li>B. Severe hypotension</li> <li>C. Seizures</li> <li>D. Malignant hyperthermia</li> </ul>                                                                                                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q28. Which induction agent is most preferred for asthma patients in present-day anesthesia practice?</b></p> <ul style="list-style-type: none"> <li>A. Ketamine</li> <li>B. Thiopentone</li> <li>C. Propofol</li> <li>D. Etomidate</li> </ul>                                                                                                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q29. Which induction agent can induce bronchoconstriction and should be avoided in asthma patients?</b></p> <ul style="list-style-type: none"> <li>A. Propofol</li> <li>B. Ketamine</li> <li>C. Thiopentone</li> <li>D. Etomidate</li> </ul>                                                                                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q30. What is the most important consideration during induction in an asthma patient?</b></p> <ul style="list-style-type: none"> <li>A. Preventing hypotension</li> <li>B. Preventing reflex stimulation of airways by laryngoscopy and intubation</li> <li>C. Using rapid sequence induction</li> <li>D. Avoiding all muscle relaxants</li> </ul>                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q31. Which inhalational agent is the agent of choice for maintenance in asthma patients?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Sevoflurane</li> <li>D. Desflurane</li> </ul>                                                                                                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q32. Why is halothane not preferred over sevoflurane for asthma despite producing slightly more bronchodilatation?</b></p> <ul style="list-style-type: none"> <li>A. It is hepatotoxic</li> <li>B. It causes increased possibility of arrhythmias</li> <li>C. It inhibits ciliary function more</li> <li>D. It has airway irritant properties</li> </ul>                                                  | <p><b>Ans: B</b></p> |

|                                                                                                                                                                                                                                                                                                                                                                                |                      |
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| <p><b>Q33. Which class of muscle relaxant is preferred in asthma patients and why?</b></p> <p>A. Benzylisoquinolines — they have rapid onset<br/> B. Steroidal class — benzylisoquinolines release histamine causing bronchospasm<br/> C. Depolarizing agents — they have the shortest duration<br/> D. Ester-based relaxants — they are metabolized fastest</p>               | <p><b>Ans: B</b></p> |
| <p><b>Q34. To avoid immediate postoperative bronchospasm in asthma patients, extubation should be done:</b></p> <p>A. When the patient is fully awake<br/> B. When the patient is deep (deep extubation)<br/> C. Only after chest X-ray confirmation<br/> D. After giving IV aminophylline</p>                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q35. What is the FIRST step before starting treatment for suspected intraoperative bronchospasm?</b></p> <p>A. Give IV steroids<br/> B. Increase depth of anesthesia with sevoflurane<br/> C. Rule out other causes like ETT obstruction, kinking, inadequate depth, pulmonary edema, aspiration, embolus, or pneumothorax<br/> D. Administer inhaled beta-2 agonist</p> | <p><b>Ans: C</b></p> |
| <p><b>Q36. Once confirmed that intraoperative bronchospasm is due to asthma, what is the first line of treatment?</b></p> <p>A. IV epinephrine<br/> B. Inhaled beta-2 agonist<br/> C. Increasing the depth of anesthesia with inhalational agent (preferably sevoflurane)<br/> D. IV aminophylline infusion</p>                                                                | <p><b>Ans: C</b></p> |
| <p><b>Q37. For refractory bronchospasm not responding to inhaled beta-2 agonists and IV steroids, which agents should be considered?</b></p> <p>A. IV magnesium and calcium channel blockers<br/> B. Epinephrine, ketamine, or aminophylline infusion<br/> C. Ipratropium and tiotropium<br/> D. Leukotriene antagonists and mast cell stabilizers</p>                         | <p><b>Ans: B</b></p> |
| <p><b>Q38. Which is the most common pulmonary disease encountered in clinical practice?</b></p> <p>A. Asthma<br/> B. Tuberculosis<br/> C. COPD<br/> D. Restrictive lung disease</p>                                                                                                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q39. Why should nitrous oxide be avoided in emphysema patients?</b></p> <p>A. It causes bronchospasm<br/> B. It can cause expansion of bullae leading to rupture and pneumothorax<br/> C. It increases pulmonary vascular resistance<br/> D. It worsens V/Q mismatch</p>                                                                                                 | <p><b>Ans: B</b></p> |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q40. In COPD patients with evidence of pulmonary hypertension (cor pulmonale), which agent should be avoided?</b></p> <ul style="list-style-type: none"> <li>A. Sevoflurane</li> <li>B. Propofol</li> <li>C. Nitrous oxide</li> <li>D. Rocuronium</li> </ul>                                                                                                | <p><b>Ans: C</b></p> |
| <p><b>Q41. The anesthetic management of COPD patients is described as similar to which group of patients with additional considerations?</b></p> <ul style="list-style-type: none"> <li>A. Restrictive lung disease patients</li> <li>B. Asthma patients</li> <li>C. Tuberculosis patients</li> <li>D. Cardiac patients</li> </ul>                                | <p><b>Ans: B</b></p> |
| <p><b>Q42. Which of the following measures is NOT mentioned as part of preoperative stabilization of COPD patients?</b></p> <ul style="list-style-type: none"> <li>A. Stopping smoking</li> <li>B. Treating infection</li> <li>C. Starting long-term oxygen therapy</li> <li>D. Correcting spasm and improving pulmonary condition by physiotherapy</li> </ul>    | <p><b>Ans: C</b></p> |
| <p><b>Q43. Patients with restrictive lung disease are more prone to hypoxemia during induction because of:</b></p> <ul style="list-style-type: none"> <li>A. Increased dead space</li> <li>B. Low FRC</li> <li>C. High airway resistance</li> <li>D. Bronchospasm</li> </ul>                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q44. What is the most important principle of ventilatory management in restrictive lung disease during general anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Using high PEEP</li> <li>B. Using high FiO<sub>2</sub></li> <li>C. Avoiding large tidal volumes and inspiratory pressures</li> <li>D. Using rapid respiratory rates</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q45. Regarding tuberculosis patients, antitubercular drugs should be continued but assessment of which organ function is a must?</b></p> <ul style="list-style-type: none"> <li>A. Kidney function</li> <li>B. Liver function</li> <li>C. Cardiac function</li> <li>D. Thyroid function</li> </ul>                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q46. In a patient with active TB undergoing emergency surgery, which equipment should be discarded after the case?</b></p> <ul style="list-style-type: none"> <li>A. Only the endotracheal tube</li> <li>B. Mask, LMA, and breathing circuits</li> <li>C. All equipment including the anesthesia machine</li> <li>D. Only disposable items</li> </ul>       | <p><b>Ans: B</b></p> |

**Q47. For how long should elective surgery be deferred in a patient with significant URTI?**

- A. 2 weeks
- B. 4 weeks
- C. 6 weeks
- D. 8 weeks

**Ans: C**

**Q48. A patient with URTI undergoing surgery has how many times increased risk of bronchospasm compared to a healthy patient?**

- A. 2 times
- B. 5 times
- C. 10 times
- D. 15 times

**Ans: C**

**Q49. In a patient with active URTI requiring general anesthesia, which airway device is preferred over endotracheal intubation?**

- A. Face mask only
- B. Laryngeal mask airway (LMA)
- C. Nasopharyngeal airway
- D. Oropharyngeal airway

**Ans: B**

**Q50. What FEV1 value is one of the operative criteria indicating need for thoracotomy/pneumonectomy?**

- A. < 1 liter
- B. < 2 liters
- C. < 3 liters
- D. < 4 liters

**Ans: B**

# Answer Key & Explanations

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## **Q1. Answer: B — Dyspnea at minimal activity and site of surgery**

The text explicitly states that 'Dyspnea at minimal activity and site of surgery are the two most important predictors of postoperative complications.'

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

## **Q2. Answer: C — 8 weeks**

The text states 'Ideally smoking should be stopped 8 weeks before surgery however stopping smoking 12 hours before may be beneficial by reducing carboxyhemoglobin levels.'

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

## **Q3. Answer: B — Carboxyhemoglobin levels**

The text states that stopping smoking 12 hours before surgery 'may be beneficial by reducing carboxyhemoglobin levels.'

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

## **Q4. Answer: C — BMI > 30**

The listed risk factors include age > 60, severity of pulmonary disease, ASA grade > 2, CHF, heavy smoking, surgery under GA, surgical factors, and serum albumin < 3.5 g/dL. BMI > 30 is not mentioned.

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

## **Q5. Answer: C — > 3 hours**

The text lists surgical factors including 'duration (> 3 hours)' as a major risk factor.

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

## **Q6. Answer: B — PFT determines severity but routine use is not recommended**

The text states 'Although pulmonary function test (PFT) do determine the severity of disease but are not always predictors of complications therefore routine use of PFT is not recommended.'

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

## **Q7. Answer: B — H2 blockade leads to unopposed H1 activation and bronchospasm**

The text states 'Premedication with H2 blockers (Ranitidine) should be avoided because H2 blockade leads to unopposed H1 activation and bronchospasm.'

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

## **Q8. Answer: C — These patients are more sensitive to respiratory depression**

The text states 'Premedication with benzodiazepines should be avoided as these patients are more sensitive to respiratory depression.'

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

## **Q9. Answer: B — It is theoretically beneficial by causing bronchodilatation but clinical advantage has not been seen**

The text states 'Theoretically premedication with anticholinergic is beneficial by causing bronchodilatation, however, clinical advantage has not been seen.'

*Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192*

**Q10. Answer: C — There is no role of prophylactic nebulization in asymptomatic patients**

The text clearly states 'there is no role of prophylactic nebulization in asymptomatic patient.'

Topic: General Considerations | Ref: Ch 19, Section: Preoperative Considerations, Page 192

**Q11. Answer: B — Regional anesthesia**

The text states 'Regional anesthesia is always preferred over general anesthesia.'

Topic: Intraoperative Considerations | Ref: Ch 19, Section: Choice of Anesthesia, Page 192

**Q12. Answer: B — T6**

The text states central neuraxial blocks are excellent if below T6. Level above T6 can cause paralysis of abdominal muscles causing significant distress, particularly in COPD patients.

Topic: Intraoperative Considerations | Ref: Ch 19, Section: Choice of Anesthesia, Page 192-193

**Q13. Answer: C — These patients have active expiration dependent on abdominal muscles which get paralyzed**

The text explains that block above T6 'can cause paralysis of abdominal muscles causing significant distress (particularly in COPD patients) because these patients have active expiration dependent on abdominal muscles.'

Topic: Intraoperative Considerations | Ref: Ch 19, Section: Choice of Anesthesia, Page 192-193

**Q14. Answer: B — Inhibiting ciliary function**

The text states 'Dry gases can inhibit ciliary function causing atelectasis in postoperative period.'

Topic: Intraoperative Considerations | Ref: Ch 19, Section: Intraoperative Considerations, Page 193

**Q15. Answer: C — Ether**

The text states 'all inhalational agents (except ether) also inhibit ciliary activity.'

Topic: Intraoperative Considerations | Ref: Ch 19, Section: Intraoperative Considerations, Page 193

**Q16. Answer: B — 6–8 mL/kg**

The text states 'Ventilation principle includes low tidal volume (6–8 mL/kg) to avoid barotrauma.'

Topic: Intraoperative Considerations | Ref: Ch 19, Section: Intraoperative Considerations, Page 193

**Q17. Answer: B — 6–10 breaths/min**

The text states a 'slow respiratory rate of (6–10 breaths/min.) to allow adequate exhalation and prevent air trapping.'

Topic: Intraoperative Considerations | Ref: Ch 19, Section: Intraoperative Considerations, Page 193

**Q18. Answer: C — High respiratory rate (16–20 breaths/min)**

The ventilation principles include low tidal volume, low I:E ratio, and slow respiratory rate (6–10 breaths/min). A high respiratory rate would prevent adequate exhalation and cause air trapping.

Topic: Intraoperative Considerations | Ref: Ch 19, Section: Intraoperative Considerations, Page 193

**Q19. Answer: C — Functional residual capacity (FRC)**

The text states 'Ideally the improvement in pulmonary functions should be guided by functional residual capacity (FRC).'

Topic: Postoperative Considerations | Ref: Ch 19, Section: Postoperative Considerations, Page 193

**Q20. Answer: C — Thoracic epidural**

The text states 'Thoracic epidural is the most preferred, most safe, most effective and most commonly used approach to achieve analgesia for thoracic and upper abdominal surgeries.'

Topic: Postoperative Considerations | Ref: Ch 19, Section: Postoperative Considerations, Page 193

**Q21. Answer: B — Delayed onset, nonavailability of probes, and very long-lasting effect (nerve recovery 1–6 months)**

The text states cryoanalgesia is 'not used commonly due to delayed onset, nonavailability of probes and very long lasting effect (nerve recovery may even take 1–6 months).'

Topic: Postoperative Considerations | Ref: Ch 19, Section: Postoperative Considerations, Page 193

**Q22. Answer: C — They cause respiratory depression**

The text states 'Parenteral opioids can cause respiratory depression therefore should be avoided as far as possible.'

Topic: Postoperative Considerations | Ref: Ch 19, Section: Postoperative Considerations, Page 193

**Q23. Answer: B — 2 levels above and below**

The text mentions 'local infiltration (2 levels above and below the thoracotomy incision site)' as one of the analgesic techniques.

Topic: Postoperative Considerations | Ref: Ch 19, Section: Postoperative Considerations, Page 193

**Q24. Answer: C — 80%**

The text states the patient 'ideally should have 80% of predicted value of peak expiratory flow rate.'

Topic: Asthma | Ref: Ch 19, Section: Asthma, Page 193

**Q25. Answer: B — Its side effects (especially vivid reactions) and propensity to increase tracheobronchial secretions**

The text states 'Ketamine in spite of its best bronchodilator property is generally avoided due to its side effect (especially vivid reactions) and propensity to increase tracheobronchial secretions.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Induction, Page 193

**Q26. Answer: B — Patients in active asthma (wheezing) undergoing life-threatening emergency surgeries**

The text states 'The use of ketamine in present day practice is restricted to patients in active asthma (wheezing) undergoing life threatening emergency surgeries.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Induction, Page 193

**Q27. Answer: C — Seizures**

The text states 'Ketamine in patient taking theophylline can precipitate seizures.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Induction, Page 193

**Q28. Answer: C — Propofol**

The text states 'Propofol is a reasonable bronchodilator devoid of side effects seen with ketamine therefore, is more preferred for asthma patients in present day anesthesia practice.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Induction, Page 193

**Q29. Answer: C — Thiopentone**

The text states 'Thiopentone can induce bronchoconstriction therefore, should be avoided.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Induction, Page 193



**Q30. Answer: B — Preventing reflex stimulation of airways by laryngoscopy and intubation**

The text states 'The most important consideration is to prevent reflex stimulation of airways by laryngoscopy and intubation.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Induction, Page 193

**Q31. Answer: C — Sevoflurane**

The text states 'Sevoflurane is the inhalational agent of choice for asthma patients.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Maintenance, Page 193

**Q32. Answer: B — It causes increased possibility of arrhythmias**

The text states 'Halothane, in spite of producing little more bronchodilatation than sevoflurane in asthma patients is not preferred because of the increase possibility of arrhythmias.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Maintenance, Page 193

**Q33. Answer: B — Steroidal class — benzylisoquinolines release histamine causing bronchospasm**

The text states 'Steroidal class of muscle relaxant is preferred. Benzylisoquinolines by release histamine can precipitate bronchospasm.'

Topic: Asthma | Ref: Ch 19, Section: Asthma – Muscle Relaxant, Page 193

**Q34. Answer: B — When the patient is deep (deep extubation)**

The text states 'To avoid immediate postoperative bronchospasm extubation should be done when patient is deep (deep extubation).'

Topic: Asthma | Ref: Ch 19, Section: Asthma, Page 194

**Q35. Answer: C — Rule out other causes like ETT obstruction, kinking, inadequate depth, pulmonary edema, aspiration, embolus, or pneumothorax**

Step 1: Before starting therapy for asthma it is must to rule out other causes of increased airway pressure/bronchospasm.

Topic: Asthma | Ref: Ch 19, Section: Management of Intraoperative Bronchospasm, Page 194

**Q36. Answer: C — Increasing the depth of anesthesia with inhalational agent (preferably sevoflurane)**

Step 2: It should first be treated by increasing the depth of anesthesia with inhalational agent (preferably sevoflurane).

Topic: Asthma | Ref: Ch 19, Section: Management of Intraoperative Bronchospasm, Page 194

**Q37. Answer: B — Epinephrine, ketamine, or aminophylline infusion**

Step 5: For refractory bronchospasm consider epinephrine, ketamine or aminophylline infusion.

Topic: Asthma | Ref: Ch 19, Section: Management of Intraoperative Bronchospasm, Page 194

**Q38. Answer: C — COPD**

The text states 'Chronic obstructive pulmonary disease (COPD) is the most common pulmonary disease encountered in clinical practice.'

Topic: COPD | Ref: Ch 19, Section: COPD, Page 194

**Q39. Answer: B — It can cause expansion of bullae leading to rupture and pneumothorax**

The text states 'Nitrous oxide should not be used in emphysema. It can cause expansion of bullae leading to rupture and pneumothorax.'

Topic: COPD | Ref: Ch 19, Section: COPD, Page 194

**Q40. Answer: C — Nitrous oxide**

The text states 'Cor pulmonale must be ruled out and if there is any evidence of pulmonary hypertension then nitrous oxide should be avoided.'

Topic: COPD | Ref: Ch 19, Section: COPD, Page 194

**Q41. Answer: B — Asthma patients**

The text states 'The anesthetic management of COPD patients is similar to asthma patient with additional emphasis on the following points.'

Topic: COPD | Ref: Ch 19, Section: COPD, Page 194

**Q42. Answer: C — Starting long-term oxygen therapy**

The text mentions stopping smoking, treating infection, correcting spasm (asthmatic bronchitis), and improve pulmonary condition by physiotherapy, but not long-term oxygen therapy.

Topic: COPD | Ref: Ch 19, Section: COPD, Page 194

**Q43. Answer: B — Low FRC**

The text states 'These patients have low FRC therefore are more prone for hypoxemia during induction.'

Topic: Restrictive Lung Disease | Ref: Ch 19, Section: Restrictive Lung Diseases, Page 194

**Q44. Answer: C — Avoiding large tidal volumes and inspiratory pressures**

The text states 'avoiding large tidal volumes and inspiratory pressures is the most important principle of ventilatory management during general anesthesia.'

Topic: Restrictive Lung Disease | Ref: Ch 19, Section: Restrictive Lung Diseases, Page 194

**Q45. Answer: B — Liver function**

The text states 'Antitubercular drugs should be continued however, assessment of liver functions is must.'

Topic: Tuberculosis | Ref: Ch 19, Section: Tuberculosis, Page 194-195

**Q46. Answer: B — Mask, LMA, and breathing circuits**

The text states 'Equipment coming in direct contact with patient like mask, laryngeal mask airway (LMA), breathing circuits should be discarded after case while other equipment and theater should be sterilized properly.'

Topic: Tuberculosis | Ref: Ch 19, Section: Tuberculosis, Page 195

**Q47. Answer: C — 6 weeks**

The text states surgery should 'ideally be deferred for 6 weeks' as 'airway remains hyperactive for 5–6 weeks after a significant URTI.'

Topic: URTI | Ref: Ch 19, Section: Respiratory Tract Infection, Page 195

**Q48. Answer: C — 10 times**

The text states patients with active URTI are prone to develop 'bronchospasm (10 times)' compared to normal.

Topic: URTI | Ref: Ch 19, Section: Respiratory Tract Infection, Page 195

**Q49. Answer: B — Laryngeal mask airway (LMA)**

The text states 'If GA is to be given then prefer to get the surgery done under laryngeal mask airway (LMA). Avoid intubation as far as possible.'

Topic: URTI | Ref: Ch 19, Section: Respiratory Tract Infection, Page 195

**Q50. Answer: B — < 2 liters**

The text lists 'FEV1 < 2 liters' as one of the operative criteria for thoracotomy/pneumectomy.

*Topic: Operative Criteria for Thoracotomy | Ref: Ch 19, Section: Operative Criteria for Thoracotomy/Pneumectomy, Page 195*

# Anesthesia for Central Nervous System Diseases – MCQ

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Anesthesia for Central Nervous System Diseases

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. In a patient with Parkinson's disease, which drug should be continued on the morning of surgery?**

- A. Anticholinergics
- B. Levodopa/Carbidopa
- C. Benzodiazepines
- D. MAO-B inhibitors

**Ans: B**

**Q2. Which anesthetic agent should be AVOIDED in Parkinson's disease due to its antidopaminergic effects?**

- A. Propofol
- B. Ketamine
- C. Droperidol
- D. Thiopentone

**Ans: C**

**Q3. Parkinson's disease patients are at increased risk of which respiratory complication?**

- A. Bronchospasm
- B. Aspiration pneumonia
- C. Pneumothorax
- D. Pulmonary embolism

**Ans: B**

**Q4. Which interaction is most dangerous in a patient taking Selegiline (MAO-B inhibitor) for Parkinson's disease?**

- A. Propofol causing bradycardia
- B. Meperidine (Pethidine) causing serotonin syndrome
- C. Succinylcholine causing hyperkalemia
- D. Atracurium causing histamine release

**Ans: B**

**Q5. In Alzheimer's disease, which enzyme inhibitor is used for treatment and may prolong the action of succinylcholine?**

- A. MAO inhibitors
- B. Anticholinesterase inhibitors (Donepezil)
- C. ACE inhibitors
- D. Proton pump inhibitors

**Ans: B**

**Q6. Which of the following is a primary anesthetic concern in Alzheimer's disease?**

- A. Malignant hyperthermia
- B. Postoperative cognitive dysfunction (POCD)
- C. Autonomic hyperreflexia
- D. Latex allergy

**Ans: B**

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. Which anticholinergic is preferred in Alzheimer's disease patients?</b></p> <p>A. Atropine<br/>B. Scopolamine (Hyoscine)<br/>C. Glycopyrrolate<br/>D. Ipratropium</p>                                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q8. The primary anesthetic concern in epilepsy is:</b></p> <p>A. Avoiding all inhalational agents<br/>B. Perioperative seizure prevention and drug interactions<br/>C. Mandatory use of regional anesthesia<br/>D. Avoiding opioids completely</p>                                           | <p><b>Ans: B</b></p> |
| <p><b>Q9. Which anesthetic agent has the HIGHEST seizure-provoking potential?</b></p> <p>A. Propofol<br/>B. Sevoflurane<br/>C. Enflurane<br/>D. Isoflurane</p>                                                                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q10. Chronic phenytoin therapy affects neuromuscular blocking agents by:</b></p> <p>A. Prolonging the action of non-depolarizing agents<br/>B. Causing resistance to non-depolarizing agents<br/>C. Prolonging the action of succinylcholine<br/>D. Having no effect on muscle relaxants</p> | <p><b>Ans: B</b></p> |
| <p><b>Q11. Which intravenous anesthetic has anticonvulsant properties and is suitable for epileptic patients?</b></p> <p>A. Ketamine<br/>B. Etomidate<br/>C. Thiopentone<br/>D. Methohexitone</p>                                                                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q12. Status epilepticus is defined as continuous seizure activity lasting more than:</b></p> <p>A. 2 minutes<br/>B. 5 minutes<br/>C. 15 minutes<br/>D. 30 minutes</p>                                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q13. In a patient with acute stroke presenting for emergency surgery, the most important anesthetic goal is:</b></p> <p>A. Controlled hypotension<br/>B. Maintenance of cerebral perfusion pressure<br/>C. Deep anesthesia to reduce CMRO<sub>2</sub><br/>D. Induced hypothermia to 32°C</p> | <p><b>Ans: B</b></p> |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q14. Elective surgery should ideally be postponed for how long after an ischemic stroke?</b></p> <p>A. 2 weeks<br/>B. 4 weeks<br/>C. 6 weeks<br/>D. 3 months</p>                                                                                                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q15. Which of the following is TRUE regarding anesthesia for carotid endarterectomy in stroke patients?</b></p> <p>A. General anesthesia is always superior to regional<br/>B. Regional anesthesia allows neurological monitoring during cross-clamp<br/>C. Hypotension is desirable during cross-clamping<br/>D. Hypothermia below 33°C is mandatory</p> | <p><b>Ans: B</b></p> |
| <p><b>Q16. In patients with chronic headache/migraine, which of the following agents may trigger an attack?</b></p> <p>A. Propofol<br/>B. Nitroglycerin (GTN)<br/>C. Fentanyl<br/>D. Lidocaine</p>                                                                                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q17. Post-dural puncture headache is characteristically:</b></p> <p>A. Worse when lying down<br/>B. Worse when sitting or standing (postural)<br/>C. Associated with fever<br/>D. Unilateral and throbbing</p>                                                                                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q18. Multiple Sclerosis is characterized by:</b></p> <p>A. Destruction of anterior horn cells<br/>B. Demyelination in the central nervous system<br/>C. Peripheral nerve demyelination<br/>D. Neuromuscular junction dysfunction</p>                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q19. In Multiple Sclerosis, which of the following may cause relapse/exacerbation?</b></p> <p>A. Hypothermia<br/>B. Hyperthermia (fever/increased body temperature)<br/>C. General anesthesia always<br/>D. Regional anesthesia always</p>                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q20. Regarding regional anesthesia in Multiple Sclerosis:</b></p> <p>A. Epidural anesthesia is absolutely contraindicated<br/>B. Spinal anesthesia may worsen symptoms due to direct exposure of demyelinated cord to local anesthetic<br/>C. Peripheral nerve blocks are contraindicated<br/>D. All regional techniques are completely safe</p>          | <p><b>Ans: B</b></p> |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <p><b>Q21. Succinylcholine should be AVOIDED in Multiple Sclerosis because of:</b></p> <ul style="list-style-type: none"> <li>A. Malignant hyperthermia risk</li> <li>B. Hyperkalemia from denervation-type receptor changes</li> <li>C. Prolonged neuromuscular block</li> <li>D. Anaphylaxis risk</li> </ul>                                 | Ans: B |
| <p><b>Q22. Syringomyelia is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. Cavity formation in the cerebral cortex</li> <li>B. Fluid-filled cavity (syrinx) within the spinal cord</li> <li>C. Demyelination of peripheral nerves</li> <li>D. Inflammation of the meninges</li> </ul>                                     | Ans: B |
| <p><b>Q23. The classic sensory finding in syringomyelia is:</b></p> <ul style="list-style-type: none"> <li>A. Glove and stocking sensory loss</li> <li>B. Dissociated sensory loss (loss of pain/temperature with preserved touch)</li> <li>C. Complete loss of all sensations below lesion</li> <li>D. Loss of proprioception only</li> </ul> | Ans: B |
| <p><b>Q24. Anesthetic concern in syringomyelia includes:</b></p> <ul style="list-style-type: none"> <li>A. Difficult airway due to limited neck extension</li> <li>B. Mandatory use of spinal anesthesia</li> <li>C. Succinylcholine is safe to use</li> <li>D. Temperature monitoring is not needed</li> </ul>                                | Ans: A |
| <p><b>Q25. Amyotrophic Lateral Sclerosis (ALS) affects:</b></p> <ul style="list-style-type: none"> <li>A. Only upper motor neurons</li> <li>B. Only lower motor neurons</li> <li>C. Both upper and lower motor neurons</li> <li>D. Only sensory neurons</li> </ul>                                                                             | Ans: C |
| <p><b>Q26. In ALS patients, succinylcholine is contraindicated because of:</b></p> <ul style="list-style-type: none"> <li>A. Risk of malignant hyperthermia</li> <li>B. Life-threatening hyperkalemia</li> <li>C. Prolonged apnea</li> <li>D. Histamine release</li> </ul>                                                                     | Ans: B |
| <p><b>Q27. The most important postoperative concern in ALS patients is:</b></p> <ul style="list-style-type: none"> <li>A. Pain management</li> <li>B. Respiratory failure requiring prolonged ventilation</li> <li>C. Urinary retention</li> <li>D. Hypothermia</li> </ul>                                                                     | Ans: B |
| <p><b>Q28. ALS patients show increased sensitivity to:</b></p> <ul style="list-style-type: none"> <li>A. Inhalational agents only</li> <li>B. Non-depolarizing neuromuscular blocking agents</li> <li>C. Local anesthetics</li> <li>D. Antiemetics</li> </ul>                                                                                  | Ans: B |



|                                                                                                                                                                                                                                                                                                                                                     |                      |
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| <p><b>Q29. Guillain-Barré Syndrome (GBS) is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. Ascending central demyelination</li> <li>B. Acute inflammatory demyelinating polyradiculoneuropathy (peripheral)</li> <li>C. Anterior horn cell destruction</li> <li>D. Neuromuscular junction antibodies</li> </ul>                | <p><b>Ans: B</b></p> |
| <p><b>Q30. The most dangerous complication of GBS requiring ICU admission is:</b></p> <ul style="list-style-type: none"> <li>A. Renal failure</li> <li>B. Respiratory failure from diaphragmatic weakness</li> <li>C. Seizures</li> <li>D. Hepatic failure</li> </ul>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q31. In GBS, succinylcholine is contraindicated due to:</b></p> <ul style="list-style-type: none"> <li>A. Prolonged block only</li> <li>B. Risk of hyperkalemia from denervation</li> <li>C. Histamine release</li> <li>D. Bradycardia</li> </ul>                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q32. Autonomic dysfunction in GBS may manifest as:</b></p> <ul style="list-style-type: none"> <li>A. Only hypertension</li> <li>B. Labile blood pressure, tachycardia, bradycardia, cardiac arrest</li> <li>C. Only bradycardia</li> <li>D. Only urinary retention</li> </ul>                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q33. Autonomic dysfunction (dysautonomia) is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. Stable hemodynamics</li> <li>B. Failure of the autonomic nervous system to regulate BP, HR, and temperature</li> <li>C. Only parasympathetic overactivity</li> <li>D. Only sympathetic overactivity</li> </ul>               | <p><b>Ans: B</b></p> |
| <p><b>Q34. In patients with autonomic dysfunction, the anesthetic concern includes:</b></p> <ul style="list-style-type: none"> <li>A. Exaggerated hypotensive response to anesthetic agents</li> <li>B. Need for higher doses of vasopressors</li> <li>C. Resistance to all anesthetic agents</li> <li>D. Increased risk of bronchospasm</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q35. Orthostatic hypotension in autonomic dysfunction is defined as a fall in systolic BP of:</b></p> <ul style="list-style-type: none"> <li>A. &gt;10 mmHg on standing</li> <li>B. &gt;20 mmHg on standing</li> <li>C. &gt;30 mmHg on standing</li> <li>D. &gt;40 mmHg on standing</li> </ul>                                                | <p><b>Ans: B</b></p> |

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| <p><b>Q36. Spinal cord transection above T6 may result in:</b></p> <ul style="list-style-type: none"> <li>A. Hypothermia only</li> <li>B. Autonomic hyperreflexia</li> <li>C. Hyperthermia only</li> <li>D. No cardiovascular changes</li> </ul>                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q37. The most common trigger for autonomic hyperreflexia is:</b></p> <ul style="list-style-type: none"> <li>A. Emotional stress</li> <li>B. Bladder distension</li> <li>C. Cold environment</li> <li>D. Exercise</li> </ul>                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q38. Treatment of autonomic hyperreflexia includes all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Remove the stimulus</li> <li>B. Sit the patient upright</li> <li>C. Administer vasopressors</li> <li>D. Give vasodilators (GTN/nifedipine)</li> </ul>                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q39. In acute spinal cord injury (spinal shock phase), the hemodynamic picture is:</b></p> <ul style="list-style-type: none"> <li>A. Hypertension and tachycardia</li> <li>B. Hypotension and bradycardia (neurogenic shock)</li> <li>C. Normal blood pressure with tachycardia</li> <li>D. Hypertension and bradycardia</li> </ul>              | <p><b>Ans: B</b></p> |
| <p><b>Q40. Succinylcholine is safe to use after spinal cord injury only within:</b></p> <ul style="list-style-type: none"> <li>A. First 24 hours</li> <li>B. First 48 hours</li> <li>C. First 72 hours</li> <li>D. First week</li> </ul>                                                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q41. Which psychiatric medication has the most significant anesthetic interaction?</b></p> <ul style="list-style-type: none"> <li>A. SSRIs</li> <li>B. MAO inhibitors</li> <li>C. Benzodiazepines</li> <li>D. Atypical antipsychotics</li> </ul>                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q42. Lithium therapy affects anesthesia by:</b></p> <ul style="list-style-type: none"> <li>A. Shortening the action of muscle relaxants</li> <li>B. Prolonging the action of both depolarizing and non-depolarizing muscle relaxants</li> <li>C. Having no effect on anesthetic drugs</li> <li>D. Reducing the MAC of volatile agents</li> </ul> | <p><b>Ans: B</b></p> |

|                                                                                                                                                                                                                                                                                                                                                    |                      |
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| <p><b>Q43. Tricyclic antidepressants (TCAs) may cause which of the following during anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Exaggerated response to indirect sympathomimetics (ephedrine)</li> <li>B. Resistance to vasopressors</li> <li>C. Bronchospasm</li> <li>D. Malignant hyperthermia</li> </ul>                      | <p><b>Ans: A</b></p> |
| <p><b>Q44. Electroconvulsive therapy (ECT) requires anesthesia with:</b></p> <ul style="list-style-type: none"> <li>A. Deep general anesthesia with intubation</li> <li>B. Brief general anesthesia with short-acting agents (methohexitone/propofol + succinylcholine)</li> <li>C. Spinal anesthesia</li> <li>D. Local anesthesia only</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q45. Neuroleptic Malignant Syndrome (NMS) is characterized by all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Hyperthermia</li> <li>B. Muscle rigidity</li> <li>C. Hypotension and hypothermia</li> <li>D. Autonomic instability</li> </ul>                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q46. In a patient with Parkinson's disease, which anesthetic technique may be preferred for lower limb surgery?</b></p> <ul style="list-style-type: none"> <li>A. General anesthesia with enflurane</li> <li>B. Regional anesthesia (spinal/epidural)</li> <li>C. Deep sedation with ketamine</li> <li>D. Only local infiltration</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q47. In Multiple Sclerosis, which factor does NOT increase the risk of perioperative relapse?</b></p> <ul style="list-style-type: none"> <li>A. Infection/fever</li> <li>B. Emotional stress</li> <li>C. Propofol administration</li> <li>D. Surgery itself</li> </ul>                                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q48. Guillain-Barré Syndrome typically presents how many weeks after a respiratory or GI infection?</b></p> <ul style="list-style-type: none"> <li>A. 1–4 weeks</li> <li>B. 6–8 weeks</li> <li>C. 3 months</li> <li>D. 6 months</li> </ul>                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q49. In autonomic dysfunction, which monitoring is ESSENTIAL during anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Only pulse oximetry</li> <li>B. Invasive arterial blood pressure monitoring</li> <li>C. Only NIBP every 15 minutes</li> <li>D. BIS monitoring</li> </ul>                                                   | <p><b>Ans: B</b></p> |

**Q50. Which anticonvulsant medication is associated with aplastic anemia, requiring preoperative blood count check?**

- A. Phenytoin
- B. Carbamazepine
- C. Sodium valproate
- D. Gabapentin

**Ans: B**

# Answer Key & Explanations

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## Q1. Answer: B — Levodopa/Carbidopa

Correct Answer: B. Levodopa/Carbidopa — Levodopa should be continued perioperatively to prevent rigidity, tremor, and difficulty with ventilation. Abrupt withdrawal can cause neuroleptic malignant-like syndrome.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q2. Answer: C — Droperidol

Correct Answer: C. Droperidol — Droperidol and other butyrophenones have antidopaminergic activity and can worsen parkinsonian symptoms. Metoclopramide should also be avoided for the same reason.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q3. Answer: B — Aspiration pneumonia

Correct Answer: B. Aspiration pneumonia — Parkinsonian patients have swallowing dysfunction and pharyngeal muscle rigidity leading to increased risk of aspiration. Rapid sequence induction may be considered.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q4. Answer: B — Meperidine (Pethidine) causing serotonin syndrome

Correct Answer: B. Meperidine (Pethidine) causing serotonin syndrome — MAO inhibitors with meperidine can cause fatal serotonin syndrome (hyperthermia, rigidity, seizures). Meperidine must be avoided in these patients.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q5. Answer: B — Anticholinesterase inhibitors (Donepezil)

Correct Answer: B. Anticholinesterase inhibitors (Donepezil) — Donepezil inhibits cholinesterase, which can prolong the duration of succinylcholine and antagonize non-depolarizing neuromuscular blocking agents.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

## Q6. Answer: B — Postoperative cognitive dysfunction (POCD)

Correct Answer: B. Postoperative cognitive dysfunction (POCD) — Alzheimer's patients are at increased risk of POCD and postoperative delirium. Centrally acting anticholinergics (atropine) should be avoided.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

## Q7. Answer: C — Glycopyrrolate

Correct Answer: C. Glycopyrrolate — Glycopyrrolate does not cross the blood-brain barrier and is preferred over atropine or scopolamine, which can worsen confusion and cognitive dysfunction in Alzheimer's patients.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

## Q8. Answer: B — Perioperative seizure prevention and drug interactions

Correct Answer: B. Perioperative seizure prevention and drug interactions — Antiepileptic drugs must be continued perioperatively. Many anticonvulsants induce hepatic enzymes, altering metabolism of anesthetic drugs.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

## Q9. Answer: C — Enflurane

Correct Answer: C. Enflurane — Enflurane at high concentrations (>2 MAC) especially with hypocapnia produces epileptiform EEG activity. It should be avoided in epileptic patients.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q10. Answer: B — Causing resistance to non-depolarizing agents**

Correct Answer: B. Causing resistance to non-depolarizing agents — Chronic anticonvulsant therapy (phenytoin, carbamazepine) causes resistance to non-depolarizing muscle relaxants due to hepatic enzyme induction and upregulation of acetylcholine receptors.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q11. Answer: C — Thiopentone**

Correct Answer: C. Thiopentone — Thiopentone and propofol have anticonvulsant properties. Methohexitone and ketamine may provoke seizures and should be avoided in epileptic patients.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q12. Answer: B — 5 minutes**

Correct Answer: B. 5 minutes — Current definition: continuous seizure activity  $\geq 5$  minutes or  $\geq 2$  seizures without full recovery between them. First-line treatment is IV benzodiazepines (diazepam/lorazepam).

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q13. Answer: B — Maintenance of cerebral perfusion pressure**

Correct Answer: B. Maintenance of cerebral perfusion pressure — In stroke patients, the ischemic penumbra depends on adequate perfusion. Avoiding hypotension and maintaining CPP is the primary anesthetic goal.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

**Q14. Answer: C — 6 weeks**

Correct Answer: C. 6 weeks — Elective surgery should be delayed at least 6 weeks after a stroke to allow stabilization of the blood-brain barrier and reduction of hemorrhagic transformation risk.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

**Q15. Answer: B — Regional anesthesia allows neurological monitoring during cross-clamp**

Correct Answer: B. Regional anesthesia allows neurological monitoring during cross-clamp — Cervical plexus block allows continuous neurological assessment during carotid cross-clamping to detect cerebral ischemia.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

**Q16. Answer: B — Nitroglycerin (GTN)**

Correct Answer: B. Nitroglycerin (GTN) — Nitroglycerin is a potent vasodilator and well-known migraine trigger. It should be used cautiously in patients with a history of migraines.

Topic: Headache | Ref: Ch 20, Page 196, Topic: Headache

**Q17. Answer: B — Worse when sitting or standing (postural)**

Correct Answer: B. Worse when sitting or standing (postural) — PDPH is a postural headache that worsens in the upright position and improves on lying flat, caused by CSF leak and reduced intracranial pressure.

Topic: Headache | Ref: Ch 20, Page 196, Topic: Headache

**Q18. Answer: B — Demyelination in the central nervous system**

Correct Answer: B. Demyelination in the central nervous system — MS is an autoimmune disease causing patchy demyelination in the brain and spinal cord, leading to relapsing-remitting neurological deficits.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q19. Answer: B — Hyperthermia (fever/increased body temperature)**

Correct Answer: B. Hyperthermia (fever/increased body temperature) — Even small increases in body temperature can worsen symptoms in MS (Uhthoff's phenomenon). Perioperative temperature control is essential.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q20. Answer: B — Spinal anesthesia may worsen symptoms due to direct exposure of demyelinated cord to local anesthetic**

Correct Answer: B. Spinal anesthesia may worsen symptoms due to direct exposure of demyelinated cord to local anesthetic — Spinal anesthesia exposes the already demyelinated cord to local anesthetic. Epidural is relatively safer; lowest effective concentration should be used.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q21. Answer: B — Hyperkalemia from denervation-type receptor changes**

Correct Answer: B. Hyperkalemia from denervation-type receptor changes — Upper motor neuron lesions in MS lead to extrajunctional acetylcholine receptor proliferation, causing hyperkalemic response to succinylcholine.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q22. Answer: B — Fluid-filled cavity (syrinx) within the spinal cord**

Correct Answer: B. Fluid-filled cavity (syrinx) within the spinal cord — Syringomyelia is a condition where a fluid-filled cyst (syrinx) forms within the spinal cord, often in the cervical region.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

**Q23. Answer: B — Dissociated sensory loss (loss of pain/temperature with preserved touch)**

Correct Answer: B. Dissociated sensory loss (loss of pain/temperature with preserved touch) — The syrinx destroys the crossing spinothalamic fibers causing cape-like loss of pain and temperature while posterior column sensation (touch, proprioception) is preserved.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

**Q24. Answer: A — Difficult airway due to limited neck extension**

Correct Answer: A. Difficult airway due to limited neck extension — Syringomyelia often coexists with Arnold-Chiari malformation causing limited neck mobility. Careful airway management with minimal neck manipulation is essential.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

**Q25. Answer: C — Both upper and lower motor neurons**

Correct Answer: C. Both upper and lower motor neurons — ALS (Motor Neuron Disease) is a progressive neurodegenerative disease affecting both upper and lower motor neurons, leading to muscle weakness, wasting, and spasticity.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

**Q26. Answer: B — Life-threatening hyperkalemia**

Correct Answer: B. Life-threatening hyperkalemia — Lower motor neuron involvement in ALS causes denervation and extrajunctional receptor proliferation, making succinylcholine extremely dangerous due to massive potassium release.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

**Q27. Answer: B — Respiratory failure requiring prolonged ventilation**

Correct Answer: B. Respiratory failure requiring prolonged ventilation — ALS patients have progressive respiratory muscle weakness. They are extremely sensitive to respiratory depressants and may require postoperative ventilatory support.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

**Q28. Answer: B — Non-depolarizing neuromuscular blocking agents**

Correct Answer: B. Non-depolarizing neuromuscular blocking agents — Due to decreased muscle mass and denervation, ALS patients are very sensitive to non-depolarizing agents. Reduced doses should be used with careful monitoring.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

**Q29. Answer: B — Acute inflammatory demyelinating polyradiculoneuropathy (peripheral)**

Correct Answer: B. Acute inflammatory demyelinating polyradiculoneuropathy (peripheral) — GBS is an autoimmune peripheral nerve demyelination causing ascending flaccid paralysis, often post-infectious.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

**Q30. Answer: B — Respiratory failure from diaphragmatic weakness**

Correct Answer: B. Respiratory failure from diaphragmatic weakness — Ascending paralysis may involve respiratory muscles. FVC <20 mL/kg, NIF <-25 cmH<sub>2</sub>O, or >30% decline in FVC indicates need for intubation.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

**Q31. Answer: B — Risk of hyperkalemia from denervation**

Correct Answer: B. Risk of hyperkalemia from denervation — Succinylcholine should be avoided from the onset of GBS due to risk of hyperkalemia from denervation-induced upregulation of acetylcholine receptors at the neuromuscular junction.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

**Q32. Answer: B — Labile blood pressure, tachycardia, bradycardia, cardiac arrest**

Correct Answer: B. Labile blood pressure, tachycardia, bradycardia, cardiac arrest — Autonomic instability in GBS can cause unpredictable hemodynamic swings. Invasive monitoring is recommended.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

**Q33. Answer: B — Failure of the autonomic nervous system to regulate BP, HR, and temperature**

Correct Answer: B. Failure of the autonomic nervous system to regulate BP, HR, and temperature — Autonomic dysfunction involves both sympathetic and parasympathetic failure, causing orthostatic hypotension, temperature dysregulation, and labile hemodynamics.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

**Q34. Answer: A — Exaggerated hypotensive response to anesthetic agents**

Correct Answer: A. Exaggerated hypotensive response to anesthetic agents — Loss of compensatory sympathetic reflexes causes profound hypotension with induction agents, neuraxial blocks, and positive pressure ventilation.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

**Q35. Answer: B — >20 mmHg on standing**

Correct Answer: B. >20 mmHg on standing — Orthostatic hypotension is a sustained reduction of ≥20 mmHg systolic or ≥10 mmHg diastolic within 3 minutes of standing, indicating autonomic failure.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

**Q36. Answer: B — Autonomic hyperreflexia**

Correct Answer: B. Autonomic hyperreflexia — Lesions above T6 can trigger autonomic hyperreflexia: massive sympathetic discharge below the lesion causing severe hypertension, reflex bradycardia, headache, and flushing above the lesion.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection



**Q37. Answer: B — Bladder distension**

Correct Answer: B. Bladder distension — Bladder distension (full catheter, bladder surgery) is the most common trigger. Other triggers include bowel distension, skin stimulation below the lesion level.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

**Q38. Answer: C — Administer vasopressors**

Correct Answer: C. Administer vasopressors — Vasopressors would worsen the hypertension. Treatment: remove stimulus, sit up patient, give vasodilators (GTN sublingual/nifedipine), deepen anesthesia if intraoperative.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

**Q39. Answer: B — Hypotension and bradycardia (neurogenic shock)**

Correct Answer: B. Hypotension and bradycardia (neurogenic shock) — Acute spinal cord injury causes loss of sympathetic tone below the lesion resulting in vasodilation (hypotension) and unopposed vagal tone (bradycardia).

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

**Q40. Answer: B — First 48 hours**

Correct Answer: B. First 48 hours — Succinylcholine is relatively safe within the first 48 hours after spinal cord injury. After that, denervation causes receptor upregulation and life-threatening hyperkalemia risk, lasting for 6-9 months or more.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

**Q41. Answer: B — MAO inhibitors**

Correct Answer: B. MAO inhibitors — MAOIs have the most dangerous interactions: hypertensive crisis with indirect sympathomimetics, serotonin syndrome with meperidine/tramadol. Traditionally stopped 2 weeks preoperatively.

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q42. Answer: B — Prolonging the action of both depolarizing and non-depolarizing muscle relaxants**

Correct Answer: B. Prolonging the action of both depolarizing and non-depolarizing muscle relaxants — Lithium inhibits acetylcholine synthesis and release, potentiating all neuromuscular blocking agents. It also reduces renal concentrating ability (diabetes insipidus-like).

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q43. Answer: A — Exaggerated response to indirect sympathomimetics (ephedrine)**

Correct Answer: A. Exaggerated response to indirect sympathomimetics (ephedrine) — TCAs block norepinephrine reuptake, potentiating the action of both direct and indirect sympathomimetics. Arrhythmias and hypertensive crisis may occur.

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q44. Answer: B — Brief general anesthesia with short-acting agents (methohexitone/propofol + succinylcholine)**

Correct Answer: B. Brief general anesthesia with short-acting agents (methohexitone/propofol + succinylcholine) — ECT requires brief GA to prevent injuries from seizure-induced muscle contraction. Methohexitone is traditional; propofol may reduce seizure duration.

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q45. Answer: C — Hypotension and hypothermia**

Correct Answer: C. Hypotension and hypothermia — NMS presents with hyperthermia, severe rigidity, altered consciousness, autonomic instability (tachycardia, labile BP), and elevated CK. Treatment includes dantrolene and dopamine agonists.

Topic: Psychiatric Disorders | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q46. Answer: B — Regional anesthesia (spinal/epidural)**

Correct Answer: B. Regional anesthesia (spinal/epidural) — Regional anesthesia avoids drug interactions, aspiration risk, and respiratory depression. It is often preferred when feasible in Parkinsonian patients.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

**Q47. Answer: C — Propofol administration**

Correct Answer: C. Propofol administration — There is no evidence that propofol triggers MS relapse. However, infection, fever, stress, and the physiological stress of surgery are known triggers for MS exacerbation.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q48. Answer: A — 1–4 weeks**

Correct Answer: A. 1–4 weeks — GBS typically develops 1–4 weeks after a viral illness (commonly *Campylobacter jejuni*, CMV, EBV). It presents as ascending symmetrical weakness with areflexia.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

**Q49. Answer: B — Invasive arterial blood pressure monitoring**

Correct Answer: B. Invasive arterial blood pressure monitoring — Due to labile hemodynamics and exaggerated responses to anesthetic drugs, continuous invasive BP monitoring is essential for titration of drugs and fluids.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

**Q50. Answer: B — Carbamazepine**

Correct Answer: B. Carbamazepine — Carbamazepine can cause aplastic anemia and agranulocytosis. Preoperative full blood count is essential. Sodium valproate can cause thrombocytopenia and platelet dysfunction.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

# Anesthesia for Central Nervous System Diseases V2 – MCQ

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Anesthesia for Central Nervous System Diseases

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Parkinson's disease occurs due to progressive loss of which neurotransmitter in the brain?**

- A. Serotonin
- B. Dopamine
- C. Acetylcholine
- D. GABA

**Ans: B**

**Q2. Why are Parkinson's disease patients prone to arrhythmias perioperatively?**

- A. Due to electrolyte imbalance
- B. Due to increased levels of dopamine in body (levodopa gets converted to dopamine)
- C. Due to vagal stimulation
- D. Due to coronary artery disease

**Ans: B**

**Q3. Parkinson's disease patients develop hypotension due to:**

- A. Blood loss
- B. Depletion of catecholamines by chronically elevated dopamine levels
- C. Adrenal insufficiency
- D. Cardiac failure

**Ans: B**

**Q4. Parkinson's disease patients are volume depleted because of:**

- A. Poor oral intake only
- B. Dopaminergic effect on kidneys
- C. Excessive sweating
- D. Diarrhea from medications

**Ans: B**

**Q5. Which anesthetic drugs should be avoided in Parkinson's disease as they increase muscle tone?**

- A. Propofol and thiopentone
- B. Ketamine and alfentanil
- C. Midazolam and fentanyl
- D. Isoflurane and sevoflurane

**Ans: B**

**Q6. Regarding Alzheimer's disease and general anesthesia, the textbook states:**

- A. General anesthesia definitely worsens dementia
- B. The rising concern that general anesthesia worsens dementia is not yet fully substantiated by studies
- C. General anesthesia improves cognitive function
- D. Only regional anesthesia must be used

**Ans: B**

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| <p><b>Q7. The major concern of Alzheimer's disease patients presenting for surgery includes all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Geriatric and dementic</li> <li>B. Non-cooperative</li> <li>C. Prone to postoperative psychosis</li> <li>D. Prone to malignant hyperthermia</li> </ul>                                                         | <p><b>Ans: D</b></p> |
| <p><b>Q8. In epileptic patients, anticonvulsants especially affect which organ system requiring preoperative assessment?</b></p> <ul style="list-style-type: none"> <li>A. Renal function</li> <li>B. Liver function</li> <li>C. Cardiac function</li> <li>D. Pulmonary function</li> </ul>                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q9. Which of the following drugs has epileptogenic potential due to its preservative?</b></p> <ul style="list-style-type: none"> <li>A. Propofol</li> <li>B. Ketamine</li> <li>C. Etomidate</li> <li>D. Thiopentone</li> </ul>                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q10. Atracurium should be used with caution in epileptics because of:</b></p> <ul style="list-style-type: none"> <li>A. Histamine release</li> <li>B. Its metabolite Laudanosine has epileptogenic potential</li> <li>C. Prolonged neuromuscular block</li> <li>D. Direct CNS stimulation</li> </ul>                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q11. Which induction agent is the agent of CHOICE for epileptic patients?</b></p> <ul style="list-style-type: none"> <li>A. Methohexitone</li> <li>B. Ketamine</li> <li>C. Thiopentone</li> <li>D. Etomidate</li> </ul>                                                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q12. Phenytoin and carbamazepine can potentiate the effect of:</b></p> <ul style="list-style-type: none"> <li>A. Depolarizing muscle relaxants</li> <li>B. Non-depolarizing muscle relaxants</li> <li>C. Opioids</li> <li>D. Benzodiazepines</li> </ul>                                                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q13. Spinal/epidural anesthesia in epileptic patients can be given safely if:</b></p> <ul style="list-style-type: none"> <li>A. Patient is on multiple anticonvulsants</li> <li>B. There are no signs and symptoms of raised intracranial tension (ICT)</li> <li>C. Patient had seizure more than 24 hours ago</li> <li>D. Patient is on phenytoin only</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q14. After a cerebrovascular accident, elective surgery should be deferred for:</b></p> <ul style="list-style-type: none"> <li>A. 3 months</li> <li>B. 6 months</li> <li>C. 9 months</li> <li>D. 12 months</li> </ul>                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q15. If elective surgery is performed within 3 months after stroke, the risk of complications is how many times higher?</b></p> <ul style="list-style-type: none"> <li>A. 3 times</li> <li>B. 5 times</li> <li>C. 9 times</li> <li>D. 14 times</li> </ul>                                               | <p><b>Ans: D</b></p> |
| <p><b>Q16. If elective surgery is performed within 6 months after stroke, the risk is how many times higher?</b></p> <ul style="list-style-type: none"> <li>A. 2 times</li> <li>B. 5 times</li> <li>C. 10 times</li> <li>D. 14 times</li> </ul>                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q17. Suxamethonium is contraindicated in stroke patients because of:</b></p> <ul style="list-style-type: none"> <li>A. Risk of malignant hyperthermia</li> <li>B. Extrajunctional receptor proliferation causing hyperkalemia</li> <li>C. Histamine release</li> <li>D. Prolonged action</li> </ul>     | <p><b>Ans: B</b></p> |
| <p><b>Q18. Patients with history of transient ischemic attacks must have which investigation before considering anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. CT brain</li> <li>B. MRI brain</li> <li>C. Carotid Doppler</li> <li>D. EEG</li> </ul>                                           | <p><b>Ans: C</b></p> |
| <p><b>Q19. In patients with chronic headache, analgesics especially affect which organ system?</b></p> <ul style="list-style-type: none"> <li>A. Liver function</li> <li>B. Renal function</li> <li>C. Cardiac function</li> <li>D. Hematologic function</li> </ul>                                           | <p><b>Ans: B</b></p> |
| <p><b>Q20. Chronic headache patients are prone to develop post-spinal headache, therefore:</b></p> <ul style="list-style-type: none"> <li>A. Use the smallest needle</li> <li>B. Spinal should be avoided</li> <li>C. Prophylactic blood patch should be given</li> <li>D. Use hyperbaric solution</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q21. In Multiple Sclerosis, which type of regional anesthesia can precipitate the disease?</b></p> <ul style="list-style-type: none"> <li>A. Epidural anesthesia</li> <li>B. Spinal anesthesia</li> <li>C. Peripheral nerve blocks</li> <li>D. Brachial plexus block</li> </ul>                                                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q22. In Multiple Sclerosis, peripheral nerve blocks:</b></p> <ul style="list-style-type: none"> <li>A. Are absolutely contraindicated</li> <li>B. Can be given safely as peripheral nerves are not involved in MS</li> <li>C. Should only be given under ultrasound guidance</li> <li>D. Require reduced local anesthetic doses</li> </ul>                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q23. In Multiple Sclerosis patients, immunosuppressant drugs especially affect which evaluation?</b></p> <ul style="list-style-type: none"> <li>A. Liver function</li> <li>B. Hematologic evaluation</li> <li>C. Cardiac evaluation</li> <li>D. Pulmonary function</li> </ul>                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q24. During general anesthesia in MS patients, which aspect is most important to prevent disease precipitation?</b></p> <ul style="list-style-type: none"> <li>A. Avoiding all opioids</li> <li>B. Blunting stress response to intubation and maintaining good depth of anesthesia</li> <li>C. Using total intravenous anesthesia only</li> <li>D. Keeping the patient hypothermic</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q25. Hyperthermia in Multiple Sclerosis patients:</b></p> <ul style="list-style-type: none"> <li>A. Has no effect on the disease</li> <li>B. Is a very important risk factor to precipitate MS and must be avoided at any cost</li> <li>C. Is beneficial for recovery</li> <li>D. Only affects patients in relapse</li> </ul>                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q26. In syringomyelia, suxamethonium must not be used because:</b></p> <ul style="list-style-type: none"> <li>A. Risk of malignant hyperthermia</li> <li>B. There is lower motor neuron (LMN) paralysis</li> <li>C. Histamine release</li> <li>D. Pseudocholinesterase deficiency</li> </ul>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q27. In syringomyelia, response to non-depolarizing muscle relaxants is:</b></p> <ul style="list-style-type: none"> <li>A. Reduced (resistance)</li> <li>B. Normal</li> <li>C. Exaggerated</li> <li>D. Unpredictable but not exaggerated</li> </ul>                                                                                                                                           | <p><b>Ans: C</b></p> |

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| <p><b>Q28. Central neuraxial blocks in syringomyelia should be avoided because of:</b></p> <ul style="list-style-type: none"> <li>A. Coagulopathy</li> <li>B. Neurological deficits and possibility of raised ICT</li> <li>C. Technical difficulty only</li> <li>D. Risk of infection</li> </ul>                              | <p><b>Ans: B</b></p> |
| <p><b>Q29. Associated scoliosis in syringomyelia can cause which problem during general anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Difficult IV access</li> <li>B. Ventilation perfusion mismatch</li> <li>C. Cardiac tamponade</li> <li>D. Pneumothorax</li> </ul>                                        | <p><b>Ans: B</b></p> |
| <p><b>Q30. In ALS (Motor Neuron Disease), central neuraxial anesthesia:</b></p> <ul style="list-style-type: none"> <li>A. Is the preferred technique</li> <li>B. Can exacerbate the disease</li> <li>C. Has no effect on the disease</li> <li>D. Is mandatory for lower limb surgery</li> </ul>                               | <p><b>Ans: B</b></p> |
| <p><b>Q31. In Guillain-Barré Syndrome, central neuraxial blocks can worsen the disease due to:</b></p> <ul style="list-style-type: none"> <li>A. Coagulopathy</li> <li>B. Double crush syndrome</li> <li>C. Autonomic dysfunction</li> <li>D. Hypotension</li> </ul>                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q32. In autonomic dysfunction, hypotension must be treated by:</b></p> <ul style="list-style-type: none"> <li>A. Ephedrine (indirect acting vasopressor)</li> <li>B. Phenylephrine (directly acting vasopressor)</li> <li>C. Fluid bolus only</li> <li>D. Atropine</li> </ul>                                           | <p><b>Ans: B</b></p> |
| <p><b>Q33. Why is ephedrine NOT the vasopressor of choice in autonomic dysfunction?</b></p> <ul style="list-style-type: none"> <li>A. It causes bradycardia</li> <li>B. It acts by releasing catecholamines which are already depleted</li> <li>C. It causes bronchospasm</li> <li>D. It has a very short duration</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q34. In acute spinal cord transection, the spinal shock phase lasts for:</b></p> <ul style="list-style-type: none"> <li>A. 24-48 hours</li> <li>B. 1-3 weeks</li> <li>C. 3-6 months</li> <li>D. 6-12 months</li> </ul>                                                                                                  | <p><b>Ans: B</b></p> |



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| <p><b>Q35. Patients with chronic spinal cord transection above T6 have no sensations below transection. Why do they still need anesthesia for surgery?</b></p> <p>A. For anxiolysis only<br/> B. Surgery is a potent stimulus for developing autonomic hyper-reflexia<br/> C. To prevent awareness<br/> D. For muscle relaxation only</p> | <p><b>Ans: B</b></p> |
| <p><b>Q36. Which anesthetic techniques can prevent autonomic hyper-reflexia in chronic spinal cord transection?</b></p> <p>A. Only general anesthesia<br/> B. Regional anesthesia and deep general anesthesia<br/> C. Only local infiltration<br/> D. Monitored anesthesia care</p>                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q37. Regarding MAO inhibitors in psychiatric patients, the CURRENT recommendation is:</b></p> <p>A. Stop 2-3 weeks before surgery<br/> B. All antidepressants including MAO inhibitors should be continued<br/> C. Stop only MAO inhibitors, continue others<br/> D. Switch to SSRIs 4 weeks before surgery</p>                     | <p><b>Ans: B</b></p> |
| <p><b>Q38. Antidepressants increase the level of brain neurotransmitters, which causes:</b></p> <p>A. Decreased anesthetic requirement<br/> B. Increased anesthetic requirement<br/> C. No change in anesthetic requirement<br/> D. Resistance to muscle relaxants</p>                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q39. Acute intake of tricyclic antidepressants (up to 3 weeks) increases which catecholamines, leading to arrhythmias?</b></p> <p>A. Dopamine and serotonin<br/> B. Epinephrine and norepinephrine<br/> C. Acetylcholine and GABA<br/> D. Histamine and bradykinin</p>                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q40. In patients on acute TCAs, which drugs should NOT be used?</b></p> <p>A. Propofol, fentanyl, atracurium<br/> B. Pancuronium, ketamine, local anesthetics with adrenaline<br/> C. Thiopentone, morphine, vecuronium<br/> D. Sevoflurane, remifentanyl, cisatracurium</p>                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q41. Chronic use of antidepressants causes catecholamine depletion, which increases the risk of:</b></p> <p>A. Hypertension<br/> B. Hypotension<br/> C. Tachycardia<br/> D. Bronchospasm</p>                                                                                                                                        | <p><b>Ans: B</b></p> |

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| <p><b>Q42. Patients on MAO inhibitors must NOT receive which opioid?</b></p> <p>A. Morphine<br/>B. Fentanyl<br/>C. Pethidine (Meperidine)<br/>D. Remifentanyl</p>                                                                                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q43. The induction agent of choice for Electroconvulsive Therapy (ECT) is:</b></p> <p>A. Thiopentone<br/>B. Propofol<br/>C. Methohexital<br/>D. Ketamine</p>                                                                                                                      | <p><b>Ans: C</b></p> |
| <p><b>Q44. During ECT, when succinylcholine is used, a tourniquet is applied on one hand before giving succinylcholine in order to:</b></p> <p>A. Prevent IV extravasation<br/>B. See the seizure activity in that hand<br/>C. Reduce the dose required<br/>D. Prevent hyperkalemia</p> | <p><b>Ans: B</b></p> |
| <p><b>Q45. Lithium, used for mania, can produce which endocrine side effect?</b></p> <p>A. Cushing's syndrome<br/>B. Diabetes insipidus and electrolyte imbalance<br/>C. Hypothyroidism only<br/>D. Addison's disease</p>                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q46. Regarding lithium and surgery, the CURRENT recommendation is:</b></p> <p>A. Stop 48 hours before surgery<br/>B. Stop 1 week before surgery<br/>C. Continue lithium<br/>D. Replace with valproate</p>                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q47. In schizophrenic patients, antipsychotic drugs make them more vulnerable for which complication during anesthesia?</b></p> <p>A. Malignant hyperthermia<br/>B. Hypotension (alpha blocking property) and QTc prolongation<br/>C. Bronchospasm<br/>D. Hyperkalemia</p>        | <p><b>Ans: B</b></p> |
| <p><b>Q48. Ketamine is contraindicated in schizophrenic patients because:</b></p> <p>A. It causes seizures<br/>B. These patients can become very violent after ketamine in the postoperative period<br/>C. It causes hypotension<br/>D. It prolongs QTc interval</p>                    | <p><b>Ans: B</b></p> |

**Q49. In drug abusers, regarding acute vs chronic drug intake and anesthetic requirement:**

- A. Both increase anesthetic requirement
- B. Acute intake decreases while chronic intake increases the anesthetic requirement
- C. Both decrease anesthetic requirement
- D. Acute intake increases while chronic intake decreases

**Ans: B**

**Q50. Alcoholic patients presenting for surgery must be evaluated for all EXCEPT:**

- A. Dilated cardiomyopathy and cirrhosis
- B. Pancreatitis and metabolic disorders (hypoglycemia)
- C. Wernicke-Korsakoff syndrome
- D. Rheumatoid arthritis

**Ans: D**

# Answer Key & Explanations

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## Q1. Answer: B — Dopamine

Correct Answer: B. Dopamine — Parkinson occurs due to progressive loss of dopamine deficiency in the brain. All medications including levodopa should be continued perioperatively.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q2. Answer: B — Due to increased levels of dopamine in body (levodopa gets converted to dopamine)

Correct Answer: B. Due to increased levels of dopamine in body (levodopa gets converted to dopamine) — PD patients are prone for arrhythmias due to increased levels of dopamine in the body as levodopa gets converted to dopamine.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q3. Answer: B — Depletion of catecholamines by chronically elevated dopamine levels

Correct Answer: B. Depletion of catecholamines by chronically elevated dopamine levels — Hypotension occurs due to depletion of catecholamines by chronically elevated dopamine levels. These patients are also volume depleted due to dopaminergic effect on kidneys.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q4. Answer: B — Dopaminergic effect on kidneys

Correct Answer: B. Dopaminergic effect on kidneys — PD patients are volume depleted due to dopaminergic effect on kidneys, therefore adequate fluid replacement is a must.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q5. Answer: B — Ketamine and alfentanil

Correct Answer: B. Ketamine and alfentanil — Anesthetic drugs which increase the muscle tone (ketamine, alfentanil) should be avoided in PD patients as rigidity is already a major concern.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

## Q6. Answer: B — The rising concern that general anesthesia worsens dementia is not yet fully substantiated by studies

Correct Answer: B. The rising concern that general anesthesia worsens dementia is not yet fully substantiated by studies — This is directly stated in the textbook regarding the relationship between GA and Alzheimer's disease.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

## Q7. Answer: D — Prone to malignant hyperthermia

Correct Answer: D. Prone to malignant hyperthermia — Alzheimer's patients are geriatric, dementic, non-cooperative, and prone to develop postoperative psychosis. Malignant hyperthermia is not a specific concern.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

## Q8. Answer: B — Liver function

Correct Answer: B. Liver function — As anticonvulsants affect the organ systems (especially liver function), evaluation of organ function is must in preoperative assessment.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q9. Answer: B — Ketamine**

Correct Answer: B. Ketamine — Ketamine has epileptogenic potential due to its preservative. Other drugs with epileptogenic potential include methohexitone, propofol (rare opisthotonus), etomidate (myoclonic activity), enflurane, sevoflurane.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q10. Answer: B — Its metabolite Laudanosine has epileptogenic potential**

Correct Answer: B. Its metabolite Laudanosine has epileptogenic potential — Atracurium's metabolite Laudanosine has epileptogenic potential and should be used with caution in epileptic patients.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q11. Answer: C — Thiopentone**

Correct Answer: C. Thiopentone — Due to its anticonvulsant property, thiopentone is the induction agent of choice for epileptics.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q12. Answer: B — Non-depolarizing muscle relaxants**

Correct Answer: B. Non-depolarizing muscle relaxants — Phenytoin and carbamazepine can potentiate the effect of non-depolarizing muscle relaxants.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q13. Answer: B — There are no signs and symptoms of raised intracranial tension (ICT)**

Correct Answer: B. There are no signs and symptoms of raised intracranial tension (ICT) — Spinal/epidural can be given safely if there are no signs and symptoms of raised intracranial tension (ICT).

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

**Q14. Answer: C — 9 months**

Correct Answer: C. 9 months — After cerebrovascular accident, elective surgery should be deferred for 9 months. The odds are stabilized at 3 times after 9 months.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

**Q15. Answer: D — 14 times**

Correct Answer: D. 14 times — The possibility of serious complications like another stroke, heart attack or even death is 14 times higher if elective surgery is performed within three months after stroke.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

**Q16. Answer: B — 5 times**

Correct Answer: B. 5 times — Risk is 5 times higher if surgery is performed within 6 months after stroke, and 14 times higher within 3 months.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

**Q17. Answer: B — Extrajunctional receptor proliferation causing hyperkalemia**

Correct Answer: B. Extrajunctional receptor proliferation causing hyperkalemia — Due to extrajunctional receptors, suxamethonium is contraindicated for stroke patients.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

**Q18. Answer: C — Carotid Doppler**

Correct Answer: C. Carotid Doppler — Patients with history of transient ischemic attacks must have at least carotid Doppler before considering for anesthesia.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

**Q19. Answer: B — Renal function**

Correct Answer: B. Renal function — As analgesics affect the organ systems (especially renal function), evaluation of organ function is must in preoperative assessment of chronic headache patients.

Topic: Headache | Ref: Ch 20, Page 196, Topic: Headache

**Q20. Answer: B — Spinal should be avoided**

Correct Answer: B. Spinal should be avoided — These patients are prone to develop post-spinal headache therefore, spinal should be avoided.

Topic: Headache | Ref: Ch 20, Page 196, Topic: Headache

**Q21. Answer: B — Spinal anesthesia**

Correct Answer: B. Spinal anesthesia — Spinal anesthesia (but not epidural) and stress can precipitate multiple sclerosis, probably because of direct exposure of local anesthetic to sheath of spinal cord.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q22. Answer: B — Can be given safely as peripheral nerves are not involved in MS**

Correct Answer: B. Can be given safely as peripheral nerves are not involved in MS — As peripheral nerves are not involved in MS, peripheral nerve blocks can be given safely.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q23. Answer: B — Hematologic evaluation**

Correct Answer: B. Hematologic evaluation — As immunosuppressant drugs affect the organ systems (especially hematologic), evaluation of organ function is must in preoperative assessment.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q24. Answer: B — Blunting stress response to intubation and maintaining good depth of anesthesia**

Correct Answer: B. Blunting stress response to intubation and maintaining good depth of anesthesia — As stress can precipitate MS, blunting stress response to intubation and maintaining good depth of anesthesia is a must.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q25. Answer: B — Is a very important risk factor to precipitate MS and must be avoided at any cost**

Correct Answer: B. Is a very important risk factor to precipitate MS and must be avoided at any cost — Hyperthermia is one of a very important risk factor to precipitate MS therefore, must be avoided at any cost.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

**Q26. Answer: B — There is lower motor neuron (LMN) paralysis**

Correct Answer: B. There is lower motor neuron (LMN) paralysis — As there is lower motor neuron (LMN) paralysis in syringomyelia, suxamethonium must not be used and exaggerated response to non-depolarizer is expected.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

**Q27. Answer: C — Exaggerated**

Correct Answer: C. Exaggerated — In syringomyelia, exaggerated response to non-depolarizer is expected due to lower motor neuron involvement.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

**Q28. Answer: B — Neurological deficits and possibility of raised ICT**

Correct Answer: B. Neurological deficits and possibility of raised ICT — Neurological deficits and possibility of raised ICT warrants the avoidance of central neuraxial blocks in syringomyelia.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

**Q29. Answer: B — Ventilation perfusion mismatch**

Correct Answer: B. Ventilation perfusion mismatch — Associated scoliosis in syringomyelia can cause ventilation perfusion mismatch during general anesthesia.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

**Q30. Answer: B — Can exacerbate the disease**

Correct Answer: B. Can exacerbate the disease — In ALS, central neuraxial can exacerbate the disease and should be avoided.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

**Q31. Answer: B — Double crush syndrome**

Correct Answer: B. Double crush syndrome — Central neuraxial blocks can worsen the already diseased nerve (double crush syndrome) in GBS, therefore should be avoided.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

**Q32. Answer: B — Phenylephrine (directly acting vasopressor)**

Correct Answer: B. Phenylephrine (directly acting vasopressor) — Hypotension must be treated by directly acting vasopressors (phenylephrine), not by indirectly acting (ephedrine) vasopressors which act by releasing catecholamines (which are already depleted).

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

**Q33. Answer: B — It acts by releasing catecholamines which are already depleted**

Correct Answer: B. It acts by releasing catecholamines which are already depleted — Indirectly acting vasopressors (ephedrine) work by releasing catecholamines, which are depleted in autonomic dysfunction, making them ineffective.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

**Q34. Answer: B — 1-3 weeks**

Correct Answer: B. 1-3 weeks — During the first 1-3 weeks patients are in spinal shock due to sympathetic cut off, and are very vulnerable for hypotension and bradycardia.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

**Q35. Answer: B — Surgery is a potent stimulus for developing autonomic hyper-reflexia**

Correct Answer: B. Surgery is a potent stimulus for developing autonomic hyper-reflexia — Although these patients have no sensations below transection, they need anesthesia because surgery is a potent stimulus for developing autonomic hyper-reflexia.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

**Q36. Answer: B — Regional anesthesia and deep general anesthesia**

Correct Answer: B. Regional anesthesia and deep general anesthesia — Regional anesthesia and deep GA can prevent autonomic hyper-reflexia in patients with chronic spinal cord transection above T6.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

**Q37. Answer: B — All antidepressants including MAO inhibitors should be continued**

Correct Answer: B. All antidepressants including MAO inhibitors should be continued — Contrary to previous recommendation of stopping MAO inhibitors 2-3 weeks prior, all antidepressants including MAO inhibitors should be continued.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q38. Answer: B — Increased anesthetic requirement**

Correct Answer: B. Increased anesthetic requirement — Antidepressants increase the level of brain neurotransmitters, increasing the anesthetic requirement.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q39. Answer: B — Epinephrine and norepinephrine**

Correct Answer: B. Epinephrine and norepinephrine — Acute intake of tricyclic antidepressants (up to 3 weeks) increases the epinephrine and norepinephrine, increasing chances of arrhythmias (particularly with halothane) and hypertensive crisis.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q40. Answer: B — Pancuronium, ketamine, local anesthetics with adrenaline**

Correct Answer: B. Pancuronium, ketamine, local anesthetics with adrenaline — Drugs like pancuronium, ketamine and local anesthetics with adrenaline should not be used due to sympathomimetic effects potentiating catecholamine excess.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q41. Answer: B — Hypotension**

Correct Answer: B. Hypotension — Chronic use causes catecholamine depletion increasing the risk of hypotension. This is opposite to acute TCA intake which causes sympathetic excess.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q42. Answer: C — Pethidine (Meperidine)**

Correct Answer: C. Pethidine (Meperidine) — Patients on MAO inhibitors must not receive pethidine due to the risk of fatal serotonin syndrome.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q43. Answer: C — Methohexital**

Correct Answer: C. Methohexital — Methohexital, being epileptogenic, is the induction agent of choice for ECT. Propofol may shorten seizure duration.

Topic: Psychiatric Disorders – ECT | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q44. Answer: B — See the seizure activity in that hand**

Correct Answer: B. See the seizure activity in that hand — If succinylcholine is used to prevent muscle contraction, tourniquet should be applied in one hand before giving succinylcholine to see the seizure activity.

Topic: Psychiatric Disorders – ECT | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders



**Q45. Answer: B — Diabetes insipidus and electrolyte imbalance**

Correct Answer: B. Diabetes insipidus and electrolyte imbalance — Patients on lithium can develop diabetes insipidus and electrolyte imbalance. Current recommendation is to continue lithium perioperatively.

Topic: Psychiatric Disorders – Mania | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q46. Answer: C — Continue lithium**

Correct Answer: C. Continue lithium — In contrary to older guidelines of stopping lithium 48 before surgery, current recommendation is to continue lithium. Lamotrigine is the other drug approved for bipolar disorder.

Topic: Psychiatric Disorders – Mania | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q47. Answer: B — Hypotension (alpha blocking property) and QTc prolongation**

Correct Answer: B. Hypotension (alpha blocking property) and QTc prolongation — Antipsychotic patients are more vulnerable for hypotension (alpha blocking property), arrhythmias (QTc prolongation and torsade de pointes).

Topic: Psychiatric Disorders – Schizophrenia | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q48. Answer: B — These patients can become very violent after ketamine in the postoperative period**

Correct Answer: B. These patients can become very violent after ketamine in the postoperative period — Ketamine is contraindicated because these patients can become very violent after ketamine in the postoperative period.

Topic: Psychiatric Disorders – Schizophrenia | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q49. Answer: B — Acute intake decreases while chronic intake increases the anesthetic requirement**

Correct Answer: B. Acute intake decreases while chronic intake increases the anesthetic requirement — Acute drug intake decreases while chronic intake increases the anesthetic requirement due to tolerance and cross-tolerance.

Topic: Psychiatric Disorders – Drug Abuse | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

**Q50. Answer: D — Rheumatoid arthritis**

Correct Answer: D. Rheumatoid arthritis — Alcoholic patients must be evaluated for dilated cardiomyopathy, cirrhosis, pancreatitis, metabolic disorders (hypoglycemia), nutritional disorders (Wernicke-Korsakoff syndrome). Rheumatoid arthritis is not associated.

Topic: Psychiatric Disorders – Drug Abuse | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

# Anesthesia for Hepatic Disease – MCQ

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Anesthesia for Hepatic Disease

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. The liver receives what percentage of total cardiac output?**

- A. 10%
- B. 25%
- C. 40%
- D. 50%

**Ans: B**

**Q2. What percentage of hepatic blood supply comes from the portal vein?**

- A. 25%
- B. 50%
- C. 75%
- D. 90%

**Ans: C**

**Q3. Child-Pugh classification assesses liver function based on all EXCEPT:**

- A. Serum bilirubin
- B. Serum albumin
- C. Serum creatinine
- D. Prothrombin time/INR

**Ans: C**

**Q4. Child-Pugh Class C (score 10–15) is associated with perioperative mortality of approximately:**

- A. 5%
- B. 10–15%
- C. 50% or higher
- D. Less than 2%

**Ans: C**

**Q5. Which coagulation factor has the shortest half-life and is first to decrease in liver disease?**

- A. Factor II
- B. Factor V
- C. Factor VII
- D. Factor X

**Ans: C**

**Q6. The most sensitive preoperative liver function test for synthetic function is:**

- A. AST/ALT
- B. Serum albumin
- C. Prothrombin time (PT/INR)
- D. Alkaline phosphatase

**Ans: C**

**Q7. Serum albumin less than what value indicates severe liver disease and increased surgical risk?**

- A. 4.0 g/dL
- B. 3.5 g/dL
- C. 3.0 g/dL
- D. 2.5 g/dL

**Ans: C**

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| <p><b>Q8. In liver disease, which of the following is TRUE about drug protein binding?</b></p> <ul style="list-style-type: none"> <li>A. Increased binding due to elevated albumin</li> <li>B. Decreased binding leading to higher free drug fraction</li> <li>C. No change in drug binding</li> <li>D. Increased binding due to elevated globulin</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q9. Preoperative correction of coagulopathy in liver disease includes all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Fresh Frozen Plasma (FFP)</li> <li>B. Vitamin K injection</li> <li>C. Platelet transfusion if count &lt;50,000</li> <li>D. Heparin infusion</li> </ul>                                                              | <p><b>Ans: D</b></p> |
| <p><b>Q10. MELD score is calculated using which three parameters?</b></p> <ul style="list-style-type: none"> <li>A. Bilirubin, Albumin, PT</li> <li>B. Bilirubin, Creatinine, INR</li> <li>C. Bilirubin, Ascites, Encephalopathy</li> <li>D. Creatinine, Albumin, PT</li> </ul>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q11. In cirrhotic patients, preoperative ascites indicates:</b></p> <ul style="list-style-type: none"> <li>A. Good prognosis for surgery</li> <li>B. Portal hypertension and increased surgical risk</li> <li>C. No impact on anesthetic management</li> <li>D. Need for immediate splenectomy</li> </ul>                                               | <p><b>Ans: B</b></p> |
| <p><b>Q12. Hepatic encephalopathy in the preoperative period is graded from:</b></p> <ul style="list-style-type: none"> <li>A. Grade I to III</li> <li>B. Grade I to IV</li> <li>C. Grade A to C</li> <li>D. Stage 1 to 3</li> </ul>                                                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q13. Preoperative optimization of hepatic disease patient includes all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Correction of electrolyte imbalances</li> <li>B. Treatment of encephalopathy with lactulose</li> <li>C. Drainage of tense ascites</li> <li>D. Starting high-dose NSAIDs for pain</li> </ul>                            | <p><b>Ans: D</b></p> |
| <p><b>Q14. Hyperdynamic circulation in cirrhosis is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. High SVR, low cardiac output</li> <li>B. Low SVR, high cardiac output, low MAP</li> <li>C. Normal SVR, normal cardiac output</li> <li>D. High SVR, high cardiac output</li> </ul>                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q15. Which volatile anesthetic agent is most preferred in liver disease?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Enflurane</li> <li>D. Desflurane</li> </ul>                                                                                                                                     | <p><b>Ans: B</b></p> |

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| <p><b>Q16. Halothane is avoided in liver disease primarily because:</b></p> <ul style="list-style-type: none"> <li>A. It causes renal failure</li> <li>B. It reduces hepatic blood flow and can cause halothane hepatitis</li> <li>C. It causes seizures</li> <li>D. It increases portal pressure</li> </ul>                             | <p><b>Ans: B</b></p> |
| <p><b>Q17. Which muscle relaxant is the agent of choice in liver disease?</b></p> <ul style="list-style-type: none"> <li>A. Pancuronium</li> <li>B. Vecuronium</li> <li>C. Atracurium/Cisatracurium</li> <li>D. Rocuronium</li> </ul>                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q18. Succinylcholine duration may be prolonged in liver disease because:</b></p> <ul style="list-style-type: none"> <li>A. Increased hepatic metabolism</li> <li>B. Reduced plasma cholinesterase (pseudocholinesterase) synthesis</li> <li>C. Increased renal excretion</li> <li>D. Enhanced neuromuscular sensitivity</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q19. Which opioid is preferred in liver disease due to minimal hepatic metabolism?</b></p> <ul style="list-style-type: none"> <li>A. Morphine</li> <li>B. Fentanyl</li> <li>C. Remifentanyl</li> <li>D. Codeine</li> </ul>                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q20. Induction agent of choice in liver disease is:</b></p> <ul style="list-style-type: none"> <li>A. Thiopentone in full dose</li> <li>B. Propofol in reduced dose</li> <li>C. Ketamine</li> <li>D. Etomidate in full dose</li> </ul>                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q21. Intraoperative monitoring in liver disease should include:</b></p> <ul style="list-style-type: none"> <li>A. Only standard ASA monitoring</li> <li>B. Invasive arterial BP, CVP, urine output, temperature, and coagulation</li> <li>C. Only pulse oximetry and ECG</li> <li>D. Only BIS monitoring</li> </ul>                | <p><b>Ans: B</b></p> |
| <p><b>Q22. Regional anesthesia in liver disease is relatively contraindicated when:</b></p> <ul style="list-style-type: none"> <li>A. Albumin is normal</li> <li>B. INR &gt;1.5 or platelets &lt;80,000</li> <li>C. Patient has mild ascites</li> <li>D. Bilirubin is mildly elevated</li> </ul>                                         | <p><b>Ans: B</b></p> |
| <p><b>Q23. Intraoperative hypoglycemia in liver disease occurs because:</b></p> <ul style="list-style-type: none"> <li>A. Excess insulin production</li> <li>B. Impaired hepatic glycogenolysis and gluconeogenesis</li> <li>C. Increased peripheral glucose utilization</li> <li>D. Renal glucose wasting</li> </ul>                    | <p><b>Ans: B</b></p> |

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| <p><b>Q24. Intraoperative fluid management in liver disease should avoid:</b></p> <ul style="list-style-type: none"> <li>A. Normal saline</li> <li>B. Lactated Ringer's solution in severe liver disease</li> <li>C. Albumin</li> <li>D. Packed red blood cells</li> </ul>                                                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q25. Which factor most decreases hepatic blood flow during anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. IPPV with high airway pressures</li> <li>B. Adequate hydration</li> <li>C. Supine position</li> <li>D. Use of isoflurane</li> </ul>                                                                                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q26. Factors that reduce hepatic blood flow intraoperatively include all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Hypotension</li> <li>B. High sympathetic stimulation</li> <li>C. IPPV</li> <li>D. Adequate intravascular volume</li> </ul>                                                                                                                                               | <p><b>Ans: D</b></p> |
| <p><b>Q27. Temperature regulation in liver disease is impaired due to:</b></p> <ul style="list-style-type: none"> <li>A. Increased basal metabolic rate</li> <li>B. Reduced hepatic heat production and vasodilation</li> <li>C. Increased shivering threshold</li> <li>D. Hypermetabolic state</li> </ul>                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q28. Benzodiazepines in liver disease have:</b></p> <ul style="list-style-type: none"> <li>A. Shorter duration of action</li> <li>B. Prolonged duration due to reduced metabolism and increased free fraction</li> <li>C. No change in pharmacokinetics</li> <li>D. Increased hepatic clearance</li> </ul>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q29. Which benzodiazepine is relatively safer in liver disease due to glucuronidation (Phase II) metabolism?</b></p> <ul style="list-style-type: none"> <li>A. Diazepam</li> <li>B. Midazolam</li> <li>C. Oxazepam/Lorazepam</li> <li>D. Chlordiazepoxide</li> </ul>                                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q30. The hepatic artery buffer response refers to:</b></p> <ul style="list-style-type: none"> <li>A. Increase in portal vein flow when hepatic artery flow decreases</li> <li>B. Compensatory increase in hepatic artery flow when portal venous flow decreases</li> <li>C. Shunting of blood through portosystemic collaterals</li> <li>D. Decrease in both hepatic artery and portal vein flow</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q31. Intraoperative management of ascites includes:</b></p> <ul style="list-style-type: none"> <li>A. Ignoring it during surgery</li> <li>B. Rapid paracentesis of all fluid at induction</li> <li>C. Careful drainage with albumin replacement to prevent cardiovascular collapse</li> <li>D. Using high-dose diuretics during surgery</li> </ul>                         | <p><b>Ans: C</b></p> |
| <p><b>Q32. Hepatorenal syndrome is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. Structural renal damage</li> <li>B. Functional renal failure due to splanchnic vasodilation and renal vasoconstriction</li> <li>C. Obstructive uropathy</li> <li>D. Glomerulonephritis</li> </ul>                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q33. Obstructive jaundice (biliary obstruction) is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. Elevated unconjugated bilirubin only</li> <li>B. Elevated conjugated bilirubin, raised ALP, pale stools, dark urine</li> <li>C. Normal bilirubin with elevated transaminases</li> <li>D. Elevated unconjugated bilirubin with normal ALP</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q34. In obstructive jaundice, Vitamin K deficiency occurs because:</b></p> <ul style="list-style-type: none"> <li>A. Increased renal excretion of Vitamin K</li> <li>B. Lack of bile salts prevents fat-soluble vitamin absorption from gut</li> <li>C. Increased hepatic metabolism of Vitamin K</li> <li>D. Dietary deficiency</li> </ul>                                | <p><b>Ans: B</b></p> |
| <p><b>Q35. In obstructive jaundice, coagulopathy from Vitamin K deficiency responds to parenteral Vitamin K within:</b></p> <ul style="list-style-type: none"> <li>A. 1-2 hours</li> <li>B. 6-12 hours</li> <li>C. 24-48 hours</li> <li>D. 5-7 days</li> </ul>                                                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q36. The major renal risk in obstructive jaundice is:</b></p> <ul style="list-style-type: none"> <li>A. Renal calculi</li> <li>B. Acute tubular necrosis (hepatorenal syndrome/bile nephropathy)</li> <li>C. Renal artery stenosis</li> <li>D. Pyelonephritis</li> </ul>                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q37. Renal protection in obstructive jaundice includes:</b></p> <ul style="list-style-type: none"> <li>A. Fluid restriction</li> <li>B. Adequate preoperative hydration and maintaining urine output</li> <li>C. Use of nephrotoxic antibiotics</li> <li>D. High-dose loop diuretics</li> </ul>                                                                            | <p><b>Ans: B</b></p> |

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| <p><b>Q38. Endotoxemia in obstructive jaundice occurs because:</b></p> <ul style="list-style-type: none"> <li>A. Excess bile production</li> <li>B. Absence of bile in gut allows bacterial overgrowth and endotoxin absorption</li> <li>C. Direct liver cell necrosis</li> <li>D. Renal failure</li> </ul>                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q39. Preoperative biliary drainage (PTBD/stenting) before surgery in obstructive jaundice is considered when:</b></p> <ul style="list-style-type: none"> <li>A. Bilirubin is mildly elevated</li> <li>B. Bilirubin &gt;10 mg/dL with planned major surgery, to reduce complications</li> <li>C. Patient is asymptomatic</li> <li>D. Only for cosmetic reasons</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q40. Anesthetic technique for patients with biliary obstruction preferably avoids:</b></p> <ul style="list-style-type: none"> <li>A. Propofol</li> <li>B. Morphine and drugs causing sphincter of Oddi spasm</li> <li>C. Fentanyl</li> <li>D. Atracurium</li> </ul>                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q41. Which drug can relieve morphine-induced sphincter of Oddi spasm?</b></p> <ul style="list-style-type: none"> <li>A. Atropine</li> <li>B. Glucagon</li> <li>C. Adrenaline</li> <li>D. Propranolol</li> </ul>                                                                                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q42. Pruritus in obstructive jaundice is caused by:</b></p> <ul style="list-style-type: none"> <li>A. Elevated bilirubin directly</li> <li>B. Deposition of bile salts in the skin</li> <li>C. Bacterial infection</li> <li>D. Vitamin K deficiency</li> </ul>                                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q43. Wound healing in obstructive jaundice is impaired due to:</b></p> <ul style="list-style-type: none"> <li>A. Excess Vitamin A</li> <li>B. Malnutrition, Vitamin K deficiency, and reduced immunity</li> <li>C. Excess collagen production</li> <li>D. Increased blood supply to wound</li> </ul>                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q44. Bradycardia is more common in jaundiced patients because:</b></p> <ul style="list-style-type: none"> <li>A. Pain</li> <li>B. Bile salts sensitize the myocardium to vagal stimulation</li> <li>C. Hypovolemia</li> <li>D. Hypoglycemia</li> </ul>                                                                                                                   | <p><b>Ans: B</b></p> |



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| <p><b>Q45. In obstructive jaundice, which antibiotic prophylaxis is important preoperatively?</b></p> <ul style="list-style-type: none"> <li>A. Not needed</li> <li>B. Broad-spectrum antibiotics covering gram-negative and anaerobic organisms</li> <li>C. Only antifungals</li> <li>D. Only gram-positive coverage</li> </ul>                      | <p><b>Ans: B</b></p> |
| <p><b>Q46. Vecuronium and rocuronium should be used cautiously in liver disease because:</b></p> <ul style="list-style-type: none"> <li>A. They cause histamine release</li> <li>B. They undergo significant hepatic metabolism and biliary excretion</li> <li>C. They worsen hepatic encephalopathy</li> <li>D. They cause liver necrosis</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q47. Postoperative jaundice in previously healthy patient may be caused by all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Blood transfusion (hemolysis)</li> <li>B. Halothane hepatitis</li> <li>C. Hypotension-related hepatic ischemia</li> <li>D. Use of atracurium</li> </ul>                                                | <p><b>Ans: D</b></p> |
| <p><b>Q48. In biliary surgery, intraoperative cholangiography may produce false positive results if which drug has been given?</b></p> <ul style="list-style-type: none"> <li>A. Propofol</li> <li>B. Morphine (sphincter of Oddi spasm mimics stone)</li> <li>C. Atracurium</li> <li>D. Sevoflurane</li> </ul>                                       | <p><b>Ans: B</b></p> |
| <p><b>Q49. Mannitol is sometimes used in obstructive jaundice surgery to:</b></p> <ul style="list-style-type: none"> <li>A. Reduce intracranial pressure</li> <li>B. Maintain renal perfusion and urine output (renal protection)</li> <li>C. Reduce portal pressure</li> <li>D. Treat hyperkalemia</li> </ul>                                        | <p><b>Ans: B</b></p> |
| <p><b>Q50. Which is the safest volatile agent regarding hepatic blood flow preservation?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Enflurane</li> <li>C. Isoflurane/Sevoflurane</li> <li>D. Desflurane at high MAC</li> </ul>                                                                                           | <p><b>Ans: C</b></p> |

# Answer Key & Explanations

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## Q1. Answer: B — 25%

Correct Answer: B. 25% — The liver receives approximately 25% of cardiac output (~1500 mL/min). Of this, 75% comes from the portal vein and 25% from the hepatic artery.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q2. Answer: C — 75%

Correct Answer: C. 75% — The portal vein provides 75% of hepatic blood flow but only 50% of oxygen supply. The hepatic artery provides 25% of blood flow but 50% of oxygen supply.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q3. Answer: C — Serum creatinine

Correct Answer: C. Serum creatinine — Child-Pugh score uses 5 parameters: Bilirubin, Albumin, Prothrombin time (INR), Ascites, and Encephalopathy grade. Creatinine is part of the MELD score, not Child-Pugh.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q4. Answer: C — 50% or higher

Correct Answer: C. 50% or higher — Child-Pugh C carries perioperative mortality >50% for major abdominal surgery. Class A: ~10%, Class B: ~30%, Class C: >50%.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q5. Answer: C — Factor VII

Correct Answer: C. Factor VII — Factor VII has the shortest half-life (~6 hours) among coagulation factors and is the first to decline in liver disease. Its level is reflected by PT/INR.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q6. Answer: C — Prothrombin time (PT/INR)

Correct Answer: C. Prothrombin time (PT/INR) — PT is the most sensitive indicator of hepatic synthetic function due to the short half-life of Factor VII. Albumin reflects chronic liver function (half-life 21 days).

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q7. Answer: C — 3.0 g/dL

Correct Answer: C. 3.0 g/dL — Albumin <3.0 g/dL indicates severe hepatic dysfunction. Low albumin causes reduced protein binding of drugs, increased free drug levels, and peripheral edema/ascites.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q8. Answer: B — Decreased binding leading to higher free drug fraction

Correct Answer: B. Decreased binding leading to higher free drug fraction — Hypoalbuminemia in liver disease results in reduced protein binding of drugs (thiopentone, diazepam), causing increased free drug levels and exaggerated clinical effects.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q9. Answer: D — Heparin infusion

Correct Answer: D. Heparin infusion — Heparin would worsen coagulopathy. Treatment includes Vitamin K (if cholestatic), FFP for active bleeding, cryoprecipitate for low fibrinogen, and platelet transfusion if <50,000/ $\mu$ L.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q10. Answer: B — Bilirubin, Creatinine, INR**

Correct Answer: B. Bilirubin, Creatinine, INR — MELD (Model for End-stage Liver Disease) uses serum bilirubin, creatinine, and INR. MELD >15 indicates significant surgical risk.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q11. Answer: B — Portal hypertension and increased surgical risk**

Correct Answer: B. Portal hypertension and increased surgical risk — Ascites reflects portal hypertension and poor hepatic function. It impairs diaphragmatic excursion, increases aspiration risk, and is associated with higher perioperative mortality.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q12. Answer: B — Grade I to IV**

Correct Answer: B. Grade I to IV — Grade I: altered mood/sleep; Grade II: drowsiness, inappropriate behavior; Grade III: stupor, confusion; Grade IV: coma. Surgery should be deferred in Grade III-IV.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q13. Answer: D — Starting high-dose NSAIDs for pain**

Correct Answer: D. Starting high-dose NSAIDs for pain — NSAIDs impair renal function (hepatorenal risk), reduce platelet function, and may precipitate GI bleeding. They should be avoided in liver disease patients.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q14. Answer: B — Low SVR, high cardiac output, low MAP**

Correct Answer: B. Low SVR, high cardiac output, low MAP — Cirrhotic patients have splanchnic vasodilation (due to nitric oxide), reduced SVR, compensatory high cardiac output, and relatively low mean arterial pressure.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q15. Answer: B — Isoflurane**

Correct Answer: B. Isoflurane — Isoflurane maintains hepatic arterial blood flow better than other agents. Halothane reduces hepatic blood flow the most and is associated with halothane hepatitis. Isoflurane and sevoflurane are preferred.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q16. Answer: B — It reduces hepatic blood flow and can cause halothane hepatitis**

Correct Answer: B. It reduces hepatic blood flow and can cause halothane hepatitis — Halothane reduces hepatic arterial flow by 25-30% and is associated with immune-mediated fulminant hepatitis (1:35,000 adults). It is contraindicated in liver disease.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q17. Answer: C — Atracurium/Cisatracurium**

Correct Answer: C. Atracurium/Cisatracurium — Atracurium undergoes Hofmann degradation (organ-independent) and ester hydrolysis. Its metabolism is independent of hepatic and renal function, making it ideal in liver disease.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q18. Answer: B — Reduced plasma cholinesterase (pseudocholinesterase) synthesis**

Correct Answer: B. Reduced plasma cholinesterase (pseudocholinesterase) synthesis — The liver synthesizes plasma cholinesterase. Severe liver disease reduces its production, prolonging succinylcholine block duration.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q19. Answer: C — Remifentanyl**

Correct Answer: C. Remifentanyl — Remifentanyl is metabolized by tissue esterases (organ-independent). Morphine has prolonged action due to reduced glucuronidation. Fentanyl metabolism is also hepatic-dependent.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q20. Answer: B — Propofol in reduced dose**

Correct Answer: B. Propofol in reduced dose — Propofol has both hepatic and extrahepatic metabolism. Dose should be reduced due to hypoalbuminemia (increased free fraction) and reduced hepatic clearance.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q21. Answer: B — Invasive arterial BP, CVP, urine output, temperature, and coagulation**

Correct Answer: B. Invasive arterial BP, CVP, urine output, temperature, and coagulation — Due to coagulopathy, hemodynamic instability, renal vulnerability, and hypothermia risk, comprehensive invasive monitoring is essential.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q22. Answer: B — INR >1.5 or platelets <80,000**

Correct Answer: B. INR >1.5 or platelets <80,000 — Neuraxial techniques carry risk of epidural/spinal hematoma in coagulopathy. Generally contraindicated if INR >1.5, platelets <80,000, or active coagulopathy.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q23. Answer: B — Impaired hepatic glycogenolysis and gluconeogenesis**

Correct Answer: B. Impaired hepatic glycogenolysis and gluconeogenesis — The liver is the primary organ for maintaining blood glucose. In severe disease, depleted glycogen stores and impaired gluconeogenesis cause hypoglycemia.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q24. Answer: B — Lactated Ringer's solution in severe liver disease**

Correct Answer: B. Lactated Ringer's solution in severe liver disease — The diseased liver cannot metabolize lactate to bicarbonate efficiently. Lactate accumulation may worsen acidosis. Normal saline or balanced crystalloids without lactate are preferred.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q25. Answer: A — IPPV with high airway pressures**

Correct Answer: A. IPPV with high airway pressures — Positive pressure ventilation increases intrathoracic pressure, reducing venous return and thereby reducing portal venous flow. High PEEP worsens this effect.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q26. Answer: D — Adequate intravascular volume**

Correct Answer: D. Adequate intravascular volume — Maintaining intravascular volume preserves hepatic perfusion. Hypotension, sympathetic stimulation (surgical stress), IPPV, and vasoconstrictors all reduce hepatic blood flow.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q27. Answer: B — Reduced hepatic heat production and vasodilation**

Correct Answer: B. Reduced hepatic heat production and vasodilation — The liver contributes significantly to heat production. Cirrhotic vasodilation increases heat loss. Active warming and temperature monitoring are essential.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q28. Answer: B — Prolonged duration due to reduced metabolism and increased free fraction**

Correct Answer: B. Prolonged duration due to reduced metabolism and increased free fraction — Hepatic oxidation of benzodiazepines is impaired in liver disease. Combined with hypoalbuminemia, this causes prolonged and exaggerated sedation.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q29. Answer: C — Oxazepam/Lorazepam**

Correct Answer: C. Oxazepam/Lorazepam — Oxazepam and lorazepam undergo Phase II conjugation (glucuronidation) which is relatively preserved in liver disease, unlike Phase I oxidation (diazepam, midazolam).

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q30. Answer: B — Compensatory increase in hepatic artery flow when portal venous flow decreases**

Correct Answer: B. Compensatory increase in hepatic artery flow when portal venous flow decreases — This semi-reciprocal mechanism maintains total hepatic blood flow and oxygen delivery when portal flow is reduced.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q31. Answer: C — Careful drainage with albumin replacement to prevent cardiovascular collapse**

Correct Answer: C. Careful drainage with albumin replacement to prevent cardiovascular collapse — Large volume paracentesis (>5L) requires albumin replacement (6-8g per liter removed) to prevent post-paracentesis circulatory dysfunction.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q32. Answer: B — Functional renal failure due to splanchnic vasodilation and renal vasoconstriction**

Correct Answer: B. Functional renal failure due to splanchnic vasodilation and renal vasoconstriction — Hepatorenal syndrome is functional (kidneys are structurally normal). Splanchnic vasodilation causes underfilling, activating RAAS and causing renal vasoconstriction.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q33. Answer: B — Elevated conjugated bilirubin, raised ALP, pale stools, dark urine**

Correct Answer: B. Elevated conjugated bilirubin, raised ALP, pale stools, dark urine — Biliary obstruction causes conjugated hyperbilirubinemia, markedly elevated alkaline phosphatase and GGT, acholic stools, and choluria.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q34. Answer: B — Lack of bile salts prevents fat-soluble vitamin absorption from gut**

Correct Answer: B. Lack of bile salts prevents fat-soluble vitamin absorption from gut — Bile salts are essential for absorption of fat-soluble vitamins (A, D, E, K). Without bile in the gut, Vitamin K malabsorption causes coagulopathy.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q35. Answer: C — 24-48 hours**

Correct Answer: C. 24-48 hours — Parenteral Vitamin K corrects coagulopathy in obstructive jaundice within 24-48 hours. If PT does not correct with Vitamin K, it suggests hepatocellular damage rather than pure obstruction.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q36. Answer: B — Acute tubular necrosis (hepatorenal syndrome/bile nephropathy)**

Correct Answer: B. Acute tubular necrosis (hepatorenal syndrome/bile nephropathy) — Obstructive jaundice predisposes to renal failure due to direct bilirubin toxicity to renal tubules, endotoxemia, and reduced renal perfusion.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q37. Answer: B — Adequate preoperative hydration and maintaining urine output**

Correct Answer: B. Adequate preoperative hydration and maintaining urine output — Preoperative IV fluids, maintaining urine output >0.5-1 mL/kg/hr, and avoiding nephrotoxic drugs are essential for renal protection in jaundiced patients.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q38. Answer: B — Absence of bile in gut allows bacterial overgrowth and endotoxin absorption**

Correct Answer: B. Absence of bile in gut allows bacterial overgrowth and endotoxin absorption — Bile salts normally prevent bacterial overgrowth and endotoxin absorption from the gut. Their absence leads to endotoxemia, predisposing to sepsis and renal failure.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q39. Answer: B — Bilirubin >10 mg/dL with planned major surgery, to reduce complications**

Correct Answer: B. Bilirubin >10 mg/dL with planned major surgery, to reduce complications — Preoperative biliary drainage may reduce renal complications, sepsis, and wound healing problems in severely jaundiced patients undergoing major procedures.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q40. Answer: B — Morphine and drugs causing sphincter of Oddi spasm**

Correct Answer: B. Morphine and drugs causing sphincter of Oddi spasm — Morphine causes sphincter of Oddi spasm (up to 50-fold increase in biliary pressure). Fentanyl has less effect; glucagon or naloxone can reverse spasm if needed.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q41. Answer: B — Glucagon**

Correct Answer: B. Glucagon — Glucagon relaxes the sphincter of Oddi smooth muscle. Naloxone (opioid reversal), nitroglycerin, and volatile anesthetics also help relieve opioid-induced biliary spasm.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q42. Answer: B — Deposition of bile salts in the skin**

Correct Answer: B. Deposition of bile salts in the skin — Retained bile salts deposit in the skin and stimulate nerve endings causing intense pruritus. Cholestyramine (bile salt binder) and rifampin may provide relief.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q43. Answer: B — Malnutrition, Vitamin K deficiency, and reduced immunity**

Correct Answer: B. Malnutrition, Vitamin K deficiency, and reduced immunity — Obstructive jaundice impairs wound healing through protein malnutrition, fat-soluble vitamin deficiency, impaired immune function, and endotoxemia.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q44. Answer: B — Bile salts sensitize the myocardium to vagal stimulation**

Correct Answer: B. Bile salts sensitize the myocardium to vagal stimulation — Elevated bile salts increase cardiac sensitivity to vagal tone and depress myocardial contractility. Atropine should be readily available.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q45. Answer: B — Broad-spectrum antibiotics covering gram-negative and anaerobic organisms**

Correct Answer: B. Broad-spectrum antibiotics covering gram-negative and anaerobic organisms — Cholangitis is common; biliary organisms include E. coli, Klebsiella, Enterococcus, and anaerobes. Prophylaxis with piperacillin-tazobactam or similar is important.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q46. Answer: B — They undergo significant hepatic metabolism and biliary excretion**

Correct Answer: B. They undergo significant hepatic metabolism and biliary excretion — Vecuronium (40-60% biliary excretion) and rocuronium (hepatic uptake) have prolonged duration in liver disease. Atracurium/cisatracurium are preferred alternatives.

*Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative*

**Q47. Answer: D — Use of atracurium**

Correct Answer: D. Use of atracurium — Atracurium does not affect the liver. Postoperative jaundice can result from blood transfusion hemolysis, drug-induced hepatitis (halothane), hepatic ischemia (hypotension), or sepsis.

*Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative*

**Q48. Answer: B — Morphine (sphincter of Oddi spasm mimics stone)**

Correct Answer: B. Morphine (sphincter of Oddi spasm mimics stone) — Morphine-induced sphincter of Oddi spasm can prevent contrast flow into the duodenum, mimicking a common bile duct stone on cholangiography.

*Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction*

**Q49. Answer: B — Maintain renal perfusion and urine output (renal protection)**

Correct Answer: B. Maintain renal perfusion and urine output (renal protection) — Mannitol (osmotic diuretic) helps maintain renal tubular flow and prevent bile pigment-induced tubular damage in jaundiced patients.

*Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction*

**Q50. Answer: C — Isoflurane/Sevoflurane**

Correct Answer: C. Isoflurane/Sevoflurane — Isoflurane and sevoflurane best preserve hepatic arterial buffer response and total hepatic blood flow. Halothane causes the greatest reduction in hepatic blood flow.

*Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative*

# Anesthesia for Hepatic Disease V2 – MCQ

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Anesthesia for Hepatic Disease

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

**Q1. What is the gold standard scoring system to assess the severity of hepatic disease and risk assessment?**

- A. MELD score
- B. Child-Pugh classification
- C. APACHE score
- D. Glasgow Coma Scale

**Ans: B**

**Q2. The aim of preoperative evaluation in hepatic patients is to assess the severity of disease AND:**

- A. Only the liver enzymes
- B. The damage caused to other organs by hepatic dysfunction
- C. Only the coagulation profile
- D. Only the nutritional status

**Ans: B**

**Q3. In Child-Pugh classification, serum bilirubin value for Group C is:**

- A. < 2 mg%
- B. 2–3 mg%
- C. > 3 mg%
- D. > 5 mg%

**Ans: C**

**Q4. In Child-Pugh classification, serum albumin for Group A is:**

- A. < 3 gm%
- B. 3–3.5 gm%
- C. > 3.5 gm%
- D. > 4.5 gm%

**Ans: C**

**Q5. Child-Pugh Class C has a mortality of:**

- A. 2–5%
- B. 10%
- C. > 50%
- D. 25%

**Ans: C**

**Q6. Hepatic patients should always be screened for:**

- A. HIV only
- B. Hepatitis B and C
- C. Hepatitis A only
- D. Tuberculosis

**Ans: B**

**Q7. When should elective surgery be deferred in acute liver disease?**

- A. For a fixed period of 6 weeks
- B. Till the liver functions are stabilized
- C. For 3 months always
- D. Only if bilirubin > 10

**Ans: B**

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| <p><b>Q8. Patients with moderate to severe chronic hepatic disease should have restoration of coagulation abnormalities by:</b></p> <ul style="list-style-type: none"> <li>A. Heparin infusion</li> <li>B. Vitamin K, cryoprecipitate or fresh frozen plasma</li> <li>C. Warfarin</li> <li>D. Aspirin discontinuation only</li> </ul>                     | <p><b>Ans: B</b></p> |
| <p><b>Q9. Premedication with benzodiazepines in hepatic patients:</b></p> <ul style="list-style-type: none"> <li>A. Is the preferred premedicant</li> <li>B. Should not be given as hepatic patients are oversensitive to sedative effects</li> <li>C. Can be given in full dose</li> <li>D. Is mandatory for anxiolysis</li> </ul>                       | <p><b>Ans: B</b></p> |
| <p><b>Q10. Why is central venous pressure (CVP) monitoring highly recommended in hepatic patients?</b></p> <ul style="list-style-type: none"> <li>A. To measure cardiac output</li> <li>B. Due to fluid loss to 3rd space and intravascular volume depletion</li> <li>C. To administer drugs centrally</li> <li>D. To monitor for air embolism</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q11. Due to high volume of distribution in hepatic patients, the initial dose of intravenous agents is:</b></p> <ul style="list-style-type: none"> <li>A. Reduced</li> <li>B. More (higher than normal)</li> <li>C. Same as normal patients</li> <li>D. Half the normal dose</li> </ul>                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q12. In hepatic patients, repeated/maintenance doses of IV agents may be prolonged because of:</b></p> <ul style="list-style-type: none"> <li>A. Increased renal excretion</li> <li>B. Decreased metabolism</li> <li>C. Increased protein binding</li> <li>D. Enhanced biliary excretion</li> </ul>                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q13. The induction agent of choice for hepatic patients is Propofol because:</b></p> <ul style="list-style-type: none"> <li>A. It has the longest duration</li> <li>B. It has short half-life and extrahepatic metabolism</li> <li>C. It is completely metabolized by the liver</li> <li>D. It increases hepatic blood flow</li> </ul>              | <p><b>Ans: B</b></p> |
| <p><b>Q14. The inhalational agent of choice for hepatic patients is:</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Sevoflurane</li> <li>D. Nitrous oxide</li> </ul>                                                                                                                                      | <p><b>Ans: C</b></p> |

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| <p><b>Q15. The most important goal during maintenance of anesthesia in hepatic patients is:</b></p> <ul style="list-style-type: none"> <li>A. Maintaining adequate depth of anesthesia</li> <li>B. Maximum perseverance of hepatic blood flow</li> <li>C. Keeping heart rate stable</li> <li>D. Avoiding awareness</li> </ul>                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q16. Halothane must not be used in hepatic patients for TWO reasons:</b></p> <ul style="list-style-type: none"> <li>A. Seizures and bronchospasm</li> <li>B. May cause hepatitis AND causes significant decrease in hepatic and portal blood flow</li> <li>C. Renal toxicity and cardiac depression</li> <li>D. Malignant hyperthermia and hyperkalemia</li> </ul>                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q17. Nitrous oxide should be avoided in hepatic patients for TWO reasons:</b></p> <ul style="list-style-type: none"> <li>A. It causes seizures and hepatitis</li> <li>B. It can increase pulmonary artery pressure (portopulmonary hypertension) AND requires higher O2 concentration (hepatopulmonary syndrome)</li> <li>C. It causes renal failure and coagulopathy</li> <li>D. It causes hypoglycemia and encephalopathy</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q18. Why are desflurane and isoflurane considered good alternatives to sevoflurane in hepatic patients?</b></p> <ul style="list-style-type: none"> <li>A. They are cheaper</li> <li>B. They preserve hepatic blood flow and produce minimal trifluoroacetic acid (hepatotoxicity is just theoretical)</li> <li>C. They have the strongest analgesic properties</li> <li>D. They do not require a vaporizer</li> </ul>                  | <p><b>Ans: B</b></p> |
| <p><b>Q19. Succinylcholine can produce prolonged block in hepatic patients because of:</b></p> <ul style="list-style-type: none"> <li>A. Increased sensitivity of receptors</li> <li>B. Decreased pseudocholinesterase</li> <li>C. Increased volume of distribution</li> <li>D. Electrolyte imbalance</li> </ul>                                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q20. The muscle relaxants of choice for hepatic patients are:</b></p> <ul style="list-style-type: none"> <li>A. Vecuronium and Pancuronium</li> <li>B. Atracurium and Cis-atracurium</li> <li>C. Rocuronium and Vecuronium</li> <li>D. Succinylcholine and Mivacurium</li> </ul>                                                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q21. Atracurium and Cis-atracurium are preferred in hepatic patients because they are metabolized by:</b></p> <ul style="list-style-type: none"> <li>A. Hepatic cytochrome P450</li> <li>B. Hoffman degradation</li> <li>C. Renal excretion</li> <li>D. Biliary excretion</li> </ul>                                                                                                                                                   | <p><b>Ans: B</b></p> |

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| <p><b>Q22. Remifentanyl is the opioid of choice in hepatic patients because:</b></p> <ul style="list-style-type: none"> <li>A. It has the longest duration</li> <li>B. It is metabolized by plasma esterases</li> <li>C. It is excreted unchanged by kidneys</li> <li>D. It has no respiratory depression</li> </ul>                                            | <p><b>Ans: B</b></p> |
| <p><b>Q23. In hepatic patients, decreased metabolism of citrate (from blood transfusion) means these patients should receive:</b></p> <ul style="list-style-type: none"> <li>A. Magnesium supplementation</li> <li>B. Calcium after blood transfusion</li> <li>C. Potassium replacement</li> <li>D. Sodium bicarbonate</li> </ul>                               | <p><b>Ans: B</b></p> |
| <p><b>Q24. Which of the following is NOT acceptable intraoperatively in hepatic patients?</b></p> <ul style="list-style-type: none"> <li>A. Normothermia</li> <li>B. Hypotension</li> <li>C. Adequate urine output</li> <li>D. Normal blood glucose</li> </ul>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q25. Albumin is the preferred colloid for hepatic patients because hypotension:</b></p> <ul style="list-style-type: none"> <li>A. Causes bronchospasm</li> <li>B. Can severely impair hepatic blood flow</li> <li>C. Increases portal pressure</li> <li>D. Causes seizures</li> </ul>                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q26. Alkalosis is dangerous in hepatic patients because it:</b></p> <ul style="list-style-type: none"> <li>A. Causes hypokalemia</li> <li>B. Increases the conversion of ammonium to ammonia which crosses blood brain barrier to precipitate hepatic encephalopathy</li> <li>C. Causes hypocalcemia only</li> <li>D. Decreases cardiac output</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q27. Hepatic disease patients are prone for hypoglycemia because:</b></p> <ul style="list-style-type: none"> <li>A. They have excessive insulin production</li> <li>B. They have impaired gluconeogenesis and exhausted glucose stores</li> <li>C. They have increased glucose utilization</li> <li>D. They have renal glucose wasting</li> </ul>         | <p><b>Ans: B</b></p> |
| <p><b>Q28. Sepsis in hepatic patients is dangerous because it can precipitate:</b></p> <ul style="list-style-type: none"> <li>A. Bronchospasm</li> <li>B. Hepatic encephalopathy</li> <li>C. Malignant hyperthermia</li> <li>D. Anaphylaxis</li> </ul>                                                                                                          | <p><b>Ans: B</b></p> |

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| <p><b>Q29. Decreased urinary output in hepatic patients can precipitate:</b></p> <ul style="list-style-type: none"> <li>A. Pulmonary edema</li> <li>B. Hepatorenal syndrome</li> <li>C. Cardiac failure</li> <li>D. Cerebral edema</li> </ul>                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q30. Postoperatively in hepatic patients, which functions should be closely monitored?</b></p> <ul style="list-style-type: none"> <li>A. Only cardiac function</li> <li>B. Renal functions and urine output</li> <li>C. Only respiratory function</li> <li>D. Only neurological function</li> </ul>                             | <p><b>Ans: B</b></p> |
| <p><b>Q31. Cirrhotic patients have intrapulmonary shunt, therefore postoperatively they are more prone for:</b></p> <ul style="list-style-type: none"> <li>A. Hyperoxia</li> <li>B. Hypoxia</li> <li>C. Hypercarbia</li> <li>D. Metabolic alkalosis</li> </ul>                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q32. Due to existing coagulopathy and intravascular volume depletion, which anesthetic technique should be avoided in hepatic patients?</b></p> <ul style="list-style-type: none"> <li>A. General anesthesia</li> <li>B. Central neuraxial blocks</li> <li>C. Peripheral nerve blocks</li> <li>D. Local infiltration</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q33. When can central neuraxial block be considered in hepatic patients?</b></p> <ul style="list-style-type: none"> <li>A. Never</li> <li>B. If there is no associated coagulopathy and hypotension can be prevented</li> <li>C. Only in Child-Pugh A</li> <li>D. Only with platelet transfusion cover</li> </ul>               | <p><b>Ans: B</b></p> |
| <p><b>Q34. Patients with biliary obstruction exhibit bradycardia because of:</b></p> <ul style="list-style-type: none"> <li>A. Vagal stimulation from pain</li> <li>B. Direct effect of bile salts on Sinoatrial (SA) node</li> <li>C. Hypothermia</li> <li>D. Beta-blocker effect of bilirubin</li> </ul>                            | <p><b>Ans: B</b></p> |
| <p><b>Q35. Just after the release of biliary obstruction, which drug should be given to wash out the bile salts?</b></p> <ul style="list-style-type: none"> <li>A. Furosemide</li> <li>B. Mannitol</li> <li>C. Dexamethasone</li> <li>D. Normal saline bolus</li> </ul>                                                               | <p><b>Ans: B</b></p> |

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| <p><b>Q36. Vecuronium should not be used in patients with biliary obstruction because:</b></p> <ul style="list-style-type: none"> <li>A. It causes histamine release</li> <li>B. It has 30–40% excretion through bile</li> <li>C. It causes bradycardia</li> <li>D. It is metabolized to toxic metabolites</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q37. Regarding opioids and sphincter of Oddi spasm, the textbook states that the incidence of opioid-induced spasm is:</b></p> <ul style="list-style-type: none"> <li>A. More than 20%</li> <li>B. Around 10%</li> <li>C. Less than 3%</li> <li>D. More than 50%</li> </ul>                                     | <p><b>Ans: C</b></p> |
| <p><b>Q38. The strict guideline of not using opioids for biliary colic has been:</b></p> <ul style="list-style-type: none"> <li>A. Reinforced by recent studies</li> <li>B. Questioned in current day practice</li> <li>C. Made into a law</li> <li>D. Unchanged for decades</li> </ul>                               | <p><b>Ans: B</b></p> |
| <p><b>Q39. In Child-Pugh classification, prothrombin time prolongation from control for Group B is:</b></p> <ul style="list-style-type: none"> <li>A. 1–4 seconds</li> <li>B. 4–6 seconds</li> <li>C. &gt; 6 seconds</li> <li>D. No prolongation</li> </ul>                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q40. In Child-Pugh classification, ascites in Group A is:</b></p> <ul style="list-style-type: none"> <li>A. Moderate</li> <li>B. Marked</li> <li>C. None</li> <li>D. Mild</li> </ul>                                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q41. Child-Pugh Group B has a surgical risk of:</b></p> <ul style="list-style-type: none"> <li>A. Minimal</li> <li>B. Moderate</li> <li>C. Marked</li> <li>D. None</li> </ul>                                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q42. Why should sevoflurane be preferred over halothane — what metabolite does sevoflurane NOT produce?</b></p> <ul style="list-style-type: none"> <li>A. Fluoride ions</li> <li>B. Trifluoroacetic acid</li> <li>C. Compound A</li> <li>D. Carbon monoxide</li> </ul>                                          | <p><b>Ans: B</b></p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
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| <p><b>Q43. Encephalopathy in hepatic patients should be corrected preoperatively by:</b></p> <ul style="list-style-type: none"> <li>A. Phenytoin and carbamazepine</li> <li>B. Lactulose and neomycin</li> <li>C. Mannitol and furosemide</li> <li>D. Dexamethasone and hypertonic saline</li> </ul>                                                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q44. If ascites is significant enough to impair respiratory functions, what should be done preoperatively?</b></p> <ul style="list-style-type: none"> <li>A. Diuretics only</li> <li>B. Tapping of ascites</li> <li>C. TIPS procedure</li> <li>D. Paracentesis is contraindicated</li> </ul>                                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q45. Child-Pugh classification includes all of the following parameters EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Encephalopathy</li> <li>B. Nutrition</li> <li>C. Creatinine</li> <li>D. Ascites</li> </ul>                                                                                                                                                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q46. Decreased albumin in hepatic patients may increase the unbound fraction of which aspect of drug pharmacology?</b></p> <ul style="list-style-type: none"> <li>A. Drug absorption</li> <li>B. Drug binding — increasing the free fraction of the drug</li> <li>C. Drug excretion</li> <li>D. Drug distribution volume only</li> </ul>                                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q47. Universal precautions intraoperatively in hepatic patients must be taken if the patient is positive for:</b></p> <ul style="list-style-type: none"> <li>A. H. pylori</li> <li>B. Hepatitis B or C</li> <li>C. CMV</li> <li>D. EBV</li> </ul>                                                                                                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q48. Preoperative evaluation of hepatic patients should rule out all of the following EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Coagulopathy and renal damage (hepatorenal syndrome)</li> <li>B. Pulmonary dysfunction (hepatopulmonary syndrome)</li> <li>C. Rheumatoid arthritis</li> <li>D. Portopulmonary hypertension and encephalopathy</li> </ul>                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q49. In the management of biliary obstruction patients, what is stated about the general principles of hepatic disease management?</b></p> <ul style="list-style-type: none"> <li>A. Completely different guidelines apply</li> <li>B. Whatever discussed for hepatic disease is applicable, with additional points to be taken care</li> <li>C. Only regional anesthesia should be used</li> <li>D. No special considerations needed</li> </ul> | <p><b>Ans: B</b></p> |

**Q50. Alcoholism should always be ruled out in hepatic patients because of associated:**

- A. Only liver damage
- B. Comorbidities including cardiomyopathy, pancreatitis, and nutritional deficiencies
- C. Only psychiatric issues
- D. Only renal damage

**Ans: B**



# Answer Key & Explanations

## Q1. Answer: B — Child-Pugh classification

Correct Answer: B. Child-Pugh classification — Child-Pugh classification is the gold standard scoring system to assess the severity of hepatic disease and hence the risk assessment (Table 21.1).

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q2. Answer: B — The damage caused to other organs by hepatic dysfunction

Correct Answer: B. The damage caused to other organs by hepatic dysfunction — The aim is not only to assess severity but also to assess the damage caused to other organs by hepatic dysfunction: coagulopathy, renal damage (hepatorenal syndrome), pulmonary dysfunction (hepatopulmonary syndrome), portopulmonary hypertension, encephalopathy, ascites, cardiac functions, electrolyte and metabolic abnormalities.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q3. Answer: C — > 3 mg%

Correct Answer: C. > 3 mg% — In Child-Pugh classification: Group A: bilirubin < 2, Group B: 2–3, Group C: > 3 mg%.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

## Q4. Answer: C — > 3.5 gm%

Correct Answer: C. > 3.5 gm% — Child-Pugh Group A: albumin > 3.5, Group B: 3–3.5, Group C: < 3 gm%.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

## Q5. Answer: C — > 50%

Correct Answer: C. > 50% — Per Table 21.1: Group A mortality 2–5%, Group B 10%, Group C > 50%.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

## Q6. Answer: B — Hepatitis B and C

Correct Answer: B. Hepatitis B and C — Always screen the hepatic patients for hepatitis B and C. Universal precautions must be taken intraoperatively if the patient is positive.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q7. Answer: B — Till the liver functions are stabilized

Correct Answer: B. Till the liver functions are stabilized — Elective surgery should be deferred in acute liver disease till the liver functions are stabilized. No need to defer in asymptomatic patients with mildly elevated liver enzymes.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Timing of Surgery

## Q8. Answer: B — Vitamin K, cryoprecipitate or fresh frozen plasma

Correct Answer: B. Vitamin K, cryoprecipitate or fresh frozen plasma — Restoration of coagulation abnormalities (by vitamin K, cryoprecipitate or fresh frozen plasma) is essential before surgery.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Timing of Surgery

## Q9. Answer: B — Should not be given as hepatic patients are oversensitive to sedative effects

Correct Answer: B. Should not be given as hepatic patients are oversensitive to sedative effects — As hepatic patients may be oversensitive to the sedative effects of benzodiazepine, premedication with benzodiazepines should not be given.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Premedication

**Q10. Answer: B — Due to fluid loss to 3rd space and intravascular volume depletion**

Correct Answer: B. Due to fluid loss to 3rd space and intravascular volume depletion — As there is fluid loss to 3rd space, the intravascular volume is depleted in these patients therefore, central venous pressure monitoring is highly recommended.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Monitoring

**Q11. Answer: B — More (higher than normal)**

Correct Answer: B. More (higher than normal) — Due to high volume of distribution the initial dose of intravenous agents is more while due to decreased metabolism the action of repeated/maintenance doses may be prolonged.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – General Anesthesia

**Q12. Answer: B — Decreased metabolism**

Correct Answer: B. Decreased metabolism — While initial doses are higher (high Vd), the action of repeated/maintenance doses may be prolonged due to decreased metabolism. Also decreased albumin may increase the unbound fraction of drugs.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – General Anesthesia

**Q13. Answer: B — It has short half-life and extrahepatic metabolism**

Correct Answer: B. It has short half-life and extrahepatic metabolism — Propofol, due to its short half-life and extrahepatic metabolism, is the induction agent of choice for hepatic patients.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Induction

**Q14. Answer: C — Sevoflurane**

Correct Answer: C. Sevoflurane — Sevoflurane is the inhalational agent of choice for two reasons: (1) Maximum perseverance of hepatic blood flow, (2) It does not produce Trifluoroacetic acid avoiding any possibility of hepatitis.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

**Q15. Answer: B — Maximum perseverance of hepatic blood flow**

Correct Answer: B. Maximum perseverance of hepatic blood flow — Maintaining hepatic blood flow should be the most important goal in these patients.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

**Q16. Answer: B — May cause hepatitis AND causes significant decrease in hepatic and portal blood flow**

Correct Answer: B. May cause hepatitis AND causes significant decrease in hepatic and portal blood flow — Halothane must not be used due to two reasons: firstly, it may cause hepatitis and secondly it causes significant decrease in hepatic as well as portal blood flow.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

**Q17. Answer: B — It can increase pulmonary artery pressure (portopulmonary hypertension) AND requires higher O2 concentration (hepatopulmonary syndrome)**

Correct Answer: B. It can increase pulmonary artery pressure (portopulmonary hypertension) AND requires higher O2 concentration (hepatopulmonary syndrome) — Nitrous oxide should be avoided for two reasons: it can increase pulmonary artery pressure (portopulmonary hypertension) and it allows only lower oxygen concentration when higher concentration is needed due to significant right to left shunt (hepatopulmonary syndrome).

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

**Q18. Answer: B — They preserve hepatic blood flow and produce minimal trifluoroacetic acid (hepatotoxicity is just theoretical)**

Correct Answer: B. They preserve hepatic blood flow and produce minimal trifluoroacetic acid (hepatotoxicity is just theoretical) — Desflurane (least metabolism) and isoflurane preserve hepatic blood flow. The amount of trifluoroacetic acid produced is too minimal that hepatotoxicity appears to be just theoretical.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

**Q19. Answer: B — Decreased pseudocholinesterase**

Correct Answer: B. Decreased pseudocholinesterase — Succinylcholine can produce prolonged block due to decreased pseudocholinesterase synthesized by the liver.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Muscle Relaxants

**Q20. Answer: B — Atracurium and Cis-atracurium**

Correct Answer: B. Atracurium and Cis-atracurium — As atracurium and cis-atracurium are metabolized by Hoffman degradation, they are the muscle relaxants of choice for hepatic patients (organ-independent metabolism).

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Muscle Relaxants

**Q21. Answer: B — Hoffman degradation**

Correct Answer: B. Hoffman degradation — Hoffman degradation is organ-independent, making atracurium and cis-atracurium ideal for patients with hepatic dysfunction.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Muscle Relaxants

**Q22. Answer: B — It is metabolized by plasma esterases**

Correct Answer: B. It is metabolized by plasma esterases — Remifentanyl is the opioid of choice because it is metabolized by plasma esterases, independent of hepatic function.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Muscle Relaxants

**Q23. Answer: B — Calcium after blood transfusion**

Correct Answer: B. Calcium after blood transfusion — As there is decreased metabolism of citrate (which can chelate calcium), these patients should receive calcium after blood transfusion.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q24. Answer: B — Hypotension**

Correct Answer: B. Hypotension — Hypotension can severely impair hepatic blood flow and is not acceptable intraoperatively. Other unacceptable conditions: alkalosis, hypoglycemia, sepsis, decreased urinary output.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

**Q25. Answer: B — Can severely impair hepatic blood flow**

Correct Answer: B. Can severely impair hepatic blood flow — Hypotension can severely impair hepatic blood flow. Albumin is the preferred colloid for a patient with compromised hepatic functions.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

**Q26. Answer: B — Increases the conversion of ammonium to ammonia which crosses blood brain barrier to precipitate hepatic encephalopathy**

Correct Answer: B. Increases the conversion of ammonium to ammonia which crosses blood brain barrier to precipitate hepatic encephalopathy — Alkalosis increases conversion of ammonium to ammonia which crosses blood brain barrier to precipitate hepatic encephalopathy.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

**Q27. Answer: B — They have impaired gluconeogenesis and exhausted glucose stores**

Correct Answer: B. They have impaired gluconeogenesis and exhausted glucose stores — Hepatic disease patients have impaired gluconeogenesis and exhausted glucose stores therefore, they are prone for hypoglycemia.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

**Q28. Answer: B — Hepatic encephalopathy**

Correct Answer: B. Hepatic encephalopathy — Sepsis can precipitate encephalopathy in hepatic patients. It is one of the conditions not acceptable intraoperatively.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

**Q29. Answer: B — Hepatorenal syndrome**

Correct Answer: B. Hepatorenal syndrome — Decreased urinary output can precipitate hepatorenal syndrome in hepatic patients and is not acceptable intraoperatively.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

**Q30. Answer: B — Renal functions and urine output**

Correct Answer: B. Renal functions and urine output — Postoperatively, renal functions and urine output should be monitored along with good antibiotic cover and oxygenation.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Postoperatively

**Q31. Answer: B — Hypoxia**

Correct Answer: B. Hypoxia — Cirrhotic patients have intrapulmonary shunt therefore, are more prone for hypoxia. Oxygenation should be closely monitored postoperatively.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Postoperatively – Oxygenation

**Q32. Answer: B — Central neuraxial blocks**

Correct Answer: B. Central neuraxial blocks — Due to the possibility of existing coagulopathy and intravascular volume depletion, central neuraxial blocks should be avoided in hepatic patients.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Choice of Anesthesia

**Q33. Answer: B — If there is no associated coagulopathy and hypotension can be prevented**

Correct Answer: B. If there is no associated coagulopathy and hypotension can be prevented — If there is no associated coagulopathy, central neuraxial block can be considered if hypotension can be prevented.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Choice of Anesthesia

**Q34. Answer: B — Direct effect of bile salts on Sinoatrial (SA) node**

Correct Answer: B. Direct effect of bile salts on Sinoatrial (SA) node — These patients exhibit bradycardia because of direct effect of bile salts on Sinoatrial (SA) node.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q35. Answer: B — Mannitol**

Correct Answer: B. Mannitol — Just after the release of obstruction, mannitol should be given to wash out the bile salts.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q36. Answer: B — It has 30–40% excretion through bile**

Correct Answer: B. It has 30–40% excretion through bile — Muscle relaxant vecuronium which has 30–40% excretion through bile should not be used in biliary obstruction patients.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q37. Answer: C — Less than 3%**

Correct Answer: C. Less than 3% — The incidence of opioid-induced spasm of sphincter of Oddi is less than 3%, therefore strict guidelines of not using opioids for biliary colic has been questioned in current day practice.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q38. Answer: B — Questioned in current day practice**

Correct Answer: B. Questioned in current day practice — As the incidence of opioid-induced sphincter of Oddi spasm is less than 3%, strict guidelines of not using opioids for biliary colic has been questioned in current day practice.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q39. Answer: B — 4–6 seconds**

Correct Answer: B. 4–6 seconds — Per Table 21.1: Group A: PT prolongation 1–4 sec, Group B: 4–6 sec, Group C: > 6 seconds.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

**Q40. Answer: C — None**

Correct Answer: C. None — Per Table 21.1: Group A: No ascites, Group B: Moderate, Group C: Marked.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

**Q41. Answer: B — Moderate**

Correct Answer: B. Moderate — Per Table 21.1: Group A: Minimal surgical risk, Group B: Moderate, Group C: Marked.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

**Q42. Answer: B — Trifluoroacetic acid**

Correct Answer: B. Trifluoroacetic acid — Sevoflurane does not produce Trifluoroacetic acid on metabolism, thereby avoiding any possibility of hepatitis (unlike halothane).

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

**Q43. Answer: B — Lactulose and neomycin**

Correct Answer: B. Lactulose and neomycin — Correction of encephalopathy (by lactulose and neomycin) should be done before taking patients for surgery.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Timing of Surgery

**Q44. Answer: B — Tapping of ascites**

Correct Answer: B. Tapping of ascites — Tapping of ascites (if significant enough to impair respiratory functions) should be done as part of preoperative optimization.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Timing of Surgery

**Q45. Answer: C — Creatinine**

Correct Answer: C. Creatinine — Child-Pugh includes: Bilirubin, Albumin, Prothrombin time, Ascites, Encephalopathy, Nutrition, Surgical risk, and Mortality. Creatinine is part of MELD score, not Child-Pugh.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

**Q46. Answer: B — Drug binding — increasing the free fraction of the drug**

Correct Answer: B. Drug binding — increasing the free fraction of the drug — Decreased albumin may increase the unbound fraction of the drug, leading to exaggerated clinical effects of protein-bound drugs.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – General Anesthesia

**Q47. Answer: B — Hepatitis B or C**

Correct Answer: B. Hepatitis B or C — Universal precautions must be taken if the patient is positive for hepatitis B or C to protect the operating room staff.

*Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative*

**Q48. Answer: C — Rheumatoid arthritis**

Correct Answer: C. Rheumatoid arthritis — Important to rule out: coagulopathy, renal damage (hepatorenal syndrome), pulmonary dysfunction (hepatopulmonary syndrome), portopulmonary hypertension, encephalopathy, ascites, gastroesophageal varices, cardiac functions, electrolyte and metabolic abnormalities.

*Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation*

**Q49. Answer: B — Whatever discussed for hepatic disease is applicable, with additional points to be taken care**

Correct Answer: B. Whatever discussed for hepatic disease is applicable, with additional points to be taken care — Whatever discussed for the management of patients with hepatic disease, the same is applicable for patients with biliary obstruction with additional points (bradycardia, mannitol, vecuronium avoidance, opioid caution).

*Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction*

**Q50. Answer: B — Comorbidities including cardiomyopathy, pancreatitis, and nutritional deficiencies**

Correct Answer: B. Comorbidities including cardiomyopathy, pancreatitis, and nutritional deficiencies — Always rule out alcoholism and associated comorbidities in hepatic patients.

*Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation*

# Anesthesia for Renal Diseases and Electrolyte Imbalances MCQ – Chapter 22

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 22)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. What is the cause of anemia in patients with renal impairment?**

- A. Iron deficiency
- B. Decreased erythropoietin or decreased intake due to anorexia
- C. Vitamin B12 deficiency
- D. Folate deficiency

**Ans: B**

**Q2. What percentage of patients with chronic renal disease exhibit hypertension?**

- A. More than 50%
- B. More than 60%
- C. More than 70%
- D. More than 80%

**Ans: D**

**Q3. What is the most common endocrine cause of chronic kidney disease?**

- A. Hypothyroidism
- B. Cushing's syndrome
- C. Diabetes
- D. Hyperparathyroidism

**Ans: C**

**Q4. Patients with acute kidney injury (AKI) should not undergo elective surgery until:**

- A. Hemoglobin normalizes
- B. The injury is fully resolved
- C. Blood pressure is controlled
- D. Potassium is less than 4 mEq/L

**Ans: B**

**Q5. What should be the target blood pressure before surgery in CKD patients?**

- A. < 120/80 mm Hg
- B. < 130/80 mm Hg
- C. < 140/90 mm Hg
- D. < 150/90 mm Hg

**Ans: B**

**Q6. How many hours before surgery should dialysis be performed in CKD patients?**

- A. 12 hours
- B. 24 hours
- C. 48 hours
- D. 6 hours

**Ans: B**

**Q7. Why should premedication with benzodiazepines be avoided in renal patients?**

- A. They are nephrotoxic
- B. They cause hyperkalemia
- C. CNS of these patients is extremely sensitive to sedatives and narcotics
- D. They increase bleeding tendency

**Ans: C**



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| <p><b>Q8. What is the minimum urine output that should be maintained intraoperatively in renal patients?</b></p> <p>A. &gt; 0.3 mL/kg/hr<br/>B. &gt; 0.5 mL/kg/hr<br/>C. &gt; 1.0 mL/kg/hr<br/>D. &gt; 2.0 mL/kg/hr</p>                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q9. Which selective dopamine-1 agonist is preferred over dopamine for maintaining urine output in major surgeries on renal patients?</b></p> <p>A. Dobutamine<br/>B. Fenoldopam<br/>C. Milrinone<br/>D. Noradrenaline</p>                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q10. Why should radial and ulnar artery be spared in renal patients requiring invasive blood pressure monitoring?</b></p> <p>A. They have high infection risk<br/>B. They may be required for hemodialysis in future<br/>C. They are calcified in renal patients<br/>D. They give inaccurate readings</p> | <p><b>Ans: B</b></p> |
| <p><b>Q11. Which artery is most often chosen for invasive BP monitoring in renal patients?</b></p> <p>A. Radial artery<br/>B. Ulnar artery<br/>C. Dorsalis pedis<br/>D. Brachial artery</p>                                                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q12. BP cuff should not be applied in the limb having:</b></p> <p>A. Peripheral neuropathy<br/>B. AV fistula<br/>C. Edema<br/>D. Cellulitis</p>                                                                                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q13. Why should central neuraxial blocks be avoided in renal patients?</b></p> <p>A. Risk of epidural abscess<br/>B. Associated uremic coagulopathy, uremic polyneuropathy, and intolerance to extra fluids<br/>C. High failure rate<br/>D. Risk of nephrotoxicity from local anesthetics</p>             | <p><b>Ans: B</b></p> |
| <p><b>Q14. What is the induction agent of choice in renal patients?</b></p> <p>A. Thiopentone<br/>B. Ketamine<br/>C. Propofol<br/>D. Etomidate</p>                                                                                                                                                              | <p><b>Ans: C</b></p> |

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| <p><b>Q15. Why should propofol be used in small doses in renal patients?</b></p> <ul style="list-style-type: none"> <li>A. It is nephrotoxic in high doses</li> <li>B. It causes hyperkalemia</li> <li>C. Exaggerated response of CNS due to disrupted blood brain barrier in uremia</li> <li>D. It causes metabolic acidosis</li> </ul>       | <p><b>Ans: C</b></p> |
| <p><b>Q16. What is the inhalational agent of choice for maintenance in renal patients?</b></p> <ul style="list-style-type: none"> <li>A. Sevoflurane</li> <li>B. Isoflurane</li> <li>C. Desflurane</li> <li>D. Halothane</li> </ul>                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q17. Why should sevoflurane not be used in renal patients?</b></p> <ul style="list-style-type: none"> <li>A. It causes bronchospasm</li> <li>B. It produces high levels of fluoride approaching renal threshold of 50 <math>\mu\text{mol/L}</math></li> <li>C. It is hepatotoxic</li> <li>D. It causes malignant hyperthermia</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q18. Why should succinylcholine not be used in renal patients?</b></p> <ul style="list-style-type: none"> <li>A. It causes malignant hyperthermia</li> <li>B. It is metabolized by kidneys</li> <li>C. These patients exhibit hyperkalemia</li> <li>D. It causes prolonged block</li> </ul>                                              | <p><b>Ans: C</b></p> |
| <p><b>Q19. Which non-depolarizing muscle relaxants are the agents of choice in renal patients?</b></p> <ul style="list-style-type: none"> <li>A. Pancuronium and Vecuronium</li> <li>B. Atracurium and Cis-atracurium</li> <li>C. Rocuronium and Vecuronium</li> <li>D. Mivacurium and Doxacurium</li> </ul>                                   | <p><b>Ans: B</b></p> |
| <p><b>Q20. What is the opioid of choice in renal patients?</b></p> <ul style="list-style-type: none"> <li>A. Morphine</li> <li>B. Pethidine</li> <li>C. Remifentanil</li> <li>D. Codeine</li> </ul>                                                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q21. Which long-acting opioids are NOT recommended in renal patients?</b></p> <ul style="list-style-type: none"> <li>A. Fentanyl and Alfentanil</li> <li>B. Morphine and Pethidine</li> <li>C. Remifentanil and Sufentanil</li> <li>D. Alfentanil and Sufentanil</li> </ul>                                                              | <p><b>Ans: B</b></p> |

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| <p><b>Q22. Why is the traditional concept of avoiding ringer lactate in renal patients no longer valid?</b></p> <p>A. Ringer lactate is cheaper<br/> B. Normal saline causes hyperchloremic metabolic acidosis producing more hyperkalemia<br/> C. Ringer lactate has less sodium<br/> D. Normal saline causes hyponatremia</p> | <p><b>Ans: B</b></p> |
| <p><b>Q23. Which fluid is preferred in renal patients for intraoperative use?</b></p> <p>A. Normal saline (0.9%)<br/> B. Ringer lactate<br/> C. 5% Dextrose<br/> D. Hetastarch</p>                                                                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q24. If ringer lactate cannot be used, which alternative fluid is preferred in renal patients?</b></p> <p>A. Normal saline 0.9%<br/> B. 0.45% saline or 0.18% saline with dextrose<br/> C. Hetastarch 6%<br/> D. 5% albumin</p>                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q25. Why should colloids (hydroxyethyl starch) be avoided in renal patients?</b></p> <p>A. They cause anaphylaxis<br/> B. They have been found to exacerbate renal injury<br/> C. They cause hypernatremia<br/> D. They are expensive</p>                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q26. If a colloid must be used in renal patients, which one is preferred?</b></p> <p>A. Hetastarch<br/> B. Dextran<br/> C. Gelatins (Gelofusine)<br/> D. Albumin 20%</p>                                                                                                                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q27. What molecular weight colloids must NOT be used in renal patients?</b></p> <p>A. &gt; 100 kDa<br/> B. &gt; 150 kDa<br/> C. &gt; 200 kDa<br/> D. &gt; 250 kDa</p>                                                                                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q28. Which drugs should be avoided postoperatively in renal patients for pain management?</b></p> <p>A. Paracetamol<br/> B. Non-steroidal anti-inflammatory drugs (NSAIDs)<br/> C. Tramadol<br/> D. Fentanyl</p>                                                                                                          | <p><b>Ans: B</b></p> |

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| <p><b>Q29. Which type of anesthesia is preferred for transurethral resection of prostate (TURP)?</b></p> <p>A. General anesthesia<br/>B. Spinal anesthesia<br/>C. Epidural anesthesia<br/>D. Local anesthesia</p>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q30. Why is spinal anesthesia preferred for TURP?</b></p> <p>A. It is cheaper<br/>B. Old patients have decreased pulmonary reserve and awake patient can tell central signs of TURP syndrome<br/>C. General anesthesia is contraindicated<br/>D. It provides better muscle relaxation</p> | <p><b>Ans: B</b></p> |
| <p><b>Q31. What is the most common irrigating fluid used in TURP?</b></p> <p>A. Normal saline<br/>B. Glycine<br/>C. Mannitol<br/>D. Sorbitol</p>                                                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q32. How much fluid is estimated to be absorbed every minute during TURP?</b></p> <p>A. 5-15 mL<br/>B. 10-30 mL<br/>C. 30-50 mL<br/>D. 50-100 mL</p>                                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q33. TURP syndrome is characterized by all of the following EXCEPT:</b></p> <p>A. Fluid overloading<br/>B. Dilutional hyponatremia<br/>C. Hypo-osmolality<br/>D. Hypernatremia</p>                                                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q34. Glycine absorbed during TURP can produce:</b></p> <p>A. Hyperglycemia and hyperammonemia<br/>B. Hypoglycemia and hyponatremia<br/>C. Hyperkalemia and acidosis<br/>D. Hypercalcemia and alkalosis</p>                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q35. Hypo-osmolality and hyponatremia in TURP syndrome can produce:</b></p> <p>A. Peripheral neuropathy<br/>B. Cerebral edema causing restlessness, confusion, or seizures<br/>C. Renal failure<br/>D. Cardiac arrest</p>                                                                 | <p><b>Ans: B</b></p> |

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| <p><b>Q36. Hyperglycemia during TURP can produce:</b></p> <ul style="list-style-type: none"> <li>A. Hypertension and tachycardia</li> <li>B. Transient visual disturbances including transient blindness</li> <li>C. Permanent blindness</li> <li>D. Hearing loss</li> </ul>                                                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q37. What is the treatment of fluid overload in TURP syndrome?</b></p> <ul style="list-style-type: none"> <li>A. Mannitol</li> <li>B. Furosemide</li> <li>C. Dialysis</li> <li>D. Fluid restriction alone</li> </ul>                                                                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q38. What is used to treat symptomatic hyponatremia in TURP syndrome?</b></p> <ul style="list-style-type: none"> <li>A. Normal saline</li> <li>B. Hypertonic saline</li> <li>C. Ringer lactate</li> <li>D. 5% Dextrose</li> </ul>                                                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q39. What serum potassium level should be corrected before taking the patient for surgery in hyperkalemia?</b></p> <ul style="list-style-type: none"> <li>A. &lt; 4.5 mEq/L</li> <li>B. &lt; 5.0 mEq/L</li> <li>C. &lt; 5.5 mEq/L</li> <li>D. &lt; 6.0 mEq/L</li> </ul>                                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q40. Which of the following is TRUE about anesthetic management of hyperkalemia?</b></p> <ul style="list-style-type: none"> <li>A. Ringer lactate is preferred</li> <li>B. Succinylcholine is safe to use</li> <li>C. Potassium-rich fluids should be avoided and succinylcholine is contraindicated</li> <li>D. Hyperventilation should be avoided</li> </ul>                       | <p><b>Ans: C</b></p> |
| <p><b>Q41. Why is mild hyperventilation beneficial during general anesthesia in hyperkalemic patients?</b></p> <ul style="list-style-type: none"> <li>A. It increases oxygen delivery</li> <li>B. It causes alkalosis which pushes potassium intracellularly</li> <li>C. It reduces anesthetic requirement</li> <li>D. It prevents bronchospasm</li> </ul>                                 | <p><b>Ans: B</b></p> |
| <p><b>Q42. What is the general consensus for mild chronic hypokalemia (serum K<sup>+</sup> &gt;3 mEq/L) before surgery?</b></p> <ul style="list-style-type: none"> <li>A. Must be corrected in all cases</li> <li>B. No need to treat if there are no ECG changes or organ dysfunction</li> <li>C. Surgery must be postponed</li> <li>D. Potassium supplementation is mandatory</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q43. In hypokalemic patients on digoxin, what serum potassium level should be achieved before surgery?</b></p> <p>A. &gt; 3 mEq/L<br/>B. &gt; 3.5 mEq/L<br/>C. &gt; 4 mEq/L<br/>D. &gt; 4.5 mEq/L</p>                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q44. In hypokalemia, the dose of non-depolarizing muscle relaxant should be:</b></p> <p>A. Increased<br/>B. Decreased as prolonged block is expected<br/>C. Unchanged<br/>D. Doubled</p>                                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q45. Why must rapid correction of hyponatremia never be done?</b></p> <p>A. It causes hyperkalemia<br/>B. It can cause pontine myelinolysis<br/>C. It causes renal failure<br/>D. It causes cardiac arrest</p>                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q46. What is the key principle in anesthetic management of hypercalcemia?</b></p> <p>A. Avoiding calcium-containing fluids<br/>B. Maintenance of hydration and urine output<br/>C. Using calcium channel blockers<br/>D. Avoiding all muscle relaxants</p>                          | <p><b>Ans: B</b></p> |
| <p><b>Q47. Hypocalcemia can produce which of the following during anesthesia?</b></p> <p>A. Hypertension and tachycardia<br/>B. Laryngospasm and hypotension<br/>C. Bronchospasm and hyperkalemia<br/>D. Malignant hyperthermia</p>                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q48. Hypermagnesemia potentiates the action of:</b></p> <p>A. Only depolarizing muscle relaxants<br/>B. Only non-depolarizing muscle relaxants<br/>C. Both depolarizing and non-depolarizing muscle relaxants<br/>D. Neither depolarizing nor non-depolarizing muscle relaxants</p> | <p><b>Ans: C</b></p> |
| <p><b>Q49. Hypomagnesemia may be associated with:</b></p> <p>A. Hyperkalemia and hypercalcemia<br/>B. Refractory hypokalemia and hypocalcemia<br/>C. Hyponatremia and metabolic alkalosis<br/>D. Polycythemia</p>                                                                         | <p><b>Ans: B</b></p> |

**Q50. What is the effect of excess fluid administration in renal patients?**

- A. It improves renal perfusion
- B. It can precipitate pulmonary edema and fluid deficiency can worsen kidney injury
- C. It has no significant effect
- D. It always improves urine output

**Ans: B**

# Answer Key & Explanations

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## **Q1. Answer: B — Decreased erythropoietin or decreased intake due to anorexia**

The anemia in renal impairment is due to decrease erythropoietin or decrease intake due to anorexia seen in uremia.

Topic: Preoperative Evaluation – Anemia | Ref: Ch 22, Page 202

## **Q2. Answer: D — More than 80%**

More than 80% patients with chronic renal disease exhibit hypertension.

Topic: Preoperative Evaluation – Hypertension | Ref: Ch 22, Page 202

## **Q3. Answer: C — Diabetes**

Diabetes is the most common cause of chronic kidney disease as an endocrine abnormality.

Topic: Preoperative Evaluation – Endocrine Abnormalities | Ref: Ch 22, Page 202

## **Q4. Answer: B — The injury is fully resolved**

Patients with acute kidney injury (AKI) should not undergo elective surgery until the injury is fully resolved.

Topic: Timing and Preparation before Surgery | Ref: Ch 22, Page 202

## **Q5. Answer: B — < 130/80 mm Hg**

Control of blood pressure target should be < 130/80 mm Hg in CKD patients before surgery.

Topic: Timing and Preparation before Surgery | Ref: Ch 22, Page 202

## **Q6. Answer: B — 24 hours**

Dialysis should be performed within 24 hours before surgery.

Topic: Timing and Preparation before Surgery | Ref: Ch 22, Page 202

## **Q7. Answer: C — CNS of these patients is extremely sensitive to sedatives and narcotics**

Premedication with benzodiazepines should be avoided as CNS of these patients is extremely sensitive to effect of sedatives and narcotics.

Topic: Premedication | Ref: Ch 22, Page 202

## **Q8. Answer: B — > 0.5 mL/kg/hr**

Aim should be to maintain urine output > 0.5 mL/kg/hr intraoperatively.

Topic: Intraoperative Monitoring – Urine Output | Ref: Ch 22, Page 202

## **Q9. Answer: B — Fenoldopam**

Fenoldopam (selective dopamine-1 agonist) is chosen over dopamine if required for very major surgeries in renal patients.

Topic: Intraoperative Monitoring – Urine Output | Ref: Ch 22, Page 202

## **Q10. Answer: B — They may be required for hemodialysis in future**

If invasive blood pressure monitoring is required then radial and ulnar artery should be spared as they may be required for hemodialysis in future.

Topic: Intraoperative Monitoring | Ref: Ch 22, Page 202-203



**Q11. Answer: C — Dorsalis pedis**

Dorsalis pedis is most often chosen in these patients for invasive BP monitoring.

Topic: Intraoperative Monitoring | Ref: Ch 22, Page 203

**Q12. Answer: B — AV fistula**

BP cuff should not be applied in the limb having AV fistula.

Topic: Intraoperative Monitoring | Ref: Ch 22, Page 203

**Q13. Answer: B — Associated uremic coagulopathy, uremic polyneuropathy, and intolerance to extra fluids**

Central neuraxial blocks should be avoided due to associated uremic coagulopathy, uremic polyneuropathy and intolerance of renal patients to extra fluids to treat hypotension.

Topic: Choice of Anesthesia | Ref: Ch 22, Page 203

**Q14. Answer: C — Propofol**

Due to shorter half-life and all inactive metabolites, propofol is the induction agent of choice in renal patients.

Topic: General Anesthesia – Induction | Ref: Ch 22, Page 203

**Q15. Answer: C — Exaggerated response of CNS due to disrupted blood brain barrier in uremia**

Propofol may be required in small doses because of exaggerated response of CNS due to disrupted blood brain barrier in uremia.

Topic: General Anesthesia – Induction | Ref: Ch 22, Page 203

**Q16. Answer: C — Desflurane**

Desflurane is the inhalational agent of choice because it does not produce fluoride.

Topic: General Anesthesia – Maintenance | Ref: Ch 22, Page 203

**Q17. Answer: B — It produces high levels of fluoride approaching renal threshold of 50 µmol/L**

Sevoflurane should not be used in renal patients as it produces high levels of fluoride approaching renal threshold of 50 µmol/L.

Topic: General Anesthesia – Maintenance | Ref: Ch 22, Page 203

**Q18. Answer: C — These patients exhibit hyperkalemia**

As these patients exhibit hyperkalemia, succinylcholine should not be used.

Topic: Muscle Relaxants | Ref: Ch 22, Page 203

**Q19. Answer: B — Atracurium and Cis-atracurium**

Due to Hoffman degradation, non-depolarizing muscle relaxants of choice are Atracurium and Cis-atracurium.

Topic: Muscle Relaxants | Ref: Ch 22, Page 203

**Q20. Answer: C — Remifentanyl**

Remifentanyl is the opioid of choice because it is rapidly metabolized in plasma by esterases.

Topic: Opioids | Ref: Ch 22, Page 203

**Q21. Answer: B — Morphine and Pethidine**

Long acting like morphine and pethidine are not recommended in renal patients.

Topic: Opioids | Ref: Ch 22, Page 203

**Q22. Answer: B — Normal saline causes hyperchloremic metabolic acidosis producing more hyperkalemia**

Recent studies have shown that normal saline causes hyperchloremic metabolic acidosis producing more hyperkalemia. Therefore, ringer lactate is preferred over normal saline with close monitoring of potassium levels.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

**Q23. Answer: B — Ringer lactate**

Ringer lactate is preferred over normal saline with close monitoring of potassium levels.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

**Q24. Answer: B — 0.45% saline or 0.18% saline with dextrose**

If ringer lactate cannot be used then 0.45% saline or 0.18% saline with dextrose is preferred over normal saline (0.9%).

Topic: Selection of Fluids | Ref: Ch 22, Page 203

**Q25. Answer: B — They have been found to exacerbate renal injury**

Colloids (hydroxyethyl starch) have been found to exacerbate renal injury.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

**Q26. Answer: C — Gelatins (Gelofusine)**

If it is mandatory to use a colloid then Gelatins (Gelofusine) is preferred.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

**Q27. Answer: C — > 200 kDa**

High molecular weight colloids (>MW 200 kDa) must not be used for renal patients.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

**Q28. Answer: B — Non-steroidal anti-inflammatory drugs (NSAIDs)**

Postoperatively, avoid non-steroidal anti-inflammatory drugs (NSAIDs) for pain management in renal patients.

Topic: Postoperative | Ref: Ch 22, Page 203

**Q29. Answer: B — Spinal anesthesia**

Spinal anesthesia is preferred over general anesthesia for TURP.

Topic: Anesthesia for TURP | Ref: Ch 22, Page 203

**Q30. Answer: B — Old patients have decreased pulmonary reserve and awake patient can tell central signs of TURP syndrome**

Spinal anesthesia is preferred because old patients have decreased pulmonary reserve and awake patient can tell central signs of TURP syndrome and bladder perforation.

Topic: Anesthesia for TURP | Ref: Ch 22, Page 203

**Q31. Answer: B — Glycine**

A large amount of irrigating fluid (most commonly glycine) is used to distend the bladder and wash away blood in TURP.

Topic: TURP Syndrome | Ref: Ch 22, Page 203

**Q32. Answer: B — 10-30 mL**

It is estimated that 10-30 mL of fluid is absorbed every minute during TURP.

Topic: TURP Syndrome | Ref: Ch 22, Page 203

**Q33. Answer: D — Hypernatremia**

TURP syndrome produces fluid overloading, dilutional hyponatremia, and hypo-osmolality — not hypernatremia.

Topic: TURP Syndrome | Ref: Ch 22, Page 203

**Q34. Answer: A — Hyperglycemia and hyperammonemia**

Glycine can also produce hyperglycemia and hyperammonemia.

Topic: TURP Syndrome | Ref: Ch 22, Page 203

**Q35. Answer: B — Cerebral edema causing restlessness, confusion, or seizures**

Hypo-osmolality and hyponatremia by producing cerebral edema produces restlessness, confusion or even seizures.

Topic: TURP Syndrome – Signs and Symptoms | Ref: Ch 22, Page 203

**Q36. Answer: B — Transient visual disturbances including transient blindness**

Hyperglycinemia can produce visual disturbances including transient blindness.

Topic: TURP Syndrome – Signs and Symptoms | Ref: Ch 22, Page 203

**Q37. Answer: B — Furosemide**

Furosemide is used to treat fluid overload in TURP syndrome.

Topic: TURP Syndrome – Treatment | Ref: Ch 22, Page 204

**Q38. Answer: B — Hypertonic saline**

Hypertonic saline may be required to treat symptomatic hyponatremia and hypo-osmolality.

Topic: TURP Syndrome – Treatment | Ref: Ch 22, Page 204

**Q39. Answer: C — < 5.5 mEq/L**

Hyperkalemia must be corrected (serum potassium should be < 5.5 mEq/L) before taking the patient for surgery.

Topic: Hyperkalemia | Ref: Ch 22, Page 204

**Q40. Answer: C — Potassium-rich fluids should be avoided and succinylcholine is contraindicated**

Potassium rich fluids should be avoided and succinylcholine is contraindicated in hyperkalemia.

Topic: Hyperkalemia – Anesthetic Considerations | Ref: Ch 22, Page 204

**Q41. Answer: B — It causes alkalosis which pushes potassium intracellularly**

Mild hyperventilation is beneficial because hyperventilation causes alkalosis which pushes the potassium intracellularly.

Topic: Hyperkalemia – Anesthetic Considerations | Ref: Ch 22, Page 204

**Q42. Answer: B — No need to treat if there are no ECG changes or organ dysfunction**

The general consensus is not to treat mild (serum potassium >3 mEq/L) chronic hypokalemia if there are no ECG changes or any organ dysfunction.

Topic: Hypokalemia | Ref: Ch 22, Page 204

**Q43. Answer: C — > 4 mEq/L**

The exception is patients on digoxin where serum potassium should be >4 mEq/L before taking them for surgery.

Topic: Hypokalemia | Ref: Ch 22, Page 204

**Q44. Answer: B — Decreased as prolonged block is expected**

Dose of non-depolarizing muscle relaxant is decreased as prolonged block is expected due to muscle weakness in hypokalemia.

Topic: Hypokalemia – Anesthetic Considerations | Ref: Ch 22, Page 204

**Q45. Answer: B — It can cause pontine myelinolysis**

Rapid correction must never be done; rapid correction can cause pontine myelinolysis.

Topic: Hyponatremia and Hypernatremia | Ref: Ch 22, Page 204

**Q46. Answer: B — Maintenance of hydration and urine output**

The key principle in hypercalcemia is maintenance of hydration and urine output.

Topic: Hypercalcemia – Anesthetic Considerations | Ref: Ch 22, Page 204

**Q47. Answer: B — Laryngospasm and hypotension**

Hypocalcemia can produce laryngospasm and hypotension.

Topic: Hypocalcemia – Anesthetic Considerations | Ref: Ch 22, Page 204

**Q48. Answer: C — Both depolarizing and non-depolarizing muscle relaxants**

Hypermagnesemia potentiates the action of both depolarizing and non-depolarizing muscle relaxants.

Topic: Hypermagnesemia – Anesthetic Considerations | Ref: Ch 22, Page 204

**Q49. Answer: B — Refractory hypokalemia and hypocalcemia**

Hypomagnesemia may be associated with refractory hypokalemia and hypocalcemia.

Topic: Hypomagnesemia – Anesthetic Considerations | Ref: Ch 22, Page 204

**Q50. Answer: B — It can precipitate pulmonary edema and fluid deficiency can worsen kidney injury**

Excess fluid can precipitate pulmonary edema and fluid deficiency can worsen kidney injury. Therefore, maintaining fluid balance is essential.

Topic: Intraoperative Monitoring – Fluid Balance | Ref: Ch 22, Page 202

# Anesthesia for Endocrinal Disorders MCQ – Chapter 23

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 23)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. What are the basic goals of preoperative evaluation in a diabetic patient?**

- A. Only checking blood sugar levels
- B. Assess severity by blood sugar & HbA1c, rule out autonomic and peripheral neuropathy, and organ damage
- C. Only ruling out nephropathy
- D. Only checking HbA1c levels

**Ans: B**

**Q2. Why should premedication against aspiration be given to diabetic patients?**

- A. They have increased gastric acid production
- B. They have delayed gastric emptying
- C. They are on multiple medications
- D. They have esophageal dysfunction

**Ans: B**

**Q3. How long before surgery should sulfonyl urea and metformin be stopped?**

- A. On the morning of surgery
- B. 12 hours before
- C. 24–48 hours prior to surgery
- D. 1 week before surgery

**Ans: C**

**Q4. In the traditional insulin regime, what fraction of the AM dose of insulin is given on the morning of surgery?**

- A. 1/4 of the AM dose
- B. 1/3 of the AM dose
- C. 1/2 of the AM dose
- D. Full AM dose

**Ans: C**

**Q5. At what rate should 5% dextrose be infused when giving insulin on the morning of surgery?**

- A. 50 mL/hour
- B. 100 mL/hour
- C. 125 mL/hour
- D. 150 mL/hour

**Ans: C**

**Q6. By how much does 1 unit of regular insulin decrease blood sugar?**

- A. 10–15 mg/dL
- B. 15–20 mg/dL
- C. 25–30 mg/dL
- D. 40–50 mg/dL

**Ans: C**

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| <p><b>Q7. What is the formula for regular insulin infusion rate (units/hour) as per current guidelines?</b></p> <p>A. Plasma glucose (mg/dL) / 100<br/> B. Plasma glucose (mg/dL) / 150<br/> C. Plasma glucose (mg/dL) / 200<br/> D. Plasma glucose (mg/dL) / 250</p>            | <p><b>Ans: B</b></p> |
| <p><b>Q8. What is the target plasma sugar level during the perioperative period?</b></p> <p>A. 100–130 mg/dL<br/> B. 130–150 mg/dL<br/> C. 150–180 mg/dL<br/> D. 180–220 mg/dL</p>                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q9. What is the most common cause of death in the perioperative period in diabetic patients?</b></p> <p>A. Hypoglycemia<br/> B. Diabetic ketoacidosis<br/> C. Myocardial ischemia<br/> D. Renal failure</p>                                                                | <p><b>Ans: C</b></p> |
| <p><b>Q10. Why may MI go silent in the postoperative period in diabetic patients?</b></p> <p>A. Due to analgesics masking pain<br/> B. Due to autonomic neuropathy<br/> C. Due to sedation<br/> D. Due to peripheral neuropathy</p>                                              | <p><b>Ans: B</b></p> |
| <p><b>Q11. Which patients with diabetes can safely receive regional anesthesia?</b></p> <p>A. All diabetic patients<br/> B. Patients not having peripheral and/or autonomic neuropathy<br/> C. Only type 2 diabetic patients<br/> D. Only insulin-dependent patients</p>         | <p><b>Ans: B</b></p> |
| <p><b>Q12. Why may difficult intubation be anticipated in diabetic patients?</b></p> <p>A. Due to obesity<br/> B. Due to involvement of atlanto-occipital and temporomandibular joint (stiff joint syndrome)<br/> C. Due to macroglossia<br/> D. Due to tracheal compression</p> | <p><b>Ans: B</b></p> |
| <p><b>Q13. What may act as a savior when hypotension and bradycardia in diabetic patients with autonomic neuropathy do not respond to atropine and ephedrine?</b></p> <p>A. Dopamine<br/> B. Noradrenaline<br/> C. Intravenous adrenaline<br/> D. Phenylephrine</p>              | <p><b>Ans: C</b></p> |

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| <p><b>Q14. Which of the following drugs should be avoided in diabetic patients because they cause sympathetic stimulation and hyperglycemia?</b></p> <p>A. Propofol, Vecuronium, Isoflurane<br/> B. Ketamine, Pancuronium, Atropine, Desflurane &gt;6%<br/> C. Thiopentone, Sevoflurane, Glycopyrrolate<br/> D. Etomidate, Atracurium, Fentanyl</p> | <p><b>Ans: B</b></p> |
| <p><b>Q15. When should elective surgery be performed in a hyperthyroid patient?</b></p> <p>A. Immediately with beta-blocker cover<br/> B. After the patient is euthyroid<br/> C. After 2 weeks of antithyroid drugs<br/> D. Only after total thyroidectomy</p>                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q16. What should be brought down to normal with beta blockers before taking a hyperthyroid patient for surgery?</b></p> <p>A. Blood pressure<br/> B. Sleeping pulse rate<br/> C. Thyroid hormone levels<br/> D. Temperature</p>                                                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q17. Why should airway assessment be done in hyperthyroid patients with large thyroid?</b></p> <p>A. Risk of vocal cord paralysis<br/> B. Large thyroid can produce tracheal compression<br/> C. Risk of aspiration<br/> D. Difficulty in mask ventilation</p>                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q18. Why should local anesthetics with adrenaline NOT be used in hyperthyroid patients?</b></p> <p>A. They are less effective<br/> B. They can cause sympathetic stimulation<br/> C. They are metabolized faster<br/> D. They cause hypothermia</p>                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q19. Why are central neuraxial blocks preferred in hyperthyroid patients?</b></p> <p>A. They provide better analgesia<br/> B. They block the sympathetic response<br/> C. They are easier to administer<br/> D. They reduce thyroid hormone levels</p>                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q20. Which is the induction agent of choice in hyperthyroid patients and why?</b></p> <p>A. Propofol – fast onset<br/> B. Thiopentone – possesses antithyroid property<br/> C. Etomidate – cardiovascular stability<br/> D. Ketamine – bronchodilator</p>                                                                                     | <p><b>Ans: B</b></p> |



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| <p><b>Q21. What is the preferred maintenance agent for hyperthyroid patients prone to arrhythmias?</b></p> <p>A. Halothane<br/>B. Sevoflurane<br/>C. Isoflurane<br/>D. Desflurane</p>                                                                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q22. Does the MAC of inhalational agents change in hyperthyroidism?</b></p> <p>A. Yes, it increases significantly<br/>B. Yes, it decreases significantly<br/>C. No, MAC remains the same<br/>D. It varies depending on the agent</p>                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q23. What is a thyroid storm and when does it most often occur?</b></p> <p>A. A benign condition occurring preoperatively<br/>B. A life-threatening condition most often occurring in postoperative period<br/>C. An allergic reaction during induction<br/>D. A transient condition during maintenance</p> | <p><b>Ans: B</b></p> |
| <p><b>Q24. Which drug is used in thyroid storm to block conversion of T4 to T3?</b></p> <p>A. Hydrocortisone<br/>B. Dexamethasone<br/>C. Prednisolone<br/>D. Methylprednisolone</p>                                                                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q25. What is the most common cause of upper airway obstruction after thyroid surgery?</b></p> <p>A. Laryngeal edema<br/>B. Tracheal compression by expanding hematoma<br/>C. Vocal cord paralysis<br/>D. Tracheomalacia</p>                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q26. What can hypoparathyroidism after thyroidectomy cause?</b></p> <p>A. Hyperkalemia<br/>B. Hypocalcemia leading to laryngospasm<br/>C. Hypernatremia<br/>D. Metabolic alkalosis</p>                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q27. What is the emergency treatment of choice for a hyperthyroid patient undergoing emergency surgery?</b></p> <p>A. IV hydrocortisone only<br/>B. IV esmolol or propranolol to settle tachycardia<br/>C. Oral antithyroid drugs only<br/>D. Plasmapheresis</p>                                            | <p><b>Ans: B</b></p> |

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| <p><b>Q28. Should elective surgery be deferred in all hypothyroid patients?</b></p> <ul style="list-style-type: none"> <li>A. Yes, in all cases</li> <li>B. No, mild hypothyroidism should not be a deterrent to postpone the surgery</li> <li>C. Only in patients above 60 years</li> <li>D. Only in patients with goiter</li> </ul>                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q29. Which of the following are associated conditions with hypothyroidism that should be ruled out preoperatively?</b></p> <ul style="list-style-type: none"> <li>A. Hypertension, tachycardia, hyperthermia</li> <li>B. Congestive cardiac failure, anemia, hypothermia, hyponatremia, hypoglycemia</li> <li>C. Hyperglycemia, hyperkalemia, polycythemia</li> <li>D. Pulmonary hypertension, renal failure</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q30. Why should premedication with benzodiazepines be avoided in hypothyroid patients?</b></p> <ul style="list-style-type: none"> <li>A. They are metabolized faster</li> <li>B. They cause hypothermia</li> <li>C. These patients are extremely sensitive to depressant drugs</li> <li>D. They cause hyponatremia</li> </ul>                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q31. What is the induction agent of choice in hypothyroid patients with normal cardiac output?</b></p> <ul style="list-style-type: none"> <li>A. Thiopentone</li> <li>B. Propofol</li> <li>C. Etomidate</li> <li>D. Ketamine</li> </ul>                                                                                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q32. Which muscle relaxants are preferred in hypothyroid patients?</b></p> <ul style="list-style-type: none"> <li>A. Pancuronium and Vecuronium</li> <li>B. Mivacurium, Atracurium/Cis-atracurium</li> <li>C. Succinylcholine</li> <li>D. Rocuronium</li> </ul>                                                                                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q33. Why should all drugs be used in minimal possible doses in hypothyroid patients?</b></p> <ul style="list-style-type: none"> <li>A. They have increased renal clearance</li> <li>B. They have increased hepatic metabolism</li> <li>C. The metabolism of drugs is reduced in hypothyroidism</li> <li>D. They have increased protein binding</li> </ul>                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q34. What additional cover is needed for emergency surgery in severely hypothyroid patients?</b></p> <ul style="list-style-type: none"> <li>A. IV antibiotics only</li> <li>B. IV T3 and T4, and steroid cover</li> <li>C. IV calcium only</li> <li>D. IV magnesium only</li> </ul>                                                                                                                                     | <p><b>Ans: B</b></p> |

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| <p><b>Q35. What drug is used to stabilize hypertension preoperatively in pheochromocytoma?</b></p> <p>A. Propranolol<br/>B. Atenolol<br/>C. Phenoxybenzamine (alpha blocker)<br/>D. Nifedipine</p>                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q36. Why must beta blockers NOT be started before alpha blockers in pheochromocytoma?</b></p> <p>A. Beta blockers are ineffective alone<br/>B. Unopposed alpha action can cause malignant hypertension<br/>C. Beta blockers cause hyperglycemia<br/>D. Alpha blockers take longer to act</p> | <p><b>Ans: B</b></p> |
| <p><b>Q37. Why is halothane absolutely contraindicated in pheochromocytoma?</b></p> <p>A. It causes bronchospasm<br/>B. It sensitizes myocardium to catecholamines<br/>C. It causes hepatotoxicity<br/>D. It causes malignant hyperthermia</p>                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q38. What can happen during tumor handling in pheochromocytoma surgery?</b></p> <p>A. Hypothermia<br/>B. Profound hypertension and life-threatening arrhythmias due to increased catecholamines<br/>C. Hypoglycemia only<br/>D. Bradycardia only</p>                                         | <p><b>Ans: B</b></p> |
| <p><b>Q39. What is the most common cause of death in the immediate postoperative period in pheochromocytoma patients?</b></p> <p>A. Arrhythmia<br/>B. Hypertensive crisis<br/>C. Hypotension<br/>D. Hypoglycemia</p>                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q40. What should be started immediately after tumor vein ligation in pheochromocytoma surgery?</b></p> <p>A. Normal saline<br/>B. Ringer lactate<br/>C. Dextrose normal saline<br/>D. Colloids</p>                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q41. For how long should hydrocortisone be continued after unilateral adrenalectomy for pheochromocytoma?</b></p> <p>A. 12 hours<br/>B. 1–2 days<br/>C. 3–6 days<br/>D. 1 week</p>                                                                                                           | <p><b>Ans: B</b></p> |

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| <p><b>Q42. What effects of cortisol must be corrected before elective surgery in Cushing syndrome?</b></p> <p>A. Only hypertension<br/>B. Hypertension, hyperglycemia, volume overload, and hypokalemia<br/>C. Only hyperglycemia<br/>D. Only hypokalemia</p>      | <p><b>Ans: B</b></p> |
| <p><b>Q43. What is the most common cause of secondary adrenal insufficiency?</b></p> <p>A. Pituitary tumor<br/>B. Autoimmune adrenalitis<br/>C. Exogenous steroids<br/>D. Tuberculosis</p>                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q44. Any patient who has taken steroids (prednisone &gt;5 mg/day) for more than how many weeks in the last 1 year must receive intraoperative steroid supplementation?</b></p> <p>A. 1 week<br/>B. 2 weeks<br/>C. 3 weeks<br/>D. 4 weeks</p>                 | <p><b>Ans: C</b></p> |
| <p><b>Q45. What is the dose of hydrocortisone supplementation for patients with adrenal insufficiency?</b></p> <p>A. 50 mg 8 hourly<br/>B. 100 mg 8 hourly<br/>C. 200 mg 12 hourly<br/>D. 50 mg 12 hourly</p>                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q46. Why is etomidate contraindicated in adrenal insufficiency?</b></p> <p>A. It causes hypotension<br/>B. It causes adrenocortical suppression<br/>C. It causes tachycardia<br/>D. It is poorly metabolized</p>                                             | <p><b>Ans: B</b></p> |
| <p><b>Q47. Why should sevoflurane be avoided in mineralocorticoid excess with hypokalemic nephropathy?</b></p> <p>A. It causes bronchospasm<br/>B. It produces fluoride that can worsen nephropathy<br/>C. It causes hyperkalemia<br/>D. It causes tachycardia</p> | <p><b>Ans: B</b></p> |
| <p><b>Q48. What is the most important concern for acromegaly patients undergoing anesthesia?</b></p> <p>A. Hyperglycemia<br/>B. Difficult airway management<br/>C. Cardiac arrhythmias<br/>D. Renal dysfunction</p>                                                | <p><b>Ans: B</b></p> |

**Q49.** How long does recovery of hypothalamic-pituitary axis (HPA) take after exogenous steroid use?

- A. 3 months
- B. 6 months
- C. 1 year
- D. 2 years

**Ans: C**

**Q50.** What can happen in hypothyroid patients in the postoperative period due to hypoventilation?

- A. Hyperglycemia
- B. Hypoxia
- C. Hypertension
- D. Tachycardia

**Ans: B**

# Answer Key & Explanations

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## **Q1. Answer: B — Assess severity by blood sugar & HbA1c, rule out autonomic and peripheral neuropathy, and organ damage**

The basic goals are to assess severity by checking blood sugar and HbA1c, rule out autonomic neuropathy, peripheral neuropathy, and organ damage especially nephropathy.

Topic: Diabetes Mellitus – Preoperative Evaluation | Ref: Ch 23, Page 206

## **Q2. Answer: B — They have delayed gastric emptying**

Diabetic patients have delayed gastric emptying, therefore premedication against aspiration should be given.

Topic: Diabetes Mellitus – Premedication | Ref: Ch 23, Page 206

## **Q3. Answer: C — 24–48 hours prior to surgery**

Due to long half-lives, sulfonyl urea and metformin should be stopped 24–48 hours prior to surgery.

Topic: Diabetes Mellitus – Management of Sugar | Ref: Ch 23, Page 206

## **Q4. Answer: C — 1/2 of the AM dose**

The most widespread traditionally employed regime is to give 1/2 of the AM dose of insulin on the morning of surgery after starting 5% dextrose infusion.

Topic: Diabetes Mellitus – Insulin Management | Ref: Ch 23, Page 206

## **Q5. Answer: C — 125 mL/hour**

5% dextrose is infused at a rate of 125 mL/hour to avoid the possibility of hypoglycemia.

Topic: Diabetes Mellitus – Insulin Management | Ref: Ch 23, Page 206

## **Q6. Answer: C — 25–30 mg/dL**

1 unit of insulin decreases blood sugar levels by 25–30 mg/dL.

Topic: Diabetes Mellitus – Insulin Management | Ref: Ch 23, Page 206

## **Q7. Answer: B — Plasma glucose (mg/dL) / 150**

Insulin is given by the formula: Units/hour = plasma glucose (mg/dL)/150.

Topic: Diabetes Mellitus – Insulin Infusion | Ref: Ch 23, Page 206

## **Q8. Answer: C — 150–180 mg/dL**

The target should be to keep plasma sugar level between 150 and 180 mg/dL.

Topic: Diabetes Mellitus – Glucose Monitoring | Ref: Ch 23, Page 206

## **Q9. Answer: C — Myocardial ischemia**

Myocardial ischemia is the most common cause of death in the perioperative period in diabetic patients.

Topic: Diabetes Mellitus – Monitoring | Ref: Ch 23, Page 206

## **Q10. Answer: B — Due to autonomic neuropathy**

The MI may go silent in postoperative period if the patient has autonomic neuropathy.

Topic: Diabetes Mellitus – Monitoring | Ref: Ch 23, Page 206

**Q11. Answer: B — Patients not having peripheral and/or autonomic neuropathy**

Patients not having peripheral and/or autonomic neuropathy can safely receive regional anesthesia.

Topic: Diabetes Mellitus – Choice of Anesthesia | Ref: Ch 23, Page 207

**Q12. Answer: B — Due to involvement of atlanto-occipital and temporomandibular joint (stiff joint syndrome)**

Difficult intubation may be anticipated due to involvement of atlanto-occipital and temporomandibular joint (stiff joint syndrome).

Topic: Diabetes Mellitus – General Anesthesia | Ref: Ch 23, Page 207

**Q13. Answer: C — Intravenous adrenaline**

Hypotension and bradycardia in patients with autonomic neuropathy may not respond to atropine and ephedrine; intravenous adrenaline may act as savior.

Topic: Diabetes Mellitus – General Anesthesia | Ref: Ch 23, Page 207

**Q14. Answer: B — Ketamine, Pancuronium, Atropine, Desflurane >6%**

Drugs causing sympathetic stimulation like ketamine, pancuronium, atropine, desflurane >6% should be avoided as they cause hyperglycemia.

Topic: Diabetes Mellitus – General Anesthesia | Ref: Ch 23, Page 207

**Q15. Answer: B — After the patient is euthyroid**

Elective surgery should be deferred till the patient is euthyroid.

Topic: Hyperthyroidism | Ref: Ch 23, Page 207

**Q16. Answer: B — Sleeping pulse rate**

Sleeping pulse rate should be brought down to normal with beta blockers before taking the patient for surgery.

Topic: Hyperthyroidism – Preoperative Evaluation | Ref: Ch 23, Page 207

**Q17. Answer: B — Large thyroid can produce tracheal compression**

As large thyroid can produce tracheal compression, airway assessment by indirect laryngoscopy, X-ray neck or CT scan is must.

Topic: Hyperthyroidism – Preoperative Evaluation | Ref: Ch 23, Page 207

**Q18. Answer: B — They can cause sympathetic stimulation**

Regional anesthesia can be safely given in hyperthyroid patients however local anesthetics with adrenaline must not be used as they cause sympathetic stimulation.

Topic: Hyperthyroidism – Choice of Anesthesia | Ref: Ch 23, Page 207

**Q19. Answer: B — They block the sympathetic response**

Central neuraxial blocks are preferred because of their capacity to block the sympathetic response.

Topic: Hyperthyroidism – Choice of Anesthesia | Ref: Ch 23, Page 207

**Q20. Answer: B — Thiopentone – possesses antithyroid property**

As thiopentone possesses antithyroid property, it is the induction agent of choice for hyperthyroid patients.

Topic: Hyperthyroidism – General Anesthesia | Ref: Ch 23, Page 207

**Q21. Answer: C — Isoflurane**

As hyperthyroid patients are more prone for arrhythmias, most cardiac stable i.e. Isoflurane should be the preferred agent.

Topic: Hyperthyroidism – General Anesthesia | Ref: Ch 23, Page 207

**Q22. Answer: C — No, MAC remains the same**

Minimum alveolar concentration (MAC) of inhalational agents remains same in hyperthyroidism.

Topic: Hyperthyroidism – General Anesthesia | Ref: Ch 23, Page 207

**Q23. Answer: B — A life-threatening condition most often occurring in postoperative period**

Thyroid storm is a life-threatening condition most often occurring in postoperative period, however can occur intraoperatively also.

Topic: Hyperthyroidism – Postoperative Complications | Ref: Ch 23, Page 207

**Q24. Answer: B — Dexamethasone**

Management of thyroid storm includes dexamethasone which blocks conversion of T4 to T3.

Topic: Hyperthyroidism – Thyroid Storm | Ref: Ch 23, Page 207

**Q25. Answer: B — Tracheal compression by expanding hematoma**

Upper airway obstruction is most often due to tracheal compression caused by expanding hematoma, treated immediately by opening the sutures.

Topic: Hyperthyroidism – Postoperative Complications | Ref: Ch 23, Page 207-208

**Q26. Answer: B — Hypocalcemia leading to laryngospasm**

Hypocalcemia caused by hypoparathyroidism can precipitate laryngospasm.

Topic: Hyperthyroidism – Postoperative Complications | Ref: Ch 23, Page 208

**Q27. Answer: B — IV esmolol or propranolol to settle tachycardia**

At least tachycardia should be settled with IV esmolol or IV propranolol. Propranolol has the added advantage of inhibiting conversion of T4 to T3.

Topic: Emergency Surgery in Hyperthyroid Patient | Ref: Ch 23, Page 208

**Q28. Answer: B — No, mild hypothyroidism should not be a deterrent to postpone the surgery**

Ideally elective surgery should be deferred till euthyroid, however mild hypothyroidism should not be a deterrent to postpone the surgery.

Topic: Hypothyroidism | Ref: Ch 23, Page 208

**Q29. Answer: B — Congestive cardiac failure, anemia, hypothermia, hyponatremia, hypoglycemia**

Associated conditions like CCF, anemia, hypothermia, hyponatremia, and hypoglycemia should be ruled out.

Topic: Hypothyroidism – Preoperative Evaluation | Ref: Ch 23, Page 208

**Q30. Answer: C — These patients are extremely sensitive to depressant drugs**

Hypothyroid patients are extremely sensitive to depressant drugs therefore premedication with benzodiazepines should be avoided.

Topic: Hypothyroidism – Preoperative Evaluation | Ref: Ch 23, Page 208



**Q31. Answer: B — Propofol**

Induction is normally done with propofol; however, if there is evidence of low cardiac output then ketamine is preferred.

Topic: Hypothyroidism – General Anesthesia | Ref: Ch 23, Page 208

**Q32. Answer: B — Mivacurium, Atracurium/Cis-atracurium**

As mivacurium, atracurium/Cis-atracurium do not have the possibility of accumulation, they are preferred muscle relaxants in hypothyroidism.

Topic: Hypothyroidism – General Anesthesia | Ref: Ch 23, Page 208

**Q33. Answer: C — The metabolism of drugs is reduced in hypothyroidism**

As the metabolism of drugs is reduced in hypothyroidism, they should be used in minimal possible doses.

Topic: Hypothyroidism – General Anesthesia | Ref: Ch 23, Page 208

**Q34. Answer: B — IV T3 and T4, and steroid cover**

IV T3 and T4 should be given, and as adrenal function is often decreased, steroid cover also becomes a must.

Topic: Emergency Surgery in Hypothyroid Patient | Ref: Ch 23, Page 208

**Q35. Answer: C — Phenoxybenzamine (alpha blocker)**

Patient's hypertension must be stabilized by alpha blocker; most often used is phenoxybenzamine.

Topic: Pheochromocytoma | Ref: Ch 23, Page 208

**Q36. Answer: B — Unopposed alpha action can cause malignant hypertension**

Beta blockers must not be started before initiating therapy with alpha blocker otherwise unopposed action can cause malignant hypertension.

Topic: Pheochromocytoma | Ref: Ch 23, Page 208

**Q37. Answer: B — It sensitizes myocardium to catecholamines**

Halothane is absolutely contraindicated because it sensitizes myocardium to catecholamines.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

**Q38. Answer: B — Profound hypertension and life-threatening arrhythmias due to increased catecholamines**

Increase in catecholamines during tumor handling can cause profound hypertension and life-threatening arrhythmias.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

**Q39. Answer: C — Hypotension**

Hypotension is the most common cause of death in immediate postoperative period in pheochromocytoma patients, caused by abrupt withdrawal of catecholamines after ligation of tumor vein.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

**Q40. Answer: C — Dextrose normal saline**

Dextrose normal saline should be started immediately after vein ligation to prevent hypotension and hypoglycemia.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

**Q41. Answer: B — 1–2 days**

Hydrocortisone (100 mg/day) is continued for 1–2 days for unilateral and 3–6 days for bilateral adrenalectomy.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

**Q42. Answer: B — Hypertension, hyperglycemia, volume overload, and hypokalemia**

Hypertension, hyperglycemia, volume overload and hypokalemia must be corrected before elective surgery.

Topic: Cushing Syndrome | Ref: Ch 23, Page 209

**Q43. Answer: C — Exogenous steroids**

Exogenous steroids are the most common cause of secondary adrenal insufficiency.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

**Q44. Answer: C — 3 weeks**

Any patient who has taken steroid for more than 3 weeks in the last 1 year must receive intraoperative supplementation with hydrocortisone.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

**Q45. Answer: B — 100 mg 8 hourly**

100 mg of hydrocortisone 8 hourly beginning on the day of surgery and continued for 24–48 hours; continuous infusion at 10 mg/hour is considered better.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

**Q46. Answer: B — It causes adrenocortical suppression**

Etomidate is contraindicated as it causes adrenocortical suppression.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

**Q47. Answer: B — It produces fluoride that can worsen nephropathy**

Sevoflurane should be avoided if there is hypokalemic nephropathy due to its fluoride production.

Topic: Mineralocorticoid Excess | Ref: Ch 23, Page 209

**Q48. Answer: B — Difficult airway management**

The most important concern for acromegaly patients is difficult airway management due to prognathism, macroglossia, overgrowth of epiglottis, and narrowed glottis.

Topic: Pituitary Dysfunction – Acromegaly | Ref: Ch 23, Page 209

**Q49. Answer: C — 1 year**

Recovery of HPA takes 1 year after exogenous steroid use.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

**Q50. Answer: B — Hypoxia**

Hypothyroid patients can go into hypoxia in postoperative period due to hypoventilation.

Topic: Hypothyroidism – Postoperative | Ref: Ch 23, Page 208

# Anesthesia for Neuromuscular Diseases MCQ – Chapter 24

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 24)

**25 Questions • 20 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Myasthenia gravis is an autoimmune disease characterized by:**

- A. Destruction of acetylcholine receptors by antibodies against these receptors, leading to weakness of muscles
- B. Antibodies against presynaptic calcium channels
- C. Abnormal ryanodine receptors
- D. Excessive acetylcholine release

**Ans: A**

**Q2. In a myasthenic patient, anticholinesterases (mainly pyridostigmine) should be:**

- A. Continued in reduced doses
- B. Stopped completely before surgery
- C. Doubled before surgery
- D. Replaced with steroids

**Ans: A**

**Q3. Why should premedication with benzodiazepines be avoided in myasthenic patients?**

- A. Because weakness of respiratory muscles makes these patients very vulnerable for respiratory depression
- B. Because they cause hypertension
- C. Because they trigger malignant hyperthermia
- D. Because they cause excessive salivation

**Ans: A**

**Q4. What is the anesthetic technique of choice for myasthenia gravis patients?**

- A. General anesthesia with muscle relaxants
- B. Local/regional anesthesia
- C. Total intravenous anesthesia
- D. Inhalational anesthesia alone

**Ans: B**

**Q5. Which induction agent is preferred in myasthenia gravis as it does not cause respiratory depression?**

- A. Etomidate
- B. Thiopentone
- C. Ketamine
- D. Propofol

**Ans: A**

**Q6. Which inhalational agent is the most preferred for maintenance in myasthenia gravis patients?**

- A. Halothane
- B. Isoflurane
- C. Desflurane
- D. Nitrous oxide

**Ans: C**

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| <p><b>Q7. Myasthenic patients exhibit what response to muscle relaxants?</b></p> <p>A. Resistant to depolarizers and hypersensitive to non-depolarizers<br/> B. Hypersensitive to depolarizers and resistant to non-depolarizers<br/> C. Hypersensitive to both<br/> D. Resistant to both</p>                                       | <p><b>Ans: A</b></p> |
| <p><b>Q8. Myasthenic syndrome (Eaton-Lambert syndrome) is an autoimmune disease in which antibodies are produced against:</b></p> <p>A. Presynaptic calcium channel, leading to decreased production of acetylcholine<br/> B. Acetylcholine receptors<br/> C. Ryanodine receptors<br/> D. Sodium channels</p>                       | <p><b>Ans: A</b></p> |
| <p><b>Q9. Reversal in myasthenic syndrome (Eaton-Lambert) requires a combination of:</b></p> <p>A. Anticholinesterase and 4-aminopyridine<br/> B. Neostigmine and glycopyrrolate only<br/> C. Sugammadex and atropine<br/> D. Edrophonium alone</p>                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q10. In hyperkalemic familial periodic paralysis, which drug is contraindicated?</b></p> <p>A. Succinylcholine<br/> B. Atracurium<br/> C. Propofol<br/> D. Etomidate</p>                                                                                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q11. In hypokalemic familial periodic paralysis (the more common type), what should be avoided?</b></p> <p>A. Hyperventilation and glucose containing solutions<br/> B. Potassium containing solutions<br/> C. Local anesthetics<br/> D. All inhalational agents</p>                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q12. Patients with muscular dystrophies are considered as high risk patients to develop:</b></p> <p>A. Malignant hyperthermia<br/> B. Anaphylaxis<br/> C. Pulmonary embolism<br/> D. Myasthenic crisis</p>                                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q13. Duchenne's dystrophy (pseudohypertrophic muscular dystrophy) is the most common type of dystrophy of childhood. Death usually occurs at 15–25 years of age due to:</b></p> <p>A. Congestive heart failure (CHF) or pneumonia<br/> B. Renal failure<br/> C. Malignant hyperthermia<br/> D. Respiratory paralysis only</p> | <p><b>Ans: A</b></p> |

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| <p><b>Q14. Which muscle relaxant is absolutely contraindicated in muscular dystrophies?</b></p> <p>A. Succinylcholine<br/>B. Mivacurium<br/>C. Atracurium<br/>D. Cis-atracurium</p>                                                                                                                                                                                                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q15. Which is the most preferred inhalational agent in Duchenne's dystrophy (as there is involvement of cardiac muscle)?</b></p> <p>A. Halothane<br/>B. Isoflurane<br/>C. Desflurane<br/>D. Sevoflurane</p>                                                                                                                                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q16. Which non-depolarizing muscle relaxants are safe and preferred in muscular dystrophies (due to increased sensitivity to relaxants)?</b></p> <p>A. Mivacurium (most preferred), atracurium and cis-atracurium<br/>B. Pancuronium and vecuronium<br/>C. Rocuronium and gallamine<br/>D. Doxacurium and tubocurare</p>                                                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q17. Which is the most common and most serious dystrophy in adults, characterized by persistent contracture after a stimulus?</b></p> <p>A. Duchenne's dystrophy<br/>B. Myotonia dystrophica<br/>C. Becker's dystrophy<br/>D. Limb-girdle dystrophy</p>                                                                                                                                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q18. In myotonia dystrophica, why must shivering be avoided at any cost?</b></p> <p>A. Because shivering can precipitate muscle contractures<br/>B. Because shivering causes hypoxia<br/>C. Because shivering triggers anaphylaxis<br/>D. Because shivering increases potassium</p>                                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q19. Why are mivacurium, atracurium and cis-atracurium preferred in myotonia dystrophica (apart from short duration)?</b></p> <p>A. Because they permit avoidance of reversal agents, which can precipitate contractures by causing muscle contraction<br/>B. Because they are long acting<br/>C. Because they cause histamine release<br/>D. Because they are eliminated by kidneys</p>                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q20. In myasthenia gravis, the anesthetic management aims to avoid muscle relaxants because:</b></p> <p>A. Myasthenia patients have weakness of muscles and inhalational agents have muscle relaxation property, allowing intubation and maintenance with inhalational agents alone<br/>B. Muscle relaxants cause malignant hyperthermia<br/>C. Muscle relaxants are contraindicated in all patients<br/>D. Muscle relaxants increase potassium dangerously</p> | <p><b>Ans: A</b></p> |

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| <p><b>Q21. Which opioids should be used in myasthenia gravis to manage respiratory depression?</b></p> <ul style="list-style-type: none"> <li>A. Only short acting opioids like remifentanyl</li> <li>B. Long acting morphine</li> <li>C. Pethidine</li> <li>D. No opioids at all</li> </ul>                                                                                                                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q22. Which myasthenic patients may require ventilation in the postoperative period?</b></p> <ul style="list-style-type: none"> <li>A. Those with significantly compromised pulmonary functions (vital capacity &lt; 4 mL/kg) or on high doses of pyridostigmine (&gt; 750 mg/day)</li> <li>B. All myasthenic patients</li> <li>C. Only those who received regional anesthesia</li> <li>D. None of them</li> </ul>                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q23. Due to weakness of laryngeal and pharyngeal reflexes in muscular dystrophy patients (prone to aspiration), they should be premedicated with:</b></p> <ul style="list-style-type: none"> <li>A. Metoclopramide and ranitidine</li> <li>B. Benzodiazepines</li> <li>C. Opioids</li> <li>D. Anticholinesterases</li> </ul>                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q24. In myasthenia gravis, why is a prolonged block expected with succinylcholine and mivacurium in patients on cholinesterase inhibitors?</b></p> <ul style="list-style-type: none"> <li>A. Because cholinesterase inhibitors inhibit pseudocholinesterase (though this prolongation is not very significant clinically)</li> <li>B. Because succinylcholine accumulates in muscle</li> <li>C. Because of increased receptor sensitivity</li> <li>D. Because of hyperkalemia</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q25. Patients with myasthenia gravis on advanced disease may be on which medications that should be continued?</b></p> <ul style="list-style-type: none"> <li>A. Steroids and immunosuppressants</li> <li>B. Beta-blockers</li> <li>C. Calcium channel blockers</li> <li>D. Anticoagulants</li> </ul>                                                                                                                                                                                    | <p><b>Ans: A</b></p> |

# Answer Key & Explanations

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## **Q1. Answer: A — Destruction of acetylcholine receptors by antibodies against these receptors, leading to weakness of muscles**

Page 211: Myasthenia gravis is an autoimmune disease characterized by destruction of acetylcholine receptors by the antibodies against these receptors leading to weakness of the muscles. Myasthenic patients most often come for thymectomy as a part of their treatment.

Topic: Myasthenia Gravis

## **Q2. Answer: A — Continued in reduced doses**

Page 211: Anticholinesterases (most commonly used is pyridostigmine) are the mainstay of treatment. Stopping these drugs can precipitate the myasthenic crisis while continuing can antagonize the effect of muscle relaxants, therefore anticholinesterases should be continued in reduced doses.

Topic: Myasthenia Gravis

## **Q3. Answer: A — Because weakness of respiratory muscles makes these patients very vulnerable for respiratory depression**

Page 211: Weakness of respiratory muscles makes these patients very vulnerable for respiratory depression, therefore premedication with benzodiazepines should be avoided. As these patients are vulnerable for aspiration, prophylaxis for aspiration should be taken.

Topic: Myasthenia Gravis

## **Q4. Answer: B — Local/regional anesthesia**

Page 211: The basic problem of myasthenic patients is that they exhibit extreme sensitivity to respiratory depression by anesthetic agents and abnormal response to muscle relaxants; therefore, the aim of anesthesia should be to avoid general anesthesia, making local/regional anesthesia the anesthetic technique of choice. However, all surgeries are not possible under regional anesthesia.

Topic: Myasthenia Gravis

## **Q5. Answer: A — Etomidate**

Page 211: As etomidate does not cause respiratory depression, it is the preferred induction agent in myasthenia gravis; however, thiopentone and propofol can be used safely. Neuromuscular monitoring is mandatory.

Topic: Myasthenia Gravis

## **Q6. Answer: C — Desflurane**

Page 212: Desflurane is the most preferred inhalational agent for maintenance because it produces maximum muscle relaxation among the currently used inhalational agents. In the absence of desflurane, sevoflurane or isoflurane can be used. Myasthenia patients have weakness of muscles and inhalational agents have muscle relaxation property, so in significant number of patients it is possible to intubate and maintain surgical relaxation with inhalational agents alone.

Topic: Myasthenia Gravis

## **Q7. Answer: A — Resistant to depolarizers and hypersensitive to non-depolarizers**

Page 212: Myasthenic patients are resistant to depolarizers therefore suxamethonium is required in high doses (2 mg/kg instead of 1–1.5 mg/kg), and hypersensitive to non-depolarizers, therefore only short acting agents with no risk of accumulation like mivacurium, atracurium and cis-atracurium must be used. Long acting like pancuronium must not be used.

Topic: Myasthenia Gravis



**Q8. Answer: A — Presynaptic calcium channel, leading to decreased production of acetylcholine**

Page 212: Myasthenic syndrome (Eaton-Lambert syndrome) is an autoimmune disease in which antibodies to presynaptic calcium channel are produced leading to decreased production of acetylcholine. These patients are sensitive to both depolarizing and non-depolarizing muscle relaxants.

Topic: Myasthenic Syndrome

**Q9. Answer: A — Anticholinesterase and 4-aminopyridine**

Page 212: In myasthenic syndrome (Eaton-Lambert), reversal requires a combination of anticholinesterase and 4-aminopyridine.

Topic: Myasthenic Syndrome

**Q10. Answer: A — Succinylcholine**

Page 212: In hyperkalemic familial periodic paralysis, potassium levels should be normalized preoperatively, potassium containing solutions should be avoided, and succinylcholine is contraindicated.

Topic: Familial Periodic Paralysis

**Q11. Answer: A — Hyperventilation and glucose containing solutions**

Page 212: Hypokalemic type is more common. Avoid anything which causes hypokalemia like hyperventilation and glucose containing solutions. Non-depolarizers will have prolonged effect.

Topic: Familial Periodic Paralysis

**Q12. Answer: A — Malignant hyperthermia**

Page 212: Muscular dystrophies patients are considered as high risk patients to develop malignant hyperthermia.

Topic: Muscular Dystrophies

**Q13. Answer: A — Congestive heart failure (CHF) or pneumonia**

Page 212: Duchenne's dystrophy (pseudohypertrophic muscular dystrophy) is the most common type of dystrophy of childhood. The involved muscles become hypertrophied due to fatty infiltration (pseudo-hypertrophy). It involves skeletal as well as cardiac muscles. Death usually occurs by 15–25 years of age due to congestive heart failure (CHF) or pneumonia.

Topic: Muscular Dystrophies

**Q14. Answer: A — Succinylcholine**

Page 212: Succinylcholine is absolutely contraindicated in muscular dystrophies. It may produce cardiac arrest due to hyperkalemia and rhabdomyolysis and can precipitate malignant hyperthermia.

Topic: Muscular Dystrophies

**Q15. Answer: B — Isoflurane**

Page 212: As there is involvement of cardiac muscle, the most cardiac safe agent, i.e. isoflurane should be the most preferred inhalational agent. Halothane is contraindicated as it can precipitate malignant hyperthermia.

Topic: Muscular Dystrophies

**Q16. Answer: A — Mivacurium (most preferred), atracurium and cis-atracurium**

Page 212: Due to increased sensitivity to muscle relaxants, short acting agents with no possibility of accumulation like mivacurium (most preferred), atracurium and cis-atracurium should be given with initial dose as 1/3 to 1/2 and further doses titrated as per neuromuscular monitoring.

Topic: Muscular Dystrophies

**Q17. Answer: B — Myotonia dystrophica**

Page 213: Myotonia dystrophica is the most common and most serious dystrophy in adults characterized by persistent contracture after a stimulus. Contracture may not be relieved by general and regional anesthesia, while local infiltration in the affected muscle may induce relaxation.

Topic: Muscular Dystrophies

**Q18. Answer: A — Because shivering can precipitate muscle contractures**

Page 213: As shivering can also precipitate muscle contractures in myotonia dystrophica, it must be avoided at any cost.

Topic: Muscular Dystrophies

**Q19. Answer: A — Because they permit avoidance of reversal agents, which can precipitate contractures by causing muscle contraction**

Page 213: Mivacurium, atracurium and cis-atracurium are not only preferred because of their short duration and no risk of accumulation, but also because they permit avoidance of reversal agents. Reversal agents can precipitate contractures by causing muscle contraction.

Topic: Muscular Dystrophies

**Q20. Answer: A — Myasthenia patients have weakness of muscles and inhalational agents have muscle relaxation property, allowing intubation and maintenance with inhalational agents alone**

Page 211: The aim should be to avoid muscle relaxants. Since myasthenia patients have weakness of muscles and inhalational agents have muscle relaxation property, in a significant number of patients it is possible to intubate and maintain the surgical relaxation with inhalational agents alone.

Topic: Myasthenia Gravis

**Q21. Answer: A — Only short acting opioids like remifentanyl**

Page 212: As these patients are very prone for respiratory depression, only short acting opioids (remifentanyl) should be used.

Topic: Myasthenia Gravis

**Q22. Answer: A — Those with significantly compromised pulmonary functions (vital capacity < 4 mL/kg) or on high doses of pyridostigmine (> 750 mg/day)**

Page 212: Many myasthenic patients, especially those who have significantly compromised pulmonary functions (vital capacity < 4 mL/kg) or on high doses of pyridostigmine (> 750 mg/day), may require ventilation in postoperative period.

Topic: Myasthenia Gravis

**Q23. Answer: A — Metoclopramide and ranitidine**

Page 212: Due to weakness of laryngeal and pharyngeal reflexes, muscular dystrophy patients are prone for aspiration, therefore they should be premedicated with metoclopramide and ranitidine. Due to respiratory muscle weakness, opioids and benzodiazepines should be avoided.

Topic: Muscular Dystrophies

**Q24. Answer: A — Because cholinesterase inhibitors inhibit pseudocholinesterase (though this prolongation is not very significant clinically)**

Page 212: A prolonged block by suxamethonium and mivacurium is expected in the patient on cholinesterase inhibitors because they inhibit pseudocholinesterase; however, this prolongation is not very significant clinically.

Topic: Myasthenia Gravis

**Q25. Answer: A — Steroids and immunosuppressants**

Page 211: Patients with advanced disease may be on steroids and immunosuppressants which should also be continued. Anticholinesterases (pyridostigmine) should be continued in reduced doses.

*Topic: Myasthenia Gravis*

# Anesthesia for Immune Mediated and Infectious Diseases MCQ – Chapter 25

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 25)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. What is the underlying pathology of rheumatoid arthritis?**

- A. Immune-mediated vasculitis
- B. Immune-mediated synovitis
- C. Degenerative joint disease
- D. Autoimmune myositis

**Ans: B**

**Q2. Which associated comorbidities should be ruled out in rheumatoid arthritis patients preoperatively?**

- A. Renal failure, hepatitis, diabetes
- B. Cardiac valvular lesions, pericarditis, and pulmonary fibrosis
- C. Thyroid dysfunction and adrenal insufficiency
- D. Peptic ulcer and esophageal varices

**Ans: B**

**Q3. What is the most important concern in anesthetic management of rheumatoid arthritis?**

- A. Drug interactions
- B. Difficult airway
- C. Renal dysfunction
- D. Coagulopathy

**Ans: B**

**Q4. Which side effects should be considered preoperatively in rheumatoid arthritis patients?**

- A. Side effects of antihypertensives
- B. Side effects of DMARDs, NSAIDs, and corticosteroids
- C. Side effects of antibiotics
- D. Side effects of anticoagulants only

**Ans: B**

**Q5. What is the preferred choice of anesthesia in rheumatoid arthritis patients?**

- A. General anesthesia
- B. Regional anesthesia
- C. Local anesthesia only
- D. Monitored anesthesia care

**Ans: B**

**Q6. Why is the lumbar spine usually suitable for spinal/epidural in rheumatoid arthritis?**

- A. It has better blood supply
- B. Lumbar spine is usually not involved in rheumatoid arthritis
- C. It is easier to access
- D. It provides better block

**Ans: B**

**Q7. Why is general anesthesia difficult and risky in rheumatoid arthritis patients?**

- A. Due to drug resistance
- B. Due to difficult intubation
- C. Due to cardiac instability
- D. Due to renal dysfunction

**Ans: B**

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| <p><b>Q8. What can happen if the neck is flexed in patients with atlantoaxial subluxation?</b></p> <p>A. Cervical disc herniation<br/> B. Odontoid process can compress spinal cord leading to quadriplegia<br/> C. Vertebral artery dissection<br/> D. Esophageal injury</p>                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q9. How should intubation be performed in rheumatoid arthritis patients with atlantoaxial subluxation?</b></p> <p>A. Rapid sequence intubation with neck flexion<br/> B. Awake intubation under topical anesthesia, or under GA with neck in extension<br/> C. Blind nasal intubation<br/> D. Tracheostomy</p>                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q10. What are the three causes of difficult intubation in rheumatoid arthritis?</b></p> <p>A. Macroglossia, prognathism, epiglottic overgrowth<br/> B. Atlantoaxial instability, stiffness of cervical spine, temporomandibular and cricoarytenoid arthritis<br/> C. Tracheal stenosis, vocal cord paralysis, subglottic edema<br/> D. Obesity, short neck, large tongue</p> | <p><b>Ans: B</b></p> |
| <p><b>Q11. Why may arterial cannulation be difficult in rheumatoid arthritis patients?</b></p> <p>A. Due to small vessel caliber<br/> B. Due to calcified radial arteries<br/> C. Due to vasospasm<br/> D. Due to thrombosed vessels</p>                                                                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q12. Why may central venous cannulation be difficult in rheumatoid arthritis?</b></p> <p>A. Due to thrombosis<br/> B. Due to cervical spine fusion<br/> C. Due to superior vena cava obstruction<br/> D. Due to coagulopathy</p>                                                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q13. Why can ankylosing spondylitis be the most difficult case from anesthesia point of view?</b></p> <p>A. Due to cardiac involvement<br/> B. Fusion of lumbar spine makes CNB impossible and fusion of cervical spine makes intubation impossible<br/> C. Due to renal involvement<br/> D. Due to coagulopathy</p>                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q14. How should intubation be performed in ankylosing spondylitis patients with cervical spine fusion?</b></p> <p>A. Direct laryngoscopy under GA<br/> B. Awake state under fiberoptic bronchoscopy<br/> C. Blind nasal intubation<br/> D. Rapid sequence intubation</p>                                                                                                     | <p><b>Ans: B</b></p> |

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| <p><b>Q15. Which cardiac conditions are common in ankylosing spondylitis?</b></p> <p>A. Mitral stenosis and atrial fibrillation<br/>B. Aortic regurgitation and bundle branch block<br/>C. Ventricular septal defect<br/>D. Pericardial effusion</p>                  | <p><b>Ans: B</b></p> |
| <p><b>Q16. What determines the perioperative morbidity in systemic lupus erythematosus (SLE)?</b></p> <p>A. Duration of disease<br/>B. Involvement of major organ functions<br/>C. Age of the patient<br/>D. Type of surgery</p>                                      | <p><b>Ans: B</b></p> |
| <p><b>Q17. What laryngeal involvement in SLE can make intubation difficult?</b></p> <p>A. Laryngeal edema only<br/>B. Cricoarytenoid arthritis and laryngeal nerve palsy<br/>C. Vocal cord polyps<br/>D. Subglottic stenosis</p>                                      | <p><b>Ans: B</b></p> |
| <p><b>Q18. Which organ involvement in scleroderma significantly affects perioperative morbidity?</b></p> <p>A. Renal failure and hepatic dysfunction<br/>B. Pulmonary fibrosis and pulmonary hypertension<br/>C. Cardiac tamponade<br/>D. Cerebrovascular disease</p> | <p><b>Ans: B</b></p> |
| <p><b>Q19. Why can intubation be difficult in scleroderma?</b></p> <p>A. Due to cervical spine fusion<br/>B. Due to small oral aperture<br/>C. Due to macroglossia<br/>D. Due to tracheal compression</p>                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q20. Why can IV cannulation be difficult in scleroderma?</b></p> <p>A. Due to vasospasm<br/>B. Due to dermal thickening<br/>C. Due to edema<br/>D. Due to calcification of vessels</p>                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q21. How are HIV positive patients classified according to ASA grading?</b></p> <p>A. ASA I<br/>B. ASA II<br/>C. ASA III<br/>D. ASA IV</p>                                                                                                                      | <p><b>Ans: B</b></p> |

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| <p><b>Q22. How are patients with AIDS classified according to ASA grading?</b></p> <p>A. ASA I/II<br/>B. ASA II/III<br/>C. ASA III/IV<br/>D. ASA IV/V</p>                                                                                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q23. Which side effects of HAART should be ruled out preoperatively?</b></p> <p>A. Hypothyroidism and osteoporosis<br/>B. Diabetes, coronary and cerebrovascular disease, deranged liver function<br/>C. Renal failure only<br/>D. Anemia only</p>                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q24. Patients who started HAART within how many months of surgery suffer more from perioperative complications?</b></p> <p>A. 3 months<br/>B. 6 months<br/>C. 9 months<br/>D. 12 months</p>                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q25. Do CD4 count and viral load increase perioperative anesthetic complications?</b></p> <p>A. Yes, significantly<br/>B. Yes, but only in emergency surgery<br/>C. No, they affect overall prognosis but not perioperative anesthetic complications<br/>D. Only viral load increases complications</p> | <p><b>Ans: C</b></p> |
| <p><b>Q26. Can regional anesthesia be given to HIV/AIDS patients?</b></p> <p>A. Never, it is contraindicated<br/>B. Yes, safely if there are no associated neuropathies<br/>C. Only epidural is safe<br/>D. Only spinal is safe</p>                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q27. What can make intubation difficult in HIV/AIDS patients?</b></p> <p>A. Cervical spine fusion<br/>B. Fat redistribution around the neck<br/>C. Tracheal compression<br/>D. Vocal cord paralysis</p>                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q28. What is vital during any procedure in HIV/AIDS patients?</b></p> <p>A. Use of specialized equipment<br/>B. Maintaining universal precautions and high degree of asepsis<br/>C. Avoiding all regional blocks<br/>D. Using only disposable equipment</p>                                             | <p><b>Ans: B</b></p> |



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| <p><b>Q29. In rheumatoid arthritis, intubation should be performed with the neck in which position?</b></p> <p>A. Neutral position<br/>B. Flexion<br/>C. Extension<br/>D. Lateral rotation</p>                                                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q30. Which joint involvement in rheumatoid arthritis contributes to difficult intubation besides atlantoaxial and cervical spine?</b></p> <p>A. Shoulder and elbow joints<br/>B. Temporomandibular and cricoarytenoid joints<br/>C. Hip and knee joints<br/>D. Sacroiliac joints</p>                                              | <p><b>Ans: B</b></p> |
| <p><b>Q31. Which of the following is TRUE about ankylosing spondylitis and anesthesia?</b></p> <p>A. Spinal anesthesia is always easy<br/>B. General anesthesia with direct laryngoscopy is preferred<br/>C. It may turn out to be the most difficult case from anesthesia point of view<br/>D. Epidural is the preferred technique</p> | <p><b>Ans: C</b></p> |
| <p><b>Q32. In ankylosing spondylitis, fusion of which spine makes central neuraxial blocks impossible?</b></p> <p>A. Cervical spine<br/>B. Thoracic spine<br/>C. Lumbar spine<br/>D. Sacral spine</p>                                                                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q33. In ankylosing spondylitis, fusion of which spine makes intubation impossible?</b></p> <p>A. Lumbar spine<br/>B. Thoracic spine<br/>C. Cervical spine<br/>D. Sacral spine</p>                                                                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q34. What type of valvular disease is commonly associated with ankylosing spondylitis?</b></p> <p>A. Mitral stenosis<br/>B. Mitral regurgitation<br/>C. Aortic regurgitation<br/>D. Aortic stenosis</p>                                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q35. What conduction abnormality is common in ankylosing spondylitis?</b></p> <p>A. Atrial fibrillation<br/>B. Bundle branch block<br/>C. Complete heart block<br/>D. Ventricular tachycardia</p>                                                                                                                                 | <p><b>Ans: B</b></p> |

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| <p><b>Q36. Which condition is characterized by immune-mediated synovitis?</b></p> <ul style="list-style-type: none"> <li>A. Ankylosing spondylitis</li> <li>B. Systemic lupus erythematosus</li> <li>C. Rheumatoid arthritis</li> <li>D. Scleroderma</li> </ul>                                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q37. In which condition can small oral aperture cause difficult intubation?</b></p> <ul style="list-style-type: none"> <li>A. Rheumatoid arthritis</li> <li>B. Ankylosing spondylitis</li> <li>C. Scleroderma</li> <li>D. SLE</li> </ul>                                                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q38. Which condition can cause cricoarytenoid arthritis leading to difficult intubation — BOTH rheumatoid arthritis AND SLE?</b></p> <ul style="list-style-type: none"> <li>A. True – both RA and SLE can cause cricoarytenoid arthritis</li> <li>B. False – only RA causes it</li> <li>C. False – only SLE causes it</li> <li>D. False – neither causes it</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q39. Which is the best method of intubation in a patient with atlantoaxial subluxation under general anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Direct laryngoscopy with neck flexion</li> <li>B. Intubation with neck in extension</li> <li>C. Blind nasal intubation</li> <li>D. Laryngeal mask airway</li> </ul>                                 | <p><b>Ans: B</b></p> |
| <p><b>Q40. Patients with atlantoaxial subluxation cannot tolerate which neck movement?</b></p> <ul style="list-style-type: none"> <li>A. Extension</li> <li>B. Flexion</li> <li>C. Lateral rotation</li> <li>D. All movements</li> </ul>                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q41. What is HAART in the context of HIV management?</b></p> <ul style="list-style-type: none"> <li>A. Highly Active Antimicrobial Therapy</li> <li>B. Highly Active Antiretroviral Therapy</li> <li>C. High Altitude Antiviral Regimen Therapy</li> <li>D. Hepatic Active Antiretroviral Treatment</li> </ul>                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q42. Which pulmonary conditions in scleroderma affect perioperative outcome?</b></p> <ul style="list-style-type: none"> <li>A. Bronchiectasis and emphysema</li> <li>B. Pulmonary fibrosis and pulmonary hypertension</li> <li>C. Pleural effusion and pneumothorax</li> <li>D. Asthma and COPD</li> </ul>                                                             | <p><b>Ans: B</b></p> |

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| <p><b>Q43. What is the awake intubation technique of choice for ankylosing spondylitis with cervical spine fusion?</b></p> <p>A. Direct laryngoscopy<br/>B. Video laryngoscopy<br/>C. Fiberoptic bronchoscopy<br/>D. Blind nasal intubation</p> | <p><b>Ans: C</b></p> |
| <p><b>Q44. Why are HIV positive patients classified as at least ASA II?</b></p> <p>A. Due to high risk of bleeding<br/>B. Due to involvement of multiple organs<br/>C. Due to immunosuppression only<br/>D. Due to drug interactions</p>        | <p><b>Ans: B</b></p> |
| <p><b>Q45. Which condition among the immune mediated diseases can cause laryngeal nerve palsy?</b></p> <p>A. Rheumatoid arthritis<br/>B. Ankylosing spondylitis<br/>C. Systemic lupus erythematosus<br/>D. Scleroderma</p>                      | <p><b>Ans: C</b></p> |
| <p><b>Q46. What is the cause of fat redistribution around the neck in HIV/AIDS patients?</b></p> <p>A. The disease itself<br/>B. Side effect of HAART (lipodystrophy)<br/>C. Malnutrition<br/>D. Opportunistic infections</p>                   | <p><b>Ans: B</b></p> |
| <p><b>Q47. Which immune mediated disease most commonly involves the cervical spine causing intubation difficulty?</b></p> <p>A. SLE<br/>B. Scleroderma<br/>C. Rheumatoid arthritis and Ankylosing spondylitis<br/>D. HIV/AIDS</p>               | <p><b>Ans: C</b></p> |
| <p><b>Q48. In which condition should neuropathies be ruled out before giving regional anesthesia?</b></p> <p>A. Scleroderma<br/>B. HIV/AIDS<br/>C. Ankylosing spondylitis<br/>D. Rheumatoid arthritis</p>                                       | <p><b>Ans: B</b></p> |
| <p><b>Q49. Pericarditis is an associated comorbidity of which immune mediated disease?</b></p> <p>A. Ankylosing spondylitis<br/>B. Scleroderma<br/>C. Rheumatoid arthritis<br/>D. HIV/AIDS</p>                                                  | <p><b>Ans: C</b></p> |

**Q50. Pulmonary fibrosis is a comorbidity associated with which TWO immune mediated diseases from this chapter?**

- A. RA and SLE
- B. Rheumatoid arthritis and Scleroderma
- C. SLE and Scleroderma
- D. Ankylosing spondylitis and Scleroderma

**Ans: B**

# Answer Key & Explanations

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## **Q1. Answer: B — Immune-mediated synovitis**

Rheumatoid arthritis is a disease characterized by immune-mediated synovitis.

Topic: Rheumatoid Arthritis | Ref: Ch 25, Page 214

## **Q2. Answer: B — Cardiac valvular lesions, pericarditis, and pulmonary fibrosis**

Associated comorbidities like cardiac valvular lesions, pericarditis, and pulmonary fibrosis should be ruled out.

Topic: Rheumatoid Arthritis – Preoperative Evaluation | Ref: Ch 25, Page 214

## **Q3. Answer: B — Difficult airway**

As difficult airway is the most important concern, airway evaluation should be done very meticulously.

Topic: Rheumatoid Arthritis – Preoperative Evaluation | Ref: Ch 25, Page 214

## **Q4. Answer: B — Side effects of DMARDs, NSAIDs, and corticosteroids**

Side effects of disease modifying antirheumatic drugs (DMARDs), NSAIDs and corticosteroids should be taken into consideration.

Topic: Rheumatoid Arthritis – Preoperative Evaluation | Ref: Ch 25, Page 214

## **Q5. Answer: B — Regional anesthesia**

Regional anesthesia is the choice. Lumbar spine is usually not involved in rheumatoid arthritis therefore spinal/epidural can be given easily.

Topic: Rheumatoid Arthritis – Choice of Anesthesia | Ref: Ch 25, Page 214

## **Q6. Answer: B — Lumbar spine is usually not involved in rheumatoid arthritis**

Lumbar spine is usually not involved in rheumatoid arthritis therefore spinal/epidural can be given easily.

Topic: Rheumatoid Arthritis – Choice of Anesthesia | Ref: Ch 25, Page 214

## **Q7. Answer: B — Due to difficult intubation**

General anesthesia is difficult and risky due to difficult intubation caused by atlantoaxial instability, stiffness of cervical spine, and temporomandibular and cricoarytenoid arthritis.

Topic: Rheumatoid Arthritis – General Anesthesia | Ref: Ch 25, Page 214

## **Q8. Answer: B — Odontoid process can compress spinal cord leading to quadriplegia**

Flexion of neck can cause odontoid process to compress spinal cord leading to quadriplegia.

Topic: Rheumatoid Arthritis – Atlantoaxial Instability | Ref: Ch 25, Page 214

## **Q9. Answer: B — Awake intubation under topical anesthesia, or under GA with neck in extension**

Intubation is best performed under topical anesthesia with awake patient. If under GA, intubation should be done with neck in extension.

Topic: Rheumatoid Arthritis – Intubation | Ref: Ch 25, Page 214

**Q10. Answer: B — Atlantoaxial instability, stiffness of cervical spine, temporomandibular and cricoarytenoid arthritis**

Difficult intubation is due to atlantoaxial instability, stiffness of cervical spine, and temporomandibular and cricoarytenoid arthritis.

Topic: Rheumatoid Arthritis – General Anesthesia | Ref: Ch 25, Page 214

**Q11. Answer: B — Due to calcified radial arteries**

Arterial cannulation may be difficult due to calcified radial arteries.

Topic: Rheumatoid Arthritis – Vascular Access | Ref: Ch 25, Page 214

**Q12. Answer: B — Due to cervical spine fusion**

Central venous cannulation may be difficult due to cervical spine fusion.

Topic: Rheumatoid Arthritis – Vascular Access | Ref: Ch 25, Page 214

**Q13. Answer: B — Fusion of lumbar spine makes CNB impossible and fusion of cervical spine makes intubation impossible**

Fusion of lumbar spine may make central neuraxial blocks impossible and fusion of cervical spine may make intubation impossible.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q14. Answer: B — Awake state under fiberoptic bronchoscopy**

Intubation in such patients should be performed in awake state under fiberoptic bronchoscopy.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q15. Answer: B — Aortic regurgitation and bundle branch block**

Aortic regurgitation and bundle branch block are common in ankylosing spondylitis.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q16. Answer: B — Involvement of major organ functions**

The involvement of major organ functions determines the perioperative morbidity in SLE.

Topic: Systemic Lupus Erythematosus | Ref: Ch 25, Page 214

**Q17. Answer: B — Cricoarytenoid arthritis and laryngeal nerve palsy**

Laryngeal involvement including cricoarytenoid arthritis and laryngeal nerve palsy can make intubation difficult in SLE.

Topic: Systemic Lupus Erythematosus | Ref: Ch 25, Page 214

**Q18. Answer: B — Pulmonary fibrosis and pulmonary hypertension**

Multiorgan involvement, particularly pulmonary fibrosis and pulmonary hypertension can significantly affect perioperative morbidity.

Topic: Scleroderma | Ref: Ch 25, Page 214

**Q19. Answer: B — Due to small oral aperture**

Small oral aperture can make intubation difficult in scleroderma.

Topic: Scleroderma | Ref: Ch 25, Page 214

**Q20. Answer: B — Due to dermal thickening**

Dermal thickening can make intravenous cannulation difficult in scleroderma.

Topic: Scleroderma | Ref: Ch 25, Page 214

**Q21. Answer: B — ASA II**

Due to involvement of multiple organs all HIV positive patients are categorized as ASA grade II.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

**Q22. Answer: C — ASA III/IV**

Patients with AIDS are categorized as ASA III/IV.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

**Q23. Answer: B — Diabetes, coronary and cerebrovascular disease, deranged liver function**

Side effects of HAART including diabetes, coronary and cerebrovascular disease, deranged liver function must be ruled out preoperatively.

Topic: HIV and AIDS – Preoperative Evaluation | Ref: Ch 25, Page 215

**Q24. Answer: B — 6 months**

Patients who have started HAART within 6 months of surgery suffer more from perioperative complications.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

**Q25. Answer: C — No, they affect overall prognosis but not perioperative anesthetic complications**

CD4 count and viral load do affect the overall prognosis after surgery but have not been found to increase perioperative anesthetic complications.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

**Q26. Answer: B — Yes, safely if there are no associated neuropathies**

Regional anesthesia can be given safely if there are no associated neuropathies.

Topic: HIV and AIDS – Choice of Anesthesia | Ref: Ch 25, Page 215

**Q27. Answer: B — Fat redistribution around the neck**

Fat redistribution around the neck can make intubation difficult in HIV/AIDS patients.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

**Q28. Answer: B — Maintaining universal precautions and high degree of asepsis**

Maintaining universal precautions and high degree of asepsis during any procedure is vital.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

**Q29. Answer: C — Extension**

Due to atlantoaxial instability, intubation in RA patients should be performed with neck in extension.

Topic: Rheumatoid Arthritis – Key Points | Ref: Ch 25, Page 214

**Q30. Answer: B — Temporomandibular and cricoarytenoid joints**

Temporomandibular and cricoarytenoid arthritis contribute to difficult intubation in rheumatoid arthritis.

Topic: Rheumatoid Arthritis – Difficult Intubation | Ref: Ch 25, Page 214

**Q31. Answer: C — It may turn out to be the most difficult case from anesthesia point of view**

From anesthesia point of view ankylosing spondylitis may turn out to be the most difficult case at times.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q32. Answer: C — Lumbar spine**

Fusion of lumbar spine may make central neuraxial blocks impossible.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q33. Answer: C — Cervical spine**

Fusion of cervical spine may make intubation impossible in ankylosing spondylitis.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q34. Answer: C — Aortic regurgitation**

Aortic regurgitation is common in ankylosing spondylitis.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q35. Answer: B — Bundle branch block**

Bundle branch block is common in ankylosing spondylitis.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q36. Answer: C — Rheumatoid arthritis**

Rheumatoid arthritis is a disease characterized by immune-mediated synovitis.

Topic: Rheumatoid Arthritis | Ref: Ch 25, Page 214

**Q37. Answer: C — Scleroderma**

In scleroderma, small oral aperture can make intubation difficult.

Topic: Scleroderma | Ref: Ch 25, Page 214

**Q38. Answer: A — True – both RA and SLE can cause cricoarytenoid arthritis**

Both rheumatoid arthritis (temporomandibular and cricoarytenoid arthritis) and SLE (laryngeal involvement including cricoarytenoid arthritis) can cause difficult intubation.

Topic: RA and SLE – Airway Management | Ref: Ch 25, Page 214

**Q39. Answer: B — Intubation with neck in extension**

If intubation has to be done under general anesthesia, it should be done with neck in extension to avoid odontoid compression of spinal cord.

Topic: Rheumatoid Arthritis – Intubation | Ref: Ch 25, Page 214

**Q40. Answer: B — Flexion**

Patients of atlantoaxial subluxation cannot tolerate flexion as it can cause odontoid process to compress spinal cord.

Topic: Rheumatoid Arthritis – Atlantoaxial Instability | Ref: Ch 25, Page 214

**Q41. Answer: B — Highly Active Antiretroviral Therapy**

HAART stands for Highly Active Antiretroviral Therapy used in HIV management.

Topic: HIV and AIDS | Ref: Ch 25, Page 215



**Q42. Answer: B — Pulmonary fibrosis and pulmonary hypertension**

Pulmonary fibrosis and pulmonary hypertension in scleroderma significantly affect perioperative morbidity.

Topic: Scleroderma | Ref: Ch 25, Page 214

**Q43. Answer: C — Fiberoptic bronchoscopy**

Intubation should be performed in awake state under fiberoptic bronchoscopy in ankylosing spondylitis patients.

Topic: Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q44. Answer: B — Due to involvement of multiple organs**

Due to involvement of multiple organs all HIV positive patients are categorized as ASA grade II.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

**Q45. Answer: C — Systemic lupus erythematosus**

In SLE, laryngeal involvement including laryngeal nerve palsy can make intubation difficult.

Topic: Systemic Lupus Erythematosus | Ref: Ch 25, Page 214

**Q46. Answer: B — Side effect of HAART (lipodystrophy)**

Fat redistribution (lipodystrophy) is a known side effect of HAART which can cause fat accumulation around the neck making intubation difficult.

Topic: HIV and AIDS | Ref: Ch 25, Page 215

**Q47. Answer: C — Rheumatoid arthritis and Ankylosing spondylitis**

Both RA (atlantoaxial instability, cervical spine stiffness) and ankylosing spondylitis (cervical spine fusion) commonly involve the cervical spine causing difficult intubation.

Topic: RA and Ankylosing Spondylitis | Ref: Ch 25, Page 214

**Q48. Answer: B — HIV/AIDS**

In HIV/AIDS patients, regional anesthesia can be given safely if there are no associated neuropathies.

Topic: HIV and AIDS – Choice of Anesthesia | Ref: Ch 25, Page 215

**Q49. Answer: C — Rheumatoid arthritis**

Associated comorbidities like cardiac valvular lesions, pericarditis, and pulmonary fibrosis should be ruled out in rheumatoid arthritis.

Topic: Rheumatoid Arthritis – Preoperative Evaluation | Ref: Ch 25, Page 214

**Q50. Answer: B — Rheumatoid arthritis and Scleroderma**

Pulmonary fibrosis is mentioned as a comorbidity of both rheumatoid arthritis (preoperative evaluation) and scleroderma (multiorgan involvement).

Topic: RA and Scleroderma | Ref: Ch 25, Page 214

# Anesthesia for Disorders of Blood MCQ – Chapter 26

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 26)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

**Q1. What is the most common type of anemia encountered in clinical practice in India?**

- A. Aplastic anemia
- B. Microcytic hypochromic anemia due to iron deficiency
- C. Hemolytic anemia
- D. Megaloblastic anemia

**Ans: B**

**Q2. What is the minimum hemoglobin level considered optimal to accept a patient for elective surgery?**

- A. 12 g/dL
- B. 6 g/dL
- C. 10 g/dL
- D. 8 g/dL

**Ans: D**

**Q3. What is the formula for calculating maximum allowable blood loss?**

- A.  $(\text{Patient Hct} - \text{Desired Hct}) / \text{Patient Hct} \times \text{Blood volume}$
- B.  $(\text{Desired Hct} - \text{Patient Hct}) / \text{Patient Hct} \times \text{Blood volume}$
- C.  $\text{Patient Hct} / \text{Desired Hct} \times \text{Blood volume}$
- D.  $(\text{Patient Hct} - \text{Desired Hct}) / \text{Desired Hct} \times \text{Blood volume}$

**Ans: A**

**Q4. What is the estimated blood volume used in the calculation for allowable blood loss?**

- A. 80-90 mL/kg
- B. 100-110 mL/kg
- C. 60-70 mL/kg
- D. 50-60 mL/kg

**Ans: A**

**Q5. What is the current day approach towards minimum use of blood products called?**

- A. Restrictive approach to transfusion
- B. Liberal transfusion policy
- C. Aggressive resuscitation
- D. Conservative approach

**Ans: A**

**Q6. When is blood transfusion generally indicated in a normal patient?**

- A. Blood loss > 20% of blood volume or Hb < 8 g%
- B. Blood loss > 30% of blood volume
- C. Blood loss > 5% of blood volume
- D. Blood loss > 10% of blood volume

**Ans: A**

**Q7. At what hemoglobin level is transfusion indicated in a patient with ischemic heart disease?**

- A. Less than 10 g%
- B. Less than 12 g%
- C. Less than 8 g%
- D. Less than 6 g%

**Ans: A**

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| <p><b>Q8. What type of anesthesia should be avoided for vitamin B12 deficiency anemia?</b></p> <ul style="list-style-type: none"> <li>A. Regional anesthesia</li> <li>B. Monitored anesthesia care</li> <li>C. General anesthesia</li> <li>D. Local anesthesia</li> </ul>                                       | <p><b>Ans: A</b></p> |
| <p><b>Q9. Which inhalational agent should be avoided in megaloblastic anemia?</b></p> <ul style="list-style-type: none"> <li>A. Sevoflurane</li> <li>B. Nitrous oxide</li> <li>C. Isoflurane</li> <li>D. Desflurane</li> </ul>                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q10. What effect does alkalosis have on the oxygen dissociation curve?</b></p> <ul style="list-style-type: none"> <li>A. No effect</li> <li>B. Flattens the curve</li> <li>C. Shifts it to the right</li> <li>D. Shifts it to the left</li> </ul>                                                         | <p><b>Ans: D</b></p> |
| <p><b>Q11. How is the uptake of inhalational agents affected in anemia patients?</b></p> <ul style="list-style-type: none"> <li>A. No change</li> <li>B. Increased uptake offset by increased cardiac output</li> <li>C. Significantly increased uptake</li> <li>D. Decreased uptake</li> </ul>                 | <p><b>Ans: B</b></p> |
| <p><b>Q12. What is the risk status of patients having sickle cell trait?</b></p> <ul style="list-style-type: none"> <li>A. Same as sickle cell disease</li> <li>B. Moderate risk</li> <li>C. No increase in morbidity and mortality</li> <li>D. High risk of complications</li> </ul>                           | <p><b>Ans: C</b></p> |
| <p><b>Q13. Which patients with sickle cell disease should NOT be taken for elective surgery?</b></p> <ul style="list-style-type: none"> <li>A. Patients on hydroxyurea</li> <li>B. Patients in acute crisis phase</li> <li>C. Patients with HbS &lt; 30%</li> <li>D. Patients with normal hematocrit</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q14. What is the target hematocrit for red cell transfusion in sickle cell patients?</b></p> <ul style="list-style-type: none"> <li>A. Hematocrit &gt; 40</li> <li>B. Hematocrit &gt; 50</li> <li>C. Hematocrit &gt; 30 (Hemoglobin &gt; 10 g%)</li> <li>D. Hematocrit &gt; 20</li> </ul>                 | <p><b>Ans: C</b></p> |
| <p><b>Q15. What is the most important perioperative goal in sickle cell patients?</b></p> <ul style="list-style-type: none"> <li>A. Maintain high hemoglobin</li> <li>B. Avoid factors that precipitate sickling</li> <li>C. Maintain hypothermia</li> <li>D. Use only general anesthesia</li> </ul>            | <p><b>Ans: B</b></p> |

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| <p><b>Q16. Which factors can precipitate sickling in sickle cell disease?</b></p> <ul style="list-style-type: none"> <li>A. Only hypoxia</li> <li>B. Alkalosis and hyperhydration</li> <li>C. Hyperoxia and warmth</li> <li>D. Hypoxia, acidosis, dehydration, hypothermia, stress and pain</li> </ul>                                                    | <p><b>Ans: D</b></p> |
| <p><b>Q17. How should dehydration be managed preoperatively in sickle cell patients?</b></p> <ul style="list-style-type: none"> <li>A. IV fluids should be started in preoperative period</li> <li>B. Fluid restriction</li> <li>C. No fluids needed</li> <li>D. Oral fluids only</li> </ul>                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q18. What premedication is recommended for sickle cell patients to avoid stress and anxiety?</b></p> <ul style="list-style-type: none"> <li>A. Benzodiazepines</li> <li>B. Antihistamines</li> <li>C. Opioids</li> <li>D. Anticholinergics</li> </ul>                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q19. Is intravenous regional anesthesia (Bier's block) safe in sickle cell patients?</b></p> <ul style="list-style-type: none"> <li>A. Only without tourniquet</li> <li>B. Only with short procedures</li> <li>C. No, it does not hold true in current day practice due to tourniquet use</li> <li>D. Yes, it is the preferred technique</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q20. Can tourniquet be used in sickle cell patients?</b></p> <ul style="list-style-type: none"> <li>A. Absolutely contraindicated</li> <li>B. Can be used safely with minimal possible pressure for minimal period</li> <li>C. Can be used without any precautions</li> <li>D. Only pneumatic tourniquet allowed</li> </ul>                         | <p><b>Ans: B</b></p> |
| <p><b>Q21. What must be avoided at any cost in sickle cell patients?</b></p> <ul style="list-style-type: none"> <li>A. Regional anesthesia</li> <li>B. Blood transfusion</li> <li>C. Inhalational agents</li> <li>D. Factors which precipitate sickling</li> </ul>                                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q22. When does acute lung syndrome usually occur postoperatively in sickle cell patients?</b></p> <ul style="list-style-type: none"> <li>A. Within 24 hours</li> <li>B. Immediately after extubation</li> <li>C. After 1 week</li> <li>D. After 48-72 hours in postoperative period</li> </ul>                                                      | <p><b>Ans: D</b></p> |

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| <p><b>Q23. What is the risk level of patients with thalassemia minor?</b></p> <p>A. Not high risk<br/>B. Very high risk<br/>C. Moderate risk<br/>D. High risk</p>                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q24. What can skeletal abnormalities of maxilla in thalassemia cause?</b></p> <p>A. Laryngospasm<br/>B. Bronchospasm<br/>C. Easy intubation<br/>D. Difficult intubation</p>                                                                                   | <p><b>Ans: D</b></p> |
| <p><b>Q25. What organ involvement should be assessed in thalassemia major patients?</b></p> <p>A. Only spleen<br/>B. Only liver<br/>C. Hepatosplenomegaly, skeletal involvement, CHF, and consequences of iron overload including cirrhosis<br/>D. Only cardiac</p> | <p><b>Ans: C</b></p> |
| <p><b>Q26. What is the target hematocrit for polycythemia patients before elective surgery?</b></p> <p>A. Less than 45<br/>B. Less than 50<br/>C. Less than 55<br/>D. Less than 35</p>                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q27. What additional bleeding risk exists in polycythemia patients?</b></p> <p>A. Vitamin K deficiency<br/>B. Increased fibrinolysis<br/>C. Decreased von Willebrand factor<br/>D. DIC</p>                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q28. What is the recommended platelet count for considering surgery in thrombocytopenia?</b></p> <p>A. &gt; 50,000/mm<sup>3</sup><br/>B. &gt; 150,000/mm<sup>3</sup><br/>C. &gt; 100,000/mm<sup>3</sup><br/>D. &gt; 25,000/mm<sup>3</sup></p>                 | <p><b>Ans: A</b></p> |
| <p><b>Q29. At what platelet count can neurosurgical patients be considered for surgery?</b></p> <p>A. &gt; 150,000<br/>B. &gt; 100,000<br/>C. &gt; 50,000<br/>D. &gt; 30,000</p>                                                                                    | <p><b>Ans: B</b></p> |

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| <p><b>Q30. Which patients need platelet transfusion before surgery regardless of platelet count?</b></p> <p>A. Asymptomatic patients<br/>B. Symptomatic patients with spontaneous bleeding<br/>C. All surgical patients<br/>D. Only cardiac patients</p> | <p><b>Ans: B</b></p> |
| <p><b>Q31. Above what platelet count can central neuraxial blocks be safely given?</b></p> <p>A. &gt; 100,000/cubic mm<br/>B. &gt; 50,000/cubic mm<br/>C. &gt; 30,000/cubic mm<br/>D. &gt; 80,000/cubic mm</p>                                           | <p><b>Ans: D</b></p> |
| <p><b>Q32. What is the relative contraindication range for platelet count regarding central neuraxial blocks?</b></p> <p>A. &lt; 30,000<br/>B. &lt; 50,000<br/>C. Between 80,000 and 100,000<br/>D. Between 50,000 and 80,000/cubic mm</p>               | <p><b>Ans: D</b></p> |
| <p><b>Q33. What should the deficit coagulation factor be increased to before considering a patient for surgery?</b></p> <p>A. At least 25% of normal<br/>B. At least 50% of normal<br/>C. At least 75% of normal<br/>D. At least 100% of normal</p>      | <p><b>Ans: B</b></p> |
| <p><b>Q34. How can coagulation factors be replenished preoperatively?</b></p> <p>A. Fresh frozen plasma, cryoprecipitate or recombinant factor<br/>B. Vitamin K only<br/>C. Platelet transfusion<br/>D. Whole blood only</p>                             | <p><b>Ans: A</b></p> |
| <p><b>Q35. What type of anesthesia is contraindicated in coagulation deficiency disorders?</b></p> <p>A. Regional anesthesia<br/>B. Local anesthesia<br/>C. General anesthesia<br/>D. IV sedation</p>                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q36. Why should intramuscular route be avoided in coagulation disorders?</b></p> <p>A. Risk of hematoma formation<br/>B. Pain<br/>C. Infection risk<br/>D. Poor absorption</p>                                                                     | <p><b>Ans: A</b></p> |

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| <p><b>Q37. Who should perform intubation in patients with massive bleeding in oral cavity from coagulation disorders?</b></p> <p>A. Expert hand with minimum trauma<br/>B. Any anesthetist<br/>C. Surgeon<br/>D. Nurse</p>                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q38. For how long should antithrombin levels be maintained &gt; 80% postoperatively in hypercoagulable disorders?</b></p> <p>A. 2 days<br/>B. 10 days<br/>C. At least 5 days<br/>D. 1 day</p>                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q39. Why are central neuraxial blocks preferred in hypercoagulable disorders?</b></p> <p>A. They decrease incidence of thromboembolic complications<br/>B. Better pain control<br/>C. Shorter duration<br/>D. Easier to perform</p>                                         | <p><b>Ans: A</b></p> |
| <p><b>Q40. When should DVT prophylaxis be started in hypercoagulable disorders?</b></p> <p>A. Only if symptomatic<br/>B. As early as possible<br/>C. After 48 hours<br/>D. Postoperatively only</p>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q41. What does the anesthetic management of G6PD deficiency patients include?</b></p> <p>A. Only regional anesthesia<br/>B. Assessing severity of anemia and avoiding drugs that trigger hemolysis<br/>C. Only monitoring hemoglobin<br/>D. Mandatory blood transfusion</p> | <p><b>Ans: B</b></p> |
| <p><b>Q42. Which benzodiazepines are safe in G6PD deficiency?</b></p> <p>A. Only lorazepam<br/>B. Diazepam and midazolam<br/>C. None are safe<br/>D. Only alprazolam</p>                                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q43. Which inhalational agents can trigger hemolysis in G6PD deficiency?</b></p> <p>A. None of them<br/>B. Isoflurane and sevoflurane<br/>C. Nitrous oxide only<br/>D. Halothane and desflurane</p>                                                                         | <p><b>Ans: B</b></p> |



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| <p><b>Q44. Which local anesthetics should be avoided in G6PD deficiency?</b></p> <p>A. All local anesthetics<br/>B. Only lidocaine<br/>C. Only bupivacaine<br/>D. Prilocaine, benzocaine, lignocaine and nitroglycerine</p>      | <p><b>Ans: D</b></p> |
| <p><b>Q45. What is the danger of administering methylene blue to G6PD deficient patients?</b></p> <p>A. It can be life threatening<br/>B. It causes hypertension<br/>C. No danger<br/>D. It causes bradycardia</p>               | <p><b>Ans: A</b></p> |
| <p><b>Q46. What conditions can trigger hemolysis in G6PD deficiency besides drugs?</b></p> <p>A. Only infection<br/>B. Only surgery<br/>C. Only acidosis<br/>D. Acidosis, hypothermia, hyperglycemia and infection</p>           | <p><b>Ans: D</b></p> |
| <p><b>Q47. Which drugs are absolutely contraindicated in porphyria?</b></p> <p>A. Suxamethonium<br/>B. Barbiturates (Thiopentone/Methohexitone)<br/>C. Propofol<br/>D. Opioids</p>                                               | <p><b>Ans: B</b></p> |
| <p><b>Q48. Which agents are considered safe in porphyria patients?</b></p> <p>A. Propofol, opioids, and suxamethonium<br/>B. Diazepam and pentazocine<br/>C. Barbiturates and etomidate<br/>D. Ketamine and muscle relaxants</p> | <p><b>Ans: A</b></p> |
| <p><b>Q49. Should regional anesthesia be used in porphyria patients with peripheral neuropathy?</b></p> <p>A. Only spinal<br/>B. Yes, it is preferred<br/>C. No, it should be avoided<br/>D. Only epidural</p>                   | <p><b>Ans: C</b></p> |
| <p><b>Q50. What can fasting, dehydration and infection trigger in porphyria patients?</b></p> <p>A. Acute crises<br/>B. Cardiac arrest<br/>C. Seizures only<br/>D. Hemolysis</p>                                                 | <p><b>Ans: A</b></p> |

# Answer Key & Explanations

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## **Q1. Answer: B — Microcytic hypochromic anemia due to iron deficiency**

The most common anemia encountered in clinical practice in India is microcytic hypochromic due to iron deficiency.

Topic: Anemia

## **Q2. Answer: D — 8 g/dL**

A minimum hemoglobin of 8 g/dL is considered optimal to accept the patient for elective surgery.

Topic: Anemia

## **Q3. Answer: A — $(\text{Patient Hct} - \text{Desired Hct}) / \text{Patient Hct} \times \text{Blood volume}$**

Maximum allowable blood loss =  $(\text{Hematocrit of patient} - \text{Desirable hematocrit}) / \text{Hematocrit of patient} \times \text{Blood volume}$ .

Topic: Anemia

## **Q4. Answer: A — 80-90 mL/kg**

Blood volume of 80-90 mL/kg is used in the calculation for maximum allowable blood loss.

Topic: Anemia

## **Q5. Answer: A — Restrictive approach to transfusion**

The approach in present day practice is towards the minimum use of blood products called restrictive approach to transfusion.

Topic: Anemia

## **Q6. Answer: A — Blood loss > 20% of blood volume or Hb < 8 g%**

Blood loss greater than 20% of blood volume or hemoglobin level less than 8 g% in a normal patient indicates transfusion.

Topic: Anemia

## **Q7. Answer: A — Less than 10 g%**

In a patient with major disease like ischemic heart disease, transfusion is indicated when Hb is less than 10 g%.

Topic: Anemia

## **Q8. Answer: A — Regional anesthesia**

Regional anesthesia should be avoided for vitamin B12 deficiency anemia due to possibility of neurological deficits.

Topic: Anemia

## **Q9. Answer: B — Nitrous oxide**

Nitrous oxide can cause megaloblastic anemia therefore should be avoided for vitamin B12 deficiency anemia.

Topic: Anemia

## **Q10. Answer: D — Shifts it to the left**

Alkalosis (especially caused by hyperventilation) shifts the oxygen dissociation curve to the left, increasing the oxygen requirement.

Topic: Anemia

**Q11. Answer: B — Increased uptake offset by increased cardiac output**

Increase uptake of inhalational agents is offset by increased cardiac output seen in anemia patients.

Topic: Anemia

**Q12. Answer: C — No increase in morbidity and mortality**

There is no increase in morbidity and mortality in patients having sickle cell trait while patients with sickle cell disease are at high risk.

Topic: Sickle Cell Disease

**Q13. Answer: B — Patients in acute crisis phase**

Patients in acute crisis phase must not be taken for elective surgery.

Topic: Sickle Cell Disease

**Q14. Answer: C — Hematocrit > 30 (Hemoglobin > 10 g%)**

The aim is to correct the anemia by giving red cell transfusion with the target hematocrit > 30 (Hemoglobin > 10 g%).

Topic: Sickle Cell Disease

**Q15. Answer: B — Avoid factors that precipitate sickling**

The most important perioperative goal is to avoid factors which can precipitate sickling: hypoxia, acidosis, dehydration, hypothermia, stress and pain.

Topic: Sickle Cell Disease

**Q16. Answer: D — Hypoxia, acidosis, dehydration, hypothermia, stress and pain**

Factors that precipitate sickling include hypoxia, acidosis, dehydration, hypothermia, stress and pain.

Topic: Sickle Cell Disease

**Q17. Answer: A — IV fluids should be started in preoperative period**

To avoid dehydration, intravenous fluids should be started in preoperative period.

Topic: Sickle Cell Disease

**Q18. Answer: A — Benzodiazepines**

To avoid stress and anxiety these patients must be premedicated with benzodiazepines.

Topic: Sickle Cell Disease

**Q19. Answer: C — No, it does not hold true in current day practice due to tourniquet use**

The old recommendation of giving IV regional anesthesia (Bier's block) to sickle cell patients does not hold true in current day practice due to tourniquet use.

Topic: Sickle Cell Disease

**Q20. Answer: B — Can be used safely with minimal possible pressure for minimal period**

Per current guidelines, tourniquet can be used safely in sickle cell patients with a little higher complications. It is recommended to use for minimal period with minimal possible pressure.

Topic: Sickle Cell Disease

**Q21. Answer: D — Factors which precipitate sickling**

The factors which precipitate sickling must be avoided at any cost.

Topic: Sickle Cell Disease

**Q22. Answer: D — After 48-72 hours in postoperative period**

Acute lung syndrome usually occurs after 48-72 hours in postoperative period.

Topic: Sickle Cell Disease

**Q23. Answer: A — Not high risk**

Patients with thalassemia minor are not high risk while those with thalassemia major are high risk patients for perioperative morbidity and mortality.

Topic: Thalassemia

**Q24. Answer: D — Difficult intubation**

Skeletal abnormalities of maxilla (maxillary overgrowth) can make intubation difficult.

Topic: Thalassemia

**Q25. Answer: C — Hepatosplenomegaly, skeletal involvement, CHF, and consequences of iron overload including cirrhosis**

It is important to assess organ involvement like hepatosplenomegaly, skeletal involvement, congestive heart failure and consequences of iron overload like cirrhosis.

Topic: Thalassemia

**Q26. Answer: A — Less than 45**

Reduce hematocrit to <45 and platelet count to less than 4 lakhs before taking the patient for elective surgery.

Topic: Polycythemia

**Q27. Answer: C — Decreased von Willebrand factor**

Patients with polycythemia are at increased risk of bleeding due to decrease von Willebrand factor seen in polycythemia.

Topic: Polycythemia

**Q28. Answer: A — > 50,000/mm<sup>3</sup>**

Platelet count of > 50,000/mm<sup>3</sup> is recommended before considering for surgery.

Topic: Thrombocytopenia

**Q29. Answer: B — > 100,000**

Neurosurgical patients cannot be considered for surgery if their platelet count is < 100,000.

Topic: Thrombocytopenia

**Q30. Answer: B — Symptomatic patients with spontaneous bleeding**

Symptomatic patients (spontaneous bleeding) irrespective of their platelet count need platelet transfusion before surgery.

Topic: Thrombocytopenia

**Q31. Answer: D — > 80,000/cubic mm**

Patients with platelet count more than 80,000/cubic mm can be safely given central neuraxial blocks.

Topic: Thrombocytopenia

**Q32. Answer: D — Between 50,000 and 80,000/cubic mm**

Platelet count between 50,000 and 80,000/cubic mm is a relative contraindication while count less than 50,000 is an absolute contraindication.

Topic: Thrombocytopenia

**Q33. Answer: B — At least 50% of normal**

The deficit factor should be increased to at least 50% of the normal before considering the patient for surgery.

Topic: Coagulation Disorders

**Q34. Answer: A — Fresh frozen plasma, cryoprecipitate or recombinant factor**

The deficit factor can be increased by administration of fresh frozen plasma, cryoprecipitate or recombinant factor.

Topic: Coagulation Disorders

**Q35. Answer: A — Regional anesthesia**

Regional anesthesia is contraindicated in coagulation deficiency disorders.

Topic: Coagulation Disorders

**Q36. Answer: A — Risk of hematoma formation**

Intramuscular route should be avoided in coagulation disorders due to risk of bleeding/hematoma.

Topic: Coagulation Disorders

**Q37. Answer: A — Expert hand with minimum trauma**

Intubation should be performed by expert hand with minimum trauma, otherwise there can be massive bleeding in oral cavity.

Topic: Coagulation Disorders

**Q38. Answer: C — At least 5 days**

Antithrombin levels should be > 80% at least for 5 days in postoperative period.

Topic: Hypercoagulable Disorders

**Q39. Answer: A — They decrease incidence of thromboembolic complications**

Central neuraxial blocks are preferred because they decrease the incidence of thromboembolic complications (due to increase blood in lower limb and pelvic veins).

Topic: Hypercoagulable Disorders

**Q40. Answer: B — As early as possible**

All prophylaxis against deep venous thrombosis (DVT) should be started as early as possible.

Topic: Hypercoagulable Disorders

**Q41. Answer: B — Assessing severity of anemia and avoiding drugs that trigger hemolysis**

The anesthetic management includes assessing the severity of anemia and avoiding drugs and conditions which can trigger hemolysis in G6PD deficient individuals.

Topic: G6PD Deficiency

**Q42. Answer: B — Diazepam and midazolam**

Diazepam; interestingly, midazolam is safe in G6PD deficiency.

Topic: G6PD Deficiency

**Q43. Answer: B — Isoflurane and sevoflurane**

Isoflurane and sevoflurane can trigger hemolysis in perioperative period in G6PD deficient patients.

*Topic: G6PD Deficiency*

**Q44. Answer: D — Prilocaine, benzocaine, lignocaine and nitroglycerine**

Prilocaine, benzocaine, lignocaine and nitroglycerine should be avoided as they can cause methemoglobinemia which can cause significant hemolysis.

*Topic: G6PD Deficiency*

**Q45. Answer: A — It can be life threatening**

Administering methylene blue to these patients can be life threatening.

*Topic: G6PD Deficiency*

**Q46. Answer: D — Acidosis, hypothermia, hyperglycemia and infection**

Acidosis, hypothermia, hyperglycemia and infection can trigger hemolysis and must be avoided throughout in these patients.

*Topic: G6PD Deficiency*

**Q47. Answer: B — Barbiturates (Thiopentone/Methohexitone)**

Barbiturates (Thiopentone/Methohexitone) are absolutely contraindicated as they induce aminolevulinic acid synthetase and precipitate all kinds of porphyrias except porphyria cutanea tarda and erythropoietic protoporphyrias.

*Topic: Porphyria*

**Q48. Answer: A — Propofol, opioids, and suxamethonium**

Propofol, opioids, suxamethonium are considered as safe agents in porphyria.

*Topic: Porphyria*

**Q49. Answer: C — No, it should be avoided**

Regional anesthesia should be avoided in the porphyria patients with peripheral neuropathy.

*Topic: Porphyria*

**Q50. Answer: A — Acute crises**

Fasting, dehydration and infection can trigger acute crises in porphyria patients.

*Topic: Porphyria*

# Neurosurgical Anesthesia MCQ – Chapter 27

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 27)

**60 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

**Q1. What is cerebral perfusion pressure (CPP)?**

- A. Sum of MAP and ICP
- B. Difference between CVP and ICP
- C. Difference between mean arterial pressure and intracranial pressure
- D. Difference between systolic and diastolic pressure

**Ans: C**

**Q2. What is the normal cerebral blood flow (CBF) per 100 g of brain tissue?**

- A. 75 mL
- B. 25 mL
- C. 50 mL
- D. 100 mL

**Ans: C**

**Q3. Between what range of mean arterial pressure is CBF maintained at constant level by autoregulation?**

- A. 50 and 100 mm Hg
- B. 60 and 150 mm Hg
- C. 80 and 180 mm Hg
- D. 40 and 120 mm Hg

**Ans: B**

**Q4. Which of the following can abolish cerebral autoregulatory mechanism?**

- A. Moderate hypocapnia
- B. Cerebral ischemia, head injury, hypercarbia, hypoxia, brain edema and inhalational agents
- C. Propofol
- D. Mild hypothermia

**Ans: B**

**Q5. By how much does each mm increase in pCO<sub>2</sub> above 40 mm Hg increase CBF?**

- A. 5-10 mL/min
- B. 15-30 mL/min
- C. 50-70 mL/min
- D. 35-50 mL/min

**Ans: B**

**Q6. When does pO<sub>2</sub> increase cerebral blood flow?**

- A. Only if pO<sub>2</sub> is below 60 mm Hg
- B. When pO<sub>2</sub> is between 60-80 mm Hg
- C. When pO<sub>2</sub> is above 100 mm Hg
- D. Always

**Ans: A**

**Q7. Which IV anesthetics decrease cerebral blood flow?**

- A. All agents decrease CBF
- B. All inhalational agents, ketamine and opioids increase CBF while barbiturates, propofol, etomidate and benzodiazepines decrease CBF
- C. Only barbiturates
- D. Ketamine and opioids

**Ans: B**



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| <p><b>Q8. What increases the cerebral metabolic rate (CMR)?</b></p> <p>A. Barbiturates<br/>B. Propofol<br/>C. Hypothermia and anesthetics<br/>D. Hyperthermia and seizures</p>                                                                                                               | <p><b>Ans: D</b></p> |
| <p><b>Q9. What is the normal intracranial tension (ICT)?</b></p> <p>A. 20-25 mm Hg<br/>B. 10-15 mm Hg<br/>C. 5-8 mm Hg<br/>D. 1-5 mm Hg</p>                                                                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q10. At what level does ICT become significant and need aggressive treatment?</b></p> <p>A. &gt; 20 mm Hg<br/>B. &gt; 15 mm Hg<br/>C. &gt; 30 mm Hg<br/>D. &gt; 25 mm Hg</p>                                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q11. What are the signs of raised ICT?</b></p> <p>A. Headache, vomiting, altered sensorium, hypertension and bradycardia (Cushing reflex), papilledema, seizures, apnea, unconsciousness and death<br/>B. Only papilledema<br/>C. Only headache<br/>D. Tachycardia and hypotension</p> | <p><b>Ans: A</b></p> |
| <p><b>Q12. What midline shift on CT scan indicates very high ICT?</b></p> <p>A. &gt; 0.5 cm<br/>B. &gt; 1.0 cm<br/>C. &gt; 1.5 cm<br/>D. &gt; 0.2 cm</p>                                                                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q13. What is the most rapid method of reducing ICT?</b></p> <p>A. Mannitol<br/>B. Steroids<br/>C. Diuretics<br/>D. Hyperventilation</p>                                                                                                                                                | <p><b>Ans: D</b></p> |
| <p><b>Q14. What is the recommended pCO<sub>2</sub> range to maintain during hyperventilation for reducing ICT?</b></p> <p>A. 25-30 mm Hg<br/>B. 35-40 mm Hg<br/>C. 30-35 mm Hg<br/>D. 20-25 mm Hg</p>                                                                                        | <p><b>Ans: A</b></p> |

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| <p><b>Q15. What is the most commonly used hyperosmotic drug for reducing ICT?</b></p> <ul style="list-style-type: none"> <li>A. Urea</li> <li>B. Mannitol</li> <li>C. Glycerol</li> <li>D. Hypertonic saline</li> </ul>                                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q16. Why should the use of mannitol in head injury and stroke be judicious?</b></p> <ul style="list-style-type: none"> <li>A. It causes hyperglycemia</li> <li>B. It has no effect</li> <li>C. If blood brain barrier is disrupted, mannitol can enter brain tissue increasing cerebral edema</li> <li>D. It causes hyponatremia</li> </ul>      | <p><b>Ans: C</b></p> |
| <p><b>Q17. Which diuretic is very effective for reducing ICT particularly when blood brain barrier is disrupted?</b></p> <ul style="list-style-type: none"> <li>A. Hydrochlorothiazide</li> <li>B. Spironolactone</li> <li>C. Mannitol</li> <li>D. Furosemide</li> </ul>                                                                               | <p><b>Ans: D</b></p> |
| <p><b>Q18. How do barbiturates (Thiopentone) reduce ICT?</b></p> <ul style="list-style-type: none"> <li>A. By increasing CMR</li> <li>B. By causing cerebral vasoconstriction which decreases CSF production</li> <li>C. By increasing CBF</li> <li>D. By causing vasodilation</li> </ul>                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q19. Which posture helps decrease ICT by increasing venous drainage?</b></p> <ul style="list-style-type: none"> <li>A. Head up position</li> <li>B. Supine position</li> <li>C. Prone position</li> <li>D. Trendelenburg position</li> </ul>                                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q20. What can increase ICT by compressing jugular veins?</b></p> <ul style="list-style-type: none"> <li>A. Neck flexion</li> <li>B. Lateral rotation</li> <li>C. Extension</li> <li>D. Head up position</li> </ul>                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q21. If ICT is not controlled by conservative measures, how should CSF be drained?</b></p> <ul style="list-style-type: none"> <li>A. Through lumbar puncture</li> <li>B. Through subdural space</li> <li>C. Through ventricles (never through lumbar puncture as brain herniation can cause death)</li> <li>D. Through epidural space</li> </ul> | <p><b>Ans: C</b></p> |

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| <p><b>Q22. What is the effect of all inhalational agents on cerebral blood flow and ICT?</b></p> <p>A. No effect<br/> B. All inhalational agents (including nitrous oxide) increase CBF and hence ICT<br/> C. All decrease CBF and ICT<br/> D. Only halothane increases CBF</p>                                        | <p><b>Ans: B</b></p> |
| <p><b>Q23. What is the order of vasodilating potency of inhalational agents?</b></p> <p>A. Halothane &gt;&gt; Desflurane ≈ Isoflurane &gt; Sevoflurane<br/> B. Sevoflurane &gt; Isoflurane &gt; Desflurane &gt; Halothane<br/> C. Desflurane &gt; Halothane &gt; Isoflurane &gt; Sevoflurane<br/> D. All are equal</p> | <p><b>Ans: A</b></p> |
| <p><b>Q24. By how much can Isoflurane and Sevoflurane decrease cerebral metabolic rate (CMR)?</b></p> <p>A. Up to 50%<br/> B. Up to 100%<br/> C. Up to 25%<br/> D. Up to 75%</p>                                                                                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q25. Which inhalational agent is preferred for neurosurgery in current day practice?</b></p> <p>A. Halothane<br/> B. Isoflurane<br/> C. Desflurane<br/> D. Sevoflurane</p>                                                                                                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q26. Which IV anesthetics are most potent in decreasing CBF, CMR, CMO<sub>2</sub> and ICT?</b></p> <p>A. Opioids<br/> B. Barbiturates<br/> C. Ketamine<br/> D. Propofol</p>                                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q27. What is the effect of opioids on CBF in neurosurgical patients?</b></p> <p>A. Decrease CMR and CMO<sub>2</sub> but increase CBF (directly acting cerebral vasodilator) and hence ICT<br/> B. Decrease CBF and ICT<br/> C. No effect<br/> D. Increase CMR</p>                                                | <p><b>Ans: A</b></p> |
| <p><b>Q28. By how much can Ketamine increase CBF and ICT?</b></p> <p>A. Up to 20%<br/> B. Up to 90%<br/> C. Up to 30%<br/> D. Up to 60%</p>                                                                                                                                                                            | <p><b>Ans: D</b></p> |

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| <p><b>Q29. What is the effect of succinylcholine on ICT?</b></p> <p>A. Increases ICT<br/>B. No effect<br/>C. Variable effect<br/>D. Decreases ICT</p>                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q30. Do non-depolarizing muscle relaxants cross the blood brain barrier?</b></p> <p>A. Only vecuronium<br/>B. No, they do not cross the blood brain barrier<br/>C. Yes, all of them<br/>D. Only pancuronium</p>                           | <p><b>Ans: B</b></p> |
| <p><b>Q31. What is the effect of lignocaine on CBF, CMR and CMO2?</b></p> <p>A. Increases all<br/>B. Decreases CBF, CMR, CMO2 (but at toxic doses can induce seizure and increase CMR)<br/>C. Only decreases CMR<br/>D. No effect</p>           | <p><b>Ans: B</b></p> |
| <p><b>Q32. How do vasopressors (Mephentermine, Ephedrine, Epinephrine) affect CBF?</b></p> <p>A. No effect on CBF<br/>B. Directly increase CBF<br/>C. Increase CBF indirectly by increasing cerebral perfusion pressure<br/>D. Decrease CBF</p> | <p><b>Ans: C</b></p> |
| <p><b>Q33. Why should glucose containing solutions be avoided in neurosurgery?</b></p> <p>A. Hyperglycemia can precipitate cerebral edema<br/>B. They cause hypernatremia<br/>C. They are too viscous<br/>D. They cause hypoglycemia</p>        | <p><b>Ans: A</b></p> |
| <p><b>Q34. What is the best policy for fluid management in neurosurgical patients?</b></p> <p>A. Only Ringer lactate<br/>B. Alternate ringer and normal saline liter by liter<br/>C. Only normal saline<br/>D. Only colloids</p>                | <p><b>Ans: B</b></p> |
| <p><b>Q35. Is routine use of hypothermia recommended in neurosurgery?</b></p> <p>A. Yes, always<br/>B. No, routine use of hypothermia is not recommended<br/>C. Only in spine surgery<br/>D. Only in posterior fossa surgery</p>                | <p><b>Ans: B</b></p> |
| <p><b>Q36. What is the optimal blood sugar range to maintain in neurosurgical patients?</b></p> <p>A. 80-120 mg%<br/>B. 100-140 mg%<br/>C. 140-180 mg%<br/>D. 180-220 mg%</p>                                                                   | <p><b>Ans: C</b></p> |

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| <p><b>Q37. What is the most important monitor for neurosurgical patients?</b></p> <p>A. Capnography<br/>B. ECG<br/>C. Temperature<br/>D. Pulse oximetry</p>                                                                                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q38. What is the beneficial range of pCO<sub>2</sub> during monitoring in neurosurgery?</b></p> <p>A. 40-45 mm Hg<br/>B. 20-25 mm Hg<br/>C. 35-40 mm Hg<br/>D. 25-30 mm Hg (moderate hypocapnia)</p>                                                                                             | <p><b>Ans: D</b></p> |
| <p><b>Q39. Why should hypoxia be avoided during pulse oximetry monitoring in neurosurgery?</b></p> <p>A. It causes hypothermia<br/>B. It causes bradycardia<br/>C. It decreases CMR<br/>D. It increases ICT by causing cerebral vasodilation</p>                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q40. What is essential for smooth emergence in neurosurgical patients?</b></p> <p>A. Emergence free of coughing, straining and hypertension<br/>B. Use of succinylcholine<br/>C. Rapid extubation<br/>D. Deep extubation always</p>                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q41. What is the best method to achieve smooth emergence in neurosurgery?</b></p> <p>A. Give succinylcholine<br/>B. Deep extubation<br/>C. Inject lignocaine before extubation, continue inhalational (at least N<sub>2</sub>O) till dressing and use beta blockers<br/>D. Only give opioids</p> | <p><b>Ans: C</b></p> |
| <p><b>Q42. What is the target CPP (MAP-ICP) to maintain during neurosurgery?</b></p> <p>A. 100-120 mm Hg<br/>B. 40-60 mm Hg<br/>C. 70-100 mm Hg<br/>D. 50-70 mm Hg</p>                                                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q43. Are central neuraxial blocks contraindicated in raised ICT?</b></p> <p>A. They are absolutely contraindicated<br/>B. They are safe<br/>C. Only epidural is contraindicated<br/>D. They are relatively contraindicated</p>                                                                   | <p><b>Ans: A</b></p> |

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| <p><b>Q44. Which induction agent is preferred over thiopentone for neurosurgery?</b></p> <ul style="list-style-type: none"> <li>A. Midazolam</li> <li>B. Propofol</li> <li>C. Etomidate</li> <li>D. Ketamine</li> </ul>                                                                                                                                                                 | <p><b>Ans: B</b></p> |
| <p><b>Q45. Why can intubation be problematic in neurosurgical patients?</b></p> <ul style="list-style-type: none"> <li>A. Intubation can significantly increase ICT due to response to laryngoscopy</li> <li>B. Vocal cord paralysis</li> <li>C. Difficult anatomy</li> <li>D. Risk of aspiration</li> </ul>                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q46. Is Suxamethonium absolutely contraindicated for neurosurgery?</b></p> <ul style="list-style-type: none"> <li>A. Only in posterior fossa surgery</li> <li>B. Yes, absolutely contraindicated</li> <li>C. Not absolutely contraindicated - can be safely used for rapid sequence induction or difficult airway</li> <li>D. Only in spine surgery</li> </ul>                    | <p><b>Ans: C</b></p> |
| <p><b>Q47. What agents should be used for maintenance if ICP is too high and surgical field too tight?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Nitrous oxide and sevoflurane</li> <li>C. High concentration inhalational agents</li> <li>D. IV agents only - inhalational agents should be avoided</li> </ul>                                           | <p><b>Ans: D</b></p> |
| <p><b>Q48. What should be assumed about all head injury patients regarding cervical injury?</b></p> <ul style="list-style-type: none"> <li>A. Only in high-velocity trauma</li> <li>B. Only in elderly patients</li> <li>C. Cervical injury is unlikely</li> <li>D. All patients should be assumed to be having cervical injury (incidence up to 15%) until proved otherwise</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q49. When should nasal intubation NOT be performed in head injury?</b></p> <ul style="list-style-type: none"> <li>A. In all head injury patients</li> <li>B. Only in pediatric patients</li> <li>C. When GCS is above 10</li> <li>D. Unless basal skull fracture has been ruled out</li> </ul>                                                                                    | <p><b>Ans: D</b></p> |
| <p><b>Q50. What is the best method for assessing neurologic status in head injury?</b></p> <ul style="list-style-type: none"> <li>A. Modified Glasgow Coma Scale</li> <li>B. MRI</li> <li>C. EEG</li> <li>D. CT scan</li> </ul>                                                                                                                                                         | <p><b>Ans: A</b></p> |

|                                                                                                                                                                                                                                                               |                      |
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| <p><b>Q51. What is the normal Glasgow Coma Scale score?</b></p> <p>A. 10/10<br/>B. 15/15<br/>C. 12/12<br/>D. 20/20</p>                                                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q52. A GCS score of 7 or less persisting for more than 6 hours is associated with what mortality?</b></p> <p>A. More than 10%<br/>B. More than 20%<br/>C. More than 35%<br/>D. More than 50%</p>                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q53. Why is DIC commonly associated with head injury?</b></p> <p>A. Due to hypoxia<br/>B. Due to blood loss<br/>C. Due to hypothermia<br/>D. Due to release of brain thromboplastin</p>                                                                 | <p><b>Ans: D</b></p> |
| <p><b>Q54. What is the most sensitive tool to detect venous air embolism (VAE)?</b></p> <p>A. CVP monitoring<br/>B. End tidal CO<sub>2</sub><br/>C. Precordial Doppler<br/>D. Transesophageal echocardiography (TEE)</p>                                      | <p><b>Ans: D</b></p> |
| <p><b>Q55. At what volume of entrained air does VAE become clinically significant?</b></p> <p>A. &gt; 25 mL<br/>B. &gt; 200 mL<br/>C. &gt; 100 mL<br/>D. &gt; 50 mL</p>                                                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q56. What is the incidence of VAE in posterior fossa craniotomies?</b></p> <p>A. 20-40%<br/>B. 5-10%<br/>C. 1-5%<br/>D. 50-60%</p>                                                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q57. What is the first step in treatment of venous air embolism?</b></p> <p>A. Start CPR<br/>B. Stop nitrous oxide and start 100% oxygen (N<sub>2</sub>O can expand the size of air embolus)<br/>C. Give vasopressors<br/>D. Trendelenburg position</p> | <p><b>Ans: B</b></p> |

**Q58. What is the sensitivity order for detection of air embolism in decreasing order?**

- A. Clinical signs > ECG > CVP > TEE
- B. ECG > CVP > TEE > Doppler
- C. Doppler > ECG > CVP > TEE
- D. TEE > Precordial Doppler > PAP, ETCO<sub>2</sub>, ET nitrogen > CVP > ECG > Clinical signs

**Ans: D**

**Q59. What are the clinical signs of venous air embolism?**

- A. Hypotension, tachycardia, cyanosis, mill wheel murmur, cardiovascular collapse
- B. Fever and tachypnea
- C. Only hypotension
- D. Hypertension and bradycardia

**Ans: A**

**Q60. After stroke surgery, when is the period of maximal vasospasm risk that should be avoided?**

- A. 4-14 days
- B. 1-3 days
- C. 15-21 days
- D. 21-30 days

**Ans: A**



# Answer Key & Explanations

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## **Q1. Answer: C — Difference between mean arterial pressure and intracranial pressure**

Cerebral perfusion pressure (CPP) is the difference between mean arterial pressure and intracranial pressure.

Topic: Cerebral Physiology

## **Q2. Answer: C — 50 mL**

CBF is 50 mL/100 g of brain tissue.

Topic: Cerebral Physiology

## **Q3. Answer: B — 60 and 150 mm Hg**

CBF is maintained at constant level by autoregulation when mean arterial pressure is between 60 mm Hg and 150 mm Hg.

Topic: Cerebral Physiology

## **Q4. Answer: B — Cerebral ischemia, head injury, hypercarbia, hypoxia, brain edema and inhalational agents**

Cerebral ischemia, head injury, hypercarbia, hypoxia, brain edema and inhalational agents can abolish this autoregulatory mechanism.

Topic: Cerebral Physiology

## **Q5. Answer: B — 15-30 mL/min**

Each mm increase in pCO<sub>2</sub> above 40 mm Hg increases CBF by 15-30 mL/min.

Topic: Cerebral Physiology

## **Q6. Answer: A — Only if pO<sub>2</sub> is below 60 mm Hg**

pO<sub>2</sub> increases CBF only if pO<sub>2</sub> is below 60 mm Hg.

Topic: Cerebral Physiology

## **Q7. Answer: B — All inhalational agents, ketamine and opioids increase CBF while barbiturates, propofol, etomidate and benzodiazepines decrease CBF**

All inhalational agents, ketamine and opioids increase CBF while propofol, etomidate and benzodiazepines decreases the CBF.

Topic: Cerebral Physiology

## **Q8. Answer: D — Hyperthermia and seizures**

Hyperthermia, seizures increase the cerebral metabolic rate (CMR) while hypothermia and anesthetics decreases the cerebral metabolic rate.

Topic: Cerebral Physiology

## **Q9. Answer: B — 10-15 mm Hg**

Normal ICT is 10-15 mm Hg.

Topic: Cerebral Physiology

## **Q10. Answer: A — > 20 mm Hg**

ICT > 20 mm Hg is significant and needs aggressive treatment.

Topic: Cerebral Physiology

**Q11. Answer: A — Headache, vomiting, altered sensorium, hypertension and bradycardia (Cushing reflex), papilledema, seizures, apnea, unconsciousness and death**

Signs include headache, vomiting, altered sensorium, hypertension and bradycardia (Cushing reflex), papilledema, seizures, apnea, unconsciousness and death.

Topic: Cerebral Physiology

**Q12. Answer: A — > 0.5 cm**

At very high ICT, midline shift > 0.5 cm on computed tomography (CT) scan is seen.

Topic: Cerebral Physiology

**Q13. Answer: D — Hyperventilation**

Hyperventilation is the most rapid method of reducing ICT. It reduces ICT by causing hypocapnia which causes cerebral vasoconstriction to decrease CSF production.

Topic: Methods to Decrease ICT

**Q14. Answer: A — 25-30 mm Hg**

It is recommended to maintain pCO<sub>2</sub> between 25-30 mm Hg. Excessive hypocapnia (< 25 mm Hg) can cause cerebral ischemia.

Topic: Methods to Decrease ICT

**Q15. Answer: B — Mannitol**

Among hyperosmotic agents, mannitol is most commonly used. These drugs are retained in intravascular compartment increasing plasma osmolality, drawing water from brain tissue.

Topic: Methods to Decrease ICT

**Q16. Answer: C — If blood brain barrier is disrupted, mannitol can enter brain tissue increasing cerebral edema**

If blood brain barrier is disrupted (head injuries, stroke) these agents can enter the brain tissue increasing the cerebral edema. Therefore, the use of mannitol in head injury and stroke should be judicious.

Topic: Methods to Decrease ICT

**Q17. Answer: D — Furosemide**

Furosemide is a very effective method of reducing ICT particularly when blood brain barrier is disrupted.

Topic: Methods to Decrease ICT

**Q18. Answer: B — By causing cerebral vasoconstriction which decreases CSF production**

Barbiturates: Thiopentone by causing cerebral vasoconstriction is effective in decreasing the production of CSF and hence ICT.

Topic: Methods to Decrease ICT

**Q19. Answer: A — Head up position**

Head up position by increasing venous drainage decreases ICT.

Topic: Methods to Decrease ICT

**Q20. Answer: A — Neck flexion**

Neck flexion by compressing jugular veins can increase ICT.

Topic: Methods to Decrease ICT

**Q21. Answer: C — Through ventricles (never through lumbar puncture as brain herniation can cause death)**

CSF drainage: If ICT is not controlled by conservative measure it should be drained through ventricles (never through lumbar puncture, brain herniation can even cause death).

Topic: Methods to Decrease ICT

**Q22. Answer: B — All inhalational agents (including nitrous oxide) increase CBF and hence ICT**

All inhalational agents (including nitrous oxide) increases the cerebral blood flow and hence ICT.

Topic: Pharmacology

**Q23. Answer: A — Halothane >> Desflurane ≈ Isoflurane > Sevoflurane**

The order of vasodilating potency is halothane >> Desflurane ≈ Isoflurane > Sevoflurane.

Topic: Pharmacology

**Q24. Answer: A — Up to 50%**

Isoflurane and Sevoflurane can decrease CMR up to 50%.

Topic: Pharmacology

**Q25. Answer: D — Sevoflurane**

Due to better preservation of autoregulation, equivalent reduction in CMR and smoother recovery, sevoflurane is preferred over isoflurane as an inhalational agent of choice for neurosurgery.

Topic: Pharmacology

**Q26. Answer: B — Barbiturates**

Barbiturates, propofol, etomidate and benzodiazepines decrease CBF, CMR, CMO<sub>2</sub> and ICT. Barbiturates are most potent in this aspect.

Topic: Pharmacology

**Q27. Answer: A — Decrease CMR and CMO<sub>2</sub> but increase CBF (directly acting cerebral vasodilator) and hence ICT**

Opioids decrease CMR and CMO<sub>2</sub> but increases CBF (directly acting cerebral vasodilator) and hence ICT. This increase in CBF is significant when blood brain barrier is disrupted.

Topic: Pharmacology

**Q28. Answer: D — Up to 60%**

Ketamine increases CBF and ICT very significantly (up to 60%). Ketamine also increases CMR and CMO<sub>2</sub>.

Topic: Pharmacology

**Q29. Answer: A — Increases ICT**

Succinylcholine increases the ICT.

Topic: Pharmacology

**Q30. Answer: B — No, they do not cross the blood brain barrier**

Non-depolarizing muscle relaxants do not cross the blood brain barrier.

Topic: Pharmacology

**Q31. Answer: B — Decreases CBF, CMR, CMO<sub>2</sub> (but at toxic doses can induce seizure and increase CMR)**

Lignocaine decreases CBF, CMR, CMO<sub>2</sub> (but at toxic doses it can induce seizure and can increase CMR).

Topic: Pharmacology

**Q32. Answer: C — Increase CBF indirectly by increasing cerebral perfusion pressure**

Vasopressors increase CBF indirectly by increasing cerebral perfusion pressure.

Topic: Pharmacology

**Q33. Answer: A — Hyperglycemia can precipitate cerebral edema**

As hyperglycemia can precipitate cerebral edema therefore, Glucose containing solutions should be avoided.

Topic: General Considerations

**Q34. Answer: B — Alternate ringer and normal saline liter by liter**

The best policy is to alternate ringer and normal saline liter by liter. Ringer lactate is hypoosmolar and large quantities can aggravate cerebral edema. Normal saline in large quantities can cause hyperchloremic metabolic acidosis.

Topic: General Considerations

**Q35. Answer: B — No, routine use of hypothermia is not recommended**

Routine use of hypothermia is not recommended.

Topic: General Considerations

**Q36. Answer: C — 140-180 mg%**

Both hyperglycemia as well as hypoglycemia can be detrimental therefore, blood sugar should be best maintained between 140 and 180 mg%.

Topic: General Considerations

**Q37. Answer: A — Capnography**

Most important is Capnography. Hypercapnia is the most potent cerebral vasodilator therefore, can significantly increase the ICT.

Topic: General Considerations

**Q38. Answer: D — 25-30 mm Hg (moderate hypocapnia)**

Moderate hypocapnia (pCO<sub>2</sub> between 25 and 30 mm Hg) is beneficial by decreasing ICT however excessive hypocapnia (pCO<sub>2</sub> < 25 mm Hg) can cause cerebral ischemia.

Topic: General Considerations

**Q39. Answer: D — It increases ICT by causing cerebral vasodilation**

Hypoxia should be avoided as it increases ICT by causing cerebral vasodilation.

Topic: General Considerations

**Q40. Answer: A — Emergence free of coughing, straining and hypertension**

Smooth emergence, i.e. emergence free of coughing, straining, and hypertension is must in neurosurgical patients as these can increase ICT and cause massive bleeding.

Topic: General Considerations

**Q41. Answer: C — Inject lignocaine before extubation, continue inhalational (at least N2O) till dressing and use beta blockers**

The best possible methods to achieve smooth emergence is to inject lignocaine before extubation, continue inhalational (at least N2O) till dressing is over and use beta blockers (Esmolol, Labetalol) if necessary.

Topic: General Considerations

**Q42. Answer: C — 70-100 mm Hg**

The aim should be to maintain CPP (MAP-ICP) between 70 and 100 mm Hg not only in intraoperative period but also in post-operative period for at least 2-3 days.

Topic: Raised ICP

**Q43. Answer: A — They are absolutely contraindicated**

Central neuraxial blocks are absolutely contraindicated in raised ICP.

Topic: Raised ICP

**Q44. Answer: B — Propofol**

Although Thiopentone has more favorable effect on CMR and ICP than propofol but still due to shorter half-life propofol is preferred over thiopentone for induction.

Topic: Raised ICP

**Q45. Answer: A — Intubation can significantly increase ICP due to response to laryngoscopy**

Intubation can significantly increase the ICP therefore, response to laryngoscopy and intubation must be blunted with lignocaine, adequate depth and beta blockers.

Topic: Raised ICP

**Q46. Answer: C — Not absolutely contraindicated - can be safely used for rapid sequence induction or difficult airway**

Suxamethonium is not absolutely contraindicated for neurosurgery and can be safely used if there is clear indication like rapid sequence induction or possibility of difficult airway.

Topic: Raised ICP

**Q47. Answer: D — IV agents only - inhalational agents should be avoided**

If ICP is too high to be controlled or surgical field is too tight that inhalational agents should be avoided and anesthesia is maintained on IV agents.

Topic: Raised ICP

**Q48. Answer: D — All patients should be assumed to be having cervical injury (incidence up to 15%) until proved otherwise**

All patients of head injury should be assumed to be having cervical injury (incidence up to 15%) until proved otherwise.

Topic: Head Injury

**Q49. Answer: D — Unless basal skull fracture has been ruled out**

Nasal intubation should not be performed unless basal skull fracture has been ruled out.

Topic: Head Injury

**Q50. Answer: A — Modified Glasgow Coma Scale**

The best method for assessing neurologic status is modified Glasgow Coma scale.

*Topic: Head Injury*

**Q51. Answer: B — 15/15**

Normal score is 15/15. Severe head injury is stated to be present if score is 7 or less.

*Topic: Head Injury*

**Q52. Answer: C — More than 35%**

Severe head injury is stated to be present if score is 7 or less and if persist for more than 6 hours is associated with more than 35% mortality.

*Topic: Head Injury*

**Q53. Answer: D — Due to release of brain thromboplastin**

DIC is commonly associated with head injury due to release of brain thromboplastin and it must be ruled out in preoperative period.

*Topic: Head Injury*

**Q54. Answer: D — Transesophageal echocardiography (TEE)**

The most sensitive tool to detect VAE is transesophageal echocardiography (TEE). It can detect as low as 0.25 mL of air.

*Topic: Posterior Fossa Surgery*

**Q55. Answer: C — > 100 mL**

VAE can become clinically significant if the volume of entertained air > 100 mL (however death has been reported with as low as 5 mL of air).

*Topic: Posterior Fossa Surgery*

**Q56. Answer: A — 20-40%**

The incidence of venous air embolism (VAE) can be as high as 20-40% in posterior fossa craniotomies however, fortunately the incidence of clinically significant VAE is very less.

*Topic: Posterior Fossa Surgery*

**Q57. Answer: B — Stop nitrous oxide and start 100% oxygen (N<sub>2</sub>O can expand the size of air embolus)**

Stop nitrous oxide and start 100% oxygen. Nitrous oxide can expand the size of air embolus.

*Topic: Posterior Fossa Surgery*

**Q58. Answer: D — TEE > Precordial Doppler > PAP, ETCO<sub>2</sub>, ET nitrogen > CVP > ECG > Clinical signs**

The sensitivity in decreasing order is: TEE > Precordial Doppler > PAP, ETCO<sub>2</sub>, ET nitrogen > CVP > ECG > clinical signs.

*Topic: Posterior Fossa Surgery*

**Q59. Answer: A — Hypotension, tachycardia, cyanosis, mill wheel murmur, cardiovascular collapse**

Clinical signs include hypotension, tachycardia, cyanosis, mill wheel murmur, cardiovascular collapse. Clinical signs appear late in venous air embolism.

*Topic: Posterior Fossa Surgery*

**Q60. Answer: A — 4-14 days**

After stroke surgery should be avoided in period of maximal vasospasm risk (4-14 days) as far as possible.

*Topic: Stroke/Cerebral Aneurysms*

# Anesthesia for Obstetrics MCQ – Chapter 28

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 28)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

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**Q1. During pregnancy, the cardiac output increases by approximately:**

- A. 40%
- B. 15%
- C. 20%
- D. 60%

**Ans: A**

**Q2. Pregnant patients are more vulnerable to cardiac failure because of:**

- A. Hyperdynamic circulation
- B. Decreased blood volume
- C. Bradycardia
- D. Hypotension

**Ans: A**

**Q3. During pregnancy, minute ventilation increases by 50% and functional residual capacity (FRC) decreases by:**

- A. 20%
- B. 40%
- C. 10%
- D. 50%

**Ans: A**

**Q4. Why is preoxygenation a must in pregnant patients?**

- A. Due to decreased FRC, ERV and increased oxygen requirement, making them more vulnerable for hypoxia
- B. Due to increased FRC
- C. Due to decreased oxygen consumption
- D. Due to bradycardia

**Ans: A**

**Q5. Progesterone decreases the minimum alveolar concentration (MAC) by:**

- A. 25–40%, making pregnant patients more susceptible to anesthetic over dosage
- B. 5–10%
- C. 50–60%
- D. It increases MAC by 25%

**Ans: A**

**Q6. To prevent high spinal in pregnant patients (due to engorged epidural veins reducing subarachnoid space), the dose of local anesthetic for spinal has to be reduced by:**

- A. 30–40%
- B. 10–15%
- C. 50–60%
- D. 5–10%

**Ans: A**

**Q7. Why are all pregnant patients considered full stomach even if fasting for more than 8 hours?**

- A. Gastric emptying is delayed (progesterone), gravid uterus makes LES incompetent, progesterone relaxes LES, and gastric contents are more acidic
- B. They eat more frequently
- C. They have increased gastric secretions only
- D. They have a smaller stomach

**Ans: A**

**Q8. What is the cause of supine hypotension syndrome (SHS) in pregnancy?**

- A. In supine position, the gravid uterus compresses the inferior vena cava and aorta, decreasing cardiac output
- B. Increased venous return
- C. Decreased blood volume
- D. Excessive vasodilatation

**Ans: A**

**Q9. To prevent supine hypotension syndrome (SHS), the pregnant patient should:**

- A. Lie in left lateral position, or a 15° wedge placed under right buttock, or table tilted 15° to left, or uterus manually displaced to left
- B. Lie flat supine
- C. Be placed in Trendelenburg position
- D. Lie in right lateral position

**Ans: A**

**Q10. What is the normal uterine blood flow in pregnancy?**

- A. 500–700 mL/min (10% of cardiac output)
- B. 100–200 mL/min
- C. 1000–1500 mL/min
- D. 250 mL/min

**Ans: A**

**Q11. All anesthetic drugs cross the placenta to fetal circulation EXCEPT:**

- A. Muscle relaxants (only gallamine has significant transfer) and glycopyrrolate
- B. Local anesthetics
- C. Opioids
- D. Inhalational agents

**Ans: A**

**Q12. Local anesthetics and opioids (weak bases) cross the placenta in unionized form but become ionized in the fetal circulation and cannot come back. This phenomenon is called:**

- A. Ion trapping
- B. Second gas effect
- C. Concentration effect
- D. Diffusion hypoxia

**Ans: A**

**Q13. Regional anesthesia is preferred over general anesthesia for cesarean section. The mortality in obstetrics during GA is:**

- A. Twice as compared to regional anesthesia
- B. Equal to regional anesthesia
- C. Half of regional anesthesia
- D. Five times regional anesthesia

**Ans: A**

|                                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |
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| <p><b>Q14. What level of block is required for cesarean section?</b></p> <p>A. Up to T4<br/>B. Up to T10<br/>C. Up to T6<br/>D. Up to T8</p>                                                                                                                                                                                                                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q15. Which vasopressor has traditionally been the vasopressor of choice in pregnancy, as it does not produce significant vasoconstriction in the uteroplacental bed?</b></p> <p>A. Ephedrine<br/>B. Phenylephrine<br/>C. Noradrenaline<br/>D. Methoxamine</p>                                                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q16. Recent human trials have shown that phenylephrine produces less fetal acidosis than ephedrine, making it:</b></p> <p>A. A preferred vasopressor over ephedrine in current day anesthetic practice<br/>B. Contraindicated in pregnancy<br/>C. Inferior to ephedrine<br/>D. Only used in PIH</p>                                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q17. Why is the onset of epidural anesthesia slow (15–20 minutes), making it reserved for elective cases or PIH?</b></p> <p>A. Onset takes a long time, so it is reserved for elective cases or specific conditions like pregnancy-induced hypertension (PIH)<br/>B. Because it requires general anesthesia first<br/>C. Because it is contraindicated in cesarean<br/>D. Because it causes high spinal</p>                | <p><b>Ans: A</b></p> |
| <p><b>Q18. Regarding propofol use in pregnancy for cesarean induction, current evidence shows that:</b></p> <p>A. Propofol does not affect the APGAR and can be safely used in pregnancy<br/>B. Propofol is absolutely contraindicated in pregnancy<br/>C. Propofol always lowers APGAR scores<br/>D. Propofol causes uterine atony</p>                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q19. Why must all pregnant patients undergoing GA for cesarean be premedicated (with metoclopramide, sodium citrate and ranitidine) and managed like rapid sequence induction?</b></p> <p>A. Because all pregnant patients must be considered full stomach irrespective of their fasting status<br/>B. Because they have low blood pressure<br/>C. Because of increased cardiac output<br/>D. Because of decreased MAC</p> | <p><b>Ans: A</b></p> |

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| <p><b>Q20. All inhalational agents can relax the uterus and cause postpartum hemorrhage (PPH). At what concentration is significant uterine relaxation NOT produced?</b></p> <p>A. At concentration less than half MAC (&lt; 0.5 MAC)<br/> B. At concentration above 1 MAC<br/> C. At any concentration<br/> D. At concentration above 2 MAC</p>       | <p><b>Ans: A</b></p> |
| <p><b>Q21. When can opioids be given during GA for cesarean section?</b></p> <p>A. Only after the delivery of the baby<br/> B. Before induction<br/> C. During induction<br/> D. Opioids are never given</p>                                                                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q22. What is the anesthetic technique of choice for cesarean section in patients with pregnancy-induced hypertension (PIH)/preeclampsia?</b></p> <p>A. Epidural<br/> B. Spinal<br/> C. General anesthesia<br/> D. Local infiltration</p>                                                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q23. Before giving central neuraxial blocks in a PIH patient, it is important to rule out which condition?</b></p> <p>A. HELLP syndrome<br/> B. Diabetes<br/> C. Anemia<br/> D. Hypothyroidism</p>                                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q24. In PIH patients on magnesium therapy, what is the anesthetic concern during general anesthesia?</b></p> <p>A. Magnesium can potentiate the action of muscle relaxants or cause respiratory or cardiac arrest at toxic levels<br/> B. Magnesium causes hypertension<br/> C. Magnesium increases MAC<br/> D. Magnesium causes tachycardia</p> | <p><b>Ans: A</b></p> |
| <p><b>Q25. Pain afferents from the uterus and cervix during labor travel up to which spinal levels?</b></p> <p>A. T10-L1<br/> B. T4-T6<br/> C. L2-L4<br/> D. S2-S4</p>                                                                                                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q26. What is the most commonly used technique for labor analgesia (painless labor)?</b></p> <p>A. Continuous lumbar epidural<br/> B. Spinal anesthesia<br/> C. Paracervical block<br/> D. Pudendal nerve block</p>                                                                                                                               | <p><b>Ans: A</b></p> |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q27. According to current recommendations, when should painless labor be started?</b></p> <ul style="list-style-type: none"> <li>A. When the patient demands, irrespective of the stage of labor</li> <li>B. Only when cervix is at least 3–4 cm dilated</li> <li>C. Only in second stage</li> <li>D. Only after full dilatation</li> </ul>                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q28. When the concentration of bupivacaine used in epidural labor analgesia is too low (0.0625%), allowing the patient to move out of bed and sit, it is called:</b></p> <ul style="list-style-type: none"> <li>A. Walking epidural</li> <li>B. Continuous spinal</li> <li>C. Saddle block</li> <li>D. Caudal block</li> </ul>                                                                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q29. Why should lignocaine with adrenaline be avoided for the test dose in a pregnant patient?</b></p> <ul style="list-style-type: none"> <li>A. Adrenaline can decrease uterine blood supply, and there is already too much variation in heart rate due to labor making it impossible to differentiate intravascular injection</li> <li>B. Adrenaline causes hypotension</li> <li>C. Adrenaline is teratogenic</li> <li>D. Adrenaline causes uterine rupture</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q30. Low concentration epidurals for labor analgesia:</b></p> <ul style="list-style-type: none"> <li>A. Neither increase the rate of cesarean section nor the rate of instrumental delivery, but can delay the duration of labor by 30–60 minutes</li> <li>B. Always increase cesarean rate</li> <li>C. Always increase instrumental delivery</li> <li>D. Shorten labor duration</li> </ul>                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q31. Which is the opioid of choice for pain relief in labor when given as a parenteral narcotic?</b></p> <ul style="list-style-type: none"> <li>A. Remifentanyl (as it is metabolized in maternal plasma, with insignificant amount reaching the fetus)</li> <li>B. Morphine</li> <li>C. Pethidine</li> <li>D. Pentazocine</li> </ul>                                                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q32. What is the anesthesia of choice for manual removal of retained placenta?</b></p> <ul style="list-style-type: none"> <li>A. General anesthesia with inhalational agents (as all inhalational agents are uterine relaxants)</li> <li>B. Spinal anesthesia</li> <li>C. Epidural anesthesia</li> <li>D. Local infiltration</li> </ul>                                                                                                                                  | <p><b>Ans: A</b></p> |

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| <p><b>Q33. For non-obstetric surgeries during pregnancy, elective surgery should be deferred until the postpartum period (6 weeks postdelivery), while urgent surgeries should be performed during:</b></p> <ul style="list-style-type: none"> <li>A. The second trimester</li> <li>B. The first trimester</li> <li>C. The third trimester</li> <li>D. Any trimester</li> </ul>                                        | <p><b>Ans: A</b></p> |
| <p><b>Q34. What is the anesthesia of choice for non-obstetric surgery during pregnancy?</b></p> <ul style="list-style-type: none"> <li>A. Regional anesthesia (the aim is to avoid general anesthesia)</li> <li>B. General anesthesia</li> <li>C. Local infiltration only</li> <li>D. Total intravenous anesthesia</li> </ul>                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q35. Pregnancy is a hypercoagulable state due to increase in which clotting factors?</b></p> <ul style="list-style-type: none"> <li>A. Factor I and factor VII</li> <li>B. Factor VIII and IX</li> <li>C. Factor V and X</li> <li>D. Factor XII and XIII</li> </ul>                                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q36. In pregnancy, plasma cholinesterase level is decreased by 25%. What is the effect on succinylcholine?</b></p> <ul style="list-style-type: none"> <li>A. Theoretically prolongs the effect of succinylcholine, but clinically this effect is insignificant</li> <li>B. Markedly prolongs succinylcholine effect</li> <li>C. Shortens succinylcholine effect</li> <li>D. Has no theoretical effect</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q37. During pregnancy, the arterial pO<sub>2</sub> and pCO<sub>2</sub> change due to hyperventilation. The pCO<sub>2</sub>:</b></p> <ul style="list-style-type: none"> <li>A. Decreases by 10 mm Hg (due to hyperventilation), with pH showing no change due to compensation</li> <li>B. Increases by 10 mm Hg</li> <li>C. Remains unchanged</li> <li>D. Decreases by 20 mm Hg with acidosis</li> </ul>          | <p><b>Ans: A</b></p> |
| <p><b>Q38. An advantage of regional anesthesia over general anesthesia for cesarean is that the awake mother can:</b></p> <ul style="list-style-type: none"> <li>A. Interact with her newborn immediately after surgery</li> <li>B. Walk immediately</li> <li>C. Eat immediately</li> <li>D. Avoid all monitoring</li> </ul>                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q39. In combined spinal epidural (CSE) for labor, the spinal component uses a very small dose of:</b></p> <ul style="list-style-type: none"> <li>A. 0.5 mL (2.5 mg) of hyperbaric bupivacaine + 10 µg of fentanyl</li> <li>B. 2 mL of 0.5% bupivacaine</li> <li>C. 5 mL of lignocaine</li> <li>D. 10 mL of ropivacaine</li> </ul>                                                                                | <p><b>Ans: A</b></p> |

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| <p><b>Q40. Which agent is given to patients who cannot receive epidural for labor analgesia, through a self-inhaler mask?</b></p> <p>A. Entonox (50% oxygen and 50% nitrous oxide)<br/> B. 100% nitrous oxide<br/> C. Sevoflurane<br/> D. Ketamine</p>                                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q41. During induction of GA for cesarean, intubation is done with which muscle relaxant while the assistant applies cricoid pressure?</b></p> <p>A. Suxamethonium<br/> B. Vecuronium<br/> C. Atracurium<br/> D. Pancuronium</p>                                                                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q42. In pregnancy, the FRC, ERV and RV decrease due to:</b></p> <p>A. Gravid uterus causing diaphragmatic elevation and basal atelectasis<br/> B. Increased respiratory rate<br/> C. Decreased blood volume<br/> D. Progesterone effect on lungs</p>                                                                                                  | <p><b>Ans: A</b></p> |
| <p><b>Q43. In cesarean for patients with heart diseases, in which conditions should central neuraxial blocks be avoided?</b></p> <p>A. Severe fixed cardiac output lesions like mitral or aortic stenosis<br/> B. Mild hypertension<br/> C. Anemia<br/> D. Diabetes</p>                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q44. A large fraction of drug reaching the fetus is metabolized by the fetal liver. What percentage of umbilical vein blood flows through the liver?</b></p> <p>A. 75%<br/> B. 50%<br/> C. 25%<br/> D. 90%</p>                                                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q45. Why is ketamine generally avoided for induction in cesarean section despite preserving upper airway reflexes?</b></p> <p>A. Because of its hallucinogenic side effects and tendency to increase uterine tone at high doses<br/> B. Because it causes hypotension<br/> C. Because it depresses the fetus<br/> D. Because it is teratogenic</p>    | <p><b>Ans: A</b></p> |
| <p><b>Q46. When GA is necessary for non-obstetric surgery in pregnancy, which drugs should be avoided as far as possible due to teratogenicity concern?</b></p> <p>A. Nitrous oxide and benzodiazepines (though the concern does not appear real in humans)<br/> B. Propofol and inhalational agents<br/> C. Muscle relaxants<br/> D. Local anesthetics</p> | <p><b>Ans: A</b></p> |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q47. In pregnancy, due to alkaline pH, which drugs can get accumulated in the fetus?</b></p> <ul style="list-style-type: none"> <li>A. Local anesthetics and opioids</li> <li>B. Muscle relaxants</li> <li>C. Inhalational agents</li> <li>D. Glycopyrrolate</li> </ul>                                                                                                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q48. In the management of accidental dural puncture during labor epidural, the preferred option is:</b></p> <ul style="list-style-type: none"> <li>A. To pass a catheter in subarachnoid space and maintain analgesia with 'continuous spinal' (at a low infusion rate of 1–3 mL/hr)</li> <li>B. To remove the needle and stop the procedure</li> <li>C. To give general anesthesia</li> <li>D. To increase the epidural infusion rate</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q49. For maintenance of GA in cesarean, which inhalational agent had been traditionally most preferred because of its cardiac stability?</b></p> <ul style="list-style-type: none"> <li>A. Isoflurane (with sevoflurane and desflurane equally good alternatives)</li> <li>B. Halothane</li> <li>C. Nitrous oxide</li> <li>D. Ether</li> </ul>                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q50. During laparoscopic surgery in pregnancy, what should the surgeon ensure?</b></p> <ul style="list-style-type: none"> <li>A. Use of minimum intra-abdominal pressure to prevent hypotension</li> <li>B. Maximum intra-abdominal pressure</li> <li>C. High concentration of nitrous oxide</li> <li>D. Use of general anesthesia only without monitoring</li> </ul>                                                                             | <p><b>Ans: A</b></p> |



# Answer Key & Explanations

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## **Q1. Answer: A — 40%**

Page 230: In pregnancy, intravascular fluid volume increases by 35%, cardiac output by 40%, heart rate by 15%, while systemic vascular resistance decreases by 15% and blood pressure decreases by 5–20% (diastolic decreases more than systolic).

*Topic: Physiological Changes in Pregnancy*

## **Q2. Answer: A — Hyperdynamic circulation**

Page 230: The clinical implication of cardiovascular changes is that pregnant patients are more vulnerable for cardiac failure due to hyperdynamic circulation.

*Topic: Physiological Changes in Pregnancy*

## **Q3. Answer: A — 20%**

Page 230: In pregnancy, tidal volume increases by 40%, respiratory rate by 10%, minute ventilation by 50%, oxygen consumption by 20%, while FRC decreases by 20% (FRC, ERV and RV decrease due to gravid uterus causing diaphragmatic elevation and basal atelectasis). Vital capacity shows no change.

*Topic: Physiological Changes in Pregnancy*

## **Q4. Answer: A — Due to decreased FRC, ERV and increased oxygen requirement, making them more vulnerable for hypoxia**

Page 230: Due to decreased FRC, ERV and increased oxygen requirement, pregnant patients are more vulnerable for hypoxia, therefore preoxygenation is a must. Due to increase in minute ventilation, induction with inhalational agents may be faster.

*Topic: Physiological Changes in Pregnancy*

## **Q5. Answer: A — 25–40%, making pregnant patients more susceptible to anesthetic over dosage**

Page 230, 235: Progesterone decreases the minimum alveolar concentration by 25–40% making pregnant patients more susceptible to anesthetic over dosage.

*Topic: Physiological Changes in Pregnancy*

## **Q6. Answer: A — 30–40%**

Page 230: As the epidural veins communicate with inferior vena cava, compression of inferior vena cava by gravid uterus causes engorgement of epidural veins, decreasing the subarachnoid space, leading the drugs to spread higher making pregnant patients more vulnerable for high spinal. Therefore, to prevent high spinal, the dose of local anesthetic for spinal has to be reduced by 30–40%.

*Topic: Physiological Changes in Pregnancy*

## **Q7. Answer: A — Gastric emptying is delayed (progesterone), gravid uterus makes LES incompetent, progesterone relaxes LES, and gastric contents are more acidic**

Page 231: Parturients are very vulnerable for aspiration (Mendelson syndrome) because gastric emptying is delayed due to progesterone, gravid uterus alters the gastro-esophageal angle making LES incompetent, progesterone relaxes the LES, and gastric contents are more acidic. Therefore, all pregnant patients should be considered full stomach even if they are fasting for more than 8 hours.

*Topic: Gastrointestinal System*

**Q8. Answer: A — In supine position, the gravid uterus compresses the inferior vena cava and aorta, decreasing cardiac output**

Page 231: In supine position, the gravid uterus can compress the inferior vena cava and aorta decreasing the cardiac output causing supine hypotension syndrome (SHS), which can cause severe hypotension or even cardiac arrest after spinal anesthesia.

Topic: Uterus / Supine Hypotension Syndrome

**Q9. Answer: A — Lie in left lateral position, or a 15° wedge placed under right buttock, or table tilted 15° to left, or uterus manually displaced to left**

Page 231: To prevent SHS, the pregnant patient should lie in left lateral position, or by putting a 15° wedge under right buttock, or by tilting the delivery table by 15° to left, or by manually displacing the uterus to left.

Topic: Uterus / Supine Hypotension Syndrome

**Q10. Answer: A — 500–700 mL/min (10% of cardiac output)**

Page 231: Uterine blood flow is 500–700 mL/min (10% of cardiac output) and placental flow is directly dependent on maternal blood flow. Anything that causes hypotension, vasoconstriction, or increases uterine tone (ketamine) can decrease maternal flow and hence placental blood flow.

Topic: Uteroplacental Circulation

**Q11. Answer: A — Muscle relaxants (only gallamine has significant transfer) and glycopyrrolate**

Page 231: All anesthetic drugs except muscle relaxants (only gallamine has significant transfer) and glycopyrrolate can be transferred to fetus from maternal circulation. Therefore, all drugs should be used in minimum concentration and dosage.

Topic: Transfer of Drugs to Fetal Circulation

**Q12. Answer: A — Ion trapping**

Page 231: Local anesthetics and opioids are bases; they cross the placenta in unionized form, become ionized in fetal circulation (due to low pH) and cannot come back to maternal circulation, leading to their trapping (called as ion trapping) and accumulation in fetus.

Topic: Transfer of Drugs to Fetal Circulation

**Q13. Answer: A — Twice as compared to regional anesthesia**

Page 232: Because of pulmonary aspiration and failed intubation, the mortality in obstetrics during general anesthesia (GA) is twice as compared to regional anesthesia. Pulmonary aspiration is 4 times more and failed intubation is 8 times more in obstetrics.

Topic: Anesthesia for Cesarean Section

**Q14. Answer: A — Up to T4**

Page 232: During cesarean there occurs handling of gut and mesentery, therefore level up to T4 is required.

Topic: Anesthesia for Cesarean Section

**Q15. Answer: A — Ephedrine**

Page 232: The disadvantage of vasopressors is that they not only produce vasoconstriction in maternal circulation but also in uteroplacental bed compromising fetal circulation — except ephedrine which does not produce significant vasoconstriction in uteroplacental bed. Therefore ephedrine had always been the vasopressor of choice in pregnancy.

Topic: Anesthesia for Cesarean Section

**Q16. Answer: A — A preferred vasopressor over ephedrine in current day anesthetic practice**

Page 232: Recent human trials have shown phenylephrine to produce less fetal acidosis than ephedrine, making phenylephrine a preferred vasopressor over ephedrine in current day anesthetic practice.

*Topic: Anesthesia for Cesarean Section*

**Q17. Answer: A — Onset takes a long time, so it is reserved for elective cases or specific conditions like pregnancy-induced hypertension (PIH)**

Page 232: Onset of epidural takes a long time (15–20 minutes), therefore it is reserved for elective cases only or for specific conditions like pregnancy-induced hypertension (PIH).

*Topic: Anesthesia for Cesarean Section*

**Q18. Answer: A — Propofol does not affect the APGAR and can be safely used in pregnancy**

Page 232: Thiopentone had been the induction agent of choice in the past as propofol safety had not been proven in newborns; however, now it is seen that at the doses used for induction, propofol does not affect the APGAR therefore can be safely used in pregnancy.

*Topic: General Anesthesia for Cesarean Section*

**Q19. Answer: A — Because all pregnant patients must be considered full stomach irrespective of their fasting status**

Page 232: All pregnant patients must be considered full stomach irrespective of their fasting status and must be managed like rapid sequence induction. Therefore they must be premedicated with metoclopramide, sodium citrate (to neutralize pH) and ranitidine.

*Topic: General Anesthesia for Cesarean Section*

**Q20. Answer: A — At concentration less than half MAC (< 0.5 MAC)**

Page 232–233: All inhalational agents in current use can relax the uterus in same magnitude and can cause postpartum hemorrhage (PPH); however, it has been seen that at concentration less than half minimum alveolar concentration (MAC) they do not produce significant uterine relaxation, therefore can be used safely in concentration < 0.5 MAC.

*Topic: General Anesthesia for Cesarean Section*

**Q21. Answer: A — Only after the delivery of the baby**

Page 233: Opioids can be given only after the delivery of baby. All muscle relaxants except gallamine can be used safely.

*Topic: General Anesthesia for Cesarean Section*

**Q22. Answer: A — Epidural**

Page 233: The anesthetic technique of choice for PIH is epidural. Difficult intubation due to laryngeal edema can make GA to become risky. As PIH patients are hypertensive they are more prone for hypotension; however, if necessary spinal can be given to PIH patients.

*Topic: Cesarean for PIH/Preeclampsia*

**Q23. Answer: A — HELLP syndrome**

Page 233: Before giving central neuraxial blocks in PIH it is important to rule out HELLP syndrome. If the patient has HELLP syndrome then epidural also becomes contraindicated and patient receives GA instead of laryngeal edema.

*Topic: Cesarean for PIH/Preeclampsia*

**Q24. Answer: A — Magnesium can potentiate the action of muscle relaxants or cause respiratory or cardiac arrest at toxic levels**

Page 233: Many PIH patients are on magnesium which can potentiate the action of muscle relaxants or can cause respiratory or cardiac arrest at toxic levels. Attenuation of cardiovascular response to intubation must be blunted, otherwise intracranial hemorrhage can occur.

Topic: Cesarean for PIH/Preeclampsia

**Q25. Answer: A — T10-L1**

Page 233: Pain afferents from uterus and cervix travel up to T10-L1. The most commonly used technique for labor analgesia (painless labor) is continuous lumbar epidural.

Topic: Labor Analgesia

**Q26. Answer: A — Continuous lumbar epidural**

Page 233: The most commonly used technique for painless labor is continuous lumbar epidural.

Topic: Labor Analgesia

**Q27. Answer: A — When the patient demands, irrespective of the stage of labor**

Page 233: Contrary to the previous recommendation that painless labor should be started once the labor is well established (cervix at least 3–4 cm dilated), the current recommendation is to start painless labor when the patient demands, irrespective of the stage of labor.

Topic: Labor Analgesia

**Q28. Answer: A — Walking epidural**

Page 233: If the concentration of bupivacaine used is too low (0.0625%), it is called as walking epidural (means patient can move out of the bed and sit in recliner).

Topic: Labor Analgesia

**Q29. Answer: A — Adrenaline can decrease uterine blood supply, and there is already too much variation in heart rate due to labor making it impossible to differentiate intravascular injection**

Page 233: Lignocaine with adrenaline should be avoided for test dose in pregnant patient due to 2 reasons: firstly, adrenaline can decrease uterine blood supply, and secondly, as there is already so much variation in heart rate due to labor, it becomes impossible to differentiate whether the change in heart rate is due to intravascular injection or labor.

Topic: Labor Analgesia

**Q30. Answer: A — Neither increase the rate of cesarean section nor the rate of instrumental delivery, but can delay the duration of labor by 30–60 minutes**

Page 234: Low concentration epidurals neither increase the rate of cesarean section nor the rate of instrumental delivery, however can delay the duration of labour by 30–60 minutes.

Topic: Labor Analgesia

**Q31. Answer: A — Remifentanyl (as it is metabolized in maternal plasma, with insignificant amount reaching the fetus)**

Page 234: As remifentanyl is metabolized in maternal plasma, insignificant amount reaches the fetus making remifentanyl a parenteral opioid of choice for pain relief in labor analgesia. Parenteral narcotics should be used only as last resort as there is always a risk of fetal as well as maternal respiratory depression.

Topic: Labor Analgesia / Parenteral Narcotics

**Q32. Answer: A — General anesthesia with inhalational agents (as all inhalational agents are uterine relaxants)**

Page 234: Patients with retained placenta may bleed profusely and need hemodynamic stability. As all inhalational agents in current use are uterine relaxants, the anesthesia of choice is GA with inhalational agents. The choice of inhalational agent is guided by hemodynamic parameters; if the patient is in shock then inhalational agent preferred will be desflurane.

*Topic: Anesthesia for Manual Removal of Placenta*

**Q33. Answer: A — The second trimester**

Page 234: Elective surgery should be deferred until postpartum period (6 weeks postdelivery) while urgent surgeries should be performed during the second trimester. In first trimester there are increased chances of abortion and congenital abnormalities, and in third trimester increased chances of preterm labor.

*Topic: Non-Obstetric Surgeries During Pregnancy*

**Q34. Answer: A — Regional anesthesia (the aim is to avoid general anesthesia)**

Page 234: The aim should be to avoid general anesthesia, therefore anesthesia of choice is regional anesthesia. Ensure that there does not occur hypotension after spinal as it can compromise fetal circulation.

*Topic: Non-Obstetric Surgeries During Pregnancy*

**Q35. Answer: A — Factor I and factor VII**

Page 231: Pregnancy is a hypercoagulable state due to increase in factor I and factor VII. Increase in plasma volume causes dilutional thrombocytopenia and physiologic anemia of pregnancy. Pregnant patients are more vulnerable for thromboembolism.

*Topic: Hematological System*

**Q36. Answer: A — Theoretically prolongs the effect of succinylcholine, but clinically this effect is insignificant**

Page 231: Plasma cholinesterase level is decreased by 25%, theoretically prolonging the effect of succinylcholine; however, clinically this effect is insignificant.

*Topic: Hepatic System*

**Q37. Answer: A — Decreases by 10 mm Hg (due to hyperventilation), with pH showing no change due to compensation**

Page 230: Blood gases in pregnancy — pO<sub>2</sub> increases by 10 mm Hg (due to hyperventilation), pCO<sub>2</sub> decreases by 10 mm Hg (due to hyperventilation), and pH shows no change (due to compensation).

*Topic: Physiological Changes in Pregnancy*

**Q38. Answer: A — Interact with her newborn immediately after surgery**

Page 232: Advantages of regional (spinal/epidural) over general anesthesia include avoiding pulmonary aspiration (4 times more in obstetrics) and failed intubation (8 times more), avoiding effect of anesthetic drugs on fetus, the awake mother can interact with her newborn immediately after surgery, and high inspired concentration of oxygen to mother can be delivered.

*Topic: Anesthesia for Cesarean Section*

**Q39. Answer: A — 0.5 mL (2.5 mg) of hyperbaric bupivacaine + 10 µg of fentanyl**

Page 234: In combined spinal epidural for labor, some clinicians give spinal with a very small dose (2.5 mg, i.e. 0.5 mL) of hyperbaric bupivacaine + 10 µg of fentanyl before giving epidural. The advantage is immediate relief of pain due to rapid onset of spinal; motor block does not occur with such a small dose of bupivacaine.

*Topic: Labor Analgesia / CSE*

**Q40. Answer: A — Entonox (50% oxygen and 50% nitrous oxide)**

Page 234: The patients who cannot receive epidural are given Entonox (combination of 50% oxygen and 50% nitrous oxide) through a self-inhaler mask.

Topic: Labor Analgesia / Entonox

**Q41. Answer: A — Suxamethonium**

Page 232: Intubation is done with suxamethonium while the assistant applies the cricoid pressure. Anticipate difficult intubation due to airway edema.

Topic: General Anesthesia for Cesarean Section

**Q42. Answer: A — Gravid uterus causing diaphragmatic elevation and basal atelectasis**

Page 230: Functional residual capacity (FRC), expiratory reserve volume (ERV) and residual volume (RV) decrease due to gravid uterus causing diaphragmatic elevation and basal atelectasis.

Topic: Physiological Changes in Pregnancy

**Q43. Answer: A — Severe fixed cardiac output lesions like mitral or aortic stenosis**

Page 232: In patients with severe fixed cardiac output lesion like mitral or aortic stenosis, central neuraxial blocks should be avoided. Pregnant patients are more vulnerable for cardiac failure than nonpregnant patient.

Topic: Cesarean in Patients with Heart Diseases

**Q44. Answer: A — 75%**

Page 231: A large fraction of drug reaching the fetus is metabolized by fetal liver (75% of umbilical vein blood flows through liver) therefore less drug reaches the vital structures like brain and heart.

Topic: Transfer of Drugs to Fetal Circulation

**Q45. Answer: A — Because of its hallucinogenic side effects and tendency to increase uterine tone at high doses**

Page 232: Ketamine appears a good choice as it preserves upper airway reflexes; however, it is generally avoided because of its hallucinogenic side effects and tendency to increase uterine tone at high doses.

Topic: General Anesthesia for Cesarean Section

**Q46. Answer: A — Nitrous oxide and benzodiazepines (though the concern does not appear real in humans)**

Page 234: The general principle is to use anesthetic drugs when absolutely necessary and in minimal possible doses. The concern of teratogenicity of nitrous oxide and benzodiazepines does not appear to be real in humans; however, they should be avoided as far as possible. Propofol can be used safely.

Topic: Non-Obstetric Surgeries During Pregnancy

**Q47. Answer: A — Local anesthetics and opioids**

Page 231, 235: Due to alkaline pH (relative to fetal circulation), local anesthetics and opioids can get accumulated in fetus (ion trapping).

Topic: Transfer of Drugs to Fetal Circulation

**Q48. Answer: A — To pass a catheter in subarachnoid space and maintain analgesia with 'continuous spinal' (at a low infusion rate of 1–3 mL/hr)**

Page 233: In accidental dural puncture, the clinician is left with 2 options. First, to remove the needle and try epidural in other space. Second, to pass a catheter in subarachnoid space and maintain analgesia with 'continuous spinal' instead of 'continuous epidural' (infusion rate is low, i.e. 1–3 mL/hr instead of 8–10 mL/hr). The second option, i.e. continuous spinal, should be preferred.

*Topic: Labor Analgesia / Complications*

**Q49. Answer: A — Isoflurane (with sevoflurane and desflurane equally good alternatives)**

Page 232–233: Traditionally isoflurane had been most preferred because of its cardiac stability; however, sevoflurane and desflurane are equally good alternatives.

*Topic: General Anesthesia for Cesarean Section*

**Q50. Answer: A — Use of minimum intra-abdominal pressure to prevent hypotension**

Page 234: During laparoscopies, ensure that the surgeon uses minimum intra-abdominal pressure to prevent hypotension.

*Topic: Non-Obstetric Surgeries During Pregnancy*

# Pediatric Anesthesia MCQ – Chapter 29

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management – Pediatric Anesthesia (Chapter 29)

**60 Questions • 40 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

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**Q1. At which cervical vertebral level is the larynx positioned in infants, compared to C6 in adults?**

- A. C4
- B. C3
- C. C5
- D. C7

**Ans: A**

**Q2. What is the narrowest part of the airway in children up to 6 years of age?**

- A. Vocal cords
- B. Subglottis (at the level of cricoid)
- C. Glottis
- D. Trachea

**Ans: B**

**Q3. Which laryngoscope blade is preferred for intubation in neonates and infants due to their anatomical airway configuration?**

- A. Curved Macintosh blade
- B. Small Macintosh blade
- C. Straight (Magill) blade
- D. Video laryngoscope blade

**Ans: C**

**Q4. What is the normal respiratory rate in a neonate?**

- A. 14–16/minute
- B. 20–25/minute
- C. 25–30/minute
- D. 35/minute

**Ans: D**

**Q5. Which respiratory parameter is the SAME (on a per kg basis) in both neonates and adults?**

- A. Tidal volume (6–8 mL/kg)
- B. Respiratory rate
- C. Alveolar ventilation
- D. Oxygen consumption

**Ans: A**

**Q6. What is the oxygen consumption rate in neonates (mL/kg/minute)?**

- A. 3 mL/kg/minute (same as adults)
- B. 6 mL/kg/minute (twice that of adults)
- C. 9 mL/kg/minute
- D. 12 mL/kg/minute

**Ans: B**

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| <p><b>Q7. The pCO<sub>2</sub> in neonates (30–36 mm Hg) is lower than in adults (35–45 mm Hg). What is the primary reason?</b></p> <p>A. Reduced CO<sub>2</sub> production by tissues<br/> B. Better CO<sub>2</sub> buffering capacity<br/> C. High respiratory rate<br/> D. Increased alveolar dead space</p>                                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q8. Due to high alveolar ventilation in children, induction with inhalational agents is:</b></p> <p>A. Slower than adults<br/> B. Same speed as adults<br/> C. Unpredictable<br/> D. Faster than adults</p>                                                                                                                                                                                                  | <p><b>Ans: D</b></p> |
| <p><b>Q9. When using cuffed endotracheal tubes in children, what leak pressure criterion must be maintained to prevent mucosal injury while avoiding barotrauma?</b></p> <p>A. A leak must be maintained at pressure &gt; 15–20 cm H<sub>2</sub>O<br/> B. No leak should be allowed at any pressure<br/> C. Leak at pressure &gt; 5–10 cm H<sub>2</sub>O<br/> D. Leak at pressure &gt; 25–30 cm H<sub>2</sub>O</p> | <p><b>Ans: A</b></p> |
| <p><b>Q10. What is the normal blood pressure in a neonate?</b></p> <p>A. 80/50 mm Hg<br/> B. 60/40 mm Hg<br/> C. 100/60 mm Hg<br/> D. 70/45 mm Hg</p>                                                                                                                                                                                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q11. What is the MOST COMMON adverse cardiac event during the perioperative period in pediatric patients?</b></p> <p>A. Ventricular tachycardia<br/> B. Supraventricular tachycardia<br/> C. Bradycardia<br/> D. Cardiac arrest</p>                                                                                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q12. In neonates, cardiac output is primarily determined by which factor (due to relatively fixed stroke volume)?</b></p> <p>A. Preload<br/> B. Afterload<br/> C. Contractility<br/> D. Heart rate</p>                                                                                                                                                                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q13. In children, extracellular fluid (ECF) constitutes what percentage of body weight (compared to 20% in adults)?</b></p> <p>A. 40%<br/> B. 30%<br/> C. 50%<br/> D. 25%</p>                                                                                                                                                                                                                                | <p><b>Ans: A</b></p> |

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| <p><b>Q14. Due to the high volume of distribution in children, the initial intravenous drug dose requirement on a per kg basis compared to adults is:</b></p> <p>A. Lower than adults<br/>B. Higher than adults<br/>C. Same as adults<br/>D. Twice the adult dose regardless of weight</p>                    | <p><b>Ans: B</b></p> |
| <p><b>Q15. Why are children comparatively more prone to hypoglycemia than adults?</b></p> <p>A. Due to increased glucose excretion by kidneys<br/>B. Due to high insulin secretion<br/>C. Due to inadequate glycogen stores and high metabolic rate<br/>D. Due to decreased hepatic gluconeogenesis alone</p> | <p><b>Ans: C</b></p> |
| <p><b>Q16. What is the recommended operating theater temperature for pediatric patients to prevent hypothermia (compared to 21°C for adults)?</b></p> <p>A. 32°C<br/>B. 30°C<br/>C. 25°C<br/>D. 28°C</p>                                                                                                      | <p><b>Ans: D</b></p> |
| <p><b>Q17. What is the ONLY mechanism for heat production available to neonates (who cannot shiver)?</b></p> <p>A. Metabolism of brown fat<br/>B. Shivering thermogenesis<br/>C. Vasoconstriction<br/>D. Increased skeletal muscle activity</p>                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q18. What is the normal hemoglobin level at birth?</b></p> <p>A. 14 g%<br/>B. 19 g%<br/>C. 12 g%<br/>D. 16 g%</p>                                                                                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q19. What is the term for the fall in hemoglobin to 10 g% that occurs physiologically at 2–3 months of age?</b></p> <p>A. Hemolytic anemia of the newborn<br/>B. Iron deficiency anemia<br/>C. Physiologic anemia of infancy<br/>D. Aplastic anemia</p>                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q20. The higher content of fetal hemoglobin in neonates shifts the oxygen dissociation curve in which direction?</b></p> <p>A. To the right (decreased oxygen affinity)<br/>B. Upward (greater oxygen-carrying capacity)<br/>C. No significant shift<br/>D. To the left (increased oxygen affinity)</p> | <p><b>Ans: D</b></p> |

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| <p><b>Q21. Why is prolonged coagulation expected in neonates?</b></p> <ul style="list-style-type: none"> <li>A. Due to vitamin K deficiency</li> <li>B. Due to immature platelet function</li> <li>C. Due to low fibrinogen levels</li> <li>D. Due to low factor VIII levels</li> </ul>                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q22. What is the correct order of MAC requirement from highest to lowest in pediatric populations?</b></p> <ul style="list-style-type: none"> <li>A. Neonates &gt; infants &gt; adults</li> <li>B. Infants &gt; neonates &gt; adults</li> <li>C. Adults &gt; infants &gt; neonates</li> <li>D. Neonates &gt; adults &gt; infants</li> </ul>                           | <p><b>Ans: B</b></p> |
| <p><b>Q23. At what age does MAC (Minimum Alveolar Concentration) peak in human beings?</b></p> <ul style="list-style-type: none"> <li>A. At birth</li> <li>B. 3 months</li> <li>C. 6 months</li> <li>D. 1 year</li> </ul>                                                                                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q24. In neonates, decreased albumin levels have what clinical implication for drug pharmacokinetics?</b></p> <ul style="list-style-type: none"> <li>A. Decreased total drug concentration in plasma</li> <li>B. Decreased drug clearance only</li> <li>C. Increased protein-bound drug fraction</li> <li>D. Increases the unbound (free) fraction of drugs</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q25. Until what age is functional maturation of the neuromuscular junction not complete, making newborns very sensitive to nondepolarizing muscle relaxants?</b></p> <ul style="list-style-type: none"> <li>A. 2 months</li> <li>B. 6 months</li> <li>C. 1 year</li> <li>D. 3 months</li> </ul>                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q26. For a child aged 6 months to 3 years, what is the recommended fasting time for milk and solid food before anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. 4 hours</li> <li>B. 6 hours</li> <li>C. 8 hours</li> <li>D. 2 hours</li> </ul>                                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q27. What is the clear liquid (water) fasting time required before anesthesia for ALL pediatric age groups?</b></p> <ul style="list-style-type: none"> <li>A. 4 hours</li> <li>B. 6 hours</li> <li>C. 2 hours</li> <li>D. 1 hour</li> </ul>                                                                                                                           | <p><b>Ans: C</b></p> |

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| <p><b>Q28. What is the simpler and more accurate intraoperative fluid replacement formula recommended for children (instead of the 4-2-1 formula)?</b></p> <p>A. 10–20 mL/kg<br/>B. 40–60 mL/kg<br/>C. The 4-2-1 formula remains the best<br/>D. 20–40 mL/kg</p>                                  | <p><b>Ans: D</b></p> |
| <p><b>Q29. What is the postoperative fluid maintenance formula used for children?</b></p> <p>A. 2-1-0.5 (0–10 kg @ 2 mL/kg, 10–20 kg @ 1 mL/kg, &gt;20 kg @ 0.5 mL/kg)<br/>B. 4-2-1 formula only<br/>C. 1 mL/kg/hr flat rate<br/>D. 3-1.5-0.75 formula</p>                                        | <p><b>Ans: A</b></p> |
| <p><b>Q30. What is the intraoperative fluid of choice for the majority of pediatric surgical patients?</b></p> <p>A. Normal saline<br/>B. Ringer lactate<br/>C. 5% Dextrose<br/>D. 5% Dextrose in 0.45% normal saline</p>                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q31. What is the most commonly used method of induction in pediatric patients (though the method of choice is always IV)?</b></p> <p>A. Intravenous induction<br/>B. Intramuscular induction<br/>C. Inhalational induction<br/>D. Rectal induction</p>                                      | <p><b>Ans: C</b></p> |
| <p><b>Q32. Sevoflurane is considered the inhalational agent of choice for induction in children. At what concentration is it used?</b></p> <p>A. 2–4%<br/>B. 4–5%<br/>C. 5–6%<br/>D. 7–8%</p>                                                                                                     | <p><b>Ans: D</b></p> |
| <p><b>Q33. Which inhalational agents must NOT be used for induction in children due to irritating properties and risk of severe laryngospasm?</b></p> <p>A. Isoflurane and desflurane<br/>B. Sevoflurane and halothane<br/>C. Halothane and nitrous oxide<br/>D. Desflurane and nitrous oxide</p> | <p><b>Ans: A</b></p> |
| <p><b>Q34. At what intramuscular dose can ketamine be given to a pediatric patient in the preoperative room with parents?</b></p> <p>A. 2 mg/kg<br/>B. 5 mg/kg<br/>C. 10 mg/kg<br/>D. 1 mg/kg</p>                                                                                                 | <p><b>Ans: B</b></p> |

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| <p><b>Q35. Which breathing circuit is preferred for children less than 6 years of age (or less than 20 kg)?</b></p> <p>A. Bains circuit<br/>B. Mapleson A circuit<br/>C. Jackson-Rees circuit<br/>D. Mapleson D circuit</p>                                                                                                                                                                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q36. Which anesthetic agents are NOT suitable for maintenance in children due to significant emergence agitation (20% and 15% incidence respectively)?</b></p> <p>A. Isoflurane and desflurane<br/>B. Desflurane and propofol<br/>C. Isoflurane and propofol<br/>D. Sevoflurane and halothane</p>                                                                                                                      | <p><b>Ans: D</b></p> |
| <p><b>Q37. Which muscle relaxants are the agents of choice in pediatric patients due to their independence from hepatic and renal metabolism?</b></p> <p>A. Atracurium and cisatracurium<br/>B. Vecuronium and rocuronium<br/>C. Succinylcholine and mivacurium<br/>D. Pancuronium and d-tubocurarine</p>                                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q38. Why should succinylcholine be avoided in children (outside of rapid sequence induction and laryngospasm treatment)?</b></p> <p>A. It causes prolonged neuromuscular blockade in children<br/>B. Due to extrajunctional receptors and possibility of undiagnosed muscular dystrophy<br/>C. It causes severe hypertension in pediatric patients<br/>D. It exclusively causes malignant hyperthermia in children</p> | <p><b>Ans: B</b></p> |
| <p><b>Q39. What drug MUST be given before administering succinylcholine in pediatric patients?</b></p> <p>A. Neostigmine<br/>B. Glycopyrrolate<br/>C. Atropine<br/>D. Adrenaline</p>                                                                                                                                                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q40. How does pain perception in pediatric patients compare to adults, according to current evidence?</b></p> <p>A. Much less than adults — pediatric patients feel less pain<br/>B. Absent in neonates due to immature nervous system<br/>C. Difficult to assess, so it is assumed to be less<br/>D. Same as adults, and even more during infancy due to underdeveloped inhibitory control</p>                        | <p><b>Ans: D</b></p> |
| <p><b>Q41. Which local anesthetic is preferred for regional nerve blocks in pediatric patients for postoperative pain management?</b></p> <p>A. Ropivacaine (due to less toxicity than bupivacaine)<br/>B. Bupivacaine<br/>C. Lidocaine<br/>D. Levobupivacaine</p>                                                                                                                                                           | <p><b>Ans: A</b></p> |

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| <p><b>Q42. What are the most commonly used regional blocks in pediatric patients (listed in order of frequency)?</b></p> <p>A. Spinal, epidural, caudal<br/> B. Caudal, ilioinguinal, penile<br/> C. Epidural, spinal, brachial plexus<br/> D. Brachial plexus, femoral, sciatic</p>                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q43. Why does the onset of regional block occur faster in pediatric patients compared to adults?</b></p> <p>A. Due to higher cardiac output increasing drug delivery to nerves<br/> B. Due to smaller nerve diameter only<br/> C. Due to immature myelin sheath allowing the drug to cross the sheath easily<br/> D. Due to higher concentrations of local anesthetic used</p>             | <p><b>Ans: C</b></p> |
| <p><b>Q44. Compared to adults, the duration of regional block in children is:</b></p> <p>A. Longer due to slower drug metabolism<br/> B. Same duration as adults<br/> C. Variable and unpredictable<br/> D. Shorter (less duration)</p>                                                                                                                                                          | <p><b>Ans: D</b></p> |
| <p><b>Q45. Why are children more vulnerable to local anesthetic toxicity than adults?</b></p> <p>A. Increased systemic absorption (increased cardiac output) and decreased metabolism (underdeveloped hepatic functions)<br/> B. Higher sensitivity of sodium channels in children<br/> C. Lower protein binding of local anesthetics<br/> D. Higher body surface area to volume ratio alone</p> | <p><b>Ans: A</b></p> |
| <p><b>Q46. Diaphragmatic hernia results from incomplete closure of the diaphragm. Which of the following is NOT a consequence of diaphragmatic hernia?</b></p> <p>A. Pulmonary hypoplasia<br/> B. Pulmonary hyperplasia<br/> C. Pulmonary hypertension<br/> D. Hypoplasia of left ventricle</p>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q47. What is the FIRST step in the anesthetic management of a neonate with diaphragmatic hernia?</b></p> <p>A. Rapid sequence induction<br/> B. Inhalational induction with sevoflurane<br/> C. Awake intubation after preoxygenation<br/> D. Spinal anesthesia</p>                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q48. What is the maximum recommended airway pressure for positive pressure ventilation in diaphragmatic hernia to avoid pneumothorax in the hypoplastic lung?</b></p> <p>A. 30 cm H<sub>2</sub>O<br/> B. 25 cm H<sub>2</sub>O<br/> C. 15 cm H<sub>2</sub>O<br/> D. &lt; 20 cm H<sub>2</sub>O</p>                                                                                           | <p><b>Ans: D</b></p> |

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| <p><b>Q49. Why is nitrous oxide absolutely contraindicated in diaphragmatic hernia surgery?</b></p> <ul style="list-style-type: none"> <li>A. It can diffuse into gut loops, causing their distension and further compression of the lung</li> <li>B. It causes severe bradycardia in neonates</li> <li>C. It causes severe pulmonary hypertension</li> <li>D. It causes cardiac depression in neonates</li> </ul>   | <p><b>Ans: A</b></p> |
| <p><b>Q50. What is the MOST COMMON type of tracheoesophageal fistula?</b></p> <ul style="list-style-type: none"> <li>A. H-type fistula with patent esophagus throughout</li> <li>B. Upper end of esophagus is blind with fistula between lower esophagus and trachea</li> <li>C. Both ends of esophagus are blind with no fistula</li> <li>D. Lower end of esophagus is blind with upper esophagus patent</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q51. What is the optimal position of the endotracheal tube in surgery for tracheoesophageal fistula?</b></p> <ul style="list-style-type: none"> <li>A. Above the fistula and above the carina</li> <li>B. At the level of the fistula</li> <li>C. Below the fistula but above the carina</li> <li>D. At the level of the carina</li> </ul>                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q52. Why is bag and mask ventilation contraindicated in tracheoesophageal fistula?</b></p> <ul style="list-style-type: none"> <li>A. It causes pulmonary hypertension</li> <li>B. Risk of pneumothorax</li> <li>C. It increases intracranial pressure dangerously</li> <li>D. Air can reach stomach through fistula causing gastric distension and aspiration</li> </ul>                                       | <p><b>Ans: D</b></p> |
| <p><b>Q53. What metabolic abnormality is typically seen in pyloric stenosis due to repeated vomiting?</b></p> <ul style="list-style-type: none"> <li>A. Hypokalemic, hypochloremic alkalosis</li> <li>B. Hyperkalemic, hyperchloremic acidosis</li> <li>C. Hyponatremic acidosis</li> <li>D. Hypernatremic alkalosis</li> </ul>                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q54. What is the MOST IMPORTANT prerequisite before taking a patient with pyloric stenosis for surgical repair?</b></p> <ul style="list-style-type: none"> <li>A. Administer prophylactic vitamin K</li> <li>B. Complete metabolic, fluid, and electrolyte correction</li> <li>C. Start inotropic support</li> <li>D. Administer broad-spectrum antibiotics</li> </ul>                                         | <p><b>Ans: B</b></p> |
| <p><b>Q55. Until what postconceptional age should elective surgeries be avoided in premature babies?</b></p> <ul style="list-style-type: none"> <li>A. 40 weeks</li> <li>B. 44 weeks</li> <li>C. 50 weeks</li> <li>D. 60 weeks</li> </ul>                                                                                                                                                                            | <p><b>Ans: C</b></p> |



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| <p><b>Q56. What is the most significant postoperative complication that premature babies are uniquely vulnerable to?</b></p> <ul style="list-style-type: none"> <li>A. Postoperative wound infection</li> <li>B. Postoperative jaundice</li> <li>C. Postoperative cardiac failure</li> <li>D. Postoperative apnea</li> </ul>                                                                                                                              | <p><b>Ans: D</b></p> |
| <p><b>Q57. Which combination of features leads to difficult airway management in children with Down syndrome?</b></p> <ul style="list-style-type: none"> <li>A. Short neck, large tongue, irregular dentition, atlanto-occipital instability, small size trachea and subglottic stenosis</li> <li>B. Macroglossia and retrognathia only</li> <li>C. Enlarged tonsils and adenoids only</li> <li>D. Micrognathia and limited mouth opening only</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q58. Which congenital heart defects are MOST commonly associated with Down syndrome?</b></p> <ul style="list-style-type: none"> <li>A. Tetralogy of Fallot and pulmonary stenosis</li> <li>B. Endocardial cushion defects and VSD</li> <li>C. Atrial septal defect and patent ductus arteriosus</li> <li>D. Coarctation of aorta and transposition of great vessels</li> </ul>                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q59. Why may Down syndrome patients exhibit hypersensitivity to muscle relaxants?</b></p> <ul style="list-style-type: none"> <li>A. Due to increased acetylcholine receptors at the NMJ</li> <li>B. Due to altered protein binding of muscle relaxants</li> <li>C. Due to hypotonia</li> <li>D. Due to immature neuromuscular junction</li> </ul>                                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q60. Why are neonates at higher risk of anesthetic agent overdoses compared to older children and adults?</b></p> <ul style="list-style-type: none"> <li>A. Due to higher MAC requirements needing larger doses</li> <li>B. Due to immature hepatic metabolism only</li> <li>C. Due to increased renal excretion of drugs</li> <li>D. Due to immature brain and poorly developed blood-brain barrier</li> </ul>                                     | <p><b>Ans: D</b></p> |

# Answer Key & Explanations

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## Q1. Answer: A — C4

Page 236: In children (particularly infants), the larynx is more anterior and high, positioned at C4 compared to C6 in adults. This anatomical difference is important for intubation technique.

Topic: Airway Anatomy

## Q2. Answer: B — Subglottis (at the level of cricoid)

Page 236–237: Subglottis (at the level of cricoid) is the narrowest part of the larynx in children up to 6 years of age. In adults the narrowest part is the vocal cords. This is why uncuffed tubes traditionally had a natural seal in children.

Topic: Airway Anatomy

## Q3. Answer: C — Straight (Magill) blade

Page 237: Due to anatomical configuration of the airway and large epiglottis, intubation is preferred with a straight blade (Magill) laryngoscope in neonates and infants. A large tongue can also obscure the view during laryngoscopy.

Topic: Airway Anatomy

## Q4. Answer: D — 35/minute

Page 236, Table 29.1: The normal respiratory rate in a neonate is 35/minute compared to 14–16/minute in adults. This higher respiratory rate (not tidal volume) is the key determinant of the neonate's high alveolar ventilation.

Topic: Respiratory Physiology

## Q5. Answer: A — Tidal volume (6–8 mL/kg)

Page 236, Table 29.1: Tidal volume is 6–8 mL/kg in both neonates and adults — it is identical. The determinant of the neonate's high alveolar ventilation (120–140 vs 60–70 mL/kg/min) is the high respiratory rate, not tidal volume.

Topic: Respiratory Physiology

## Q6. Answer: B — 6 mL/kg/minute (twice that of adults)

Page 236, Table 29.1: Oxygen consumption in neonates is 6 mL/kg/minute compared to 3 mL/kg/minute in adults — twice that of adults. Calorie requirement in neonates is 100 kcal/kg vs 30 kcal/kg in adults (three times).

Topic: Respiratory Physiology

## Q7. Answer: C — High respiratory rate

Page 236, Table 29.1: The comment column in Table 29.1 states that the lower pCO<sub>2</sub> in neonates is “Due to high respiratory rate.” This results in more CO<sub>2</sub> being blown off per minute.

Topic: Respiratory Physiology

## Q8. Answer: D — Faster than adults

Page 237: Because of high alveolar ventilation, induction with inhalational agents is faster in children compared to adults. This is an important anesthetic implication of the respiratory physiology of children.

Topic: Respiratory – Anesthetic Implications

## Q9. Answer: A — A leak must be maintained at pressure > 15–20 cm H<sub>2</sub>O

Page 237: A leak must be maintained around the cuff (inflated or not inflated) at a pressure of more than 15–20 cm H<sub>2</sub>O. This prevents tracheal mucosal injury but also prevents excessive leak that can cause hypoventilation and aspiration risk.

Topic: Airway Management

**Q10. Answer: B — 60/40 mm Hg**

Page 237: Blood pressure in a neonate is 60/40 mm Hg — 50% of adult value. Therefore, cardiac output is mainly determined by heart rate, which is 120–140/minute at birth.

Topic: Cardiovascular System

**Q11. Answer: C — Bradycardia**

Page 237: Bradycardia is the most common adverse cardiac event during the perioperative period in pediatric patients. The key point states “Bradycardia is not acceptable at any cost.” Cardiac arrests are also more common in pediatric age groups.

Topic: Cardiovascular System

**Q12. Answer: D — Heart rate**

Page 237: Cardiac output in neonates is mainly determined by heart rate, which is 120–140/minute at birth. Since stroke volume is relatively fixed, bradycardia directly reduces cardiac output, which is why it is not acceptable at any cost.

Topic: Cardiovascular System

**Q13. Answer: A — 40%**

Page 237: Extracellular fluid (ECF) constitutes 40% of body weight in children, compared to 20% in adults. This high volume of distribution dilutes initial intravenous drugs, making initial dose requirements higher on a per kg basis.

Topic: Body Water

**Q14. Answer: B — Higher than adults**

Page 237: The high volume of distribution dilutes the initial intravenous drug; therefore the initial dose requirement (on per kg weight basis) is higher than adults. However, excessive fluid leads to fluid overload as the heart is less compliant.

Topic: Body Water

**Q15. Answer: C — Due to inadequate glycogen stores and high metabolic rate**

Page 237: Due to inadequate glycogen stores and high metabolic rate, children are comparatively more prone to hypoglycemia. Despite this, Ringer lactate is still sufficient for most children as they are not so prone to hypoglycemia that dextrose is always required.

Topic: Metabolic

**Q16. Answer: D — 28°C**

Page 237: To prevent hypothermia, the operation theater temperature should be maintained at 28°C for pediatric patients (usually maintained at 21°C for adults). Children are very prone to hypothermia due to decreased ability to produce and conserve heat and increased heat loss due to large body surface area.

Topic: Thermoregulation

**Q17. Answer: A — Metabolism of brown fat**

Page 237: The only mechanism for heat production in neonates is metabolism of brown fat, which is present in the posterior neck, interscapular and vertebral areas, and around the kidneys and adrenal glands. Neonates cannot shiver.

Topic: Thermoregulation

**Q18. Answer: B — 19 g%**

Page 237: Hemoglobin at birth is 19 g%, which falls to 10 g% at 2–3 months (physiologic anemia of infancy), and thereafter steadily increases to adult value by 12–13 years.

Topic: Hematology

**Q19. Answer: C — Physiologic anemia of infancy**

Page 237: The fall in hemoglobin from 19 g% at birth to 10 g% at 2–3 months is called the physiologic anemia of infancy. It is a normal physiological process, after which Hgb steadily increases to adult value by 12–13 years.

Topic: Hematology

**Q20. Answer: D — To the left (increased oxygen affinity)**

Page 237: Neonates have a higher content of fetal hemoglobin, which shifts the oxygen dissociation curve to the left. This means higher oxygen affinity but decreased oxygen release to peripheral tissues.

Topic: Hematology

**Q21. Answer: A — Due to vitamin K deficiency**

Page 237: Prolonged coagulation is expected in neonates due to vitamin K deficiency. This has important clinical implications for surgical procedures and bleeding risk in the neonatal period.

Topic: Hematology

**Q22. Answer: B — Infants > neonates > adults**

Page 238: MAC requirement in decreasing order is Infants > neonates > adults. Maximum MAC peaks in infancy at 6 months. Neonates have less MAC compared to infants due to the immature CNS.

Topic: CNS and MAC

**Q23. Answer: C — 6 months**

Page 238: Maximum MAC in human beings occurs in infancy, peaking at the age of 6 months. Thereafter it decreases steadily throughout life, except for a slight increase at puberty.

Topic: CNS and MAC

**Q24. Answer: D — Increases the unbound (free) fraction of drugs**

Page 238: Decreased albumin increases the unbound fraction of drugs. More free drug is available in the plasma, which can lead to toxicity at lower doses than expected.

Topic: Hepatic and Renal

**Q25. Answer: A — 2 months**

Page 238: Functional maturation of the neuromuscular junction is not complete until 2 months of age; therefore, newborns are very sensitive to nondepolarizing muscle relaxants and must be dosed with extreme caution.

Topic: Neuromuscular Junction

**Q26. Answer: B — 6 hours**

Page 238: Fasting recommendations — 6 months to 3 years: milk and solid food 6 hours, clear liquid (water) 2 hours. For < 6 months: breast milk 4 hours, clear liquid 2 hours. For > 3 years: 8 hours for milk/solid, 2 hours for clear liquid.

Topic: Fasting Recommendations

**Q27. Answer: C — 2 hours**

Page 238: The clear liquid (water) fasting time is uniformly 2 hours for all pediatric age groups (< 6 months, 6 months–3 years, and > 3 years). This is consistent across all age groups.

Topic: Fasting Recommendations

**Q28. Answer: D — 20–40 mL/kg**

Page 238: A simpler and probably more accurate method is to give 20–40 mL/kg of fluid during the intraoperative period. Thereafter in the postoperative period the fluid is given by the 2-1-0.5 formula.

Topic: Fluid Management

**Q29. Answer: A — 2-1-0.5 (0–10 kg @ 2 mL/kg, 10–20 kg @ 1 mL/kg, >20 kg @ 0.5 mL/kg)**

Page 238: In the postoperative period the fluid is given by the formula of 2-1-0.5: 0–10 kg @ 2 mL/kg, 10–20 @ 1 mL/kg, and >20 kg @ 0.5 mL/kg. If the fluid has to be continued after 12 hours, 5% dextrose with 0.45% normal saline by 4-2-1 formula is used.

Topic: Fluid Management

**Q30. Answer: B — Ringer lactate**

Page 238: As children are not so prone to hypoglycemia, Ringer lactate may be the only fluid required for the majority of children. If there is concern of hypoglycemia, 5% dextrose in 0.45% normal saline is added as a piggyback (not as a replacement fluid).

Topic: Fluid Management

**Q31. Answer: C — Inhalational induction**

Page 238: Inhalational induction is the most commonly used technique as children will not allow putting IV lines. However, the method of choice for induction is always IV. An important precaution is to put an IV line once the child falls asleep before proceeding to a deeper plane.

Topic: Anesthetic Induction

**Q32. Answer: D — 7–8%**

Page 238: Due to smooth and rapid induction, sevoflurane (7–8%) is considered the inhalational agent of choice for induction in children. Halothane is still commonly used in India due to lower cost.

Topic: Anesthetic Induction

**Q33. Answer: A — Isoflurane and desflurane**

Page 238–239: Isoflurane and desflurane have irritating induction and can produce severe laryngospasm, therefore they must not be used for induction. Conversely, sevoflurane and halothane are excellent for induction but not suitable for maintenance.

Topic: Anesthetic Induction

**Q34. Answer: B — 5 mg/kg**

Page 239: Ketamine 5 mg/kg can be given IM when the child is in the preoperative room with parents. This is useful when IV access is not available and inhalational induction is not feasible or contraindicated.

Topic: Anesthetic Induction

**Q35. Answer: C — Jackson-Rees circuit**

Page 239: Jackson-Rees circuit is used for children less than 6 years (or < 20 kg) and Bain's circuit for children more than 6 years (or > 20 kg). However, the preference should be the pediatric circle system (pediatric closed circuit).

Topic: Airway Circuit

**Q36. Answer: D — Sevoflurane and halothane**

Page 239: Sevoflurane (20% emergence agitation) and halothane (15%) are not suitable for maintenance. Anesthesia is best maintained with desflurane or isoflurane. To summarize: sevoflurane/halothane are excellent for induction; desflurane/isoflurane are excellent for maintenance.

Topic: Anesthetic Maintenance

**Q37. Answer: A — Atracurium and cisatracurium**

Page 239: Atracurium and cisatracurium are the relaxants of choice in pediatric patients as they are not dependent on hepatic and renal functions. This is important given the underdeveloped hepatic and renal systems in children.

Topic: Muscle Relaxants

**Q38. Answer: B — Due to extrajunctional receptors and possibility of undiagnosed muscular dystrophy**

Page 239: Succinylcholine should be avoided in children due to the presence of extrajunctional receptors and the possibility of undiagnosed muscular dystrophy. Its use should be reserved for rapid sequence induction and treating laryngospasm.

Topic: Muscle Relaxants

**Q39. Answer: C — Atropine**

Page 239: Pediatric age group is very vulnerable to bradycardia after succinylcholine, therefore atropine must be given before injecting suxamethonium. This is an important safety precaution unique to the pediatric population.

Topic: Muscle Relaxants

**Q40. Answer: D — Same as adults, and even more during infancy due to underdeveloped inhibitory control**

Page 239: It has been well proven that pain perception in pediatric patients is same as adults; rather, due to underdevelopment of inhibitory control pathways it is more during infancy. Pain management considerations are therefore equally important for children.

Topic: Postoperative Pain Management

**Q41. Answer: A — Ropivacaine (due to less toxicity than bupivacaine)**

Page 239: Regional nerve block is always a preferred option. Ropivacaine is preferred over bupivacaine due to less toxicity. If regional anesthesia is not possible, short-acting opioids (fentanyl, remifentanyl) are preferred over long-acting (morphine).

Topic: Postoperative Pain Management

**Q42. Answer: B — Caudal, ilioinguinal, penile**

Page 239–240: Most commonly used blocks in pediatric patients are caudal, followed by ilioinguinal and penile blocks. Regional anesthesia in children is always given after GA to decrease anesthetic requirement and achieve postoperative analgesia.

Topic: Regional Anesthesia

**Q43. Answer: C — Due to immature myelin sheath allowing the drug to cross the sheath easily**

Page 239–240: Onset of block is faster in pediatric patients due to immature myelin sheath letting the drug to cross the sheath easily. This also explains why lower doses of local anesthetic are required.

Topic: Regional Anesthesia

**Q44. Answer: D — Shorter (less duration)**

Page 240: Duration of block is less in children compared to adults. This is one of the key differences from adult regional anesthesia. Children also have increased vulnerability to local anesthetic toxicity.

Topic: Regional Anesthesia

**Q45. Answer: A — Increased systemic absorption (increased cardiac output) and decreased metabolism (underdeveloped hepatic functions)**

Page 240: Children are more vulnerable to toxicity due to increased systemic absorption (because of increased cardiac output) and decreased metabolism (due to underdeveloped hepatic functions).

Topic: Regional Anesthesia

**Q46. Answer: B — Pulmonary hyperplasia**

Page 240: Diaphragmatic hernia leads to herniation of abdominal contents in thorax resulting in pulmonary HYPOPLASIA (not hyperplasia), pulmonary hypertension, and hypoplasia of the left ventricle. The prognosis is very poor.

Topic: Diaphragmatic Hernia

**Q47. Answer: C — Awake intubation after preoxygenation**

Page 240: After preoxygenation, awake intubation is done in diaphragmatic hernia. Bag and mask ventilation is contraindicated as it will increase distension of bowels, leading to more compression of the lung.

Topic: Diaphragmatic Hernia

**Q48. Answer: D — < 20 cm H<sub>2</sub>O**

Page 240: Positive pressure ventilation should be done with airway pressure < 20 cm H<sub>2</sub>O, otherwise pneumothorax can occur in the hypoplastic lung. If PPV fails, high frequency oscillatory ventilation is used.

Topic: Diaphragmatic Hernia

**Q49. Answer: A — It can diffuse into gut loops, causing their distension and further compression of the lung**

Page 240: Nitrous oxide is contraindicated in diaphragmatic hernia as it can diffuse into gut loops, causing their distension and further compression of the already hypoplastic lung. Anesthesia is maintained on oxygen and low-dose volatile anesthetics or opioids like fentanyl.

Topic: Diaphragmatic Hernia

**Q50. Answer: B — Upper end of esophagus is blind with fistula between lower esophagus and trachea**

Page 240: Tracheoesophageal fistula is of 5 types. The most common type has the upper end of the esophagus blind (blind pouch) with a fistula between the lower esophagus and the trachea.

Topic: Tracheoesophageal Fistula

**Q51. Answer: C — Below the fistula but above the carina**

Page 240: The position of the tube is most important in TEF; it should be below the fistula but above the carina. Positioning the tube below the fistula prevents air from going into the stomach through the fistula, while staying above the carina allows bilateral lung ventilation.

Topic: Tracheoesophageal Fistula

**Q52. Answer: D — Air can reach stomach through fistula causing gastric distension and aspiration**

Page 240: Ventilation with bag and mask is contraindicated in TEF as air can reach the stomach through the fistula, causing gastric distension and aspiration. These patients are also very vulnerable to hypoxia as there is significant V/Q mismatch in the lateral position.

Topic: Tracheoesophageal Fistula

**Q53. Answer: A — Hypokalemic, hypochloremic alkalosis**

Page 240: Due to repeated vomiting, patients with pyloric stenosis are dehydrated with hypokalemic, hypochloremic alkalosis. Therefore, metabolic, fluid and electrolyte correction must be done before taking the patient for surgery.

Topic: Pyloric Stenosis

**Q54. Answer: B — Complete metabolic, fluid, and electrolyte correction**

Page 240: Metabolic, fluid and electrolyte correction must be done before taking the patient with pyloric stenosis for surgery. Either awake intubation or rapid sequence intubation (crash intubation with Sellick's maneuver) is performed. Bag and mask ventilation is contraindicated.

Topic: Pyloric Stenosis

**Q55. Answer: C — 50 weeks**

Page 240: Avoid elective surgeries till 50 weeks of postconception in premature babies. If the baby had developed hyaline membrane disease, defer elective surgery for 6 months. Premature babies are very vulnerable to postoperative apnea.

*Topic: Anesthetic Considerations for Prematures*

**Q56. Answer: D — Postoperative apnea**

Page 240: Premature babies are very vulnerable to postoperative apnea. This is a key reason why elective surgeries are avoided until 50 weeks of postconception age and why these patients require close monitoring after anesthesia.

*Topic: Anesthetic Considerations for Prematures*

**Q57. Answer: A — Short neck, large tongue, irregular dentition, atlanto-occipital instability, small size trachea and subglottic stenosis**

Page 240: Children with Down syndrome have a difficult airway due to short neck, large tongue, irregular dentition, atlanto-occipital instability (avoid unnecessary flexion and extension of neck), small size trachea, and subglottic stenosis.

*Topic: Down Syndrome*

**Q58. Answer: B — Endocardial cushion defects and VSD**

Page 240: Down syndrome patients have associated congenital heart diseases, mainly endocardial cushion defects and VSD. Other associated anomalies include tracheoesophageal fistula, mental retardation, and seizures, which can increase morbidity and mortality.

*Topic: Down Syndrome*

**Q59. Answer: C — Due to hypotonia**

Page 240: Down syndrome patients may exhibit hypersensitivity to muscle relaxants due to hypotonia. This is an important consideration during anesthetic management, and lower doses should be anticipated.

*Topic: Down Syndrome*

**Q60. Answer: D — Due to immature brain and poorly developed blood-brain barrier**

Page 238: Neonates have an immature brain and a poorly developed blood-brain barrier, putting them at higher risk of overdoses. Combined with reduced albumin (increased free drug fraction) and underdeveloped hepatic/renal functions, careful dose titration is essential.

*Topic: CNS and Blood-Brain Barrier*



# Geriatric Anesthesia MCQ – Chapter 30

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 30)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. What is the most common cardiac finding in old age?**

- A. Systolic dysfunction
- B. Diastolic dysfunction
- C. Atrial fibrillation
- D. Congestive heart failure

**Ans: B**

**Q2. What cardiac function is usually preserved in old age despite other cardiovascular changes?**

- A. Heart rate reserve
- B. Cardiac output at rest
- C. Diastolic function
- D. Systolic function

**Ans: D**

**Q3. What happens to blood pressure with advancing age?**

- A. Decreases due to reduced cardiac output
- B. Decreases due to vasodilation
- C. Remains unchanged
- D. Blood pressure increases with age

**Ans: D**

**Q4. Which respiratory change in old age leads to V/Q mismatch and decreased pO<sub>2</sub>?**

- A. Decreased functional residual capacity
- B. Decreased residual volume
- C. Increased respiratory muscle function
- D. Increased closing capacity

**Ans: D**

**Q5. Which spinal deformities can cause difficulty in ventilation in geriatric patients?**

- A. Kyphosis and lordosis
- B. Lordosis and scoliosis
- C. Spondylosis only
- D. Kyphosis and scoliosis

**Ans: D**

**Q6. Why is the cough reflex decreased in old age patients?**

- A. Decreased respiratory muscle function decreases the cough reflex
- B. Due to increased mucus production
- C. Due to CNS depression
- D. Due to increased airway sensitivity

**Ans: A**

**Q7. What is the CNS consequence of loss of neurons in old age?**

- A. Increased cognitive function
- B. Improved neuroplasticity
- C. Increased brain reserve
- D. Decreased brain reserve leading to dementia

**Ans: D**

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| <p><b>Q8. In which age group is perioperative delirium and postoperative cognitive dysfunction (POCD) far more common?</b></p> <p>A. Pediatric patients<br/>B. Middle-aged patients<br/>C. Young adults<br/>D. Old age people</p>                                                                                          | <p><b>Ans: D</b></p> |
| <p><b>Q9. What thermoregulation risk is the geriatric population more prone to?</b></p> <p>A. Hypothermia<br/>B. Malignant hyperthermia<br/>C. Hyperthermia<br/>D. Hyperpyrexia</p>                                                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q10. Is age a contraindication for surgery?</b></p> <p>A. Age is not a contraindication for any surgery<br/>B. Age &gt; 75 requires special approval<br/>C. Yes, age &gt; 70 is a relative contraindication<br/>D. Yes, age &gt; 80 is contraindication</p>                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q11. Which preoperative assessment is a MUST for all geriatric patients?</b></p> <p>A. Echocardiography<br/>B. Renal function tests<br/>C. Assessment of cognitive function<br/>D. PFT (Pulmonary Function Tests)</p>                                                                                                | <p><b>Ans: C</b></p> |
| <p><b>Q12. What is the significance of preoperative dementia in geriatric patients?</b></p> <p>A. Has no perioperative impact<br/>B. Patients with preoperative dementia have higher incidence of postoperative delirium<br/>C. Requires ECT pre-treatment<br/>D. Contraindicates surgery</p>                              | <p><b>Ans: B</b></p> |
| <p><b>Q13. In geriatric patients, what determines when investigations should be done preoperatively?</b></p> <p>A. All patients need a complete battery of tests<br/>B. Investigations should be guided by associated comorbidity<br/>C. Patient's weight<br/>D. Age above 65 years</p>                                    | <p><b>Ans: B</b></p> |
| <p><b>Q14. Why must a detailed record of medication be obtained in geriatric patients preoperatively?</b></p> <p>A. To identify controlled substances<br/>B. To calculate drug costs<br/>C. Many old age people are on polypharmacy and appropriate instructions must be given<br/>D. To determine drug allergies only</p> | <p><b>Ans: C</b></p> |

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| <p><b>Q15. Which type of anesthesia is PREFERRED over general anesthesia in geriatric patients?</b></p> <p>A. Regional anesthesia<br/>B. Ketamine-based anesthesia<br/>C. Total IV anesthesia (TIVA)<br/>D. Monitored anesthesia care (MAC)</p>                                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q16. Which benefit of regional anesthesia is related to reduced postoperative venous thrombosis?</b></p> <p>A. Faster recovery<br/>B. Reduced PONV<br/>C. Reduced drug toxicity<br/>D. Central neuraxial blocks decrease the risk of thromboembolism</p>                                                                                                    | <p><b>Ans: D</b></p> |
| <p><b>Q17. Why may central neuraxial blocks be technically difficult in geriatric patients?</b></p> <p>A. Due to increased blood pressure<br/>B. Due to patient non-cooperation<br/>C. Due to degenerative changes in spine<br/>D. Due to osteoporosis only</p>                                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q18. Why do old age patients achieve a higher level of spinal and epidural block?</b></p> <p>A. Due to increased drug sensitivity only<br/>B. Due to narrowing of spaces in the spine<br/>C. Due to increased CSF volume<br/>D. Due to decreased blood pressure</p>                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q19. By how much should benzodiazepine premedication requirement be reduced in geriatric patients?</b></p> <p>A. 50%<br/>B. 25%<br/>C. 75%<br/>D. 100% (completely avoided)</p>                                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q20. Why is the dose of ALL IV induction agents decreased in geriatric patients?</b></p> <p>A. Due to smaller blood volume<br/>B. Due to reduced weight only<br/>C. Only due to reduced cardiac output<br/>D. Increased brain sensitivity, decreased volume of distribution, decreased unbound fraction (low albumin), reduced metabolism and clearance</p> | <p><b>Ans: D</b></p> |
| <p><b>Q21. Why is Propofol preferred as an induction agent in geriatric patients?</b></p> <p>A. It has the longest duration of action<br/>B. It is cheapest<br/>C. It has antiemetic properties<br/>D. Propofol because of shorter half-life is preferred</p>                                                                                                     | <p><b>Ans: D</b></p> |

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| <p><b>Q22. What is the clinical significance of decreased pseudocholinesterase levels in geriatric patients?</b></p> <p>A. Requires avoidance of all muscle relaxants<br/> B. Prolongs the effect of suxamethonium clinically<br/> C. Theoretically prolongs suxamethonium effect but clinically no prolongation is seen<br/> D. Has no effect on any drug</p> | <p><b>Ans: C</b></p> |
| <p><b>Q23. Why may intubation and mask ventilation become difficult in geriatric patients?</b></p> <p>A. Due to increased secretions<br/> B. Due to edentulous jaw and decreased neck movements from RA and osteoarthritis<br/> C. Due to macroglossia<br/> D. Due to obesity</p>                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q24. What happens to stress response during surgery and anesthesia in geriatric patients?</b></p> <p>A. Remains unchanged<br/> B. Stress response to cope up with hemodynamic changes decreases<br/> C. Only hormonal response is affected<br/> D. Increases significantly</p>                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q25. By how much does the MAC of inhalational agents decrease per decade in geriatric patients?</b></p> <p>A. 1-2%<br/> B. 2-3%<br/> C. 4-6%<br/> D. 10-15%</p>                                                                                                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q26. Which is the preferred inhalational agent for geriatric patients and why?</b></p> <p>A. Isoflurane - due to low cost<br/> B. Desflurane - because of rapid emergence and least metabolism<br/> C. Sevoflurane - due to non-pungency<br/> D. Halothane - due to cardiac stability</p>                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q27. By how much should opioid doses be reduced in geriatric patients?</b></p> <p>A. No dose change required<br/> B. Reduced by half (50%)<br/> C. 25%<br/> D. 75%</p>                                                                                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q28. Which is the most preferred opioid for geriatric patients?</b></p> <p>A. Remifentanyl<br/> B. Morphine<br/> C. Fentanyl<br/> D. Meperidine</p>                                                                                                                                                                                                      | <p><b>Ans: A</b></p> |

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| <p><b>Q29. What property of Atracurium and Cis-atracurium makes them the muscle relaxants of choice for old age patients?</b></p> <p>A. Plasma cholinesterase metabolism<br/>B. Renal elimination<br/>C. Hepatic metabolism<br/>D. Hoffman degradation</p>                                                                               | <p><b>Ans: D</b></p> |
| <p><b>Q30. Why can decreased cardiac output delay the onset of muscle relaxants in geriatric patients?</b></p> <p>A. Due to larger volume of distribution<br/>B. Decreased cardiac output delays drug delivery to the neuromuscular junction<br/>C. Due to increased receptor density<br/>D. Due to increased plasma protein binding</p> | <p><b>Ans: B</b></p> |
| <p><b>Q31. What change in pain perception occurs in the postoperative period in geriatric patients?</b></p> <p>A. Studies show age-related decrease in pain perception<br/>B. Increased pain sensitivity<br/>C. Hyperpathia is common<br/>D. No change in pain sensitivity</p>                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q32. What is the relationship between geriatric patients and opioid sensitivity postoperatively?</b></p> <p>A. They require higher opioid doses<br/>B. Sensitivity remains unchanged<br/>C. They are less sensitive to opioids<br/>D. They are oversensitive to opioids and therefore opioids should be used very carefully</p>    | <p><b>Ans: D</b></p> |
| <p><b>Q33. Which postoperative complications are geriatric patients more prone to?</b></p> <p>A. Hypoglycemia and hypertension<br/>B. DVT and pulmonary embolism only<br/>C. Hyperthermia and urinary incontinence<br/>D. Hypoxia and urinary retention in postoperative period</p>                                                      | <p><b>Ans: D</b></p> |
| <p><b>Q34. What is the overall prevalence of postoperative delirium in older patients?</b></p> <p>A. 10%<br/>B. 50%<br/>C. 5%<br/>D. 25%</p>                                                                                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q35. To what percentage can postoperative delirium incidence increase in patients shifted to ICU?</b></p> <p>A. 15-20%<br/>B. 30-35%<br/>C. 25%<br/>D. More than 50%</p>                                                                                                                                                           | <p><b>Ans: D</b></p> |

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| <p><b>Q36. Which age is cited as an important predisposing factor for postoperative delirium?</b></p> <p>A. Age &gt; 50 years<br/>B. Age &gt; 60 years<br/>C. Age &gt; 75 years<br/>D. Age &gt; 65 years</p>                                                                                                                                                                             | <p><b>Ans: D</b></p> |
| <p><b>Q37. Which type of surgery is a predisposing factor for postoperative delirium?</b></p> <p>A. Ophthalmic surgery<br/>B. Laparoscopic surgery only<br/>C. Major surgery<br/>D. Minor elective surgery</p>                                                                                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q38. Which analgesic drug is listed as predisposing to postoperative delirium?</b></p> <p>A. Fentanyl<br/>B. Tramadol<br/>C. Morphine<br/>D. Meperidine (pethidine)</p>                                                                                                                                                                                                            | <p><b>Ans: D</b></p> |
| <p><b>Q39. Which of the following is NOT listed as a predisposing factor for postoperative delirium?</b></p> <p>A. Sleep deprivation<br/>B. Polypharmacy<br/>C. Female gender<br/>D. Electrolyte imbalance</p>                                                                                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q40. What are the short-term changes after surgery related to POCD in geriatric patients?</b></p> <p>A. Only affects patients &gt;80 years<br/>B. Short-term (up to 3 months) changes in cognitive function well-documented in 10-15% of geriatric population after surgery<br/>C. Short-term changes in 30-40% of patients<br/>D. Permanent cognitive decline in all patients</p> | <p><b>Ans: B</b></p> |
| <p><b>Q41. What percentage of patients can have long-term (&gt;3 months) cognitive impairment after surgery?</b></p> <p>A. 5%<br/>B. 10%<br/>C. 15%<br/>D. 1%</p>                                                                                                                                                                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q42. What is the relationship between POCD and the type of surgery/anesthesia?</b></p> <p>A. POCD is more common with general anesthesia<br/>B. POCD is more common with regional anesthesia<br/>C. The incidence of POCD is independent of both surgery and anesthesia<br/>D. POCD is more common with major surgery</p>                                                          | <p><b>Ans: C</b></p> |

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| <p><b>Q43. What is the current treatment for POCD?</b></p> <ul style="list-style-type: none"> <li>A. Cognitive rehabilitation therapy</li> <li>B. Cholinesterase inhibitors</li> <li>C. At present there is no specific treatment available for POCD</li> <li>D. NSAID prophylaxis</li> </ul>                                                                                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q44. What change in albumin levels occurs in old age and what is its pharmacological significance?</b></p> <ul style="list-style-type: none"> <li>A. No change in albumin levels</li> <li>B. Albumin decreases, leading to decreased unbound fraction of drug, decreasing dose requirements but increasing toxicity</li> <li>C. Albumin increases, so more drug is protein-bound</li> <li>D. Decreased albumin leads to decreased unbound fraction</li> </ul>   | <p><b>Ans: B</b></p> |
| <p><b>Q45. What happens to the volume of distribution of drugs in geriatric patients?</b></p> <ul style="list-style-type: none"> <li>A. Only water-soluble drug distribution changes</li> <li>B. Increases for all drugs</li> <li>C. Decrease in volume of distribution</li> <li>D. Volume of distribution is unchanged</li> </ul>                                                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q46. Why does general anesthesia have a higher incidence of postoperative delirium compared to regional anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Due to increased stress response under GA</li> <li>B. General anesthesia (compared to regional anesthesia) is an important predisposing factor for postoperative delirium</li> <li>C. Due to higher opioid use under GA</li> <li>D. Due to longer duration of surgery under GA</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q47. What happens to residual volume and functional residual capacity in old age due to decreased lung elasticity?</b></p> <ul style="list-style-type: none"> <li>A. Decreased elasticity leads to alveolar distension and increase in residual volume and functional residual capacity</li> <li>B. Residual volume decreases, FRC increases</li> <li>C. Both decrease</li> <li>D. Both remain unchanged</li> </ul>                                             | <p><b>Ans: A</b></p> |
| <p><b>Q48. What happens to the adrenergic response with age and what is its clinical significance?</b></p> <ul style="list-style-type: none"> <li>A. Only alpha-adrenergic response is affected</li> <li>B. Increased adrenergic response causing hypertension</li> <li>C. Decreased adrenergic response, contributing to increased incidence of cardiac complications</li> <li>D. No change in adrenergic response</li> </ul>                                        | <p><b>Ans: C</b></p> |



**Q49. Which pulmonary complication associated with general anesthesia can be avoided by using regional anesthesia in geriatric patients?**

- A. Aspiration pneumonitis only
- B. Pulmonary edema only
- C. Pulmonary complications associated with general anesthesia may be avoided with regional anesthesia
- D. Pneumothorax only

**Ans: C**

**Q50. What is the key statement regarding chronological age and surgery/investigations in geriatric patients?**

- A. Age > 70 requires full workup before surgery
- B. Chronological age is neither a contraindication for any surgery nor a criterion to do battery of routine investigations
- C. Chronological age is a contraindication for high-risk surgery
- D. Age > 65 requires cardiac clearance

**Ans: B**

# Answer Key & Explanations

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## **Q1. Answer: B — Diastolic dysfunction**

Page 242: Decreased ventricular compliance leading to diastolic dysfunction is the most common finding in old age. It is also a significant determinant of morbidity.

*Topic: Physiological Changes - Cardiovascular*

## **Q2. Answer: D — Systolic function**

Page 242: Systolic function usually remains preserved in old age, whereas diastolic dysfunction is the most common finding due to decreased ventricular compliance.

*Topic: Physiological Changes - Cardiovascular*

## **Q3. Answer: D — Blood pressure increases with age**

Page 242: Blood pressure increases with age. This along with decreased heart rate, decreased adrenergic response, and decreased cardiac reserve increase the incidence of cardiac complications.

*Topic: Physiological Changes - Cardiovascular*

## **Q4. Answer: D — Increased closing capacity**

Page 242: Increase in closing capacity increases the V/Q mismatch decreasing the pO<sub>2</sub>. This is due to decreased elasticity leading to alveolar distension.

*Topic: Physiological Changes - Respiratory*

## **Q5. Answer: D — Kyphosis and scoliosis**

Page 242: Kyphosis and scoliosis can cause difficulty in ventilation in geriatric patients.

*Topic: Physiological Changes - Respiratory*

## **Q6. Answer: A — Decreased respiratory muscle function decreases the cough reflex**

Page 242: Decreased respiratory muscle function decreases the cough reflex in geriatric patients, making them more prone to aspiration pneumonia.

*Topic: Physiological Changes - Respiratory*

## **Q7. Answer: D — Decreased brain reserve leading to dementia**

Page 242: Loss of neurons decreases the brain reserve leading to dementia. There also occurs increased sensitivity to anesthetic medications.

*Topic: Physiological Changes - CNS*

## **Q8. Answer: D — Old age people**

Page 242: The risk of perioperative delirium and postoperative cognitive dysfunction (POCD) is far more common in old age people due to decreased brain reserve.

*Topic: Physiological Changes - CNS*

## **Q9. Answer: A — Hypothermia**

Page 242: Geriatric population is more prone for hypothermia due to impaired thermoregulation mechanisms.

*Topic: Physiological Changes - Thermoregulation*

**Q10. Answer: A — Age is not a contraindication for any surgery**

Page 242: Age is not a contraindication for any surgery. Chronological age is neither a contraindication for any surgery nor a criterion to do battery of routine investigations.

*Topic: Anesthetic Management - General*

**Q11. Answer: C — Assessment of cognitive function**

Page 242: Assessment of cognitive function is must. Patients with preoperative dementia have higher incidence of postoperative delirium.

*Topic: Anesthetic Management - Preoperative*

**Q12. Answer: B — Patients with preoperative dementia have higher incidence of postoperative delirium**

Page 242: Patients with preoperative dementia have higher incidence of postoperative delirium. Moreover, cognitive assessment can be used to compare if patient develops cognitive function in the postoperative period.

*Topic: Anesthetic Management - Preoperative*

**Q13. Answer: B — Investigations should be guided by associated comorbidity**

Page 242: Age is not a criteria to do battery of routine investigations. Investigations should be guided by associated comorbidity.

*Topic: Anesthetic Management - Preoperative*

**Q14. Answer: C — Many old age people are on polypharmacy and appropriate instructions must be given**

Page 242: Many of the old age people are on polypharmacy. A detailed record of medication should be obtained and instructions are given accordingly.

*Topic: Anesthetic Management - Preoperative*

**Q15. Answer: A — Regional anesthesia**

Page 242: Regional anesthesia is preferred over general anesthesia due to increased sensitivity of brain to anesthetic agents, decreased metabolism/excretion, reduced chances of drug toxicity, decreased thromboembolism risk, fewer pulmonary complications, and less incidence of postoperative delirium.

*Topic: Anesthetic Management - Choice of Anesthesia*

**Q16. Answer: D — Central neuraxial blocks decrease the risk of thromboembolism**

Page 243: Central neuraxial blocks decrease the risk of thromboembolism. This is one reason regional anesthesia is preferred over general anesthesia in geriatric patients.

*Topic: Anesthetic Management - Choice of Anesthesia*

**Q17. Answer: C — Due to degenerative changes in spine**

Page 243: Regional anesthesia, particularly central neuraxial blocks, may be technically difficult due to degenerative changes in spine.

*Topic: Anesthetic Management - Choice of Anesthesia*

**Q18. Answer: B — Due to narrowing of spaces in the spine**

Page 243: Old age patients achieve higher level of spinal (and epidural) due to narrowing of spaces. Geriatric population is also more sensitive to the effect of local anesthetics during nerve blocks.

*Topic: Anesthetic Management - Choice of Anesthesia*

**Q19. Answer: A — 50%**

Page 243: As the requirement of benzodiazepines may be reduced by 50%, premedication with benzodiazepines should be done cautiously in geriatric patients.

Topic: Anesthetic Management - General Anesthesia

**Q20. Answer: D — Increased brain sensitivity, decreased volume of distribution, decreased unbound fraction (low albumin), reduced metabolism and clearance**

Page 243: Increase sensitivity of brain, decrease volume of distribution, decreased unbound fraction (due to decreased albumin levels) and reduced metabolism and clearance not only decreases the dose of all IV agents but also increases toxicity.

Topic: Anesthetic Management - General Anesthesia

**Q21. Answer: D — Propofol because of shorter half-life is preferred**

Page 243: Propofol because of shorter half-life is preferred. Decreased cardiac output may slow the onset with IV agents.

Topic: Anesthetic Management - General Anesthesia

**Q22. Answer: C — Theoretically prolongs suxamethonium effect but clinically no prolongation is seen**

Page 243: Decreased pseudocholinesterase levels due to decreased hepatic functions may theoretically prolong the effect of suxamethonium, however clinically no prolongation has been seen.

Topic: Anesthetic Management - General Anesthesia

**Q23. Answer: B — Due to edentulous jaw and decreased neck movements from RA and osteoarthritis**

Page 243: Intubation and mask ventilation may become difficult due to edentulous jaw and decreased neck movements (due to rheumatoid arthritis and osteoarthritis).

Topic: Anesthetic Management - General Anesthesia

**Q24. Answer: B — Stress response to cope up with hemodynamic changes decreases**

Page 243: Stress response to cope up with hemodynamic changes during surgery and anesthesia decreases in geriatric patients.

Topic: Anesthetic Management - General Anesthesia

**Q25. Answer: C — 4-6%**

Page 243: The minimum alveolar concentration (MAC) of inhalational agents decreases by 4-6% per decade in geriatric patients.

Topic: Anesthetic Management - General Anesthesia

**Q26. Answer: B — Desflurane - because of rapid emergence and least metabolism**

Page 243: Desflurane because of its rapid emergence and least metabolism is the preferred inhalational agent for geriatric patients.

Topic: Anesthetic Management - General Anesthesia

**Q27. Answer: B — Reduced by half (50%)**

Page 243: Doses of opioids should be reduced by half in geriatric patients due to age-related decrease in drug metabolism and excretion.

Topic: Anesthetic Management - General Anesthesia

**Q28. Answer: A — Remifentanyl**

Page 243: Remifentanyl because of its ultra-short life is the most preferred opioid for old age patients.

Topic: Anesthetic Management - General Anesthesia

**Q29. Answer: D — Hoffman degradation**

Page 243: Hoffman degradation makes Atracurium and Cis-atracurium to be the muscle relaxants of choice for old age patients as it is independent of hepatic and renal function.

Topic: Anesthetic Management - General Anesthesia

**Q30. Answer: B — Decreased cardiac output delays drug delivery to the neuromuscular junction**

Page 243: Decrease cardiac output can delay the onset of muscle relaxants in geriatric patients as the drug takes longer to reach the neuromuscular junction.

Topic: Anesthetic Management - General Anesthesia

**Q31. Answer: A — Studies show age-related decrease in pain perception**

Page 243: Studies have shown that there occurs age related decrease in pain perception in geriatric patients in the postoperative period.

Topic: Anesthetic Management - Postoperative

**Q32. Answer: D — They are oversensitive to opioids and therefore opioids should be used very carefully**

Page 243: Geriatric patients are oversensitive to opioids, therefore opioids should be used very carefully in the postoperative period.

Topic: Anesthetic Management - Postoperative

**Q33. Answer: D — Hypoxia and urinary retention in postoperative period**

Page 243: Geriatric population is more prone for hypoxia and urinary retention in the postoperative period.

Topic: Anesthetic Management - Postoperative

**Q34. Answer: A — 10%**

Page 243: The overall prevalence of postoperative delirium in older patients is estimated to be 10%.

Topic: Postoperative Delirium

**Q35. Answer: D — More than 50%**

Page 243: The incidence of delirium can increase to more than 50% in patients shifted to ICU.

Topic: Postoperative Delirium

**Q36. Answer: D — Age > 65 years**

Page 243: Important predisposing factors are age >65 years, male gender, major surgery, general anesthesia (compared to regional anesthesia), and other factors like drugs and comorbidities.

Topic: Postoperative Delirium

**Q37. Answer: C — Major surgery**

Page 243: Important predisposing factors include major surgery (compared to minor), general anesthesia (compared to regional anesthesia), age >65 years and male gender.

Topic: Postoperative Delirium

**Q38. Answer: D — Meperidine (pethidine)**

Page 243: Drugs like meperidine (pethidine) are listed as predisposing factors for postoperative delirium in geriatric patients.

Topic: Postoperative Delirium

**Q39. Answer: C — Female gender**

Page 243: Male gender (not female gender) is listed as a predisposing factor. Other predisposing factors include age >65 years, major surgery, GA, meperidine, cognitive impairment, dementia, comorbidities, polypharmacy, electrolyte imbalance, pain, and sleep deprivation.

Topic: Postoperative Delirium

**Q40. Answer: B — Short-term (up to 3 months) changes in cognitive function well-documented in 10-15% of geriatric population after surgery**

Page 243: Short-term (up to 3 months) changes in cognitive function have been very well-documented in 10-15% of geriatric population after surgery.

Topic: Postoperative Cognitive Dysfunction (POCD)

**Q41. Answer: D — 1%**

Page 243: Long-term (>3 months) cognitive impairment can also occur in 1% of the patients.

Topic: Postoperative Cognitive Dysfunction (POCD)

**Q42. Answer: C — The incidence of POCD is independent of both surgery and anesthesia**

Page 243: The incidence of postoperative cognitive dysfunction (POCD) is independent of both surgery and anesthesia. Studies have shown no difference in the incidence of POCD in minor versus major surgery and regional versus general anesthesia.

Topic: Postoperative Cognitive Dysfunction (POCD)

**Q43. Answer: C — At present there is no specific treatment available for POCD**

Page 243: At present there is no specific treatment available for POCD.

Topic: Postoperative Cognitive Dysfunction (POCD)

**Q44. Answer: B — Albumin decreases, leading to decreased unbound fraction of drug, decreasing dose requirements but increasing toxicity**

Page 243: Decreased albumin levels in old age lead to decreased unbound fraction of the drug. This along with reduced metabolism and clearance decreases the dose of all IV agents but also increases toxicity.

Topic: Anesthetic Management - Pharmacology

**Q45. Answer: C — Decrease in volume of distribution**

Page 243: Decrease volume of distribution is one of the reasons why dose of all IV agents decreases in geriatric patients, along with increased brain sensitivity, decreased albumin, and reduced metabolism.

Topic: Anesthetic Management - Pharmacology

**Q46. Answer: B — General anesthesia (compared to regional anesthesia) is an important predisposing factor for postoperative delirium**

Page 243: General anesthesia compared to regional anesthesia is listed as an important predisposing factor for postoperative delirium in geriatric patients. This is also why regional anesthesia is preferred.

Topic: Postoperative Delirium

**Q47. Answer: A — Decreased elasticity leads to alveolar distension and increase in residual volume and functional residual capacity**

Page 242: Decreased elasticity leads to alveolar distension and increase in residual volume and functional residual capacity in old age.

*Topic: Physiological Changes - Respiratory*

**Q48. Answer: C — Decreased adrenergic response, contributing to increased incidence of cardiac complications**

Page 242: Decrease in adrenergic response along with decreased heart rate and decreased cardiac reserve increase the incidence of cardiac complications in old age.

*Topic: Physiological Changes - Cardiovascular*

**Q49. Answer: C — Pulmonary complications associated with general anesthesia may be avoided with regional anesthesia**

Page 243: Pulmonary complications associated with general anesthesia (GA) may be avoided with the use of regional anesthesia in geriatric patients. This is one of the advantages of regional anesthesia.

*Topic: Anesthetic Management - Choice of Anesthesia*

**Q50. Answer: B — Chronological age is neither a contraindication for any surgery nor a criterion to do battery of routine investigations**

Page 244 (Key Points): Chronological age is neither a contraindication for any surgery nor a criterion to do battery of routine investigations. Investigations should be guided by associated comorbidity.

*Topic: Key Points*

# Anesthesia for Obese Patients (Bariatric Anesthesia) MCQ – Chapter 31

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 31)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

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**Q1. What is the definition of obesity based on BMI?**

- A. BMI more than 20%
- B. BMI more than 25%
- C. BMI more than 30%
- D. BMI more than 40%

**Ans: C**

**Q2. What BMI defines extreme/superobese?**

- A. More than 60%
- B. More than 50%
- C. More than 30%
- D. More than 40%

**Ans: B**

**Q3. Why are obese patients one of the most challenging for anesthesiologists?**

- A. Due to positioning only
- B. Due to long procedures only
- C. Due to associated comorbidities and difficult airway management
- D. Due to cost issues

**Ans: C**

**Q4. What cardiovascular conditions may obese patients have?**

- A. Only hypotension
- B. Only hypertension
- C. Only bradycardia
- D. Hypertension, coronary artery disease, LVH, RVH, and cor pulmonale

**Ans: D**

**Q5. How much does cardiac output increase per kg of fat to perfuse extra fat?**

- A. 0.5 L/min
- B. 0.1 L/min
- C. 0.05 L/min
- D. 0.2 L/min

**Ans: B**

**Q6. Why do obese patients develop right ventricular hypertrophy?**

- A. Due to mitral regurgitation
- B. Due to fluid overload
- C. Due to aortic stenosis
- D. Resulting from pulmonary hypertension which is because of persistent hypoxia

**Ans: D**

**Q7. What causes cor pulmonale in obese patients?**

- A. Left ventricular failure
- B. Aortic stenosis
- C. Mitral stenosis
- D. Right ventricular strain

**Ans: D**

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| <p><b>Q8. What respiratory changes are seen in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Increased FRC</li> <li>B. Reduced ERV, FRC, vital capacity, increased airway resistance, and reduced oxygen saturation</li> <li>C. Normal lung function</li> <li>D. Increased vital capacity</li> </ul>                    | <p><b>Ans: B</b></p> |
| <p><b>Q9. What is obstructive sleep apnea hypopnea syndrome (OSAHS) and its significance?</b></p> <ul style="list-style-type: none"> <li>A. Patients with OSAHS are at increased risk of morbidity and mortality</li> <li>B. It has no anesthetic significance</li> <li>C. A benign condition</li> <li>D. It only causes snoring</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q10. What GI complications may obese patients have?</b></p> <ul style="list-style-type: none"> <li>A. Only liver disease</li> <li>B. Only diarrhea</li> <li>C. Only constipation</li> <li>D. Hiatus hernia, gastroesophageal reflux, delayed gastric emptying and fatty infiltration of liver</li> </ul>                              | <p><b>Ans: D</b></p> |
| <p><b>Q11. Why may obese patients have polycythemia?</b></p> <ul style="list-style-type: none"> <li>A. Due to dehydration</li> <li>B. Due to EPO injections</li> <li>C. Due to iron overload</li> <li>D. Because of chronic hypoxia</li> </ul>                                                                                              | <p><b>Ans: D</b></p> |
| <p><b>Q12. What are obese patients very prone to in postoperative period?</b></p> <ul style="list-style-type: none"> <li>A. Thromboembolism</li> <li>B. Hyperthermia</li> <li>C. Bradycardia</li> <li>D. Hypothermia</li> </ul>                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q13. Why do obese patients often have difficult airway?</b></p> <ul style="list-style-type: none"> <li>A. Short neck (decreased thyromental distance), restricted mouth opening, and restricted neck movement</li> <li>B. Due to dental issues</li> <li>C. Due to large tongue only</li> <li>D. Due to small mouth only</li> </ul>    | <p><b>Ans: A</b></p> |
| <p><b>Q14. Is obesity per se the criteria to order routine tests?</b></p> <ul style="list-style-type: none"> <li>A. No, investigation should be based on clinical assessment and associated comorbidities</li> <li>B. Only blood tests are needed</li> <li>C. Yes, all tests must be done</li> <li>D. Only chest X-ray is needed</li> </ul> | <p><b>Ans: A</b></p> |

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| <p><b>Q15. What must be checked by pulse oximetry as baseline in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Temperature</li> <li>B. Heart rate</li> <li>C. Oxygen saturation</li> <li>D. Blood pressure</li> </ul>                                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q16. In which obese patients should polysomnography be strongly considered?</b></p> <ul style="list-style-type: none"> <li>A. Morbidly obese patients or patients with a history of obstructive sleep apnea</li> <li>B. All obese patients</li> <li>C. Only those with hypertension</li> <li>D. Only super-obese patients</li> </ul>                                       | <p><b>Ans: A</b></p> |
| <p><b>Q17. Why should respiratory depressant drugs like benzodiazepines be avoided for premedication in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Due to risk of respiratory depression in patients already prone to hypoxia and sleep apnea</li> <li>B. They cause nausea</li> <li>C. They are expensive</li> <li>D. They cause hypertension</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q18. What premedication should be considered for obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Benzodiazepines</li> <li>B. Antihistamines only</li> <li>C. Opioids</li> <li>D. Metoclopramide and ranitidine</li> </ul>                                                                                                                                | <p><b>Ans: D</b></p> |
| <p><b>Q19. What size BP cuff should be used for monitoring obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Appropriate size BP cuff</li> <li>B. Standard adult cuff</li> <li>C. Pediatric cuff</li> <li>D. Any available cuff</li> </ul>                                                                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q20. What is the anesthesia of choice for obese patients due to difficult airway?</b></p> <ul style="list-style-type: none"> <li>A. Regional anesthesia</li> <li>B. IV sedation</li> <li>C. General anesthesia</li> <li>D. Local anesthesia only</li> </ul>                                                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q21. Why is regional anesthesia often difficult in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Due to infection risk</li> <li>B. Due to bleeding risk</li> <li>C. Due to reduced epidural and subarachnoid space making them vulnerable for high spinals</li> <li>D. Due to patient refusal</li> </ul>                                               | <p><b>Ans: C</b></p> |

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| <p><b>Q22. What has been found beneficial during preoxygenation in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Nebulization</li> <li>B. High flow nasal oxygen</li> <li>C. Adding 10 cm H<sub>2</sub>O of CPAP during preoxygenation</li> <li>D. Incentive spirometry</li> </ul>                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q23. Why do lipid soluble drugs (IV induction agents, opioids) have increased volume of distribution in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Due to increased renal clearance</li> <li>B. Due to increased fat stores</li> <li>C. Due to hepatic enzyme induction</li> <li>D. Due to decreased protein binding</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q24. On what weight basis should lipid soluble drugs be given in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Adjusted body weight</li> <li>B. Actual body weight</li> <li>C. Lean body weight</li> <li>D. Ideal body weight</li> </ul>                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q25. Is the increased volume of distribution effect as marked on propofol as on thiopentone?</b></p> <ul style="list-style-type: none"> <li>A. More marked on propofol</li> <li>B. No, this effect is not so marked on propofol as on thiopentone</li> <li>C. Neither is affected</li> <li>D. Yes, equally marked</li> </ul>                           | <p><b>Ans: B</b></p> |
| <p><b>Q26. What can achieving IV lines in obese patients become?</b></p> <ul style="list-style-type: none"> <li>A. Impossible always</li> <li>B. An arduous task</li> <li>C. A simple procedure</li> <li>D. Very easy</li> </ul>                                                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q27. What is the most challenging part in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Positioning</li> <li>B. Monitoring</li> <li>C. Airway management</li> <li>D. Drug dosing</li> </ul>                                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q28. How much more is the incidence of failed intubation in bariatric population?</b></p> <ul style="list-style-type: none"> <li>A. 5 times more</li> <li>B. 10 times more</li> <li>C. 3 times more</li> <li>D. 2 times more</li> </ul>                                                                                                                | <p><b>Ans: C</b></p> |

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| <p><b>Q29. What is the preferred position for intubation in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Ramped position (ear to sternum position)</li> <li>B. Trendelenburg position</li> <li>C. Lateral position</li> <li>D. Supine position</li> </ul>                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q30. How is the ramp position achieved?</b></p> <ul style="list-style-type: none"> <li>A. By flexing the neck only</li> <li>B. By extending the head only</li> <li>C. By tilting the table</li> <li>D. By elevating the upper back using sheets/blankets so external auditory meatus lies in horizontal line with sternal notch</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q31. What should be considered if intubation is anticipated to be difficult in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Tracheostomy first</li> <li>B. LMA only</li> <li>C. Awake intubation with fiberoptic bronchoscopy</li> <li>D. Blind nasal intubation</li> </ul>                                           | <p><b>Ans: C</b></p> |
| <p><b>Q32. How must confirmation of intubation be done in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. By capnography as breath sounds may not be clearly audible on auscultation</li> <li>B. By auscultation only</li> <li>C. By tube length marking</li> <li>D. By chest rise only</li> </ul>                             | <p><b>Ans: A</b></p> |
| <p><b>Q33. Why are inhalational agents more extensively metabolized in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Due to increased fat stores and hepatic blood flow</li> <li>B. Due to decreased protein binding</li> <li>C. Due to increased renal clearance</li> <li>D. Due to enzyme deficiency</li> </ul>            | <p><b>Ans: A</b></p> |
| <p><b>Q34. Which is the inhalational agent of choice for maintenance in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Sevoflurane</li> <li>D. Desflurane</li> </ul>                                                                                                             | <p><b>Ans: D</b></p> |
| <p><b>Q35. Why should halothane be avoided in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Increased metabolism of halothane makes obese patients more vulnerable for halothane hepatitis</li> <li>B. It causes arrhythmias</li> <li>C. It is not potent enough</li> <li>D. It causes bronchospasm</li> </ul>               | <p><b>Ans: A</b></p> |

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| <p><b>Q36. On what weight basis should water soluble drugs (nondepolarizing muscle relaxants) be given?</b></p> <p>A. Actual body weight<br/>B. Ideal body weight<br/>C. Total body weight<br/>D. Adjusted body weight</p>                                           | <p><b>Ans: B</b></p> |
| <p><b>Q37. What does application of 10 cm H<sub>2</sub>O PEEP during mechanical ventilation do in obese patients?</b></p> <p>A. Decreases oxygenation<br/>B. Improves oxygenation by preventing alveolar collapse<br/>C. Causes hypotension<br/>D. Has no effect</p> | <p><b>Ans: B</b></p> |
| <p><b>Q38. What position improves oxygenation particularly during laparoscopic surgeries in obese patients?</b></p> <p>A. Lateral decubitus<br/>B. Supine flat<br/>C. 30 degree reverse Trendelenburg<br/>D. Trendelenburg</p>                                       | <p><b>Ans: C</b></p> |
| <p><b>Q39. What are obese patients very prone to in the postoperative period?</b></p> <p>A. Hypothermia<br/>B. Hypoxia<br/>C. Bradycardia<br/>D. Hyperglycemia only</p>                                                                                              | <p><b>Ans: B</b></p> |
| <p><b>Q40. In what position should obese patients be nursed postoperatively?</b></p> <p>A. Trendelenburg<br/>B. Supine flat<br/>C. 45 degree sitting (semi-reclined position) with supplemental oxygen<br/>D. Prone</p>                                              | <p><b>Ans: C</b></p> |
| <p><b>Q41. When should CPAP or BiPAP be applied in postoperative obese patients?</b></p> <p>A. Only after 24 hours<br/>B. If patient is not maintaining oxygen saturation on supplemental oxygen<br/>C. Never needed<br/>D. Always routinely</p>                     | <p><b>Ans: B</b></p> |
| <p><b>Q42. What must be taken to prevent thromboembolism postoperatively in obese patients?</b></p> <p>A. Every precaution (DVT prophylaxis)<br/>B. Only aspirin<br/>C. Only compression stockings<br/>D. Only bed rest</p>                                          | <p><b>Ans: A</b></p> |

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| <p><b>Q43. Why is good antibiotic cover a must in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Due to poor hygiene only</li> <li>B. Due to diabetes only</li> <li>C. Due to immunosuppression</li> <li>D. Because these patients are prone for wound infection</li> </ul>                                                  | <p><b>Ans: D</b></p> |
| <p><b>Q44. Why are obese patients in high cardiac output state?</b></p> <ul style="list-style-type: none"> <li>A. To perfuse extra fat tissue</li> <li>B. Due to fever</li> <li>C. Due to anxiety</li> <li>D. Due to exercise</li> </ul>                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q45. What restrictive pattern do obese patients have in their pulmonary functions?</b></p> <ul style="list-style-type: none"> <li>A. Mixed pattern</li> <li>B. Normal pattern</li> <li>C. Extra parenchymal restrictive pattern</li> <li>D. Obstructive pattern</li> </ul>                                                                | <p><b>Ans: C</b></p> |
| <p><b>Q46. What does the ramped position provide for laryngoscopy?</b></p> <ul style="list-style-type: none"> <li>A. No improvement</li> <li>B. Superior view</li> <li>C. Lateral view</li> <li>D. Inferior view</li> </ul>                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q47. What improves oxygenation in obese patients through application of CPAP and PEEP?</b></p> <ul style="list-style-type: none"> <li>A. Decreasing dead space</li> <li>B. Decreasing respiratory rate</li> <li>C. Preventing alveolar collapse and improving functional residual capacity</li> <li>D. Increasing tidal volume</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q48. Why is decreased thyromental distance a concern in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. It indicates thyroid disease</li> <li>B. It indicates short neck making intubation difficult</li> <li>C. It indicates dental problems</li> <li>D. It has no significance</li> </ul>                             | <p><b>Ans: B</b></p> |
| <p><b>Q49. What causes reduced oxygen saturation in obese patients?</b></p> <ul style="list-style-type: none"> <li>A. Pulmonary fibrosis</li> <li>B. Anemia</li> <li>C. Obesity itself and obstructive sleep apnea</li> <li>D. Heart failure only</li> </ul>                                                                                    | <p><b>Ans: C</b></p> |

**Q50. Why is the ramp position also called ear to sternum position?**

- A. Because neck is hyperextended
- B. Because ear touches the sternum
- C. Because external auditory meatus lies in horizontal line with sternal notch
- D. Because head is below sternum

**Ans: C**



# Answer Key & Explanations

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**Q1. Answer: C — BMI more than 30%**

Obesity is defined as body mass index (BMI) more than 30% and extreme/superobese as BMI more than 50%.

**Q2. Answer: B — More than 50%**

Extreme/superobese is defined as BMI more than 50%.

**Q3. Answer: C — Due to associated comorbidities and difficult airway management**

Obese patients are one of the most challenging patients for anesthesiologist especially due to their associated comorbidities and difficult airway management.

**Q4. Answer: D — Hypertension, coronary artery disease, LVH, RVH, and cor pulmonale**

Obese patients may have hypertension, coronary artery disease, left ventricular hypertrophy, right ventricular hypertrophy from pulmonary hypertension, and cor pulmonale.

**Q5. Answer: B — 0.1 L/min**

Increase in cardiac output by 0.1 L/min to perfuse each kg of fat leads to increased stroke volume and hence left ventricular hypertrophy.

**Q6. Answer: D — Resulting from pulmonary hypertension which is because of persistent hypoxia**

Right ventricular hypertrophy resulting from pulmonary hypertension which in turn is because of persistent hypoxia.

**Q7. Answer: D — Right ventricular strain**

Cor pulmonale because of right ventricular strain.

**Q8. Answer: B — Reduced ERV, FRC, vital capacity, increased airway resistance, and reduced oxygen saturation**

Obese patients may have reduced expiratory reserve volume (ERV), functional residual capacity (FRC) and vital capacity, increased airway resistance, and reduced oxygen saturation.

**Q9. Answer: A — Patients with OSAHS are at increased risk of morbidity and mortality**

Patients with OSAHS are at increased risk of morbidity and mortality.

**Q10. Answer: D — Hiatus hernia, gastroesophageal reflux, delayed gastric emptying and fatty infiltration of liver**

Obese patients may have associated hiatus hernia, gastroesophageal reflux, delayed gastric emptying and fatty infiltration of liver (may compromise liver functions).

**Q11. Answer: D — Because of chronic hypoxia**

There may be polycythemia because of chronic hypoxia.

**Q12. Answer: A — Thromboembolism**

Obese patients are very prone for thromboembolism in postoperative period.

**Q13. Answer: A — Short neck (decreased thyromental distance), restricted mouth opening, and restricted neck movement**

Obese patients often have difficult airway due to short neck (decreased thyromental distance), restricted mouth opening due to increased adipose tissue and decreased mobility of temporomandibular joint, and restricted neck movement because of decreased mobility of atlanto-occipital joint.

**Q14. Answer: A — No, investigation should be based on clinical assessment and associated comorbidities**

Obesity per se is not the criteria to order a battery of routine tests, therefore investigation should be based on clinical assessment and associated comorbidities.

**Q15. Answer: C — Oxygen saturation**

Baseline oxygen saturation must be checked by pulse oximetry (if abnormal, then arterial blood gases should be done).

**Q16. Answer: A — Morbidly obese patients or patients with a history of obstructive sleep apnea**

Polysomnography should be strongly considered in morbidly obese patients or in patients with a history of obstructive sleep apnea.

**Q17. Answer: A — Due to risk of respiratory depression in patients already prone to hypoxia and sleep apnea**

Premedication with respiratory depressant drugs like benzodiazepines should be avoided.

**Q18. Answer: D — Metoclopramide and ranitidine**

Premedication with metoclopramide and ranitidine should be considered (due to hiatus hernia and GERD risk).

**Q19. Answer: A — Appropriate size BP cuff**

Appropriate size BP cuff should be chosen for monitoring.

**Q20. Answer: A — Regional anesthesia**

Due to difficult airway management, anesthesia of choice is regional anesthesia.

**Q21. Answer: C — Due to reduced epidural and subarachnoid space making them vulnerable for high spinals**

Unfortunately regional anesthesia is often difficult in obese patients. Due to reduced epidural and subarachnoid space obese patients are more vulnerable for high spinals.

**Q22. Answer: C — Adding 10 cm H<sub>2</sub>O of CPAP during preoxygenation**

Adding 10 cm H<sub>2</sub>O of continuous positive airway pressure (CPAP) during preoxygenation has found to be beneficial in maintaining oxygenation in the perioperative period.

**Q23. Answer: B — Due to increased fat stores**

Lipid soluble drugs (IV induction agents, opioids) have increased volume of distribution, therefore given on actual body weight basis.

**Q24. Answer: B — Actual body weight**

Lipid soluble drugs have increased volume of distribution, therefore given on actual body weight basis.

**Q25. Answer: B — No, this effect is not so marked on propofol as on thiopentone**

Interestingly, this effect is not so marked on propofol as on thiopentone.

**Q26. Answer: B — An arduous task**

Achieving IV lines sometimes can become an arduous task.

**Q27. Answer: C — Airway management**

Airway management is the most challenging part in obese patients.

**Q28. Answer: C — 3 times more**

The incidence of failed intubation is 3 times more in bariatric population.

**Q29. Answer: A — Ramped position (ear to sternum position)**

The Ramped position provides a superior view of laryngoscopy and is the preferred position for intubation for obese patients.

**Q30. Answer: D — By elevating the upper back using sheets/blankets so external auditory meatus lies in horizontal line with sternal notch**

A ramp is created by using sheets and blankets to elevate the upper back of the patient to the level when external auditory meatus lie in horizontal line with the sternal notch (ear to sternum position).

**Q31. Answer: C — Awake intubation with fiberoptic bronchoscopy**

If intubation is anticipated to be difficult awake intubation with fiberoptic bronchoscopy should be considered.

**Q32. Answer: A — By capnography as breath sounds may not be clearly audible on auscultation**

Confirmation of intubation must be done by capnography as breath sounds may not be clearly audible on auscultation.

**Q33. Answer: A — Due to increased fat stores and hepatic blood flow**

Inhalational agents are more extensively metabolized in obese patients.

**Q34. Answer: D — Desflurane**

Desflurane is the inhalational agent of choice because it is the agent with least metabolism.

**Q35. Answer: A — Increased metabolism of halothane makes obese patients more vulnerable for halothane hepatitis**

Increased metabolism of halothane makes obese patients more vulnerable for halothane hepatitis.

**Q36. Answer: B — Ideal body weight**

Water soluble drugs (nondepolarizing muscle relaxants) have limited volume of distribution therefore should be given on ideal body weight basis.

**Q37. Answer: B — Improves oxygenation by preventing alveolar collapse**

Application of 10 cm H<sub>2</sub>O of positive end expiratory pressure (PEEP) during mechanical ventilation improves oxygenation by preventing alveolar collapse.

**Q38. Answer: C — 30 degree reverse Trendelenburg**

30 degree reverse Trendelenburg improves oxygenation particularly during laparoscopic surgeries.

**Q39. Answer: B — Hypoxia**

Obese patients are very prone to hypoxia in postoperative period.

**Q40. Answer: C — 45 degree sitting (semi-reclined position) with supplemental oxygen**

They should be nursed in 45 degree sitting (semi-reclined position) with supplemental oxygen.

**Q41. Answer: B — If patient is not maintaining oxygen saturation on supplemental oxygen**

CPAP or BiPAP should be applied if the patient is not maintaining oxygen saturation on supplemental oxygen.

**Q42. Answer: A — Every precaution (DVT prophylaxis)**

Every precaution must be taken to prevent thromboembolism.

**Q43. Answer: D — Because these patients are prone for wound infection**

Good antibiotic cover is must as these patients are prone for wound infection.

**Q44. Answer: A — To perfuse extra fat tissue**

To perfuse extra fat obese patients are in high cardiac output state.

**Q45. Answer: C — Extra parenchymal restrictive pattern**

Obese patients may have extra parenchymal restrictive pattern in their pulmonary functions.

**Q46. Answer: B — Superior view**

The ramped position provides a superior view of laryngoscopy.

**Q47. Answer: C — Preventing alveolar collapse and improving functional residual capacity**

Application of CPAP and PEEP improves oxygenation in obese patients by preventing alveolar collapse.

**Q48. Answer: B — It indicates short neck making intubation difficult**

Short neck (decreased thyromental distance) contributes to difficult airway in obese patients.

**Q49. Answer: C — Obesity itself and obstructive sleep apnea**

Reduced oxygen saturation because of obesity and obstructive sleep apnea.

**Q50. Answer: C — Because external auditory meatus lies in horizontal line with sternal notch**

It is also called ear to sternum position because external auditory meatus lie in horizontal line with the sternal notch.

# Anesthesia for Laparoscopy MCQ – Chapter 32

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 32)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Why have laparoscopic surgeries been on the rise over a decade?**

- A. Due to better cosmesis only
- B. Due to shorter hospital stay, avoidance of big surgical incision, less tissue trauma and less postoperative pain
- C. Due to lower cost only
- D. Due to faster anesthesia

**Ans: B**

**Q2. Which gas is most frequently used for creating pneumoperitoneum?**

- A. Helium
- B. Carbon dioxide
- C. Argon
- D. Nitrogen

**Ans: B**

**Q3. Which gas is considered the most ideal for laparoscopy but not used due to cost?**

- A. Nitrogen
- B. Argon
- C. Helium
- D. Nitrous oxide

**Ans: B**

**Q4. What respiratory effect does increased intra-abdominal pressure cause during pneumoperitoneum?**

- A. Increased FRC
- B. Improved ventilation
- C. Increased vital capacity
- D. Pushes diaphragm upwards causing basal atelectasis, V/Q mismatch and decreased FRC

**Ans: D**

**Q5. By how much may thoracopulmonary compliance be decreased during pneumoperitoneum?**

- A. Up to 50%
- B. Up to 25%
- C. Up to 10%
- D. Up to 75%

**Ans: A**

**Q6. How does increased intra-abdominal pressure affect cardiac output?**

- A. Compresses inferior vena cava leading to decreased venous return and cardiac output
- B. No effect
- C. Increases it
- D. Only affects heart rate

**Ans: A**

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| <p><b>Q7. By how much can intra-abdominal pressure of 12-14 mm Hg decrease cardiac output?</b></p> <p>A. 10-15%<br/>B. 5-10%<br/>C. 50-60%<br/>D. 30-35%</p>                                                                                                                                             | <p><b>Ans: D</b></p> |
| <p><b>Q8. Why does peripheral vascular resistance (PVR) increase during laparoscopy?</b></p> <p>A. Due to patient positioning only<br/>B. Due to hypothermia<br/>C. Due to blood loss<br/>D. Due to high pCO<sub>2</sub> and increased catecholamines, vasopressin and prostaglandins</p>                | <p><b>Ans: D</b></p> |
| <p><b>Q9. What type of arrhythmias are more frequent during laparoscopy due to stretching of peritoneum?</b></p> <p>A. Ventricular tachycardia<br/>B. Atrial fibrillation<br/>C. Bradyarrhythmias<br/>D. Tachyarrhythmias</p>                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q10. What effect does CO<sub>2</sub> have on the central nervous system during laparoscopy?</b></p> <p>A. CO<sub>2</sub> by producing cerebral vasodilation can increase the intracranial tension<br/>B. Causes CNS depression<br/>C. No CNS effect<br/>D. Decreases intracranial tension</p>      | <p><b>Ans: A</b></p> |
| <p><b>Q11. Why is the risk of aspiration increased during laparoscopy?</b></p> <p>A. Due to opioid use<br/>B. Increased intra-abdominal pressure distorts the gastroesophageal junction making patient more vulnerable<br/>C. Due to fasting violation<br/>D. Due to gas insufflation into stomach</p>   | <p><b>Ans: B</b></p> |
| <p><b>Q12. How can carbon dioxide from peritoneal cavity cause pneumothorax?</b></p> <p>A. Through the chest wall<br/>B. Through the diaphragm directly<br/>C. Through embryonic remnants between pleural and peritoneal cavity<br/>D. Through the esophagus</p>                                         | <p><b>Ans: C</b></p> |
| <p><b>Q13. What are the diagnostic signs of pneumothorax during laparoscopy?</b></p> <p>A. Only chest pain<br/>B. Only decreased SpO<sub>2</sub><br/>C. Increased airway resistance, decreased SpO<sub>2</sub>, decreased air entry, X-ray chest, and subcutaneous emphysema<br/>D. Only hypotension</p> | <p><b>Ans: C</b></p> |

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| <p><b>Q14. What is the FIRST step in managing pneumothorax during laparoscopy?</b></p> <p>A. Immediately stop nitrous oxide<br/>B. Adjust ventilator<br/>C. Start 100% oxygen<br/>D. Insert chest tube</p>                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q15. Should chest tube drainage be used routinely for capnothorax?</b></p> <p>A. Only with bilateral pneumothorax<br/>B. No, reserved only for cases not responding to conservative measures or life-threatening ones<br/>C. Yes, always<br/>D. Only in pediatric patients</p> | <p><b>Ans: B</b></p> |
| <p><b>Q16. In how many minutes does capnothorax usually resolve by itself?</b></p> <p>A. 10-20 minutes<br/>B. 5-10 minutes<br/>C. 60-90 minutes<br/>D. 30-60 minutes</p>                                                                                                             | <p><b>Ans: D</b></p> |
| <p><b>Q17. Where can CO2 enter to cause pneumomediastinum and subcutaneous emphysema?</b></p> <p>A. Through the trachea<br/>B. Through esophageal, vena cava or aortic hiatus into mediastinum and pericardium<br/>C. Through the pleura only<br/>D. Through the chest wall</p>      | <p><b>Ans: B</b></p> |
| <p><b>Q18. What is the mortality rate of CO2 embolism during laparoscopy?</b></p> <p>A. 5%<br/>B. 30%<br/>C. 50%<br/>D. 10%</p>                                                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q19. How does CO2 embolism occur during laparoscopy?</b></p> <p>A. Through arterial line<br/>B. Through pulmonary veins<br/>C. Through IV line<br/>D. Due to accidental placement of needle into vessel during insufflation</p>                                                | <p><b>Ans: D</b></p> |
| <p><b>Q20. What can large volume of gas cause if it enters the inferior vena cava?</b></p> <p>A. Pneumomediastinum<br/>B. Subcutaneous emphysema<br/>C. Pneumothorax<br/>D. Gas lock leading to cardiovascular collapse</p>                                                          | <p><b>Ans: D</b></p> |



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| <p><b>Q21. What is the most sensitive test to diagnose CO2 embolism?</b></p> <ul style="list-style-type: none"> <li>A. Transesophageal echocardiography</li> <li>B. Capnography</li> <li>C. Pulmonary artery pressure</li> <li>D. Doppler</li> </ul>                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q22. What is the capnography response pattern in CO2 embolism?</b></p> <ul style="list-style-type: none"> <li>A. No change in ETCO2</li> <li>B. Biphasic response - initial rise in end tidal CO2 followed by fall</li> <li>C. Steady decrease in ETCO2</li> <li>D. Steady increase in ETCO2</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q23. What is diagnostic for CO2 embolism when aspirated from central catheter?</b></p> <ul style="list-style-type: none"> <li>A. Blood</li> <li>B. Clear fluid</li> <li>C. Gas or foamy blood</li> <li>D. Saline</li> </ul>                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q24. What clinical signs indicate CO2 embolism?</b></p> <ul style="list-style-type: none"> <li>A. Tachycardia, hypotension, right heart strain on ECG and Mill wheel murmur</li> <li>B. Hypertension and bradycardia</li> <li>C. Only cyanosis</li> <li>D. Only decreased SpO2</li> </ul>               | <p><b>Ans: A</b></p> |
| <p><b>Q25. What must the surgeon ascertain before CO2 insufflation to prevent embolism?</b></p> <ul style="list-style-type: none"> <li>A. Oxygen saturation</li> <li>B. Patient's heart rate</li> <li>C. Patient's blood pressure</li> <li>D. Position of Veress needle</li> </ul>                            | <p><b>Ans: D</b></p> |
| <p><b>Q26. What should the rate of insufflation NOT exceed to prevent CO2 embolism?</b></p> <ul style="list-style-type: none"> <li>A. 1 L/min</li> <li>B. 2 L/min</li> <li>C. 3 L/min</li> <li>D. 0.5 L/min</li> </ul>                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q27. In what position should the patient be placed for curative management of CO2 embolism?</b></p> <ul style="list-style-type: none"> <li>A. Supine</li> <li>B. Trendelenburg</li> <li>C. Durant position (left lateral with Trendelenburg)</li> <li>D. Reverse Trendelenburg</li> </ul>               | <p><b>Ans: C</b></p> |

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| <p><b>Q28. When may cardiopulmonary bypass be required for CO2 embolism?</b></p> <ul style="list-style-type: none"> <li>A. Never required</li> <li>B. Only for pediatric patients</li> <li>C. For massive embolism</li> <li>D. For all cases</li> </ul>                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q29. What positions are most laparoscopic procedures performed in?</b></p> <ul style="list-style-type: none"> <li>A. Prone and sitting</li> <li>B. Trendelenburg or anti-Trendelenburg</li> <li>C. Supine and lateral</li> <li>D. Lithotomy only</li> </ul>                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q30. What abdominal organ trauma can occur during laparoscopy?</b></p> <ul style="list-style-type: none"> <li>A. Only splenic injury</li> <li>B. Only bladder injury</li> <li>C. Intestines and stomach may be perforated by needle or trocar with undetected bleeding and visceral trauma</li> <li>D. Only liver injury</li> </ul>   | <p><b>Ans: C</b></p> |
| <p><b>Q31. Why can increase in pCO2 during laparoscopy be dangerous in cardiac patients?</b></p> <ul style="list-style-type: none"> <li>A. It causes hypotension only</li> <li>B. It has no effect on cardiac patients</li> <li>C. It causes hypothermia</li> <li>D. It can produce arrhythmias and ischemia in cardiac patients</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q32. At what intra-abdominal pressure can cardiac output decrease by more than two-thirds?</b></p> <ul style="list-style-type: none"> <li>A. 15 mm Hg</li> <li>B. 12-14 mm Hg</li> <li>C. 10 mm Hg</li> <li>D. &gt; 20 mm Hg</li> </ul>                                                                                               | <p><b>Ans: D</b></p> |
| <p><b>Q33. What is the recommended range for intra-abdominal pressure during laparoscopy?</b></p> <ul style="list-style-type: none"> <li>A. 20-25 mm Hg</li> <li>B. 12-14 mm Hg</li> <li>C. 15-20 mm Hg</li> <li>D. 8-10 mm Hg</li> </ul>                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q34. What effect can intra-abdominal pressure &gt; 20 mm Hg have on abdominal organs?</b></p> <ul style="list-style-type: none"> <li>A. Significantly impair renal, mesenteric and intestinal blood flow</li> <li>B. Only affects kidneys</li> <li>C. Only affects liver</li> <li>D. No effect</li> </ul>                             | <p><b>Ans: A</b></p> |

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| <p><b>Q35. What is the most common postoperative complication of laparoscopic surgeries?</b></p> <p>A. Infection<br/>B. Bleeding<br/>C. Nausea and vomiting (incidence 40-75%)<br/>D. Pain</p>                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q36. What causes shoulder pain in the postoperative period after laparoscopy?</b></p> <p>A. Positioning on table<br/>B. Surgical retraction<br/>C. IV line placement<br/>D. Sub-diaphragmatic CO2 irritating diaphragm</p>                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q37. Under what type of anesthesia can very short duration laparoscopic surgeries (&lt;15 minutes) be performed?</b></p> <p>A. Local anesthesia<br/>B. Regional anesthesia<br/>C. General anesthesia<br/>D. Spinal anesthesia</p>                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q38. Why does CO2 limit the use of central neuraxial blocks for laparoscopy?</b></p> <p>A. Due to nausea<br/>B. Due to increased intracranial tension<br/>C. Due to diaphragmatic irritation by CO2 causing patient discomfort (shoulder pain)<br/>D. Due to hypotension risk</p> | <p><b>Ans: C</b></p> |
| <p><b>Q39. What is the most commonly used technique for most laparoscopic surgeries?</b></p> <p>A. Epidural anesthesia<br/>B. Spinal anesthesia with sedation<br/>C. Local anesthesia<br/>D. General anesthesia with endotracheal intubation and mild hyperventilation</p>              | <p><b>Ans: D</b></p> |
| <p><b>Q40. When can second generation LMA (I-gel, Proseal) be considered for laparoscopy?</b></p> <p>A. Only in emergency<br/>B. Never<br/>C. If there is no risk of aspiration<br/>D. Always</p>                                                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q41. At what level should ETCO2 be maintained during laparoscopy?</b></p> <p>A. 35 mm Hg<br/>B. 30 mm Hg<br/>C. 45 mm Hg<br/>D. 25 mm Hg</p>                                                                                                                                      | <p><b>Ans: A</b></p> |

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| <p><b>Q42. Why should inhalational agents that depress cardiac function (like halothane) be avoided?</b></p> <p>A. They cause bronchospasm<br/> B. Because cardiac output is already compromised by pneumoperitoneum<br/> C. They are expensive<br/> D. They cause nausea</p> | <p><b>Ans: B</b></p> |
| <p><b>Q43. Why is propofol the induction agent of choice for laparoscopy?</b></p> <p>A. Because of its antiemetic properties<br/> B. It has fastest onset<br/> C. It is cheapest<br/> D. It has longest duration</p>                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q44. Which of the following is an ABSOLUTE contraindication for laparoscopy?</b></p> <p>A. Pregnancy<br/> B. Controlled coronary insufficiency<br/> C. Congestive heart failure<br/> D. Obesity</p>                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q45. Why is right-to-left cardiac shunt an absolute contraindication for laparoscopy?</b></p> <p>A. Due to arrhythmia<br/> B. Risk of paradoxical embolism<br/> C. Due to bleeding risk<br/> D. Due to hypotension</p>                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q46. What is the relative contraindication for laparoscopy?</b></p> <p>A. Coronary insufficiency<br/> B. Obesity<br/> C. Pregnancy<br/> D. Ventriculoperitoneal (VP) or peritoneojugular (PJ) shunt</p>                                                                 | <p><b>Ans: D</b></p> |
| <p><b>Q47. Is laparoscopy considered safe during pregnancy?</b></p> <p>A. Yes, laparoscopy is considered safe during pregnancy<br/> B. Absolutely contraindicated<br/> C. Only in first trimester<br/> D. Only in third trimester</p>                                         | <p><b>Ans: A</b></p> |
| <p><b>Q48. Is laparoscopy now preferred over open surgery in obese patients?</b></p> <p>A. It is equally preferred<br/> B. Yes, it has become the preferred method over open surgeries<br/> C. Only for minor procedures<br/> D. No, it is contraindicated</p>                | <p><b>Ans: B</b></p> |

**Q49. Can a patient with controlled ischemic heart disease be safely considered for laparoscopy?**

- A. Only with CABG first
- B. Yes, with enhanced perioperative cardiovascular monitoring
- C. Only for short procedures
- D. Absolutely contraindicated

**Ans: B**

**Q50. Are respiratory complications after laparoscopic surgery higher or lower than open surgery?**

- A. Significantly lower
- B. Only slightly lower
- C. Same
- D. Higher

**Ans: A**

# Answer Key & Explanations

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## **Q1. Answer: B — Due to shorter hospital stay, avoidance of big surgical incision, less tissue trauma and less postoperative pain**

Page 248: Laparoscopic surgeries have been on rise over a decade due to shorter hospital stay, avoidance of big surgical incision, less tissue trauma and less postoperative pain.

Topic: General

## **Q2. Answer: B — Carbon dioxide**

Page 248: Carbon dioxide is most frequently used gas for laparoscopy because it is more soluble in blood leading to its rapid elimination from the body in cases of pneumothorax and accidental injection into vascular compartment.

Topic: Gases

## **Q3. Answer: B — Argon**

Page 248: Since argon is an inert gas with no side effects it is considered as most ideal gas for laparoscopy, however because of its very high cost it is not used.

Topic: Gases

## **Q4. Answer: D — Pushes diaphragm upwards causing basal atelectasis, V/Q mismatch and decreased FRC**

Page 248: Increase intra-abdominal pressure due to pneumoperitoneum pushes the diaphragm upwards, leading to basal atelectasis producing ventilation perfusion (V/Q) mismatch and decrease in functional residual capacity (FRC).

Topic: Pathophysiology - Respiratory

## **Q5. Answer: A — Up to 50%**

Page 248: Thoracopulmonary compliance may be decreased by up to 50% during pneumoperitoneum.

Topic: Pathophysiology - Respiratory

## **Q6. Answer: A — Compresses inferior vena cava leading to decreased venous return and cardiac output**

Page 248: Increase intra-abdominal pressure compresses the inferior vena cava leading to decreased venous return and cardiac output.

Topic: Pathophysiology - Cardiovascular

## **Q7. Answer: D — 30-35%**

Page 248: Intra-abdominal pressure of 12-14 mm Hg can decrease the cardiac output by 30-35%.

Topic: Pathophysiology - Cardiovascular

## **Q8. Answer: D — Due to high pCO<sub>2</sub> and increased catecholamines, vasopressin and prostaglandins**

Page 249: Increase in PVR occurs because of high pCO<sub>2</sub>, and more importantly, because of increased catecholamines, vasopressin and prostaglandins.

Topic: Pathophysiology - Cardiovascular

## **Q9. Answer: C — Bradyarrhythmias**

Page 249: CO<sub>2</sub> by producing coronary vasoconstriction may produce any arrhythmia however, bradyarrhythmias occurring because of stretching of the peritoneum are more frequent.

Topic: Pathophysiology - Cardiovascular

**Q10. Answer: A — CO2 by producing cerebral vasodilation can increase the intracranial tension**

Page 248: CO2 by producing cerebral vasodilation can increase the intracranial tension.

Topic: Pathophysiology - CNS

**Q11. Answer: B — Increased intra-abdominal pressure distorts the gastroesophageal junction making patient more vulnerable**

Page 248: Increased Intra-abdominal pressure distorts the gastroesophageal junction making patient more vulnerable for aspiration.

Topic: Pathophysiology - GI

**Q12. Answer: C — Through embryonic remnants between pleural and peritoneal cavity**

Page 249: Carbon dioxide from peritoneal cavity can diffuse into the pleural cavity through embryonic remnants present between pleural and peritoneal cavity which gets open by increased intra-abdominal pressure or through pleural tears during surgery.

Topic: Complications - Pneumothorax

**Q13. Answer: C — Increased airway resistance, decreased SpO2, decreased air entry, X-ray chest, and subcutaneous emphysema**

Page 249: Diagnosis includes increased airway resistance on mechanical ventilation, decreased oxygen saturation (SpO2), decreased air entry on auscultation, X-ray chest, abnormal movement of diaphragm, and associated subcutaneous emphysema of neck and chest.

Topic: Complications - Pneumothorax

**Q14. Answer: A — Immediately stop nitrous oxide**

Page 249: Management - Immediately stop nitrous oxide (it can expand the pneumothorax).

Topic: Complications - Pneumothorax

**Q15. Answer: B — No, reserved only for cases not responding to conservative measures or life-threatening ones**

Page 249: Chest tube drainage should be reserved only for the cases not responding to conservative measures or the Pneumothorax is life-threatening.

Topic: Complications - Pneumothorax

**Q16. Answer: D — 30-60 minutes**

Page 249: Usually capnothorax resolves by itself in 30-60 minutes (because of high diffusibility of CO2).

Topic: Complications - Pneumothorax

**Q17. Answer: B — Through esophageal, vena cava or aortic hiatus into mediastinum and pericardium**

Page 249: The CO2 can enter mediastinum and pericardium through esophageal, vena cava or aortic hiatus. Gas from mediastinum may diffuse cephalad to produce subcutaneous emphysema of face and neck.

Topic: Complications - Pneumomediastinum

**Q18. Answer: B — 30%**

Page 249: Gas embolism can be a fatal complication of laparoscopy; it carries a mortality of 30%.

Topic: Complications - CO2 Embolism

**Q19. Answer: D — Due to accidental placement of needle into vessel during insufflation**

Page 249: CO2 embolism occurs because of accidental placement of a needle into the vessel during insufflation.

Topic: Complications - CO2 Embolism

**Q20. Answer: D — Gas lock leading to cardiovascular collapse**

Page 249: Large volume of gas can cause 'gas lock' in the inferior vena cava leading to cardiovascular collapse.

Topic: Complications - CO2 Embolism

**Q21. Answer: A — Transesophageal echocardiography**

Page 249: Most sensitive test to diagnose CO2 embolism is transesophageal echocardiography.

Topic: Complications - CO2 Embolism

**Q22. Answer: B — Biphasic response - initial rise in end tidal CO2 followed by fall**

Page 249: In CO2 embolism, there is biphasic response, i.e. initial rise in end tidal CO2 (due to direct injection of CO2 in vessels) followed by fall.

Topic: Complications - CO2 Embolism

**Q23. Answer: C — Gas or foamy blood**

Page 249: Aspiration of gas or foamy blood from central catheter is diagnostic.

Topic: Complications - CO2 Embolism

**Q24. Answer: A — Tachycardia, hypotension, right heart strain on ECG and Mill wheel murmur**

Page 249: Clinical signs like tachycardia, hypotension, right heart strain on ECG and Mill wheel murmur.

Topic: Complications - CO2 Embolism

**Q25. Answer: D — Position of Veress needle**

Page 249: The surgeon must ascertain the position of Veress needle before CO2 insufflation.

Topic: Complications - CO2 Embolism

**Q26. Answer: A — 1 L/min**

Page 249: Rate of insufflation should not be more than 1 L/min.

Topic: Complications - CO2 Embolism

**Q27. Answer: C — Durant position (left lateral with Trendelenburg)**

Page 249: If possible place the patient in Durant position (left lateral with Trendelenburg) so that gas get remains in right ventricle and can be aspirated.

Topic: Complications - CO2 Embolism

**Q28. Answer: C — For massive embolism**

Page 249: For massive embolism cardiopulmonary bypass may be required.

Topic: Complications - CO2 Embolism

**Q29. Answer: B — Trendelenburg or anti-Trendelenburg**

Page 250: Majority of the laparoscopic procedures are performed in either Trendelenburg or anti-Trendelenburg position.

Topic: Complications - Positioning



**Q30. Answer: C — Intestines and stomach may be perforated by needle or trocar with undetected bleeding and visceral trauma**

Page 250: Abdominal organs, especially intestines and stomach may be perforated by the needle or trocar. Moreover, bleeding and visceral trauma may go undetected.

Topic: Complications - Organ Trauma

**Q31. Answer: D — It can produce arrhythmias and ischemia in cardiac patients**

Page 250: Increase in pCO<sub>2</sub> is well-tolerated by young individuals, but can produce arrhythmias and ischemia in cardiac patients.

Topic: Complications - pCO<sub>2</sub>

**Q32. Answer: D — > 20 mm Hg**

Page 250: Intra-abdominal pressure of > 20 mm Hg can decrease the cardiac output by more than two-thirds producing even cardiac arrest.

Topic: Complications - IAP

**Q33. Answer: B — 12-14 mm Hg**

Page 250: It is highly recommended that intra-abdominal pressure should be kept between 12-14 mm Hg.

Topic: Complications - IAP

**Q34. Answer: A — Significantly impair renal, mesenteric and intestinal blood flow**

Page 250: Intra-abdominal pressure of > 20 mm Hg can significantly impair renal, mesenteric and intestinal blood flow.

Topic: Complications - IAP

**Q35. Answer: C — Nausea and vomiting (incidence 40-75%)**

Page 250: Nausea and vomiting is the most common postoperative complication of laparoscopic surgeries; incidence can be as high as 40-75%.

Topic: Complications - PONV

**Q36. Answer: D — Sub-diaphragmatic CO<sub>2</sub> irritating diaphragm**

Page 250: Shoulder pain: Sub-diaphragmatic CO<sub>2</sub> irritating diaphragm causing shoulder pain in postoperative period is commonly complained by many patients.

Topic: Complications - Shoulder Pain

**Q37. Answer: A — Local anesthesia**

Page 250: Laparoscopic surgeries of very short duration (<15 minutes) can only be considered under local anesthesia. Laparoscopic tubal ligation is the most common surgery performed under local anesthesia.

Topic: Anesthetic Management

**Q38. Answer: C — Due to diaphragmatic irritation by CO<sub>2</sub> causing patient discomfort (shoulder pain)**

Page 250: Patient discomfort (shoulder pain due to diaphragmatic irritation by CO<sub>2</sub>) limits the use of central neuraxial blocks (spinal/epidural) for laparoscopy.

Topic: Anesthetic Management

**Q39. Answer: D — General anesthesia with endotracheal intubation and mild hyperventilation**

Page 250: General anesthesia with endotracheal intubation and mild hyperventilation is most commonly used technique for most of the laparoscopic surgeries.

Topic: Anesthetic Management

**Q40. Answer: C — If there is no risk of aspiration**

Page 250: General anesthesia with second generation laryngeal mask airway (I-gel, proseal) can be considered if there is no risk of aspiration.

Topic: Anesthetic Management

**Q41. Answer: A — 35 mm Hg**

Page 250: ETCO<sub>2</sub> should be maintained at 35 mm Hg.

Topic: Anesthetic Management

**Q42. Answer: B — Because cardiac output is already compromised by pneumoperitoneum**

Page 250: Inhalational agents which depress cardiac function (like halothane) should be avoided.

Topic: Anesthetic Management

**Q43. Answer: A — Because of its antiemetic properties**

Page 250: Propofol because of its antiemetic properties is the induction agent of choice.

Topic: Anesthetic Management

**Q44. Answer: C — Congestive heart failure**

Page 250: Absolute contraindications include: Increased intracranial tension, congestive heart failure, severe hypovolemia and hypotension, right-to-left cardiac shunts or patent foramen ovale, generalized abdominal peritonitis, and retinal detachment.

Topic: Contraindications

**Q45. Answer: B — Risk of paradoxical embolism**

Page 250: Right-to-left cardiac shunts or patent foramen ovale (risk of paradoxical embolism) is an absolute contraindication.

Topic: Contraindications

**Q46. Answer: D — Ventriculoperitoneal (VP) or peritoneojugular (PJ) shunt**

Page 250: Relative contraindication - Ventriculoperitoneal (VP) or peritoneojugular (PJ) shunt: Insufflation in peritoneum can displace the shunt.

Topic: Contraindications

**Q47. Answer: A — Yes, laparoscopy is considered safe during pregnancy**

Page 250: Although pneumoperitoneum may cause transient fetal acidosis, clinical studies suggest that additional risks to be very low for both mother and fetus. Therefore, laparoscopy is considered safe during pregnancy.

Topic: Contraindications - Past

**Q48. Answer: B — Yes, it has become the preferred method over open surgeries**

Page 250: Laparoscopy once used to be considered as contraindication in obese patients has now become the preferred method over open surgeries.

Topic: Contraindications - Past

**Q49. Answer: B — Yes, with enhanced perioperative cardiovascular monitoring**

Page 250: A patient with the controlled ischemic heart disease can be safely considered for laparoscopic surgeries however, with enhanced perioperative cardiovascular monitoring.

Topic: Contraindications - Past

**Q50. Answer: A — Significantly lower**

Page 250: Respiratory complications after laparoscopic surgery are significantly lower than open surgery.

*Topic: Contraindications - Past*

# Anesthesia for Ophthalmic Surgery MCQ – Chapter 33

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 33)

**25 Questions • 20 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Patients posted for ocular surgeries are usually of which age groups?**

- A. Pediatric or geriatric
- B. Only adults
- C. Only young adults
- D. Only middle-aged

**Ans: A**

**Q2. Which topical eye medication can prolong the effect of suxamethonium by inhibiting pseudocholinesterase?**

- A. Echothiophate
- B. Phenylephrine
- C. Timolol
- D. Acetazolamide

**Ans: A**

**Q3. What is the normal range of intraocular pressure (IOP), the maintenance of which is one of the most important goals during eye surgery?**

- A. 10–20 mm Hg
- B. 5–10 mm Hg
- C. 20–30 mm Hg
- D. 30–40 mm Hg

**Ans: A**

**Q4. The oculocardiac reflex is triggered by pressure on the globe or by traction on extraocular muscle (especially the medial rectus) during squint surgery. What is its afferent and efferent pathway?**

- A. Afferent trigeminal, efferent vagus
- B. Afferent vagus, efferent trigeminal
- C. Afferent facial, efferent vagus
- D. Afferent optic, efferent oculomotor

**Ans: A**

**Q5. What is the treatment of the oculocardiac reflex?**

- A. Immediate cessation of manipulation and IV atropine if not reversed by cessation of manipulation
- B. Increasing the depth of anesthesia
- C. Giving adrenaline
- D. Giving suxamethonium

**Ans: A**

**Q6. Under what type of anesthesia are most ocular surgeries performed?**

- A. Regional anesthesia (topical, peribulbar, retrobulbar and sub-tenon blocks)
- B. General anesthesia
- C. Local infiltration only
- D. Total intravenous anesthesia

**Ans: A**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. Which block has now almost completely replaced the retrobulbar block?</b></p> <ul style="list-style-type: none"> <li>A. Peribulbar block</li> <li>B. Sub-tenon block</li> <li>C. Topical anesthesia</li> <li>D. General anesthesia</li> </ul>                                                                                                                                                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q8. In a peribulbar block, the needle is never inserted beyond how much depth (because large vessels and optic nerve are encountered with deeper penetration)?</b></p> <ul style="list-style-type: none"> <li>A. 25 mm</li> <li>B. 10 mm</li> <li>C. 40 mm</li> <li>D. 50 mm</li> </ul>                                                                                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q9. In retrobulbar block, a serious complication where the drug travels along the optic nerve sheath can enter the central nervous system, causing:</b></p> <ul style="list-style-type: none"> <li>A. Brain stem anesthesia</li> <li>B. Retinal detachment</li> <li>C. Ecchymosis</li> <li>D. Oculocardiac reflex</li> </ul>                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q10. All IV induction agents reduce the IOP EXCEPT which one (which increases the IOP and should be avoided in ocular surgeries)?</b></p> <ul style="list-style-type: none"> <li>A. Ketamine</li> <li>B. Propofol</li> <li>C. Thiopentone</li> <li>D. Etomidate</li> </ul>                                                                                                                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q11. Intubation and laryngoscopy can increase the IOP. How should the response be blunted?</b></p> <ul style="list-style-type: none"> <li>A. By IV lignocaine and fentanyl</li> <li>B. By increasing depth of inhalational agent only</li> <li>C. By giving ketamine</li> <li>D. By suxamethonium</li> </ul>                                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q12. Why should nitrous oxide be avoided in retinal detachment surgery if an air, sulfur hexafluoride or perfluoropropane bubble is used for correction?</b></p> <ul style="list-style-type: none"> <li>A. Nitrous oxide can expand the bubble size and raise the IOP significantly</li> <li>B. Nitrous oxide decreases the bubble size</li> <li>C. Nitrous oxide causes retinal artery occlusion</li> <li>D. Nitrous oxide causes bradycardia</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q13. Succinylcholine increases the IOP, but the rise is transient (5–10 minutes). When is it safer to use succinylcholine?</b></p> <ul style="list-style-type: none"> <li>A. If the patient is not fasting and risk of pulmonary aspiration is high</li> <li>B. In all ocular surgeries</li> <li>C. In retinal detachment surgery</li> <li>D. When the eye is open</li> </ul>                                                                             | <p><b>Ans: A</b></p> |

|                                                                                                                                                                                                                                                                                                                                                                                                            |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q14. During reversal in ocular surgery, anticholinergics (atropine/glycopyrrolate) given IV or IM:</b></p> <ul style="list-style-type: none"> <li>A. Do not raise the IOP significantly, therefore can be used safely with anticholinesterases</li> <li>B. Significantly raise the IOP and should be avoided</li> <li>C. Cause retinal detachment</li> <li>D. Trigger oculocardiac reflex</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q15. Which patient position decreases the IOP by reducing the venous pressure?</b></p> <ul style="list-style-type: none"> <li>A. Head up by 10–15°</li> <li>B. Head down by 15°</li> <li>C. Supine flat</li> <li>D. Lateral position</li> </ul>                                                                                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q16. Postoperative nausea and vomiting incidence is very high after which ocular surgery?</b></p> <ul style="list-style-type: none"> <li>A. Strabismus surgery</li> <li>B. Cataract surgery</li> <li>C. Keratoplasty</li> <li>D. Retinal detachment surgery</li> </ul>                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q17. Cataract surgery can be performed in patients receiving warfarin. For other ocular surgeries, warfarin should be stopped how many days before?</b></p> <ul style="list-style-type: none"> <li>A. 4 days</li> <li>B. 1 day</li> <li>C. 7 days</li> <li>D. 10 days</li> </ul>                                                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q18. Which local anesthetic is used for rapid onset in eye blocks, and which for prolonged effect?</b></p> <ul style="list-style-type: none"> <li>A. 2% xylocaine for rapid onset; 0.5% bupivacaine for prolonged effect</li> <li>B. 0.5% bupivacaine for rapid onset; 2% xylocaine for prolonged effect</li> <li>C. Procaine for both</li> <li>D. Ropivacaine for rapid onset</li> </ul>            | <p><b>Ans: A</b></p> |
| <p><b>Q19. In the sub-tenon (episcleral) block, the drug is injected beneath the Tenon's fascia. Compared to retrobulbar and peribulbar blocks, the complications are:</b></p> <ul style="list-style-type: none"> <li>A. Reported to be less</li> <li>B. Reported to be more</li> <li>C. Equal</li> <li>D. Much more severe</li> </ul>                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q20. Cataract surgery (phaco-emulsification) is possible under which type of anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Topical anesthesia of the eye</li> <li>B. General anesthesia only</li> <li>C. Retrobulbar block only</li> <li>D. Spinal anesthesia</li> </ul>                                                                                                             | <p><b>Ans: A</b></p> |

**Q21. What are the usual indications for general anesthesia in ophthalmic surgery?**

- A. Pediatric patients, uncooperative patients not allowing regional anesthesia, retinal detachment surgeries and keratoplasties
- B. All cataract surgeries
- C. Only diabetic patients
- D. Only elderly patients

**Ans: A**

**Q22. Although benzodiazepines reduce the intraocular pressure, in which patients should they be used cautiously?**

- A. Old age patients
- B. Pediatric patients
- C. Diabetic patients
- D. Hypertensive patients

**Ans: A**

**Q23. A disadvantage of the peribulbar block compared to the retrobulbar block is:**

- A. Slower onset and more chances of ecchymosis
- B. More serious complications
- C. Need to block facial nerve separately
- D. More painful

**Ans: A**

**Q24. In the peribulbar block technique, the patient is asked to look in which direction?**

- A. Straight ahead (neutral gaze)
- B. Upward and inward
- C. Downward
- D. To the side

**Ans: A**

**Q25. Which agent is most often used for induction in ophthalmic general anesthesia (as it reduces IOP)?**

- A. Propofol
- B. Ketamine
- C. Etomidate
- D. Thiopentone

**Ans: A**



# Answer Key & Explanations

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## **Q1. Answer: A — Pediatric or geriatric**

Page 252: The patients usually posted for ocular surgeries are either pediatric or geriatric. Therefore, proper preoperative evaluation should be done in geriatric population to rule out coexisting medical diseases.

*Topic: Preoperative Evaluation*

## **Q2. Answer: A — Echothiophate**

Page 252: Topical eye medications can have systemic effects. For example, echothiophate by inhibiting pseudocholinesterase can prolong the effect of suxamethonium, phenylephrine can cause hypertension, timolol can cause bradycardia and bronchospasm, and acetazolamide can cause hypokalemia.

*Topic: Preoperative Evaluation*

## **Q3. Answer: A — 10–20 mm Hg**

Page 252: One of the most important goals during eye surgery is to reduce or at least maintain the intraocular pressure within normal range of 10–20 mm Hg. Factors that increase IOP — hypertension, coughing, straining, hypercapnia, hypoxia, ketamine, suxamethonium and light anesthesia — must be avoided.

*Topic: Intraoperative*

## **Q4. Answer: A — Afferent trigeminal, efferent vagus**

Page 252: Prevention of oculocardiac reflex — it is triggered by pressure on globe or by traction on extraocular muscle (especially the medial rectus) during squint surgery. The afferents are carried by trigeminal and efferent by vagus. It is manifested as bradycardia, atrioventricular (AV) block or even asystole.

*Topic: Intraoperative*

## **Q5. Answer: A — Immediate cessation of manipulation and IV atropine if not reversed by cessation of manipulation**

Page 252: Treatment of oculocardiac reflex includes immediate cessation of manipulation and intravenous (IV) atropine if not reversed by cessation of manipulation. Pretreatment with atropine may be indicated in patients with a history of conduction blocks and  $\beta$ -blocker therapy.

*Topic: Intraoperative*

## **Q6. Answer: A — Regional anesthesia (topical, peribulbar, retrobulbar and sub-tenon blocks)**

Page 252: Most of the ocular surgeries are performed under regional anesthesia (topical, peribulbar, retrobulbar and sub-tenon blocks). Eye blocks are most often given by ophthalmologist themselves.

*Topic: Anesthesia*

## **Q7. Answer: A — Peribulbar block**

Page 252: Peribulbar block has now almost completely replaced retrobulbar block because of its less serious complications and avoidance of separate facial block to provide ocular akinesia.

*Topic: Peribulbar Block*

## **Q8. Answer: A — 25 mm**

Page 252: In peribulbar block, the needle is never inserted beyond 25 mm because large vessels and optic nerve are encountered with deeper penetration. 6–10 mL of drug is injected. Orbital compression with Hanon ball or compression balloon (30–40 mm Hg) facilitates the block and decreases the IOP.

*Topic: Peribulbar Block*

**Q9. Answer: A — Brain stem anesthesia**

Page 253: In retrobulbar block, complications include oculocardiac reflex, retrobulbar hemorrhage, puncture of globe (leading to retinal detachment), optic nerve injury, intraocular injection, brain stem anesthesia (drug traveling along optic nerve sheath can enter central nervous system), retinal artery occlusion and blindness, and death.

Topic: Retrobulbar Block

**Q10. Answer: A — Ketamine**

Page 253: All IV agents, except ketamine, reduce the IOP therefore can be safely used for maintenance. Ketamine increases the IOP, therefore should be avoided in ocular surgeries. Propofol is most often used for induction.

Topic: General Anesthesia

**Q11. Answer: A — By IV lignocaine and fentanyl**

Page 253: Intubation and laryngoscopy can increase the IOP, therefore the response should be blunted by IV lignocaine and fentanyl.

Topic: General Anesthesia

**Q12. Answer: A — Nitrous oxide can expand the bubble size and raise the IOP significantly**

Page 253: Nitrous oxide should be avoided in retinal detachment surgery if air, sulfur hexafluoride or perfluoropropane bubble is used for correction. Nitrous oxide can expand the bubble size and thus can raise the IOP significantly. Nitrous oxide should not be used for 5 days after air, 10 days after sulfur hexafluoride and 30 days after perfluoropropane.

Topic: General Anesthesia

**Q13. Answer: A — If the patient is not fasting and risk of pulmonary aspiration is high**

Page 253: Succinylcholine increases the IOP; however, the rise is transient (5–10 minutes), therefore if the patient is not fasting and risk of pulmonary aspiration is high, it is safer to use succinylcholine. Non-depolarizing agents decrease the IOP.

Topic: General Anesthesia

**Q14. Answer: A — Do not raise the IOP significantly, therefore can be used safely with anticholinesterases**

Page 253: Anticholinergics (atropine/glycopyrrolate) when given IV or IM do not raise IOP significantly, therefore can be used safely with anticholinesterases. Recovery should be very smooth — coughing and straining during extubation can be disastrous.

Topic: General Anesthesia

**Q15. Answer: A — Head up by 10–15°**

Page 254: Position of patient — head up by 10–15° decreases the IOP by reducing the venous pressure.

Topic: General Anesthesia

**Q16. Answer: A — Strabismus surgery**

Page 254: The incidence of postoperative nausea and vomiting is very high after strabismus surgery.

Topic: Postoperative

**Q17. Answer: A — 4 days**

Page 252: Cataract surgery can be performed in patients receiving warfarin. For other ocular surgeries, warfarin should be stopped 4 days before.

Topic: Preoperative Evaluation

**Q18. Answer: A — 2% xylocaine for rapid onset; 0.5% bupivacaine for prolonged effect**

Page 253: A solution for eye blocks consists of 2% xylocaine for rapid onset, 0.5% bupivacaine for prolonged effect, 1 in 200000 adrenaline for vasoconstriction, and hyaluronidase for better spread.

Topic: Local Anesthetics for Blocks

**Q19. Answer: A — Reported to be less**

Page 253: In sub-tenon block, the drug is injected beneath the Tenon's fascia. Complications are reported to be less than retrobulbar and peribulbar block.

Topic: Sub-tenon Block

**Q20. Answer: A — Topical anesthesia of the eye**

Page 253: The majority of the cataract surgeries (phaco-emulsification) is possible under topical anaesthesia of the eye.

Topic: Topical Anesthesia

**Q21. Answer: A — Pediatric patients, uncooperative patients not allowing regional anesthesia, retinal detachment surgeries and keratoplasties**

Page 253: Usual indications for general anesthesia are pediatric patients, uncooperative patients not allowing regional anesthesia, retinal detachment surgeries and keratoplasties.

Topic: General Anesthesia

**Q22. Answer: A — Old age patients**

Page 253: Although benzodiazepines reduce the intraocular pressure, however they should be used cautiously in old age patients.

Topic: General Anesthesia

**Q23. Answer: A — Slower onset and more chances of ecchymosis**

Page 253: Disadvantages of peribulbar block are slower onset and more chances of ecchymosis. Advantages include less serious complications, facial nerve need not be blocked separately, and it is less painful.

Topic: Peribulbar Block

**Q24. Answer: A — Straight ahead (neutral gaze)**

Page 252: In peribulbar block, ask the patient to look straight ahead (neutral gaze). A dull short beveled needle is entered through the inferior fornix of conjunctiva at the junction of lateral 1/3rd and medial 2/3rd.

Topic: Peribulbar Block

**Q25. Answer: A — Propofol**

Page 253: All IV agents except ketamine reduce the IOP. Propofol is most often used for induction. Ketamine increases the IOP, therefore should be avoided in ocular surgeries.

Topic: General Anesthesia

# Anesthesia for ENT Surgery MCQ – Chapter 34

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 34)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Why is ENT anesthesia considered one of the most difficult?**

- A. Due to patient age
- B. Due to lengthy procedures
- C. Because airway is often difficult, surgeon and anesthetist share the common field, and complications are related to airway
- D. Due to bleeding only

**Ans: C**

**Q2. What does panendoscopy involve?**

- A. Only esophagoscopy
- B. Laryngoscopy, bronchoscopy, and esophagoscopy (also called triple endoscopy)
- C. Only laryngoscopy
- D. Only bronchoscopy

**Ans: B**

**Q3. What happens to the glottic opening due to laryngeal growth in panendoscopy patients?**

- A. It remains normal
- B. It may be reduced to the extent that even small tube insertion becomes impossible
- C. It becomes wider
- D. No change occurs

**Ans: B**

**Q4. What is the incidence of myocardial ischemia and arrhythmias due to laryngeal manipulation?**

- A. 10-15%
- B. Less than 1%
- C. 2-4%
- D. More than 20%

**Ans: C**

**Q5. What must be done during preoperative evaluation for panendoscopy patients?**

- A. Only chest X-ray
- B. Assessment of airway by preoperative endoscopy or indirect laryngoscopy is must
- C. Only blood tests
- D. Only ECG

**Ans: B**

**Q6. Why are IV lignocaine, IV fentanyl and topical anesthesia of the larynx important for panendoscopy?**

- A. To reduce bleeding
- B. To prevent laryngeal reflexes
- C. To reduce secretions
- D. To reduce postoperative pain

**Ans: B**

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| <p><b>Q7. What are MLS tubes?</b></p> <p>A. Double lumen tubes<br/> B. Tracheostomy tubes<br/> C. Adult size ETT without cuff<br/> D. Pediatric size tubes with cuff available in 5, 5.5 and 6 mm size</p>                                                                                                         | <p><b>Ans: D</b></p> |
| <p><b>Q8. In what percentage of patients is it possible to get surgery done with MLS tube?</b></p> <p>A. 50%<br/> B. 25%<br/> C. &gt;95%<br/> D. 75%</p>                                                                                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q9. What is the disadvantage of MLS tubes?</b></p> <p>A. High ventilatory pressures are required to overcome the narrow size of tube<br/> B. They cause more bleeding<br/> C. They cannot be used in adults<br/> D. They are expensive</p>                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q10. In High Frequency Positive Pressure Ventilation (HFPPV), what is the frequency of breaths delivered?</b></p> <p>A. 20-40 breaths/min<br/> B. 10-20 breaths/min<br/> C. 60-100 breaths/min<br/> D. 120-150 breaths/min</p>                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q11. What is the diameter of the catheter used in HFPPV?</b></p> <p>A. 3.5-4.0 mm<br/> B. 1.0-1.5 mm<br/> C. 5.0-6.0 mm<br/> D. 2.0-2.5 mm</p>                                                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q12. In jet ventilation, where is the tip of the jet kept?</b></p> <p>A. Within the laryngoscope to avoid direct barotrauma to trachea<br/> B. Beyond the vocal cords<br/> C. In the trachea directly<br/> D. In the pharynx</p>                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q13. What is the apnea technique in ventilation for panendoscopy?</b></p> <p>A. Mask ventilation only<br/> B. High flow nasal oxygen<br/> C. Intubation is done, adequate ventilation given, tube removed for 1-2 minutes for surgery, then reintubated if needed<br/> D. Patient breathes spontaneously</p> | <p><b>Ans: C</b></p> |

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| <p><b>Q14. How is anesthesia maintained during panendoscopy?</b></p> <p>A. With inhalational agent and repeated doses of succinylcholine for small procedures or nondepolarizing agent for prolonged surgery<br/> B. Only regional anesthesia<br/> C. Only IV agents<br/> D. Only propofol infusion</p> | <p><b>Ans: A</b></p> |
| <p><b>Q15. Why should nitrous oxide be avoided in laryngeal surgeries?</b></p> <p>A. It is not potent enough<br/> B. It causes bleeding<br/> C. It causes laryngospasm<br/> D. It can increase postsurgical edema by causing expansion of the cuff of endotracheal tube</p>                             | <p><b>Ans: D</b></p> |
| <p><b>Q16. When is Heliox administration strongly considered in ENT?</b></p> <p>A. In patients with stridor<br/> B. In all ENT surgeries<br/> C. Only in pediatric patients<br/> D. Only in bronchoscopy</p>                                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q17. What is the most important complication of laser surgery?</b></p> <p>A. Infection<br/> B. Fires<br/> C. Bleeding<br/> D. Nerve injury</p>                                                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q18. What FiO2 should be used during laser surgery to prevent fire?</b></p> <p>A. Less than 30%<br/> B. 50-60%<br/> C. 40-50%<br/> D. 100%</p>                                                                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q19. Why should N2O be avoided during laser surgery?</b></p> <p>A. It is not potent<br/> B. It can support fire<br/> C. It causes nausea<br/> D. It causes laryngospasm</p>                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q20. What special kind of endotracheal tube is used for laser surgery?</b></p> <p>A. Laser resistant endotracheal tube<br/> B. Double lumen tube<br/> C. Uncuffed tube<br/> D. Reinforced tube</p>                                                                                                | <p><b>Ans: A</b></p> |

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| <p><b>Q21. What should the tube cuff be filled with during laser surgery instead of air?</b></p> <p>A. Helium<br/>B. Saline<br/>C. Oxygen<br/>D. Nitrous oxide</p>                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q22. What is the FIRST step if fire occurs during laser surgery?</b></p> <p>A. Call for help<br/>B. Stop oxygen<br/>C. Immediately remove tube and submerge in water<br/>D. Flush pharynx with saline</p>                                       | <p><b>Ans: C</b></p> |
| <p><b>Q23. What is the second step in the fire protocol after removing the tube?</b></p> <p>A. Flush with saline<br/>B. Give steroids<br/>C. Reintubate<br/>D. Immediately stop oxygen (and nitrous oxide) and start ventilation with air by mask</p> | <p><b>Ans: D</b></p> |
| <p><b>Q24. What volume of cold saline should be used to flush the pharynx after airway fire?</b></p> <p>A. 10-20 mL<br/>B. 50-60 mL<br/>C. 100-200 mL<br/>D. 500 mL</p>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q25. When may tracheostomy be required after an airway fire?</b></p> <p>A. Always<br/>B. Never needed<br/>C. Only in children<br/>D. If damage is severe enough not to allow intubation</p>                                                     | <p><b>Ans: D</b></p> |
| <p><b>Q26. Under what type of anesthesia is flexible bronchoscopy for diagnostic biopsies usually performed?</b></p> <p>A. Regional anesthesia<br/>B. Topical anesthesia<br/>C. Spinal anesthesia<br/>D. General anesthesia</p>                       | <p><b>Ans: B</b></p> |
| <p><b>Q27. What is rigid bronchoscopy usually done for?</b></p> <p>A. Foreign body removal under general anesthesia<br/>B. Lavage only<br/>C. Diagnostic biopsies<br/>D. Stent placement only</p>                                                     | <p><b>Ans: A</b></p> |



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| <p><b>Q28. What are the ventilation methods for rigid bronchoscopy?</b></p> <ul style="list-style-type: none"> <li>A. High frequency jet ventilation, insufflation, and apnea technique</li> <li>B. Only spontaneous breathing</li> <li>C. Only IPPV</li> <li>D. Only mask ventilation</li> </ul>                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q29. What is the majority of patients undergoing adenotonsillectomy?</b></p> <ul style="list-style-type: none"> <li>A. Pediatric</li> <li>B. Adolescents</li> <li>C. Adults</li> <li>D. Elderly</li> </ul>                                                                                                                                                    | <p><b>Ans: A</b></p> |
| <p><b>Q30. If there are no adenoids, what route should be preferred for nasal intubation?</b></p> <ul style="list-style-type: none"> <li>A. Nasal is always preferred</li> <li>B. Only oral RAE tube</li> <li>C. Nasal intubation is always safe</li> <li>D. Nasal intubation should preferably be avoided due to increased chances of bleeding</li> </ul>          | <p><b>Ans: D</b></p> |
| <p><b>Q31. What type of endotracheal tubes are most frequently used for tonsillectomy?</b></p> <ul style="list-style-type: none"> <li>A. Nasal RAE tubes</li> <li>B. Oral Ring-Adair-Elwyn (RAE) preformed tubes</li> <li>C. Standard ETT</li> <li>D. Reinforced tubes</li> </ul>                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q32. Why should the throat be packed during tonsillectomy?</b></p> <ul style="list-style-type: none"> <li>A. To prevent aspiration of blood</li> <li>B. To reduce swelling</li> <li>C. To improve ventilation</li> <li>D. To keep the tube in place</li> </ul>                                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q33. In what position should tonsillectomy patients be nursed postoperatively?</b></p> <ul style="list-style-type: none"> <li>A. Sitting upright</li> <li>B. Prone position</li> <li>C. Tonsillar position - on one side (preferably left lateral) with head low</li> <li>D. Supine with head elevated</li> </ul>                                             | <p><b>Ans: C</b></p> |
| <p><b>Q34. How should a patient with post-tonsillectomy bleeding requiring re-exploration be managed?</b></p> <ul style="list-style-type: none"> <li>A. As a full stomach patient (patient has swallowed blood causing significant aspiration risk)</li> <li>B. Only with sedation</li> <li>C. Only under local anesthesia</li> <li>D. As a routine case</li> </ul> | <p><b>Ans: A</b></p> |

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| <p><b>Q35. How should peritonsillar abscess and Ludwig angina ideally be drained?</b></p> <p>A. Under regional block<br/> B. Under local anesthesia (as intubation may be impossible due to trismus and aspiration risk)<br/> C. Under general anesthesia always<br/> D. Under IV sedation</p> | <p><b>Ans: B</b></p> |
| <p><b>Q36. If GA is required for peritonsillar abscess, what should be done before induction?</b></p> <p>A. Rapid sequence induction<br/> B. Blind nasal intubation<br/> C. Awake nasal intubation with fiberoptic bronchoscopy<br/> D. LMA insertion</p>                                      | <p><b>Ans: C</b></p> |
| <p><b>Q37. How many times more soluble is nitrous oxide compared to nitrogen?</b></p> <p>A. 10 times<br/> B. 35 times<br/> C. 20 times<br/> D. 50 times</p>                                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q38. By how much can nitrous oxide increase middle ear pressure if Eustachian tube is blocked?</b></p> <p>A. 375 mm H<sub>2</sub>O within half an hour<br/> B. 200 mm H<sub>2</sub>O in 2 hours<br/> C. 50 mm H<sub>2</sub>O in 15 minutes<br/> D. 100 mm H<sub>2</sub>O in 1 hour</p>   | <p><b>Ans: A</b></p> |
| <p><b>Q39. In which ear conditions should nitrous oxide NOT be used?</b></p> <p>A. Only otitis media<br/> B. Patients with blocked Eustachian tubes (upper respiratory tract infection, adenoids, sinusitis and otitis media)<br/> C. All ear surgeries<br/> D. Only sinusitis</p>             | <p><b>Ans: B</b></p> |
| <p><b>Q40. Why is nitrous oxide contraindicated for tympanoplasties?</b></p> <p>A. Increase in middle ear pressure can displace the graft<br/> B. It causes nausea<br/> C. It causes infection<br/> D. It causes bleeding</p>                                                                  | <p><b>Ans: A</b></p> |
| <p><b>Q41. When is increased middle ear pressure from N<sub>2</sub>O beneficial in ear surgery?</b></p> <p>A. Never<br/> B. In mastoidectomy<br/> C. In all tympanoplasties<br/> D. When surgeon is using 'inlay graft' (helps hold the graft)</p>                                             | <p><b>Ans: D</b></p> |

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| <p><b>Q42. What is very high in incidence after ear surgeries requiring prophylaxis?</b></p> <ul style="list-style-type: none"> <li>A. Pain</li> <li>B. Postoperative nausea and vomiting</li> <li>C. Bleeding</li> <li>D. Infection</li> </ul>                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q43. What is the Samter triad that must be ruled out in nasal surgery patients?</b></p> <ul style="list-style-type: none"> <li>A. Deviated septum, sinusitis, rhinitis</li> <li>B. Hypertension, diabetes, obesity</li> <li>C. Bleeding, infection, pain</li> <li>D. Nasal polyps, asthma, and sensitivity to aspirin/NSAIDs</li> </ul>                     | <p><b>Ans: D</b></p> |
| <p><b>Q44. Why is throat pack a must in nasal surgery?</b></p> <ul style="list-style-type: none"> <li>A. To improve ventilation</li> <li>B. To reduce swelling</li> <li>C. To prevent aspiration of blood</li> <li>D. To stabilize the tube</li> </ul>                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q45. Why must nasal surgery patients be counseled to breathe from mouth preoperatively?</b></p> <ul style="list-style-type: none"> <li>A. For better oxygenation</li> <li>B. To reduce anxiety</li> <li>C. Because they can become very restless in postoperative period after nasal packing if not trained</li> <li>D. To improve lung function</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q46. Why may muscle relaxants be deterred in parotid surgery?</b></p> <ul style="list-style-type: none"> <li>A. Due to patient positioning</li> <li>B. Due to risk of bleeding</li> <li>C. Due to risk of aspiration</li> <li>D. Because continuous monitoring of facial nerve function is needed</li> </ul>                                                | <p><b>Ans: D</b></p> |
| <p><b>Q47. Why is obstructive sleep apnea surgery very challenging for anesthesiologists?</b></p> <ul style="list-style-type: none"> <li>A. Due to bleeding only</li> <li>B. Due to obesity only</li> <li>C. All concerns related to obese patients especially difficult airway management</li> <li>D. Due to long duration only</li> </ul>                       | <p><b>Ans: C</b></p> |
| <p><b>Q48. What does limited mouth opening in TMJ surgeries mandate?</b></p> <ul style="list-style-type: none"> <li>A. Oral intubation under GA</li> <li>B. LMA use</li> <li>C. Tracheostomy</li> <li>D. Awake nasal intubation</li> </ul>                                                                                                                        | <p><b>Ans: D</b></p> |

**Q49. Are laser resistant endotracheal tubes 100% protective against laser fires?**

- A. No, they are not 100% protective against laser fires
- B. Yes, completely protective
- C. Only metal tubes are protective
- D. Only silicone tubes are protective

**Ans: A**

**Q50. What laryngeal complication can stimulation during panendoscopy induce?**

- A. Vocal cord paralysis
- B. Laryngeal edema
- C. Subglottic stenosis
- D. Laryngospasm

**Ans: D**

# Answer Key & Explanations

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**Q1. Answer: C — Because airway is often difficult, surgeon and anesthetist share the common field, and complications are related to airway**

ENT anesthesia is one of the most difficult because quite often airway is difficult, surgeon and anesthetist share the common field and complications are related to airway.

**Q2. Answer: B — Laryngoscopy, bronchoscopy, and esophagoscopy (also called triple endoscopy)**

Panendoscopy involves laryngoscopy, bronchoscopy, and esophagoscopy. It is also called as triple endoscopy.

**Q3. Answer: B — It may be reduced to the extent that even small tube insertion becomes impossible**

Due to the growth, the glottic opening may be reduced and sometimes to an extent that even insertion of the small size tube becomes impossible.

**Q4. Answer: C — 2-4%**

Laryngeal manipulation causes sympathetic stimulation. Incidence of myocardial ischemia and arrhythmias is significant (2-4%).

**Q5. Answer: B — Assessment of airway by preoperative endoscopy or indirect laryngoscopy is must**

During preoperative evaluation, assessment of airway by preoperative endoscopy or indirect laryngoscopy is must.

**Q6. Answer: B — To prevent laryngeal reflexes**

Intravenous (IV) lignocaine, IV fentanyl and topical anesthesia of the larynx are very important to prevent laryngeal reflexes.

**Q7. Answer: D — Pediatric size tubes with cuff available in 5, 5.5 and 6 mm size**

Microlaryngeal surgery (MLS) tubes are pediatric size tubes with cuff. They are available in 5, 5.5 and 6 mm size.

**Q8. Answer: C — >95%**

In the majority (>95%) of the patients it is possible to get the surgery done with MLS tube.

**Q9. Answer: A — High ventilatory pressures are required to overcome the narrow size of tube**

The disadvantage is that high ventilatory pressures are required to overcome the narrow size of tube.

**Q10. Answer: C — 60-100 breaths/min**

In HFPPV, small tidal volumes are delivered at high frequency (60-100 breaths/min) at a high pressure (60 psi) through a stiff 3.5-4.0 mm catheter passed through the vocal cord.

**Q11. Answer: A — 3.5-4.0 mm**

A stiff 3.5-4.0 mm catheter is passed through the vocal cord for HFPPV.

**Q12. Answer: A — Within the laryngoscope to avoid direct barotrauma to trachea**

The tip of the jet is kept within the laryngoscope to avoid direct barotrauma to the trachea and gas is delivered at high pressure (50-60 psi).

**Q13. Answer: C — Intubation is done, adequate ventilation given, tube removed for 1-2 minutes for surgery, then reintubated if needed**

In apnea technique intubation is done and after adequate ventilation tube is removed for short period (1-2 minutes or till oxygen saturation starts falling). In this period surgeon performs the procedure.

**Q14. Answer: A — With inhalational agent and repeated doses of succinylcholine for small procedures or nondepolarizing agent for prolonged surgery**

Anesthesia is maintained with inhalational agent and repeated doses of succinylcholine for small procedures or nondepolarizing agent for prolonged surgery.

**Q15. Answer: D — It can increase postsurgical edema by causing expansion of the cuff of endotracheal tube**

Nitrous oxide should be avoided because it can increase the postsurgical edema by causing expansion of the cuff of endotracheal tube.

**Q16. Answer: A — In patients with stridor**

Administration of Heliox is strongly considered in patients with stridor.

**Q17. Answer: B — Fires**

The most important complication of laser surgery is fires, therefore the most important aim should be to prevent fire.

**Q18. Answer: A — Less than 30%**

Use low FiO<sub>2</sub> (<30%) to prevent fire during laser surgery.

**Q19. Answer: B — It can support fire**

Avoid N<sub>2</sub>O as it can support fire.

**Q20. Answer: A — Laser resistant endotracheal tube**

Use special kind of laser resistant endotracheal tube.

**Q21. Answer: B — Saline**

Fill the tube cuff with saline (instead of air) during laser surgery.

**Q22. Answer: C — Immediately remove tube and submerge in water**

If fire occurs: immediately remove tube and submerge in water.

**Q23. Answer: D — Immediately stop oxygen (and nitrous oxide) and start ventilation with air by mask**

Immediately stop oxygen (and nitrous oxide) and start ventilation of the patient with air by mask.

**Q24. Answer: B — 50-60 mL**

Flush pharynx with 50-60 mL of cold saline.

**Q25. Answer: D — If damage is severe enough not to allow intubation**

Tracheostomy may be required if damage is severe enough not to allow intubation.

**Q26. Answer: B — Topical anesthesia**

Flexible bronchoscopy, usually done for diagnostic biopsies, can be performed under topical anesthesia.

**Q27. Answer: A — Foreign body removal under general anesthesia**

Rigid bronchoscopy, usually done for foreign body removal is performed under general anesthesia.

**Q28. Answer: A — High frequency jet ventilation, insufflation, and apnea technique**

Ventilation during rigid bronchoscopy is achieved with: high frequency jet ventilation through the oxygen port, insufflation, and apnea technique.

**Q29. Answer: A — Pediatric**

Majority of the patients for adenotonsillectomy/tonsillectomy are pediatric.

**Q30. Answer: D — Nasal intubation should preferably be avoided due to increased chances of bleeding**

If there are no adenoids nasal intubation can be done, however due to increased chances of bleeding nasal intubation should preferably be avoided.

**Q31. Answer: B — Oral Ring-Adair-Elwyn (RAE) preformed tubes**

Oral Ring-Adair-Elwyn (RAE) preformed tubes are most frequently used.

**Q32. Answer: A — To prevent aspiration of blood**

Throat should be packed to prevent aspiration.

**Q33. Answer: C — Tonsillar position - on one side (preferably left lateral) with head low**

These patients are nursed in tonsillar position, i.e. on one side (preferably left lateral) with head low to prevent aspiration and laryngospasm.

**Q34. Answer: A — As a full stomach patient (patient has swallowed blood causing significant aspiration risk)**

Such patient should be resuscitated and managed like full stomach patient considering that patient has swallowed enough blood which can cause significant aspiration.

**Q35. Answer: B — Under local anesthesia (as intubation may be impossible due to trismus and aspiration risk)**

General anesthesia should be avoided as intubation at times may be impossible due to trismus and patient can aspirate the pus. Therefore, the abscesses should be drained under local anesthesia.

**Q36. Answer: C — Awake nasal intubation with fiberoptic bronchoscopy**

If GA is required than awake nasal intubation with fiberoptic bronchoscopy should be achieved before induction of anesthesia.

**Q37. Answer: B — 35 times**

Nitrous oxide is 35 times more soluble than nitrogen.

**Q38. Answer: A — 375 mm H2O within half an hour**

If the Eustachian tube is blocked, nitrous oxide can increase the middle ear pressure to 375 mm H2O within half an hour.

**Q39. Answer: B — Patients with blocked Eustachian tubes (upper respiratory tract infection, adenoids, sinusitis and otitis media)**

Nitrous oxide should not be used in patients with blocked Eustachian tubes (upper respiratory tract infection, adenoids, sinusitis and otitis media).

**Q40. Answer: A — Increase in middle ear pressure can displace the graft**

Increase middle ear pressure by nitrous oxide can displace the graft making nitrous oxide to be contraindicated for Tympanoplasties.

**Q41. Answer: D — When surgeon is using 'inlay graft' (helps hold the graft)**

If the surgeon is using 'inlay graft' then increase in middle ear pressure is beneficial in holding the graft.

**Q42. Answer: B — Postoperative nausea and vomiting**

The incidence of postoperative nausea and vomiting is very high in ear surgeries; therefore prophylaxis with antiemetics is must.

**Q43. Answer: D — Nasal polyps, asthma, and sensitivity to aspirin/NSAIDs**

In the preoperative evaluation always rule out the Samter triad (nasal polyps, asthma, and sensitivity to aspirin/NSAIDs).

**Q44. Answer: C — To prevent aspiration of blood**

Throat pack is must to prevent aspiration of blood.

**Q45. Answer: C — Because they can become very restless in postoperative period after nasal packing if not trained**

The patients must be counseled and trained in the preoperative period to take breathing from mouth otherwise they can become very restless in the postoperative period after nasal packing.

**Q46. Answer: D — Because continuous monitoring of facial nerve function is needed**

Continuous monitoring of facial nerve function may deter the anesthetist to use muscle relaxants.

**Q47. Answer: C — All concerns related to obese patients especially difficult airway management**

All concerns related to obese patients especially difficult airway management makes obstruction sleep apnea (OSA) surgeries (uvulopalatoplasties) to be very challenging to anesthetist.

**Q48. Answer: D — Awake nasal intubation**

Due to the limited mouth opening intubation mandates awake nasal intubation.

**Q49. Answer: A — No, they are not 100% protective against laser fires**

Laser resistant endotracheal tubes are not 100% protective against laser fires.

**Q50. Answer: D — Laryngospasm**

Laryngeal stimulation can induce laryngospasm.



# Anesthesia for Trauma and Burns MCQ – Chapter 35

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 35)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Why is management of trauma patients uniquely challenging?**

- A. Inadequate medical history, not fasting, intoxication, multiple injuries, shock and difficult airway
- B. Only due to blood loss
- C. Only due to intoxication
- D. Only due to multiple injuries

**Ans: A**

**Q2. Why is adequate medical history often unavailable in trauma patients?**

- A. Language barrier only
- B. Patient refuses to share
- C. Hospital policy
- D. Attendant may not be present and patient may not be fit enough to give history

**Ans: D**

**Q3. What is the minimum systolic blood pressure recommended before induction in trauma patients?**

- A. 80-90 mm Hg
- B. 120 mm Hg
- C. 60-70 mm Hg
- D. 100-110 mm Hg

**Ans: A**

**Q4. When may damage control surgery warrant induction at very low blood pressure?**

- A. When surgery is the only method to stop bleeding
- B. When patient requests it
- C. For elective procedures
- D. For cosmetic surgery

**Ans: A**

**Q5. What major injuries must be ruled out in trauma patients?**

- A. Only abdominal injuries
- B. Chest injury (pneumo/hemothorax), head injury, spine injuries, abdominal and pelvic injuries
- C. Only spine injuries
- D. Only chest injuries

**Ans: B**

**Q6. Why is head injury significant for choice of anesthesia technique?**

- A. It makes GA easier
- B. Central neuraxial blocks may be absolutely contraindicated with elevated intracranial tension (ICT)
- C. It has no effect
- D. It only affects monitoring

**Ans: B**

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| <p><b>Q7. What devastating complications can improper movements of spine injury patients cause?</b></p> <p>A. Only numbness<br/>B. Only pain<br/>C. Quadriplegia, paraplegia or even death<br/>D. Only temporary weakness</p>                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q8. If fasting status cannot be obtained in trauma, what premedication should be given?</b></p> <p>A. Only omeprazole<br/>B. Only antacids<br/>C. Metoclopramide and ranitidine/sodium citrate<br/>D. Only ondansetron</p>                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q9. Why is continuous blood pressure by invasive monitoring recommended in trauma?</b></p> <p>A. Although difficult to cannulate artery in low output state, every effort should be made as trauma patients are prone to hypotension<br/>B. It is cheaper<br/>C. For research purposes<br/>D. It is more convenient</p> | <p><b>Ans: A</b></p> |
| <p><b>Q10. Why is temperature monitoring essential in trauma patients?</b></p> <p>A. For billing purposes<br/>B. To calculate drug doses<br/>C. Because trauma patients are very prone to hypothermia<br/>D. For documentation</p>                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q11. How should a trauma patient always be considered regarding stomach contents?</b></p> <p>A. As safe for LMA<br/>B. As having a full stomach until proved otherwise, managed like rapid sequence induction<br/>C. As partially fasted<br/>D. As having an empty stomach</p>                                          | <p><b>Ans: B</b></p> |
| <p><b>Q12. What is the choice of anesthesia for majority of trauma surgeries?</b></p> <p>A. Monitored anesthesia care<br/>B. Local anesthesia<br/>C. General anesthesia (except for limb saving surgeries which may be done under regional)<br/>D. Regional anesthesia</p>                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q13. Why is selection of induction agent difficult in trauma patients?</b></p> <p>A. Only availability issues<br/>B. Only cost issues<br/>C. IV induction agents may produce profound hypotension or even cardiac arrest in shock patients<br/>D. All agents are equally safe</p>                                       | <p><b>Ans: C</b></p> |

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| <p><b>Q14. How is the dose requirement of IV induction agents in hypotensive patients?</b></p> <ul style="list-style-type: none"> <li>A. Doubled</li> <li>B. Increased</li> <li>C. Same as normal</li> <li>D. Drastically reduced</li> </ul>                                                                                                                         | <p><b>Ans: D</b></p> |
| <p><b>Q15. Which agents are often selected for induction in trauma patients and why?</b></p> <ul style="list-style-type: none"> <li>A. Etomidate (cardiovascular stability) and ketamine (sympathetic system stimulating property)</li> <li>B. Propofol for its rapid onset</li> <li>C. Midazolam for its amnesia</li> <li>D. Thiopentone for its potency</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q16. Why can etomidate still cause hypotension in catecholamine-depleted shock patients?</b></p> <ul style="list-style-type: none"> <li>A. It causes direct cardiac depression</li> <li>B. It causes adrenocortical suppression</li> <li>C. It causes histamine release</li> <li>D. It is a vasodilator</li> </ul>                                             | <p><b>Ans: B</b></p> |
| <p><b>Q17. Why can ketamine cause hypotension in catecholamine-depleted patients?</b></p> <ul style="list-style-type: none"> <li>A. It causes histamine release</li> <li>B. It causes direct myocardial depression in the absence of catecholamine</li> <li>C. It blocks sympathetic ganglia</li> <li>D. It causes vasodilation</li> </ul>                           | <p><b>Ans: B</b></p> |
| <p><b>Q18. What should be assumed about ALL trauma patients regarding cervical spine?</b></p> <ul style="list-style-type: none"> <li>A. Cervical spine is safe</li> <li>B. Only elderly patients</li> <li>C. All should be assumed to have cervical spine injury until proved otherwise</li> <li>D. Only high-velocity trauma patients</li> </ul>                    | <p><b>Ans: C</b></p> |
| <p><b>Q19. How should trauma patients with uncleared cervical spine be intubated?</b></p> <ul style="list-style-type: none"> <li>A. Without any precautions</li> <li>B. Only with tracheostomy</li> <li>C. Blind nasal intubation</li> <li>D. With in-line manual stabilization</li> </ul>                                                                           | <p><b>Ans: D</b></p> |
| <p><b>Q20. When should nasal route NOT be acquired for intubation in trauma?</b></p> <ul style="list-style-type: none"> <li>A. In all cases</li> <li>B. Until basal skull fracture is ruled out</li> <li>C. In mandibular fractures</li> <li>D. In neck trauma only</li> </ul>                                                                                       | <p><b>Ans: B</b></p> |

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| <p><b>Q21. What equipment must be ready for difficult airway in trauma including face/neck trauma?</b></p> <p>A. Only LMA<br/>B. Only bougie<br/>C. All equipment for difficult airway including emergency cricothyrotomy<br/>D. Only extra ETT</p>                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q22. Through which route can patients with maxillary fractures most often be easily intubated?</b></p> <p>A. Oral route<br/>B. Tracheostomy only<br/>C. LMA only<br/>D. Nasal route</p>                                                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q23. Why are patients with mandibular fracture often intubated nasally?</b></p> <p>A. For better surgical access only<br/>B. Oral intubation is preferred<br/>C. Trismus prevents oral intubation and maxillofacial surgeons do dental wiring to close mouth after surgery<br/>D. Nasal route is easier</p> | <p><b>Ans: C</b></p> |
| <p><b>Q24. What may patients with both mandibular and maxillary fracture require?</b></p> <p>A. Nasal intubation<br/>B. Tracheostomy<br/>C. Oral intubation<br/>D. LMA</p>                                                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q25. What is the best solution for pharyngeal and laryngeal trauma where intubation can worsen the injury?</b></p> <p>A. A low tracheostomy<br/>B. LMA<br/>C. Oral intubation<br/>D. Nasal intubation</p>                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q26. Is application of cricoid pressure contraindicated in any trauma situation?</b></p> <p>A. Only in cervical spine injury<br/>B. Only in facial fractures<br/>C. Yes, it is contraindicated in laryngeal trauma<br/>D. Never contraindicated</p>                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q27. Why must a chest tube be inserted before intubation if there is pneumothorax?</b></p> <p>A. Because positive pressure ventilation can convert a small pneumothorax into tension pneumothorax<br/>B. To prevent bleeding<br/>C. To improve oxygenation<br/>D. To allow nasal intubation</p>             | <p><b>Ans: A</b></p> |

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| <p><b>Q28. Which inhalational agent is most preferred for maintenance in trauma patients?</b></p> <p>A. Halothane<br/>B. Sevoflurane<br/>C. Isoflurane<br/>D. Desflurane</p>                                                                                                                                                                                                                                         | <p><b>Ans: D</b></p> |
| <p><b>Q29. Why may patients in shock exhibit extreme sensitivity to inhalational agents?</b></p> <p>A. Like IV agents, patients in shock exhibit extreme sensitivity to inhalational agents<br/>B. Due to hepatic failure<br/>C. Due to hypothermia only<br/>D. Due to renal failure</p>                                                                                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q30. Despite causing hyperkalemia in crush injury, why does succinylcholine remain the muscle relaxant of choice in trauma?</b></p> <p>A. Because it remains the choice for intubation in difficult airway and high risk of aspiration - consequences of failed intubation and aspiration can be devastating<br/>B. It causes no side effects<br/>C. It is the cheapest<br/>D. It has the longest duration</p> | <p><b>Ans: A</b></p> |
| <p><b>Q31. Which non-depolarizing muscle relaxants are preferred in trauma?</b></p> <p>A. Rocuronium only<br/>B. Vecuronium and pancuronium<br/>C. Atracurium and Cis-atracurium (due to their independency on hepatic and renal functions)<br/>D. Mivacurium only</p>                                                                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q32. What type of IV access should be secured in trauma patients?</b></p> <p>A. Only central line<br/>B. Only intraosseous<br/>C. Small peripheral cannula<br/>D. Large bore cannulas; if peripheral access not possible then central access (most often subclavian) or intraosseous access</p>                                                                                                                | <p><b>Ans: D</b></p> |
| <p><b>Q33. How should hypothermia be prevented in trauma patients?</b></p> <p>A. Only warm fluids<br/>B. Only warming blankets<br/>C. OT temperature should be maintained and fluids/blood should be warmed to room temperature before infusion<br/>D. No measures needed</p>                                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q34. Why is extubation tricky in trauma patients?</b></p> <p>A. Only due to pain<br/>B. Deep extubation can cause aspiration while light extubation can increase intraocular/intracranial pressure and restart hemorrhage<br/>C. Only due to swelling<br/>D. Only due to bleeding</p>                                                                                                                          | <p><b>Ans: B</b></p> |

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| <p><b>Q35. Is it uncommon to see trauma patients getting electively ventilated postoperatively?</b></p> <p>A. Only in burns<br/>B. Only in head injury<br/>C. No, it is not uncommon<br/>D. Yes, very uncommon</p>                                                                        | Ans: C |
| <p><b>Q36. What is the most common immediate cause of death if burns occur in closed space?</b></p> <p>A. Hypovolemia<br/>B. Hypothermia<br/>C. Carbon monoxide poisoning<br/>D. Sepsis</p>                                                                                               | Ans: C |
| <p><b>Q37. What is the immediate treatment for carbon monoxide poisoning in burns?</b></p> <p>A. IV fluids<br/>B. Oxygen or hyperbaric oxygen (in severe cases)<br/>C. Intubation<br/>D. Steroids</p>                                                                                     | Ans: B |
| <p><b>Q38. What formula is used for fluid replacement in burn patients?</b></p> <p>A. Holliday formula<br/>B. 4-2-1 rule<br/>C. Parkland formula (4 mL/kg/% burn)<br/>D. Holiday-Segar formula</p>                                                                                        | Ans: C |
| <p><b>Q39. How should the Parkland formula fluid be distributed?</b></p> <p>A. All in first 4 hours<br/>B. Equal over 24 hours<br/>C. All in first 8 hours<br/>D. 50% in first 8 hours and next 50% in next 16 hours</p>                                                                  | Ans: D |
| <p><b>Q40. What is the current recommendation for fluid management in burns?</b></p> <p>A. Restrict fluids<br/>B. Give only colloids<br/>C. Start fluid by Parkland formula but immediately start tapering once urine output becomes &gt;0.5-1 mL/kg/hr<br/>D. Give only crystalloids</p> | Ans: C |
| <p><b>Q41. What is the fluid of choice in burn resuscitation?</b></p> <p>A. Dextrose<br/>B. Normal saline<br/>C. Ringer lactate<br/>D. Colloids</p>                                                                                                                                       | Ans: C |

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| <p><b>Q42. Can colloids be given in acute phase of burns?</b></p> <p>A. Yes, if necessary can be given in acute phase (contrary to previous recommendation of not giving colloid for first 48 hours)<br/> B. Absolutely contraindicated<br/> C. Only albumin<br/> D. Only after 72 hours</p>    | <p><b>Ans: A</b></p> |
| <p><b>Q43. What cardiac complications can electrical burns present with?</b></p> <p>A. Only hypertension<br/> B. Cardiac arrhythmias or even cardiac arrest<br/> C. Only bradycardia<br/> D. Only tachycardia</p>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q44. Why is intraoperative ECG monitoring mandatory in electrical burns?</b></p> <p>A. It is not mandatory<br/> B. For billing<br/> C. For research<br/> D. Because there are high chances of arrhythmias</p>                                                                             | <p><b>Ans: D</b></p> |
| <p><b>Q45. What is the usual surgery done in burn patients?</b></p> <p>A. Amputation<br/> B. Release of post-burn contractures and split skin grafts<br/> C. Debridement only<br/> D. Fasciotomy only</p>                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q46. Why is succinylcholine contraindicated in burn patients and for how long?</b></p> <p>A. For one year due to presence of extrajunctional receptors causing hyperkalemia<br/> B. For 6 months due to bradycardia<br/> C. For 2 weeks due to pain<br/> D. It is not contraindicated</p> | <p><b>Ans: A</b></p> |
| <p><b>Q47. Why may the dose of IV induction agent be slightly higher in burn patients?</b></p> <p>A. Due to renal failure<br/> B. Due to hepatic enzyme induction<br/> C. Due to large volume of distribution (hyperdynamic circulation) and increased metabolism<br/> D. Due to tolerance</p>  | <p><b>Ans: C</b></p> |
| <p><b>Q48. Why may airway be difficult in burn patients?</b></p> <p>A. Due to trismus<br/> B. Due to laryngeal injury only<br/> C. Due to edema only<br/> D. Due to contracture of face and neck</p>                                                                                            | <p><b>Ans: D</b></p> |



**Q49. If fiberoptic intubation is not possible in burn contracture patients, what alternative is used?**

- A. LMA only
- B. Tracheostomy always
- C. Release contracture under IV ketamine and then intubation is done
- D. Blind nasal intubation

**Ans: C**

**Q50. What should be evaluated in patients with electrical burns before giving regional anesthesia?**

- A. Neurological deficits (as electrical burns can have neurological injuries)
- B. Blood pressure
- C. Skin condition
- D. Temperature

**Ans: A**

# Answer Key & Explanations

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## **Q1. Answer: A — Inadequate medical history, not fasting, intoxication, multiple injuries, shock and difficult airway**

Management of trauma patients poses unique challenges due to inadequate medical history, most patients are not fasting, may be intoxicated, multiple injuries make anesthesia complex, airway may be difficult due to facial/neck injuries, and presence of shock.

*Topic: Trauma General*

## **Q2. Answer: D — Attendant may not be present and patient may not be fit enough to give history**

Inadequate medical history: Attendant may not be present and patient may not be fit enough to give his/her medical history.

*Topic: Trauma General*

## **Q3. Answer: A — 80-90 mm Hg**

A systolic blood pressure of at least 80-90 mm Hg is recommended before induction.

*Topic: Preoperative*

## **Q4. Answer: A — When surgery is the only method to stop bleeding**

Sometimes the situation like damage control surgery (surgery is the only method to stop bleeding) warrants the induction at very low or even nonrecordable blood pressure.

*Topic: Preoperative*

## **Q5. Answer: B — Chest injury (pneumo/hemothorax), head injury, spine injuries, abdominal and pelvic injuries**

It is very important to rule out major injuries especially chest injury like pneumo/hemothorax, head injury, spine injuries, and abdominal/pelvic injuries.

*Topic: Preoperative*

## **Q6. Answer: B — Central neuraxial blocks may be absolutely contraindicated with elevated intracranial tension (ICT)**

Head injury—presence of head injury can alter the anesthesia technique. For example, central neuraxial blocks may be absolutely contraindicated with elevated intracranial tension (ICT).

*Topic: Preoperative*

## **Q7. Answer: C — Quadriplegia, paraplegia or even death**

Spine injuries: Improper movements can produce devastating complications, including quadriplegia, paraplegia or even death.

*Topic: Preoperative*

## **Q8. Answer: C — Metoclopramide and ranitidine/sodium citrate**

If fasting status cannot be obtained, pre-medicate the patient with metoclopramide and ranitidine/sodium citrate.

*Topic: Preoperative*

**Q9. Answer: A — Although difficult to cannulate artery in low output state, every effort should be made as trauma patients are prone to hypotension**

Although it may be difficult to cannulate artery in low output state but still every effort should be made to measure continuous blood pressure by invasive monitoring.

Topic: Monitoring

**Q10. Answer: C — Because trauma patients are very prone to hypothermia**

As trauma patients are very prone for hypothermia, temperature monitoring becomes must.

Topic: Monitoring

**Q11. Answer: B — As having a full stomach until proved otherwise, managed like rapid sequence induction**

A trauma patient should always be considered full stomach until proved otherwise and should be managed like rapid sequence induction.

Topic: Choice of Anesthesia

**Q12. Answer: C — General anesthesia (except for limb saving surgeries which may be done under regional)**

Except for limb saving surgeries which may be performed in regional anesthesia, majority of the trauma surgeries are done in general anesthesia.

Topic: Choice of Anesthesia

**Q13. Answer: C — IV induction agents may produce profound hypotension or even cardiac arrest in shock patients**

Selection of the induction agent is difficult in trauma patients. IV induction agents may produce profound hypotension (or even cardiac arrest) in shock patients.

Topic: Induction

**Q14. Answer: D — Drastically reduced**

Requirement of doses is drastically reduced in hypotensive patients.

Topic: Induction

**Q15. Answer: A — Etomidate (cardiovascular stability) and ketamine (sympathetic system stimulating property)**

Etomidate (because of its cardiovascular stability) and ketamine (because of its sympathetic system stimulating property) are the agents often selected for induction.

Topic: Induction

**Q16. Answer: B — It causes adrenocortical suppression**

Etomidate can produce hypotension in shock patients who are already catecholamine depleted; etomidate by causing adrenocortical suppression.

Topic: Induction

**Q17. Answer: B — It causes direct myocardial depression in the absence of catecholamine**

Ketamine can produce hypotension in shock patients who are already catecholamine depleted; ketamine by causing direct myocardial depression in the absence of catecholamine.

Topic: Induction

**Q18. Answer: C — All should be assumed to have cervical spine injury until proved otherwise**

All trauma patients should be assumed to be having cervical spine injury until proved otherwise.

Topic: Intubation

**Q19. Answer: D — With in-line manual stabilization**

If cervical spine injury is not cleared they should be intubated with in-line manual stabilization.

Topic: Intubation

**Q20. Answer: B — Until basal skull fracture is ruled out**

Nasal route should not be acquired for intubation until basal skull fracture is ruled out.

Topic: Intubation

**Q21. Answer: C — All equipment for difficult airway including emergency cricothyrotomy**

All equipment for difficult airway (including emergency cricothyrotomy) must be ready hand available because intubating a patient with face and neck trauma can become a big challenge.

Topic: Intubation

**Q22. Answer: A — Oral route**

Patients with maxillary fractures should not be intubated nasally; however most often can be easily intubated through oral route.

Topic: Intubation

**Q23. Answer: C — Trismus prevents oral intubation and maxillofacial surgeons do dental wiring to close mouth after surgery**

Patients with mandibular fracture are often intubated nasally due to two reasons: trismus prevents oral intubation, and most often maxillofacial surgeons do the dental wiring to close the mouth after mandibular surgery.

Topic: Intubation

**Q24. Answer: B — Tracheostomy**

Patients with both mandibular and maxillary fracture may require tracheostomy.

Topic: Intubation

**Q25. Answer: A — A low tracheostomy**

Intubation in pharyngeal and laryngeal trauma can worsen the injury or even lead to complete loss of airway. A low tracheostomy is the best solution in most of the cases.

Topic: Intubation

**Q26. Answer: C — Yes, it is contraindicated in laryngeal trauma**

Application of cricoid pressure is also contraindicated in laryngeal trauma.

Topic: Intubation

**Q27. Answer: A — Because positive pressure ventilation can convert a small pneumothorax into tension pneumothorax**

Chest tube must be inserted before intubation if there is pneumothorax because positive pressure ventilation can convert a small pneumothorax into tension pneumothorax.

Topic: Intubation

**Q28. Answer: D — Desflurane**

Desflurane is often the most preferred inhalational agent because of its ability to maintain blood pressure by causing sympathetic stimulation and least metabolism.

Topic: Maintenance

**Q29. Answer: A — Like IV agents, patients in shock exhibit extreme sensitivity to inhalational agents**

Like IV agents patients in shock may exhibit extreme sensitivity to inhalational agents.

Topic: Maintenance

**Q30. Answer: A — Because it remains the choice for intubation in difficult airway and high risk of aspiration - consequences of failed intubation and aspiration can be devastating**

In spite of succinylcholine causing hyperkalemia in crush injury and increasing intraocular and intracranial pressure, it remains the muscle relaxant of choice for intubation in trauma patients with difficult airway and high risk of aspiration.

Topic: Muscle Relaxants

**Q31. Answer: C — Atracurium and Cis-atracurium (due to their independency on hepatic and renal functions)**

Atracurium and Cis-atracurium are preferred non depolarizers because of their independency on hepatic and renal functions.

Topic: Muscle Relaxants

**Q32. Answer: D — Large bore cannulas; if peripheral access not possible then central access (most often subclavian) or intraosseous access**

Secure IV lines with large bore cannulas. If peripheral access is not possible, then obtain central access (most often subclavian) or intraosseous access (if central access is also not possible).

Topic: Intraoperative

**Q33. Answer: C — OT temperature should be maintained and fluids/blood should be warmed to room temperature before infusion**

As trauma patients are very prone for hypothermia OT temperature should be maintained and fluids/blood should be warmed to room temperature before infusion.

Topic: Intraoperative

**Q34. Answer: B — Deep extubation can cause aspiration while light extubation can increase intraocular/intracranial pressure and restart hemorrhage**

Extubation in trauma patients is a tricky exercise. Deep extubation can cause aspiration while extubation under light anesthesia can increase intraocular, intracranial pressure and restart the hemorrhage by increasing blood pressure.

Topic: Postoperative

**Q35. Answer: C — No, it is not uncommon**

It is not uncommon to see trauma patients getting electively ventilated in postoperative period.

Topic: Postoperative

**Q36. Answer: C — Carbon monoxide poisoning**

Carbon monoxide poisoning is the most common immediate cause of death if burns occur in closed space.

Topic: Burns

**Q37. Answer: B — Oxygen or hyperbaric oxygen (in severe cases)**

Immediate treatment with oxygen or hyperbaric oxygen (in severe cases) may be required.

Topic: Burns

**Q38. Answer: C — Parkland formula (4 mL/kg/% burn)**

Fluid should be given by the standard Parkland formula which recommends fluid replacement at a rate of 4 mL/kg/% burn.

Topic: Burns

**Q39. Answer: D — 50% in first 8 hours and next 50% in next 16 hours**

50% of this fluid is given in first 8 hours and next 50% in next 16 hours.

Topic: Burns

**Q40. Answer: C — Start fluid by Parkland formula but immediately start tapering once urine output becomes >0.5-1 mL/kg/hr**

Current recommendation is to start fluid by Parkland formula but immediately start tapering once urine output becomes >0.5-1 mL/kg/hr.

Topic: Burns

**Q41. Answer: C — Ringer lactate**

Ringer lactate is the fluid of choice.

Topic: Burns

**Q42. Answer: A — Yes, if necessary can be given in acute phase (contrary to previous recommendation of not giving colloid for first 48 hours)**

Colloids if necessary can be given in acute phase (contrary to previous recommendation of not giving colloid for first 48 hours).

Topic: Burns

**Q43. Answer: B — Cardiac arrhythmias or even cardiac arrest**

Patients with electrical burns can present with cardiac arrhythmias or even cardiac arrest.

Topic: Burns

**Q44. Answer: D — Because there are high chances of arrhythmias**

Intraoperative monitoring of ECG is mandatory in electric burns as there are high chances of arrhythmias.

Topic: Burns

**Q45. Answer: B — Release of post-burn contractures and split skin grafts**

Usual surgeries done in burn patients are release of post-burn contractures and split skin grafts.

Topic: Burns

**Q46. Answer: A — For one year due to presence of extrajunctional receptors causing hyperkalemia**

Presence of extrajunctional receptors contraindicates the use of succinylcholine for one year and decreases the sensitivity (increase dose requirement) of nondepolarizers.

Topic: Burns

**Q47. Answer: C — Due to large volume of distribution (hyperdynamic circulation) and increased metabolism**

Dose of intravenous induction agent may be slightly higher because of the large volume of distribution (hyperdynamic circulation) and increased metabolism.

*Topic: Burns*

**Q48. Answer: D — Due to contracture of face and neck**

Airway may be difficult due to contracture of face and neck. Awake intubation under fiberoptic bronchoscopy may be required.

*Topic: Burns*

**Q49. Answer: C — Release contracture under IV ketamine and then intubation is done**

If that is not possible then contracture is released under intravenous ketamine and then intubation is done.

*Topic: Burns*

**Q50. Answer: A — Neurological deficits (as electrical burns can have neurological injuries)**

Patients with electrical burn can have neurological injuries. They should be evaluated for neurological deficits before giving regional anesthesia.

*Topic: Burns*

# Anesthesia for Orthopedics MCQ – Chapter 36

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 36)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.



# Questions

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**Q1. What special problems does orthopedic anesthesia pose?**

- A. Only blood loss
- B. Only tourniquet problems
- C. Lengthy procedures, increased blood loss, positioning, tourniquet-related problems, fat embolism and DVT
- D. Only positioning issues

**Ans: C**

**Q2. What complications are associated with tourniquet during inflation?**

- A. Hypothermia
- B. Hypertension, increased cardiac output and increased pulmonary artery pressure
- C. Hypotension and decreased cardiac output
- D. Bradycardia

**Ans: B**

**Q3. What metabolic change occurs within 6 minutes of tourniquet inflation?**

- A. Local tissue ischemia leads to acidosis (mitochondrial pO<sub>2</sub> becomes zero)
- B. No metabolic change
- C. Alkalosis
- D. Hyperglycemia

**Ans: A**

**Q4. How is pain during tourniquet inflation relieved?**

- A. Only by deflation (not relieved by IV analgesics)
- B. By local anesthetics
- C. By IV analgesics
- D. By increasing tourniquet pressure

**Ans: A**

**Q5. What causes hypotension during tourniquet deflation?**

- A. Vasovagal reflex
- B. Sudden release of accumulated metabolites like thromboxane and substance P
- C. Blood loss
- D. Cardiac depression

**Ans: B**

**Q6. Why does hypothermia occur during tourniquet deflation?**

- A. Due to heat loss as acid metabolites are vasodilators
- B. Due to blood transfusion
- C. Due to air conditioning
- D. Due to cold fluids

**Ans: A**

**Q7. What happens to end tidal CO<sub>2</sub> during tourniquet deflation?**

- A. No change
- B. Decreases
- C. Becomes zero
- D. Transient increase (CO<sub>2</sub> accumulated in limb is released)

**Ans: D**

|                                                                                                                                                                                                                                                |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <p><b>Q8. Which is the most crucial period during tourniquet use?</b></p> <p>A. Inflation<br/>B. Pre-inflation<br/>C. During surgery<br/>D. Deflation</p>                                                                                      | Ans: D |
| <p><b>Q9. What is the maximum duration for tourniquet pressure for upper and lower limbs?</b></p> <p>A. 4 hours for lower limbs<br/>B. Not more than 2 hours for both upper and lower limbs<br/>C. 3 hours<br/>D. 1 hour</p>                   | Ans: B |
| <p><b>Q10. If re-inflation of tourniquet is needed, how long should it be deflated first?</b></p> <p>A. 1 hour<br/>B. 10 minutes<br/>C. 5 minutes<br/>D. At least 30 minutes</p>                                                               | Ans: D |
| <p><b>Q11. What is the maximum tourniquet pressure above systolic?</b></p> <p>A. 150 mm Hg above systolic<br/>B. Not more than 100 mm Hg above systolic (maximum 250 mm Hg)<br/>C. 200 mm Hg above systolic<br/>D. 50 mm Hg above systolic</p> | Ans: B |
| <p><b>Q12. What complication may occur if tourniquet pressure is high or time is prolonged?</b></p> <p>A. Infection<br/>B. Thrombosis<br/>C. Neuropraxia<br/>D. Compartment syndrome</p>                                                       | Ans: C |
| <p><b>Q13. When is fat embolism syndrome (FES) usually seen?</b></p> <p>A. After cardiac surgery<br/>B. After spinal surgery<br/>C. After fracture of long bones and pelvis<br/>D. After abdominal surgery</p>                                 | Ans: C |
| <p><b>Q14. When does fat embolism classically present after fracture?</b></p> <p>A. Immediately<br/>B. After 1 week<br/>C. Within 24-72 hours<br/>D. Within 6 hours</p>                                                                        | Ans: C |
| <p><b>Q15. Is the incidence of FES higher in nailing or plating?</b></p> <p>A. Higher in plating<br/>B. Higher in nailing compared to plating<br/>C. Same in both<br/>D. Neither causes FES</p>                                                | Ans: B |

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| <p><b>Q16. What is the classic triad of fat embolism syndrome?</b></p> <p>A. Chest pain, cough, hemoptysis<br/> B. Headache, vomiting, seizures<br/> C. Dyspnea, confusion and petechial hemorrhages<br/> D. Fever, tachycardia, hypotension</p>                                                                                                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q17. What is pathognomonic of fat embolism syndrome?</b></p> <p>A. Confusion<br/> B. Dyspnea<br/> C. Tachycardia<br/> D. Petechial rash</p>                                                                                                                                                                                                                                      | <p><b>Ans: D</b></p> |
| <p><b>Q18. What are the signs and symptoms of fat embolism?</b></p> <p>A. Only neurological symptoms<br/> B. Dyspnea, chest discomfort, tachycardia, hypotension, mental clouding, restlessness, seizures, delirium, petechial hemorrhage on upper chest, axilla and conjunctiva, hypoxia and finally coma and death<br/> C. Only petechial rash<br/> D. Only respiratory symptoms</p> | <p><b>Ans: B</b></p> |
| <p><b>Q19. What do fat globules in urine indicate regarding fat embolism?</b></p> <p>A. No significance<br/> B. Definitive diagnosis of FES<br/> C. Helpful in diagnosis but may not necessarily indicate progression to FES<br/> D. Rules out FES</p>                                                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q20. What does X-ray chest show in fat embolism?</b></p> <p>A. Consolidation<br/> B. Pleural effusion<br/> C. Diffuse interstitial infiltrates<br/> D. Pneumothorax</p>                                                                                                                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q21. What does capnography show in fat embolism?</b></p> <p>A. Increase in end tidal CO<sub>2</sub><br/> B. Fluctuating ETCO<sub>2</sub><br/> C. No change<br/> D. Fall in end tidal CO<sub>2</sub> (ETCO<sub>2</sub> may become zero if embolus is large enough to block main pulmonary artery)</p>                                                                             | <p><b>Ans: D</b></p> |
| <p><b>Q22. What ECG finding is seen in fat embolism?</b></p> <p>A. Atrial fibrillation<br/> B. Right ventricular strain<br/> C. ST elevation<br/> D. Left ventricular strain</p>                                                                                                                                                                                                       | <p><b>Ans: B</b></p> |

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| <p><b>Q23. What happens to serum lipase in fat embolism?</b></p> <p>A. Increased (but has no relation to severity of disease)<br/> B. Absent<br/> C. Normal<br/> D. Decreased</p>                                                                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q24. What is the first step in treatment of fat embolism?</b></p> <p>A. Heparin therapy<br/> B. Ethanol infusion<br/> C. Early stabilization of fracture (may prevent fat embolism)<br/> D. Corticosteroids</p>                                                                                                  | <p><b>Ans: C</b></p> |
| <p><b>Q25. What ventilatory support is used if hypoxia persists on oxygen therapy in fat embolism?</b></p> <p>A. CPAP only<br/> B. Non-invasive ventilation only<br/> C. High flow nasal cannula only<br/> D. Intermittent positive pressure ventilation (IPPV) with positive end expiratory pressure (PEEP)</p>       | <p><b>Ans: D</b></p> |
| <p><b>Q26. What is the pathophysiology of fat embolism in the lungs?</b></p> <p>A. Causes pleural effusion<br/> B. Fat emboli cause endothelial injury which can precipitate ARDS and cerebral edema<br/> C. Causes bronchospasm<br/> D. Causes pneumothorax</p>                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q27. Are corticosteroids, heparin and ethanol considered effective treatments for fat embolism currently?</b></p> <p>A. Only corticosteroids are effective<br/> B. No, they are now considered as outdated treatment modalities<br/> C. Only heparin is effective<br/> D. Yes, they are first-line treatment</p> | <p><b>Ans: B</b></p> |
| <p><b>Q28. What is the incidence of fatal pulmonary embolism (PE) after hip surgery?</b></p> <p>A. 5-10%<br/> B. Less than 0.5%<br/> C. 1-3%<br/> D. 10-15%</p>                                                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q29. In which orthopedic patients is the incidence of DVT and PE highest?</b></p> <p>A. Upper limb surgery patients<br/> B. Arthroscopy patients<br/> C. Hand surgery patients<br/> D. Hip surgery patients particularly after prolonged immobilization</p>                                                      | <p><b>Ans: D</b></p> |

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| <p><b>Q30. When should low molecular weight heparin be started for DVT prophylaxis in hip surgery?</b></p> <p>A. Immediately before surgery<br/>B. 24 hours before surgery<br/>C. Only postoperatively<br/>D. 12 hours or more in preoperative period and restart 12 hours after surgery</p>                                                                     | <p><b>Ans: D</b></p> |
| <p><b>Q31. What serious complications have been reported in shoulder surgeries done in beach chair position?</b></p> <p>A. Stroke or even brain death due to ischemia<br/>B. Only nerve injury<br/>C. Only hypotension<br/>D. Only bleeding</p>                                                                                                                  | <p><b>Ans: A</b></p> |
| <p><b>Q32. Where should the arterial pressure transducer be zeroed for shoulder surgeries in beach chair position?</b></p> <p>A. At the level of the hip<br/>B. At the level of external auditory meatus to denote cerebral perfusion pressure<br/>C. At the level of the arm<br/>D. At the level of the heart</p>                                               | <p><b>Ans: B</b></p> |
| <p><b>Q33. What is cement implantation syndrome (CIS)?</b></p> <p>A. Hypoxia, profound hypotension or even cardiac arrest occurring during cemented prosthesis in total hip replacement<br/>B. Allergic reaction to cement<br/>C. Infection due to cement<br/>D. Local tissue necrosis</p>                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q34. What causes cement implantation syndrome?</b></p> <p>A. Blood loss<br/>B. Infection<br/>C. Allergic reaction only<br/>D. Embolization of bone marrow debris, microemboli or fat due to increased intramedullary pressure or reaction to cement (methyl methacrylate)</p>                                                                              | <p><b>Ans: D</b></p> |
| <p><b>Q35. What is the treatment for cement implantation syndrome?</b></p> <p>A. Oxygenation, adequate hydration and adrenaline for severe reactions<br/>B. Antihistamines<br/>C. Only steroids<br/>D. Only IV fluids</p>                                                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q36. Why is regional anesthesia always preferred over general anesthesia for orthopedic surgeries?</b></p> <p>A. Due to decreased incidence of thromboembolism, decreased blood loss, fewer respiratory complications and better control of pain<br/>B. Only because of cost<br/>C. Only because of convenience<br/>D. Only because of faster recovery</p> | <p><b>Ans: A</b></p> |

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| <p><b>Q37. What type of regional block is used for upper limb surgeries?</b></p> <p>A. Femoral nerve block<br/>B. Brachial plexus block<br/>C. Epidural<br/>D. Spinal block</p>                                                                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q38. What block can be used for small procedures lasting less than 30-45 minutes?</b></p> <p>A. Brachial plexus block<br/>B. Bier's block (IV regional anesthesia)<br/>C. Spinal anesthesia<br/>D. General anesthesia</p>                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q39. What is the most preferred technique for lower limb surgeries?</b></p> <p>A. General anesthesia<br/>B. Bier's block<br/>C. Spinal/epidural or combined spinal epidural (most preferred technique)<br/>D. Femoral nerve block alone</p>                                                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q40. When is general anesthesia required for orthopedic surgery?</b></p> <p>A. Only for hip replacement<br/>B. For all surgeries<br/>C. Only for spine surgery<br/>D. For very prolonged surgeries of upper limb, shoulder or spine surgeries or when regional is not possible</p>                                                                            | <p><b>Ans: D</b></p> |
| <p><b>Q41. What nerve blocks are frequently utilized for postoperative analgesia in orthopedics?</b></p> <p>A. Only local infiltration<br/>B. Only femoral nerve block<br/>C. Continuous adductor canal block for knee, fascia iliaca/femoral nerve block for hip surgeries, brachial plexus block for upper limb<br/>D. Only epidural</p>                          | <p><b>Ans: C</b></p> |
| <p><b>Q42. In what position is spine surgery performed?</b></p> <p>A. Supine<br/>B. Lateral<br/>C. Sitting<br/>D. Prone position under general anesthesia</p>                                                                                                                                                                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q43. What are the complications related to prone position in spine surgery?</b></p> <p>A. Endotracheal tube dislodgement, airway edema, injury to brachial plexus and peroneal nerve, injury to genitalia and breast, ocular injury (postoperative vision loss), hypotension<br/>B. Only hypotension<br/>C. Only nerve injury<br/>D. Only airway problems</p> | <p><b>Ans: A</b></p> |

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| <p><b>Q44. What monitoring is required for major spine surgeries like scoliosis?</b></p> <p>A. Monitoring of spinal cord function by evoked responses like SSEP, MEP and EMG<br/> B. Only routine monitoring<br/> C. Only blood pressure<br/> D. Only pulse oximetry</p>                     | <p><b>Ans: A</b></p> |
| <p><b>Q45. Is wake up test done in current day practice for spinal cord monitoring?</b></p> <p>A. No, it is not performed in current day practice<br/> B. Yes, it is the standard<br/> C. Only in pediatric patients<br/> D. Only in cervical spine surgery</p>                              | <p><b>Ans: A</b></p> |
| <p><b>Q46. What should be the concentration of inhalational agents during evoked response monitoring?</b></p> <p>A. Half minimum alveolar concentration (MAC) as inhalational agents can depress evoked responses<br/> B. Full MAC<br/> C. 1.5 MAC<br/> D. No inhalational agents at all</p> | <p><b>Ans: A</b></p> |
| <p><b>Q47. Do IV agents preserve evoked responses?</b></p> <p>A. Only opioids preserve them<br/> B. No, they suppress them completely<br/> C. Only ketamine preserves them<br/> D. Yes, IV agents preserve evoked responses except a little inhibition seen with Propofol</p>                | <p><b>Ans: D</b></p> |
| <p><b>Q48. Can muscle relaxants be used during MEP monitoring?</b></p> <p>A. Only non-depolarizing agents<br/> B. Only succinylcholine<br/> C. Yes, all muscle relaxants can be used<br/> D. No, muscle relaxants cannot be used during MEP monitoring</p>                                   | <p><b>Ans: D</b></p> |
| <p><b>Q49. What can happen at the time of tourniquet deflation that is most dangerous?</b></p> <p>A. Only hypothermia<br/> B. Cardiac arrest<br/> C. Only pain<br/> D. Only hypotension</p>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q50. What coagulation changes are seen in fat embolism?</b></p> <p>A. Decreased clotting time<br/> B. Only increased platelets<br/> C. Increased clotting time and decreased platelets<br/> D. No coagulation changes</p>                                                              | <p><b>Ans: C</b></p> |

# Answer Key & Explanations

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**Q1. Answer: C — Lengthy procedures, increased blood loss, positioning, tourniquet-related problems, fat embolism and DVT**

Orthopedic anesthesia poses special problems due to lengthy procedures, increase blood loss, positioning, tourniquet-related problems, complications like fat embolism and deep vein thrombosis.

*Topic: General*

**Q2. Answer: B — Hypertension, increased cardiac output and increased pulmonary artery pressure**

During inflation: Hypertension, increased cardiac output and increased pulmonary artery pressure. This is due to increased blood volume associated with exsanguination.

*Topic: Tourniquet Problems*

**Q3. Answer: A — Local tissue ischemia leads to acidosis (mitochondrial pO<sub>2</sub> becomes zero)**

Metabolic changes: Local tissue ischemia leads to acidosis (mitochondrial pO<sub>2</sub> becomes zero within 6 minutes of tourniquet inflation).

*Topic: Tourniquet Problems*

**Q4. Answer: A — Only by deflation (not relieved by IV analgesics)**

Pain is due to cellular acidosis, therefore may not be relieved by intravenous analgesics. It is relieved only by deflation.

*Topic: Tourniquet Problems*

**Q5. Answer: B — Sudden release of accumulated metabolites like thromboxane and substance P**

Hypotension because of sudden release of accumulated metabolites like thromboxane and substance P.

*Topic: Tourniquet Problems*

**Q6. Answer: A — Due to heat loss as acid metabolites are vasodilators**

Hypothermia due to heat loss as acid metabolites are vasodilators.

*Topic: Tourniquet Problems*

**Q7. Answer: D — Transient increase (CO<sub>2</sub> accumulated in limb is released)**

Increase in end tidal CO<sub>2</sub> (which is accumulated in limb).

*Topic: Tourniquet Problems*

**Q8. Answer: D — Deflation**

Deflation is the most crucial period. Cardiac arrest may occur at the time of deflation due to hypotension and release of acid metabolites.

*Topic: Tourniquet Problems*

**Q9. Answer: B — Not more than 2 hours for both upper and lower limbs**

Pressure in tourniquet should not exceed more than 2 hours for both upper and lower limbs.

*Topic: Tourniquet Problems*

**Q10. Answer: D — At least 30 minutes**

If re-inflation is needed, then it should be deflated for at least 30 minutes before re-inflating.

*Topic: Tourniquet Problems*



**Q11. Answer: B — Not more than 100 mm Hg above systolic (maximum 250 mm Hg)**

Pressure in tourniquet should not exceed more than 100 mm Hg above systolic (maximum-250 mm Hg).

*Topic: Tourniquet Problems*

**Q12. Answer: C — Neuropraxia**

Post-tourniquet: Neuropraxia may occur if tourniquet pressure is high or time is prolonged.

*Topic: Tourniquet Problems*

**Q13. Answer: C — After fracture of long bones and pelvis**

Usually seen after fracture of long bones and pelvis. Incidence of significant embolism after long bone and pelvic fractures may be as high as 15%.

*Topic: Fat Embolism*

**Q14. Answer: C — Within 24-72 hours**

Fat embolism classically presents within 24-72 hours of fracture.

*Topic: Fat Embolism*

**Q15. Answer: B — Higher in nailing compared to plating**

Due to rimming the incidence of fat embolism syndrome (FES), is higher in nailing as compared to plating.

*Topic: Fat Embolism*

**Q16. Answer: C — Dyspnea, confusion and petechial hemorrhages**

The triad of fat embolism syndrome is dyspnea, confusion and petechial hemorrhages.

*Topic: Fat Embolism*

**Q17. Answer: D — Petechial rash**

Petechial rash is pathognomonic of FES.

*Topic: Fat Embolism*

**Q18. Answer: B — Dyspnea, chest discomfort, tachycardia, hypotension, mental clouding, restlessness, seizures, delirium, petechial hemorrhage on upper chest, axilla and conjunctiva, hypoxia and finally coma and death**

Signs include dyspnea, chest discomfort, tachycardia, hypotension, mental clouding, restlessness, seizures, delirium, petechial hemorrhage on upper chest, axilla and conjunctiva, hypoxia and finally coma and death.

*Topic: Fat Embolism*

**Q19. Answer: C — Helpful in diagnosis but may not necessarily indicate progression to FES**

The fat globules in urine are helpful in diagnosis, but may not necessarily indicate the progression to fat embolism syndrome.

*Topic: Fat Embolism*

**Q20. Answer: C — Diffuse interstitial infiltrates**

X-ray chest shows diffuse interstitial infiltrates.

*Topic: Fat Embolism*

**Q21. Answer: D — Fall in end tidal CO<sub>2</sub> (ETCO<sub>2</sub> may become zero if embolus is large enough to block main pulmonary artery)**

Capnography shows fall in end tidal CO<sub>2</sub> (ETCO<sub>2</sub> may become zero if embolus is large enough to block the main pulmonary artery).

Topic: Fat Embolism

**Q22. Answer: B — Right ventricular strain**

ECG shows the right ventricular strain.

Topic: Fat Embolism

**Q23. Answer: A — Increased (but has no relation to severity of disease)**

Serum lipase is increased (but has no relation to the severity of disease).

Topic: Fat Embolism

**Q24. Answer: C — Early stabilization of fracture (may prevent fat embolism)**

Stabilization of fracture: Early stabilization of fracture may prevent fat embolism.

Topic: Fat Embolism

**Q25. Answer: D — Intermittent positive pressure ventilation (IPPV) with positive end expiratory pressure (PEEP)**

If hypoxia persists on oxygen therapy consider intermittent positive pressure ventilation (IPPV) with positive end expiratory pressure (PEEP).

Topic: Fat Embolism

**Q26. Answer: B — Fat emboli cause endothelial injury which can precipitate ARDS and cerebral edema**

Fat emboli released from long bone causes endothelial injury in the lungs and brain, which can precipitate acute respiratory distress syndrome (ARDS) and cerebral edema respectively.

Topic: Fat Embolism

**Q27. Answer: B — No, they are now considered as outdated treatment modalities**

Corticosteroids, heparin (which stimulates circulating lipase that breaks neutral fat), ethanol and hypertonic saline are now considered as outdated treatment modalities.

Topic: Fat Embolism

**Q28. Answer: C — 1-3%**

The incidence of fatal PE after hip surgery can be as high as 1-3%.

Topic: DVT and PE

**Q29. Answer: D — Hip surgery patients particularly after prolonged immobilization**

The incidence of deep vein thrombosis (DVT) and pulmonary embolism (PE) is highest in orthopedic patients particularly after hip surgery due to prolonged immobilization.

Topic: DVT and PE

**Q30. Answer: D — 12 hours or more in preoperative period and restart 12 hours after surgery**

Many centers start low molecular weight heparin before 12 hours or more in preoperative period and restart 12 hours after surgery.

Topic: DVT and PE

**Q31. Answer: A — Stroke or even brain death due to ischemia**

There have been case reports of stroke or even brain death in shoulder surgeries done in beach chair position due to ischemia.

*Topic: Positioning*

**Q32. Answer: B — At the level of external auditory meatus to denote cerebral perfusion pressure**

Invasive blood pressure is highly recommended for shoulder surgeries done in the beach chair position with transducer zeroed at the level of external auditory meatus to denote cerebral perfusion pressure.

*Topic: Positioning*

**Q33. Answer: A — Hypoxia, profound hypotension or even cardiac arrest occurring during cemented prosthesis in total hip replacement**

Cement implantation syndrome (CIS) occurs more commonly in total hip replacement where cemented prosthesis is used. There occurs hypoxia, profound hypotension or even cardiac arrest.

*Topic: Cement Implantation*

**Q34. Answer: D — Embolization of bone marrow debris, microemboli or fat due to increased intramedullary pressure or reaction to cement (methyl methacrylate)**

It may occur because of embolization of bone marrow debris, microemboli or fat due to increased intramedullary pressure created by cement or due to reaction to cement (methyl methacrylate).

*Topic: Cement Implantation*

**Q35. Answer: A — Oxygenation, adequate hydration and adrenaline for severe reactions**

The treatment includes Oxygenation, adequate hydration and adrenaline for severe reactions.

*Topic: Cement Implantation*

**Q36. Answer: A — Due to decreased incidence of thromboembolism, decreased blood loss, fewer respiratory complications and better control of pain**

Regional anesthesia is always preferred over general anesthesia due to decrease incidence of thromboembolism, decrease blood loss, respiratory complications and better control of pain.

*Topic: Choice of Anesthesia*

**Q37. Answer: B — Brachial plexus block**

Upper limb surgeries can be performed under brachial plexus block.

*Topic: Choice of Anesthesia*

**Q38. Answer: B — Bier's block (IV regional anesthesia)**

Small procedures lasting less than 30-45 minutes can be performed under Bier's block.

*Topic: Choice of Anesthesia*

**Q39. Answer: C — Spinal/epidural or combined spinal epidural (most preferred technique)**

Lower limb surgeries are performed under spinal/epidural or combined spinal epidural (most preferred technique).

*Topic: Choice of Anesthesia*

**Q40. Answer: D — For very prolonged surgeries of upper limb, shoulder or spine surgeries or when regional is not possible**

General anesthesia is required either for very prolonged surgeries of upper limb, shoulder or spine surgeries or when regional is not possible.

Topic: Choice of Anesthesia

**Q41. Answer: C — Continuous adductor canal block for knee, fascia iliaca/femoral nerve block for hip surgeries, brachial plexus block for upper limb**

Continuous adductor canal block for knee replacement, fascia iliaca/femoral nerve block for hip surgeries, brachial plexus block for upper limb surgeries are frequently utilized for postoperative analgesia.

Topic: Pain Management

**Q42. Answer: D — Prone position under general anesthesia**

Spine surgery is performed under general anesthesia in prone position.

Topic: Spine Surgery

**Q43. Answer: A — Endotracheal tube dislodgement, airway edema, injury to brachial plexus and peroneal nerve, injury to genitalia and breast, ocular injury (postoperative vision loss), hypotension**

Complications related to prone position include endotracheal tube dislodgement, airway edema, injury to brachial plexus and peroneal nerve, injury to genitalia and breast, ocular injury (post-operative vision loss), hypotension.

Topic: Spine Surgery

**Q44. Answer: A — Monitoring of spinal cord function by evoked responses like SSEP, MEP and EMG**

Monitoring of spinal cord function is required for major spine surgeries like scoliosis. The function of the spinal cord is monitored by evoked responses like somatosensory evoked potential (SSEP), motor evoked potential (MEP) and electromyogram (EMG).

Topic: Spine Surgery

**Q45. Answer: A — No, it is not performed in current day practice**

Wake up test done in the past is not performed in current day practice.

Topic: Spine Surgery

**Q46. Answer: A — Half minimum alveolar concentration (MAC) as inhalational agents can depress evoked responses**

As inhalational agents can depress these evoked responses their concentration should be kept at half minimum alveolar concentration (MAC).

Topic: Spine Surgery

**Q47. Answer: D — Yes, IV agents preserve evoked responses except a little inhibition seen with Propofol**

IV agents preserve evoked responses except a little inhibition seen with Propofol.

Topic: Spine Surgery

**Q48. Answer: D — No, muscle relaxants cannot be used during MEP monitoring**

Muscle relaxants cannot be used during MEP monitoring.

Topic: Spine Surgery

**Q49. Answer: B — Cardiac arrest**

Cardiac arrest can occur at the time of tourniquet deflation.

*Topic: Tourniquet Problems*

**Q50. Answer: C — Increased clotting time and decreased platelets**

Coagulation abnormalities like increased clotting time and decreased platelets are seen in fat embolism.

*Topic: Fat Embolism*

# Anesthesia at Remote Locations MCQ – Chapter 37

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 37)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

**Q1. What chronic condition do patients at high altitudes (low barometric pressure) have?**

- A. Chronic hypercapnia
- B. Chronic alkalosis
- C. Chronic hypoxia
- D. Chronic hyperoxia

**Ans: C**

**Q2. What complications can chronic hypoxia at high altitude lead to?**

- A. Only respiratory alkalosis
- B. Only polycythemia
- C. Only pulmonary hypertension
- D. Polycythemia, pulmonary hypertension, decreased pCO<sub>2</sub> and lower bicarbonate concentration

**Ans: D**

**Q3. What happens to the potency of anesthetic gases at high altitudes?**

- A. Decreases (proportional to partial pressure which decreases at high altitude)
- B. Remains same
- C. Becomes unpredictable
- D. Increases

**Ans: A**

**Q4. At what altitude is the potency of nitrous oxide reduced to 50%?**

- A. More than 3,300 meters
- B. 2,000 meters
- C. 5,000 meters
- D. 1,500 meters

**Ans: A**

**Q5. At what altitude should nitrous oxide not be used?**

- A. More than 3,000 meters
- B. More than 2,000 meters
- C. More than 4,000 meters
- D. More than 1,000 meters

**Ans: B**

**Q6. What happens to the density of gases at high altitudes and how does it affect the rotameter?**

- A. Density decreases, rotameter under reads the actual flow rate
- B. Density increases, rotameter over-reads
- C. Density decreases, rotameter over-reads
- D. No change

**Ans: A**

**Q7. By how much can the rotameter error be at high altitudes?**

- A. Up to 1 liter/minute
- B. Up to 4 liters/minute
- C. Up to 2 liters/minute
- D. Up to 6 liters/minute

**Ans: B**

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q8. At altered barometric pressure, do all vaporizers deliver fixed partial pressure of vapor?</b></p> <p>A. Only sevoflurane vaporizer<br/>B. All except desflurane deliver fixed partial pressure<br/>C. Yes, all vaporizers<br/>D. None of them</p> | <p><b>Ans: B</b></p> |
| <p><b>Q9. Which vaporizer needs upward adjustment at high altitude?</b></p> <p>A. Sevoflurane<br/>B. Desflurane<br/>C. Isoflurane<br/>D. Halothane</p>                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q10. Why do Venturi type devices deliver higher oxygen at high altitude?</b></p> <p>A. Due to increased humidity<br/>B. Due to higher flow rates<br/>C. Because of decreased density and resistance<br/>D. Due to increased pressure</p>               | <p><b>Ans: C</b></p> |
| <p><b>Q11. Why is blood loss during surgery more at high altitude?</b></p> <p>A. Due to coagulopathy<br/>B. Due to high venous pressure and blood volume<br/>C. Due to polycythemia only<br/>D. Due to hypothermia</p>                                       | <p><b>Ans: B</b></p> |
| <p><b>Q12. Why are chances of postoperative hypoxemia high at high altitude?</b></p> <p>A. Due to low ambient partial pressure of oxygen<br/>B. Due to metabolic alkalosis<br/>C. Due to anemia<br/>D. Due to hyperventilation</p>                           | <p><b>Ans: A</b></p> |
| <p><b>Q13. What is the incidence of postspinal headache at high altitude under regional anesthesia?</b></p> <p>A. Very rare<br/>B. High (due to increased CSF pressure)<br/>C. Low<br/>D. Same as sea level</p>                                              | <p><b>Ans: B</b></p> |
| <p><b>Q14. In hyperbaric chamber, what can happen to tissues with nitrous oxide?</b></p> <p>A. No effect<br/>B. Tissues become dehydrated<br/>C. Tissues can become supersaturated causing significant diffusion hypoxia<br/>D. Tissues become acidotic</p>  | <p><b>Ans: C</b></p> |



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| <p><b>Q15. Why should nitrous oxide not be used in helium atmosphere?</b></p> <ul style="list-style-type: none"> <li>A. It causes explosion</li> <li>B. It reacts with helium</li> <li>C. The effect of supersaturation is more evident in helium atmosphere</li> <li>D. It becomes toxic</li> </ul>                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q16. Why does the rotameter falsely give high values in hyperbaric chamber?</b></p> <ul style="list-style-type: none"> <li>A. Due to humidity</li> <li>B. Due to temperature changes</li> <li>C. Due to low pressure</li> <li>D. Due to increased gas density</li> </ul>                                                                               | <p><b>Ans: D</b></p> |
| <p><b>Q17. What should the endotracheal tube cuff be filled with in hyperbaric chamber?</b></p> <ul style="list-style-type: none"> <li>A. Helium</li> <li>B. Water or saline</li> <li>C. Air</li> <li>D. Nitrous oxide</li> </ul>                                                                                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q18. What happens to the vaporizer output in hyperbaric chamber?</b></p> <ul style="list-style-type: none"> <li>A. Remains same</li> <li>B. Becomes unpredictable</li> <li>C. Becomes less</li> <li>D. Increases</li> </ul>                                                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q19. Is regional anesthesia safe in hyperbaric chamber?</b></p> <ul style="list-style-type: none"> <li>A. Yes, regional anesthesia is safe</li> <li>B. No, it is contraindicated</li> <li>C. Only epidural is safe</li> <li>D. Only spinal is safe</li> </ul>                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q20. What does anesthesia outside operating theater mean?</b></p> <ul style="list-style-type: none"> <li>A. Only emergency procedures</li> <li>B. Only dental procedures</li> <li>C. Field anesthesia only</li> <li>D. All procedures done outside the operation theater (radiology, MRI, CT scan, gastroscopy, cardiology, neurology etc.)</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q21. What are the challenges of procedure rooms outside OT?</b></p> <ul style="list-style-type: none"> <li>A. Only staffing issues</li> <li>B. Small and crowded, making placement of anesthesia machine, drug trolley, emergency cart and patient access difficult</li> <li>C. Only equipment issues</li> <li>D. Only space issues</li> </ul>         | <p><b>Ans: B</b></p> |

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| <p><b>Q22. What equipment-related challenge exists in nonoperating room anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Only monitor issues</li> <li>B. Too much equipment</li> <li>C. Equipment is too modern</li> <li>D. Nonavailability of adequate equipment, monitors, technical support and help in case of emergency</li> </ul>                                               | <p><b>Ans: D</b></p> |
| <p><b>Q23. In MRI suites, why does the anesthetist stand far away from the patient?</b></p> <ul style="list-style-type: none"> <li>A. Due to radiation</li> <li>B. Due to noise</li> <li>C. Due to temperature</li> <li>D. Due to the magnetic field (MRI suites require special non-ferrous equipment)</li> </ul>                                                                                 | <p><b>Ans: D</b></p> |
| <p><b>Q24. Why do medical proceduralists sometimes view anesthesia demand as obstructive?</b></p> <ul style="list-style-type: none"> <li>A. Unlike surgeons, many medical proceduralists do not understand the intricacies and importance of anesthesia</li> <li>B. Due to space constraints</li> <li>C. Due to scheduling conflicts</li> <li>D. Due to cost</li> </ul>                            | <p><b>Ans: A</b></p> |
| <p><b>Q25. How do patients undergoing medical procedures compare to surgical patients in terms of health status?</b></p> <ul style="list-style-type: none"> <li>A. They are the same</li> <li>B. They are always well-prepared</li> <li>C. They are healthier</li> <li>D. They are often even sicker than surgical patients and seldom evaluated and medically optimized preoperatively</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q26. What is the risk associated with radiology procedures requiring anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Risk of bleeding</li> <li>B. Risk of exposure to radiation</li> <li>C. Risk of infection</li> <li>D. Risk of hypothermia</li> </ul>                                                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q27. What complication can occur in radiology procedures?</b></p> <ul style="list-style-type: none"> <li>A. Contrast-related anaphylaxis</li> <li>B. Only bleeding</li> <li>C. Only infection</li> <li>D. Only pain</li> </ul>                                                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q28. Why may patients for nonoperating room procedures not be adequately fasting?</b></p> <ul style="list-style-type: none"> <li>A. They refuse to fast</li> <li>B. Fasting is not required</li> <li>C. They are always admitted</li> <li>D. Cases often come as unbooked, disturbing the schedule, and patients may not be adequately fasting</li> </ul>                                    | <p><b>Ans: D</b></p> |

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| <p><b>Q29. What is the full fasting recommendation for nonoperating room anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. 8 hours for everything</li> <li>B. 4 hours for all</li> <li>C. 6 hours for light food, 8 hours for full meal, and 2 hours for water</li> <li>D. No fasting required</li> </ul>                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q30. Why is inadequate sedation a concern in nonoperating room procedures?</b></p> <ul style="list-style-type: none"> <li>A. It causes awareness</li> <li>B. It can lead to conversion to general anesthesia, so full fasting is mandatory</li> <li>C. It causes pain only</li> <li>D. It is never a concern</li> </ul>                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q31. What monitoring is highly desirable for spontaneously breathing patients in nonoperating room anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Only ECG</li> <li>B. Capnography (nasal capnography for spontaneously breathing patients) - can pick up life-threatening complications very early</li> <li>C. Only pulse oximetry</li> <li>D. Only blood pressure</li> </ul>                 | <p><b>Ans: B</b></p> |
| <p><b>Q32. What anesthesia technique is most often used for nonoperating room procedures?</b></p> <ul style="list-style-type: none"> <li>A. Either sedation with midazolam or total intravenous anesthesia (TIVA) with Propofol + Remifentanyl</li> <li>B. Spinal anesthesia</li> <li>C. Only local anesthesia</li> <li>D. General anesthesia with intubation</li> </ul>                                            | <p><b>Ans: A</b></p> |
| <p><b>Q33. In India, what is used instead of Remifentanyl for TIVA in nonoperating room procedures?</b></p> <ul style="list-style-type: none"> <li>A. Sufentanyl</li> <li>B. Alfentanyl</li> <li>C. Morphine</li> <li>D. Fentanyl (due to nonavailability of remifentanyl)</li> </ul>                                                                                                                               | <p><b>Ans: D</b></p> |
| <p><b>Q34. What must all equipment in MRI suites be made of?</b></p> <ul style="list-style-type: none"> <li>A. Stainless steel</li> <li>B. Copper</li> <li>C. Aluminum (MRI compatible, nonferrous)</li> <li>D. Plastic only</li> </ul>                                                                                                                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q35. Why is nonoperating room anesthesia considered very challenging?</b></p> <ul style="list-style-type: none"> <li>A. Due to long procedures only</li> <li>B. Due to patient refusal</li> <li>C. Due to small crowded space, inadequate equipment, inadequate preoperative evaluation, and inadequate knowledge of anesthesia intricacies by proceduralists</li> <li>D. Due to complex surgeries</li> </ul> | <p><b>Ans: C</b></p> |

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| <p><b>Q36. Is nonoperating room anesthesia preferred by most anesthetists?</b></p> <ul style="list-style-type: none"> <li>A. Yes, most preferred</li> <li>B. Only preferred by junior anesthetists</li> <li>C. No, it is the least preferred by majority of anesthetists</li> <li>D. Equally preferred</li> </ul>                                                      | <p><b>Ans: C</b></p> |
| <p><b>Q37. What is the compensation for anesthetists working outside OT compared to surgical cases?</b></p> <ul style="list-style-type: none"> <li>A. No compensation</li> <li>B. Same compensation</li> <li>C. Almost always undercompensated financially compared to surgical cases</li> <li>D. Better compensation</li> </ul>                                       | <p><b>Ans: C</b></p> |
| <p><b>Q38. Why is pCO<sub>2</sub> decreased at high altitude?</b></p> <ul style="list-style-type: none"> <li>A. Due to increased CO<sub>2</sub> production</li> <li>B. Due to renal compensation</li> <li>C. Due to decreased metabolism</li> <li>D. Due to hyperventilation (compensatory response to chronic hypoxia)</li> </ul>                                     | <p><b>Ans: D</b></p> |
| <p><b>Q39. Why is bicarbonate concentration lower at high altitude?</b></p> <ul style="list-style-type: none"> <li>A. Due to dehydration</li> <li>B. Due to metabolic acidosis</li> <li>C. As a compensatory mechanism to low pCO<sub>2</sub> (renal compensation for respiratory alkalosis)</li> <li>D. Due to increased renal excretion of CO<sub>2</sub></li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q40. At what altitude is it recommended to not use nitrous oxide?</b></p> <ul style="list-style-type: none"> <li>A. More than 4,000 meters</li> <li>B. More than 2,000 meters</li> <li>C. More than 1,000 meters</li> <li>D. More than 500 meters</li> </ul>                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q41. What is the effect of high altitude on pulmonary circulation?</b></p> <ul style="list-style-type: none"> <li>A. Pulmonary hypotension</li> <li>B. Pulmonary hypertension</li> <li>C. Pulmonary vasodilation</li> <li>D. No effect</li> </ul>                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q42. What happens to all vaporizers except desflurane at altered barometric pressure?</b></p> <ul style="list-style-type: none"> <li>A. They deliver fixed partial pressure of vapor so settings need not be changed</li> <li>B. They malfunction</li> <li>C. They deliver higher concentration</li> <li>D. They need recalibration</li> </ul>                   | <p><b>Ans: A</b></p> |

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| <p><b>Q43. What is the fasting duration for water before nonoperating room anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. 6 hours</li> <li>B. 1 hour</li> <li>C. 4 hours</li> <li>D. 2 hours</li> </ul>                                                                                                                                           | <p><b>Ans: D</b></p> |
| <p><b>Q44. What is the fasting duration for a full meal before nonoperating room anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. 8 hours</li> <li>B. 10 hours</li> <li>C. 6 hours</li> <li>D. 4 hours</li> </ul>                                                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q45. What is the fasting duration for light food before nonoperating room anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. 2 hours</li> <li>B. 6 hours</li> <li>C. 8 hours</li> <li>D. 4 hours</li> </ul>                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q46. Can capnography detect life-threatening complications early in nonoperating room anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. Yes, capnography can pick up life-threatening complications very early</li> <li>B. Only for cardiac complications</li> <li>C. Only in intubated patients</li> <li>D. No, it has limited use</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q47. Why must oxygen cylinders in MRI suites be made of aluminum?</b></p> <ul style="list-style-type: none"> <li>A. Aluminum stores more oxygen</li> <li>B. They must be MRI compatible (nonferrous) - ferrous materials are attracted by the magnetic field</li> <li>C. Aluminum is lighter</li> <li>D. Aluminum is cheaper</li> </ul>                    | <p><b>Ans: B</b></p> |
| <p><b>Q48. What type of polycythemia develops at high altitude?</b></p> <ul style="list-style-type: none"> <li>A. Polycythemia vera</li> <li>B. Relative polycythemia</li> <li>C. Secondary polycythemia (due to chronic hypoxia stimulating erythropoietin)</li> <li>D. Primary polycythemia</li> </ul>                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q49. What is the main reason nonoperating room anesthesia cases are challenging regarding scheduling?</b></p> <ul style="list-style-type: none"> <li>A. Too few cases</li> <li>B. Complex procedures</li> <li>C. Too many cases</li> <li>D. Majority of cases come as unbooked, disturbing the schedule</li> </ul>                                         | <p><b>Ans: D</b></p> |

**Q50. Are full fasting recommendations required for non-operating room anesthesia?**

- A. No, shorter fasting is acceptable
- B. Only for pediatric patients
- C. Only for GA
- D. Yes, full fasting recommendations are required

**Ans: D**

# Answer Key & Explanations

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## **Q1. Answer: C — Chronic hypoxia**

The patients at high altitudes (low barometric pressure) have chronic hypoxia which can lead to polycythemia, pulmonary hypertension, decreased pCO<sub>2</sub> and lower bicarbonate concentration.

*Topic: High Altitude*

## **Q2. Answer: D — Polycythemia, pulmonary hypertension, decreased pCO<sub>2</sub> and lower bicarbonate concentration**

Chronic hypoxia can lead to polycythemia, pulmonary hypertension, decreased pCO<sub>2</sub> (due to hyperventilation) and lower bicarbonate concentration (as a compensatory mechanism to low pCO<sub>2</sub>).

*Topic: High Altitude*

## **Q3. Answer: A — Decreases (proportional to partial pressure which decreases at high altitude)**

Potency of anesthetic gases is proportional to their partial pressure. At high altitudes where barometric pressure is decreased, the potency also decreases.

*Topic: High Altitude*

## **Q4. Answer: A — More than 3,300 meters**

The potency of nitrous oxide is reduced to 50% at altitudes >3,300 meters.

*Topic: High Altitude*

## **Q5. Answer: B — More than 2,000 meters**

It is recommended that nitrous oxide should not be used at altitudes more than 2,000 meters.

*Topic: High Altitude*

## **Q6. Answer: A — Density decreases, rotameter under reads the actual flow rate**

Density of gases decreases at high altitudes, therefore, rotameter under reads the actual flow rate and this error may be up to 4 liters/minute.

*Topic: High Altitude*

## **Q7. Answer: B — Up to 4 liters/minute**

The rotameter under reads the actual flow rate and this error may be up to 4 liters/minute.

*Topic: High Altitude*

## **Q8. Answer: B — All except desflurane deliver fixed partial pressure**

At altered barometric pressure, all vaporizers (except desflurane) deliver fixed partial pressure of vapor. Therefore, vaporizer settings need not be changed at low or high barometric pressure except for desflurane.

*Topic: High Altitude*

## **Q9. Answer: B — Desflurane**

Vaporizer settings need not be changed at low or high barometric pressure except for desflurane, which needs an upward adjustment at high altitude.

*Topic: High Altitude*

**Q10. Answer: C — Because of decreased density and resistance**

Venturi type devices deliver higher oxygen (because of decreased density and resistance).

Topic: High Altitude

**Q11. Answer: B — Due to high venous pressure and blood volume**

Blood loss during surgery is more due to high venous pressure and blood volume.

Topic: High Altitude

**Q12. Answer: A — Due to low ambient partial pressure of oxygen**

In the postoperative period, chances of hypoxemia are high due to low ambient partial pressure of oxygen.

Topic: High Altitude

**Q13. Answer: B — High (due to increased CSF pressure)**

The incidence of postspinal headache is high due to increased CSF pressure.

Topic: High Altitude

**Q14. Answer: C — Tissues can become supersaturated causing significant diffusion hypoxia**

Tissues can become supersaturated with nitrous oxide causing significant diffusion hypoxia.

Topic: Hyperbaric Chamber

**Q15. Answer: C — The effect of supersaturation is more evident in helium atmosphere**

The effect of supersaturation is more evident in helium atmosphere, therefore nitrous oxide should not be used in helium atmosphere.

Topic: Hyperbaric Chamber

**Q16. Answer: D — Due to increased gas density**

Rotameter will falsely give high values due to increased gas density.

Topic: Hyperbaric Chamber

**Q17. Answer: B — Water or saline**

Endotracheal tube cuff should be filled with water or saline.

Topic: Hyperbaric Chamber

**Q18. Answer: C — Becomes less**

The vaporizer output becomes less.

Topic: Hyperbaric Chamber

**Q19. Answer: A — Yes, regional anesthesia is safe**

Regional anesthesia is safe.

Topic: Hyperbaric Chamber

**Q20. Answer: D — All procedures done outside the operation theater (radiology, MRI, CT scan, gastroscopy, cardiology, neurology etc.)**

Nonoperating room anesthesia means all procedures done outside the operation theater. For example, anesthesia services utilized in radiology (MRI, CT scan), gastroscopy suites, intervention cardiology and neurology, etc.

Topic: Outside OT



**Q21. Answer: B — Small and crowded, making placement of anesthesia machine, drug trolley, emergency cart and patient access difficult**

Procedure rooms are often small and crowded making placement of anesthesia machine, drug trolley, emergency cart and patient access difficult.

Topic: Outside OT

**Q22. Answer: D — Nonavailability of adequate equipment, monitors, technical support and help in case of emergency**

Nonavailability of adequate equipment, monitors, technical support and help in case of emergency.

Topic: Outside OT

**Q23. Answer: D — Due to the magnetic field (MRI suites require special non-ferrous equipment)**

Many times, like MRI suites, anesthetist stands far away from the patient.

Topic: Outside OT

**Q24. Answer: A — Unlike surgeons, many medical proceduralists do not understand the intricacies and importance of anesthesia**

Unlike surgeons, many of the medical proceduralists do not understand the intricacies and importance of anesthesia. Any demand by anesthetist is often considered as obstructive by medical specialist.

Topic: Outside OT

**Q25. Answer: D — They are often even sicker than surgical patients and seldom evaluated and medically optimized preoperatively**

Majority of the patients undergoing medical procedures are even sicker than surgical patients. In spite of that, they are seldom evaluated and medically optimized in preoperative period.

Topic: Outside OT

**Q26. Answer: B — Risk of exposure to radiation**

Risk of exposure to radiation.

Topic: Outside OT

**Q27. Answer: A — Contrast-related anaphylaxis**

Contrast-related anaphylaxis can occur in radiology procedures.

Topic: Outside OT

**Q28. Answer: D — Cases often come as unbooked, disturbing the schedule, and patients may not be adequately fasting**

Majority of the times, cases come as unbooked and so, disturbing the schedule. Patients may not be adequately fasting.

Topic: Outside OT

**Q29. Answer: C — 6 hours for light food, 8 hours for full meal, and 2 hours for water**

Full fasting recommendation, i.e. 6 hours for light food, 8 hours for a full meal, and 2 hours for water is mandatory.

Topic: Outside OT

**Q30. Answer: B — It can lead to conversion to general anesthesia, so full fasting is mandatory**

Inadequate sedation leading to conversion to general anesthesia is very common, therefore, full fasting recommendation is mandatory.

Topic: Outside OT

**Q31. Answer: B — Capnography (nasal capnography for spontaneously breathing patients) - can pick up life-threatening complications very early**

Other than pulse oximetry, ECG and BP, capnography (nasal capnography for spontaneously breathing patients) is highly desirable; capnography can pick up life-threatening complications very early.

Topic: Outside OT

**Q32. Answer: A — Either sedation with midazolam or total intravenous anesthesia (TIVA) with Propofol + Remifentanyl**

Most often either sedation with midazolam or total intravenous anesthesia (TIVA) with Propofol + Remifentanyl is utilized for nonoperating room procedures.

Topic: Outside OT

**Q33. Answer: D — Fentanyl (due to nonavailability of remifentanyl)**

In India fentanyl, due to nonavailability of remifentanyl, is utilized for nonoperating room procedures.

Topic: Outside OT

**Q34. Answer: C — Aluminum (MRI compatible, nonferrous)**

For MRI suites, everything including anesthesia machine, monitors, oxygen cylinders must be MRI compatible, i.e. nonferrous. The equipments used in MRI suites are made up of aluminum.

Topic: Outside OT

**Q35. Answer: C — Due to small crowded space, inadequate equipment, inadequate preoperative evaluation, and inadequate knowledge of anesthesia intricacies by proceduralists**

Small and crowded space, nonavailability of adequate equipment, monitors, technical support and help in case of emergency, inadequate preoperative evaluation of patients and inadequate knowledge of anesthesia intricacies by medical proceduralists make nonoperating room anesthesia very challenging.

Topic: Outside OT

**Q36. Answer: C — No, it is the least preferred by majority of anesthetists**

Nonoperating room anesthesia is very challenging and least preferred by majority of anesthetists.

Topic: Outside OT

**Q37. Answer: C — Almost always undercompensated financially compared to surgical cases**

In spite of the high risk and adverse working circumstances, anesthetists are almost always undercompensated financially as compared to surgical cases.

Topic: Outside OT

**Q38. Answer: D — Due to hyperventilation (compensatory response to chronic hypoxia)**

Decreased pCO<sub>2</sub> is due to hyperventilation which is a compensatory response to chronic hypoxia at high altitude.

Topic: High Altitude

**Q39. Answer: C — As a compensatory mechanism to low pCO<sub>2</sub> (renal compensation for respiratory alkalosis)**

Lower bicarbonate concentration is a compensatory mechanism to low pCO<sub>2</sub>.

Topic: High Altitude

**Q40. Answer: B — More than 2,000 meters**

It is recommended that nitrous oxide should not be used at altitudes more than 2,000 meters.

Topic: High Altitude

**Q41. Answer: B — Pulmonary hypertension**

Chronic hypoxia at high altitude can lead to pulmonary hypertension.

Topic: High Altitude

**Q42. Answer: A — They deliver fixed partial pressure of vapor so settings need not be changed**

At altered barometric pressure, all vaporizers (except desflurane) deliver fixed partial pressure of vapor. Therefore, vaporizer settings need not be changed.

Topic: High Altitude

**Q43. Answer: D — 2 hours**

Full fasting recommendation: 6 hours for light food, 8 hours for a full meal, and 2 hours for water.

Topic: Outside OT

**Q44. Answer: A — 8 hours**

Full fasting recommendation: 8 hours for a full meal.

Topic: Outside OT

**Q45. Answer: B — 6 hours**

Full fasting recommendation: 6 hours for light food.

Topic: Outside OT

**Q46. Answer: A — Yes, capnography can pick up life-threatening complications very early**

Capnography can pick up life-threatening complications very early. It is highly desirable for nonoperating room anesthesia.

Topic: Outside OT

**Q47. Answer: B — They must be MRI compatible (nonferrous) - ferrous materials are attracted by the magnetic field**

For MRI suites, everything including oxygen cylinders must be MRI compatible, i.e. nonferrous. The equipments used in MRI suites are made up of aluminum.

Topic: Outside OT

**Q48. Answer: C — Secondary polycythemia (due to chronic hypoxia stimulating erythropoietin)**

Chronic hypoxia at high altitude leads to polycythemia as a compensatory mechanism to increase oxygen carrying capacity.

Topic: High Altitude

**Q49. Answer: D — Majority of cases come as unbooked, disturbing the schedule**

Majority of the times, cases come as unbooked and so, disturbing the schedule.

*Topic: Outside OT*

**Q50. Answer: D — Yes, full fasting recommendations are required**

Full fasting recommendations are required for non-operating room anesthesia.

*Topic: Outside OT*

# Anesthesia for Day-care Surgery MCQ – Chapter 38

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 38)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. When a patient undergoing a day-care procedure stays in the hospital for one night, the surgery is referred to as:**

- A. Short-stay surgery/procedure
- B. Extended day-care surgery
- C. Ambulatory surgery
- D. Semi-elective surgery

**Ans: A**

**Q2. Early ambulation in day-care (ambulatory) anesthesia is specifically stated in the textbook to decrease which postoperative complication?**

- A. Deep vein thrombosis
- B. Pulmonary embolism
- C. Wound infection
- D. Urinary retention

**Ans: B**

**Q3. Which of the following procedures is NOT suitable for day-care surgery?**

- A. A procedure in a 90-year-old non-obese, non-OSA patient
- B. A procedure in an ASA grade II patient
- C. A procedure requiring prolonged parenteral or epidural analgesics postoperatively
- D. A 3-hour surgery in an otherwise healthy patient

**Ans: C**

**Q4. Which of the following statements about ASA grading and day-care surgery is CORRECT?**

- A. Only ASA grade I patients can undergo day-care surgery
- B. ASA grade III patients are absolutely excluded from day-care surgery
- C. ASA grading is an absolute parameter for exclusion from day-care
- D. ASA grading is NOT an absolute parameter; usually ASA I, II, and III patients are considered for day-care

**Ans: D**

**Q5. What is the ONLY age-related absolute exclusion for day-care surgery according to the textbook?**

- A. Prematures up to 50th postconception week
- B. Neonates less than 1 week old
- C. Infants less than 6 months old
- D. Patients older than 85 years

**Ans: A**

**Q6. What is the MOST IMPORTANT goal of premedication for day-care patients?**

- A. To provide preoperative analgesia
- B. To relieve anxiety
- C. To prevent aspiration
- D. To reduce oropharyngeal secretions

**Ans: B**

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| <p><b>Q7. What is the PREFERRED method of relieving anxiety in day-care surgery patients according to the textbook?</b></p> <p>A. Oral benzodiazepines<br/>B. Intravenous midazolam at induction<br/>C. Non-pharmacological approach (taking the patient into detail to clear his/her doubts)<br/>D. Oral opioid analgesics</p>                                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q8. Which benzodiazepine is the drug of choice for premedication in day-care surgery patients?</b></p> <p>A. Diazepam<br/>B. Lorazepam<br/>C. Alprazolam<br/>D. Midazolam</p>                                                                                                                                                                                                                                                                     | <p><b>Ans: D</b></p> |
| <p><b>Q9. Why are benzodiazepines used with caution for premedication in day-care surgery patients?</b></p> <p>A. They can delay the recovery<br/>B. They cause severe respiratory depression<br/>C. They produce paradoxical excitation in elderly patients<br/>D. They cannot be antagonized if needed</p>                                                                                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q10. What is the required fasting (nil orally) duration for clear fluids (water) before day-care surgery?</b></p> <p>A. 4 hours<br/>B. 2 hours<br/>C. 6 hours<br/>D. 8 hours</p>                                                                                                                                                                                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q11. What is the required fasting duration before day-care surgery for a light meal?</b></p> <p>A. 2 hours<br/>B. 4 hours<br/>C. 6 hours<br/>D. 8 hours</p>                                                                                                                                                                                                                                                                                       | <p><b>Ans: C</b></p> |
| <p><b>Q12. What is the required fasting duration before day-care surgery for a full meal?</b></p> <p>A. 2 hours<br/>B. 4 hours<br/>C. 6 hours<br/>D. 8 hours</p>                                                                                                                                                                                                                                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q13. Which statement best describes the role of Bispectral Index (BIS) monitoring in day-care surgery?</b></p> <p>A. It is suggested for measuring depth of anesthesia but considered optional due to high cost and insignificant difference in awake time<br/>B. It is mandatory for all day-care surgeries<br/>C. It significantly reduces awake time and is therefore recommended<br/>D. It is used exclusively for TIVA cases in day-care</p> | <p><b>Ans: A</b></p> |

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| <p><b>Q14. Which airway device is ALWAYS preferred over endotracheal intubation for general anesthesia in day-care surgery?</b></p> <p>A. Oropharyngeal airway (Guedel)<br/> B. Laryngeal mask airway (LMA)<br/> C. Nasopharyngeal airway<br/> D. I-gel supraglottic device</p>                                                                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q15. What is the PRIMARY reason for preferring LMA over endotracheal intubation in day-care surgery?</b></p> <p>A. Easier to insert in all patients<br/> B. Reduced risk of laryngospasm during insertion<br/> C. To avoid complications of intubation, especially postoperative sore throat<br/> D. LMA provides better ventilation during laparoscopic procedures</p>                           | <p><b>Ans: C</b></p> |
| <p><b>Q16. What combination of properties makes propofol the induction agent of choice for day-care surgery?</b></p> <p>A. Long duration of action and analgesic properties<br/> B. Absence of cardiovascular side effects and low cost<br/> C. Bronchodilatory properties and ease of titration<br/> D. Rapid, smooth recovery and antiemetic property</p>                                             | <p><b>Ans: D</b></p> |
| <p><b>Q17. In the propofol autoco-induction technique for day-care surgery, what initial dose of propofol is administered?</b></p> <p>A. 30 mg<br/> B. 50 mg<br/> C. 100 mg<br/> D. 1 mg/kg</p>                                                                                                                                                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q18. Which inhalational agent is the agent of choice for MAINTENANCE of anesthesia in day-care surgery?</b></p> <p>A. Isoflurane<br/> B. Desflurane<br/> C. Sevoflurane<br/> D. Halothane</p>                                                                                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q19. What pharmacological property of desflurane makes it the preferred maintenance agent for day-care surgery?</b></p> <p>A. Highest blood gas coefficient among volatile agents<br/> B. Non-pungent odor suitable for inhalational induction<br/> C. Lowest blood gas coefficient, leading to earliest induction and recovery<br/> D. No cardiovascular side effects at high concentrations</p> | <p><b>Ans: C</b></p> |
| <p><b>Q20. Which inhalational agent serves as an EXCELLENT ALTERNATIVE to desflurane when desflurane is unavailable for day-care surgery maintenance?</b></p> <p>A. Isoflurane<br/> B. Halothane<br/> C. Enflurane<br/> D. Sevoflurane</p>                                                                                                                                                              | <p><b>Ans: D</b></p> |



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| <p><b>Q21. How quickly is nitrous oxide eliminated from the body after discontinuation, contributing to faster recovery in day-care surgery?</b></p> <p>A. 3–5 minutes<br/>B. 10–15 minutes<br/>C. 20–30 minutes<br/>D. 60 minutes</p>                                                                                                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q22. Which opioid is the MOST PREFERRED for intravenous analgesia in day-care surgery patients?</b></p> <p>A. Morphine<br/>B. Remifentanil<br/>C. Fentanyl<br/>D. Pethidine</p>                                                                                                                                                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q23. What is the half-life of remifentanil that makes it suitable for day-care surgery?</b></p> <p>A. 2 minutes<br/>B. 5 minutes<br/>C. 10 minutes<br/>D. 30 minutes</p>                                                                                                                                                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q24. Why is succinylcholine NOT preferred as a muscle relaxant for day-care surgery?</b></p> <p>A. It has a very long duration of action requiring reversal<br/>B. It causes significant bronchospasm<br/>C. It cannot be antagonized with neostigmine<br/>D. High incidence of postoperative myalgia</p>                                                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q25. Which feature of mivacurium makes it the PREFERRED muscle relaxant for day-care surgery?</b></p> <p>A. Short duration of action (10 minutes)<br/>B. No requirement for reversal agents<br/>C. Can be administered orally<br/>D. Does not require neuromuscular monitoring</p>                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q26. What is the combination of choice for Total Intravenous Anesthesia (TIVA) in day-care surgery?</b></p> <p>A. Ketamine + Midazolam<br/>B. Propofol + Remifentanil<br/>C. Propofol + Fentanyl<br/>D. Thiopentone + Morphine</p>                                                                                                                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q27. Which statement regarding Total Intravenous Anesthesia (TIVA) in day-care surgery is CORRECT?</b></p> <p>A. TIVA includes both inhalational and intravenous anesthetic agents<br/>B. Muscle relaxants are routinely included in TIVA protocols<br/>C. TIVA uses only intravenous anesthetics with no inhalational agents and no muscle relaxants<br/>D. Nitrous oxide is an essential component of TIVA</p> | <p><b>Ans: C</b></p> |

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| <p><b>Q28. Which regional anesthesia technique is FREQUENTLY used for day-care surgery patients?</b></p> <ul style="list-style-type: none"> <li>A. Epidural anesthesia</li> <li>B. Caudal block</li> <li>C. Brachial plexus block</li> <li>D. Spinal anesthesia</li> </ul>                                                                                                                                                                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q29. Which local anesthetic agents are PREFERRED for spinal anesthesia in day-care surgery?</b></p> <ul style="list-style-type: none"> <li>A. Prilocaine or chloroprocaine (short/intermediate-acting agents)</li> <li>B. Bupivacaine (long-acting agent)</li> <li>C. Lignocaine</li> <li>D. Ropivacaine</li> </ul>                                                                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q30. Why is lignocaine NOT preferred for spinal anesthesia in day-care surgery patients?</b></p> <ul style="list-style-type: none"> <li>A. It has too short a duration of action for most procedures</li> <li>B. Possibility of transient neurological symptoms (TNS)</li> <li>C. High risk of total spinal anesthesia</li> <li>D. Significant cardiovascular toxicity at spinal doses</li> </ul>                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q31. What is 'Selective Spinal Anesthesia (SSA)' as described in the context of day-care surgery?</b></p> <ul style="list-style-type: none"> <li>A. Spinal anesthesia administered exclusively in the sitting position</li> <li>B. Use of hyperbaric solutions to target specific dermatomes</li> <li>C. Use of isobaric local anesthetic to ensure symmetrical spread</li> <li>D. Giving minimal doses of local anesthetic to block only the required segments</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q32. What is the MAXIMUM dose of bupivacaine recommended for spinal anesthesia in day-care surgery to avoid the need for extended monitoring?</b></p> <ul style="list-style-type: none"> <li>A. 5 mg</li> <li>B. 10 mg</li> <li>C. 15 mg</li> <li>D. 7 mg</li> </ul>                                                                                                                                                                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q33. What is the MOST PREFERRED drug for sedation during Monitored Anesthesia Care (MAC) in day-care surgery?</b></p> <ul style="list-style-type: none"> <li>A. Midazolam</li> <li>B. Ketamine</li> <li>C. Dexmedetomidine</li> <li>D. Fentanyl</li> </ul>                                                                                                                                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q34. What is the MOST COMMON postoperative problem seen in day-care surgery patients?</b></p> <ul style="list-style-type: none"> <li>A. Nausea and vomiting</li> <li>B. Pain</li> <li>C. Sedation</li> <li>D. Bleeding</li> </ul>                                                                                                                                                                                                                                          | <p><b>Ans: B</b></p> |

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| <p><b>Q35. What is the SECOND MOST COMMON postoperative problem in day-care surgery patients?</b></p> <p>A. Pain<br/>B. Bleeding<br/>C. Nausea and vomiting<br/>D. Myalgia</p>                                                                                                                                                                                                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q36. What does 'fast-track discharge' mean in the context of day-care surgery?</b></p> <p>A. Discharging the patient directly home without any recovery room observation<br/>B. Discharging the patient within 30 minutes of completing surgery<br/>C. Transferring the patient from Phase I PACU directly to the ward for overnight stay<br/>D. Shifting the patient directly from operation theater to Phase II (step down) postanesthesia care unit (PACU)</p> | <p><b>Ans: D</b></p> |
| <p><b>Q37. What is the reported incidence of unexpected admission (unable to discharge) after day-care surgery?</b></p> <p>A. 1% to 6%<br/>B. 10% to 20%<br/>C. Less than 0.5%<br/>D. 7% to 10%</p>                                                                                                                                                                                                                                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q38. What are the TWO MOST COMMON causes of unexpected admission after day-care surgery?</b></p> <p>A. Surgical site infection and wound bleeding<br/>B. Uncontrolled pain and nausea and vomiting<br/>C. Excessive sedation and respiratory depression<br/>D. Urinary retention and uncontrolled hypertension</p>                                                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q39. According to the home readiness criteria, for how long must a patient's vital signs be stable before discharge from day-care surgery?</b></p> <p>A. 30 minutes<br/>B. 45 minutes<br/>C. 60 minutes<br/>D. 120 minutes</p>                                                                                                                                                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q40. According to the criteria for home readiness, for how long must a patient be free from nausea (or have only minimal nausea) before discharge?</b></p> <p>A. 60 minutes<br/>B. 45 minutes<br/>C. 20 minutes<br/>D. 30 minutes</p>                                                                                                                                                                                                                             | <p><b>Ans: D</b></p> |
| <p><b>Q41. Regarding oral fluid tolerance after day-care surgery, which statement is CORRECT?</b></p> <p>A. Ability to accept liquids orally without vomiting is NOT a mandatory criterion for discharge<br/>B. Patient must tolerate oral liquids for 30 minutes before discharge<br/>C. Oral fluid tolerance is mandatory only after spinal anesthesia<br/>D. Patient must consume at least 100 mL of fluids before discharge</p>                                     | <p><b>Ans: A</b></p> |

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| <p><b>Q42. Which statement about the ability to void urine as a discharge criterion in day-care surgery is CORRECT?</b></p> <ul style="list-style-type: none"> <li>A. Passing urine is mandatory for ALL patients before discharge</li> <li>B. Passing urine is NOT mandatory except for specific high-risk patients (urology, urogynecology, inguinal/perianal surgery, age &gt;70, urinary dysfunction, bupivacaine &gt;7 mg)</li> <li>C. Passing urine is mandatory only after spinal anesthesia</li> <li>D. Passing urine is mandatory only for patients over 60 years of age</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q43. For which of the following patients is the ability to pass urine considered a MANDATORY discharge criterion after day-care surgery?</b></p> <ul style="list-style-type: none"> <li>A. All patients who received spinal anesthesia regardless of other factors</li> <li>B. All patients with ASA grade II or III status</li> <li>C. Patients with history of urinary dysfunction or bupivacaine dose &gt;7 mg used for spinal</li> <li>D. All patients above 50 years of age</li> </ul>                                                                                            | <p><b>Ans: C</b></p> |
| <p><b>Q44. For how long should patients with obstructive sleep apnea (OSA) be monitored after day-care surgery if there is EVEN A SINGLE episode of desaturation on room air?</b></p> <ul style="list-style-type: none"> <li>A. More than 1 hour</li> <li>B. More than 3 hours</li> <li>C. More than 5 hours</li> <li>D. More than 7 hours</li> </ul>                                                                                                                                                                                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q45. After day-care surgery, patients are instructed not to perform important and skillful work such as driving or operating machinery for how long?</b></p> <ul style="list-style-type: none"> <li>A. 24 hours</li> <li>B. 12 hours</li> <li>C. 48 hours</li> <li>D. 6 hours</li> </ul>                                                                                                                                                                                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q46. What is the incidence of re-admission or emergency visit after day-care surgery?</b></p> <ul style="list-style-type: none"> <li>A. Less than 1%</li> <li>B. 2–3%</li> <li>C. 5–8%</li> <li>D. 10–15%</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q47. What is the MOST COMMON cause of RE-ADMISSION after day-care surgery according to the textbook?</b></p> <ul style="list-style-type: none"> <li>A. Nausea and vomiting</li> <li>B. Uncontrolled pain</li> <li>C. Surgical complications such as surgical site infection</li> <li>D. Respiratory depression</li> </ul>                                                                                                                                                                                                                                                              | <p><b>Ans: C</b></p> |
| <p><b>Q48. Which group of patients CANNOT be considered for day-care surgery regardless of other clinical criteria?</b></p> <ul style="list-style-type: none"> <li>A. Patients with ASA grade III status</li> <li>B. Patients above 70 years of age</li> <li>C. Patients with mild obesity (BMI 30–35 kg/m<sup>2</sup>)</li> <li>D. Patients not accompanied by a responsible attendant</li> </ul>                                                                                                                                                                                           | <p><b>Ans: D</b></p> |

**Q49. Which form of obesity is stated as a BARRIER to day-care procedures according to the textbook?**

- A. Morbid obesity with obstructive sleep apnea
- B. Simple obesity with BMI 30–40 kg/m<sup>2</sup>
- C. Obesity in patients above 60 years
- D. Obesity in ASA grade II patients

**Ans: A**

**Q50. What is the policy on preoperative investigations for day-care patients according to the textbook?**

- A. Full blood count, electrolytes, and ECG are mandatory for all day-care patients
- B. No routine investigations; investigations should be guided by existing comorbidity and nature of surgery
- C. Chest X-ray and ECG mandatory for all patients above 40 years
- D. Echocardiography required for all ASA grade II patients

**Ans: B**

# Answer Key & Explanations

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## **Q1. Answer: A — Short-stay surgery/procedure**

Page 267 (Chapter 38 intro): If the patient stays for one night after a day-care procedure, it is called a 'short-stay surgery/procedure'. Day-care surgery proper means the patient is admitted, procedure done, and discharged on the same calendar day.

*Topic: Introduction to Day-care Surgery*

## **Q2. Answer: B — Pulmonary embolism**

Page 267: One of the reasons for increased day-care surgeries is that early ambulation (that's why called ambulatory anesthesia) decreases the complications like pulmonary embolism. This is explicitly mentioned in the textbook.

*Topic: Introduction to Day-care Surgery*

## **Q3. Answer: C — A procedure requiring prolonged parenteral or epidural analgesics postoperatively**

Page 267: Surgeries associated with major hemodynamic alterations, postoperative complications, requiring prolonged immobilization and parenteral/epidural analgesics are NOT considered for day-care surgeries. Duration of surgery is no longer a parameter for exclusion.

*Topic: Selection of Surgery and Patients*

## **Q4. Answer: D — ASA grading is NOT an absolute parameter; usually ASA I, II, and III patients are considered for day-care**

Page 267: Although the American Society of Anesthesiologists (ASA) grading is not an absolute parameter, however, usually ASA grade I, II and III (with their disease well controlled) patients are considered for day-care procedures.

*Topic: Selection of Surgery and Patients*

## **Q5. Answer: A — Prematures up to 50th postconception week**

Page 267: Age is no bar for day-care surgery; even a newborn or 100-year-old patient can be considered. The ONLY exclusion is prematures up to 50th postconception week because prematures are at high risk of apnea up to 50th postconception week.

*Topic: Selection of Surgery and Patients*

## **Q6. Answer: B — To relieve anxiety**

Page 267: Like indoor patients, the most important goal for premedication for day-care patients is to relieve the anxiety. This is explicitly stated as the primary goal of premedication in day-care settings.

*Topic: Preoperative Assessment and Premedication*

## **Q7. Answer: C — Non-pharmacological approach (taking the patient into detail to clear his/her doubts)**

Page 267: As benzodiazepines can delay the recovery, the preferred method of relieving anxiety should be non-pharmacological (taking to the patient in detail to clear his/her doubts). Pharmacological methods should only be used when deemed necessary.

*Topic: Preoperative Assessment and Premedication*

## **Q8. Answer: D — Midazolam**

Page 267: Because of its shorter half-life, midazolam is the benzodiazepine of choice for day-care patients. If pharmacological premedication is deemed necessary, only benzodiazepines should be used, and midazolam is specifically preferred.

*Topic: Preoperative Assessment and Premedication*

**Q9. Answer: A — They can delay the recovery**

Page 267: As benzodiazepines can delay the recovery, the preferred method of relieving anxiety should be non-pharmacological. If pharmacological premedication is required, only benzodiazepines with shorter half-life (midazolam) should be used to minimize delay in recovery.

*Topic: Preoperative Assessment and Premedication*

**Q10. Answer: B — 2 hours**

Page 267: Nil orally guidelines are same (as for indoor patients), i.e. 6 hours for light meal, 8 hours for full meal and 2 hours for clear fluids (water). These guidelines apply equally to day-care patients.

*Topic: Preoperative Assessment and Premedication*

**Q11. Answer: C — 6 hours**

Page 267: Nil orally guidelines are same as for indoor patients: 6 hours for light meal, 8 hours for full meal, and 2 hours for clear fluids (water). This applies equally to day-care surgery patients.

*Topic: Preoperative Assessment and Premedication*

**Q12. Answer: D — 8 hours**

Page 267: Nil orally guidelines are same as for indoor patients: 6 hours for light meal, 8 hours for full meal, and 2 hours for clear fluids (water). A full meal requires the longest fasting period of 8 hours.

*Topic: Preoperative Assessment and Premedication*

**Q13. Answer: A — It is suggested for measuring depth of anesthesia but considered optional due to high cost and insignificant difference in awake time**

Page 268: BIS monitor to measure the depth of anesthesia is suggested for day-care surgery, however, still considered optional because of its high cost and insignificant difference in awake time in patients whom BIS was used vs. in whom BIS was not used.

*Topic: Monitoring*

**Q14. Answer: B — Laryngeal mask airway (LMA)**

Page 268: To avoid the complications of intubation (especially postoperative sore throat), general anesthesia (GA) with laryngeal mask airway (LMA) is always preferred over intubation. It is now considered safe to do even laparoscopic surgeries in LMA patients who are adequately fasting and do not have a risk factor for aspiration.

*Topic: General Anesthesia*

**Q15. Answer: C — To avoid complications of intubation, especially postoperative sore throat**

Page 268: GA with LMA is always preferred over intubation to avoid the complications of intubation, especially postoperative sore throat. This is a key consideration in day-care surgery where patient comfort and rapid recovery are paramount.

*Topic: General Anesthesia*

**Q16. Answer: D — Rapid, smooth recovery and antiemetic property**

Page 268: Propofol is the agent of choice for induction because of its rapid, smooth recovery and antiemetic property. These three properties make it uniquely suited for day-care surgery where quick, complication-free recovery is essential.

*Topic: General Anesthesia*

**Q17. Answer: A — 30 mg**

Page 268: Propofol autoco-induction, the technique in which 30 mg of propofol is given as an initial dose, has been found to reduce the requirement of propofol, hastening the recovery. This subinduction dose sensitizes the patient to subsequent propofol doses.

*Topic: General Anesthesia*

**Q18. Answer: B — Desflurane**

Page 268: Due to lowest blood gas coefficient, Desflurane has earliest induction and recovery, therefore, is the inhalational agent of choice for day-care patients. In case of non-availability of desflurane, sevoflurane serves as an excellent alternative.

Topic: General Anesthesia

**Q19. Answer: C — Lowest blood gas coefficient, leading to earliest induction and recovery**

Page 268: Due to lowest blood gas coefficient, Desflurane has earliest induction and recovery. The blood-gas coefficient determines how quickly an agent equilibrates between the blood and alveoli; a lower coefficient means faster wash-in and wash-out, thus earliest recovery.

Topic: General Anesthesia

**Q20. Answer: D — Sevoflurane**

Page 268: In case of non-availability of desflurane, sevoflurane serves as an excellent alternative for maintenance of anesthesia in day-care patients. Sevoflurane also has a relatively low blood-gas coefficient, providing reasonably quick recovery.

Topic: General Anesthesia

**Q21. Answer: A — 3–5 minutes**

Page 268: Nitrous oxide decreases the requirement of other anesthetic agents hence fastening the recovery (nitrous oxide itself is eliminated from body in 3–5 minutes). This rapid elimination is a significant advantage for day-care surgery.

Topic: General Anesthesia

**Q22. Answer: B — Remifentanyl**

Page 268: Opioids should be avoided if possible. Remifentanyl because of its ultra-short half-life (10 minutes) is the most preferred opioid for day-care surgery. Its rapid elimination minimizes postoperative sedation and respiratory depression.

Topic: General Anesthesia

**Q23. Answer: C — 10 minutes**

Page 268: Remifentanyl because of its ultra-short half-life (10 minutes) is the most preferred opioid. This ultra-short half-life ensures rapid offset of opioid effects, minimizing postoperative sedation and respiratory depression in day-care patients.

Topic: General Anesthesia

**Q24. Answer: D — High incidence of postoperative myalgia**

Page 268: Succinylcholine, in spite of having short duration, is usually not preferred due to high incidence of postoperative myalgia. In day-care surgery, postoperative myalgia is particularly undesirable as patients are discharged the same day.

Topic: General Anesthesia

**Q25. Answer: A — Short duration of action (10 minutes)**

Page 268: Short duration of action (10 minutes) makes mivacurium as preferred muscle relaxant for day-care surgery. Its brief duration of action aligns well with the short surgical procedures typically performed in day-care settings.

Topic: General Anesthesia

**Q26. Answer: B — Propofol + Remifentanyl**

Page 268: The combination of choice for total intravenous anesthesia (TIVA) is Propofol + Remifentanyl. Both agents have rapid, predictable offset of action making them ideal for TIVA in day-care surgery. This is also listed in the Key Points of the chapter.

Topic: Total Intravenous Anesthesia



**Q27. Answer: C — TIVA uses only intravenous anesthetics with no inhalational agents and no muscle relaxants**

Page 268: TIVA is the technique where only intravenous anesthetics are used (no inhalational, no muscle relaxant). The combination of choice for TIVA is Propofol + Remifentanyl. The exclusion of inhalational agents and muscle relaxants is a defining characteristic of TIVA.

*Topic: Total Intravenous Anesthesia*

**Q28. Answer: D — Spinal anesthesia**

Page 268: Regional nerve blocks are simple and safe techniques for day-care surgery. In current-day practice, spinal anesthesia is frequently used for day-care patients. Epidural is hardly used for day-care. Spinal anesthesia with appropriate agents provides predictable, titratable block duration.

*Topic: Regional Anesthesia*

**Q29. Answer: A — Prilocaine or chlorprocaine (short/intermediate-acting agents)**

Page 268: Spinal anesthesia is performed with small gauge pencil tip needles and short/intermediate-acting agents like prilocaine or chlorprocaine. These agents provide a block of appropriate duration for day-care procedures without prolonged motor block that would delay discharge.

*Topic: Regional Anesthesia*

**Q30. Answer: B — Possibility of transient neurological symptoms (TNS)**

Page 268: Lignocaine is not preferred due to the possibility of transient neurologic symptoms (TNS). TNS manifests as back pain radiating to the lower limbs, which would be particularly distressing for patients being discharged on the same day.

*Topic: Regional Anesthesia*

**Q31. Answer: D — Giving minimal doses of local anesthetic to block only the required segments**

Page 268: If a long-acting agent like bupivacaine is to be used, then the spinal should be 'selective spinal anesthesia (SSA)' which entails giving minimal doses to block only the required segments. This minimizes the extent of motor and sympathetic block, facilitating early discharge.

*Topic: Regional Anesthesia*

**Q32. Answer: D — 7 mg**

Page 269: Patients with obstructive sleep apnea should be monitored if the dysfunction or dose of bupivacaine used for spinal is > 7 mg. Also, the ability to pass urine is not mandatory except when bupivacaine dose >7 mg is used. Doses above 7 mg indicate a denser block requiring longer observation.

*Topic: Regional Anesthesia*

**Q33. Answer: A — Midazolam**

Page 268: Most preferred drug for sedation (in monitored anesthesia care) is midazolam. Sometimes, propofol may be required in low (sedative) doses. Midazolam's anxiolytic, amnestic and sedative properties make it ideal for MAC in day-care patients.

*Topic: Monitored Anesthesia Care*

**Q34. Answer: B — Pain**

Page 268: The common problems seen in the postoperative period are pain (most common), nausea and vomiting (2nd most common), sedation, myalgia, weakness and bleeding. Pain is the most common postoperative problem and one of the most common causes of unexpected admission.

*Topic: Postoperative Period*

**Q35. Answer: C — Nausea and vomiting**

Page 268: The common problems seen in the postoperative period are pain (most common), nausea and vomiting (2nd most common), sedation, myalgia, weakness and bleeding. Nausea and vomiting together are the second most common complication and one of the two leading causes of unexpected admission.

Topic: Postoperative Period

**Q36. Answer: D — Shifting the patient directly from operation theater to Phase II (step down) postanesthesia care unit (PACU)**

Page 268: The aim should be fast-track discharge, which means shifting the patient directly from operation theater to Phase II (step down) postanesthesia care unit (PACU). This will decrease the cost and facilitate early discharge by bypassing Phase I PACU.

Topic: Discharge

**Q37. Answer: A — 1% to 6%**

Page 268: The incidence of unexpected admission (unable to discharge) varies from 1% to 6%. The two most common causes of unexpected admission are uncontrolled pain and nausea and vomiting.

Topic: Discharge

**Q38. Answer: B — Uncontrolled pain and nausea and vomiting**

Page 268 (Key Points): The two most common causes of unexpected admission are uncontrolled pain and nausea and vomiting. This is also specifically highlighted in the Key Points section of Chapter 38, emphasizing the importance of adequate analgesic and antiemetic protocols.

Topic: Discharge

**Q39. Answer: C — 60 minutes**

Page 269: Criteria for home readiness include: (i) Vital signs are stable for more than 60 minutes. This ensures that the patient has had adequate time for hemodynamic stabilization after the procedure before being discharged.

Topic: Discharge Criteria

**Q40. Answer: D — 30 minutes**

Page 269: Criteria for home readiness include: (ii) There is no nausea (or minimal nausea) for at least 30 minutes. This criterion ensures the patient is comfortable and unlikely to develop significant nausea and vomiting during the journey home.

Topic: Discharge Criteria

**Q41. Answer: A — Ability to accept liquids orally without vomiting is NOT a mandatory criterion for discharge**

Page 269: Able to accept liquids orally without vomiting is not a mandatory criterion in current-day practice. This is a change from older discharge criteria, and is highlighted in the Key Points: 'Able to accept liquids and pass urine is not considered as mandatory criteria for discharging a patient after day-care surgery.'

Topic: Discharge Criteria

**Q42. Answer: B — Passing urine is NOT mandatory except for specific high-risk patients (urology, urogynecology, inguinal/perianal surgery, age >70, urinary dysfunction, bupivacaine >7 mg)**

Page 269: Able to pass urine is also not considered mandatory criteria even after spinal, EXCEPT for high-risk patients like urology/urogynecology, inguinal or perianal surgery, age >70, history of urinary dysfunction, or bupivacaine dose >7 mg used for spinal. This nuanced approach avoids unnecessary delays in discharge.

Topic: Discharge Criteria

**Q43. Answer: C — Patients with history of urinary dysfunction or bupivacaine dose >7 mg used for spinal**

Page 269: Able to pass urine IS mandatory for high-risk patients including those with: urology/urogynecology procedures, inguinal or perianal surgery, age >70, history of urinary dysfunction, or bupivacaine dose >7 mg. These patients are at higher risk of urinary retention and require confirmed voiding before discharge.

*Topic: Discharge Criteria*

**Q44. Answer: D — More than 7 hours**

Page 269: Patients with obstructive sleep apnea should be monitored for more than 3 hours and more than 7 hours, if there is even a single episode of desaturation on room air. A single desaturation episode warrants the longer 7-hour monitoring period to ensure patient safety.

*Topic: Discharge Criteria*

**Q45. Answer: A — 24 hours**

Page 269: The patients should be instructed not to do important and skillful work like driving, operating a machine for at least 24 hours. This is a standard postoperative instruction for day-care surgery patients to ensure safety during the recovery period.

*Topic: Postoperative Instructions*

**Q46. Answer: B — 2–3%**

Page 269: Although > 50% of the patients complain of little drowsiness after discharge, the incidence of re-admission (or emergency visit) is 2–3%. This is distinct from the incidence of unexpected admission (unable to discharge), which is 1–6%.

*Topic: Postoperative Instructions*

**Q47. Answer: C — Surgical complications such as surgical site infection**

Page 269: Most often the re-admission is due to surgical complications, such as surgical site infection. This is highlighted in the Key Points section. Note the distinction: while uncontrolled pain and PONV are the most common causes of INABILITY TO DISCHARGE, surgical complications are the most common cause of RE-ADMISSION.

*Topic: Postoperative Instructions*

**Q48. Answer: D — Patients not accompanied by a responsible attendant**

Page 267: Patients not accompanied by responsible attendant cannot be considered for day-care surgery. This is an absolute criterion because post-discharge supervision is essential for safety. Additionally, patients must be able to stay in the vicinity of the hospital for the first 24 hours.

*Topic: Selection of Surgery and Patients*

**Q49. Answer: A — Morbid obesity with obstructive sleep apnea**

Page 267: Obesity (not even the morbid obesity with obstructive sleep apnea) is a barrier to day-care procedures. This means simple obesity alone is NOT a barrier; only morbid obesity combined with obstructive sleep apnea is considered a barrier to day-care surgery.

*Topic: Selection of Surgery and Patients*

**Q50. Answer: B — No routine investigations; investigations should be guided by existing comorbidity and nature of surgery**

Page 267: Investigations: As for other patients, there are no routine investigations for day-care patients too; investigations should be guided by the existing comorbidity and the nature of surgery. This rational, individualized approach avoids unnecessary tests and reduces costs without compromising safety.

*Topic: Preoperative Assessment and Premedication*

# Pain Management MCQ – Chapter 39

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 39)

**50 Questions • 35 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Pain has been designated as which vital sign?**

- A. The fifth vital sign
- B. The fourth vital sign
- C. The sixth vital sign
- D. The third vital sign

**Ans: A**

**Q2. Which is the most commonly used scale to assess pain?**

- A. Visual analog scale
- B. McGill pain questionnaire
- C. Smiley scale
- D. CHEOPS scale

**Ans: A**

**Q3. For younger children (usually less than 3–4 years) who cannot identify the faces, which scale is most commonly used to objectively assess pain?**

- A. Children's Hospital of Eastern Ontario Pain Scale (CHEOPS)
- B. Visual analog scale
- C. McGill pain questionnaire
- D. Smiley scale

**Ans: A**

**Q4. Postoperative pain management should follow which approach?**

- A. Multimodal approach (combination of different modalities)
- B. Opioids only
- C. NSAIDs only
- D. Regional blocks only

**Ans: A**

**Q5. NSAIDs in multimodal analgesia can reduce the dose requirement of opioids by approximately:**

- A. One third
- B. One half
- C. Two thirds
- D. Three quarters

**Ans: A**

**Q6. Although opioids remain the mainstay of treatment for postoperative analgesia, which is the most preferred route?**

- A. Epidural route
- B. Intramuscular route
- C. Subcutaneous route
- D. Oral route

**Ans: A**

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| <p><b>Q7. What is the most frequently used combination for controlling postoperative pain via the epidural route?</b></p> <ul style="list-style-type: none"> <li>A. Bupivacaine (0.125–0.25%) or ropivacaine (0.2%) with/without fentanyl</li> <li>B. Morphine alone</li> <li>C. Lignocaine with adrenaline</li> <li>D. Ketamine and midazolam</li> </ul>                                                                                                                   | <p><b>Ans: A</b></p> |
| <p><b>Q8. In case of non-feasibility of epidural, the IV route is most often utilized. Which is the method of choice for postoperative analgesia if opioids have to be given by the parenteral route?</b></p> <ul style="list-style-type: none"> <li>A. Patient-controlled analgesia (PCA)</li> <li>B. Continuous IV infusion</li> <li>C. Intermittent IV boluses by nurses</li> <li>D. Intramuscular injection</li> </ul>                                                  | <p><b>Ans: A</b></p> |
| <p><b>Q9. Regional nerve blocks given before skin incision are called pre-emptive analgesia. What is their benefit?</b></p> <ul style="list-style-type: none"> <li>A. They decrease the stress response, the requirement of opioids, and decrease/abolish the requirement and incidence of acute pain getting converted into chronic pain</li> <li>B. They cause more pain</li> <li>C. They increase opioid requirement</li> <li>D. They only work after surgery</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q10. Which block is very effective to produce analgesia for abdominal surgeries such as cesarean section and hernia repair, given in the fascial sheath between internal oblique and transversus abdominis muscle?</b></p> <ul style="list-style-type: none"> <li>A. Transversus abdominis plane (TAP) block</li> <li>B. Thoracic paravertebral block</li> <li>C. Lumbar plexus block</li> <li>D. Brachial plexus block</li> </ul>                                    | <p><b>Ans: A</b></p> |
| <p><b>Q11. What is the incidence of chronic pain?</b></p> <ul style="list-style-type: none"> <li>A. 25–30%</li> <li>B. 5–10%</li> <li>C. 50–60%</li> <li>D. 10–15%</li> </ul>                                                                                                                                                                                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q12. In chronic pain management, neurolysis can be done with chemical agents or radiofrequency. Which is always preferred?</b></p> <ul style="list-style-type: none"> <li>A. Radiofrequency ablation (RFA) is always preferred over chemical neurolysis</li> <li>B. Chemical neurolysis is always preferred</li> <li>C. Both are equal</li> <li>D. Surgical intervention is always preferred</li> </ul>                                                               | <p><b>Ans: A</b></p> |

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| <p><b>Q13. In epidural steroid administration for low backache, only 40–50% of the drug reaches the nerve roots. Which route is therefore preferred nowadays?</b></p> <p>A. Transforaminal route (drug injected as the nerve exits through the foramina)<br/> B. Caudal route<br/> C. Intrathecal route<br/> D. Interlaminar route</p> | <p><b>Ans: A</b></p> |
| <p><b>Q14. What is the most definitive modality of treatment for facet arthropathy?</b></p> <p>A. Radiofrequency (RF) ablation of medial branch of posterior ramus of spinal nerve<br/> B. Epidural steroids<br/> C. Sacroiliac joint injection<br/> D. Vertebroplasty</p>                                                             | <p><b>Ans: A</b></p> |
| <p><b>Q15. What is the most common neuropathy encountered in clinical practice?</b></p> <p>A. Diabetic neuropathy<br/> B. Trigeminal neuralgia<br/> C. Postherpetic neuralgia<br/> D. Carpal tunnel syndrome</p>                                                                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q16. What is the drug of choice for neuropathic pain (other than trigeminal neuralgia)?</b></p> <p>A. Pregabalin (followed by gabapentin)<br/> B. Carbamazepine<br/> C. Morphine<br/> D. Amitriptyline</p>                                                                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q17. In trigeminal neuralgia, the pain is most often in which division, and what is the drug of choice?</b></p> <p>A. Maxillary division; carbamazepine<br/> B. Mandibular division; pregabalin<br/> C. Ophthalmic division; morphine<br/> D. Maxillary division; gabapentin</p>                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q18. Complex regional pain syndrome (CRPS) Type II, which includes causalgia (neuralgic pain after direct nerve injury), is also called:</b></p> <p>A. Causalgia<br/> B. Reflex sympathetic dystrophy<br/> C. Sudeck's dystrophy<br/> D. Phantom limb</p>                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q19. What is the hallmark of CRPS?</b></p> <p>A. Extremely painful limb (hyperalgesia, allodynia, hyperpathia, anesthesia dolorosa)<br/> B. Complete numbness of the limb<br/> C. Paralysis of the limb<br/> D. Swelling without pain</p>                                                                                        | <p><b>Ans: A</b></p> |

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| <p><b>Q20. What is the mainstay of treatment of CRPS? For upper limb CRPS, which block is performed?</b></p> <p>A. Sympathetic blocks; stellate ganglion block for upper limb<br/> B. Opioids; epidural block for upper limb<br/> C. NSAIDs; brachial plexus block<br/> D. Steroids; lumbar sympathetic block</p> | <p><b>Ans: A</b></p> |
| <p><b>Q21. What is pathognomonic of fibromyalgia?</b></p> <p>A. Multiple trigger points<br/> B. A single trigger point<br/> C. Muscle paralysis<br/> D. Joint swelling</p>                                                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q22. The stellate ganglion is formed by the fusion of:</b></p> <p>A. Lower cervical and first thoracic ganglion<br/> B. Upper cervical and second thoracic ganglion<br/> C. First and second thoracic ganglion<br/> D. Lower cervical ganglion only</p>                                                     | <p><b>Ans: A</b></p> |
| <p><b>Q23. Which is considered the first sign of a successful stellate ganglion block?</b></p> <p>A. Conjunctival congestion<br/> B. Nasal stuffiness (Guttman's sign)<br/> C. Ptosis<br/> D. Miosis</p>                                                                                                          | <p><b>Ans: A</b></p> |
| <p><b>Q24. The maximum systemic absorption / highest blood level of local anesthetic per volume of drug injected is achieved after which block?</b></p> <p>A. Intercostal nerve block<br/> B. Stellate ganglion block<br/> C. Celiac plexus block<br/> D. Trigeminal nerve block</p>                              | <p><b>Ans: A</b></p> |
| <p><b>Q25. What is the most common complication of celiac plexus block (usually given for gastric and pancreatic malignancy)?</b></p> <p>A. Hypotension<br/> B. Pneumothorax<br/> C. Paraplegia<br/> D. Hematoma</p>                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q26. What does the WHO recommend for control of cancer pain?</b></p> <p>A. A step ladder (non-opioids for mild, weak opioids for moderate, strong opioids for severe pain)<br/> B. Strong opioids for all cancer pain<br/> C. NSAIDs only<br/> D. Surgical intervention first</p>                           | <p><b>Ans: A</b></p> |



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| <p><b>Q27. What remains the mainstay of treatment for cancer pain?</b></p> <ul style="list-style-type: none"> <li>A. Opioids</li> <li>B. NSAIDs</li> <li>C. Neurolytic blocks</li> <li>D. Surgical intervention</li> </ul>                                                                                                                                                                                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q28. For cancer pain, the transdermal fentanyl patch provides analgesia for how long, and the butorphanol patch for how long?</b></p> <ul style="list-style-type: none"> <li>A. Fentanyl patch 48–72 hours; butorphanol patch 6–7 days</li> <li>B. Fentanyl patch 24 hours; butorphanol patch 3 days</li> <li>C. Fentanyl patch 7 days; butorphanol patch 48 hours</li> <li>D. Both for 24 hours</li> </ul>                                                                                                        | <p><b>Ans: A</b></p> |
| <p><b>Q29. Inadequate control of acute (postoperative) pain can elicit stress responses. Which of the following is one such response?</b></p> <ul style="list-style-type: none"> <li>A. Immunosuppression, poor wound healing, increased myocardial oxygen demand, decreased GI motility and chronic postsurgical pain</li> <li>B. Increased immunity</li> <li>C. Faster wound healing</li> <li>D. Decreased myocardial oxygen demand</li> </ul>                                                                         | <p><b>Ans: A</b></p> |
| <p><b>Q30. Acetaminophen (paracetamol) along with NSAIDs or regional nerve blocks may omit the need for opioids. It may serve as an alternative to NSAIDs in patients where:</b></p> <ul style="list-style-type: none"> <li>A. NSAIDs are contraindicated</li> <li>B. Opioids are required</li> <li>C. Regional blocks have failed</li> <li>D. PCA is being used</li> </ul>                                                                                                                                              | <p><b>Ans: A</b></p> |
| <p><b>Q31. Why may analgesics become less effective with PCA devices when delays occur in the delivery of analgesics by nurses?</b></p> <ul style="list-style-type: none"> <li>A. Delays lead to escalation of pain, making analgesic to become less effective; pain is best controlled when the analgesic is delivered before or just at the beginning of pain</li> <li>B. Because nurses give too much</li> <li>C. Because PCA cannot deliver opioids</li> <li>D. Because the lockout interval is too short</li> </ul> | <p><b>Ans: A</b></p> |
| <p><b>Q32. Which regional block is used to provide analgesia for thoracic, breast and upper abdominal surgeries?</b></p> <ul style="list-style-type: none"> <li>A. Thoracic paravertebral block</li> <li>B. Transversus abdominis plane (TAP) block</li> <li>C. Femoral nerve block</li> <li>D. Brachial plexus block</li> </ul>                                                                                                                                                                                         | <p><b>Ans: A</b></p> |

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| <p><b>Q33. Which other techniques are occasionally utilized to achieve postoperative analgesia?</b></p> <p>A. TENS, acupuncture, cryoanalgesia and ketamine infusion<br/> B. General anesthesia<br/> C. Hypothermia<br/> D. Surgical sympathectomy</p>                                                                                            | <p><b>Ans: A</b></p> |
| <p><b>Q34. In postherpetic neuralgia, which treatments are used?</b></p> <p>A. Tricyclic antidepressants, intercostal nerve blocks, transdermal lignocaine patch / EMLA cream, and topical capsaicin cream or patch<br/> B. Carbamazepine only<br/> C. Morphine only<br/> D. Surgical excision</p>                                                | <p><b>Ans: A</b></p> |
| <p><b>Q35. Which lumbar block is used to provide analgesia for hip and knee surgery?</b></p> <p>A. Lumbar plexus block (psoas compartment block)<br/> B. Celiac plexus block<br/> C. Stellate ganglion block<br/> D. Superior hypogastric block</p>                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q36. Which block is performed for pain relief of the perineal area?</b></p> <p>A. Ganglion impar (ganglion of Walther) block<br/> B. Superior hypogastric block<br/> C. Celiac plexus block<br/> D. Lumbar sympathetic chain block</p>                                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q37. The lumbar sympathetic chain block is utilized for CRPS of the lower limb (especially Berger's disease). Where is it blocked?</b></p> <p>A. Anterior to lumbar vertebral bodies<br/> B. Posterior to lumbar vertebral bodies<br/> C. At the sacral hiatus<br/> D. At the celiac plexus</p>                                             | <p><b>Ans: A</b></p> |
| <p><b>Q38. Which is an indication for the stellate ganglion block, apart from CRPS of the upper limb?</b></p> <p>A. Intra-arterial thiopentone injection and hand pallor (ischemia) following radial artery cannulation for invasive blood pressure monitoring<br/> B. Pelvic malignancy<br/> C. Pancreatic cancer pain<br/> D. Perineal pain</p> | <p><b>Ans: A</b></p> |
| <p><b>Q39. Which complication of stellate ganglion block, though it occurs, is not bothersome (as it only reduces the vital capacity by 20–25%)?</b></p> <p>A. Unilateral phrenic nerve block<br/> B. Brachial plexus block<br/> C. Pneumothorax<br/> D. Esophageal perforation</p>                                                               | <p><b>Ans: A</b></p> |

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| <p><b>Q40. The trigeminal nerve has three main branches. Which are they?</b></p> <p>A. Ophthalmic, maxillary and mandibular<br/> B. Buccal, lingual and inferior alveolar<br/> C. Facial, glossopharyngeal and vagus<br/> D. Frontal, nasal and zygomatic</p>                                                                                      | <p><b>Ans: A</b></p> |
| <p><b>Q41. Treatment of myofascial pain syndrome (characterized by muscle pain and spasm with typical trigger points) includes:</b></p> <p>A. Physiotherapy, TENS, acupuncture, topical cooling of muscle, and trigger point injection with local anesthetic<br/> B. Opioids only<br/> C. Surgical excision of muscle<br/> D. Neurolytic block</p> | <p><b>Ans: A</b></p> |
| <p><b>Q42. What is the usual indication for epidural steroids in low backache?</b></p> <p>A. Chronic disc prolapse with nerve root compression<br/> B. Facet arthropathy<br/> C. Sacroiliitis<br/> D. Vertebral prolapse</p>                                                                                                                       | <p><b>Ans: A</b></p> |
| <p><b>Q43. Before treating low backache with neurosurgical and/or orthopedic consultation, what should always be done?</b></p> <p>A. Rule out any surgical problem<br/> B. Start opioids immediately<br/> C. Perform a nerve block first<br/> D. Give epidural steroids first</p>                                                                  | <p><b>Ans: A</b></p> |
| <p><b>Q44. Which adjuvant therapy for cancer pain is a directly acting calcium channel blocker?</b></p> <p>A. Ziconotide<br/> B. Pregabalin<br/> C. Gabapentin<br/> D. Amitriptyline</p>                                                                                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q45. For severe intractable cancer pain not relieved by other means, which surgical intervention should be considered?</b></p> <p>A. Cordotomy<br/> B. Sympathectomy<br/> C. Laminectomy<br/> D. Rhizotomy</p>                                                                                                                               | <p><b>Ans: A</b></p> |
| <p><b>Q46. The McGill pain questionnaire (MPQ) contains 20 sets of descriptive words. It determines:</b></p> <p>A. Not only pain intensity but also effective and cognitive components of pain<br/> B. Only the pain intensity<br/> C. Only the location of pain<br/> D. Only the duration of pain</p>                                             | <p><b>Ans: A</b></p> |

**Q47. For cancer pain, in which patients is the neuraxial route (preservative-free morphine through epidural/spinal catheter) used, and how can it be given long-term?**

- A. If epidural/spinal is effective in relieving pain, the spinal catheter may be surgically implanted for long-term use
- B. It is never used for cancer pain
- C. It is used only for mild pain
- D. It is given orally

**Ans: A**

**Q48. Intravenous regional sympathetic block (chemical sympathectomy) with guanethidine for CRPS:**

- A. Was popular in the past, but is not recommended in current day practice (though intravenous regional block with local anesthetic is still performed occasionally)
- B. Is the treatment of choice currently
- C. Has never been used
- D. Is the first-line treatment

**Ans: A**

**Q49. Which regional blocks are used for pelvic malignancies (for chronic/cancer pain)?**

- A. Celiac plexus block, hypogastric block
- B. Stellate ganglion block
- C. Brachial plexus block
- D. Femoral nerve block

**Ans: A**

**Q50. For the maxillary and mandibular nerve block, an 8 cm long 22 G needle is inserted at the inferior edge of coronoid notch till the lateral pterygoid plate is encountered (usually at how much depth)?**

- A. 5 cm
- B. 2 cm
- C. 8 cm
- D. 10 cm

**Ans: A**

# Answer Key & Explanations

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## **Q1. Answer: A — The fifth vital sign**

Page 270: Pain has been designated as the fifth vital sign. Pain is a subjective feeling; there may be up to 10-fold variation in pain perception among individuals.

*Topic: Assessment of Pain*

## **Q2. Answer: A — Visual analog scale**

Page 270: The visual analog scale is the most commonly used scale. It is a horizontal line divided into 10 equal points (0–10). At one end (0 end), the label is no pain; at the other end (10 end) the label is worst pain imaginable. The patient is asked to mark the point where the pain lies.

*Topic: Assessment of Pain*

## **Q3. Answer: A — Children's Hospital of Eastern Ontario Pain Scale (CHEOPS)**

Page 270: For younger children (usually less than 3–4 years), the most commonly used scale to objectively assess the pain is Children's Hospital of Eastern Ontario Pain Scale (CHEOPS). It includes cry, facial expression, verbal response, torso (body position), touch (touching wound) and legs position.

*Topic: Assessment of Pain*

## **Q4. Answer: A — Multimodal approach (combination of different modalities)**

Page 270: Postoperative pain management should have multimodal approach, which means combination of different modalities to control postoperative pain. These include NSAIDs, acetaminophen, opioids, regional nerve blocks and other techniques.

*Topic: Method of Relief*

## **Q5. Answer: A — One third**

Page 270: NSAIDs or cyclo-oxygenase-2 (COX-2) inhibitors should always be the part of multimodal analgesia because of their anti-inflammatory property. They can reduce the dose requirement of opioids by almost one third.

*Topic: Method of Relief*

## **Q6. Answer: A — Epidural route**

Page 270: Opioids still remain the mainstay of treatment for postoperative analgesia. Although opioids can be given by almost all routes, epidural route is the most preferred route for postoperative analgesia.

*Topic: Method of Relief*

## **Q7. Answer: A — Bupivacaine (0.125–0.25%) or ropivacaine (0.2%) with/without fentanyl**

Page 270: Bupivacaine (0.125–0.25%) or ropivacaine (0.2%) with/without fentanyl is the most frequently used combination for controlling postoperative pain.

*Topic: Method of Relief*

## **Q8. Answer: A — Patient-controlled analgesia (PCA)**

Page 270–271: Excellent patient satisfaction seen with patient-controlled analgesia (PCA) has made it a method of choice for postoperative analgesia if opioids have to be given by parenteral route. PCA is an automatic system connected to an IV line; the patient presses a button which delivers a prefixed amount of opioid (usually morphine). A lockout interval is set to avoid over dosage.

*Topic: Method of Relief*

**Q9. Answer: A — They decrease the stress response, the requirement of opioids, and decrease/abolish the requirement and incidence of acute pain getting converted into chronic pain**

Page 271: Regional nerve blocks with local anesthetics as a single shot or continuous infusion provide excellent postoperative analgesia. The nerve block given before skin incision (called as pre-emptive analgesia) not only decreases the stress response and requirement of opioids but also decreases (or abolishes) the requirement and incidence of acute pain getting converted into chronic pain.

*Topic: Method of Relief*

**Q10. Answer: A — Transversus abdominis plane (TAP) block**

Page 271: The transversus abdominis plane (TAP) block is very effective to produce analgesia for abdominal surgeries such as cesarean section, hernia repair, etc. TAP block is given in the fascial sheath deep to the internal oblique muscle and superficial to the transversus abdominis muscle.

*Topic: Method of Relief*

**Q11. Answer: A — 25–30%**

Page 271: Incidence of chronic pain can be as high as 25–30%. Chronic pain has most tangled and complicated pathophysiology. Chronic pain not only causes physical pain but damages the patient psychologically and emotionally also.

*Topic: Chronic Pain Management*

**Q12. Answer: A — Radiofrequency ablation (RFA) is always preferred over chemical neurolysis**

Page 271: Neurolysis can be done with chemical agents (5% phenol in glycerin or absolute alcohol) or with radiofrequency probes, which ablate the nerve by burning at 60–90°C. As chemical neurolysis can damage adjacent structures, radiofrequency ablation (RFA) is always preferred over chemical neurolysis. Neurolytic blocks usually provide analgesia for 6 months to 2 years.

*Topic: Chronic Pain Management*

**Q13. Answer: A — Transforaminal route (drug injected as the nerve exits through the foramina)**

Page 271: Studies have shown that only 40–50% of drug reaches the nerve roots if steroids are given through epidural route, therefore nowadays the transforaminal route (drug injected as the nerve exits through the foramina) is always preferred over epidural route.

*Topic: Low Backache*

**Q14. Answer: A — Radiofrequency (RF) ablation of medial branch of posterior ramus of spinal nerve**

Page 272: Radiofrequency (RF) ablation of medial branch of posterior ramus of spinal nerve (supplying the facet joint) is the most definitive modality of treatment for facet arthropathy.

*Topic: Low Backache*

**Q15. Answer: A — Diabetic neuropathy**

Page 272: The most common neuropathy encountered in clinical practice is diabetic neuropathy. The drug of choice for neuropathic pain is pregabalin (except trigeminal neuralgia which still responds best to carbamazepine).

*Topic: Neuropathic Pain*

**Q16. Answer: A — Pregabalin (followed by gabapentin)**

Page 272: The drug of choice for neuropathic pain is pregabalin (followed by gabapentin), except trigeminal neuralgia which still responds best to carbamazepine.

*Topic: Neuropathic Pain*

**Q17. Answer: A — Maxillary division; carbamazepine**

Page 272: In trigeminal neuralgia, the pain is most often in the maxillary division. The drug of choice is carbamazepine. Patients not responding to carbamazepine are subjected to radiofrequency ablation of Gasserian ganglion or neurolytic block of involved segment of trigeminal nerve.

Topic: Trigeminal Neuralgia

**Q18. Answer: A — Causalgia**

Page 272: CRPS is classified as Type I, which includes reflex sympathetic dystrophies such as Raynaud's phenomenon/disease, Berger's disease, phantom limb, Sudeck's dystrophy; and Type II, which includes causalgia (neuralgic pain after direct nerve injury is called as causalgia).

Topic: Complex Regional Pain Syndrome

**Q19. Answer: A — Extremely painful limb (hyperalgesia, allodynia, hyperpathia, anesthesia dolorosa)**

Page 272: The hallmark of CRPS is extremely painful limb. The limb becomes extremely sensitive to pain (hyperalgesia); even a light touch can produce pain (allodynia), pain may persist after the end of stimulus (hyperpathia), or patient may feel pain in anesthetized area (anesthesia dolorosa).

Topic: Complex Regional Pain Syndrome

**Q20. Answer: A — Sympathetic blocks; stellate ganglion block for upper limb**

Page 272: The mainstay of treatment of CRPS is sympathetic blocks. For upper limb CRPS, stellate ganglion block is performed; and for lower limb dystrophies, lumbar sympathetic chain block is performed. Usually 3–7 blocks are required to break the cycle of pain.

Topic: Complex Regional Pain Syndrome

**Q21. Answer: A — Multiple trigger points**

Page 272: Fibromyalgia is a very common pain disorder characterized by pain in different groups of muscles, easy fatigability and sleep deprivation. The multiple trigger points are pathognomonic of fibromyalgia.

Topic: Fibromyalgia

**Q22. Answer: A — Lower cervical and first thoracic ganglion**

Page 273: Stellate ganglion is formed by the fusion of lower cervical and first thoracic ganglion. It is blocked anterior to the tubercle of transverse process of C6 vertebra called as Chassaignac tubercle.

Topic: Stellate Ganglion Block

**Q23. Answer: A — Conjunctival congestion**

Page 273: Signs of successful stellate ganglion block include Horner syndrome (miosis, enophthalmos, ptosis, anhydrosis), flushing of face, conjunctival congestion (most often the first sign), nasal stuffiness (Guttman's sign), congestion of tympanic membrane (Mueller's syndrome), increased skin temperature and lacrimation.

Topic: Stellate Ganglion Block

**Q24. Answer: A — Intercostal nerve block**

Page 274: After intercostal block, the highest blood level of local anesthetic is achieved per volume of drug injected. Complications include pneumothorax.

Topic: Intercostal Nerve Block

**Q25. Answer: A — Hypotension**

Page 274: Celiac plexus block is usually given for pain relief in gastric and pancreatic malignancy. The most common complication is hypotension.

Topic: Celiac Plexus Block

**Q26. Answer: A — A step ladder (non-opioids for mild, weak opioids for moderate, strong opioids for severe pain)**

Page 272: For cancer pain, the WHO recommends step ladder for control: non-opioids like NSAIDs, paracetamol for mild pain; weak opioids like codeine for moderate pain; strong opioids like morphine for severe pain. Opioid remains the mainstay of treatment for cancer pain.

Topic: Cancer Pain

**Q27. Answer: A — Opioids**

Page 272: Opioid remains the mainstay of treatment for cancer pain. Depending on the severity and feasibility, opioids are given through many routes (oral, transdermal patch, parenteral, neuraxial, regional blocks).

Topic: Cancer Pain

**Q28. Answer: A — Fentanyl patch 48–72 hours; butorphanol patch 6–7 days**

Page 272: Transdermal fentanyl/butorphanol patch — the patch is a very good alternative to oral route. Fentanyl patch provides analgesia for 48–72 hours while butorphanol patch provides analgesia for 6–7 days.

Topic: Cancer Pain

**Q29. Answer: A — Immunosuppression, poor wound healing, increased myocardial oxygen demand, decreased GI motility and chronic postsurgical pain**

Page 270: Inadequate pain control in postoperative period can elicit the stress responses which can manifest as immunosuppression, poor wound healing, increased myocardial oxygen demand, decreased gastrointestinal motility and chronic postsurgical pain.

Topic: Acute Pain Management

**Q30. Answer: A — NSAIDs are contraindicated**

Page 270: Acetaminophen (paracetamol) along with the NSAIDs or regional nerve blocks may omit the need for opioids. It may serve as an alternative to NSAIDs in the patients where NSAIDs are contraindicated.

Topic: Method of Relief

**Q31. Answer: A — Delays lead to escalation of pain, making analgesic to become less effective; pain is best controlled when the analgesic is delivered before or just at the beginning of pain**

Page 271: The rationale for the success of PCA is that pain is best controlled when the analgesic is delivered either before or just at the beginning of pain. The delays in the delivery of analgesics by nurses lead to escalation of pain (making analgesic to become less effective) which can be avoided with PCA devices.

Topic: Method of Relief

**Q32. Answer: A — Thoracic paravertebral block**

Page 271: Thoracic paravertebral block is used to provide analgesia for thoracic, breast and upper abdominal surgeries.

Topic: Method of Relief

**Q33. Answer: A — TENS, acupuncture, cryoanalgesia and ketamine infusion**

Page 271: Other techniques occasionally utilized to achieve postoperative analgesia are transcutaneous electrical nerve stimulation (TENS), acupuncture, cryoanalgesia (produced by cryoprobes using compressed CO<sub>2</sub> or nitrous oxide, cooling peripheral nerves to –5 to –20°C), and ketamine infusion in analgesic doses.

Topic: Method of Relief



**Q34. Answer: A — Tricyclic antidepressants, intercostal nerve blocks, transdermal lignocaine patch / EMLA cream, and topical capsaicin cream or patch**

Page 272: Treatment of postherpetic neuralgia includes tricyclic antidepressants, intercostal nerve blocks, transdermal lignocaine patch / topical application of eutectic mixture of lignocaine and prilocaine (EMLA) cream, and topical application of capsaicin cream or patch.

*Topic: Postherpetic Neuralgia*

**Q35. Answer: A — Lumbar plexus block (psoas compartment block)**

Page 271: Lumbar plexus block (also known as psoas compartment block) is given to provide analgesia for hip and knee surgery. Femoral nerve block and fascia iliaca (modified femoral) block are utilized for hip fractures and hip surgeries.

*Topic: Method of Relief*

**Q36. Answer: A — Ganglion impar (ganglion of Walther) block**

Page 274: Ganglion impar (ganglion of Walther) block is used for pain relief of perineal area. Superior hypogastric block is performed for pain relief for pelvic malignancies.

*Topic: Commonly Used Pain Blocks*

**Q37. Answer: A — Anterior to lumbar vertebral bodies**

Page 274: Lumbar sympathetic chain block is usually utilized for CRPS of lower limb, especially Berger's disease. It is blocked anterior to lumbar vertebral bodies.

*Topic: Commonly Used Pain Blocks*

**Q38. Answer: A — Intra-arterial thiopentone injection and hand pallor (ischemia) following radial artery cannulation for invasive blood pressure monitoring**

Page 273: Stellate ganglion block is usually employed for complex regional pain syndromes (CRPS) of upper limb. Other indications may be intra-arterial thiopentone injection and hand pallor (ischemia) following radial artery cannulation for invasive blood pressure monitoring.

*Topic: Stellate Ganglion Block*

**Q39. Answer: A — Unilateral phrenic nerve block**

Page 273: Among complications of stellate ganglion block, phrenic nerve block, however, unilateral phrenic nerve block is not bothersome; it just reduces the vital capacity by 20–25%. Other complications include brachial plexus block, recurrent laryngeal nerve block (hoarseness), pneumothorax, esophageal perforation, hypotension and bradycardia.

*Topic: Stellate Ganglion Block*

**Q40. Answer: A — Ophthalmic, maxillary and mandibular**

Page 273: Trigeminal nerve has three main branches — ophthalmic, maxillary and mandibular. The principal indication of trigeminal nerve block is trigeminal neuralgia. Depending on the site of involvement, block may be given at Gasserian ganglion level (at foramen ovale) or at one of the branches.

*Topic: Trigeminal Nerve Block*

**Q41. Answer: A — Physiotherapy, TENS, acupuncture, topical cooling of muscle, and trigger point injection with local anesthetic**

Page 272: Myofascial pain syndrome is a syndrome characterized by muscle pain and muscle spasm. Painful points (trigger points) are typical of myofascial pain syndromes. Treatment includes physiotherapy, transcutaneous electrical nerve stimulation (TENS), acupuncture, topical cooling of muscle and trigger point injection with local anesthetic.

*Topic: Myofascial Pain Syndrome*

**Q42. Answer: A — Chronic disc prolapse with nerve root compression**

Page 271: The usual indication for epidural steroids in low backache is chronic disc prolapse with nerve root compression.

Topic: Low Backache

**Q43. Answer: A — Rule out any surgical problem**

Page 271: Low backache is one of the most commonly encountered conditions in clinical practice. Before treating low backache, neurosurgical and/or orthopedic consultation should always be sought to rule out any surgical problem.

Topic: Low Backache

**Q44. Answer: A — Ziconotide**

Page 273: Adjuvant therapies for cancer pain like antidepressants, pregabalin/gabapentin and ziconotide (directly acting calcium channel blocker) are used. For intractable pain, surgical intervention (cordotomy) should be considered.

Topic: Cancer Pain

**Q45. Answer: A — Cordotomy**

Page 273: For intractable cancer pain, surgical intervention (cordotomy) should be considered. If a block with local anesthetic is successful, then only neurolytic block should be done.

Topic: Cancer Pain

**Q46. Answer: A — Not only pain intensity but also effective and cognitive components of pain**

Page 270: McGill pain questionnaire (MPQ) contains 20 sets of descriptive words. It not only determines pain intensity but also effective and cognitive components of pain.

Topic: Assessment of Pain

**Q47. Answer: A — If epidural/spinal is effective in relieving pain, the spinal catheter may be surgically implanted for long-term use**

Page 273: Neuraxial route — preservative-free morphine is used through epidural/spinal catheter. If this is effective in relieving pain, the spinal catheter may be surgically implanted for long-term use.

Topic: Cancer Pain

**Q48. Answer: A — Was popular in the past, but is not recommended in current day practice (though intravenous regional block with local anesthetic is still performed occasionally)**

Page 272: Intravenous regional sympathetic block (chemical sympathectomy) with guanethidine, which was popular in past, is not recommended in current-day practice; however, intravenous regional block with local anesthetic is still performed occasionally.

Topic: Complex Regional Pain Syndrome

**Q49. Answer: A — Celiac plexus block, hypogastric block**

Page 273: Regional blocks like celiac plexus block for pancreatic and stomach malignancy, and hypogastric block for pelvic malignancies, are used in cancer pain management.

Topic: Cancer Pain

**Q50. Answer: A — 5 cm**

Page 274: For the maxillary and mandibular nerve block, an 8 cm long 22 G needle is inserted at the inferior edge of coronoid notch till lateral pterygoid plate is encountered (usually at 5 cm). The needle is walked off anteriorly for 0.5 cm to block the maxillary nerve and posteriorly to block mandibular nerve.

Topic: Trigeminal Nerve Block

# Intensive Care Management MCQ – Chapter 40

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 9: Cardiorespiratory Care (Chapter 40)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

**Q1. The first intensive care unit (ICU) was started in which year?**

- A. 1945
- B. 1950
- C. 1953
- D. 1960

**Ans: C**

**Q2. What percentage of hospital beds should be allocated to the ICU?**

- A. 5%
- B. 10%
- C. 15%
- D. 20%

**Ans: B**

**Q3. Acute respiratory failure is considered to be present when  $pO_2$  is less than:**

- A. 40 mm Hg
- B. 50 mm Hg
- C. 60 mm Hg
- D. 70 mm Hg

**Ans: C**

**Q4. Elevated  $pCO_2$  in the presence of normal pH constitutes:**

- A. Acute respiratory failure
- B. Chronic respiratory failure
- C. Combined respiratory failure
- D. Metabolic acidosis

**Ans: B**

**Q5. Type II respiratory failure is characterized by all of the following EXCEPT:**

- A. Increased  $pCO_2$  (>50 mm Hg)
- B.  $pO_2$  is usually normal
- C. Increased  $pAO_2 - paO_2$  gradient
- D. Normal  $pAO_2 - paO_2$  gradient

**Ans: C**

**Q6. Which of the following is a cause of Type II (hypercapnic ventilatory) respiratory failure?**

- A. Pulmonary embolism
- B. Pulmonary edema
- C. Guillain-Barré syndrome
- D. Interstitial lung disease

**Ans: C**

**Q7. Oxygen therapy is helpful in improving oxygen saturation in all types of hypoxia EXCEPT:**

- A. Hypoxic hypoxia
- B. Anemic hypoxia
- C. Stagnant hypoxia
- D. Histotoxic hypoxia

**Ans: D**

|                                                                                                                                                                                                                                                                                                 |                      |
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| <p><b>Q8. Ideally, inspired oxygen concentration (FiO<sub>2</sub>) should not exceed more than what percentage to avoid oxygen toxicity?</b></p> <p>A. 30%<br/>B. 40%<br/>C. 50%<br/>D. 60%</p>                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q9. Which of the following is an indication for mechanical ventilation based on pulmonary function?</b></p> <p>A. Vital capacity &gt;15 mL/kg<br/>B. Respiratory rate &gt;35/minute<br/>C. Dead space volume (VD/VT) &lt;0.6<br/>D. Tidal volume &gt;5 mL/kg</p>                          | <p><b>Ans: B</b></p> |
| <p><b>Q10. On the basis of blood gas analysis, mechanical ventilation is indicated when pO<sub>2</sub>/FiO<sub>2</sub> is less than:</b></p> <p>A. 150 mm Hg<br/>B. 200 mm Hg<br/>C. 250 mm Hg<br/>D. 300 mm Hg</p>                                                                             | <p><b>Ans: C</b></p> |
| <p><b>Q11. The chief advantage of volume preset (volume controlled) ventilation is:</b></p> <p>A. Decreased chances of barotrauma<br/>B. Assurance of delivery of preset tidal volume<br/>C. Maintenance of preset airway pressure<br/>D. Better patient comfort</p>                            | <p><b>Ans: B</b></p> |
| <p><b>Q12. The usual initial tidal volume setting on a ventilator is:</b></p> <p>A. 4–5 mL/kg<br/>B. 6–8 mL/kg<br/>C. 10–12 mL/kg<br/>D. 12–15 mL/kg</p>                                                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q13. The normal inspiratory to expiratory (I:E) ratio set on a ventilator is:</b></p> <p>A. 1:1<br/>B. 1:2<br/>C. 2:1<br/>D. 1:3</p>                                                                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q14. The major disadvantage of controlled-mode ventilation (CMV) is:</b></p> <p>A. Hyperventilation<br/>B. Increased work of breathing<br/>C. Intrathoracic pressure always remains positive, decreasing venous return and cardiac output<br/>D. Cannot deliver adequate tidal volume</p> | <p><b>Ans: C</b></p> |

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| <p><b>Q15. In Assist-Control (A/C) ventilation, if the preset rate is 10 breaths/min and the patient's spontaneous rate is 12, the ventilator will:</b></p> <p>A. Deliver 10 breaths in controlled mode<br/> B. Assist 10 breaths and deliver 2 in control mode<br/> C. Assist all 12 breaths<br/> D. Deliver 12 breaths in controlled mode</p> | <p><b>Ans: C</b></p> |
| <p><b>Q16. Which of the following is an advantage of SIMV over CMV?</b></p> <p>A. Less work of breathing<br/> B. Less hypotension due to spontaneous breaths allowing good venous return<br/> C. Better control of tidal volume<br/> D. Less hyperventilation</p>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q17. In Inverse Ratio Ventilation (IRV), the I:E ratio is reversed to:</b></p> <p>A. 1:1<br/> B. 1:3<br/> C. 2:1<br/> D. 3:1</p>                                                                                                                                                                                                          | <p><b>Ans: C</b></p> |
| <p><b>Q18. In Pressure Support Ventilation (PSV), the ventilator cycles to expiration when the predetermined flow falls below:</b></p> <p>A. 10%<br/> B. 15%<br/> C. 20%<br/> D. 25%</p>                                                                                                                                                        | <p><b>Ans: D</b></p> |
| <p><b>Q19. The most commonly used dual mode of ventilation in India is:</b></p> <p>A. Adaptive support ventilation (ASV)<br/> B. Pressure-regulated volume control (PRVC)<br/> C. Volume-assured pressure support (VAPS)<br/> D. Automode ventilation</p>                                                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q20. High-frequency jet ventilation operates at a frequency of:</b></p> <p>A. 60–120 cycles/min<br/> B. 100–300 cycles/min<br/> C. 300–600 cycles/min<br/> D. 600–3000 cycles/min</p>                                                                                                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q21. Which is the most commonly used mode for high-frequency ventilation?</b></p> <p>A. High-frequency positive-pressure ventilation<br/> B. High-frequency jet ventilation<br/> C. High-frequency oscillation<br/> D. High-frequency percussive ventilation</p>                                                                          | <p><b>Ans: B</b></p> |

**Q22. The ideal PEEP is defined as the PEEP which maintains oxygen saturation >90% without decreasing the cardiac output significantly. The usual range is:**

- A. 2–5 cm H<sub>2</sub>O
- B. 5–12 cm H<sub>2</sub>O
- C. 12–20 cm H<sub>2</sub>O
- D. 20–25 cm H<sub>2</sub>O

**Ans: B**

**Q23. CPAP (Continuous Positive Airway Pressure) is essentially:**

- A. A mode of mechanical ventilation
- B. PEEP given to a nonintubated patient
- C. Pressure support ventilation through a mask
- D. Bilevel positive airway pressure

**Ans: B**

**Q24. In BIPAP, the typical inspiratory positive airway pressure (IPAP) setting is:**

- A. 2–5 cm H<sub>2</sub>O
- B. 5–10 cm H<sub>2</sub>O
- C. 8–20 cm H<sub>2</sub>O
- D. 20–30 cm H<sub>2</sub>O

**Ans: C**

**Q25. What is the most common indication of NIPPV in the acute setting?**

- A. Sleep apnea syndrome
- B. Acute exacerbation of COPD
- C. Cardiogenic pulmonary edema
- D. Pneumothorax

**Ans: B**

**Q26. What is the major problem during NIPPV?**

- A. Aspiration
- B. Claustrophobia
- C. Leakage
- D. Skin breakdown

**Ans: C**

**Q27. For weaning from ventilator, the Rapid Shallow Breathing Index (RSBI) should be:**

- A. <50
- B. <80
- C. <100
- D. <150

**Ans: C**

**Q28. Weaning from ventilator is possible in all modes of ventilation EXCEPT:**

- A. Assist-Control mode
- B. SIMV mode
- C. Controlled-mode ventilation
- D. Pressure Support Ventilation

**Ans: C**

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| <p><b>Q29. The incidence of pulmonary barotrauma during mechanical ventilation is:</b></p> <ul style="list-style-type: none"> <li>A. 1–3%</li> <li>B. 5–7%</li> <li>C. 7–10%</li> <li>D. 15–20%</li> </ul>                                                                                                                                       | Ans: C |
| <p><b>Q30. The most common source of nosocomial infections in ICU is:</b></p> <ul style="list-style-type: none"> <li>A. Pulmonary infections</li> <li>B. Urinary tract infections</li> <li>C. Bloodstream infections</li> <li>D. Wound infections</li> </ul>                                                                                     | Ans: B |
| <p><b>Q31. Ventilator-associated pneumonia (VAP) is defined as pneumonia developing in patients on ventilators for more than:</b></p> <ul style="list-style-type: none"> <li>A. 24 hours</li> <li>B. 48 hours</li> <li>C. 72 hours</li> <li>D. 96 hours</li> </ul>                                                                               | Ans: B |
| <p><b>Q32. The simplest and most cost-effective method to decrease ventilator-associated pneumonia is:</b></p> <ul style="list-style-type: none"> <li>A. Use of subglottic suctioning endotracheal tubes</li> <li>B. Oral hygiene with chlorhexidine</li> <li>C. Positioning head of bed at 30 degrees</li> <li>D. Early tracheostomy</li> </ul> | Ans: C |
| <p><b>Q33. Shock is classified into how many major types?</b></p> <ul style="list-style-type: none"> <li>A. Two</li> <li>B. Three</li> <li>C. Four</li> <li>D. Five</li> </ul>                                                                                                                                                                   | Ans: C |
| <p><b>Q34. In cardiogenic shock, the pulmonary capillary wedge pressure (PCWP) in left ventricular failure is typically:</b></p> <ul style="list-style-type: none"> <li>A. &gt;15 mm Hg</li> <li>B. &gt;18 mm Hg</li> <li>C. &gt;25 mm Hg</li> <li>D. &gt;30 mm Hg</li> </ul>                                                                    | Ans: C |
| <p><b>Q35. The hallmark hemodynamic finding in distributive shock is:</b></p> <ul style="list-style-type: none"> <li>A. Increased CVP</li> <li>B. Increased PCWP</li> <li>C. Decreased SVR</li> <li>D. Decreased cardiac output</li> </ul>                                                                                                       | Ans: C |



|                                                                                                                                                                                                                                                                                                                                  |                      |
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| <p><b>Q36. In hemorrhagic shock, the current guidelines recommend maintaining systolic blood pressure (SBP) between:</b></p> <ul style="list-style-type: none"> <li>A. 60–80 mm Hg</li> <li>B. 80–100 mm Hg</li> <li>C. 100–120 mm Hg</li> <li>D. 120–140 mm Hg</li> </ul>                                                       | <p><b>Ans: B</b></p> |
| <p><b>Q37. The concept of combining hypotensive resuscitation and hemostatic resuscitation is called:</b></p> <ul style="list-style-type: none"> <li>A. Goal-directed therapy</li> <li>B. Balanced resuscitation</li> <li>C. Damage control resuscitation</li> <li>D. Permissive hypotension</li> </ul>                          | <p><b>Ans: B</b></p> |
| <p><b>Q38. Which parameter is considered the most reliable to assess fluid status and titrate fluid therapy?</b></p> <ul style="list-style-type: none"> <li>A. Central venous pressure (CVP)</li> <li>B. Pulse pressure variation</li> <li>C. Stroke volume variation</li> <li>D. Pulmonary artery occlusion pressure</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q39. The most preferred crystalloid for fluid resuscitation in shock is:</b></p> <ul style="list-style-type: none"> <li>A. Normal saline</li> <li>B. Ringer's lactate</li> <li>C. 5% dextrose</li> <li>D. Dextrose-normal saline</li> </ul>                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q40. The vasopressor of choice for septicemic shock is:</b></p> <ul style="list-style-type: none"> <li>A. Phenylephrine</li> <li>B. Ephedrine</li> <li>C. Norepinephrine (noradrenaline)</li> <li>D. Vasopressin</li> </ul>                                                                                                | <p><b>Ans: C</b></p> |
| <p><b>Q41. Steroids are contraindicated in which type of shock?</b></p> <ul style="list-style-type: none"> <li>A. Septicemic shock</li> <li>B. Hypovolemic shock</li> <li>C. Cardiogenic shock</li> <li>D. Neurogenic shock</li> </ul>                                                                                           | <p><b>Ans: C</b></p> |
| <p><b>Q42. The vein of choice for parenteral nutrition is:</b></p> <ul style="list-style-type: none"> <li>A. Internal jugular vein</li> <li>B. Subclavian vein</li> <li>C. Femoral vein</li> <li>D. Antecubital vein</li> </ul>                                                                                                  | <p><b>Ans: B</b></p> |

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| <p><b>Q43. Out of total energy requirement, what percentage should be provided by carbohydrates?</b></p> <p>A. 40–50%<br/>B. 50–60%<br/>C. 60–70%<br/>D. 70–80%</p>                                                                                                                                                                                                                                                                                                                 | Ans: C |
| <p><b>Q44. Sepsis is the leading cause of death in ICU. The most common cause of death in septicemia is:</b></p> <p>A. Cardiogenic shock<br/>B. ARDS and renal failure<br/>C. Disseminated intravascular coagulation<br/>D. Hepatic failure</p>                                                                                                                                                                                                                                     | Ans: B |
| <p><b>Q45. Pulmonary edema develops when pulmonary artery occlusion pressure (left atrial pressure) rises to more than:</b></p> <p>A. 15 mm Hg<br/>B. 18 mm Hg<br/>C. 25 mm Hg<br/>D. 35 mm Hg</p>                                                                                                                                                                                                                                                                                  | Ans: C |
| <p><b>Q46. According to the Berlin definition, severe ARDS is defined as <math>\text{PaO}_2/\text{FiO}_2</math>:</b></p> <p>A. <math>\leq 300</math> mm Hg with PEEP <math>\geq 5</math> cm H<math>_2</math>O<br/>B. <math>\leq 200</math> mm Hg with PEEP <math>\geq 5</math> cm H<math>_2</math>O<br/>C. <math>\leq 100</math> mm Hg with PEEP <math>\geq 5</math> cm H<math>_2</math>O<br/>D. <math>\leq 50</math> mm Hg with PEEP <math>\geq 10</math> cm H<math>_2</math>O</p> | Ans: C |
| <p><b>Q47. The most common cause of ARDS is:</b></p> <p>A. Polytrauma<br/>B. Pancreatitis<br/>C. Sepsis<br/>D. Aspiration of gastric contents</p>                                                                                                                                                                                                                                                                                                                                   | Ans: C |
| <p><b>Q48. Carbon monoxide poisoning is usually defined as carboxyhemoglobin greater than what percentage in blood?</b></p> <p>A. 10%<br/>B. 15%<br/>C. 20%<br/>D. 30%</p>                                                                                                                                                                                                                                                                                                          | Ans: C |
| <p><b>Q49. The normal anion gap is:</b></p> <p>A. 4–8 mM/L<br/>B. 8–16 mM/L<br/>C. 16–24 mM/L<br/>D. 24–32 mM/L</p>                                                                                                                                                                                                                                                                                                                                                                 | Ans: B |

**Q50. The most common cause of metabolic acidosis in anesthesia is:**

- A. Renal failure
- B. Diabetic ketoacidosis
- C. Lactic acidosis due to tissue hypoxia
- D. Diarrhea with bicarbonate loss

**Ans: C**

# Answer Key & Explanations

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## Q1. Answer: C — 1953

According to the textbook, the first intensive care unit (ICU) was started in 1953.

Topic: ICU Setup | Ref: Ch 40, Section: Intensive Care Unit, Page 277

## Q2. Answer: B — 10%

Total number of beds in ICU should be 10% of hospital beds, with each bed having 20 m<sup>2</sup> floor area.

Topic: ICU Setup | Ref: Ch 40, Section: Intensive Care Unit, Page 277

## Q3. Answer: C — 60 mm Hg

Acute respiratory failure is considered to be present if pO<sub>2</sub> is less than 60 mm Hg, pCO<sub>2</sub> over 50 mm Hg and pH less than 7.30.

Topic: Acute Respiratory Failure | Ref: Ch 40, Section: Acute Respiratory Failure – Definition, Page 277

## Q4. Answer: B — Chronic respiratory failure

Elevated pCO<sub>2</sub> in presence of normal pH constitutes chronic respiratory failure, as pH is compensated by renal tubular reabsorption of bicarbonate.

Topic: Acute Respiratory Failure | Ref: Ch 40, Section: Acute Respiratory Failure – Definition, Page 277

## Q5. Answer: C — Increased pAO<sub>2</sub>–paO<sub>2</sub> gradient

In Type II (hypercapnic ventilatory failure), pCO<sub>2</sub> is >50 mm Hg, pO<sub>2</sub> is usually normal, and pAO<sub>2</sub>–paO<sub>2</sub> gradient is normal. Increased pAO<sub>2</sub>–paO<sub>2</sub> gradient is a feature of Type I failure.

Topic: Classification of Respiratory Failure | Ref: Ch 40, Section: Classification, Page 277–278

## Q6. Answer: C — Guillain-Barré syndrome

Guillain-Barré syndrome is a disorder affecting signal transmission to respiratory muscles and is a cause of Type II failure. The other options are causes of Type I failure.

Topic: Classification of Respiratory Failure | Ref: Ch 40, Section: Type II Failure, Page 277–278

## Q7. Answer: D — Histotoxic hypoxia

Oxygen therapy is helpful in improving oxygen saturation in all types of hypoxia except histotoxic hypoxia. Maximum benefit is seen in hypoxic hypoxia.

Topic: Supplemental Oxygen | Ref: Ch 40, Section: Supplemental Oxygen, Page 278

## Q8. Answer: C — 50%

Ideally, inspired oxygen concentration (FiO<sub>2</sub>) should not exceed more than 50%, otherwise oxygen toxicity can occur.

Topic: Supplemental Oxygen | Ref: Ch 40, Section: Supplemental Oxygen, Page 278

## Q9. Answer: B — Respiratory rate >35/minute

Indications for mechanical ventilation based on pulmonary functions include respiratory rate >35/min, vital capacity <15 mL/kg, VD/VT >0.6, and tidal volume <5 mL/kg.

Topic: Mechanical Ventilation | Ref: Ch 40, Section: Mechanical Ventilation – Indications, Page 278

**Q10. Answer: C — 250 mm Hg**

Mechanical ventilation is indicated when  $pO_2/FiO_2 < 250$  mm Hg (normal  $> 400$ ).

Topic: Mechanical Ventilation | Ref: Ch 40, Section: Mechanical Ventilation – Indications, Page 278

**Q11. Answer: B — Assurance of delivery of preset tidal volume**

The chief advantage of volume preset breath is assurance of delivery of preset tidal volume and hence decreased chances of hypoventilation; however, the chief disadvantage is increased chances of barotrauma.

Topic: Ventilators | Ref: Ch 40, Section: Ventilators, Page 279

**Q12. Answer: B — 6–8 mL/kg**

The usual initial tidal volume set on ventilator is 6–8 mL/kg.

Topic: Ventilator Settings | Ref: Ch 40, Section: Initial Setting of Ventilator, Page 279

**Q13. Answer: B — 1:2**

The usual initial I:E ratio set on ventilator is 1:2.

Topic: Ventilator Settings | Ref: Ch 40, Section: Initial Setting of Ventilator, Page 279

**Q14. Answer: C — Intrathoracic pressure always remains positive, decreasing venous return and cardiac output**

The major disadvantage of CMV is that the intrathoracic pressure always remains positive, decreasing the venous return and cardiac output.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – CMV, Page 279

**Q15. Answer: C — Assist all 12 breaths**

In A/C mode, if the patient's spontaneous rate exceeds the set rate (e.g., 12 vs set rate of 10), the ventilator will assist all 12 breaths, working in assist mode only.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – A/C, Page 280

**Q16. Answer: B — Less hypotension due to spontaneous breaths allowing good venous return**

Advantages of SIMV over CMV include less hypotension (as good venous return occurs during spontaneous breaths), no need for heavy sedation/muscle relaxants, and less sense of breathlessness.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – SIMV, Page 280

**Q17. Answer: C — 2:1**

In IRV, the normal I:E ratio of 1:2 is reversed to 2:1. Doubling the inspiratory time doubles the gas exchange time.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – IRV, Page 280

**Q18. Answer: D — 25%**

In PSV, the ventilator will cycle to expiration once the predetermined flow falls below 25% (flow cycled).

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – PSV, Page 280

**Q19. Answer: B — Pressure-regulated volume control (PRVC)**

The most commonly used dual mode in India is pressure-regulated volume control (PRVC). PRVC can be considered as the mode of choice for ventilation in current day critical care practice.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Dual Mode Ventilation, Page 281

**Q20. Answer: B — 100–300 cycles/min**

High-frequency jet ventilation operates at a frequency of 100–300 cycles/min with gases at high pressure of 60 psi. It is the most commonly used mode for high frequency ventilation.

Topic: High-Frequency Ventilation | Ref: Ch 40, Section: High-Frequency Ventilation, Page 281

**Q21. Answer: B — High-frequency jet ventilation**

High-frequency jet ventilation is the most commonly used mode for high frequency ventilation, with frequency of 100–300 cycles/min.

Topic: High-Frequency Ventilation | Ref: Ch 40, Section: High-Frequency Ventilation, Page 281

**Q22. Answer: B — 5–12 cm H<sub>2</sub>O**

The ideal PEEP which maintains oxygen saturation >90% without decreasing the cardiac output significantly is usually 5–12 cm H<sub>2</sub>O.

Topic: PEEP | Ref: Ch 40, Section: PEEP, Page 282

**Q23. Answer: B — PEEP given to a nonintubated patient**

CPAP is described as a misnomer. It is just PEEP given to a nonintubated patient, i.e., the noninvasive method of giving PEEP through a tight fitting mask.

Topic: CPAP | Ref: Ch 40, Section: CPAP, Page 282

**Q24. Answer: C — 8–20 cm H<sub>2</sub>O**

Typical BIPAP settings are 8–20 cm H<sub>2</sub>O positive pressure during inspiration (IPAP) and 5 cm H<sub>2</sub>O during expiration (EPAP).

Topic: BIPAP | Ref: Ch 40, Section: BIPAP, Page 282

**Q25. Answer: B — Acute exacerbation of COPD**

In the acute setting, NIPPV is most frequently applied for acute exacerbation of COPD. In the chronic setting, it is most effectively used for sleep apnea syndrome.

Topic: NIPPV | Ref: Ch 40, Section: NIPPV – Common Indications, Page 282

**Q26. Answer: C — Leakage**

Leakage is a major problem during NIPPV. Significant leaks can cause hypoventilation and are best avoided by using tight fitting masks.

Topic: NIPPV | Ref: Ch 40, Section: NIPPV – Complications, Page 283

**Q27. Answer: C — <100**

RSBI = Respiratory rate (breaths/min) / Tidal volume (in liters). RSBI should be <100 for successful weaning.

Topic: Weaning from Ventilator | Ref: Ch 40, Section: Weaning from Ventilator, Page 283

**Q28. Answer: C — Controlled-mode ventilation**

It is possible to wean a patient in any mode of ventilation except control mode ventilation.

Topic: Weaning from Ventilator | Ref: Ch 40, Section: Method of Weaning, Page 283

**Q29. Answer: C — 7–10%**

The incidence of barotrauma is 7–10%. It may manifest as pneumothorax, pneumomediastinum, pneumopericardium, pneumoperitoneum, bronchopleural fistula and air embolism.

Topic: Complications of Mechanical Ventilation | Ref: Ch 40, Section: Complications of Mechanical Ventilation, Page 283–284

**Q30. Answer: B — Urinary tract infections**

The most common source of nosocomial infections in ICU are urinary tract infections accounting for 35–40% of total ICU infections, followed by pulmonary (25–30%) and bloodstream infections (20%).

Topic: ICU Infections | Ref: Ch 40, Section: Complications – Infections, Page 284

**Q31. Answer: B — 48 hours**

VAP is defined as pneumonia developing in patients on ventilators for >48 hours with no evidence of preintubation pneumonia. The incidence of VAP is around 20%.

Topic: VAP | Ref: Ch 40, Section: Complications – VAP, Page 284

**Q32. Answer: C — Positioning head of bed at 30 degrees**

Positioning the head of the bed at 30 degrees decreases gastroesophageal reflux and proves to be the simplest and most cost-effective method to decrease VAP.

Topic: VAP Prevention | Ref: Ch 40, Section: Complications – VAP, Page 284

**Q33. Answer: C — Four**

Shock is classified into four major types: hypovolemic, cardiogenic, distributive, and obstructive.

Topic: Shock Classification | Ref: Ch 40, Section: Shock – Classification, Page 285

**Q34. Answer: C — >25 mm Hg**

In cardiogenic shock, poor pump function leads to increased CVP and increased PCWP. In LVF, PCWP is >25 mm Hg.

Topic: Shock Hemodynamics | Ref: Ch 40, Section: Shock – Hemodynamic Parameters, Page 285

**Q35. Answer: C — Decreased SVR**

In distributive shock, poor vascular tone (decreased SVR) is the hallmark. Decreased SVR decreases the afterload, increasing the cardiac output.

Topic: Shock Hemodynamics | Ref: Ch 40, Section: Shock – Hemodynamic Parameters, Page 285

**Q36. Answer: B — 80–100 mm Hg**

Current guidelines favor hypotensive resuscitation, maintaining SBP between 80–100 mm Hg and MAP between 60–65 mm Hg to prevent disruption of clots and increased bleeding.

Topic: Shock Management | Ref: Ch 40, Section: Management of Shock – Early Goals, Page 286

**Q37. Answer: B — Balanced resuscitation**

The newer concept of hypotensive and hemostatic resuscitation is called balanced resuscitation.

Topic: Shock Management | Ref: Ch 40, Section: Management of Shock – Early Goals, Page 286

**Q38. Answer: C — Stroke volume variation**

Stroke volume variation is considered most reliable to assess fluid status and titrate fluid therapy; however, due to technical feasibility and cost, CVP is still most commonly used.

Topic: Shock Management – Fluids | Ref: Ch 40, Section: Treatment – Fluids, Page 286

**Q39. Answer: B — Ringer's lactate**

Ringer is the most preferred crystalloid as it is more physiological, except for hypochloremic alkalosis and brain injury where normal saline is preferred.

Topic: Shock Management – Fluids | Ref: Ch 40, Section: Treatment – Choice of Fluids, Page 287

**Q40. Answer: C — Norepinephrine (noradrenaline)**

Noradrenaline is the vasopressor of choice for septicemic shock. Phenylephrine is first-line for neurogenic shock and ephedrine is most preferred for spinal shock.

Topic: Shock Management – Drugs | Ref: Ch 40, Section: Treatment – Drugs, Page 287

**Q41. Answer: C — Cardiogenic shock**

Steroids are contraindicated in cardiogenic shock as they can alter the healing process of myocardium. They are effective for septicemic shock refractory to fluids and vasopressors.

Topic: Shock Management – Drugs | Ref: Ch 40, Section: Treatment – Drugs, Page 287

**Q42. Answer: B — Subclavian vein**

Due to large diameter (and hence less chances of thrombophlebitis due to lipid emulsions), subclavian is the vein of choice for parenteral nutrition.

Topic: Nutritional Support | Ref: Ch 40, Section: Nutritional Support – Parenteral Route, Page 288

**Q43. Answer: C — 60–70%**

Out of total energy required, 60–70% should be provided by carbohydrates (dextrose) and 30–40% by fats. Amino acids are given as supplement, never as a substrate for source of energy.

Topic: Nutritional Support | Ref: Ch 40, Section: Substrates for Providing Energy, Page 288

**Q44. Answer: B — ARDS and renal failure**

ARDS and renal failure are the most common causes of death in septicemia.

Topic: Sepsis | Ref: Ch 40, Section: Sepsis, Page 290

**Q45. Answer: C — 25 mm Hg**

Cardiogenic pulmonary edema develops when pulmonary artery occlusion pressure (left atrial pressure) rises to more than 25 mm Hg (or 30 cm H<sub>2</sub>O).

Topic: Pulmonary Edema | Ref: Ch 40, Section: Pulmonary Edema – Hemodynamic Type, Page 290

**Q46. Answer: C — ≤100 mm Hg with PEEP ≥5 cm H<sub>2</sub>O**

Berlin definition classifies ARDS as: Mild ( $\text{PaO}_2/\text{FiO}_2 \leq 300$ ), Moderate ( $\leq 200$ ), and Severe ( $\leq 100$  mm Hg), all with PEEP or CPAP  $\geq 5$  cm H<sub>2</sub>O.

Topic: ARDS | Ref: Ch 40, Section: ARDS – Berlin Definition, Page 291

**Q47. Answer: C — Sepsis**

Sepsis is the most common cause of ARDS.

Topic: ARDS | Ref: Ch 40, Section: Causes of ARDS, Page 291

**Q48. Answer: C — 20%**

Carbon monoxide poisoning is usually defined as carboxyhemoglobin  $>20\%$  in blood. The affinity of CO to bind hemoglobin is 240 times that of oxygen.

Topic: Carbon Monoxide Poisoning | Ref: Ch 40, Section: Carbon Monoxide Poisoning, Page 292

**Q49. Answer: B — 8–16 mM/L**

Normal anion gap = 8–16 mM/L (average 12 mM/L). Anion gap = Sodium concentration – (chloride + bicarbonate concentration). It is mainly constituted by plasma albumin.

Topic: Acid-Base Management | Ref: Ch 40, Section: Anion Gap, Page 294



**Q50. Answer: C — Lactic acidosis due to tissue hypoxia**

Lactic acidosis is the most common cause of metabolic acidosis in anesthesia and is due to tissue hypoxia. It is associated with increased anion gap.

*Topic: Acid-Base Management | Ref: Ch 40, Section: Metabolic Acidosis – Increased Anion Gap, Page 295*

# Cardiopulmonary & Cerebral Resuscitation MCQ – Chapter 41

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 9: Cardiorespiratory Care (Chapter 41)

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. Cardiac arrest in adults is usually of which origin?**

- A. Cardiac
- B. Metabolic
- C. Neurological
- D. Respiratory

**Ans: A**

**Q2. Unwitnessed cardiac arrest in children should be considered due to which rhythm until proved otherwise?**

- A. Pulseless electrical activity
- B. Asystole
- C. Ventricular tachycardia
- D. Ventricular fibrillation

**Ans: B**

**Q3. Every minute delay in initiating CPR decreases prognosis by what percentage?**

- A. 7–10%
- B. 5–7%
- C. 1–3%
- D. 15–20%

**Ans: A**

**Q4. The standard CPR sequence was changed from A-B-C to C-A-B in which year's guidelines?**

- A. 2015
- B. 2020
- C. 2005
- D. 2010

**Ans: D**

**Q5. According to newer studies, what is the more important cause of airway obstruction in unconscious patients?**

- A. Epiglottitis
- B. Laryngospasm
- C. Foreign body
- D. Tongue fall

**Ans: A**

**Q6. In a patient with suspected cervical spine fracture, what is the preferred method of airway management?**

- A. Jaw thrust
- B. Neck extension
- C. Head tilt and chin lift
- D. Cricothyroidotomy

**Ans: A**

|                                                                                                                                                                                                                                                                                                      |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q7. Which device, being usable by lay rescuers, is part of Basic Life Support?</b></p> <ul style="list-style-type: none"> <li>A. Transcutaneous pacemaker</li> <li>B. Synchronized cardioverter</li> <li>C. Manual defibrillator</li> <li>D. Automatic external defibrillator (AED)</li> </ul> | <p><b>Ans: D</b></p> |
| <p><b>Q8. Mouth to mouth rescue breathing is recommended for whom?</b></p> <ul style="list-style-type: none"> <li>A. Only medical doctors</li> <li>B. Only paramedics</li> <li>C. Only trained rescuers</li> <li>D. All lay rescuers</li> </ul>                                                      | <p><b>Ans: C</b></p> |
| <p><b>Q9. What is the most definitive method of airway management during CPR?</b></p> <ul style="list-style-type: none"> <li>A. Laryngeal mask airway</li> <li>B. Guedel airway</li> <li>C. Combitube</li> <li>D. Endotracheal tube</li> </ul>                                                       | <p><b>Ans: D</b></p> |
| <p><b>Q10. What is the Heimlich maneuver used for?</b></p> <ul style="list-style-type: none"> <li>A. Cervical spine stabilization</li> <li>B. Foreign body airway obstruction in adults and older children</li> <li>C. Defibrillation technique</li> <li>D. Cardiac massage in infants</li> </ul>    | <p><b>Ans: B</b></p> |
| <p><b>Q11. For cervical spine fracture patients, which type of neck stabilization is preferred during intubation?</b></p> <ul style="list-style-type: none"> <li>A. Manual stabilization</li> <li>B. Halo vest</li> <li>C. Hard cervical collar</li> <li>D. Soft cervical collar</li> </ul>          | <p><b>Ans: A</b></p> |
| <p><b>Q12. During cardiac massage, the sternum should be depressed by how much?</b></p> <ul style="list-style-type: none"> <li>A. 2 inches (5 cm) but not more than 2.4 inches (6 cm)</li> <li>B. 1 inch (2.5 cm)</li> <li>C. 3 inches (7.5 cm)</li> <li>D. 1.5 inches (4 cm)</li> </ul>             | <p><b>Ans: A</b></p> |
| <p><b>Q13. What is the recommended rate of chest compressions during CPR?</b></p> <ul style="list-style-type: none"> <li>A. 120–140/minute</li> <li>B. 80–100/minute</li> <li>C. 100–120/minute</li> <li>D. 60–80/minute</li> </ul>                                                                  | <p><b>Ans: C</b></p> |

|                                                                                                                                                                                                                                  |                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q14. Effective cardiac output generated by successful cardiac massage is what percentage of normal?</b></p> <p>A. 30%<br/>B. 50%<br/>C. 10%<br/>D. 20%</p>                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q15. What is the compression to ventilation ratio for adults WITHOUT advanced airway, irrespective of the number of rescuers?</b></p> <p>A. 15:2<br/>B. 15:1<br/>C. 30:2<br/>D. 5:1</p>                                    | <p><b>Ans: C</b></p> |
| <p><b>Q16. With advanced airway in place, what is the ventilation rate during CPR?</b></p> <p>A. 8 breaths/minute<br/>B. 10 breaths/minute<br/>C. 6 breaths/minute<br/>D. 12 breaths/minute</p>                                  | <p><b>Ans: B</b></p> |
| <p><b>Q17. What is the minimum ETCO<sub>2</sub> that should be produced by successful cardiac massage?</b></p> <p>A. 20 mm Hg<br/>B. 15 mm Hg<br/>C. 10 mm Hg<br/>D. 30 mm Hg</p>                                                | <p><b>Ans: A</b></p> |
| <p><b>Q18. ETCO<sub>2</sub> becoming normal (40 mm Hg) during CPR is the earliest indicator of what?</b></p> <p>A. Adequate ventilation<br/>B. Myocardial recovery<br/>C. Return of normal circulation<br/>D. Brain death</p>    | <p><b>Ans: C</b></p> |
| <p><b>Q19. Which of the following is NOT a shockable rhythm?</b></p> <p>A. Asystole<br/>B. Ventricular fibrillation<br/>C. Polymorphic ventricular tachycardia (Torsade de Pointes)<br/>D. Pulseless ventricular tachycardia</p> | <p><b>Ans: A</b></p> |
| <p><b>Q20. What is the recommended energy for monophasic defibrillation?</b></p> <p>A. 120 Joules<br/>B. 200 Joules<br/>C. 300 Joules<br/>D. 360 Joules</p>                                                                      | <p><b>Ans: D</b></p> |

|                                                                                                                                                                                                                                                                                                                                                       |                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q21. What is the recommended defibrillation paddle size for adults?</b></p> <ul style="list-style-type: none"> <li>A. 4.5 cm</li> <li>B. 13 cm</li> <li>C. 15 cm</li> <li>D. 8 cm</li> </ul>                                                                                                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q22. After delivering a defibrillation shock, what should be done immediately?</b></p> <ul style="list-style-type: none"> <li>A. Check the rhythm immediately</li> <li>B. Give 5 cycles of 30:2 or 2 minutes of CPR before checking rhythm</li> <li>C. Administer adrenaline immediately</li> <li>D. Wait for 5 minutes and reassess</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q23. In monitored in-hospital patients, defibrillation should ideally be done within how many minutes?</b></p> <ul style="list-style-type: none"> <li>A. 3 minutes</li> <li>B. 5 minutes</li> <li>C. 1 minute</li> <li>D. 10 minutes</li> </ul>                                                                                                 | <p><b>Ans: A</b></p> |
| <p><b>Q24. What is the most common cause of pulseless electrical activity (PEA)?</b></p> <ul style="list-style-type: none"> <li>A. Tension pneumothorax</li> <li>B. Hypovolemia</li> <li>C. Hypoxia</li> <li>D. Hypothermia</li> </ul>                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q25. Which of the following is NOT one of the 5H causes of PEA?</b></p> <ul style="list-style-type: none"> <li>A. Hydrogen ion (acidosis)</li> <li>B. Hypovolemia</li> <li>C. Hypothermia</li> <li>D. Hypernatremia</li> </ul>                                                                                                                  | <p><b>Ans: D</b></p> |
| <p><b>Q26. Which of the following is one of the 5T causes of PEA?</b></p> <ul style="list-style-type: none"> <li>A. Tracheomalacia</li> <li>B. Thyrotoxicosis</li> <li>C. Tamponade (cardiac)</li> <li>D. Tachycardia</li> </ul>                                                                                                                      | <p><b>Ans: C</b></p> |
| <p><b>Q27. Asystole is described as:</b></p> <ul style="list-style-type: none"> <li>A. The terminal event of all arrhythmias with very poor prognosis</li> <li>B. The most common initial rhythm in adults</li> <li>C. The rhythm most responsive to adrenaline</li> <li>D. A shockable rhythm</li> </ul>                                             | <p><b>Ans: A</b></p> |

|                                                                                                                                                                                                                                                                                                               |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q28. Cough CPR is applicable in which situation?</b></p> <ul style="list-style-type: none"> <li>A. Pediatric patient with PEA</li> <li>B. Pregnant patient with cardiac arrest</li> <li>C. Conscious patient having ventricular fibrillation</li> <li>D. Unconscious patient with asystole</li> </ul>   | <p><b>Ans: C</b></p> |
| <p><b>Q29. What is the target core body temperature range in targeted temperature management (TTM) after cardiac arrest?</b></p> <ul style="list-style-type: none"> <li>A. 36–38°C for 48 hours</li> <li>B. 32–34°C for 12 hours</li> <li>C. 32–36°C for 24 hours</li> <li>D. 28–30°C for 12 hours</li> </ul> | <p><b>Ans: C</b></p> |
| <p><b>Q30. What is the target SpO2 range in post-cardiac arrest care?</b></p> <ul style="list-style-type: none"> <li>A. 88–92%</li> <li>B. 90–94%</li> <li>C. 98–100%</li> <li>D. 94–98%</li> </ul>                                                                                                           | <p><b>Ans: D</b></p> |
| <p><b>Q31. What blood sugar range (mg/dL) should be maintained in post-resuscitation care?</b></p> <ul style="list-style-type: none"> <li>A. 180–220 mg/dL</li> <li>B. 144–180 mg/dL</li> <li>C. 80–110 mg/dL</li> <li>D. 110–140 mg/dL</li> </ul>                                                            | <p><b>Ans: B</b></p> |
| <p><b>Q32. Emergency PCI for STEMI patients after cardiac arrest should preferably be performed within what time?</b></p> <ul style="list-style-type: none"> <li>A. 30 minutes</li> <li>B. 90 minutes</li> <li>C. 60 minutes</li> <li>D. 120 minutes</li> </ul>                                               | <p><b>Ans: B</b></p> |
| <p><b>Q33. When should neurologic prognostication be performed after ROSC in patients NOT given TTM?</b></p> <ul style="list-style-type: none"> <li>A. 24 hours</li> <li>B. 48 hours</li> <li>C. 96 hours</li> <li>D. 72 hours</li> </ul>                                                                     | <p><b>Ans: D</b></p> |
| <p><b>Q34. From a CPR point of view, what age group defines an 'infant'?</b></p> <ul style="list-style-type: none"> <li>A. 1 year to 5 years</li> <li>B. 1 year till puberty</li> <li>C. 4 weeks to 1 year</li> <li>D. Birth to 4 weeks</li> </ul>                                                            | <p><b>Ans: C</b></p> |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q35. What is the compression to ventilation ratio for a CHILD with TWO rescuers and NO advanced airway?</b></p> <p>A. 3:1<br/>B. 5:1<br/>C. 30:2<br/>D. 15:2</p>                                                                                                                                                          | <p><b>Ans: D</b></p> |
| <p><b>Q36. What is the pulse check site for infants during CPR?</b></p> <p>A. Brachial/Femoral<br/>B. Radial<br/>C. Dorsalis pedis<br/>D. Carotid</p>                                                                                                                                                                           | <p><b>Ans: A</b></p> |
| <p><b>Q37. What heart rate in a pediatric patient with signs of poor perfusion is an indication to start chest compressions?</b></p> <p>A. &lt;80/min<br/>B. &lt;40/min<br/>C. &lt;100/min<br/>D. &lt;60/min</p>                                                                                                                | <p><b>Ans: D</b></p> |
| <p><b>Q38. What is the first shock energy for pediatric manual defibrillation?</b></p> <p>A. 1 J/kg<br/>B. 10 J/kg<br/>C. 4 J/kg<br/>D. 2 J/kg</p>                                                                                                                                                                              | <p><b>Ans: D</b></p> |
| <p><b>Q39. What is the compression to ventilation ratio for NEWBORN resuscitation?</b></p> <p>A. 15:2<br/>B. 30:2<br/>C. 3:1<br/>D. 5:1</p>                                                                                                                                                                                     | <p><b>Ans: C</b></p> |
| <p><b>Q40. What is the indication for starting chest compressions in a newborn?</b></p> <p>A. HR &lt;60/min despite adequate ventilation with 100% O2 for 30 seconds<br/>B. HR &lt;100/min despite adequate ventilation<br/>C. HR &lt;80/min despite adequate ventilation<br/>D. HR &lt;40/min despite adequate ventilation</p> | <p><b>Ans: A</b></p> |
| <p><b>Q41. During CPR in pregnancy, how should aortocaval compression be relieved?</b></p> <p>A. Trendelenburg position<br/>B. Left lateral tilt<br/>C. Right lateral tilt<br/>D. Manual uterine displacement</p>                                                                                                               | <p><b>Ans: D</b></p> |



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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Q42. Emergency cesarean section during maternal cardiac arrest should be considered if fetus is &gt;25 weeks and can be performed within how many minutes?</b></p> <p>A. 10 minutes<br/>B. 2 minutes<br/>C. 4 minutes<br/>D. 6 minutes</p>                                                                                                 | <p><b>Ans: C</b></p> |
| <p><b>Q43. What is the fluid of choice during CPR?</b></p> <p>A. Dextrose-saline<br/>B. 5% dextrose<br/>C. Normal saline<br/>D. Ringer lactate</p>                                                                                                                                                                                               | <p><b>Ans: C</b></p> |
| <p><b>Q44. What is the dose of adrenaline by endotracheal route compared to intravenous dose?</b></p> <p>A. 1.5 times IV dose<br/>B. 5 times IV dose<br/>C. Same as IV dose<br/>D. 2.5 times IV dose</p>                                                                                                                                         | <p><b>Ans: D</b></p> |
| <p><b>Q45. Which of the following drugs can NOT be administered via endotracheal route?</b></p> <p>A. Lignocaine<br/>B. Adrenaline<br/>C. Amiodarone<br/>D. Naloxone</p>                                                                                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q46. Routine use of sodium bicarbonate without proven acidosis during CPR is harmful because of all EXCEPT:</b></p> <p>A. Is hypertonic and increases plasma osmolarity<br/>B. Causes direct myocardial depression<br/>C. Produces CO<sub>2</sub> which can worsen neuronal injury<br/>D. Can cause paradoxical intracellular acidosis</p> | <p><b>Ans: B</b></p> |
| <p><b>Q47. The only recommended modality for global cerebral ischemia during CPR is:</b></p> <p>A. Barbiturates (Thiopentone)<br/>B. Targeted temperature management<br/>C. Calcium channel blockers<br/>D. Propofol infusion</p>                                                                                                                | <p><b>Ans: B</b></p> |
| <p><b>Q48. In the criteria for brain death, no spontaneous respiration should occur even if pCO<sub>2</sub> increases to:</b></p> <p>A. 50 mm Hg<br/>B. 60 mm Hg<br/>C. 70 mm Hg<br/>D. 40 mm Hg</p>                                                                                                                                             | <p><b>Ans: B</b></p> |

**Q49. What is the IV dose of adrenaline during CPR and how frequently is it repeated?**

- A. 5 mg once only
- B. 1 mg every 3–5 minutes
- C. 0.5 mg every 1–2 minutes
- D. 2 mg every 5–10 minutes

**Ans: B**

**Q50. What is the first dose of amiodarone during CPR for ventricular tachycardia?**

- A. 150 mg IV
- B. 300 mg IV
- C. 450 mg IV
- D. 100 mg IV

**Ans: B**

# Answer Key & Explanations

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## Q1. Answer: A — Cardiac

As per the textbook, cardiac arrest in adults is usually of cardiac origin while in children it is usually of respiratory origin.

Topic: CPR Basics | Ref: Ch 41, Section: CPR Basics, Page 297

## Q2. Answer: B — Asystole

Unwitnessed cardiac arrest in children should be considered due to asystole until proved otherwise, while in adults it should be considered due to ventricular fibrillation.

Topic: CPR Basics | Ref: Ch 41, Section: CPR Basics, Page 297

## Q3. Answer: A — 7–10%

As per AHA chain of survival, every minute delay in CPR decreases prognosis by 7–10%.

Topic: Chain of Survival | Ref: Ch 41, Section: Management of CPR, Page 297

## Q4. Answer: D — 2010

The standard sequence A (Airway) → B (Breathing) → C (Circulation) was changed to C → A → B in the 2010 CPR guidelines.

Topic: CPR Sequence | Ref: Ch 41, Section: Management of CPR, Page 297

## Q5. Answer: A — Epiglottis

As per newer studies, epiglottis rather than tongue fall has been found to be the more important cause of airway obstruction in unconscious patients.

Topic: Airway Management | Ref: Ch 41, Section: Airway Management, Page 298

## Q6. Answer: A — Jaw thrust

In patients with suspected cervical spine fracture, head tilt and chin lift should be avoided and airway should be managed only by jaw thrust. However, if airway is not manageable by jaw thrust alone, head tilt and chin lift can be given as life takes priority.

Topic: Airway Management | Ref: Ch 41, Section: Airway Management, Page 298

## Q7. Answer: D — Automatic external defibrillator (AED)

AED can be used by lay rescuers and therefore is part of basic life support. AEDs automatically detect rhythm and deliver shock if rhythm is shockable.

Topic: Basic Life Support | Ref: Ch 41, Section: Basic Life Support, Page 298

## Q8. Answer: C — Only trained rescuers

Rescue breath by mouth to mouth is reserved only for trained rescuers (medical as well as non-medical). For untrained lay rescuers, CPR means 'hands only' i.e. only cardiac massage.

Topic: Breathing | Ref: Ch 41, Section: Breathing (Ventilation), Page 300

## Q9. Answer: D — Endotracheal tube

Endotracheal intubation is the most definitive and best method for ventilation during CPR.

Topic: Airway Management | Ref: Ch 41, Section: Breathing (Ventilation), Page 300

**Q10. Answer: B — Foreign body airway obstruction in adults and older children**

Heimlich maneuver involves abdominal compression 6–10 times with the patient standing and rescuer behind. It is used for adults and older children with foreign body airway obstruction.

Topic: Airway Management | Ref: Ch 41, Section: Management of Airway Obstruction due to Foreign Body, Page 299

**Q11. Answer: A — Manual stabilization**

As intubation often requires removal of cervical collar, manual stabilization is always preferred over collar stabilization.

Topic: Cervical Spine Management | Ref: Ch 41, Section: Airway and Breathing for Patients with Cervical Spine, Page 300

**Q12. Answer: A — 2 inches (5 cm) but not more than 2.4 inches (6 cm)**

The force generated during massage should be able to depress sternum by 2 inches (5 cm) but not more than 2.4 inches (6 cm) to avoid complications due to excessive pressure.

Topic: Circulation | Ref: Ch 41, Section: Circulation, Page 300

**Q13. Answer: C — 100–120/minute**

The rate of compression should be 100–120 compressions/minute but not more than 120, because at rates >120/min, compression depth decreases significantly.

Topic: Circulation | Ref: Ch 41, Section: Circulation, Page 301

**Q14. Answer: A — 30%**

Effective cardiac output generated by successful massage is only 30% of normal, therefore all efforts should be directed to establish spontaneous circulation.

Topic: Circulation | Ref: Ch 41, Section: Physiological Considerations of Cardiac Massage, Page 301

**Q15. Answer: C — 30:2**

Without advanced airway, the ratio for adults should be 30:2 (30 compressions followed by 2 breaths) irrespective of number of resuscitators.

Topic: Compression to Ventilation Ratio | Ref: Ch 41, Section: Compression to Ventilation Ratio, Page 301

**Q16. Answer: B — 10 breaths/minute**

With advanced airway, breathing is at a rate of 10 breaths/minute (1 breath every 6 seconds) without any synchronization with compressions.

Topic: Compression to Ventilation Ratio | Ref: Ch 41, Section: Compression to Ventilation Ratio, Page 301

**Q17. Answer: A — 20 mm Hg**

Successful cardiac massage should be able to produce ET<sub>CO2</sub> of at least 20 mm Hg. If ET<sub>CO2</sub> is less than 10 mm Hg, cardiac massage should be considered grossly abnormal.

Topic: Monitoring of CPR | Ref: Ch 41, Section: Monitoring of CPR, Page 301

**Q18. Answer: C — Return of normal circulation**

ET<sub>CO2</sub> becoming normal (40 mm Hg) is the earliest indicator of return of normal circulation during CPR.

Topic: Monitoring of CPR | Ref: Ch 41, Section: Monitoring of CPR, Page 301

**Q19. Answer: A — Asystole**

Shockable rhythms include VF, pulseless VT and polymorphic VT (Torsade de Pointes). Non-shockable rhythms include asystole and pulseless electrical activity (PEA).

Topic: Arrhythmia Management | Ref: Ch 41, Section: Management of Arrhythmias Seen During Cardiac Arrest, Page 301

**Q20. Answer: D — 360 Joules**

With monophasic defibrillators, all shocks should be of 360 Joules. With biphasic defibrillators, 150–200 J (exponential waveform) or 120 J (rectilinear waveform).

Topic: Defibrillation | Ref: Ch 41, Section: Defibrillation, Page 302

**Q21. Answer: B — 13 cm**

Paddle sizes: Adults – 13 cm, Children – 8 cm, Infants – 4.5 cm.

Topic: Defibrillation | Ref: Ch 41, Section: Defibrillation, Page 302

**Q22. Answer: B — Give 5 cycles of 30:2 or 2 minutes of CPR before checking rhythm**

Immediately after giving shock, give 5 cycles of 30:2 (without advanced airway) or 2 minutes of CPR (with advanced airway) before checking rhythm, as it takes ~2 minutes for heart to generate cardiac output even if rhythm becomes sinus.

Topic: Defibrillation | Ref: Ch 41, Section: Defibrillation, Page 302

**Q23. Answer: A — 3 minutes**

Ideally, in monitored patients, defibrillation should be done within 3 minutes.

Topic: Defibrillation | Ref: Ch 41, Section: Defibrillation, Page 302

**Q24. Answer: B — Hypovolemia**

Hypovolemia (shock) is the most common cause of PEA as stated in the textbook.

Topic: Nonshockable Rhythms | Ref: Ch 41, Section: Pulseless Electrical Activity (PEA), Page 303

**Q25. Answer: D — Hypernatremia**

The 5H causes of PEA are: Hypovolemia, Hypokalemia/Hyperkalemia, Hypothermia, Hypoxia, and Hydrogen ion (acidosis). Hypernatremia is not included.

Topic: Nonshockable Rhythms | Ref: Ch 41, Section: Causes of PEA, Page 304

**Q26. Answer: C — Tamponade (cardiac)**

The 5T causes of PEA are: Tension Pneumothorax, Tamponade (cardiac), Thrombosis (coronary/pulmonary), Tablets (drug overdoses)/Toxins, and Trauma.

Topic: Nonshockable Rhythms | Ref: Ch 41, Section: Causes of PEA, Page 304

**Q27. Answer: A — The terminal event of all arrhythmias with very poor prognosis**

Asystole is the terminal event of all arrhythmias. The prognosis of asystole is very poor as compared to PEA and ventricular fibrillation.

Topic: Nonshockable Rhythms | Ref: Ch 41, Section: Asystole, Page 304

**Q28. Answer: C — Conscious patient having ventricular fibrillation**

Cough CPR is applicable to a conscious patient having VF. Coughing during VF increases intrathoracic pressure and can maintain cerebral perfusion (consciousness) for 90 seconds.

Topic: Newer Techniques in CPR | Ref: Ch 41, Section: Newer Techniques/Variations in CPR, Page 305

**Q29. Answer: C — 32–36°C for 24 hours**

TTM means maintaining core body temperature between 32–36°C for 24 hours. This was changed from the 2010 guideline of 32–34°C for 12 hours.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

**Q30. Answer: D — 94–98%**

Hyperoxia can produce complications including ARDS, therefore use minimum FiO<sub>2</sub> which maintains target SpO<sub>2</sub> between 94 and 98%.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

**Q31. Answer: B — 144–180 mg/dL**

Moderate blood sugar of 144–180 mg/dL (8–10 mmol) should be maintained in the post-resuscitation period, as both hypoglycemia and hyperglycemia are unacceptable.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

**Q32. Answer: B — 90 minutes**

Emergency (preferably within 90 minutes) coronary angiography/angioplasty is recommended for all patients with STEMI.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

**Q33. Answer: D — 72 hours**

Prognostication is recommended 72 hours after return of spontaneous circulation for those who have not been given TTM.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

**Q34. Answer: C — 4 weeks to 1 year**

From CPR point of view: Neonate = first 4 weeks, Infant = 4 weeks to 1 year, Child = 1 year till puberty, Adult = beyond puberty.

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 305

**Q35. Answer: D — 15:2**

For children without advanced airway, the ratio is 30:2 for one rescuer and 15:2 for two rescuers (while for adults it is 30:2 irrespective of rescuers).

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 306

**Q36. Answer: A — Brachial/Femoral**

Pulse check sites: Infants – Brachial/Femoral, Children – Carotid, Adults – Carotid/Femoral.

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 306

**Q37. Answer: D — <60/min**

Heart rate <60/min with signs of poor perfusion (pallor/cyanosis) is the indication to start chest compressions in the pediatric age group.

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 306

**Q38. Answer: D — 2 J/kg**

Energy selection for shock (manual) in pediatric patients is 2 J/kg for first shock and 4 J/kg subsequently.

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 306

**Q39. Answer: C — 3:1**

Compression ventilation ratio for newborns is 3:1 (90 compressions with 30 breaths) and is synchronized.

Topic: Newborn CPR | Ref: Ch 41, Section: Newborn CPR, Page 306

**Q40. Answer: A — HR <60/min despite adequate ventilation with 100% O2 for 30 seconds**

In newborns, indication for chest compression is HR <60/min in spite of adequate ventilation with 100% oxygen for 30 seconds.

Topic: Newborn CPR | Ref: Ch 41, Section: Newborn CPR, Page 306

**Q41. Answer: D — Manual uterine displacement**

Lateral tilt has been eliminated from new CPR guidelines as it compromises CPR quality. Aortocaval compression should be achieved by uterine displacement.

Topic: CPR in Pregnancy | Ref: Ch 41, Section: CPR in Pregnancy, Page 307

**Q42. Answer: C — 4 minutes**

Emergency LSCS should be considered if fetus >25 weeks and LSCS can be performed within 4 minutes of maternal arrest.

Topic: CPR in Pregnancy | Ref: Ch 41, Section: CPR in Pregnancy, Page 307

**Q43. Answer: C — Normal saline**

Normal saline is the fluid of choice during CPR. Glucose-containing solutions increase cerebral edema, and Ringer lactate should be avoided as calcium present can cause neuronal injury.

Topic: Important Considerations in CPR | Ref: Ch 41, Section: Fluid During CPR, Page 307

**Q44. Answer: D — 2.5 times IV dose**

The dose by endotracheal route is 2.5 times of intravenous dose and the drug should be diluted with 10 mL of saline in adults and 5 mL in children.

Topic: Important Considerations in CPR | Ref: Ch 41, Section: Endotracheal (ET), Page 307

**Q45. Answer: C — Amiodarone**

Drugs that can be delivered by ET route are: Adrenaline, Atropine, Lignocaine (Xylocard), and Naloxone. Amiodarone is not in this list.

Topic: Important Considerations in CPR | Ref: Ch 41, Section: Endotracheal (ET), Page 307

**Q46. Answer: B — Causes direct myocardial depression**

The textbook lists harms of sodium bicarbonate as: CO<sub>2</sub> production worsening neuronal injury, increased intracranial tension, paradoxical intracellular acidosis, and hypertonic effect increasing plasma osmolarity. Direct myocardial depression is not mentioned.

Topic: Sodium Bicarbonate in CPR | Ref: Ch 41, Section: Use of Sodium Bicarbonate, Page 308

**Q47. Answer: B — Targeted temperature management**

The only recommended modality for global cerebral ischemia seen in CPR is targeted temperature management. Barbiturates, Propofol, and calcium channel blockers are useful for focal ischemia but their role in global ischemia is not proven.

Topic: Cerebral Protection | Ref: Ch 41, Section: Cerebral Protection, Page 308

**Q48. Answer: B — 60 mm Hg**

One of the clinical tests for brain death is no spontaneous respiration even if pCO<sub>2</sub> increases to 60 mm Hg.

Topic: Brain Death | Ref: Ch 41, Section: Brain Death, Page 308

**Q49. Answer: B — 1 mg every 3–5 minutes**

The CPR dose of adrenaline is 1 mg IV to be repeated every 3–5 minutes in concentration of 1:10,000.

Topic: Drugs for CPR | Ref: Ch 41, Section: Adrenaline (Epinephrine), Page 308

**Q50. Answer: B — 300 mg IV**

Amiodarone is the drug of choice for ventricular tachycardia. The first dose is 300 mg IV and subsequent doses are 150 mg.

*Topic: Drugs for CPR | Ref: Ch 41, Section: Amiodarone, Page 308*